

Finite-time Navier–Stokes construction: reconstruction and validation in progress

9 September 2026

Abstract

This reader records independent derivations of the released classical three-dimensional construction, retaining its coordinates, signs, constants, domains and displayed remainder terms. The source is the 165-page paper [FINITE TIME BLOWUP FOR NAVIER–STOKES](#), SHA-256 8c8a94ad9ac824c8b605b9827cadf7beaca48bd10b380de3cfc872a2c37afa81. The reader remains under independent validation; imported dependencies and the Section 9 assembly are recorded in their ledgers. No prize claim is made.

Contents

1	Scope and provenance	6
2	Validation ledger	7
3	Correction and continuation in this edition	7
4	Exact leading-field maps and radial primitives	7
4.1	1. Coordinates and their exact Jacobian	7
4.2	2. Cartesian axis extension and incompressibility	8
4.3	3. Exact transport and stress integration	8
4.4	4. The five moments and their exact matching map	10
4.5	5. Cone equivalence and finite-dimensional solve	10
4.6	6. Direct heat-exterior verification	11
5	Analytic axis construction and endpoint inequalities	12
5.1	1. Axis data and the complementary parameter regions	12
5.2	2. The coefficient space, with all derivative losses accounted for	13
5.3	3. Exact rescaling and contraction	13
5.4	4. Positivity and endpoint alternative with the original constants	15
6	Continuation, exact moment repair, and assembly of the annular profile	16
6.1	1. The reference continuation and its estimates	16
6.2	2. Activation of stress, with division by the flat factor proved	17
6.3	3. Continuation to 110 and preservation of parameter analyticity	18
6.4	4. Transition length fixed before amplitude	18
6.5	5. The five matching equations, including all scaling factors	19
6.6	6. The finite-frequency assembly in Section 4	20
6.7	7. Endpoint, weight and total-moment conclusions	21
7	The physical coordinate map and the exact coefficient equations	22
8	The analytic radial recursion on an interval independent of the order	23
9	Power moment maps and exact corrections	25
9.1	An invertible moment map and its quadratic inverse	25
9.2	The five positive-order moments, with all row signs	25

10	Weighted residual integrals and compactly supported stresses	26
11	Finite residuals, exact divergence after summation, and uniformity	27
12	The complete outer schedule and its exact moment closure	29
12.1	Definitions of all stages and the adjustable coefficients	29
12.2	Pulse corrections, angular correction, and amplitude root	30
13	Pressure datum and the cone margins on every outer stage	31
13.1	Exact pressure and its parameter analyticity	31
13.2	The exact quantities used for the cone test	32
13.3	Reference and axial-decrease stages	32
13.4	Intermediate power law and the complete pulse	33
13.5	Interpolation and exterior transitions	34
14	The heat factor, its exact equation, and the compensated replacement	34
14.1	A positive solution of the radial swirl heat equation	34
14.2	The three heat discrepancies and their exact cancellation	35
15	Exterior stress, exact flat factors, and the order-one edge estimate	36
15.1	Convergent moment subtraction and backward stress formulas	36
15.2	An exact flat integral map	36
15.3	Positive angular stress and the limiting direction	37
15.4	The remaining positive-order edge term	38
16	Audit conclusion and exact dependency boundary	38
17	Auxiliary geometry and coefficient algebra: independent derivation	38
17.1	Coordinate maps and exact derivative intertwining	38
17.2	Fourier inversion and descent	39
17.3	Separation of all interacting labels	40
17.4	Common torus, lifts, integration paths, and label counts	40
17.5	Weighted coefficient classes and complete product rules	41
18	Phase, constrained evolution, and every fixed derivative	42
18.1	Cone coordinates and the phase gradient	42
18.2	Exact pressure elimination and plane isomorphism	43
18.3	Envelope and uniform forward inverse	45
18.4	The real homogeneous pulse and energy transfer	46
19	Covariance, signed stress increments, exact curls, and cutoff residuals	48
19.1	Two columns and their positive coefficients	48
19.2	Signed differential inverse and assembly across bands	50
19.3	Curl identity with all cylindrical terms	51
19.4	Exact cutoff residual and terminal flatness at each fixed stage	52
19.5	Actual covariance increment with every remainder retained	53
20	Admissible shear loop, radial modulation, and moment restoration	53
20.1	Scalar cone and its four gaps	53
20.2	A smooth loop whose average is exactly the original shear	54
20.3	Exact insertion into the unchanged original radial coordinate	55
20.4	All five moments and the dependence of the integrated stress	56
20.5	Nonlinear moment correction with an exact endpoint map	57
20.6	Edge direction and the common weight	58

21 Independent audit of compactly supported mean corrections	58
21.1 Original coordinates, operators, and spaces	59
21.2 Exact radial range, inverse, obstruction, and projection	60
21.3 Primitive estimates and the full flatness argument	61
21.4 Mean momentum equations derived in the cylindrical frame	62
21.5 Pressure, solenoidal axial realization, and dimensions	62
21.6 Integrated residuals and both covariance targets	63
21.7 Fast-time inverse and chart agreement	64
21.8 Explicit five-dimensional moment correction	65
21.9 Every nonlinear change and its exponent	66
21.10 Recalculation, endpoint scope, and audit disposition	67
22 Residual improvement and the local field (independent reconstruction)	68
22.1 Curl identities and the linear residual	68
22.2 Nonlinear products and finite-stage source decomposition	69
22.3 Finite correction state and initialization	70
22.4 One correction cycle	70
22.5 Common domain and Cartesian derivative budgets	73
22.6 Realization by a diagonal single-stage sum	73
23 The terminal force, the comparison argument, and the two official domains	75
23.1 The local representatives and the localization map	76
23.2 All terminal derivatives, including the exterior pressure	77
23.3 A concrete smooth extension of the prescribed force	78
23.4 The energy estimate retains the dissipation	78
23.5 Comparison without a pressure condition at infinity	79
23.6 The actual path of velocity growth	80
23.7 Exact viscosity and parabolic maps	81
23.8 The unit-periodic construction, with periodic pressure	81
24 The exact constructed objects and the order of the argument	82
25 The outer pressure is the actual axis datum	83
26 One finite choice of radius, amplitude, and joining widths	84
27 Every row of both five-moment repairs and the exterior map	85
28 Applying the outer endpoint calculation to the actual join	86
29 A single finite loop insertion and restoration	87
30 Closed leading-profile conclusion	88
31 Exact leading-field maps and radial primitives	89
31.1 1. Coordinates and their exact Jacobian	89
31.2 2. Cartesian axis extension and incompressibility	90
31.3 3. Exact transport and stress integration	91
31.4 4. The five moments and their exact matching map	92
31.5 5. Cone equivalence and finite-dimensional solve	92
31.6 6. Direct heat-exterior verification	93
32 Power moment maps and exact corrections	94
32.1 An invertible moment map and its quadratic inverse	94
33 The complete outer schedule and its exact moment closure	94
33.1 Definitions of all stages and the adjustable coefficients	94
33.2 Pulse corrections, angular correction, and amplitude root	95

34 Pressure datum and the cone margins on every outer stage	97
34.1 Exact pressure and its parameter analyticity	97
34.2 The exact quantities used for the cone test	97
34.3 Reference and axial-decrease stages	98
34.4 Intermediate power law and the complete pulse	98
34.5 Interpolation and exterior transitions	99
35 The heat factor, its exact equation, and the compensated replacement	100
35.1 A positive solution of the radial swirl heat equation	100
35.2 The three heat discrepancies and their exact cancellation	101
36 Analytic axis construction and endpoint inequalities	101
36.1 1. Axis data and the complementary parameter regions	101
36.2 2. The coefficient space, with all derivative losses accounted for	102
36.3 3. Exact rescaling and contraction	103
36.4 4. Positivity and endpoint alternative with the original constants	104
37 Continuation, exact moment repair, and assembly of the annular profile	105
37.1 1. The reference continuation and its estimates	105
37.2 2. Activation of stress, with division by the flat factor proved	107
37.3 3. Continuation to 110 and preservation of parameter analyticity	108
37.4 4. Transition length fixed before amplitude	108
37.5 5. The five matching equations, including all scaling factors	109
38 Exterior stress, exact flat factors, and the order-one edge estimate	110
38.1 Convergent moment subtraction and backward stress formulas	110
38.2 An exact flat integral map	111
38.3 Positive angular stress and the limiting direction	111
39 Admissible shear loop, radial modulation, and moment restoration	112
39.1 Scalar cone and its four gaps	112
39.2 A smooth loop whose average is exactly the original shear	113
39.3 Exact signed square root at zero variance	114
39.4 Exact insertion into the unchanged original radial coordinate	115
39.5 All five moments and the dependence of the integrated stress	115
39.6 Nonlinear moment correction with an exact endpoint map	116
39.7 Edge direction and the common weight	117
40 Exact coordinate, evaluation, and covering maps	118
40.1 The global q map and its boundary	118
40.2 Chart-to-profile maps with fixed Q	119
40.3 Physical evaluation as a differential algebra map	120
40.4 Integer coverings, differential intertwining, and Haar measure	120
40.5 The original scaled operators and their threshold	121
40.6 Common torus, local band range, and exact pulse path	122
40.7 Separation for all products used in covariance assembly	123
40.8 Averaging, evaluation, and the preserved covariance target	124
41 Phase, constrained evolution, and every fixed derivative	124
41.1 Cone coordinates and the phase gradient	124
41.2 Exact pressure elimination and plane isomorphism	126
41.3 Envelope and uniform forward inverse	127
41.4 The real homogeneous pulse and energy transfer	128
42 The complete radial obstruction and the five-equation correction	130
42.1 Coordinates, domains, and fixed parameters	131
42.2 The full radial exact sequence and its specified splitting	132
42.3 Weighted primitive bounds with moving shell edges	133
42.4 Every derivative of the projector is flat after Haar cancellation	133
42.5 Pressure, temporal inversion, and the actual divergence-free field	134

42.6	The original five rows and every profile-to-chart factor	135
42.7	Why the Borel base retains the exact reserved power law	137
42.8	Exact nonlinear recomputation after the five rows	139
42.9	Use in the initial state and the fixed-domain conclusion	139
42.10	Signed covariance and original positive-amplitude audit	140
42.10.1	1. Original objects and input provenance	140
42.10.2	2. Exact column integral and its constants	141
42.10.3	3. Column directions and the exact determinant	142
42.10.4	4. A uniform threshold that proves positivity of the original amplitudes	143
42.10.5	5. Derivatives and smooth zero extension with the original shell weight	144
42.10.6	6. Exact signed differential inverse, including its full quadratic remainder	145
42.10.7	7. Physical assembly, chart maps, and the missing domain qualification	146
42.10.8	8. Actual divergence-free covariance increment and every exponent	147
42.10.9	9. Audit disposition	148
43	The actual initial finite state	148
43.1	The base expansion supplies the precise chart input	148
43.2	The complete primary curl and its residual	149
43.3	Initial full mean equations and the improved Haar average	150
43.4	Temporal cancellation with the entire reconstructed field	150
43.5	The five exact moment rows and the resulting budgets	151
44	One fixed domain and the actual residual source class	152
45	Shrinking-cutoff summation and local realization: exact source audit	154
45.1	Result and coverage	154
45.2	1. Original objects, types and derivative convention	155
45.3	2. Exact similarity map proves the required mixed derivatives of q	156
45.4	3. Cutoff derivatives, all curl terms and an explicit fixed loss	156
45.5	4. Diagonal scales and exact tail estimate	157
45.6	5. Divergence and the coordinate-level representative map	158
45.7	6. Full differential-polynomial residual comparison	158
45.8	7. Check of the physical derivative losses in (9.17)	160
45.9	8. Exact Navier–Stokes specialization and truncation choice	161
45.10	9. The actual representation (9.21), initialization and remaining scope	162
45.11	10. Bounded formal-source correspondence and the initial cutoff	162
45.11.1	10.1 Exact half-gain accounting	163
45.11.2	10.2 Exact map between cut stage zero and an uncut base	164
45.12	Transcription repair record	164
46	The exact quotient of smooth terminal force extensions	164
47	Compact radial stress reconstruction with every moment, derivative and chart factor	167
47.1	1. Original parameters and supported spaces	167
47.2	2. Exact moment complement and radial inverse	168
47.3	3. Every moving- q profile derivative	168
47.4	4. Full stress derivatives and finite operator estimates	169
47.5	5. Full symmetric tensor and all three force components	170
47.6	6. Exact physical-to-chart correspondence with Q fixed	171
47.7	7. The original flat edge weight is preserved by the radial integral	172
47.8	8. Exact role in the source correction and remaining dependencies	174
48	Four-step correction-cycle gain and invariant audit	174
48.1	Finding and exact scope	174
48.2	1. Original state, parameters and class interpretation	175
48.3	2. Full residual and exact recomputation rule	176
48.4	3. Step 1: particular supported harmonics	176
48.5	4. Step 2: signed stress-amplitude change and its exact covariance	177
48.6	5. Step 3: nonconstant auxiliary means, with the full axial remainder	178
48.7	6. Step 4: two exact moment constraints and three nonlinear residual defects	180

48.8	7. Exact invariant checks and the completed exponent ledger	181
48.9	8. Flat remainders and the use in the physical estimate (9.18)	182
48.10	9. Verification boundary	183
48.11	10. Static formal-source correspondence	183
48.11.1	Frozen source identities	185
49	Constructed leading profiles, complete background and pulse inputs	186
49.1	1. Exact input and its provenance	186
49.2	2. The common-radius analytic solution map	186
49.3	3. Exact extension, five rows, and common support	188
49.4	4. Exact integrated residual map and the edge term	188
49.5	5. Exact Borel map and all fixed derivative bounds	189
49.6	6. The exact chart morphism into the pulse hypotheses	190
49.7	7. Exact preservation of the reserved interval	191
49.8	8. Bounded disposition	191
50	Full phase, pulse, inverse and covariance audit	191
50.1	1. Frozen labels and rounded carrier	191
50.2	2. Exact gradient and transport defect	192
50.3	3. Pressure and the exact plane morphism	193
50.4	4. Forward inverse and every fixed derivative	193
50.5	5. Support, smoothness, and descent	194
50.6	6. Homogeneous lower bound and exact energy production	195
50.7	7. Exact transfer of the lower bound to covariance mass	196
51	Actual shear and the full force-pressure correspondence	196
51.1	1. Original chart and the two contributions to the shear difference	196
51.2	2. Complete first-jet coordinate map and both inverses	197
51.3	3. The same actual phase in the two background operators	198
51.4	4. Exact force–pressure bijection on the full constrained solution data	198
51.5	5. Exact real energy production and an explicit finite positivity threshold	199
51.6	6. Quantitative production for the actual homogeneous pulse	199
51.7	7. The derivative bound for the radial transport product	200
52	Complete downstream closure of the constructed finite states	200
52.1	1. The exact physical mean-stress domain is attained	200
52.2	2. Covariance inversion and the actual residual gain	202
52.3	3. The finite stage residual is the physical residual used by summation	203
52.4	4. The exact representative sum and terminal residual	204
52.5	5. Representatives, endpoint jets, localization, and the classical equation	205
52.6	6. Audited conclusion and remaining integration record	206
53	Complete ambient projection and both directions of the coordinate equation	207
54	Combined analytic interface disposition	207

1 Scope and provenance

The target is $\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f$, $\nabla \cdot u = 0$ on $\mathbb{R}^3$ and, after the proved periodization map, $\mathbb{T}^3$. The source construction uses zero initial velocity, compact support, a smooth compactly supported force, finite energy, and velocity divergence at the terminal time. Every map is written on the full triple (u, p, f) .

The included bodies are the independently written axis/profile, base/heat, oscillation/curl, mean-correction, finite-stage correction-cycle, and terminal localization/comparison derivations. They are assembled with their exact definitions and dependencies. The corrected finite-stage calculation retains the full phase vector and the longitudinal part of the complete curl amplitude. A newer 166-page source revision has also been obtained. The component derivations remain explicitly pinned to the 165-page source hash above; later source audits identify their own capture. A completed revision audit records exact page/glyph correspondences and citation changes, with its finite comparison scope stated in the accompanying JSON.

2 Validation ledger

The static audit of the pinned formal repository found 580 local Navier–Stokes modules and matched 26 of 26 official statement declarations. That audit did not run Lean and does not certify a kernel proof. Actual import compilation is a separate ongoing check. The accompanying JSON contains the component proof source and check programs. The source-hash comparisons, exact algebra checks, numerical evaluations, and structural text checks are identified separately in their receipts; none alone certifies an analytic estimate or the full theorem. The terminal chapter was read against source pages 117–126.

3 Correction and continuation in this edition

The earlier 162-page reader remains available as an immutable publication at [doi:10.5281/zenodo.22667379](https://doi.org/10.5281/zenodo.22667379). This edition corrects the energy interpretation in the pulse calculation. With the original actual shear $g = (RF_R, G_R)$, frozen leading shear $g_0 = |g_0|N$, and flux $T = t_r(t_\theta, t_z) = T_N N + T_K K$, the exact identity is

$$-g \cdot T = -|g_0|T_N - (g - g_0) \cdot T.$$

The remainder is retained throughout the production estimate. The displayed pulse energy equation already uses the actual shear and remains unchanged; viscous damping remains part of that equation. The complete proof below gives the positive production bound with the actual pulse and original constants. The transport-defect calculation also now explicitly retains the polynomial S_* factors in coefficient derivatives of $bn_{\Phi, r}$. The constrained plane-coordinate inverse is proved in both directions on its stated domain.

The additional chapters give the complete radial stress reconstruction, correction-cycle estimates, and cross-component analytic derivations. These chapters preserve the original fields and fixed positive viscosity. They address the forced whole-space and periodic formulations. The reconstruction JSON records this work and its reproducible checks; it is not an original discovery conversation. Independent formal verification remains in progress. No prize claim or claim to discovery priority is made.

The continuation chapters contain the complete additional derivations named in their headings. They retain the separately identified 165-page and 166-page source captures. Their inclusion records those calculations; the independent whole-construction and formal checks remain in progress.

4 Exact leading-field maps and radial primitives

This is an independent derivation of the mathematical identities used on printed pp. 24-31 and 41-45 of *Finite Time Blowup for Navier-Stokes*, source SHA-256 8c8a94ad9ac824c8b605b9827caddf7beaca48bd10b380de3cfc872a2c37afa81. The symbols below retain their source values and orientations. Equations are mathematical identities; the accompanying arguments are independently written. The construction of the complete outer profile and the wave cone is recorded separately in the dependency ledger, not assumed to have been validated by this file.

The completed source construction chooses and fixes $0 < h < 1/100$. That restriction is retained throughout its use here. The coordinate lemma itself has the larger validity range $0 < h < 1/2$, which is stated below solely to record its precise domain; it does not enlarge the construction’s chosen parameter range.

4.1 1. Coordinates and their exact Jacobian

Take right-handed cylindrical coordinates

$$x = (r \cos \theta, r \sin \theta, z), \quad e_\theta = (-\sin \theta, \cos \theta, 0), \quad \theta \in \mathbb{R}/(2\pi\mathbb{Z}), \quad s = r^2/2.$$

Put

$$\begin{aligned} \tau &= 1 - t > 0, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad 0 < h < \frac{1}{2}, \\ z &= q^D \eta, \quad \tau = q(1 - \eta^2), \quad X = s/q, \\ d &= 1 - \eta^2, \quad L = 1 - 2h\eta^2. \end{aligned}$$

For fixed $\tau > 0, z \in \mathbb{R}$, eliminating η gives $g(q) = q - z^2 q^{2h} = \tau$. At $q_* = |z|^{1/D}$, $g(q_*) = 0$; for $z = 0$ take $q_* = 0$. For $q > q_*$, $z^2 q^{2h-1} < 1$, and

$$g'(q) = 1 - 2hz^2 q^{2h-1} \geq 1 - 2h > 0.$$

Moreover $g(q) \rightarrow \infty$. Thus exactly one $q > q_*$ exists, $|\eta| < 1$, and the implicit function theorem makes q, η smooth for $t < 1$. The values $\eta = \pm 1$ used for profiles are one-sided parameter endpoints; they are not additional points with $\tau > 0$.

Differentiation at fixed physical coordinates gives

$$q_t = -L^{-1}, \quad \eta_t = \frac{D\eta}{qL}, \quad X_t = \frac{X}{qL}, \quad q_z = \frac{2\eta q^{1-D}}{L}, \quad \eta_z = \frac{d}{q^D L}, \quad X_z = -\frac{2\eta X}{q^D L}.$$

The formula for η_z uses the exact equality $L - 2D\eta^2 = d$. Consequently, for every real b and smooth profile f ,

$$\begin{aligned} \partial_t(q^b f) &= q^{b-1} L^{-1}(-bf + D\eta f_\eta + X f_X), \\ \partial_z(q^b f) &= q^{b-D} L^{-1}(2b\eta f + df_\eta - 2\eta X f_X). \end{aligned}$$

No constant or power of q is discarded in these maps.

4.2 2. Cartesian axis extension and incompressibility

Let $C > 1$, $F = \phi/C$, $E = \sqrt{2X}F$, and prescribe

$$u_\theta = q^{-A}E, \quad u_z = q^{-A}U, \quad ru_r = V_0 = Xv_0, \quad p = q^{-2A}\Pi.$$

For F, U, v_0, Π smooth on $[0, R] \times [-1, 1]$, the Cartesian components are exactly

$$u_1 = \frac{v_0}{2q}x_1 - q^{-A-1/2}Fx_2, \quad u_2 = \frac{v_0}{2q}x_2 + q^{-A-1/2}Fx_1, \quad u_3 = q^{-A}U.$$

Since $X = (x_1^2 + x_2^2)/(2q)$, these formulas prove smoothness across $r = 0$ for each $t < 1$. Smoothness is imposed on F , not on E as a function of X . This explicitly proves the map from scalar profile regularity to Cartesian velocity regularity.

Incompressibility is $\partial_s(ru_r) + \partial_z u_z = 0$. Because $A + D = 1$, its exact scalar equation is

$$(V_0)_X = L^{-1}(2A\eta U - dU_\eta + 2\eta X U_X).$$

Integrate with $V_0(0, \eta) = 0$. Writing $M(X, \eta) = \int_0^X U(x, \eta) dx$, the result is

$$V_0 = \frac{2\eta XU - 2D\eta M - dM_\eta}{L}.$$

The extension of M/X at zero is smooth because $M/X = \int_0^1 U(vX, \eta) dv$; differentiating this integral proves every mixed derivative assertion. Thus the formula also proves smoothness of V_0/X .

The radial derivative of pressure and centrifugal term have the common factor $q^{-2A-1/2}$, with respective coefficients $\sqrt{2X}\Pi_X$ and $E^2/\sqrt{2X}$. Their exact equality is

$$\Pi_X = \frac{E^2}{2X} = F^2.$$

The other radial terms are still present in the full Navier-Stokes residual. Radial time differentiation, both material-transport contributions and radial viscosity have scale $q^{-3/2} = q^{2h}q^{-2A-1/2}$. Axial viscosity of the radial velocity has scale $q^{-1/2-2D} = q^{-3/2+2h} = q^{4h}q^{-2A-1/2}$. Axial viscosity has scale $q^{-A-2D} = q^{2h}q^{-A-1}$ in the tangential equations. These equalities identify every remaining type of term without setting any of them to zero.

4.3 3. Exact transport and stress integration

All residual identities here have viscosity one, as defined on source p. 6. Define

$$H = \sqrt{2X}E = 2XF, \quad l = X\partial_X \log H, \quad W = 1 - \frac{2D\eta M + dM_\eta}{X}, \quad H_c = D\eta + dU.$$

Applying the preceding derivative maps and $u_r = V_0/r$ gives, for arbitrary b, f ,

$$(\partial_t + u_r \partial_r + u_z \partial_z)(q^b f) = \frac{q^{b-1}}{L} \{WX f_X + H_c f_\eta - b(1 - 2\eta U)f\}.$$

Indeed the coefficient of Xf_X before substituting V_0 is $1 + LV_0/X - 2\eta U$, exactly W . Angular momentum is $ru_\theta = q^{-h}H$. Hence the angular material derivative equals $-q^{-h-1}HS_q/L$, where

$$S_q = -Wl - h(1 - 2\eta U) - H_c(\log E)_\eta.$$

For the axial velocity, add $\partial_z p$. Since $-2A - D = -A - 1$, the result is $-q^{-A-1}S_n/L$, with

$$S_n = -WXU_X - A(1 - 2\eta U)U - H_cU_\eta - d\Pi_\eta + 4A\eta\Pi + 2\eta X\Pi_X.$$

Let $D_X = X\partial_X$. Solve

$$D_X Q_s + (1 + l)Q_s = S_q, \quad D_X N_s + N_s = S_n.$$

Their integrating factors are XH and X . For $\phi > 0$ smooth at the axis, $H = 2X\phi/C$, and any nonzero integration constant produces an X^{-2} singularity in Q_s or an X^{-1} singularity in N_s . Therefore the unique regular choices are

$$Q_s = \frac{\int_0^X HS_q dx}{XH}, \quad N_s = \frac{1}{X} \int_0^X S_n dx.$$

In particular,

$$Q_s = \frac{1}{F(X, \eta)} \int_0^1 vF(vX, \eta)S_q(vX, \eta) dv, \quad N_s = \int_0^1 S_n(vX, \eta) dv,$$

which prove the scalar smooth extensions and give $Q_s(0) = S_q(0)/2$, $N_s(0) = S_n(0)$.

Define

$$p_s = \left(\frac{XQ_s}{L}, \frac{XN_s}{LE} \right), \quad a = 1 - 2D_X \log E = 2 - 2l, \\ b_s = \frac{2D_X U}{E}, \quad s_{\text{shear}} = (a, -b_s), \quad T_0 = F(p_s - s_{\text{shear}}).$$

Here s_{shear} is only a typographical disambiguation from the unchanged coordinate $s = r^2/2$; it is exactly the manuscript's vector s . The stress components remain ordered $(r\theta, rz)$.

For any profile G , at fixed t, z ,

$$(\partial_r + k/r)(q^{-A-1/2}G) = q^{-A-1} \left(\sqrt{2X} G_X + \frac{k}{\sqrt{2X}} G \right).$$

Substituting $G = FXQ_s/L = HQ_s/(2L)$, $k = 2$, produces $q^{-A-1}ES_q/L$. Substituting $G = FXN_s/(LE) = \sqrt{X/2}N_s/L$, $k = 1$, produces $q^{-A-1}S_n/L$. The remaining stress terms are exactly

$$\partial_r u_\theta - u_\theta/r = -q^{-A-1/2}Fa, \quad \partial_r u_z = q^{-A-1/2}Fb_s.$$

The differential identities

$$(\partial_r + 2/r)(\partial_r u_\theta - u_\theta/r) = \partial_{rr} u_\theta + r^{-1}\partial_r u_\theta - r^{-2}u_\theta,$$

$$(\partial_r + 1/r)\partial_r u_z = \partial_{rr} u_z + r^{-1}\partial_r u_z$$

now prove, including both signs,

$$R_\theta^{(0)} = -(\partial_r + 2/r)(q^{-A-1/2}T_{0,\theta}), \quad R_z^{(0)} = -(\partial_r + 1/r)(q^{-A-1/2}T_{0,z}),$$

where $R_j^{(0)} = \mathcal{R}(u, p) \cdot e_j + \partial_{zz} u_j$. Thus only axial viscosity has been removed from these exact identities.

Zero stress is equivalent to $Q_s = -2L\phi_X/\phi$ and $N_s = -2LU_X$. Applying the two radial operators and canceling the two opposite quadratic logarithmic-derivative terms gives exactly

$$-2L \frac{X\phi_{XX} + 2\phi_X}{\phi} = S_q, \quad -2L(XU_{XX} + U_X) = S_n.$$

Conversely, these equations and regularity imply the same Q_s, N_s , by uniqueness of the regular radial primitives, hence zero stress. The first displayed fraction was checked directly against PDF p. 27.

4.4 4. The five moments and their exact matching map

Retain all five quantities

$$\begin{aligned} M &= \int_0^X U \, dx, \quad I = \int_0^X H \, dx, \quad J = \int_0^X UH \, dx, \\ S &= \int_0^X (U^2 - E^2/2) \, dx, \quad C_p = \int_0^X \frac{E^2}{2x} \, dx, \quad \Pi = \Pi_{\text{ax}} + C_p. \end{aligned}$$

Since $(XW)_X = 1 - 2D\eta U - dU_\eta$, integration by parts in $-WXH_X$ gives $-XWH$ and the remaining integrand $(1-h)H - D\eta H_\eta - d(UH)_\eta + 2(h-D)\eta UH$. Hence

$$Q_s = -W + \frac{(1-h)I - D\eta I_\eta - dJ_\eta + 2(h-D)\eta J}{XH}.$$

For the axial primitive, integrate $-WXU_X$ in the same way. Its velocity terms become $D(U - \eta U_\eta) + 4h\eta U^2 - d(U^2)_\eta$. Pressure contributes

$$X(4A\eta\Pi - d\Pi_\eta) - 2h\eta \int_0^X E^2 \, dx + \frac{d}{2}\partial_\eta \int_0^X E^2 \, dx,$$

because $\int_0^X \Pi \, dx = X\Pi - \frac{1}{2} \int_0^X E^2 \, dx$. Thus

$$N_s = -WU + \frac{D(M - \eta M_\eta) + 4h\eta S - dS_\eta}{X} + 4A\eta\Pi - d\Pi_\eta.$$

The independent exact check differentiates both right-hand primitives, retains $\Pi_X = H^2/(4X^2)$, and obtains precisely HS_q and S_n .

If two pairs (U_i, E_i) agree for $X \geq X_h > 0$ and their five moment values at X_h agree as functions of η , all five moment differences are identically zero beyond X_h : their radial derivatives are zero and their initial values are zero. All parameter derivatives also agree. With the same Π_{ax} , the explicit formulas then give identical Π, V_0, Q_s, N_s, p_s . Agreement of profiles on an interval gives agreement of radial derivatives, hence of a, b_s, T_0 . This proves the complete morphism from matching data to the exterior field and stress; pressure matching alone would not suffice.

On a compact X -rectangle bounded away from zero and with $E_i \geq e_{\min} > 0$, the displayed formulas are compositions of addition, multiplication and reciprocal on denominators X, L, E, H bounded away from zero. For example $H_2^{-1} - H_1^{-1} = -(H_2 - H_1)/(H_1 H_2)$. The product rule through k parameter derivatives bounds the output difference in (Q_s, N_s, p_s) by the input differences in (U, E, M, I, J, S, C_p) through order $k + 1$. This uses no radial derivative of the difference. The one extra parameter derivative is necessary because W contains M_η and the primitive formulas contain moment derivatives.

Under the exact coordinate substitution $X = \rho x$, use

$$\begin{aligned} \widehat{U}(x, \eta) &= U(\rho x, \eta), \quad \widehat{E}(x, \eta) = E(\rho x, \eta), \quad H = \rho^{1/2} \widehat{H}, \\ (M, I, J, S, C_p) &= (\rho \widehat{M}, \rho^{3/2} \widehat{I}, \rho^{3/2} \widehat{J}, \rho \widehat{S}, \widehat{C}_p), \quad \widehat{Q}_s = Q_s, \quad \widehat{N}_s = N_s, \quad \widehat{p}_s = \rho^{-1} p_s. \end{aligned}$$

Direct substitution proves that precisely the same rational formulas hold on the fixed x -rectangle. This is an invertible change of variables retaining all factors, not a replacement of the original moment data.

4.5 5. Cone equivalence and finite-dimensional solve

For $a > 0$, define

$$t_s = -b_s/a, \quad v_s = a(1 + t_s^2), \quad P_c = p_{s,1} + t_s p_{s,2}, \quad J_c = p_{s,2} - t_s p_{s,1}.$$

The roots in v of $2(P_c - v)^2 - (v - 2)J_c^2$ are

$$v_\pm = P_c + J_c^2/4 \pm |J_c| \sqrt{(P_c - 2)/2 + J_c^2/16}.$$

For $P_c > 2$, evaluate the polynomial at 2 and P_c : it is positive at 2, nonpositive at P_c , and has its vertex at or above P_c . Consequently $2 < v_- \leq P_c$. This includes $J_c = 0$, where the two roots are P_c . Therefore for $v_s > 2$, the admissible inequality $v_s < v_-$ is exactly

$$P_c > v_s, \quad (v_s - 2)J_c^2 < 2(P_c - v_s)^2.$$

In particular $v_s \leq 2$, $P_c > 2$ always implies the strict relaxed inequality $v_s < v_-$.

For $p_{s,2} = wp_{s,1}$, put $c = 1 - b_s w/a$, $j = w + b_s/a$, $v = a + b_s^2/a$. Direct expansion gives

$$(v-2)j^2 - 2c^2 = (1 + b_s^2/a^2)((a-2)w^2 + 2b_s w + b_s^2/a - 2).$$

On the compact set specified in Lemma 4.5, choose actual minima $c_K > 0$, $\gamma_K = \min(2c^2 - (v-2)j^2) > 0$, and $B_K \geq \max\{1, v, cv\}$. For

$$p_{s,1} \geq P_K = \max\{(B_K + 2)/c_K, 8B_K/\gamma_K\},$$

one has $P_c > \max\{2, v\}$ and

$$\frac{2(P_c - v)^2 - (v-2)J_c^2}{p_{s,1}^2} \geq \gamma_K - 4B_K/p_{s,1} \geq \gamma_K/2.$$

These are exactly the source constants, with the endpoint $v = 2$ covered by the preceding root calculation.

Stress coordinates satisfy, by direct scalar products,

$$T_{0,\theta} + t_s T_{0,z} = F(P_c - v_s), \quad T_{0,z} - t_s T_{0,\theta} = FJ_c.$$

Thus a positive margin for the two stress inequalities proves nonvanishing and controls direction independently of magnitude.

For completeness, the moment matrix invertibility used later is proved as follows. Distinct real exponents α_i form a Chebyshev family on $(0, \infty)$. For m exponents, divide a nonzero combination by its smallest power and differentiate. If the original combination had m distinct positive zeros, Rolle's theorem would give $m-1$ zeros of a combination of $m-1$ distinct powers, contradicting the induction starting with one nonzero power. Therefore the determinant $\det[x_j^{\alpha_i}]$ never vanishes for ordered $x_1 < \dots < x_m$, and its sign is constant on that connected set. Multilinearity expresses the bump moment determinant as the integral of this determinant against $\prod_j \beta_j(x_j) \geq 0$. Ordered disjoint nontrivial supports make the integral nonzero. Compact smooth parameter families then have bounded inverse matrices and bounded fixed derivatives by the adjugate formula.

For the exact quadratic moment equation $Bc + Q(c, c) = d$, multiplication by B^{-1} has norm ν_k , and Q has bilinear norm q_k on C^k . On the ball of radius $r_k = 2\nu_k \|d\|_{C^k}$, the map $c \mapsto B^{-1}(d - Q(c, c))$ has image norm at most $r_k/2 + \nu_k q_k r_k^2$ and Lipschitz constant at most $2\nu_k q_k r_k$. The source condition $8\nu_k^2 q_k \|d\|_{C^k} \leq 1$ gives invariance and Lipschitz constant at most $1/2$. Iteration from zero in C^0 and C^k gives the same fixed point. Its pointwise derivative matrix is $B + Q(c, \cdot) + Q(\cdot, c)$; the corresponding B^{-1} -perturbation has norm at most $1/2$, proving invertibility. The finite-dimensional implicit function theorem therefore gives all higher fixed parameter derivatives without imposing infinitely many smallness conditions.

4.6 6. Direct heat-exterior verification

The function named $\mathcal{H}$ here is exactly the source's $H(Z)$ in (4.29), distinguished typographically from angular momentum $H(X, \eta)$:

$$\mathcal{H}(Z) = \frac{1}{\Gamma(1+h)} \int_0^\infty e^{-v} v^h (1+Zv)^{-h} dv, \quad Z \geq 0.$$

For each k , its k -th derivative is the integral with factor $(-1)^k (h)_k v^k (1+Zv)^{-h-k}$. The integrable majorant is $(h)_k e^{-v} v^{h+k}$, proving smooth one-sided derivatives at $Z = 0$ and positivity. Differentiating $e^{-v} v^{h+1} (1+Zv)^{-h-1}$ with respect to v shows

$$Z^2 \mathcal{H}'' + ((2+2h)Z + 1) \mathcal{H}' + h(1+h) \mathcal{H} = 0.$$

In fact the numerator of the ODE integrand after taking the common denominator $(1+Zv)^{h+2}$ is $h(h+1-v-Zv^2)$, exactly h times that derivative numerator; both boundary values vanish. This is a full integration-by-parts proof.

For $K = c_\infty s^{-A} \mathcal{H}(2\tau/s)$, let $Z = 2\tau/s$. Then

$$\partial_t K = -2c_\infty s^{-A-1} \mathcal{H}', \quad \partial_s K = c_\infty s^{-A-1} (-A\mathcal{H} - Z\mathcal{H}'),$$

$$(\partial_{rr} + r^{-1} \partial_r - r^{-2}) K = c_\infty s^{-A-1} \{(2A^2 - \frac{1}{2})\mathcal{H} + (4A+2)Z\mathcal{H}' + 2Z^2 \mathcal{H}''\}.$$

Using $2A^2 - 1/2 = 2h(1+h)$ and the preceding ODE proves the exact radial heat equation. Also

$$A\mathcal{H} + Z\mathcal{H}' = \frac{1}{\Gamma(1+h)} \int_0^\infty e^{-v} v^h \frac{A + Zv/2}{(1+Zv)^{h+1}} dv > 0,$$

so $K_r = rK_s < 0$ for $r > 0$. Finally $q^{-A} c_\infty X^{-A} \mathcal{H}(2d/X) = c_\infty s^{-A} \mathcal{H}(2\tau/s)$ exactly; the exterior has no residual hidden by a coordinate substitution.

5 Analytic axis construction and endpoint inequalities

This independent proof calculation covers source pp. 144-150 (B.1-B.4 through Proposition B.3). It uses the exact pressure function Π_0 prepared in Appendix A. The separate outer-profile audit must establish that preparation; the present file proves the axis operations on that specific datum and records the dependency instead of pretending to have proved Appendix A in this lane.

5.1 1. Axis data and the complementary parameter regions

Keep $A = 1/2 + h$, $D = 1/2 - h$, $d = 1 - \eta^2$, $L = 1 - 2h\eta^2$, and the already fixed $0 < h \leq 10^{-2}$. The imported datum is holomorphic near $[-1, 1]$, real and even on that interval, and has

$$\Pi_0 \leq -\frac{5}{2}P_*^2(1 + \eta^2)^{-2}, \quad \eta\Pi'_0(\eta) > 0 \quad (\eta \neq 0).$$

Choose $0 < j_0 \leq 0.05$ and set exactly

$$U_* = 4\eta + j_0, \quad H_* = D\eta + dU_*, \quad W_* = 1 - dU'_* - 2D\eta U_*,$$

$$Z_* = -A(1 - 2\eta U_*)U_* - H_*U'_* - d\Pi'_0 + 4A\eta\Pi_0.$$

For $-1 < \eta < 1$,

$$\frac{H_*}{d} = \frac{D\eta}{1 - \eta^2} + 4\eta + j_0, \quad \partial_\eta(H_*/d) = D\frac{1 + \eta^2}{(1 - \eta^2)^2} + 4 > 0.$$

Its endpoint limits are $-\infty, +\infty$, so it has exactly one zero η_0 . Its value at zero is $j_0 > 0$, hence $\eta_0 < 0$. The root equation yields

$$|\eta_0| = \frac{j_0}{4 + D/(1 - \eta_0^2)}.$$

First this is at most $j_0/4 \leq 1/80$. Substitution back gives the useful explicit bounds $j_0/5 < |\eta_0| < j_0/4$. Further,

$$-W_* = 3 - 8h\eta^2 + (1 - 2h)j_0\eta \geq 3 - 0.08 - 0.05 = 2.87 > 2.8.$$

At the zero, $H_*U'_* = 0$ and $-d\Pi'_0 \geq 0$. The term $4A\eta_0\Pi_0$ is positive. Using $(1 + \eta_0^2)^{-2} \geq 1/4$, it is at least $(A/2)j_0P_*^2$. The remaining negative term has magnitude bounded by a fixed multiple of j_0 , since $U_* = -D\eta_0/d$ at the zero. Therefore the already large fixed P_* gives $Z_*(\eta_0) > 0$, quantitatively at least a positive constant times $j_0P_*^2$.

Continuity gives an open neighborhood V of η_0 on which Z_* is bounded below by a positive number. Choose $\delta_* > 0$ smaller than half that lower bound. The compact set $K_\delta = \{\eta : |Z_*(\eta)| \leq \delta_*\}$ avoids V , hence $|H_*| \geq m > 0$ there (if this set is empty, the following restriction is vacuous). Choose $\sigma_* > 0$ with $\sigma_*^2 < m^2/99$. Then

$$\chi = \frac{H_*^2}{H_*^2 + \sigma_*^2} > 0.99 \quad \text{on } K_\delta.$$

Thus $\chi \leq 0.99$ implies $|Z_*| > \delta_*$, exactly the later endpoint alternative. Strict margins allow the same choices on a slightly larger compact interval $I \supset [-1, 1]$. There is a bounded simply connected complex neighborhood Ω of I on whose closure L and $H_*^2 + \sigma_*^2$ are nonzero and Π_0 is holomorphic: take a sufficiently thin neighborhood of the compact real interval, avoiding the closed zero sets.

Define

$$\zeta_* = -\frac{LH_*}{H_*^2 + \sigma_*^2}, \quad \xi_0 = \Lambda\zeta_*, \quad \phi_*(\eta) = \exp\left(\Lambda \int_0^\eta \zeta_*(w) dw\right).$$

The primitive is single-valued on Ω , and $\phi_* > 0$ on I . The exact identity needed below is $H_*\zeta_*/L = -\chi$. After choosing Λ , impose

$$C \geq \sup_{\eta \in \Omega} |\phi_*(\eta)|.$$

This supremum is finite after choosing Ω with compact closure inside the holomorphic domain. Then $g = \phi_*/C$ has a complex bound independent of both later large parameters. A real-interval bound alone would not establish the required parameter derivative bounds.

5.2 2. The coefficient space, with all derivative losses accounted for

Set $Y = \Lambda X$. For coefficient sequences $F = \sum_{\alpha \geq 0} F_\alpha(\eta) Y^\alpha$, use precisely

$$a_{\alpha\beta} = \frac{20^{-\alpha} \rho^{-\beta} \beta! \binom{\alpha+\beta}{\beta}}{(\alpha+1)^2 (\beta+1)^2}, \quad \|F\|_\rho = \sup_{\alpha, \beta \geq 0} \sup_{\eta \in I} \frac{|\partial_\eta^\beta F_\alpha(\eta)|}{a_{\alpha\beta}}.$$

Completeness follows coefficient by coefficient: a norm-Cauchy sequence converges uniformly for every derivative, and the fundamental theorem of calculus identifies each derivative limit with the derivative of its predecessor. The supremum bound passes to the limit, so this is a Banach space.

Let $C_{\text{sq}} = 8 \sum_{n \geq 1} n^{-2}$. For $0 \leq i \leq N/2$, $(N+1)^2/(N-i+1)^2 \leq 4$, while the other half has the analogous bound after $i \mapsto N-i$. Therefore

$$\sum_{i=0}^N \frac{(N+1)^2}{(i+1)^2 (N-i+1)^2} \leq C_{\text{sq}}.$$

In a coefficient product and its β -th derivative, the Leibniz factor $\binom{\beta}{k}$ cancels the factorials in the two input weights. The remaining binomial quotient is at most one because

$$\binom{i+k}{k} \binom{\alpha-i+\beta-k}{\beta-k} \leq \binom{\alpha+\beta}{\beta}.$$

It counts subsets selecting specified numbers from the two fixed blocks. Applying the preceding convolution bound once in the radial and once in the parameter index proves $\|FG\|_\rho \leq C_{\text{sq}}^2 \|F\|_\rho \|G\|_\rho$.

Define $\mathcal{I}F(Y) = \int_0^Y F(v) dv$, $\mathcal{A}F = Y^{-1} \mathcal{I}F$, and let $\mathcal{J}_\nu F$, $\nu = 1, 2$, be the regular solution with zero value at $Y = 0$ of $Y G_{YY} + \nu G_Y = F$. Comparing coefficients gives

$$(\mathcal{J}_\nu F)_{\alpha+1} = \frac{F_\alpha}{(\alpha+1)(\alpha+\nu)}, \quad (\mathcal{J}_\nu F)_0 = 0.$$

The exact weight ratios are

$$\begin{aligned} \frac{a_{\alpha\beta}}{a_{\alpha+1,\beta}} &= 20 \frac{(\alpha+2)^2}{(\alpha+1)^2} \frac{\alpha+1}{\alpha+\beta+1} \leq 80, \\ \frac{a_{i,k+1}}{a_{i+1,k}} &= \frac{20}{\rho} (i+1) \frac{(i+2)^2}{(i+1)^2} \frac{(k+1)^2}{(k+2)^2} \leq \frac{80}{\rho} (i+1). \end{aligned}$$

These prove boundedness of multiplication by Y , $\mathcal{I}$, $\mathcal{J}_\nu$, averaging (which divides coefficient α by $\alpha+1$), and $\partial_\eta \mathcal{I}$. The last assertion uses its integration divisor $i+1$ to cancel the last factor in the second ratio.

For $\mathcal{J}_\nu[(\partial_\eta F)(Y \partial_Y G)]$, consider output degree $N = \alpha+1$, split the input radial degrees as i and $\alpha-i$, and allocate the added degree to the differentiated F -factor. Put $I_1 = i+1$, $J_1 = \alpha-i$. The weight comparison and radial inverse give, after dividing by the output weight, at most

$$\frac{80}{\rho} \|F\|_\rho \|G\|_\rho \sum_{i=0}^\alpha \sum_{k=0}^\beta \frac{I_1 J_1}{N(\alpha+\nu)} \frac{(N+1)^2}{(I_1+1)^2 (J_1+1)^2} \frac{(\beta+1)^2}{(k+1)^2 (\beta-k+1)^2}.$$

The first fraction is at most one, including the zero endpoints; the two sums are bounded by C_{sq} . If the logarithmic derivative is absent, replace J_1 by one, still giving a fraction at most one. If F is averaged, its divisor $i+1$ cancels I_1 . If the parameter derivative is absent, use the first weight ratio instead. Inserting further undifferentiated factors adds the same convolution bounds with unchanged degree allocation. This proves every nonlinear operator estimate invoked below, including the case where both differentiated factors are copies of the same unknown.

A holomorphic coefficient $b(\eta)$ bounded on a larger neighborhood belongs as degree zero to this space: Cauchy's estimate gives $|b^{(\beta)}| \leq M \beta! r^{-\beta}$, and choosing $\rho < r$ bounds $(\beta+1)^2 (\rho/r)^\beta$. This applies uniformly to g , since its complex supremum is at most one.

5.3 3. Exact rescaling and contraction

Write the original fields as

$$\phi = \phi_* \Phi, \quad U = U_* + \Lambda^{-1} u, \quad \Pi = \Pi_0 + \Lambda^{-1} p, \quad p = \mathcal{I}(g^2 \Phi^2), \quad \Phi(0) = 1, \quad u(0) = 0.$$

Thus $\Pi_X = g^2\Phi^2 = \phi^2/C^2$ exactly. Set

$$B = -2D\eta Au - d\partial_\eta Au, \quad W = W_* + \Lambda^{-1}B, \quad H_c = H_* + \Lambda^{-1}du.$$

Substitute these into the two stress-free equations in `core_exact.md`. The unchanged equations are

$$2(Y\Phi_{YY} + 2\Phi_Y) = -\chi\Phi + \Lambda^{-1}R_1, \quad 2(Yu_{YY} + u_Y) = -Z_*/L + \Lambda^{-1}R_2,$$

where

$$\begin{aligned} LR_1 &= [W + h(1 - 2\eta U) + du\zeta_*]\Phi + WY\Phi_Y + H_c\Phi_\eta, \\ LR_2 &= [A(1 - 4\eta U_*) + dU_*']u - 2A\eta\Lambda^{-1}u^2 + WYu_Y + H_*u_\eta + d\Lambda^{-1}uu_\eta - 4A\eta p + d\partial_\eta p - 2\eta Yp_Y. \end{aligned}$$

Both identities have been checked as exact rational expressions by `exact_checks.py`, before any estimate. In particular the pressure terms retain all three signs and coefficients. The angular $-\chi\Phi$ comes from $H_*\zeta_*/L = -\chi$; this term is order one and must be inverted, not treated as small.

The only products with both an unintegrated parameter derivative and a logarithmic radial derivative are $(\partial_\eta Au)Y\Phi_Y$ and $(\partial_\eta Au)Yu_Y$. They have the precise structure proved above. The term $\partial_\eta p = \partial_\eta \mathcal{I}(g^2\Phi^2)$ is bounded, and $Yp_Y = Yg^2\Phi^2$. Every other monomial is controlled by multiplication and a radial inverse, with at most one of the two derivative types. Subtracting two monomials by replacing their factors one at a time proves the corresponding local Lipschitz estimates. All constants are finite on a fixed ball, uniformly for $\Lambda \geq 1$ and C satisfying the complex bound.

Let M_χ be the bounded multiplier norm of χ , and set $T = \mathcal{J}_2\chi/2$, multiplication preceding integration. A sequence vanishing below degree b obeys

$$\|\mathcal{J}_2 F\|_\rho \leq \frac{80}{(b+1)(b+2)}\|F\|_\rho.$$

Multiplication by χ preserves that vanishing order and $\mathcal{J}_2$ increases it by one. Therefore

$$\|T^k\| \leq \frac{(40M_\chi)^k}{k!(k+1)!}.$$

The absolutely convergent series $\sum_{k \geq 0} (-T)^k$ is a two-sided inverse of $1 + T$: multiply finite partial sums by $1 + T$ and use $\|T^{k+1}\| \rightarrow 0$. No smallness of χ is required.

The center of the contraction is

$$\Phi_0 = f_0(Y\chi), \quad u_0 = -\frac{YZ_*}{2L}, \quad f_0(z) = \sum_{\alpha \geq 0} \frac{(-z/2)^\alpha}{\alpha!(\alpha+1)!}.$$

Its recurrence verifies $2(zf_0'' + 2f_0') = -f_0$, $f_0(0) = 1$, so $\Phi_0 = (1 + T)^{-1}1$. On the closed radius-one ball about (Φ_0, u_0) , let M bound the pair $((1 + T)^{-1}\mathcal{J}_2R_1/2, \mathcal{J}_1R_2/2)$, and let K be its Lipschitz constant. These are the finite constants supplied by the proved estimates. Choosing $\Lambda \geq \max\{1, 2M, 2K\}$ makes the map

$$(\Phi, u) \mapsto \left(\Phi_0 + \frac{(1 + T)^{-1}\mathcal{J}_2R_1}{2\Lambda}, u_0 + \frac{\mathcal{J}_1R_2}{2\Lambda} \right)$$

invariant and a contraction with factor at most $1/2$. Its unique fixed point has norm distance at most M/Λ . This explicitly supplies the existential large-parameter threshold from finite operator constants.

To recover analytic functions, for $R < 20$ use

$$\sum_{\alpha \geq 0} \binom{\alpha + \beta}{\beta} (R/20)^\alpha = (1 - R/20)^{-\beta-1}.$$

The coefficient norm therefore bounds parameter derivatives on $|Y| \leq R$ by a constant times $\beta![\rho(1 - R/20)]^{-\beta}$. It gives analytic continuation in a common positive parameter radius. Choose $4.1 < R_1 < R_2 < 20$; Cauchy's formula on the larger radial disk and a slightly smaller parameter neighborhood gives, for every fixed r, s ,

$$\sup_{[0, 4.1] \times I} |\partial_Y^r \partial_\eta^s (\Phi - f_0(Y\chi), u + YZ_*/(2L))| \leq C_{r,s}/\Lambda.$$

The same parameter domain serves every fixed radial derivative because those derivatives use radial Cauchy estimates on the fixed larger disk. For original radial derivatives the factor is exactly $\partial_X^r = \Lambda^r \partial_Y^r$, giving $C_{r,s}\Lambda^{r-1}$. Real coefficients and uniqueness imply real profiles on the real rectangle.

5.4 4. Positivity and endpoint alternative with the original constants

For $0 \leq z \leq 4.1$, put $t = z/2 \leq 2.05$. The magnitude ratio of consecutive terms of f_0 is $t/[(n+1)(n+2)]$, so the alternating tail after the cubic has decreasing magnitudes. Consequently

$$f_0(z) \geq 1 - t/2 + t^2/12 - t^3/144 \geq 305719/1152000 > 0.265.$$

The cubic derivative is $-1/2 + t/6 - t^2/48 < 0$: its discriminant is negative and its leading coefficient is negative. Its minimum is thus the displayed exact endpoint value. Grouping the alternating series after the constant term gives $f_0 \leq 1$. The proved $O(\Lambda^{-1})$ bound gives $\Phi \geq c_0 > 0$ when Λ is sufficiently large. In particular division by ϕ in the original equation is justified. Uniform complex derivative bounds then give a smaller common complex domain with no zeros; derivatives of $\log \Phi$ are bounded there.

Differentiate $\chi = H_*^2/(H_*^2 + \sigma_*^2)$:

$$|H_* \chi'| = 2|H_*'| \frac{H_*^2 \sigma_*^2}{(H_*^2 + \sigma_*^2)^2} \leq 2\|H_*'\|_\infty \chi.$$

Since f_0 is positive and has bounded fixed derivatives on the stated compact range,

$$|D_X \log \Phi| + |H_* \partial_\eta \log \Phi| \leq C\chi + C/\Lambda.$$

Here $D_X = Y \partial_Y$ and no division by χ is used. Also

$$-H_c \xi_0 = \Lambda L \chi - du \zeta_*, \quad |du \zeta_*| \leq C_{\sigma_*} \sqrt{\chi},$$

because $|H_*|/(H_*^2 + \sigma_*^2) \leq \sigma_*^{-1} \sqrt{\chi}$. Thus the full source has the lower bound

$$S_q \geq \Lambda L \chi - W_* - 10h - C\chi - C_{\sigma_*} \sqrt{\chi} - C/\Lambda.$$

For any fixed small $\varepsilon > 0$, Young's inequality in the exact form $K\sqrt{\chi} \leq \varepsilon \Lambda L \chi + K^2/(4\varepsilon \Lambda L)$ absorbs the square-root term. Increase Λ so the linear $C\chi$ term uses the remaining part of a total $0.05\Lambda L \chi$ allowance. Since $-W_* > 2.8$, $10h \leq 0.1$, and the remaining constant error can be made at most 0.2, this gives

$$S_q \geq 2.5 + 0.95\Lambda L \chi.$$

The integral formula

$$Q_s(X, \eta) = \frac{\int_0^X x \Phi(\Lambda x, \eta) S_q(x, \eta) dx}{X^2 \Phi(\Lambda X, \eta)}$$

proves $p_1/X = Q_s/L \geq c_1 > 0$. Stress freedom gives the exact identities

$$p_1 = \frac{X Q_s}{L} = a = -2Y \Phi_Y / \Phi, \quad n_s = N_s/L = -2U_X = -2u_Y = Z_*/L + O(\Lambda^{-1}), \quad p_2 = X n_s / E.$$

All fixed parameter derivatives of $\Phi, \log \Phi, u, p_1, n_s$ are bounded uniformly in subsequent C , with Λ fixed; the radial scaling factor has already been accounted for.

At $X_0 = 4/\Lambda$, first take $\chi > 0.99$, so $t = 2\chi \in (1.98, 2]$. The series for $f_0(z) + z f'_0(z)$ has terms $(-t)^n/(n!)^2$. Its alternating remainder beyond degree four is negative, giving

$$f_0 + z f'_0 \leq 1 - t + t^2/4 - t^3/36 + t^4/576.$$

The derivative is $-1 + t/2 - t^2/12 + t^3/144$; on $[1.98, 2]$ its first two terms are nonpositive and its last two have sum $-t^2(12 - t)/144 < 0$. Thus the polynomial decreases, and its value at 1.98 is exactly $-75535511/400000000 < -0.188 < -0.18$. Since $f_0 \leq 1$,

$$-2z f'_0/f_0 = 2 - 2(f_0 + z f'_0)/f_0 > 2.36.$$

Taking Λ large gives $p_1 > 2.3$.

On $\chi \leq 0.99$, $|Z_*| > \delta_*$. Since $L \leq 1$ on $[-1, 1]$, the preceding error estimate gives $|n_s| \geq \delta_*/2$. At the same endpoint, exactly

$$|p_2| = C \sqrt{2/\Lambda} \frac{|n_s|}{\phi_* \Phi}.$$

For fixed Λ , $\phi_* \Phi$ has a finite positive upper bound independent of subsequent C ; p_1 has a finite upper bound independent of C . Thus choosing C large makes $p_2^2/p_1 > 2.3$ uniformly on this complementary compact set. The two alternatives yield

$$p_1 + p_2^2/p_1 > 2 + 0.2 \quad \text{at } X_0 = 4/\Lambda,$$

retaining the source's allowed $c_{\text{ex}} = 0.2$. The proof uses distinct regions only after constructing their exact covering through χ and Z_* ; no missing case at $H_* = 0$ remains.

6 Continuation, exact moment repair, and assembly of the annular profile

This file gives the independent continuation derivation for source pp. 150-157 and checks the assembly on pp. 36-45. Original source parameters remain $0 < h < 1/100$, $X_0 = X_a = 4/\Lambda$, $X_b^{\text{ref}} = 100$, $X_i = 110$. The notation X_b^{ref} distinguishes the temporary number 100 in Appendix B from the final outer annular edge $X_b = e^3 X_{\text{tail}}$ in Section 4; neither numerical value or role is changed. Refer to `core_exact.md` and `axis_exact.md` for all field definitions and proved identities.

6.1 1. The reference continuation and its estimates

The smooth step is exactly

$$\sigma(y) = \frac{e^{-1/y^2}}{e^{-1/y^2} + e^{-1/(1-y)^2}} \quad (0 < y < 1), \quad \sigma = 0 \ (y \leq 0), \quad \sigma = 1 \ (y \geq 1).$$

It is increasing: the ratio of its second denominator term to its first has logarithmic derivative $-2/(1-y)^3 - 2/y^3 < 0$. It is smooth and flat at both endpoints by the corresponding exponential formulas. Choose fixed $\bar{t} > 0$ with $4e^{2\bar{t}} < 4.1$. Set $y = \log(X/X_0)$. For $0 < t_1 \leq \bar{t}$, retain the analytic axis profile through $y = t_1$, then prescribe on $(t_1, 2t_1)$

$$\partial_y \log \phi_r = (1 - \sigma((y - t_1)/t_1)) D_X \log \phi_{\text{nat}}, \quad \partial_y U_r = (1 - \sigma((y - t_1)/t_1)) D_X U_{\text{nat}}.$$

After $2t_1$ keep ϕ_r, U_r constant in y . All endpoint jets match because the step is flat. Pressure and the integrated coefficients continue to be computed from the axis by the original integral definitions.

The axis approximation bounds $U_{\text{nat}} - U_*$ and its fixed parameter derivatives by $O(\Lambda^{-1})$; its cutoff variation integrates over an interval of length t_1 and is $O(t_1/\Lambda)$. Thus $U_r - U_*$ and its radial average remain small uniformly through X_i : explicitly

$$\|\mathcal{A}_X U_r - U_*\|_{C_\eta^k} \leq \sup_{0 \leq X \leq X_i} \|U_r - U_*\|_{C_\eta^k}.$$

No factor equal to the long logarithmic interval appears. Likewise integrate the bounded natural logarithmic slope to bound $\log \phi_r - \log \phi_*$. Its leading parameter derivative remains ξ_0 , with the $O(\chi) + O(\Lambda^{-1})$ weighted-gradient errors proved for the natural solution and an additional error tending to zero with t_1 after Λ is fixed. In particular $CE_r = \sqrt{2X} \phi_r$, its reciprocal on $[X_0, X_i]$, and all their fixed parameter derivatives have finite bounds independent of subsequent large C and sufficiently small t_1 .

Pressure satisfies the exact identity and estimate

$$\Pi_r - \Pi_0 = C^{-2} \int_0^X \phi_r(x, \eta)^2 dx, \quad \sup_{X \leq X_i} \|\Pi_r - \Pi_0\|_{C_\eta^k} \leq K_k C^{-2}.$$

The axis factor makes this integral finite at zero. The natural shear has $a = p_1 > 0$; cutting its nonpositive $D_X \log \phi$ to zero proves $l_r \leq 1$. In the angular source the unchanged large term is $-H_* \xi_0 = \Lambda L \chi$. Replacing H_* by $H_{c,r}$ costs $K_{\sigma_*} \sqrt{\chi} + o_{t_1}(1)$; the other weighted gradients cost $K \chi + K/\Lambda$. Young's inequality as in `axis_exact.md`, now allocating a total $0.06 \Lambda L \chi$, and the strict $-W_* > 2.8$ margin give

$$S_{q,r} \geq 0.94 \Lambda L \chi + 2.4.$$

Here one first enlarges Λ , then C , and then decreases t_1 ; all residual constant errors have a strictly positive allowance. The axial source is

$$S_{n,r} = Z_* + O(\Lambda^{-1}) + o_{t_1}(1) + O(C^{-2})$$

uniformly on the finite interval. Indeed all changes of U and of its parameter derivative are small, the logarithmic radial derivative is $O(\Lambda^{-1})$ on the cutoff and zero later, and pressure is controlled by the preceding exact integral.

The defining primitive equations imply

$$D_X p_{1,r} = X S_{q,r} / L - l_r p_{1,r}, \quad D_X n_{s,r} + n_{s,r} = S_{n,r} / L.$$

Positive regular initial data and $l_r \leq 1$ imply $p_{1,r} > 0$. Multiplying the first inequality by the integrating factor e^y , keeping the two positive source terms separately, gives

$$\begin{aligned} p_{1,r}(y) &\geq p_{1,r}(0)e^{-y} + 1.88\chi(e^y - e^{-y}), \\ p_{1,r}(X) &\geq p_{1,r}(X_0)X_0/X + 1.2(X - X_0^2/X). \end{aligned}$$

For the first, $X = X_0e^y = (4/\Lambda)e^y$ supplies the exact factor $0.94\Lambda X_0/2 = 1.88$. For the second, use $L \leq 1$, integrate $2.4X$ against the same factor, and convert back to X . On $\chi > 0.99$, the first lower bound remains above 2.3: replace its initial value by 2.3; its derivative is positive for $y \geq 0$, since $1.88\chi > 2.3/2$. On $\chi \leq 0.99$, $|Z_*| > \delta_*$, and the exact primitive for $n_{s,r}$ retains its sign and a fixed positive lower bound when the displayed errors are small. Because $p_{2,r} = Xn_{s,r}/E_r$, increasing C then makes $p_{2,r}^2/p_{1,r}$ large uniformly. This proves

$$v_r := p_{1,r} + p_{2,r}^2/p_{1,r} > 2 + c$$

on the entire reference interval, for a fixed $c > 0$. The second comparison yields $p_{1,r} > 3$ on $[100, 110]$, since $X_0 \leq 4$. Upper bounds for $p_{1,r}, n_{s,r}$ and their fixed parameter derivatives follow directly from their integral representations and the finite coefficient bounds. They are independent of subsequent C and of the small reference widths.

6.2 2. Activation of stress, with division by the flat factor proved

Fix a sufficiently large finite C . Choose $t_* > 0$ so the preceding bounds hold for every $0 < t_1 \leq t_*$. On this family, v_r has a finite common upper bound $V_{\max}$; it may depend on C , which has already been fixed. Choose $0 < \kappa_0 < 1/2$ later small enough relative to this bound. On $0 < y < t_1$ put

$$e_a = (1 - \kappa_0)\sigma(y/t_1), \quad \kappa = 1 - e_a, \quad a = \kappa p_{1,r}, \quad D_X U = -\kappa X n_{s,r}/2, \quad \partial_y \log \phi = -\kappa p_{1,r}/2.$$

The natural reference is stress-free here. Therefore the exact differences obey

$$\partial_y (\log \phi - \log \phi_r) = e_a p_{1,r}/2, \quad \partial_y (U - U_r) = e_a X n_{s,r}/2.$$

For a fixed parameter derivative, the integral from zero is bounded by a constant times $ye_a(y)$, since e_a is increasing. The constants are uniform in smaller t_1, κ_0 after Λ, C are fixed. The five moment differences are integrals of these value differences, with fixed smooth weights on the activation interval; their parameter derivatives have the same bound. The proved moment map in `core_exact.md`, using one further parameter derivative, then gives

$$p_s - p_{s,r} = O(ye_a), \quad E_r/E = 1 + O(ye_a).$$

The actual shear vector is exactly

$$s_{\text{shear}} = \kappa(p_{1,r}, p_{2,r}E_r/E).$$

Because κ cancels before forming its slope, there is no inverse power of κ_0 in

$$t_s = (p_{2,r}/p_{1,r})(E_r/E), \quad v_s = \kappa v_r + O(ye_a), \quad P_c = v_r + O(ye_a), \quad J_c = O(ye_a).$$

Consequently $P_c - v_s = e_a v_r + O(ye_a) \geq ce_a$ for small fixed t_1 , and $P_c > 2 + c'$. The ratio $(v_s - 2)_+ J_c^2 / (P_c - v_s)^2$ is at most a fixed constant times y^2 , hence below 2 for small t_1 . The exact cone equivalence proves admissibility wherever $v_s > 2$ and the relaxed condition everywhere else. Since $v_s(0) = v_r(0) > 2 + c$, continuity on the compact parameter interval supplies a first collar of positive width with $v_s > 2 + c/2$.

The stronger endpoint assertion requires more than an $O(ye_a)$ estimate. Here is the exact flat-integral calculation. For $c > 0$ and smooth b , substitute $u = y/\sqrt{1 + y^2v}$ to obtain

$$\int_0^y e^{-c/u^2} u^{-j} b(u, \eta) du = \frac{1}{2} e^{-c/y^2} y^{3-j} \int_0^\infty e^{-cv} (1 + y^2v)^{(j-3)/2} b\left(\frac{y}{\sqrt{1 + y^2v}}, \eta\right) dv.$$

For each fixed derivative the final integrand is bounded by e^{-cv} times a polynomial in v , uniformly on the compact parameter set; denominators $1 + y^2v$ are at least one. Differentiation under the integral proves a smooth coefficient through $y = 0$, with limiting value $b(0, \eta)/c$. This proves the needed result from Appendix A.9 directly.

Near zero, the specific step gives $e_a = e^{-t_1^2/y^2} g_a(y)$, where

$$g_a(y) = \frac{1 - \kappa_0}{e^{-t_1^2/y^2} + e^{-1/(1-y/t_1)^2}}$$

extends smoothly with $g_a(0) = (1 - \kappa_0)e > 0$. Taking $j = 0$ in the flat-integral formula proves

$$\log \phi - \log \phi_r, \quad U - U_r \in e_a y^3 C^\infty.$$

This class is preserved by smooth compositions, products with smooth functions, and division by nonvanishing fields: for example $(e^{e_a y^3 b} - 1)/(e_a y^3) = b \int_0^1 e^{v e_a y^3 b} dv$ is smooth. Moment and pressure differences acquire a second integration and belong to $e_a y^6 C^\infty$. Substitution into the exact moment formulas therefore shows

$$T_0 = e_a B_0(y, \eta), \quad B_0 \in C^\infty, \quad B_0(0, \eta) = F(X_0, \eta) p_{s,r}(X_0, \eta) \neq 0.$$

At the endpoint the latter vector is a positive multiple of the limiting shear. Its component along $(1, t_s)$ is positive, and that along $(-t_s, 1)$ is zero. Both cone-direction margins consequently stay positive on a smaller collar. Every derivative of the exponential adds only finitely many powers of y^{-1} , so for every fixed mixed derivative

$$|T_0| \geq c e^{-t_1^2/y^2}, \quad |\partial^\alpha T_0| \leq C_\alpha e^{-t_1^2/y^2} y^{-N_\alpha}.$$

Since $X_0 > 0$, converting y -derivatives to X -derivatives changes only fixed smooth coefficients. Thus the vector direction is smooth through the zero-stress edge, not merely convergent along individual paths.

6.3 3. Continuation to 110 and preservation of parameter analyticity

After activation retain the same shear prescription with $\kappa = \kappa_0$ through the reference cutoff and then through $X = 100$. The natural reference slopes are nonzero only in its initial interval of length $2t_1$. Integrating the difference on that interval gives $O(t_1)$. Beyond it, the reference slopes vanish, and the new slopes have size at most κ_0 times the bounded functions $p_{1,r}/2$, $X n_{s,r}/2$. Integration on the fixed interval gives $O(\kappa_0)$. Thus field values, logarithms and the required parameter derivatives differ from reference by $O(t_1 + \kappa_0)$. Their integrated coefficients obey the same estimate with finite constants depending on the already fixed C, Λ .

Choose κ_0 small enough that $\kappa_0 V_{\max} < 1/4$ and that its comparison errors use at most one quarter of the minimum $v_r - 2$. Then choose t_1 small for the other error quarters and for the activation inequalities. The same exact slope formula yields $P_c > 2 + c/2$, $v_s < 1$. The relaxed cone therefore holds regardless of J_c . This order works for the entire family of reference widths, avoiding a circular choice of κ_0 from a width that itself depends on κ_0 .

Starting at $X = 100$, multiply the axial prescription by a smooth β from one to zero on a short logarithmic interval. The integrated coefficients change by arbitrarily small errors when the interval is short, while

$$P_c = p_{1,r} + \beta p_{2,r}^2/p_{1,r} + o(1) > 2, \quad v_s < 1,$$

using $p_{1,r} > 3$ there. Then hold $D_X U = 0$ and interpolate a convexly from $\kappa_0 p_{1,r}$ to 0.8, after arranging $\kappa_0 \sup p_{1,r} < 0.8$. During this second interval $v_s = a \leq 0.8$, $P_c = p_1 > 2$ by continuity and its strictly positive initial margin. Both intervals can fit strictly between 100 and 110.

On the final interval, $a = 0.8$, $l = 0.6$, $D_X U = 0$. The unchanged leading gradient and the already small field changes give $S_q > 1$: its constant source contribution is $0.6(-W_*) > 1.68$, while the large $\Lambda L \chi$ term absorbs the previously identified weighted errors. At a hypothetical first downward crossing $p_1 = 2$,

$$D_X p_1 = X S_q / L - 0.6 p_1 > X - 1.2 > 0,$$

a contradiction. Hence $p_1 > 2$ persists through $X_i = 110$.

On a small first collar, all reference coefficients arise from the analytic axis solution, its derivatives and forward integrals. The only cutoffs depend on y , not on η . Choose a compact complex neighborhood on which the natural Φ , L , and the relevant denominators do not vanish. Integration in the real radial variable preserves holomorphy with uniform bounds on that neighborhood. Exponentiating the prescribed logarithm preserves nonvanishing. Radial differentiation changes the cutoff bounds but does not change the holomorphic parameter domain. A smaller fixed $t_c \in (0, t_1)$, with $4e^{t_c} < 4.1$, therefore gives the common parameter domain for $F, U, V_0/X, \Pi$ and every fixed radial derivative throughout $[0, X_0 e^{t_c}]$. Subsequent changes have later support, so their forward primitives leave this rectangle exactly unchanged.

6.4 4. Transition length fixed before amplitude

Let $\ell_i = \log(CE(X_i, \eta))$, $G_i = U(X_i, \eta)$, and $L_0 = \log(X_i/X_0)$. Write M_k, N_k for the reference bounds on $p_{1,r}, n_{s,r}$, which are independent of subsequent C . Before the final interpolation the logarithmic ϕ slope is $-\kappa p_{1,r}/2$; during it the slope is a convex interpolation with -0.4 ; after it the slope is exactly -0.4 . Hence

$$\|\log \phi(X_i)\|_{C_\eta^k} \leq \|\log \phi_* + \log \Phi(4, \cdot)\|_{C_\eta^k} + \frac{1}{2}(M_k + 0.8)L_0.$$

Adding $\frac{1}{2} \log(2X_i)$ proves $\|\ell_i\|_{C_\eta^k} \leq B_k$, where B_k is independent of subsequent C and smaller transition widths. The axial change on activation is at most $X_0 e^{\bar{t}} N_k t_1/2$; the remaining small-shear part contributes at most $X_i N_k \kappa_0/2$. The natural-axis difference is $O(\Lambda^{-1})$ with a constant independent of Λ . Therefore

$$\|G_i - U_*\|_{C_\eta^k} \leq C_k^{\text{mat}}/\Lambda + C_k^{\text{join}}(\Lambda)(t_1 + \kappa_0 + \omega_{\text{fin}}),$$

where ω_{fin} is the sum of the two final logarithmic widths. No inverse cutoff width occurs in these integrals.

With $f = (1 + \eta^2)^{-1}$, choose a finite positive T_{sh} satisfying exactly the source lower bound

$$T_{\text{sh}} \geq 20\|\sigma'\|_\infty(B_0 + \|\log f\|_\infty).$$

For $y_i = \log(X/X_i)$, prescribe

$$\log E = -\log C + y_i/10 + (1 - \sigma(y_i/T_{\text{sh}}))\ell_i + \sigma(y_i/T_{\text{sh}})\log f, \quad U = G_i.$$

At the left endpoint this matches $E(X_i)$ and its radial jet, because both original and new profiles have $D_X \log E = 0.1$ there and the step is flat. At the right endpoint it matches $C^{-1}f(X/X_i)^{1/10}$ with its entire radial jet. Direct differentiation gives

$$l = 0.6 + T_{\text{sh}}^{-1}\sigma'(y_i/T_{\text{sh}})(\log f - \ell_i) \in [0.55, 0.65], \quad a \in [0.7, 0.9], \quad b_s = 0.$$

The old parameter gradient satisfies $-H_c \ell'_i \geq \Lambda L\chi - K\chi - K_{\sigma_*}\sqrt{\chi} - K/\Lambda - o(1)$. The new one is

$$-H_c(\log f)' = \frac{2\eta H_c}{1 + \eta^2} \geq -Kj_0^2 - K/\Lambda - o(1).$$

For the latter bound, the exact identity $\eta H_* = (D + 4d)\eta^2 + dj_0\eta$ and completion of the square bound its negative part by Kj_0^2 ; the $U - U_*$ error is already controlled. These gradients are combined with nonnegative coefficients summing to one. The old large positive term absorbs its correspondingly weighted errors, and the new negative part is small. Since $-W$ is close to $-W_* > 2.8$ and $l \geq 0.55$, the constant contribution exceeds 1.54 before the small h, j_0, Λ^{-1} errors. Thus $S_q > 1$. The same crossing argument at $p_1 = 2$, now $l \leq 0.65$, gives $D_X p_1 > X - 1.3 > 0$. Since $b_s = 0$, this proves $P_c = p_1 > 2$, $v_s = a < 1$, hence the relaxed cone, along the full transition and the following ideal-angular interval.

6.5 5. The five matching equations, including all scaling factors

Set

$$X_R = 110(CP_*)^{10}, \quad x = X/X_R, \quad X_{\text{sep}} = 110e^{T_{\text{sh}}}, \quad x_{\text{sep}} = \frac{e^{T_{\text{sh}}}}{(CP_*)^{10}}.$$

Here X_{sep} is fixed before C . The ideal pair is $U_0 = 4\eta$, $E_0 = P_* f x^{1/10}$. The exact normalized moments are

$$\begin{aligned} \widehat{M}_0 &= 4\eta x, \quad \widehat{I}_0 = \frac{5\sqrt{2}}{8}P_* f x^{8/5}, \quad \widehat{J}_0 = 4\eta \widehat{I}_0, \\ \widehat{S}_0 &= 16\eta^2 x - \frac{5}{12}P_*^2 f^2 x^{6/5}, \quad C_{p,0} = \frac{5}{2}P_*^2 f^2 x^{1/5}. \end{aligned}$$

These follow by direct integration, retaining the factors $(M, I, J, S, C_p) = (X_R \widehat{M}, X_R^{3/2} \widehat{I}, X_R^{3/2} \widehat{J}, X_R \widehat{S}, C_p)$.

On $[0, X_{\text{sep}}]$, fixed parameter derivatives of U and CE are bounded independently of later C ; near zero $CE = O(\sqrt{X})$. Thus

$$\|\widehat{M}\|_{C^k} + \|\widehat{S}\|_{C^k} \leq K_k x_{\text{sep}}, \quad \|\widehat{I}\|_{C^k} + \|\widehat{J}\|_{C^k} \leq K_k C^{-1} x_{\text{sep}}^{3/2}, \quad \|C_p\|_{C^k} \leq K_k C^{-2}$$

at $x = x_{\text{sep}}$. For pressure the proof uses the physical integral $C^{-2} \int_0^{X_{\text{sep}}} (CE)^2/(2X) dX$, finite because $CE = O(\sqrt{X})$. All constants are fixed before increasing C . The ideal moments also tend to zero with the displayed exact powers. Their pressure increment is $O(C^{-2})$, because $x_{\text{sep}}^{1/5} = e^{T_{\text{sh}}/5}(CP_*)^{-2}$.

Beyond X_{sep} , the angular profile is exactly ideal, since $C^{-1}f(X/X_i)^{1/10} = P_* f x^{1/10}$. Choose C so $x_{\text{sep}} < e^{-8}$. Restore G_i to 4η by a fixed smooth step on $-8 < \log x < -7$, retaining E_0 . The size of this change, and its fixed logarithmic radial and parameter derivatives, is bounded by fixed constants times $\|G_i - 4\eta\|$. The moment errors accumulated to the correction interval are bounded by

$$K_k \|G_i - 4\eta\|_{C_\eta^{k+1}} + o_C(1).$$

Every relevant radial weight is integrable at zero: their actual exponents are $0, 3/5, 1/5$ and $-4/5$ for the pressure integrand, all greater than -1 . No divergent contribution is discarded.

On $-6 < \log x < -5$, choose two fixed ordered nonnegative bumps for U , three for E . At the ideal pair make the invertible row operations $J \mapsto J - 4\eta I$, $S \mapsto S - 8\eta M$, with 4η held at its base value. The linearized axial block has weights $1, fx^{3/5}$; the azimuthal block has weights $x^{1/2}, fx^{1/10}, fx^{-9/10}$, retaining fixed nonzero factors $P_*, \sqrt{2}$ in the matrix. The exponents in each block are distinct. The Chebyshev determinant proof and quadratic contraction in `core_exact.md` therefore solve all five moment equations exactly, with small smooth coefficients. The row operations do not impose that the corrected U remain constant; the full nonlinear differences still include $\delta U \delta E$, $(\delta U)^2$, and $(\delta E)^2$.

To verify one tolerance can precede C , put $R = [e^{-8}, e^{-5}]$, $Q_{\min} = \min Q_{s,0} > 0$, $e_* = \min E_0 > 0$, and $w_* = 1 + \max |N_{s,0}/(E_0 Q_{s,0})|$ on $R \times [-1, 1]$. The ideal source is positive directly: for this pair

$$S_{q,0} = \frac{9}{5} - h + \frac{16}{5}h\eta^2 + \frac{2\eta^2(D+4d)}{1+\eta^2} > 0,$$

as $W = -3 + 8h\eta^2$, $l = 3/5$, $H_c = (D+4d)\eta$. Its regular weighted primitive gives $Q_{s,0} > 0$. The compact minima thus exist without importing the positivity of a separate abbreviated formula. The exact moment map has no X_R in Q_s, N_s , so a fixed small tolerance in the fields and normalized moments through one extra parameter derivative ensures

$$Q_s \geq Q_{\min}/2, \quad E \geq e_*/2, \quad |a - 0.8| \leq 0.1, \quad |N_s/(EQ_s)| \leq w_*, \quad |b_s| \leq 0.1/(1+w_*).$$

Then, retaining the source numbers,

$$v_s \leq 0.9 + 0.01/0.7 < 1, \quad G := Q_s - b_s N_s/(aE) \geq (1 - 1/7)Q_s \geq 3Q_{\min}/7, \quad P_c = X_R x G/L.$$

Choosing X_R large now gives $P_c > 2$ without shrinking the tolerance. First choose j_0 , then Λ , so $j_0 + C_k^{\text{mat}}/\Lambda$ fits the tolerance. Next fix the B_k and T_{sh} , then enlarge C for the vanishing moment contributions. Finally decrease $\kappa_0, t_1, \omega_{\text{fin}}$ for the remaining G_i error. These choices prove the exact order

$$(M_d, T_d, P_*, \lambda, h) \rightarrow \text{moment tolerance}, j_0 \rightarrow \delta_*, \sigma_*, \Lambda \rightarrow (B_k), T_{\text{sh}} \rightarrow C, X_R \rightarrow \kappa_0, t_1, \text{final widths}.$$

Only finitely many parameter derivative orders need to be small. Smooth implicit differentiation supplies all remaining fixed orders after the choices. At $X_h = X_R e^{-5}$, all five moments and both fields agree with the ideal pair, with the same axis pressure Π_0 , so the exact exterior-preservation map from `core_exact.md` applies.

6.6 6. The finite-frequency assembly in Section 4

The outer construction in Appendix A supplies the separate profile, its exact total moments, pressure datum and reserved intervals. The loop construction in Appendix C supplies the periodic shear family. These two existence constructions are dependencies allocated to the other audit lanes. The operations performed with their outputs in Section 4 can be checked exactly here.

The join at X_h is smooth because both pairs coincide with the same ideal pair on a neighborhood, not only at one radius. Their five matching moments and common axis pressure imply exact agreement of every integrated exterior field. In particular all total moments and the outer heat field persist. The perturbation interval $I = [X_-, X_+]$ lies after the reserved analytic collar and before the first repair interval; its endpoints already have the admissible cone.

For the loop input t_ℓ, ρ_ℓ with mean t_s , variance ρ_ℓ/a , and $v_\ell = v_s + \rho_\ell > 2$, Section 4 changes the angular parameter by

$$\frac{d\varphi}{d\theta'} = \frac{a(1+t_\ell^2)}{2\pi v_\ell}, \quad (a_L, -b_L) = \frac{v_\ell(1, t_\ell)}{1+t_\ell^2}.$$

The derivative is positive and its integral over $[0, 2\pi]$ is $a(1+t_s^2+\rho_\ell/a)/v_\ell = 1$. Thus it defines an orientation-preserving smooth circle coordinate of period one. Its shear mean is

$$\int_0^1 (a_L, -b_L) d\varphi = \frac{a}{2\pi} \int_0^{2\pi} (1, t_\ell) d\theta' = (a, -b_s).$$

The compact positive derivative gives a smooth inverse including all endpoint parameter derivatives. This establishes exactly the reparameterization asserted in Lemma 4.11; the existence of t_ℓ, ρ_ℓ itself remains the separately audited Appendix C construction.

Take zero-mean periodic primitives $\mathcal{A}, \mathcal{B}$ with

$$\partial_\varphi \mathcal{A} = -(a_L - a_0)/2, \quad \partial_\varphi \mathcal{B} = E_0(b_L - b_0)/2.$$

They exist because the two means vanish, and they vanish near both endpoints where the loop is constant. Set

$$E_N = E_0 \exp(\mathcal{A}(X, \eta, N \log X)/N), \quad U_N = U_0 + \mathcal{B}(X, \eta, N \log X)/N.$$

Holding phase fixed in the following slow derivatives, direct differentiation gives exactly

$$a_N = a_L - 2D_X \mathcal{A}/N, \quad b_N = e^{-\mathcal{A}/N} (b_L + 2D_X \mathcal{B}/(NE_0)).$$

Since the phase has no η -dependence and all intervals are fixed, field differences and fixed parameter derivatives are $O(N^{-1})$, while the actual shear differs from the desired loop by $O(N^{-1})$. The five exact integrands of the changes, writing $u_N = U_N - U_0$, $e_N = E_N - E_0$, are

$$u_N, \quad \sqrt{2X}e_N, \quad H_0 u_N + U_0 \sqrt{2X}e_N + \sqrt{2X}u_N e_N,$$

$$2U_0 u_N + u_N^2 - E_0 e_N - e_N^2/2, \quad E_0 e_N/X + e_N^2/(2X).$$

Integration on the fixed interval preserves $O(N^{-1})$ bounds. The moment map, using one extra parameter derivative, gives $p_{s,N} - p_{s,0} = O(N^{-1})$ without assuming small radial derivatives of the field changes.

The four defining gaps are the components of the smooth map

$$\Psi(a, b, p) = (a, v - 2, c - v, 2(c - v)^2 - (v - 2)^2), \quad v = a + b^2/a, \quad c = p_1 - bp_2/a, \quad j = p_2 + bp_1/a.$$

The loop and unchanged admissible part have positive compact minima μ_L, μ_R . Fix a compact neighborhood with $a > 0$, and let C_Ψ be the supremum norm of its derivative. Choosing N so the preceding errors are below $\min(\mu_L, \mu_R)/(2C_\Psi)$ leaves each relevant gap at least half its original minimum. This is a finite uniform choice after every radial scale and loop has been fixed.

On a later repair interval $(Y_0, Y_1) \subset I_1$, the unchanged pair is $U_0 = 0$, $E_0 = K(\eta)X^{-1/2-\lambda}$, $K = c_{\text{patch}}(1 + \eta^2)^{-1} > 0$. Two U bumps and three E bumps have the exact quadratic moment changes, in order (M, J, I, S, C_p) ,

$$\begin{aligned} & \int u_c, \quad \int (H_0 u_c + \sqrt{2X}u_c e_c), \quad \int \sqrt{2X}e_c, \\ & \int (u_c^2 - E_0 e_c - e_c^2/2), \quad \int (E_0 e_c/X + e_c^2/(2X)). \end{aligned}$$

The linear blocks have weights $(1, X^{-\lambda})$ and $(X^{1/2}, X^{-1/2-\lambda}, X^{-3/2-\lambda})$ with their fixed positive $K, \sqrt{2}$ factors. Distinctness follows from the actual $\lambda > 0$. The determinant and quadratic-solve proofs in `core_exact.md` apply. With $\|d_N\|_{C^k} \leq D_k/N$, the source's explicit requirement

$$N \geq 8 \max_{k=0,2} \nu_k^2 q_k D_k$$

gives a smooth correction with $\|c\|_{C^2} \leq 2\nu_2 D_2/N$. Fixed bump shapes also make its first radial derivative $O(N^{-1})$. Increasing N so $N \geq 4C_\Psi C_{\text{rep}}/\mu_R$ leaves the repair gaps at least $\mu_R/4$. At Y_1 every moment is restored exactly. Below the first perturbation and above Y_1 , the full field and stress are therefore exactly the original joined ones. No later modification reaches the analytic collar or the reserved intervals I_3, I_4 .

6.7 7. Endpoint, weight and total-moment conclusions

The exact inner factor already proved gives $T_0 = e^{-t_1^2/y_a^2} g_a B_a$, where $y_a = \log(X/X_a)$, $g_a(0) > 0$, and $B_a(0) = F_{s_{\text{shear}}} \neq 0$. The imported outer endpoint computation has the exact factors

$$T_{0,\theta} = e^{-4/y_b^2} y_b^{-3} b_\theta, \quad T_{0,z} = e^{-4/y_b^2} y_b^3 b_z, \quad y_b = \log(X_b/X), \quad b_\theta(0) > 0.$$

Consequently the unit direction is $B_a/|B_a|$ near the inner edge and $(b_\theta, y_b^6 b_z)/(b_\theta^2 + y_b^{12} b_z^2)^{1/2}$ near the outer edge. Both extend smoothly. The endpoint directions are respectively $(1, t_s)/(1 + t_s^2)^{1/2}$ and $(1, 0)$, with $t_s = 0$ at the latter edge.

Let $n = T_0/|T_0|$, $A_n = n_\theta + t_s n_z$, $B_n = n_z - t_s n_\theta$. In the interior,

$$A_n = (P_c - v_s)/|p_s - s_{\text{shear}}| > 0, \quad G_n := 2 - (v_s - 2)B_n^2/A_n^2 > 0.$$

At either edge $B_n = 0$, while $A_n > 0$, so $G_n = 2$. Compactness of the closed annulus gives positive minima for $A_n, G_n, F, a, v_s - 2$. The choice $\kappa = \min\{1, \min A_n, \min G_n\} > 0$ proves exactly the two directional bounds in source (4.26), with $\kappa < 2$.

The exact weight

$$\zeta(X) = \exp(-t_1^2/y_a^2 - 4/y_b^2) \quad (X_a < X < X_b), \quad \zeta = 0 \text{ otherwise}$$

is smooth and flat at both edges. On either fixed collar its ratio to the local exponential has positive upper and lower bounds. The nonzero vector coefficients and finite powers introduced by differentiation therefore prove

$$|T_0| \geq c\zeta, \quad |\partial^\alpha T_0| \leq C_\alpha \zeta \min\{1, y_a, y_b\}^{-m_\alpha}.$$

On the compact region between the collars this follows from positivity and smoothness directly. Thus all six conclusions concerning stress direction and weight follow from the actual endpoint factors, not from a claim that every flat function has a smooth normalized direction.

Every completed join restores all five cumulative integrals at a finite radius. Therefore total M, J, S remain zero, $C_p(\infty) = -\Pi_0$, and the renormalized angular momentum remains unchanged because the perturbation of H is compactly supported and has total integral zero. The pressure is hence exactly

$$\Pi(X) = \Pi_0 + C_p(X) = C_p(X) - C_p(\infty) = - \int_X^\infty E(x)^2/(2x) dx.$$

Above the retained X_v , the outer data have $U = M = J = 0$; the formula $V_0 = (2\eta XU - 2D\eta M - dM_\eta)/L$ then gives $V_0 = 0$ exactly. The heat formula was proved directly in `core_exact.md`. The compensation used only I_2 , the last repair only I_1 ; therefore $I_{\text{pos}} = I_3$ and $I_{\text{mean}} = I_4$ retain their exact specified profiles. These are dependency-preserving assembly statements; they do not by themselves validate any later residual summation or wave realization.

This is a bounded mathematical audit of Section 5 and Appendix A of *Finite Time Blowup for Navier-Stokes*, cited as [NS]. The source is the 165-page [source PDF](#). Its SHA-256, written on two lines without an intervening character, is

8c8a94ad9ac824c8b605b9827cadf7be
aca48bd10b380de3cfc872a2c37afa81

Page numbers below are both printed and PDF page numbers. Every proof in pp. 45–62 and pp. 126–144 was read. Definitions and proofs in (4.1)–(4.23) were read as dependencies. The existence and gluing of the leading axis profile in Appendix B, the cone realization in Appendix C, and the wave and final localization arguments are outside this lane. Their conclusions are identified at the exact points where they enter; this document does not assert a global Navier–Stokes theorem.

The derivations below are independent calculations, not a reproduction of the source prose. Symbols, physical coefficients, signs, and original parameter choices are retained. We write $\mathcal{H}$ for the heat factor called H in (A.32) solely to distinguish it from the angular-momentum profile $H = \sqrt{2}XE$; the exact map between them is $E_{\text{heat}} = c_\infty X^{-A} \mathcal{H}(2(1 - \eta^2)/X)$.

7 The physical coordinate map and the exact coefficient equations

Set

$$\begin{aligned} A &= \tfrac{1}{2} + h, & D &= \tfrac{1}{2} - h, & 0 < h < \tfrac{1}{2}, & \tau &= 1 - t, \\ z &= q^D \eta, & \tau &= q(1 - \eta^2), & s &= \tfrac{1}{2} r^2, & X &= s/q, \end{aligned}$$

and $d = 1 - \eta^2$, $L = 1 - 2h\eta^2$. For $\tau > 0$, the equation for q is $q - z^2 q^{2h} = \tau$ on $q > |z|^{1/D}$. Its derivative is $1 - 2hz^2 q^{2h-1} = L \geq 1 - 2h > 0$, it starts at zero and tends to infinity. It consequently defines exactly one q , with $|\eta| < 1$. Direct differentiation, with the other physical variables fixed, gives

$$\begin{aligned} q_t &= -L^{-1}, & \eta_t &= D\eta/(qL), & X_t &= X/(qL), \\ q_z &= 2\eta q^{1-D}/L, & \eta_z &= d/(q^D L), & X_z &= -2\eta X/(q^D L). \end{aligned}$$

Thus, for every real b and every differentiable profile f ,

$$\begin{aligned} \partial_t(q^b f) &= q^{b-1} T_b f, & T_b f &= L^{-1}(-bf + D\eta f_\eta + X f_X), \\ \partial_z(q^b f) &= q^{b-D} Z_b f, & Z_b f &= L^{-1}(2b\eta f + df_\eta - 2\eta X f_X). \end{aligned}$$

In particular two axial derivatives mean $Z_{b-D}Z_b$, in that order. There is no division by d , including at $\eta = \pm 1$. These are the operators of [NS, (4.2), pp. 24–25].

Retain the fixed constant $C > 1$. Put $\lambda_n = 2nh$, $E_n = \sqrt{2X}\phi_n/C$, and

$$u_{\theta,n} = rq^{-A-1/2+\lambda_n}\phi_n/C = q^{-A+\lambda_n}E_n, \quad u_{z,n} = q^{-A+\lambda_n}U_n, \quad ru_{r,n} = q^{\lambda_n}V_n, \quad p_n = q^{-2A+\lambda_n}\Pi_n.$$

The zero-index profiles are the leading profiles. All negative-index profiles are zero. Since $\partial_s(ru_r) + \partial_z u_z = 0$ and $A + D = 1$,

$$(V_n)_X = -Z_{-A+\lambda_n}U_n.$$

For $F_n = \int_0^X U_n(x, \eta) dx = X\mathcal{A}_X U_n$ and the regular axis value $V_n(0) = 0$, integration of $x(U_n)_x$ gives

$$V_n = \frac{2\eta XU_n - 2\eta(D + \lambda_n)F_n - d(F_n)_\eta}{L}. \quad (1)$$

This calculation fixes both coefficients of the moment terms; replacing $D + \lambda_n$ by D would change incompressibility.

The angular vector Laplacian and the axial scalar Laplacian have different axis coefficients. Directly using $\partial_r X = r/q$ gives

$$(\partial_{rr} + r^{-1}\partial_r - r^{-2})(r\phi(X)) = \frac{2r}{q}(X\phi_{XX} + 2\phi_X),$$

$$(\partial_{rr} + r^{-1}\partial_r)U(X) = \frac{2}{q}(XU_{XX} + U_X), \quad (\partial_{rr} + r^{-1}\partial_r - r^{-2})\frac{V(X)}{r} = \frac{2X}{qr}V_{XX}.$$

Angular transport is $u_r(\partial_r + r^{-1})u_\theta$; after removing $rq^{-A-3/2+\lambda_n}/C$ its coefficient is $V_i((\phi_j)_X + \phi_j/X)$. Axial radial transport gives $V_i(U_j)_X$. Multiplication by u_z adds the correct Z operator because $A + D = 1$. Finally $1 - 2D = 2h$: axial viscosity of index $n - 1$ occurs at index n . These identities give the following exact coefficient equations, with $b = -A - 1/2$, $c = -A$:

$$2(X\phi_{n,XX} + 2\phi_{n,X}) = T_{b+\lambda_n}\phi_n + \sum_{i+j=n} \{V_i(\phi_{j,X} + \phi_j/X) + U_i Z_{b+\lambda_j}\phi_j\} - Z_{b+\lambda_{n-1}-D}Z_{b+\lambda_{n-1}}\phi_{n-1}, \quad (2)$$

$$2(XU_{n,XX} + U_{n,X}) = T_{c+\lambda_n}U_n + \sum_{i+j=n} \{V_i U_{j,X} + U_i Z_{c+\lambda_j}U_j\} + Z_{-2A+\lambda_n}\Pi_n - Z_{c+\lambda_{n-1}-D}Z_{c+\lambda_{n-1}}U_{n-1}, \quad (3)$$

$$\Pi_{n,X} = C^{-2} \sum_{i+j=n} \phi_i \phi_j - \Omega_{n-1}/(2X), \quad (4)$$

$$\Omega_k = T_{\lambda_k}V_k + \sum_{i+j=k} \{V_i(V_{j,X} - V_j/(2X)) + U_i Z_{\lambda_j}V_j\} - 2XV_{k,XX} - Z_{\lambda_{k-1}-D}Z_{\lambda_{k-1}}V_{k-1}. \quad (5)$$

For the last two lines multiply the physical radial equation by r . The pressure term is $2X\Pi_{n,X}$ and the centrifugal term is $-\sum E_i E_j$. All other radial terms carry $q^{-1+\lambda_k}$, whereas these carry $q^{-2A+\lambda_n}$; equality requires $k = n - 1$. This proves the sign $-\Omega_{n-1}/(2X)$ rather than choosing it by convention. The terms in (5) all contain a factor X when $V_j = Xv_j$. For example $V_i(V_{j,X} - V_j/(2X)) = Xv_i(v_j/2 + Xv_{j,X})$. The T and Z operators preserve divisibility by X , as does XV_{XX} . Therefore the quotient in (4) is a smooth axis source. Equations (1)–(5) verify [NS, (5.2)–(5.6), pp. 46–47].

8 The analytic radial recursion on an interval independent of the order

The precise input here is the leading axis solution and its first collar from [NS, Theorem 4.6(i), Proposition B.2]: there is a fixed interval $0 \leq \xi \leq a$, $X = \xi^2$, where the leading coefficients are smooth radially and holomorphic in a complex neighborhood of $[-1, 1]$. The collar is also the location from which one chooses the retained radius X_- . This input is owned by the profiles audit; no existence of that leading solution is inferred here.

At positive order the sums in (2)–(4) are linear in the current profiles. Indeed their only current-index pairs are $(0, n)$ and $(n, 0)$. For a completely local formulation define

$$K_n = \mathcal{A}_X U_n - U_n, \quad W_n = (\phi_n, U_n, K_n, \Pi_n, \partial_\xi \phi_n, \partial_\xi U_n)^T.$$

Then $K_{n,\xi} + 2K_n/\xi = -U_{n,\xi}$. Formula (1) becomes

$$\frac{V_n}{X} = \frac{2\eta(A - \lambda_n)U_n - 2\eta(D + \lambda_n)K_n - d(U_{n,\eta} + K_{n,\eta})}{L}.$$

Also $2(X\partial_{XX} + 2\partial_X) = \frac{1}{2}(\partial_{\xi\xi} + 3\xi^{-1}\partial_\xi)$ and $2(X\partial_{XX} + \partial_X) = \frac{1}{2}(\partial_{\xi\xi} + \xi^{-1}\partial_\xi)$. The pressure row is $\Pi_{n,\xi} = 4\xi C^{-2}\phi_0\phi_n + f_{n,4}$, with known smooth $f_{n,4}$. Substituting this row into the term $X\Pi_{n,X}$ in the axial equation gives

$$W_{n,\xi} + \xi^{-1} \text{diag}(0, 0, 2, 0, 3, 1)W_n = A_0 W_n + A_1 \partial_\eta W_n + f_n, \quad W_n(0) = 0. \quad (6)$$

There are no singular entries left in A_0 or A_1 : $V_0/\xi = \xi v_0$, $V_n\phi_0/X = v_n\phi_0$, and Ω_{n-1}/X were explicitly resolved above. The first four equations and (2)–(3) specify every entry of A_0 exactly. More usefully for convergence, its derivative matrix has precisely the following potentially nonzero entries, with $H_c = D\eta + dU_0$:

$$\begin{aligned} (A_1)_{51} &= 2H_c/L, & (A_1)_{52} &= (A_1)_{53} = -2d(\phi_0 + X\phi_{0,X})/L, \\ (A_1)_{62} &= 2(H_c - dXU_{0,X})/L, & (A_1)_{63} &= -2dXU_{0,X}/L, & (A_1)_{64} &= 2d/L. \end{aligned}$$

For example the last entry follows from the $d\Pi_{n,\eta}/L$ term and the factor two introduced when converting the axial radial operator. These entries verify the sparse matrix in [NS, p. 48], including its factors of two.

Here is a direct convergence proof that allows the order-dependent norms to be arbitrarily large. For $c_i = (0, 0, 2, 0, 3, 1)_i$ let

$$(Gg)_i(\xi) = \int_0^\xi (s/\xi)^{c_i} g_i(s) ds, \quad K = G(A_0 + A_1 \partial_\eta).$$

The kernel is diagonal, has norm at most one, and is independent of η . Each $\partial_\eta^j A_1$ has range in coordinates 5, 6 and annihilates coordinates 5, 6. Thus $A_1(\xi)D_0(\partial_\eta^j A_1)(s) = 0$ for every intervening diagonal kernel D_0 . Expanding a word of k factors K , any adjacent choice of two derivative factors is zero, including the term where the left derivative lands on the right matrix. Consequently a surviving word contains at most $p_k = \lceil k/2 \rceil$ derivatives in η .

Choose $S_\rho = \{\eta : \text{dist}(\eta, [-1, 1]) < \rho\}$ with no zero of L , on which coefficients and source are bounded uniformly in ξ . For a smaller $S_{\rho'}$, distribute the gap $\Delta = \rho - \rho'$ equally among the at most p_k differentiations. Repeated Cauchy estimates bound their cost by $(\max(1, p_k/\Delta))^{p_k}$. The $k+1$ ordered radial integrals have volume $a^{k+1}/(k+1)!$. The at most 2^k word choices can be absorbed into a constant C_n^{k+1} . Therefore

$$\|K^k Gf_n\|_{S_{\rho'}} \leq \frac{C_n^{k+1} a^{k+1}}{(k+1)!} (\max(1, p_k/\Delta))^{p_k}.$$

The logarithm divided by k is $-\frac{1}{2} \log k + O(1)$, so the series $W_n = \sum_{k \geq 0} K^k Gf_n$ converges for this entire fixed a . For a homogeneous bounded analytic solution, $W = K^k W$ gives the same estimate with k ordered integrals; it tends to zero. This proves uniqueness in the stated analytic class without a bound uniform in n .

To obtain axis regularity, use the nonsingular identities

$$(Gg)_i = \xi \int_0^1 t^{c_i} g_i(t\xi) dt, \quad \partial_\xi (Gg)_i = g_i(\xi) - c_i \int_0^1 t^{c_i} g_i(t\xi) dt.$$

Start with continuous W holomorphic in η ; Cauchy estimates give continuous η derivatives on a smaller neighborhood, so the right sides give a radial derivative. Repeating this argument gives every fixed mixed derivative on some smaller neighborhood. The equations are invariant under reflection of ξ with the first four components even and the last two odd. Uniqueness of the reflected integral equation imposes these parities. An even C^∞ function of ξ is C^∞ as a function of $X = \xi^2 \geq 0$: Taylor's formula to order $2m+2$ contains only even powers and gives the needed remainders after each of the first m X derivatives. This works for every m , with the same parameter regularity at that finite order. It proves the local conclusion of [NS, Lemma 5.1], with the leading-profile existence kept as its exact input.

9 Power moment maps and exact corrections

9.1 An invertible moment map and its quadratic inverse

For distinct real exponents $\alpha_1, \dots, \alpha_m$, a nonzero sum $\sum c_i x^{\alpha_i}$ has at most $m - 1$ positive zeros. To prove this by induction, order the exponents and divide by the smallest power. If the result had m zeros, its derivative would have $m - 1$ zeros by Rolle's theorem. That derivative is either zero, in which case the original result is constant, or is a nonzero combination of $m - 1$ distinct powers, which has at most $m - 2$ zeros by induction. Both alternatives exclude m zeros.

For ordered disjoint intervals $I_1 < \dots < I_m$ and nonnegative nonzero $\beta_j \in C_c^\infty(I_j)$, the determinant of $(x_j^{\alpha_i})$ is nonzero at every $x_1 < \dots < x_m$, and its continuous sign is fixed on that connected set. Multilinearity and Fubini give

$$\det \left(\int x^{\alpha_i} \beta_j(x) dx \right) = \int_{I_1 \times \dots \times I_m} \det(x_j^{\alpha_i}) \prod_j \beta_j(x_j) dx_1 \dots dx_m \neq 0.$$

Smooth families on compact parameter sets have bounded inverse derivatives, by the determinant formula and $\partial B^{-1} = -B^{-1}(\partial B)B^{-1}$. Under $x = e^y$, this is also the exponential-weight statement in [NS, Lemma A.1]. For two identical translated logarithmic bumps at centers separated by $\Delta > 0$, after dividing row i by $e^{s_i c_1}$ the determinant is $b(s_1)b(s_2)(e^{s_2 \Delta} - e^{s_1 \Delta})$, where $b(s) = \int e^{st} \beta(t) dt > 0$. On a fixed compact slope interval, the mean value theorem gives an inverse bound $O(|s_1 - s_2|^{-1})$. In particular the near-coincident weights in this construction cost exactly an allowed $O(\lambda^{-1})$ upper bound, not a constant independent of λ .

For completeness the finite nonlinear solve can be made fully quantitative. Let $F_\eta(c) = B(\eta)c + Q_\eta(c, c)$, with Q bilinear. In a chosen $C^k([-1, 1])$ norm let multiplication by B^{-1} have norm β_k and the bilinear map have norm κ_k . If $8\beta_k^2 \kappa_k \|d\|_{C^k} \leq 1$, the map $\Phi(c) = B^{-1}(d - Q(c, c))$ preserves the ball of radius $r = 2\beta_k \|d\|_{C^k}$, since

$$\|\Phi(c)\| \leq \beta_k \|d\| + \beta_k \kappa_k r^2 \leq r/2 + r/4 < r.$$

Its Lipschitz constant is at most $2\beta_k \kappa_k r \leq 1/2$. Iteration from zero therefore converges to the unique solution in this ball. At $k = 0$, $\|B^{-1} D_c Q(c, c)\| \leq 1/2$, so the Jacobian is invertible. Pointwise implicit differentiation proves smoothness. Solving in any fixed C^k ball gives the same branch because the iterates are identical. This proves [NS, Lemma A.2] and also specifies the branch selected in every use.

The five original leading moments are

$$\begin{aligned} M &= \int_0^X U dx, & I &= \int_0^X \sqrt{2x} E dx, & J &= \int_0^X U \sqrt{2x} E dx, \\ S &= \int_0^X (U^2 - E^2/2) dx, & C_p &= \int_0^X E^2/(2x) dx, & \Pi &= \Pi_{\text{ax}} + C_p. \end{aligned}$$

The substitution $X = \rho x$ is an exact map: the factors of these five moments are $(\rho, \rho^{3/2}, \rho^{3/2}, \rho, 1)$, with $H = \rho^{1/2} \sqrt{2x} E$. No factor is discarded. At $U = u_c(\eta)$ and $E = e_* f_*(\eta) x^\alpha$, take the differential of the rows $J - u_c I$ and $S - 2u_c M$, where u_c in the row operation stays at its original value. These differentials are $\int \sqrt{2x} E \delta U dx$ and $-\int E \delta E dx$. Together with M, I, C_p , they give the two blocks of weights

$$(1, x^{\alpha+1/2}), \quad (x^{1/2}, x^\alpha, x^{\alpha-1}),$$

with the factors $e_* f_*$ and $\sqrt{2}$ retained in their row multipliers. The exponents are distinct precisely when $\alpha \notin \{-1/2, 1/2, 3/2\}$. The functionals are degree at most two, so their unlisted remainder is exactly a quadratic bilinear map. This derives [NS, Corollary A.3].

9.2 The five positive-order moments, with all row signs

Let $R = \sqrt{2X}$, so $dX = R dR$. Choose fixed radii $X_- < X_{\text{keep}} < X_{\text{cut}} < a^2 < \inf I_{\text{pos}}$, with X_- in the leading inner collar. Take a smooth radial function κ , equal to one through X_{keep} and zero from X_{cut} onward. It may be constructed from the step σ defined in Section 6 by affine rescaling. The preliminary extension is $\tilde{\phi}_n = \kappa \phi_n^{\text{in}}$, $\tilde{U}_n = \kappa U_n^{\text{in}}$, extended by zero after a^2 , and $\tilde{E}_n = R \tilde{\phi}_n / C$. Add two U bumps and three E bumps in the reserved interval I_{pos} : $U_n = \tilde{U}_n + \sum_{j=1}^2 \alpha_j b_j^U(R)$, $E_n = \tilde{E}_n + \sum_{j=1}^3 \beta_j b_j^E(R)$, and $\phi_n = C E_n / R$ by its smooth axis extension. Choose the bumps nonnegative, of unit integral in dR , with

disjoint ordered supports. Here the exact leading values are $U_0 = 0$, $E_0 = e_* f(\eta) R^{-1-2\lambda}$, $e_* > 0$, $f > 0$. The moments to prescribe are

$$\begin{aligned} m_1 &= \int_0^\infty R U_n dR, & m_2 &= \int_0^\infty R^2 E_n dR, & m_3 &= \int_0^\infty \Pi_{n,R} dR, \\ m_4 &= \int_0^\infty R^2 \sum_{i+j=n} U_i E_j dR, & m_5 &= \int_0^\infty \left(R \sum_{i+j=n} U_i U_j - \frac{1}{2} R^2 \Pi_{n,R} \right) dR. \end{aligned}$$

Reconstruct F_n, V_n by integration and (1), and reconstruct pressure by (4) with zero axis value. The already finalized lower orders are fixed. Forward integration and the common zero axis values make the new profiles agree exactly with the inner solution through X_- . Since κ is independent of η and all bumps lie past a^2 , the entire interval $[0, a^2]$ retains parameter analyticity, even where the radial cutoff has changed the coefficient. Thus the next recursion uses finalized lower orders on the same analytic rectangle. Then

$$\begin{aligned} \Pi_{n,R} &= \frac{2E_0 E_n + \sum_{i=1}^{n-1} E_i E_{n-i} - \Omega_{n-1}}{R}, \\ m_5 &= \int_0^\infty \left(\sum U_i U_j - \frac{1}{2} \sum E_i E_j + \frac{1}{2} \Omega_{n-1} \right) dX. \end{aligned}$$

Hence the exact changes under $\delta U_n, \delta E_n$ on this patch are

$$\begin{aligned} \delta m_1 &= \int R \delta U_n dR, & \delta m_4 &= e_* f \int R^{1-2\lambda} \delta U_n dR, \\ \delta m_2 &= \int R^2 \delta E_n dR, & \delta m_3 &= 2e_* f \int R^{-2-2\lambda} \delta E_n dR, & \delta m_5 &= -e_* f \int R^{-2\lambda} \delta E_n dR. \end{aligned}$$

There are no current-order quadratic terms because $n > 0$ and $i + j = n$. For the unit-integral ordered bumps, define B_U with row powers $(1, 1 - 2\lambda)$ and B_E with powers $(2, -2 - 2\lambda, -2\lambda)$. All powers in each block are distinct for $\lambda > 0$. If m^0 are the moments before these five additions, the exact solution is

$$\alpha = -B_U^{-1} \begin{pmatrix} m_1^0 \\ m_4^0/(e_* f) \end{pmatrix}, \quad \beta = -B_E^{-1} \begin{pmatrix} m_2^0 \\ m_3^0/(2e_* f) \\ -m_5^0/(e_* f) \end{pmatrix}.$$

Thus the correction works for arbitrary finite discrepancies; no smallness or nonlinear inversion is needed at positive order. It is smooth in η because f is bounded away from zero. This independently verifies (5.14)–(5.16).

Beyond all these velocity bumps, $F_n = m_1 = 0$, hence $V_n = 0$ by (1). Choose X_+ also beyond the leading axial pulse, so $V_0 = U_0 = 0$ there. Every term of every Ω_k then vanishes, including its shifted viscosity term. Since a product $\phi_i \phi_j$ of positive total order has a positive-order factor, $\Pi_{n,X} = 0$ there and $\Pi_n = m_3 = 0$. This proves the common support of all five reconstructed profiles. It is essential to include the leading axial pulse in this choice: Ω_0 need not vanish immediately after the positive-order correction patch.

On the separate mean patch, the positive-order velocities already vanish: $E_n = U_n = 0$ beyond their bump supports, and $F_n = 0$ there by $m_1 = 0$. Therefore $V_n = 0$ as well. This assertion concerns the velocities, and does not assert that positive-order pressure has vanished at this earlier patch.

10 Weighted residual integrals and compactly supported stresses

For any smooth axisymmetric divergence-free field, expand $\partial_r(r u_r) + r \partial_z u_z = 0$ in the two tangential momentum equations. One obtains the exact conservative identities

$$\begin{aligned} r^2 \mathcal{R}_\theta &= \partial_t(r^2 u_\theta) + \partial_r(r^2 u_r u_\theta) + \partial_z(r^2 u_z u_\theta) - \partial_r(r^2 u_{\theta,r} - r u_\theta) - \partial_{zz}(r^2 u_\theta), \\ r \mathcal{R}_z &= \partial_t(r u_z) + \partial_r(r u_r u_z) + \partial_z(r(u_z^2 + p)) - \partial_r(r u_{z,r}) - \partial_{zz}(r u_z). \end{aligned}$$

For example the two surplus terms from the angular flux derivatives are $2r u_r u_\theta + r^2 u_\theta (\partial_r u_r + \partial_z u_z) = r u_r u_\theta$, precisely the cylindrical-basis term.

At index $n > 0$, the moments before physical differentiation are

$$\begin{aligned} \int r^2 u_{\theta,n} dr &= q^{3/2-A+\lambda_n} m_2, & \int r^2 \sum u_{z,i} u_{\theta,j} dr &= q^{3/2-2A+\lambda_n} m_4, \\ \int r u_{z,n} dr &= q^{1-A+\lambda_n} m_1, & \int r \left(\sum u_{z,i} u_{z,j} + p_n \right) dr &= q^{1-2A+\lambda_n} m_5. \end{aligned}$$

For the last equality, $\int R\Pi_n dR = -\frac{1}{2} \int R^2\Pi_{n,R} dR$, with zero boundary terms because pressure is smooth at zero and compactly supported. All four moments vanish as identities in η . Their t, z derivatives therefore vanish before any coefficient powers are removed. Axis parity and compact positive-order supports eliminate the radial boundary terms.

The angular axial-viscosity term at $n = 1$ needs a separate exact integral, proved below from the heat construction:

$$\int_0^\infty r^2(u_{\theta,0} - c_\infty(r^2/2)^{-A}) dr = 0. \quad (7)$$

The pure power is independent of z . The second z derivative of this convergent integral is legitimate since the exterior heat field is independent of z ; on compact physical parameter neighborhoods all remaining z derivatives have bounded radial support. For $n \geq 2$ the lower angular moment is $m_{n-1,2} = 0$. The lower axial-viscosity moment is zero also for $n = 1$ because the leading $M(\infty)$ vanishes. Consequently

$$\int_0^\infty R^2 r_{\theta,n} dR = 0, \quad \int_0^\infty R r_{z,n} dR = 0.$$

Here $\mathcal{R}_{j,n} = q^{-A-1+\lambda_n} r_{j,n}$; the shifted viscous terms have this same physical power, as checked in Section 1.

Define

$$\begin{aligned} T_{n,\theta} &= -R^{-2} \int_0^R \rho^2 r_{\theta,n} d\rho = R^{-2} \int_R^\infty \rho^2 r_{\theta,n} d\rho, \\ T_{n,z} &= -R^{-1} \int_0^R \rho r_{z,n} d\rho = R^{-1} \int_R^\infty \rho r_{z,n} d\rho. \end{aligned}$$

Then $(\partial_R + 2/R)T_{n,\theta} = -r_{\theta,n}$ and $(\partial_R + 1/R)T_{n,z} = -r_{z,n}$, including their signs. The inner equations force zero stress through X_- ; for $n \geq 2$ all exterior sources vanish past X_+ , giving common interior stress support. For $n = 1$ only angular axial viscosity of the leading field can remain beyond X_+ . The exterior is z independent past X_b , so that stress still vanishes there. Its quantitative edge bound is proved after the flat-integral calculation below.

11 Finite residuals, exact divergence after summation, and uniformity

For a smooth Cartesian profile $g(x_\perp/\sqrt{q}, \eta)$, direct repeated use of the coordinate derivatives gives

$$|\partial_{x_\perp}^\beta \partial_z^k \partial_t^\ell (q^b g)| \leq C_m (1 + |b|)^m \|g\|_{C^m} q^{b-|\beta|/2-Dk-\ell}, \quad m = |\beta| + k + \ell.$$

The factors produced by differentiating b enter the constant, whereas the power loss depends only on derivative orders. Cartesian axis regularity follows from $E_n = \sqrt{2X}\phi_n/C$ and $V_n = Xv_n$:

$$(u_{1,n}, u_{2,n}) = \frac{1}{2} q^{-1+\lambda_n} v_n(x_1, x_2) + q^{-A-1/2+\lambda_n} (\phi_n/C)(-x_2, x_1).$$

This equality is componentwise, with the same scalar profiles evaluated at (X, η) . It eliminates apparent r^{-1} singularities at the axis. On stress support, $X \geq X_a > 0$, each Cartesian derivative of r^{-1} costs a fixed further negative power of q .

For a truncation through N , equations in Sections 1–4 cancel every retained coefficient of

$$F_{\text{slow}}(u, p, T) = \mathcal{R}(u, p) + (\partial_r + 2/r)T_\theta e_\theta + (\partial_r + 1/r)T_z e_z.$$

The remaining indices are at least $N + 1$. Tangential terms have base power q^{-A-1} and radial terms base power $q^{-2A-1/2}$; the latter is the worse of the two because $A - 1/2 = h > 0$. Thus one valid explicit loss in (5.25) is

$$K_m = 2A + \frac{1}{2} + m, \quad |F_{\text{slow}}(U^{[N]})|_m \leq C_{N,m} q^{2h(N+1)-K_m}, \quad 0 < q \leq 1,$$

on each compact profile rectangle. This is independent of N as required.

The exact streamfunction and potential for a positive-order velocity are

$$S_n = q^{1-A+\lambda_n} F_n, \quad \mathcal{A}_n = (S_n/r)e_\theta = \frac{1}{2} q^{-A+\lambda_n} (F_n/X)(-x_2, x_1, 0).$$

They satisfy $u_{z,n} = \partial_s S_n$ and $ru_{r,n} = -\partial_z S_n$. For the latter identity use $\partial_z(q^{1-A+\lambda_n} F_n)$ and $1 - A = D$: the resulting bracket is exactly the negative of (1). Because $F_n/X = \int_0^1 U_n(tX, \eta) dt$, this potential is smooth at the axis.

We give the summation argument with the quantifiers needed for uniform residual bounds. Suppose potentials and remaining tuple entries satisfy $|Z_j|_m \leq C_{j,m} q^{g_j - \ell_m} (1 + |\log q|)^{P_{j,m}}$, with $g_j > 0$ nondecreasing to infinity and ℓ_m independent of j . In the present coefficient construction one may take $g_j = 2jh$ and $\ell_m = 2A + m$. For a cutoff $\chi(aq)$, equal to one on $aq \leq 1/2$ and zero on $aq \geq 1$, every differentiated term has the form $a^k \chi^{(k)}(aq) \prod_{i=1}^k \partial_z^{\alpha_i} \partial_t^{\beta_i} q$. The bound $|\partial_z^{\alpha} \partial_t^{\beta} q| \leq C_{\alpha,\beta} q^{1-D\alpha-\beta}$ supplies q^k to pair with a^k . On support of a nonzero differentiated cutoff, $1/2 \leq aq \leq 1$. Thus its derivatives cost only $q^{-D \sum \alpha_i - \sum \beta_i}$, uniformly in a .

Apply these cutoffs to the potentials, not just to their curls:

$$\text{curl}(\chi(a_j q) \mathcal{A}_j) = \chi(a_j q) \text{curl} \mathcal{A}_j + a_j \chi'(a_j q) \nabla q \times \mathcal{A}_j.$$

At each j , choose $a_j \geq 2a_{j-1}$ sufficiently large to ensure

$$\widehat{C}_{j,m} (1 + |\log q|)^{\widehat{P}_{j,m}} q^{g_j/2} \leq 2^{-j}, \quad 0 < q \leq a_j^{-1}, \quad 0 \leq m \leq j.$$

Only finitely many inequalities are imposed at this step and each holds near zero because $g_j > 0$. Each cutoff increment then has bound $2^{-j} q^{g_j/2 - \ell'_m}$, with ℓ'_m independent of j and accounting for the extra curl derivative. For $q < (2a_J)^{-1}$ the first J cutoffs equal one, so

$$|U - U^{[J]}|_m \leq 2^{-J} q^{g_{J+1}/2 - \ell'_m}, \quad J \geq \max(1, m). \quad (8)$$

The sum is locally finite because q is bounded below on compact subsets. Every curl is divergence free, and an axisymmetric angular field $b(r, z, t) e_\theta$ remains divergence free when multiplied by $\chi(a_j q)$. Consequently the resulting velocity is exactly divergence free throughout every cutoff transition.

For a fixed differential polynomial F of order s and degree d_F , write the difference of each monomial as the sum obtained by changing one factor at a time. Leibniz's rule and polynomial bounds on its fixed coefficients give, for $e = U - U^{[J]}$,

$$|F(U) - F(U^{[J]})|_m \leq C_m q^{-H_m} |e|_{m+s} (1 + |U^{[J]}|_{m+s} + |e|_{m+s})^{d_F-1}.$$

The loss H_m is independent of J . The finite partial sum has a bound $|U^{[J]}|_k \leq C_{J,k} q^{-K_k} (1 + |\log q|)^{P_{J,k}}$ with K_k independent of J . Fix m and the desired power N first. Choose J so large that the exponent in (8), after the losses in this displayed comparison, exceeds N , and the finite residual exponent exceeds $N + 1$. Keep this J fixed and decrease the neighborhood of $q = 0$ so its fixed constants and logarithms are bounded by one additional q^{-1} factor. Then the comparison is $O(q^N)$ and so is the fixed finite residual. Any flat remainder at that fixed stage remains $O(q^N)$; there is no exchange of limits between an unknown infinite family of remainders. This proves [NS, Lemma 5.4].

For the actual coefficient potentials, the extra radial velocity is exactly

$$ru_{r,n}^{\text{cut}} = q^{2nh} \left[\chi(c_n q) V_n - \frac{2\eta}{L} (c_n q) \chi'(c_n q) F_n \right], \quad u_{z,n}^{\text{cut}} = q^{-A+2nh} \chi(c_n q) U_n. \quad (9)$$

This follows by differentiating $\chi(c_n q) S_n$ in z ; its minus sign is fixed by $ru_r = -S_z$. It shows explicitly that the cutoff correction has the same radial support and vanishes on the reserved mean patch.

The diagonal choice can also impose, through order n , the finite collection of $q \partial_q, \partial_X, \partial_\eta$ seminorms of every normalized coefficient, of the bracket in (9), and for $n \geq 2$ of T_n/ζ . It can therefore make these terms at most $2^{-n} q^{nh}$. At a fixed derivative order retain the finitely many early terms; they all have their original power at least q^{2h} . Sum the tail starting at an index at least two and above the derivative order to obtain $O(q^{2h})$. It follows that the three differences in the first line of (5.42) are $O(q^{2h})$. In the common q^A velocity scale the leading radial component is $O(q^h)$ and its positive-order difference is $O(q^{3h})$. For stress the only noncompact interior quotient is the single $n = 1$ edge term established below; this yields precisely $|\mathcal{D}^I(\widehat{T} - T_0)| \leq C_I q^{2h} \zeta \delta^{-N_I}$.

For a full expansion after order N , take a tail index $J \geq \max(2N + 2, m + 1)$. The intermediate finite block retains the original orders at least $q^{2h(N+1)}$, whereas

$$\sum_{n \geq J} 2^{-n} q^{nh} \leq 2^{1-J} q^{Jh} \leq 2^{1-J} q^{2h(N+1)}.$$

Physical derivatives introduce only the already fixed losses. Thus no asymptotic coefficient is changed by using half-powers for the tail estimates. The construction is simultaneous on every compact profile range: all positive velocity and pressure corrections are supported below the single fixed X_+ , and all stresses below X_b . Outside them the exact leading heat solution has zero augmented residual. A cutoff sequence selected on one rectangle containing these supports therefore works for every larger rectangle as well.

On any compact set with $q > 0$, only finitely many positive-order profiles occur. Their parameter neighborhoods can be intersected. The leading heat factor has the one-sided smoothness proved below, and $L \geq 1 - 2h$ transfers every fixed physical derivative continuously to $\eta = \pm 1$. The resulting residual is flat as $q \downarrow 0$, uniformly for all $|\eta| \leq 1$ on compact profile ranges. This is the uniformity asserted in [NS, Proposition 5.5]; it does not assert that the singular velocity itself extends smoothly at $q = 0$.

12 The complete outer schedule and its exact moment closure

12.1 Definitions of all stages and the adjustable coefficients

Use the fixed step

$$\sigma(y) = \frac{e^{-1/y^2}}{e^{-1/y^2} + e^{-1/(1-y)^2}} \quad (0 < y < 1), \quad \sigma = 0 \quad (y \leq 0), \quad \sigma = 1 \quad (y \geq 1).$$

The schedule chooses, in order, M_d large, $T_d = e^{M_d} + 10$, $P_* > e^{T_d}$, $\lambda > 0$ small depending on these choices, and $h > 0$ sufficiently small relative to both λ and e^{-T_d} . Only after these choices is X_R made large. All estimates denoted C_{pre} below depend on the first three choices and the fixed steps, not on the later X_R or amplitude factor C . Smallness requirements use finitely many derivatives, and parameters can be chosen to meet all of them.

Write $x = X/X_R$ and, initially, $y = \log x$. The reference fields for $y \leq 0$ are $U = 4\eta$, $E = P_* f e^{y/10}$, $f = (1 + \eta^2)^{-1}$, $J_0 = \log(1 + \eta^2)$. In each next stage y is reset to zero at that stage's left endpoint, and $l = X \partial_X \log(\sqrt{2} X E)$. The fields are fully specified by the following data, with values at endpoints matched by integration of $\partial_y \log E = l - 1/2$:

1. On a unit interval, $U = 4\eta$, $l = (3/5)(1 - \sigma(y))$. On the following interval $0 \leq y \leq T_d$, set $l = 0$, $U = k(y)\eta$, $k = 4[1 - \sigma(\log(1+y)/M_d)]$. Thus k vanishes for its last eleven units, and $|k'| \leq e_d/(1+y)$ with $e_d = 4\|\sigma'\|_\infty/M_d$. The swirl on this interval is $E = P_1 f e^{-y/2}$, where $P_1 \asymp P_*$.
2. On a unit interval set $U = 0$, $l = -\lambda\sigma(y)$, and then hold $l = -\lambda$ for $T_w = 60 \log(1/\lambda)$. Reserve $(T_w - 25, T_w - 20)$, $(T_w - 20, T_w - 15)$, $(T_w - 14, T_w - 9)$, and $(T_w - 8, T_w - 3)$ for the four later corrections.
3. At $X = X_p$ start a pulse with $0 \leq y \leq 13/\lambda$, $E = e_b f e^{-(1/2+\lambda)y}$, $U = E R_b$, $\xi_b = \lambda y$. Define $\varphi_b(\xi) = \int_0^\xi \sigma(v/.02) dv$ for $\xi \geq 0$, extended by zero to negative arguments, and $R_0(\xi) = \varphi_b(\xi)[1 - \sigma(\xi - 10)]$. Set $R_b = \text{Amp}(\eta) R_0(\lambda y) + c_1 \beta_1(y) + c_2 \beta_2(y)$, where the two identical width .3 bumps have centers $13/\lambda - 3$ and $13/\lambda - 1$. Choose c_1, c_2 to make $M = J = 0$ at the pulse end.
4. From the pulse endpoint $E = e_{\text{end}} f$, set $U = 0$ permanently and

$$\log E = \log e_{\text{end}} - (1/2 + \lambda)y - \vartheta_f(y)J_0 - (1 - \vartheta_f(y)) \log 2, \quad \vartheta_f = 1 - \sigma(y/T_f).$$

A fixed sufficiently large T_f ensures $-\lambda - .1 \leq l \leq -\lambda$. Next retain the now η independent exponential for $30 \log(1/\lambda)$. Two relative swirl bumps of width .3, three and one units before its end, set $I = XH/(1 - \lambda)$ there and make their total integral $\int (E^2 - E_{\text{unedited}}^2) dy$ zero.

5. Starting at X_{rel} , vary l from $-\lambda$ to -1 in a unit interval, hold $l = -1$ for $4 \log(1/h)$, and vary from -1 to $-h$ in a unit interval, all using σ . The terminal multiplier is $f_o(y) = 1 - \rho_o \psi_o(y)$ on $0 \leq y \leq 3$, $\psi_o = 1 - \sigma((y-1)/2)$, $\rho_o = c_o h > 0$ with c_o fixed so that $0 \leq f'_o/f_o < h/4$. Define

$$Q_p = \int_0^3 \exp\left(\int_0^v (1 + l(s)) ds\right) \frac{f'_o(v)}{f_o(v)} dv, \quad l = -h + f'_o/f_o$$

on this terminal interval. Hold $l = -h$ before it until Q_s reaches Q_p . Then apply this terminal interval, and retain $l = -h$ thereafter. On and after the terminal interval the field is exactly $E_{\text{pow}} f_o$, $E_{\text{pow}} = c_\infty X^{-A}$, with c_∞ independent of η .

All stage endpoints divided by X_R are constants independent of η . The following calculations prove that the apparently adjustable stages have the stated finite solutions, and determine the pulse amplitude.

12.2 Pulse corrections, angular correction, and amplitude root

During the long interval T_w , the quantities XE and XHE grow at rates $1/2 - \lambda$ and $1/2 - 2\lambda$, respectively, whereas XE^2 decays at rate 2λ . The preceding M, J are constant once $U = 0$. Since $e^{-T_w/2} = \lambda^{30}$ and $\lambda T_w = 60\lambda \log(1/\lambda) \rightarrow 0$, it follows, with one η derivative as well, that

$$e_b \leq C_{\text{pre}} \lambda^{30}, \quad \left\| \frac{M(X_p)}{X_p e_b f} \right\|_{C^1} + \left\| \frac{J(X_p)}{X_p H(X_p) e_b f} \right\|_{C^1} \leq C_{\text{pre}} \lambda^{29},$$

$$\left\| \frac{S(X_p)}{X_p e_b^2 f^2} \right\|_{C^1} \leq C_{\text{pre}} (1 + \log(1/\lambda)).$$

For the last inequality, express S as an integral of $X(U^2 - E^2/2)$ in logarithmic coordinates. The finite preceding stages contribute a fixed constant in this scale and the long $U = 0$ stage contributes its length times a bounded factor $e^{-2\lambda y}$. These statements prove (A.14) with all scale factors retained.

The exact two equations for the pulse correction are

$$\begin{pmatrix} \int e^{s_1 y} \beta_1 dy & \int e^{s_1 y} \beta_2 dy \\ \int e^{s_2 y} \beta_1 dy & \int e^{s_2 y} \beta_2 dy \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = - \begin{pmatrix} M(X_p)/(X_p e_b f) + \text{Amp} \int e^{s_1 y} R_0(\lambda y) dy \\ J(X_p)/(X_p H(X_p) e_b f) + \text{Amp} \int e^{s_2 y} R_0(\lambda y) dy \end{pmatrix},$$

where $s_1 = 1/2 - \lambda$, $s_2 = 1/2 - 2\lambda$. Divide each row by its exponential value at the first bump center. The main pulse is zero after $11/\lambda$, leaving a gap at least $2/\lambda - 3$; both slopes exceed .4 for small λ . Its normalized moments are at most a polynomial in λ^{-1} times $(1 + |\text{Amp}|)e^{-.4(2/\lambda - 3)}$. The prior moments have an even larger exponential gap. The matrix inverse costs $O(\lambda^{-1})$. Reducing the positive exponential constant absorbs every such fixed polynomial. Hence the exact affine solution $c = c^{(0)} + \text{Amp} c^{(1)}$ satisfies

$$\|c^{(0)}\|_{C^1} + \|c^{(1)}\|_{C^1} \leq C_{\text{pre}} e^{-c/\lambda}.$$

The same argument applies to any fixed derivative, with additional powers of λ^{-1} for ξ_b derivatives. This proves exact $M = J = 0$ and (A.15).

For angular momentum put $r_I = I/(XH)$. Since $I_X = H$, $r'_I + (1 + l)r_I = 1$. During the uniform interval following interpolation, $r_I - (1 - \lambda)^{-1}$ decays at rate $1 - \lambda$. At the first swirl bump center its discrepancy is $O_{C^1}(C_{\text{pre}} \lambda^{28})$; the exponent 28 is a valid relaxed bound for $e^{-(1-\lambda)(30 \log(1/\lambda) - 3)}$. For the relative edit $E \mapsto E(1 + c_1 \beta_1 + c_2 \beta_2)$, the linear angular row has logarithmic weight $e^{(1-\lambda)y}$ and the linear pressure row has weight $2e^{(-1-2\lambda)y}$. Their slope separation tends to 2, so their normalized inverse is bounded. The pressure row alone has its exact quadratic remainder. The contraction proved above yields coefficients $O_{C^1}(C_{\text{pre}} \lambda^{28})$, with both exact targets met. All fixed radial derivatives of these bumps are of the same size, so they preserve $E > 0$ and $l \leq -\lambda/2$.

After the correction, $I = XH/(1 - \lambda)$ and $U = M = J = 0$ at the start of the exterior transition. The exact radial identities of (4.16) give

$$Q_s = -1 + (1 - h)I/(XH) = (\lambda - h)/(1 - \lambda).$$

Subsequently $Q'_s + (1 + l)Q_s = -l - h$. During the first transition the nonnegative source and bounded integrating factor retain a positive $c\lambda$ lower bound. The $l = -1$ hold gives $Q'_s = 1 - h$, so its end value is comparable to $\log(1/h)$. The next unit transition changes this by bounded factors. On the other hand $Q_p \asymp \rho_o \asymp h$: f_o is bounded above and below, the interval has fixed length, and $\int f'_o = \rho_o$. The intervening $l = -h$ solution is $Q_{\text{in}} e^{-(1-h)y}$; the exact required length is $(1 - h)^{-1} \log(Q_{\text{in}}/Q_p) > 0$ for sufficiently small h . On the terminal interval the resulting solution is

$$Q_s(y) = \int_y^3 \exp \left(\int_y^v (1 + l(s)) ds \right) \frac{f'_o(v)}{f_o(v)} dv.$$

Differentiating this expression proves the equation and its zero right endpoint. Thus $Q_s = 0$ afterward and $I = XH_{\text{pow}}/(1 - h)$. Since $\int_0^X H_{\text{pow}} dx = XH_{\text{pow}}/(1 - h)$ for $h < 1$, this is exactly $\int_0^\infty (H - H_{\text{pow}}) dX = 0$. The subtraction is integrable at zero, and is identically zero at large X at this pre-heat stage.

For the other moment, note the exact identities on the $l = -1$ hold:

$$\frac{(XE^2)_{\text{end}}}{(XE^2)_{\text{start}}} = h^8, \quad \frac{E_{\text{end}}}{E_{\text{start}}} = h^6.$$

They follow by integrating $(\log(XE^2))' = 2l = -2$ and $(\log E)' = -3/2$ over $4 \log(1/h)$. The first transition and hold have total $\int XE^2 dy \leq CX_{\text{rel}} e_{\text{rel}}^2$. After the hold $l \leq -3h/4$, so the remaining integral is at most

$CX_{\text{rel}}e_{\text{rel}}^2h^8\int_0^\infty e^{-3hy/2}dy \leq CX_{\text{rel}}e_{\text{rel}}^2h^7$. This proves the h uniform bound (A.18). Interpolation and its next hold cost at most $C(1+\log(1/\lambda))X_{\text{end}}e_{\text{end}}^2$; their first parameter derivatives satisfy the same bounds because f^ϑ and its derivatives are bounded and the exterior is parameter independent.

There is also an exact algebraic specification of the amplitude root. Substitute the affine c above and write $R_b = R^{(0)} + \text{Amp} R^{(1)}$. In the full $S(\infty)$, every term outside $\int_{\text{pulse}} E^2 R_b^2 dX$ is independent of Amp . Consequently this moment is an explicitly defined quadratic polynomial $a(\eta)\text{Amp}^2 + b(\eta)\text{Amp} + c(\eta)$, where, without dropping any end bumps,

$$\begin{aligned} a &= \int_{\text{pulse}} E^2 (R^{(1)})^2 dX > 0, & b &= 2 \int_{\text{pulse}} E^2 R^{(0)} R^{(1)} dX, \\ c &= S_{\text{no axial pulse}}(\infty) + \int_{\text{pulse}} E^2 (R^{(0)})^2 dX. \end{aligned}$$

Changing variables $\xi_b = \lambda y$ in this polynomial gives

$$\frac{\lambda S(\infty)}{X_p e_b^2 f^2} = \text{Amp}^2 K_b - \frac{1 - e^{-26}}{4} + \mathcal{E}(\text{Amp}, \eta), \quad K_b = \int_0^{13} e^{-2\xi} R_0(\xi)^2 d\xi,$$

$$\|\mathcal{E}\|_{C^1([.9, 1.2] \times [-1, 1])} \leq C_{\text{pre}} \lambda (1 + \log(1/\lambda)).$$

Indeed the prior S and the post-pulse negative $E^2/2$ integral have precisely the bounds just proved after multiplication by λ , whereas the affine end bumps are exponentially small; all cross terms are retained in $\mathcal{E}$. On $.02 \leq \xi \leq 9$, $R_0 \geq \xi - .02$, and globally $0 \leq R_0 \leq \xi$. Therefore

$$\frac{e^{-.04}}{4} [1 - e^{-17.96}(1 + 17.96 + 2(8.98)^2)] \leq K_b \leq \frac{1}{4},$$

whose lower bound exceeds .20. At $\text{Amp} = .9$ the principal value is below $-.047$; at 1.2 it exceeds .038; its derivative is at least .36. For sufficiently small λ the exact polynomial has a unique root on this interval. It is the positive branch $(-b + \sqrt{b^2 - 4ac})/(2a)$, with its nonzero derivative already proved by the bracket estimate. Implicit differentiation gives $|\partial_\eta \text{Amp}| \leq C_{\text{pre}} \lambda (1 + \log(1/\lambda))$. This constructs the smooth amplitude and proves both required moments, including all terms in (A.19)–(A.20).

13 Pressure datum and the cone margins on every outer stage

13.1 Exact pressure and its parameter analyticity

Omit only the swirl bumps whose total $\int E^2 dy$ was just proved to be zero. The remaining scheduled field has the form $E = c(y)f(\eta)^{\vartheta(y)}$, $0 \leq \vartheta \leq 1$, with c, ϑ and all interval lengths independent of η and X_R . On one simply connected complex neighborhood of $[-1, 1]$ avoiding $\pm i$, choose the logarithm of f . The holomorphic functions $\exp(\vartheta \log f)$ are uniformly bounded for $0 \leq \vartheta \leq 1$. The squared integrand is bounded by $Ce^{y/5}$ as $y \rightarrow -\infty$ and $Ce^{-(1+2h)y}$ as $y \rightarrow \infty$. Uniform convergence of the integral on compact complex subsets therefore defines a holomorphic function

$$\Pi_{\text{ax}}(\eta) = -\frac{1}{2} \int_{-\infty}^{\infty} E_{\text{sched}}(y, \eta)^2 dy.$$

It is even, independent of X_R , and its derivative is an integral of $\vartheta J_0' c^2 f^{2\vartheta}$. This has the sign of η and is strict for $\eta \neq 0$ because the reference interval has $\vartheta = 1$. The reference interval contributes exactly $-\frac{1}{2} P_*^2 f^2 \int_{-\infty}^0 e^{y/5} dy = -\frac{5}{2} P_*^2 f^2$. Restoring the pressure-neutral bumps leaves the total unchanged. Hence the actual pressure is exactly

$$\Pi(X, \eta) = -\frac{1}{2} \int_{\log(X/X_R)}^{\infty} E(v, \eta)^2 dv.$$

This proves all assertions of [NS, Lemma A.5]. It also proves the precise preservation map: an edit of zero total $\int \delta(E^2)/(2X) dX$ keeps forward pressure unchanged before and after its support; it need not keep pressure unchanged inside that support.

13.2 The exact quantities used for the cone test

Retain

$$\begin{aligned} W &= 1 - 2D\eta M/X - dM_\eta/X, \quad H_c = D\eta + dU, \quad a = 2 - 2l, \quad b_s = 2XU_X/E, \\ Q_s &= -W + \frac{(1-h)I - D\eta I_\eta - dJ_\eta + 2(h-D)\eta J}{XH}, \\ N_s &= -WU + \frac{D(M - \eta M_\eta) + 4h\eta S - dS_\eta}{X} + 4A\eta\Pi - d\Pi_\eta. \end{aligned}$$

These formulas follow by integrating the source equations

$$Q'_s + (1+l)Q_s = -Wl - h(1-2\eta U) - H_c(\log E)_\eta,$$

$$N'_s + N_s = -WU' - A(1-2\eta U)U - H_cU_\eta - d\Pi_\eta + 4A\eta\Pi + 2\eta X\Pi_X.$$

For example $(XW)_X = 1 - 2D\eta U - dU_\eta$; integration by parts of $-WXH_X$ produces $-XWH$ and the displayed I, J terms. In the pressure terms use $\int_0^X \Pi, dx = X\Pi - \int_0^X E^2/2, dx$. These operations produce the exact S and M terms above, and establish the map from the five original moments to stress coefficients in [NS, (4.16)].

Set $w = N_s/(EQ_s)$ where $Q_s > 0$, and $p_{s,1} = XQ_s/L$, $p_{s,2} = wp_{s,1}$. To explain precisely the sufficient test used below, put

$$c = 1 - b_s w/a, \quad j = w + b_s/a, \quad v_s = a + b_s^2/a, \quad P_c = p_{s,1}c, \quad J_c = p_{s,1}j.$$

Direct expansion gives

$$2c^2 - (v_s - 2)j^2 = \left(1 + \frac{b_s^2}{a^2}\right) \left[2 - 2b_s w - \frac{b_s^2}{a} - (a-2)w^2\right].$$

Thus on a compact set with $a > 0$, $a - b_s w > 0$, and the bracket positive, there are constants $c_* > 0$, $G_* > 0$, and $B < \infty$ bounding these expressions away from zero and bounding v_s, cv_s . For $p_{s,1} \geq \max((B+2)/c_*, 8B/G_*)$, $P_c > \max(2, v_s)$ and

$$\frac{2(P_c - v_s)^2 - (v_s - 2)J_c^2}{p_{s,1}^2} \geq G_* - 4B/p_{s,1} \geq G_*/2.$$

For $v_s > 2$ these are exactly the admissible cone inequalities; for $v_s \leq 2$ they imply the relaxed cone. Equivalence follows by solving the quadratic $2(P_c - v)^2 - (v - 2)J_c^2 = 0$, whose smaller root is the function in (4.21), between 2 and P_c . This verifies the finite threshold used in Lemma 4.5.

13.3 Reference and axial-decrease stages

On the ideal reference interval $U = 4\eta$, $W = 1 - 4L$, $l = 3/5$, so

$$Q_s = \frac{(3/5)(4L - 1) - h(1 - 8\eta^2) + (D + 4d)\eta J'_0}{8/5} > c > 0$$

for the chosen small h . Here $a = 4/5$, $b_s = 0$. Its N_s geometric source is $O(|\eta|)$ and its pressure source is $O(P_*^2|\eta|)$; these estimates follow by substituting the constant U and the bounded backward pressure integral above. The vanishing at zero follows from even pressure and odd U , and one parameter derivative removes the $|\eta|$ factor. The $N'_s + N_s$ convolution has finite bounds on each prescribed reference interval $x \geq x_- > 0$.

During the first unit transition, $b_s = 0$, $.8 \leq a \leq 2$, and the angular source is at least $-h$ since $W < 0$, $l \geq 0$, and $H_c J'_0 \geq 0$. The initial positive lower bound and variation of constants retain $Q_s > c$ for small h . With $b_s = 0$ and $a \leq 2$, the second sufficient cone bracket is automatically at least 2.

During axial decrease, $l = 0$, $a = 2$, $b_s = 2k'\eta/E$. The angular source is exactly $-h(1 - 2k\eta^2) + (D + dk)\eta J'_0$. Since $\eta J'_0 = 2\eta^2/(1 + \eta^2) \asymp \eta^2$, convolution with e^{-y} and $h \ll e^{-T_d}$ gives $Q_s \geq c(\eta^2 + e^{-y})$ for $0 \leq y \leq T_d$. The geometric axial source is $O(|\eta|)$. The pressure source is $O(E^2|\eta|)$ because subsequent scheduled swirl decays at least $e^{-\Delta y/2}$ and $|\partial_\eta \log E| \leq |J'_0|$; the later pressure-neutral angular bumps may be omitted from the backward integral. Solving the N_s equation yields

$$|N_s|/E^2 \leq C|\eta|(e^{T_d}/P_*^2 + 1 + y) \leq C|\eta|(1 + y).$$

Consequently $|b_s w| \leq C|k'\eta^2(1 + y)/(\eta^2 + e^{-y}) \leq Ce_d$ and $b_s^2 \leq Ce_d^2 e^{T_d}/P_*^2$. Choose M_d so Ce_d is small, then $P_* > e^{T_d}$ sufficiently large. Both required strict margins follow, uniformly in η and this whole stage.

13.4 Intermediate power law and the complete pulse

Write $M/X = \bar{k}\eta$ during axial decrease. Then $\bar{k}' = k - \bar{k}$. Because $0 \leq k \leq 4$ and its final eleven units are zero, $L\bar{k} < 1/2$ at the next transition. When $U = 0$ this gives $W \geq 1/2$. Write $c_\eta = D\eta J'_0 \asymp \eta^2$. The angular source is $(-l)W - h + c_\eta$. During the unit transition $0 \geq l \geq -\lambda$, the prior positive lower bound persists; w has a finite bound depending only on earlier choices. Thus $\sqrt{\lambda}|w|$ is small. The cone is admissible as soon as $l < 0$, and relaxed at its initial endpoint.

On the constant-slope interval, $\bar{k}(y) = \bar{k}(0)e^{-y}$ and

$$Q'_s + (1 - \lambda)Q_s = \lambda - h + c_\eta - \lambda L\bar{k}(0)e^{-y}.$$

The right side is at least $\lambda/4 + c_\eta$ for small h/λ . Its explicit convolution therefore gives

$$Q_s \geq c_{\text{pre}}(\eta^2 + \lambda + e^{-(1-\lambda)y}), \quad Q_s - \frac{\lambda - h + c_\eta}{1 - \lambda} = O_{C^1}(C_{\text{pre}}(1 + y)e^{-(1-\lambda)y}).$$

There is no geometric N_s source when $U = 0$; the pressure source is $O(E^2|\eta|)$. Its convolution gives $|N_s/E| \leq C_{\text{pre}}|\eta|(1 + y)e^{-(1/2-\lambda)y}$. Using $|\eta|/(\eta^2 + e^{-(1-\lambda)y}) \leq \frac{1}{2}e^{(1-\lambda)y/2}$ proves

$$\sqrt{\lambda}|w| \leq C_{\text{pre}}\sqrt{\lambda}(1 + y)e^{\lambda y/2} = o(1) \quad (0 \leq y \leq 60 \log(1/\lambda)).$$

Since $a = 2 + 2\lambda$, $b_s = 0$, this supplies the strict cone bracket.

During the pulse set $m = (M/X)/E$ and $\beta = 1/2 - \lambda$. The exact identity $m' + \beta m = R_b$ follows by differentiating M/X and E . With the pulse extended flatly to the left, write its convolution as $\int_0^\infty e^{-\beta v} R_b(\lambda(y - v)) dv$ plus $m(0)e^{-\beta y}$. Taylor's formula under this integral gives

$$m = R_b/\beta - \lambda \partial_{\xi_b} R_b/\beta^2 + O(\lambda^2) + m(0)e^{-\beta y},$$

because the integral of $v^2 e^{-\beta v}$ is finite uniformly for $\beta > .4$. The end bumps have exponentially small fixed ξ_b derivatives. The same estimate holds after one η derivative.

Since $E \leq C_{\text{pre}}\lambda^{30}$, all terms from $W = 1 - 2D\eta Em - d(Em)_\eta$ and from dU in the angular source have size $O(C_{\text{pre}}E)$, including one parameter derivative. The initial constant-slope estimate and exponential convolution give

$$Q_s = \frac{\lambda - h + c_\eta}{1 - \lambda} + O(C_{\text{pre}}\lambda^{28}).$$

The backward pressure and its first derivative are $O(E^2)$. For S , integration over $0 \leq \lambda y \leq 13$ bounds $S/(XE^2)$ by $C_{\text{pre}}e^{26}/\lambda$ in C^1 . The numerator accumulates at most a fixed integral in ξ_b divided by λ and the denominator loses at most e^{-26} ; this explains the constant e^{26}/λ , not $e^{26/\lambda}$. Substituting $M/X = Em$ in the exact moment formula for N_s gives

$$N_s/E = -R_b + (D + c_\eta)m - D\eta m_\eta + O(C_{\text{pre}}\lambda^{27}).$$

For example $D(M - \eta M_\eta)/(XE) = D(m - \eta m_\eta + \eta J'_0 m)$, giving exactly $(D + c_\eta)m$. The S and pressure remainders are at most $C E e^{26}/\lambda$ in this quotient, which is smaller than the stated bound for sufficiently small λ . The difference $W - 1$ similarly costs $O(E)$.

The amplitude derivative and the initial-moment bounds show $|m_\eta| \leq C_{\text{pre}}\lambda(1 + \log(1/\lambda))$. Since $|\eta|/(\lambda + c_\eta) \leq C/\sqrt{\lambda}$, the quotient of ηm_η by Q_s is $o(1)$. The preceding formulas therefore imply, uniformly throughout the pulse,

$$w = 2R_b - C_d \partial_{\xi_b} R_b + o(1), \quad C_d = \frac{\lambda(D + c_\eta)(1 - \lambda)}{\beta^2(\lambda - h + c_\eta)} \leq 2 + o(1).$$

The coefficient of R_b before this last replacement is exactly $(1 - \lambda)/\beta$, since $D + c_\eta - \beta = \lambda - h + c_\eta$. The quotient defining C_d is maximal at $c_\eta = 0$, because its derivative in c_η has sign $\lambda - h - D < 0$; $h/\lambda \rightarrow 0$ gives the bound. Also $b_s = -(1 + 2\lambda)R_b + 2\lambda \partial_{\xi_b} R_b = -R_b + O(\lambda)$. Apart from exponentially small end bumps, $R_b \geq 0$ and $\partial_{\xi_b} R_b \leq 1.2$, since the declining cutoff has a nonpositive contribution. The exact upper bounds for the two leading quadratics are

$$\sup_{R \geq 0} (-2R^2 + 2.41R) = 2.41^2/8 < .74,$$

$$\sup_{R \geq 0} (-3.5R^2 + 4.82R) = 4.82^2/14 < 1.68.$$

The uniformly $o(1)$ remainders can be made smaller than the strict numerical gaps. Hence $a - b_s w > 1$ and $2b_s w + b_s^2/a + (a - 2)w^2 < 2$ on the complete pulse, including its descending and correction parts. Here $v_s \geq a = 2 + 2\lambda > 2$.

13.5 Interpolation and exterior transitions

After the pulse $U = M = J = 0$, so $W = 1$. The angular source during interpolation is $-l - h + \vartheta_f c_\eta$, giving $Q_s \geq c\lambda$. The tiny angular correction preserves this bound. Since the complete amplitude solve gives $S(\infty) = 0$,

$$S(X) = \frac{1}{2} \int_X^\infty E(X')^2 dX'.$$

Before exterior release, $l \leq -\lambda/2$ and the later release has the h uniform integral bound proved above. Therefore $(|S| + |S_\eta|)/X \leq CE^2/\lambda$ and $|\Pi| + |\Pi_\eta| \leq CE^2$. The exact moment formula yields $|w| \leq CE/\lambda^2 \ll 1$, because pulse end reduces E exponentially in $1/\lambda$. Here $b_s = 0$ and $2 < a \leq 2 + 2\lambda + .2 + o(1)$, so both cone margins remain strict.

On the parameter-independent release and tail, the same backward integrals give the exact expression

$$N_s = 2\eta \int_y^\infty (he^{v-y} - A)E(v)^2 dv.$$

Since $l \leq -3h/4$, $E(v)^2 \leq E(y)^2 e^{-(1+3h/2)(v-y)}$. Thus the he^{v-y} integral is at most $\frac{2}{3}E(y)^2$ and the other at most $E(y)^2/(1+3h/2)$. In particular $|N_s| \leq CE^2$ with a constant independent of small h . Before the steep hold, $Q_s \geq c\lambda$ and the amplitude is already small relative to λ . During the hold it remains above this bound. After the hold, $E \leq Ce_{\text{rel}}h^6$, while $Q_s \geq ch$ up to terminal coordinate $y = 1/2$. The latter follows from its explicit positive terminal integral and its intervening monotone decay to $Q_p \asymp h$. Consequently $|w| \leq CE/Q_s \ll 1$, even if $h \ll e_{\text{rel}}$. Here $b_s = 0$, $2 < a \leq 4$, so the sufficient cone test has uniform strict slack.

All these intervals are compact once the sequential parameters are fixed. Under the exact moment scaling $X = X_R x$, every X_R factor cancels in Q_s, N_s , while $p_{s,1} = X_R x Q_s / L$ grows linearly with X_R . Thus one finite increase of X_R meets the threshold proved above on the whole designated interval. This completes the local reconstruction of [NS, Proposition A.4], including each part of its cone verification.

14 The heat factor, its exact equation, and the compensated replacement

14.1 A positive solution of the radial swirl heat equation

Define for $Z \geq 0$,

$$\mathcal{H}(Z) = \frac{1}{\Gamma(1+h)} \int_0^\infty e^{-v} v^h (1+Zv)^{-h} dv.$$

Dominated differentiation gives for every nonnegative integer m

$$\mathcal{H}^{(m)}(Z) = \frac{(-1)^m (h)_m}{\Gamma(1+h)} \int_0^\infty e^{-v} v^{h+m} (1+Zv)^{-h-m} dv, \quad \mathcal{H}^{(m)}(0) = (-1)^m (h)_m (1+h)_m.$$

For fixed m the integrand after deleting the last factor is integrable uniformly for $Z \geq 0$. Hence $\mathcal{H}$ is positive, smooth up to zero from the right, and has bounded fixed derivatives on every bounded nonnegative interval (in fact on the entire half-line for absolute derivatives). For $0 < h \leq h_0 < 1/2$ and fixed $m \geq 1$, the last factorial product is at most $C_m h$. Taylor's formula therefore proves

$$\mathcal{H}(Z) = 1 - h(1+h)Z + O_h(Z^2), \quad |\mathcal{H}(Z) - 1| + |Z\mathcal{H}'(Z)| \leq ChZ$$

on a fixed bounded nonnegative interval.

The integral solves the required ODE by an exact polynomial calculation. Multiply $Z^2\mathcal{H}'' + (1+2(1+h)Z)\mathcal{H}' + h(1+h)\mathcal{H}$ by $\Gamma(1+h)/h$, and use common integrand $e^{-v}v^h(1+Zv)^{-h-2}$. Its remaining polynomial is

$$(1+h)Z^2v^2 - (1+2(1+h)Z)v(1+Zv) + (1+h)(1+Zv)^2 = (1+h) - v - Zv^2.$$

This is precisely the polynomial from $\partial_v[e^{-v}v^{1+h}(1+Zv)^{-1-h}]$ in the same common factor. The boundary values at zero and infinity are both zero. Thus

$$Z^2\mathcal{H}'' + (1+2(1+h)Z)\mathcal{H}' + h(1+h)\mathcal{H} = 0. \tag{10}$$

Set $K(r, t) = c_\infty s^{-A} \mathcal{H}(2\tau/s)$, $s = r^2/2$, $Z = 2\tau/s$. Direct differentiation gives

$$K_t = -2c_\infty s^{-A-1} \mathcal{H}',$$

$$(\partial_{rr} + r^{-1} \partial_r - r^{-2})K = 2c_\infty s^{-A-1} \{Z^2 \mathcal{H}'' + (2A+1)Z \mathcal{H}' + (A^2 - 1/4)\mathcal{H}\}.$$

Since $2A+1 = 2(1+h)$ and $A^2 - 1/4 = h(1+h)$, (10) proves the heat equation, with its swirl term $-r^{-2}K$.

Normalize the positive integrand defining $\mathcal{H}$ to a probability density. Then $-Z\mathcal{H}'/\mathcal{H}$ is its average of $hZv/(1+Zv)$, so $0 \leq -Z\mathcal{H}'/\mathcal{H} < h$. It follows that $rK_r/(2K) = -A - Z\mathcal{H}'/\mathcal{H} < -1/2$, in particular $K_r < 0$. At $Z = 0$ this is still strict because $h > 0$. Finally $Z = 2(1-\eta^2)/X$ gives only factors $-4\eta/X$ and $-4/X$ when differentiated in η , never a denominator $1-\eta^2$. Consequently all fixed mixed X, η derivatives have one-sided limits at $\eta = \pm 1$, and for each fixed logarithmic radial and parameter derivative,

$$\frac{E_{\text{heat}}}{E_{\text{pow}}} = \mathcal{H}(2d/X) = 1 - \frac{2h(1+h)d}{X} + O_{h,j,m}(X^{-2}).$$

This proves [NS, Lemma A.6] on its stated nonnegative domain. No analytic continuation through negative Z is used or required.

14.2 The three heat discrepancies and their exact cancellation

Let $y = \log(X/X_{\text{tail}})$, $X_K = e^2 X_{\text{tail}}$, $X_b = e^3 X_{\text{tail}}$, and choose $\chi_K = 0$ for $y \leq .2$, $\chi_K = 1$ for $y \geq .5$. Replace the reference swirl E_{cl} by

$$E_{\text{cl}}[1 + \chi_K(y)(\mathcal{H}(2d/X) - 1)].$$

For $x = X/X_K \geq 1$ and its reference amplitude e_K , the preceding derivative bounds give exactly

$$|(X\partial_X)^j \partial_\eta^m (E - E_{\text{cl}})| \leq C_{j,m} e_K X_K^{-1} x^{-A-1}.$$

The profile is comparable to $e_K x^{-A}$ with constants depending on the fixed schedule. Thus the pressure, S , and subtracted angular discrepancies obey, after m parameter derivatives,

$$|\Delta C_p|_m \leq C_m \frac{e_K^2}{X_K} \int_1^\infty x^{-2A-2} dx, \quad |\Delta S|_m \leq C_m e_K^2 \int_1^\infty x^{-2A-1} dx,$$

$$\left| \Delta \int (H - H_{\text{pow}}) dX \right|_m \leq C_m e_K X_K^{1/2} \int_1^\infty x^{-A-1/2} dx = C_m e_K X_K^{1/2} / h.$$

The powers arise respectively from $\delta(E^2)dX/(2X)$, $-\delta(E^2)dX/2$, and $\sqrt{2X}\delta E, dX$. The final $1/h$ is retained: h has already been fixed before X_R grows.

At the earlier second reserved patch write $x_* = X/X_*$, $E = e_* f x_*^{-1/2-\lambda}$, $U = 0$, and add $\delta E = e_* \sum_{i=1}^3 c_i(\eta) \beta_i(x_*)$. Divide the three rows by e_*^2 , $X_* e_*^2$, and $X_*^{3/2} e_*$. The exact linear weights are then

$$f x_*^{-3/2-\lambda}, \quad -f x_*^{-1/2-\lambda}, \quad \sqrt{2} x_*^{1/2}.$$

The first two rows have exact quadratic terms and the third is linear. The moment determinant is nonzero by Section 3; its inverse is uniform in η . Since X_*/X_K and e_*/e_K are fixed, all three normalized discrepancies are $O_m(X_K^{-1})$. The contraction gives the exact correction with $\|c\|_{C^m} \leq C_m X_K^{-1}$ for any prescribed finite m , after choosing X_R large. More generally its higher fixed derivative bounds follow by implicit differentiation with the same small forcing: the Jacobian stays invertible and every differentiated equation has only lower derivative products and an $O_m(X_K^{-1})$ inhomogeneity. This justifies the stated O_m estimates without selecting infinitely many smallness thresholds at once.

Both edits have $U = 0$, so M, J are unchanged exactly. All three other total moments are restored by this solve. The total pressure integral is therefore unchanged, so the analytic axis pressure and every profile before the patch remain unchanged. On the compact logarithmic interval up to $y = .5$, the exact moment formulas transfer the $O(X_K^{-1})$ changes of profiles and moments to Q_s, N_s , costing one more η derivative. Indeed substituting $I = X_R^{3/2} \hat{I}$, $J = X_R^{3/2} \hat{J}$, $M = X_R \hat{M}$, $S = X_R \hat{S}$, $H = X_R^{1/2} \hat{H}$ cancels every X_R in those formulas. Positive lower bounds for E, Q_s and the already strict cone margins persist by continuity, uniformly on this fixed compact interval. This proves [NS, Proposition A.7] and its preservation claims; the remaining endpoint is treated next rather than being inferred from a compact-interior estimate.

15 Exterior stress, exact flat factors, and the order-one edge estimate

15.1 Convergent moment subtraction and backward stress formulas

The leading regular axis and exact matching of the five moments are inputs from the profiles lane at this stage, precisely as in [NS, Lemma A.8]. Once supplied, $M(\infty) = J(\infty) = S(\infty) = 0$ and the compensated subtracted angular moment gives

$$I(X) = \frac{XH_{\text{pow}}(X)}{1-h} - \int_X^\infty (H - H_{\text{pow}})dx, \quad S(X) = \frac{1}{2} \int_X^\infty E^2 dx.$$

The exact pressure is the backward integral already established. These identities, substituted into the moment formulas, determine the same forward stress everywhere in the tail.

To prove the backward physical stress representation rather than assume its boundary constants vanish, put $K_{\text{pow}} = c_\infty(r^2/2)^{-A}$. For $r \rightarrow \infty$, uniformly on bounded $\tau \geq 0$ intervals, the heat expansion gives

$$K - K_{\text{pow}} = O_h(\tau r^{-3-2h}), \quad K_t = O_h(r^{-3-2h}), \quad r^2 K_r - rK = O_h(r^{-2h}).$$

Thus the weighted difference and its time derivative are dominated at infinity by Cr^{-1-2h} , whose integral from R_* is $R_*^{-2h}/(2h)$. At zero the subtracted pure power has weight r^{1-2h} , integrable for $h < 1$, while the regular field is integrable by axis smoothness. The substitution $r = \sqrt{q}R$, $dX = R dR$ proves exactly

$$\int_0^\infty r^2(u_\theta - K_{\text{pow}})dr = q^{3/2-A} \int_0^\infty (H - H_{\text{pow}})dX = 0.$$

Time differentiation is justified by the displayed integrable majorant on compact physical parameter neighborhoods, proving (7) and the zero integrated time term. The angular transport flux has zero axial integral because it is $q^{3/2-2A}J(\infty) = 0$; its radial boundary vanishes by regularity at zero and $U = M = V_0 = 0$ in the exterior. The radial viscous boundary is zero by the r^{-2h} estimate above.

For the axial component $\int ru_z dr = q^{1-A}M(\infty) = 0$. Fubini with the absolutely convergent positive pressure integrand yields

$$\int_0^\infty rp(r)dr = - \int_0^\infty \frac{u_\theta(r')^2}{r'} \left(\int_0^{r'} r dr \right) dr' = -\frac{1}{2} \int_0^\infty r' u_\theta(r')^2 dr'.$$

The pressure has decay $O(r^{-2-4h})$, hence $r^2 p \rightarrow 0$, and its contribution to the integrated axial flux is exactly $q^{1-2A}S(\infty) = 0$. All axial radial transport and viscosity boundaries vanish as well. Therefore both leading residuals have zero weighted totals, so the stresses defined with a minus sign by integration from the axis are exactly

$$T_\theta(r) = r^{-2} \int_r^\infty r'^2 \mathcal{R}_\theta^{(0)}(r') dr', \quad T_z(r) = r^{-1} \int_r^\infty r' \mathcal{R}_z^{(0)}(r') dr'. \quad (11)$$

For $X \geq X_b$ the field is $(u_r, u_\theta, u_z) = (0, K, 0)$ and pressure is its z independent backward radial integral. The heat equation makes both leading residuals zero there, proving zero exterior stress.

15.2 An exact flat integral map

For $c > 0$, integer $j \geq 0$, and a smooth $b(u, \eta)$ on a closed compact parameter rectangle, substitute $u = \delta(1 + \delta^2 v)^{-1/2}$. Then $du = -\frac{1}{2}\delta^3(1 + \delta^2 v)^{-3/2}dv$ and $c/u^2 = c/\delta^2 + cv$. Hence

$$\int_0^\delta e^{-c/u^2} u^{-j} b(u, \eta) du = \frac{1}{2} e^{-c/\delta^2} \delta^{3-j} B(\delta, \eta), \quad (12)$$

where the coefficient is explicitly

$$B(\delta, \eta) = \int_0^\infty e^{-cv} (1 + \delta^2 v)^{(j-3)/2} b(\delta/\sqrt{1 + \delta^2 v}, \eta) dv.$$

Every fixed derivative of the integrand is bounded by e^{-cv} times a polynomial in v , uniformly in the compact parameters and $0 \leq \delta \leq \delta_0$. This follows directly from differentiating $\delta(1 + \delta^2 v)^{-1/2}$; its denominators are at least one and all its numerators are polynomials in δ, v . Dominated differentiation proves smoothness of B with all fixed derivatives bounded, and $B(0, \eta) = b(0, \eta)/c$. This proves the exact map in [NS, Lemma A.9], including the factor $1/2$.

15.3 Positive angular stress and the limiting direction

For terminal $y \geq .5$, the heat cutoff is one, so $u_\theta = Kf$, $f = f_o(y)$, and $u_r = u_z = 0$. At fixed z, t , $y_r = 2/r$, while $y_t = (qL)^{-1}$, $y_z = -2\eta/(q^D L)$. The leading angular residual, which omits axial viscosity, is exactly

$$\mathcal{R}_\theta^{(0)} = Kf'/(qL) - K(f_{rr} + r^{-1}f_r) - 2K_r f_r.$$

Differentiate the physical pressure $p = -\int_r^\infty K(r')^2 f(r')^2 dr'/r'$. Since $K_z = 0$ and $dr'/r' = dy'/2$, this gives the sign and factor

$$p_z(r) = \frac{2\eta}{q^D L} \int_y^3 K(y')^2 f(y') f'(y') dy'.$$

Inserting the angular residual into (11) and integrating its viscous terms by parts gives

$$T_\theta = Kf_r + \frac{1}{r^2} \int_r^\infty (r'K - r'^2 K_{r'}) f_{r'} dr' + \frac{1}{qLr^2} \int_r^\infty r'^2 K f' dr'. \quad (13)$$

To check the integration sign, the term $-\int r'^2 K f_{r'} dr'$ contributes

$$r^2 K f_r + \int (2r'K + r'^2 K_{r'}) f_{r'} dr'.$$

The two remaining viscous terms leave exactly $r'K - r'^2 K_{r'}$ in this integral. Every term in (13) is nonnegative since $K > 0$, $K_r < 0$, $f' \geq 0$. The remaining interval has logarithmic length at most 2.5; radii and heat factors there have uniformly bounded ratios. Since $\int_y^3 f' dv = \rho_o \psi_o(y)$, its last term implies

$$T_\theta \geq c \frac{rK}{qL} \rho_o \psi_o(y) > 0 \quad (y < 3).$$

The pressure formula and its next radial integral similarly give

$$|p_z| \leq Cq^{-D} K^2 \rho_o \psi_o, \quad |T_z| \leq Crq^{-D} K^2 \rho_o \psi_o, \quad |T_z/T_\theta| \leq Cq^{1-D} K = CE_{\text{pow}} \mathcal{H}(2d/X).$$

The final equality uses $1 - D = A$. The h^6 amplitude reduction makes this ratio small. Furthermore, since $\partial_y Z = -Z$,

$$a = 2 + 2h + 2Z\mathcal{H}'/\mathcal{H} - 2f'/f.$$

For sufficiently large X_R , $-Z\mathcal{H}'/\mathcal{H} < h/4$ uniformly in η ; also $f'/f < h/4$ by construction. Hence $2 + h < a \leq 2 + 2h$ and $b_s = 0$. The exact stress identities are $P_c - a = T_{0,\theta}/F > 0$, $J_c = T_{0,z}/F$ with $F = E/\sqrt{2X} > 0$. Thus the cone is exactly $(a - 2)(T_z/T_\theta)^2 < 2$, with uniform strict slack throughout the terminal interval. This also covers the plateau $.5 \leq y \leq 1$: the positive backward time integral persists there even though $f'(y) = 0$.

Near $\delta = 3 - y = 0$, the specified step has the exact factorization

$$\psi_o = e^{-4/\delta^2} g(\delta), \quad g(\delta) = \frac{1}{e^{-(1-\delta/2)^{-2}} + e^{-4/\delta^2}}, \quad g(0) = e,$$

$$f' = \rho_o e^{-4/\delta^2} \delta^{-3} [8g(\delta) + \delta^3 g'(\delta)].$$

The normalized boundary term in (13) is

$$q^{A+1/2} K f_r = \frac{2E_{\text{pow}}(X) \mathcal{H}(2d/X)}{\sqrt{2X}} f'.$$

The two remaining angular integrals contain f' , so (12) with $c = 4, j = 3$ makes each e^{-4/δ^2} times a smooth coefficient. Relative to the boundary factor $e^{-4/\delta^2} \delta^{-3}$ they vanish to at least order δ^3 . Therefore

$$T_{0,\theta} = e^{-4/\delta^2} \delta^{-3} b_\theta(\delta, \eta), \quad b_\theta(0, \eta) = \frac{16\rho_o E_{\text{pow}}(X_b)}{\sqrt{2X_b}} \mathcal{H}(2d/X_b) g(0) > 0.$$

The heat factor has a positive minimum on $[0, 2/X_b]$, so b_θ has a positive lower bound on a sufficiently short closed collar uniformly for $|\eta| \leq 1$. The pressure integral contains one f' factor; the same substitution gives $p_z = q^{-D-2A} e^{-4/\delta^2}$ times a smooth coefficient. Its further radial integration for T_z uses (12) with $j = 0$, giving

$$T_{0,z} = e^{-4/\delta^2} \delta^3 b_z(\delta, \eta).$$

The powers of q cancel exactly: $q^{A+1/2}q^{1/2}q^{-D-2A} = q^{1-D-A} = 1$. The remaining coefficients depend only on $X, \eta, \mathcal{H}(2d/X), L^{-1}$, with their already proved closed-parameter smoothness. Consequently

$$T_{0,z}/T_{0,\theta} = \delta^6 b_z/b_\theta \longrightarrow 0$$

smoothly, and the stress direction extends to $(1, 0)$ at the outer edge. Every fixed profile derivative costs only a finite inverse power of δ , and $|T_0| \geq ce^{-4/\delta^2}\delta^{-3}$ there. This proves (A.48)–(A.51) including the precise leading coefficient and the independence from the physical scale q .

15.4 The remaining positive-order edge term

At order $n = 1$, on the final collar the only exterior source is $-\partial_{zz}(Kf_o)$ in the angular equation. With $a_z = -2\eta/L$ and $b_z^{\text{coord}} = (-2D\eta a_z + d\partial_\eta a_z)/L$, direct differentiation of $y_z = q^{-D}a_z$ gives

$$\partial_{zz}(Kf_o) = Kq^{-2D}\{a_z^2\partial_{yy}f_o + b_z^{\text{coord}}\partial_yf_o\}.$$

All parameter derivatives of the coefficients are bounded because $L > 0$. The two profile terms have factors $e^{-4/\delta^2}\delta^{-6}$ and $e^{-4/\delta^2}\delta^{-3}$ times smooth functions. The backward stress integral gains three powers by (12), producing $|T_1| \leq Ce^{-4/\delta^2}\delta^{-3}$ and, for every fixed derivative, $|\partial^I T_1| \leq C_I e^{-4/\delta^2}\delta^{-N_I}$. The physical exponent after radial integration is $q^{-A-2D+1/2} = q^{-A-1/2+2h}$, exactly the order-one stress scale. Near this outer edge the other factor in ζ is a positive smooth bounded factor, so this is the weighted estimate required in (5.13) and (5.43). Near the inner edge T_1 is identically zero; for $n \geq 2$ the common compact interior support makes every fixed derivative of T_n/ζ bounded.

16 Audit conclusion and exact dependency boundary

The calculations above verify the coordinate operators; every coefficient and sign in the order recursion; the sparse analytic Volterra mechanism; the five positive-order moment equations and common supports; conservative residual cancellation including the subtracted leading angular moment; divergence under the cutoff curl; the order-independent derivative losses and flat Borel residual; all outer schedule moment corrections and amplitude selection; the pressure datum; each outer cone estimate; the heat ODE and endpoint regularity; the compensation map; and the two outer stress factors.

No mathematical contradiction was found in this assigned subsystem after these derivations. Two extraction ambiguities required visual resolution: the pulse estimate is $C_{\text{pre}}e^{26}/\lambda$, and the endpoint angular coefficient has denominator $\sqrt{2X_b}$. Neither is a correction to the source PDF. The proof of simultaneous compact-range Borel bounds is made explicit above using the common correction supports; it fills in the domain choice without changing the statement or the construction.

This result has an exact dependency boundary. The common analytic axis/collar data used in the positive-order recursion and the final regular-axis matching used in the backward stress identity are the outputs of [NS, Appendix B and Theorem 4.6]. Their independent proof is outside this document, as is the oscillatory realization of the admissible stress and the final exact Navier–Stokes construction. No acceptance of those parts, or of a claimed global blowup theorem, follows from this bounded audit.

17 Auxiliary geometry and coefficient algebra: independent derivation

Source: supplied *Finite Time Blowup for Navier-Stokes*, Section 6, pp. 62–72. Source identity and audit boundary are recorded in LOGBOOK.md and READ_SCOPE.md. The formulas below use the source’s original cylindrical coordinate orientation, chart scales, integer covering, and support domains. The geometric and algebraic assertions are proved here directly.

17.1 Coordinate maps and exact derivative intertwining

Let $A = 1/2 + h$, $D = 1/2 - h$, $0 < h < 1/100$, $Q = 2^{-\ell}$, $\varepsilon = Q^h$, $S_* = \ell^2$, and

$$R = r/Q^{1/2}, \quad Z = z/Q^D, \quad T = (1-t)/Q.$$

The profile radius is $R_{\text{profile}} = r/q^{1/2}$, so the exact transition is $R = \sqrt{q/Q} R_{\text{profile}}$. No equality of these different radius coordinates is used. The original similarity variables obey $z = q^D\eta$, $1-t = q(1-\eta^2)$, with

$L = 1 - 2h\eta^2 > 0$. On a dyadic shell $q/Q \in [1/2, 2]$, q/Q and η are smooth functions of Z, T , including one-sided derivatives at $T = 0$. Indeed $T = s - Z^2 s^{2h}$, $s = q/Q$, has derivative $1 - 2hZ^2 s^{2h-1} = L \geq 1 - 2h$. Repeated implicit differentiation bounds each fixed chart derivative on the compact shell. Thus the ordinary chart and profile derivatives used below are related by finite sums with uniformly bounded smooth coefficients.

Set

$$J_g = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad b_g = \sqrt{2} - 1, \quad v_r = (1, -b_g), \quad v_t = (b_g, 1),$$

$$\Lambda_g = 4 - \sqrt{2}, \quad T_g = 4 + \sqrt{2}, \quad \rho_g = \frac{\log \Lambda_g}{\log T_g}, \quad \kappa_s = 10^{-5}, \quad d_r = 2((1+h)\rho_g - h\kappa_s).$$

Direct multiplication gives $J_g v_r = \Lambda_g v_r$, $J_g v_t = T_g v_t$, $v_r \cdot v_t = 0$, and $\det J_g = 14$. Also $\rho_g > 0$ and $\rho_g > \kappa_s$, hence $d_r > 0$. For example $\Lambda_g > 2$, $T_g < 6$ imply $\rho_g > \log 2 / \log 6 > 1/3 > \kappa_s$.

On the extended domain with independent $Y \in \mathbb{R}^2 / \mathbb{Z}^2$, define

$$\iota(r, \theta, z, t) = (r, \theta, z, t, v_r r^{d_r} + v_t t \bmod \mathbb{Z}^2).$$

For every smooth extended scalar F , the ordinary chain rule proves the exact identities

$$\partial_t(\iota^* F) = \iota^*((\partial_t + v_t \cdot \partial_Y)F), \quad \partial_r(\iota^* F) = \iota^*((\partial_r + d_r r^{d_r-1} v_r \cdot \partial_Y)F).$$

The z, θ identities have no extra term. These are morphisms of differential algebras: pullback preserves sums and products and intertwines the four stated derivations. Their commutators vanish because the only variable coefficient depends on r , the extra directions are constant and independent of Y, t, z, θ , and the radial derivation is the only derivation differentiating that coefficient. The fields depending on Y vanish in a neighborhood of the axis at each $q > 0$, so no differentiability of r^{d_r} at $r = 0$ is invoked.

For $i = \lfloor \log_{T_g}(Q^{-1-h}/S_*) \rfloor \geq 0$, let $\pi_i(Y) = J_g^i Y \bmod \mathbb{Z}^2$. This is a surjective group homomorphism of tori with finite kernel of cardinality 14^i . On a pullback $f \circ \pi_i$, the two directional operators are exactly $T_g^i N_i f$ and $\Lambda_g^i L_i f$, where $N_i = v_t \cdot \partial_{Y_i}$, $L_i = v_r \cdot \partial_{Y_i}$. Thus

$$\begin{aligned} \mathbf{t}_* &= Q^{1+h} \mathbf{t} = -\varepsilon \partial_T + c_i N_i, \quad D_r = \sqrt{Q} \mathbf{r} = \partial_R + M_i d_r R^{d_r-1} L_i, \\ c_i &= T_g^i Q^{1+h}, \quad M_i = \Lambda_g^i Q^{d_r/2}, \quad D_z = \sqrt{Q} \partial_z = \varepsilon \partial_Z, \quad D_\theta = R^{-1} \partial_\theta. \end{aligned}$$

The floor inequality is strict on its left:

$$T_g^{-1} Q^{-1-h} / S_* < T_g^i \leq Q^{-1-h} / S_*.$$

It follows that $T_g^{-1} S_*^{-1} < c_i \leq S_*^{-1}$, and, using $\Lambda_g^i = (T_g^i)^{\rho_g}$,

$$T_g^{-\rho_g} \varepsilon^{-\kappa_s} S_*^{-\rho_g} < M_i \leq \varepsilon^{-\kappa_s} S_*^{-\rho_g}.$$

Every radial derivative of $d_r R^{d_r-1}$ is bounded on the fixed positive radial interval. This proves the claimed additional radial cost $\varepsilon^{-\kappa_s}$, with its retained power of S_* .

For physical velocity $Q^{-A} u_*$, pressure $Q^{-2A} p_*$, and residual multiplier $Q^{2A+1/2}$, the time, spatial first-derivative, and Laplacian coefficients are respectively

$$Q^{2A+1/2-A} = Q^{1+h}, \quad Q^{1/2}, \quad Q^{2A+1/2-A-1} = Q^h = \varepsilon.$$

These identities retain the source's viscosity-one chart convention; scaling to arbitrary viscosity is outside this lane.

17.2 Fourier inversion and descent

For $n = (n_1, n_2) \neq 0$,

$$v_r \cdot n = (n_1 + n_2) - \sqrt{2} n_2.$$

Its product with the algebraic conjugate is the nonzero integer $(n_1 + n_2)^2 - 2n_2^2$. It cannot be zero unless $n_2 = 0 = n_1$, by irrationality of $\sqrt{2}$. The conjugate has absolute value at most $C|n|$, so $|v_r \cdot n| \geq 1/(C|n|)$. For $v_t \cdot n = (n_2 - n_1) + \sqrt{2} n_1$, the same argument uses $(n_2 - n_1)^2 - 2n_1^2$. Therefore both directional inverses on zero-mean smooth torus functions have Fourier multipliers bounded by $C(1 + |n|)$, retaining the Fourier derivative factor $2\pi i$: the inverse of $v \cdot \partial_Y$ multiplies the n -coefficient by $(2\pi i v \cdot n)^{-1}$. Rapid

Fourier decay is preserved. In Sobolev norms this gives $\|L^{-1}f\|_{H^s} \leq C_s\|f\|_{H^{s+1}}$, with the zero coefficient defined to be zero. Taking a further fixed Sobolev embedding loss gives all fixed classical derivatives. No infinite derivative loss occurs.

If f descends through π_i , its Fourier support lies in $(J_g^i)^T \mathbb{Z}^2$. Translation multiplies each coefficient by a character, averaging deletes nonzero coefficients, and a directional Fourier inverse multiplies them by scalar values. Each operation preserves this sublattice. Radial integration at fixed z, t preserves deck invariance pointwise in the integral. These are the precise maps proving the descent claims.

17.3 Separation of all interacting labels

Use the dyadic squared partition $\sum_\ell \chi_\ell(q)^2 = 1$, the product grid squared partitions of mesh S_*^{-3} , and the duplicated signs in the source's labels $\gamma = (\ell, a, \sigma)$. On the active annulus select every grid support meeting the active shell and a representative in that intersection. Then $\eta_\gamma = \chi_\ell \chi_{\ell, a}$ gives $\sum_{\beta=(\ell, a)} \eta_\beta^2 = 1$, counting each box once and the two signs separately only when assigning rectangles.

Overlapping dyadic supports imply $|\ell - \ell'| \leq 2$. Join labels whenever the fixed-factor enlarged physical grid boxes intersect and the band difference is at most four, and join the two signs of each box. A chart scale ratio for bands differing by four is bounded above and below; $\ell^2/(\ell')^2$ has the same property for $\ell, \ell' \geq \ell_0 > 4$. A grid box thus meets a uniformly bounded number of boxes in each of finitely many bands. The countable graph has maximum degree $d < \infty$. Enumerating its vertices and assigning the first unused color in $\{1, \dots, d+1\}$ gives a proper coloring.

Writing $f(\ell) = ((1+h)\ell \log 2 - 2 \log \ell) / \log T_g$, the mean-value bound and the floor operation give

$$|i(\ell) - i(\ell')| \leq 1 + \frac{4(1+h) \log 2 + 8/\ell_0}{\log T_g}.$$

Fix an integer upper bound $\Delta_{\max}$. There are only finitely many ordered color constraints

$$c_\mu \neq J_g^\Delta c_\nu \pmod{\mathbb{Z}^2}, \quad 0 \leq \Delta \leq \Delta_{\max},$$

excluding only the tautological constraint $(\Delta, \mu, \nu) = (0, \nu, \nu)$. For different colors, the forbidden equality is a proper closed subset with empty interior of the finite product of tori. For a self-pair with $\Delta > 0$, $J_g^\Delta - I$ is nonsingular because its eigenvalues are $\Lambda_g^\Delta - 1, T_g^\Delta - 1 > 0$; its kernel on the torus is finite, so this forbidden subset also has empty interior. A finite union of these closed sets has empty interior. The complement is open and nonempty, and rational tuples are dense, giving the rational centers specified in the source. There is consequently a strictly positive minimum distance d_c of every forbidden difference from the lattice.

Let $C_J = \max_{0 \leq \Delta \leq \Delta_{\max}} \|J_g^\Delta\|$ and $C_v = \|v_r\| + \|v_t\|$. Choose $r_0 > 0$ sufficiently small that all enlarged rectangles are injective and $2r_0 C_v (C_J + 1) < d_c$. If two absolute preimages of enlarged rectangles at levels $i, i + \Delta$ intersect, their common point produces

$$c_\mu - J_g^\Delta c_\nu = J_g^\Delta e_\nu - e_\mu \pmod{\mathbb{Z}^2}, \quad |e_\nu|, |e_\mu| \leq 2r_0 C_v.$$

The right side has distance from zero less than d_c , a contradiction for joined labels. All labels whose slow supports meet were joined. Therefore their enlarged auxiliary preimages are disjoint. For smoothly extended fields supported in these products, the support of every derivative is contained in the same closed support, so all cross-label derivative products vanish before and after ι^* . This proves the required nonlinear support assertion, including boundaries.

On a band rectangle, write $Y_i - c_\gamma - k = \xi_g v_r + \eta_g v_t$ and $v = (\eta_g + r_0)/c_i$, $L_s = 2r_0/c_i$. Orthogonality gives $L_i \eta_g = N_i \xi_g = 0$, $L_i \xi_g = N_i \eta_g = 1$. Hence $D_r v = D_z v = 0$, $t_* v = 1$. This is the exact reason spatial differentiation of the pulse does not incur a time-direction torus frequency.

17.4 Common torus, lifts, integration paths, and label counts

At a fixed (z_0, t_0) , put $q_{\text{loc}} = q(z_0, t_0) > 0$. By continuity choose a neighborhood U with $q(U) \subset (q_{\text{loc}}/2, 2q_{\text{loc}})$. An active dyadic band obeys $q/2 \leq Q_\ell \leq 2q$. Every band meeting U therefore has $q_{\text{loc}}/4 < Q_\ell < 4q_{\text{loc}}$, so its dyadic indices lie in a finite set $\mathcal{L}(U)$ with span at most four. The previously proved four-band bound on $i(\ell) - i(\ell')$ therefore applies to this entire set. Take $i_0 = \min_{\ell \in \mathcal{L}(U)} i(\ell)$, $H = J_g^{i_0} Y$. Then $Y_i = J_g^\Delta H$, $0 \leq \Delta \leq \Delta_{\max}$. The preimage of an injective rectangle consists exactly of the 14^Δ images

$$J_g^{-\Delta}(c_\gamma + k + \xi_g v_r + \eta_g v_t) \pmod{\mathbb{Z}^2}, \quad [k] \in \mathbb{Z}^2 / J_g^\Delta \mathbb{Z}^2.$$

Different cosets give disjoint copies: an overlap would imply that their k -difference is in $J_g^\Delta \mathbb{Z}^2$, after using injectivity in the band rectangle. This also proves there are no omitted copies. Derivative factors of J_g^Δ are uniformly bounded. Surjectivity of π_{i_0} transfers the previously proved support separation to H .

For a torus character $e^{2\pi i n \cdot Y_i}$, pullback has frequency $(J_g^\Delta)^T n$, zero exactly when $n = 0$. Integrating a Fourier series therefore proves exact preservation of normalized Haar measure, not an approximation depending on frequency. This proves agreement of the absolute, common, and band averages. On overlaps the representatives pull back to the identical extended field; their compatibility is equality after the respective covering maps. Those maps are independent of R, Z, T , so slow derivatives preserve compatibility. The normalized fast derivatives are the same absolute operators $Q^{1+h} N_{\text{abs}}$ and $Q^{d_r/2} L_{\text{abs}}$; they also preserve it.

A mesh of spacing S_*^{-3} gives $O(1)$ overlap at a point, $O(S_*^9)$ boxes in a compact three-dimensional chart set, and $O(S_*^3)$ boxes along a bounded radial line. The two signs and bounded covering multiplicities contribute fixed factors. Radial integration fixes $q(z, t)$; it can collect radial boxes but no additional bands. The collected sum therefore introduces only a polynomial in S_* .

Define $\lambda_t(w) = v_t \cdot w / (1 + b_g^2)$. In a lifted common rectangle, the exact path holding slow and transverse coordinates fixed is

$$H_w = H + T_g^{-\Delta} c_i (w - v(H)) v_t.$$

Because $\lambda_t J_g^\Delta = T_g^\Delta \lambda_t$,

$$D_H v = T_g^\Delta \lambda_t / c_i, \quad D_H H_w = I - v_t \lambda_t,$$

and all higher derivatives of these affine expressions vanish. The first is $O(S_*)$, the second uniformly bounded. No phase-averaging operation is hidden in this path: values at distinct preimages remain distinct. Deck translations commute with the path and merely reindex the rectangle copy. A uniquely prescribed initial-value solution therefore descends to the common torus.

17.5 Weighted coefficient classes and complete product rules

On $X_a < X < X_b$, let

$$\zeta = \exp \left(-\frac{a_a}{\log^2(X/X_a)} - \frac{a_b}{\log^2(X_b/X)} \right), \quad \delta = \min\{1, \log(X/X_a), \log(X_b/X)\},$$

with fixed $a_a, a_b > 0$, and extend ζ by zero. For any $b > 0, N \geq 0$, $\zeta^b \delta^{-N} \rightarrow 0$ faster than every power at either edge: put $x = 1/\delta \rightarrow \infty$ and use $x^N e^{-cx^2} \rightarrow 0$, proved for example by comparing $N \log x - cx^2$ with $-cx^2/2$ for large x . Repeated derivatives of the explicit exponential have the same exponential times a finite sum of inverse edge-distance powers. Thus all stated derivative bounds imply smooth zero extension at radial boundaries.

The mean class $\mathcal{M}_\alpha$ consists of angular-independent compatible coefficients with derivatives bounded by $C_I \varepsilon^\alpha S_*^{b_I} \zeta \delta^{-d_I}$, smooth zero extension outside the active shell, and no rectangle or envelope condition. The wave class $\mathcal{W}_\alpha$ uses the same derivative variables (R, Z, T, H_1, H_2) , retains a label and nonzero harmonic, has the prescribed closed slow/transverse/shell supports, and bounds $C_I \varepsilon^\alpha S_*^{b_I} \sqrt{\zeta} \delta^{-d_I} P(v)$. Actual cutoff coefficients also have compact pulse support and smooth extension there. Equal label-harmonic contributions must be added before testing the bounds. $P(v)$ is a pointwise weight; these definitions already require all coefficient derivatives to obey it, so no differentiation of a mere inequality for P is being assumed.

Leibniz's formula has finitely many terms at each order. The retained products of weights obey

$$\zeta^2 \leq \zeta, \quad \zeta^{3/2} P \leq \sqrt{\zeta} P, \quad \zeta P^2 \leq \zeta, \quad \zeta P^2 \leq \sqrt{\zeta} P$$

because $0 \leq \zeta, P \leq 1$. These prove $\mathcal{M}_\alpha \mathcal{M}_\beta \subset \mathcal{M}_{\alpha+\beta}$, $\mathcal{M}_\alpha \mathcal{W}_\beta \subset \mathcal{W}_{\alpha+\beta}$, and the two same-label wave-product rules. A zero-harmonic product extends to a mean after temporal cutoff. Before cutoff it is only a mean-bounded local coefficient. Multiplication by a mean cannot enlarge the wave support; products of different actual labels are zero by separation. Finite sums retain the same exponent and suitable maxima of the polynomial degrees.

For a nonzero angular frequency $kp_\gamma \in \mathbb{Z} \setminus \{0\}$, angular integration of $e^{ik(m+m')\Phi_\gamma}$ is zero unless $m + m' = 0$, in which case it is one. This proves exactly which product coefficients survive. It does not identify angular averaging with auxiliary averaging. Products add harmonic integers and differentiation preserves them; a finite sequence of such operations leaves a finite band-independent set at each finite stage.

Ordinary coefficient derivatives shift the derivative index in the defining estimate, preserving α . Multiplication by $M_i d_r R^{d_r-1}$, with all its derivatives bounded above, proves $D_r : \mathcal{C}_\alpha \rightarrow \mathcal{C}_{\alpha-\kappa_s}$, $\mathcal{C} = \mathcal{M}, \mathcal{W}$. The other exact maps are $D_z = \varepsilon \partial_z : \mathcal{C}_\alpha \rightarrow \mathcal{C}_{\alpha+1}$, $-\varepsilon \partial_T : \mathcal{C}_\alpha \rightarrow \mathcal{C}_{\alpha+1}$, and $c_{i_0} N_{i_0} : \mathcal{C}_\alpha \rightarrow \mathcal{C}_\alpha$. Here $c_{i_0} = T_g^{-\Delta} c_i$ and $M_{i_0} = \Lambda_g^{-\Delta} M_i$, so a change to a common torus changes constants, not exponents. All these claims concern coefficients with the exponential factored out. Differentiating the exponential itself produces its exact $ikm \nabla_* \Phi$ or $ikm t_* \Phi$ factor and is accounted for separately in the pulse analysis.

Finally, for $f_* = Q^{a_f} f_{\text{phys}}$, substitution $r = \sqrt{Q} R$ gives the exact moment map

$$\int R^e \langle f_* \rangle_Y dR = Q^{a_f - (e+1)/2} \int r^e \langle f_{\text{phys}} \rangle_Y dr.$$

The measure and power weight are both retained. Normalized Haar compatibility proves that the average may be computed in any of the three torus representations.

18 Phase, constrained evolution, and every fixed derivative

This file independently derives the claims of source pp. 73-81, Lemma 7.1, Proposition 7.2, Corollary 7.3, and Lemma 7.4. The exact auxiliary maps and operators are proved in the preceding file. The imported background assertion is Proposition 5.5: on the fixed enlarged annulus its chart radial component b is $O(\varepsilon)$, its tangential components V, G differ from their leading profiles by $O(\varepsilon^2)$, and these statements hold at every fixed slow derivative. The construction of that background belongs to another audit lane. Here the consequences for the pulse system are established explicitly.

18.1 Cone coordinates and the phase gradient

Write $F = V/R$, $g = (RF_R, G_R)$ in component order (θ, z) . At the fixed representative of a slow box, the leading shear is

$$g_0 = F_0(-a, b_s), \quad N = g_0/|g_0|, \quad K = (-N_z, N_\theta).$$

Let $t_s = -b_s/a$ and $v_s = a(1+t_s^2)$. Since $a, F_0 > 0$,

$$N = -\frac{(1, t_s)}{\sqrt{1+t_s^2}}, \quad K = \frac{(t_s, -1)}{\sqrt{1+t_s^2}}, \quad |g_0| = aF_0\sqrt{1+t_s^2}.$$

Retain the source's

$$\lambda_0^2 = -2F_0 N_\theta (2F_0 N_\theta + |g_0|), \quad c_0 = \lambda_0 / (2F_0 N_\theta).$$

Substitution gives

$$\lambda_0^2 = 2F_0^2 \left(a - \frac{2}{1+t_s^2} \right) = 2aF_0^2(1-2/v_s) > 0, \quad c_0^2 = (v_s - 2)/2, \quad c_0 < 0.$$

The positive lower bounds on $F_0, a, v_s - 2$ and compact upper bounds give positive lower bounds on $\lambda_0, |c_0|$, and finite upper bounds.

At the same, unfrozen profile point, $T_0 = F(p_s - (a, -b_s))$, $P_c = p_{s,1} + t_s p_{s,2}$, $J_c = p_{s,2} - t_s p_{s,1}$. Consequently

$$T_0 \cdot N = -\frac{F(P_c - v_s)}{\sqrt{1+t_s^2}}, \quad T_0 \cdot K = -\frac{F J_c}{\sqrt{1+t_s^2}}.$$

Multiplication of T_0 by the positive chart factor $(Q/q)^{A+1/2}$ changes neither sign nor quotient. Thus the two stress cone inequalities are exactly $T_{0,*} \cdot N < 0$ and

$$\left| c_0 \frac{T_{0,*} \cdot K}{T_{0,*} \cdot N} \right| < 1.$$

The weighted direction at a vanishing edge is used, not a division of zero values. Its smooth extension and strict compact margin give a supremum $\mu < 1$. Choosing $u_* > 0$ with $u_*/\sqrt{1+u_*^2} > \mu$ is possible because that function increases to one. All quantities now frozen at representatives are kept fixed under differentiation. Uniform continuity in the compact chart domain makes the same strict cone margin valid on every enlarged box of diameter CS_*^{-3} , after one lower bound on ℓ .

Retain $k = \lceil \varepsilon^{-1/2} \rceil$, so for $0 < \varepsilon \leq 1$,

$$1 \leq \varepsilon k^2 \leq (1 + \sqrt{\varepsilon})^2 \leq 4, \quad k^{-1} \leq \sqrt{\varepsilon}.$$

Define

$$B_s^2 = \frac{\lambda_0}{\varepsilon k^2 (1 + u_*^2)^{3/2}}, \quad s(v) = \sigma(u_*/2 + u_* v/L_s), \quad x_0 = \sigma B_s u_*/2.$$

The B_s have fixed positive lower and upper bounds. Set

$$(\tilde{p}/R_0, p_z) = B_s \left(K - \frac{\sigma u_* g_0}{L_s |g_0|^2} \right).$$

Choose kp as the specified nearest nonzero integer to $k\tilde{p}$. The error $|p - \tilde{p}| \leq k^{-1}$ suffices, including when the nearest unrestricted integer would have been zero. The integer is constant with the label. Put

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v H_\Phi, \quad H_\Phi = pF + p_z G.$$

Because $D_r v = D_z v = 0$ and the base is auxiliary-independent,

$$n_\Phi = \nabla_* \Phi = (x_0 - v(H_\Phi)_R, p/R, p_z - \varepsilon v(H_\Phi)_Z).$$

At the representative, before rounding, $(\tilde{p}/R_0, p_z) \cdot g_0 = -\sigma B_s u_*/L_s$. Variation across the slow box changes this by $O(S_*^{-3})$; the background change contributes $O(\varepsilon^2)$; rounding contributes $O(k^{-1})$. Hence

$$|(H_\Phi)_R + \sigma B_s u_*/L_s| \leq C(S_*^{-3} + \varepsilon^2 + k^{-1}).$$

Multiplication by $v \leq L_s = O(S_*)$ gives the radial comparison with $B_s s(v)$. The tangential comparison has contributions $O(S_*^{-1})$ from the deliberate tilt, $O(S_*^{-3} + \varepsilon^2 + k^{-1})$ from freezing and rounding, and $O(\varepsilon S_*)$ from the axial phase term. A single sufficiently large ℓ_* satisfies

$$S_*^2(\varepsilon + \varepsilon^2 + k^{-1}) \leq 1$$

for every $\ell \geq \ell_*$, since $S_* = \ell^2$ and $\varepsilon = 2^{-h\ell}$. Therefore

$$|n_\Phi - B_s(s(v), K)| \leq C/S_*.$$

Differentiation in v gives $n'_\Phi = (-(H_\Phi)_R, 0, -\varepsilon(H_\Phi)_Z)$, also $O(S_*^{-1})$. Each fixed number of slow derivatives of these coefficients is bounded by a polynomial in S_* : the only growing factors in these formulas are v , ε^{-1} in the phase itself, and the fixed chart derivatives. The ε^{-1} term of Φ disappears in n_Φ ; it is not discarded from the full exponential when physical derivatives are later taken.

The exact transport defect follows term by term:

$$\mathbf{t}_* \Phi = -H_\Phi + \varepsilon v(H_\Phi)_T, \quad F\partial_\theta \Phi + G D_z \Phi = H_\Phi - \varepsilon v G(H_\Phi)_Z.$$

Thus

$$E_{\text{ik}} := (\mathbf{t}_* + b D_r + F\partial_\theta + G D_z) \Phi = \varepsilon v((H_\Phi)_T - G(H_\Phi)_Z) + b n_{\Phi,r}.$$

The background satisfies $D^I b = O_I(\varepsilon)$, while every fixed coefficient derivative of $n_{\Phi,r}$ is polynomially bounded in S_* . The exact product rule therefore gives

$$D^I(b n_{\Phi,r}) = \sum_{J \leq I} \binom{I}{J} (D^J b)(D^{I-J} n_{\Phi,r}) = O_I(\varepsilon S_*^{d_I}).$$

For example, at fixed pulse coordinate, $\partial_R n_{\Phi,r} = -v(H_\Phi)_{RR}$, whose bound can contain a factor S_* . Auxiliary coefficient derivatives of v are $O(S_*)$, and derivatives of the fixed affine path of order two or greater vanish. Applying the same finite product rule to $\varepsilon v((H_\Phi)_T - G(H_\Phi)_Z)$ proves $D^I E_{\text{ik}} = O_I(\varepsilon S_*^{b_I})$ for every fixed I , on the same domain. This introduces no smallness condition on the higher derivatives.

18.2 Exact pressure elimination and plane isomorphism

Set

$$\mathcal{K} = \begin{pmatrix} 0 & -2F & 0 \\ 2F + R F_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad d = \varepsilon k^2 |n_\Phi|^2, \quad \mathcal{A}_\Phi = -\mathcal{K} + \frac{n_\Phi(n_\Phi^T \mathcal{K} - (n'_\Phi)^T)}{|n_\Phi|^2}.$$

For the original equation

$$t'_m + \mathcal{K} t_m + m^2 dt_m + i k m n_\Phi \pi_m = -f_m, \quad n_\Phi \cdot t_m = 0,$$

differentiate its constraint to get $n_\Phi \cdot t'_m = -n'_\Phi \cdot t_m$. Taking its normal component then forces

$$\pi_m = \frac{i}{km} \frac{n_\Phi \cdot \mathcal{K}t_m - n'_\Phi \cdot t_m + n_\Phi \cdot f_m}{|n_\Phi|^2}.$$

The factor has sign $+i/(km)$: multiplying it by $ikmn_\Phi$ gives the negative of the required normal vector. Substituting gives the projected equation

$$t'_m = \mathcal{A}_\Phi t_m - m^2 dt_m - \text{proj}_{n_\Phi^\perp} f_m.$$

Conversely this projected equation and the displayed pressure imply the original equation provided the constraint holds. For any solution of the projected equation,

$$(n_\Phi \cdot t_m)' = -m^2 d(n_\Phi \cdot t_m).$$

An initially zero constraint remains zero, proving equivalence, existence, and uniqueness once the finite-dimensional initial-value equation is solved.

Since $n_{\Phi, \tan}$ stays close to $B_s K$, its norm has a uniform positive lower bound. The nonzero p/R also prevents it from being literally zero. Define

$$K_a = n_{\Phi, \tan} / |n_{\Phi, \tan}|, \quad N_a = ((K_a)_z, -(K_a)_\theta), \quad s_a = n_{\Phi, r} / |n_{\Phi, \tan}|.$$

Then $N_a \perp K_a$, $|N_a| = |K_a| = 1$, and

$$U = [e_r - s_a K_a, N_a], \quad M = \begin{pmatrix} 1 & 1 \\ c_0 \sqrt{1+s^2} & -c_0 \sqrt{1+s^2} \end{pmatrix}, \quad B = UM$$

maps $\mathbb{C}^2$ isomorphically onto the actual complex plane $n_\Phi^\perp$. The exact left inverse is

$$B^\ell = M^{-1} \begin{pmatrix} e_r^T \\ N_a^T \end{pmatrix}$$

because the last displayed row matrix times U is I_2 . The determinant of M is $-2c_0 \sqrt{1+s^2}$, bounded away from zero. All factors and their inverses are uniformly bounded in value and polynomially bounded in each fixed coefficient derivative.

For clarity the ambient map also has an exact formula. Regard K_a as the three-vector with zero radial component, and let $W_a = (e_r^T; N_a^T)$. Direct multiplication gives $W_a U = I_2$. For any $w = w_r e_r + w_K K_a + w_N N_a \in \mathbb{C}^3$,

$$BB^\ell w = UW_a w = w_r(e_r - s_a K_a) + w_N N_a = w - K_a(w_K + s_a w_r).$$

Since $n_\Phi = |n_{\Phi, \tan}|(s_a e_r + K_a)$, this proves

$$B^\ell B = I_2, \quad BB^\ell = I_3 - \frac{K_a n_\Phi^T}{|n_{\Phi, \tan}|}.$$

Thus BB^ℓ is the projection onto $n_\Phi^\perp$ along $\text{span}_{\mathbb{C}}\{K_a\}$. Its restriction to $n_\Phi^\perp$ is the identity; its action on K_a is zero. In particular the inverse assertion is not made on all of $\mathbb{C}^3$.

The moving-plane differential equation has the exact coordinate form

$$z'_m = B^\ell(\mathcal{A}_\Phi B - B')z_m - m^2 dz_m - B^\ell \text{proj}_{n_\Phi^\perp} f_m.$$

For its converse, differentiate $n_\Phi^T B = 0$ to obtain $n_\Phi^T B' = -(n'_\Phi)^T B$. The definition of $\mathcal{A}_\Phi$ gives $n_\Phi^T \mathcal{A}_\Phi B = -(n'_\Phi)^T B$. Therefore each column of $\mathcal{A}_\Phi B - B'$ lies in $n_\Phi^\perp$. Both Bz_m and the projected force lie there as well. Multiplying the coordinate equation by B and using $BB^\ell = I$ on these three terms yields

$$Bz'_m = (\mathcal{A}_\Phi B - B')z_m - m^2 dBz_m - \text{proj}_{n_\Phi^\perp} f_m.$$

Adding $B'z_m$ proves the projected equation for $t_m = Bz_m$, and $n_\Phi^T t_m = 0$ holds by construction. The displayed pressure then recovers the full equation with its actual normal forcing. Conversely, a constrained solution has $z_m = B^\ell t_m$ and $t_m = Bz_m$. Differentiating the latter identity and multiplying the projected equation by B^ℓ gives exactly the coordinate equation above. Both inverse compositions, including their initial data and the unique pressure, are therefore proved.

For the reference normal $n_{\text{ref}} = B_s(s, K)$, use $e = e_r - sK$ and N as an orthogonal plane basis, of lengths $\sqrt{1+s^2}$ and 1. The actual source matrix $\mathcal{K}_0$ gives

$$\mathcal{K}_0 e = (2F_0 s K_\theta, 2F_0 e_\theta + g_0), \quad \mathcal{K}_0 N = (-2F_0 N_\theta, 0).$$

Consequently

$$\begin{aligned} e \cdot (-\mathcal{K}_0 e) &= sK \cdot g_0 = 0, & e \cdot (-\mathcal{K}_0 N) &= 2F_0 N_\theta, \\ N \cdot (-\mathcal{K}_0 e) &= -(2F_0 N_\theta + |g_0|), & N \cdot (-\mathcal{K}_0 N) &= 0. \end{aligned}$$

Orthogonal projection onto $n_{\text{ref}}^\perp$ leaves these scalar products unchanged. Dividing the first coordinate by $1+s^2$ produces

$$\mathcal{D} = \begin{pmatrix} 0 & 2F_0 N_\theta / (1+s^2) \\ -(2F_0 N_\theta + |g_0|) & 0 \end{pmatrix}.$$

Its two eigenvectors are exactly the columns of M , with eigenvalues $\lambda = \lambda_0 / \sqrt{1+s^2}$ and $-\lambda$, since $c_0 = \lambda_0 / (2F_0 N_\theta)$ and $\lambda_0^2 = -2F_0 N_\theta (2F_0 N_\theta + |g_0|)$. This verifies the orientation and signs of the growing coordinate.

Replacing the background and normal by the actual ones changes the algebraic projected matrix by $O(S_*^{-1})$. This follows from the explicit rational expression for the projection, the uniform positive denominators, the value bounds already proved, and the mean-value formula on their common compact range. Also $K_a - K, s_a - s = O(S_*^{-1})$, while their v -derivatives and s' are $O(S_*^{-1})$. Hence $B' = O(S_*^{-1})$ and the normal-motion term in $\mathcal{A}_\Phi$ is $O(S_*^{-1})$. Therefore

$$B^\ell(\mathcal{A}_\Phi B - B') = \text{diag}(\lambda, -\lambda) + E, \quad |E| \leq C/S_*.$$

Every fixed derivative of E is polynomially bounded; no derivative-smallness claim is needed. The damping comparison is

$$d_{\text{ref}} = \varepsilon k^2 B_s^2(1+s^2), \quad |d - d_{\text{ref}}| \leq C/S_*,$$

because εk^2 is bounded and $|n_\Phi|^2 - B_s^2(1+s^2)$ is $O(S_*^{-1})$.

18.3 Envelope and uniform forward inverse

Let

$$a_{\text{net}}(y) = \frac{\lambda_0}{\sqrt{1+y^2}} - \frac{\lambda_0(1+y^2)}{(1+u_*^2)^{3/2}}, \quad y = |s(v)| = u_*/2 + u_*v/L_s.$$

It is zero at $v = L_s/2$, and direct differentiation gives

$$\frac{d}{dv} a_{\text{net}}(|s(v)|) = -\frac{u_* \lambda_0 |s(v)|}{L_s} \left((1+s(v)^2)^{-3/2} + 2(1+u_*^2)^{-3/2} \right).$$

Since $u_*/2 \leq |s| \leq 3u_*/2$, its value lies between $-C/L_s$ and $-c/L_s$, for fixed positive c, C . Define $P(v) = \exp(\int_{L_s/2}^v a_{\text{net}}(|s(w)|) dw)$. Integrating the derivative inequality on either side of the midpoint, keeping the direction of each integral, gives

$$-C_1(v - L_s/2)^2/L_s \leq \log P(v) \leq -c_1(v - L_s/2)^2/L_s.$$

This proves the two Gaussian bounds and $0 < P \leq 1$ with the source's integral normalization.

Writing $t_m = Bz_m$ gives

$$z'_m = A_m z_m + g_m, \quad A_m = \text{diag}(\lambda, -\lambda) + E - m^2 dI_2, \quad g_m = -B^\ell \text{proj}_{n_\Phi^\perp} f_m.$$

For the complex Euclidean norm, taking the real part of $z^* A_1 z$ bounds the upper right derivative of $|z|$ by $(\lambda - d_{\text{ref}} + C/S_*)|z|$. Integration gives

$$\|V_1(v, w)\| \leq \exp(C(v-w)/S_*) P(v)/P(w) \leq C' P(v)/P(w).$$

The factor C' is fixed because $0 \leq v - w \leq L_s \asymp S_*$. Since the difference of A_m from A_1 is the scalar matrix $-(m^2 - 1)dI_2$, differentiation directly verifies

$$V_m(v, w) = \exp\left(-(m^2 - 1) \int_w^v d(a) da\right) V_1(v, w).$$

For every nonzero integer m , $m^2 - 1 \geq 0$ and $d > 0$, so the same value bound is valid for all such harmonics. This is the exact mechanism preventing a new small-domain requirement when the finite harmonic set increases.

If the source obeys

$$|D^I f_m| \leq C_I \varepsilon^\alpha S_*^{b_I} \sqrt{\zeta} \delta^{-a_I} P(v),$$

the zero-data Duhamel integral gives

$$z_m(v) = \int_0^v V_m(v, w) g_m(w) dw, \quad |z_m(v)| \leq C \varepsilon^\alpha S_*^{b+1} \sqrt{\zeta} \delta^{-a} P(v).$$

The source's $P(w)$ cancels the propagator's denominator exactly. The shell weights are constant along the integration path because X is constant there.

For completeness, parameter differentiation can be performed without differentiating a possibly exponentially large propagator in isolation. Hold v fixed and use slow variables and the initial transverse coordinate as parameters. They commute with ∂_v ; the zero initial data and all their parameter derivatives vanish. For each multi-index I ,

$$(\partial^I z_m)' = A_m \partial^I z_m + \partial^I g_m + \sum_{0 < J \leq I} \binom{I}{J} (\partial^J A_m) \partial^{I-J} z_m.$$

At fixed finite harmonic bound M , each $\partial^J A_m$ is bounded by $C_{J,M} S_*^{c_J}$. Induction on $|I|$, using the already proved propagator bound, shows that every source term on the right has the same $\varepsilon^\alpha \sqrt{\zeta} P$ weight with a larger inverse-distance exponent and polynomial degree. A further time integral contributes only L_s . One may choose the degree at step I to exceed

$$1 + \max\{b_I, \max_{0 < J \leq I} (c_J + b'_{I-J})\}.$$

Pulse derivatives follow from the differential equation, and mixed pulse/parameter derivatives follow by repeated differentiation of that equation. Returning to the current H uses $D_H v = O(S_*)$, $D_H H_w = I - v_t \lambda_t$, with higher affine derivatives zero, and therefore introduces only additional polynomial powers. Multiplying by B yields the stated bound for t_m . The explicit pressure has the extra factor $(km)^{-1} = O(\varepsilon^{1/2})$; its other factors have the proved derivative bounds. Thus the exponent for pressure is $\alpha + 1/2$.

The support argument also follows from this equation. If a fixed slow/transverse point is outside the source's closed support, the entire source path is zero; uniqueness gives a zero solution. At a boundary all source jets vanish by the prescribed smooth zero extension, so the differentiated equations inductively give zero solution jets. Across a shell edge the retained $\sqrt{\zeta} \delta^{-a_I}$ bounds give the same conclusion. Deck translations preserve the path and the prescribed data, giving common-torus descent by uniqueness. Smooth extension to the rectangle enlargement follows from smooth finite-dimensional ODE dependence on its coefficients and data; no temporal zero extension is asserted before cutoff. On compact sets with $q > 0$, one-sided background/source derivatives at $\tau = 0$ pass through the same parameter equations and pressure formula.

Only the lower bounds for denominators and finitely many value comparisons determine q_* . The induction above changes constants and polynomial degrees for each fixed M, I ; it does not impose further smallness inequalities. This proves the quantifier statement of Corollary 7.3.

18.4 The real homogeneous pulse and energy transfer

For $m = 1, f = 0$, specify $z_+(0) = P(0) > 0, z_-(0) = 0$. While $z_+ > 0$, the ratio $r = z_-/z_+$ obeys exactly

$$r' = E_{21} + (-2\lambda + E_{22} - E_{11})r - E_{12}r^2.$$

Let $c_\lambda = \inf \lambda > 0$. On $r = \pm K_b/S_*$, the outward derivative is at most

$$\frac{-2c_\lambda K_b + C(1 + K_b/S_* + K_b^2/S_*^2)}{S_*}.$$

Choose fixed $K_b > C/c_\lambda$, then enlarge the fixed band threshold to make the numerator negative. The closed interval $[-K_b/S_*, K_b/S_*]$ is invariant until any putative first zero of z_+ . On this interval,

$$(\log(z_+/P))' = -(d - d_{\text{ref}}) + E_{11} + E_{12}r = O(S_*^{-1}).$$

Since $z_+(0) = P(0)$ and L_s/S_* is bounded, integration proves $cP \leq z_+ \leq CP$, ruling out that putative zero. For all v ,

$$x = z_+ + z_- = z_+(1+r), \quad y = c_0\sqrt{1+s^2}(z_+ - z_-),$$

so

$$cP \leq x \leq CP, \quad y/x = c_0\sqrt{1+s^2} + O(S_*^{-1}).$$

The same differentiated equation used for the inverse proves every fixed derivative bound, with the nonzero initial value retained: it is fixed with the label, so all slow/transverse derivatives of that datum vanish. The value solution contributes $\|V_1(v, 0)\|P(0) \leq CP(v)$. Pulse derivatives are obtained from the equation. Pressure again has the extra $\varepsilon^{1/2}$.

For the real homogeneous amplitude t , $n_\Phi \cdot t = 0$ makes the normal component of $\mathcal{A}_\Phi t$ perpendicular to t . Direct multiplication, keeping both cylindrical $2F$ terms, gives

$$t \cdot \mathcal{K}t = -2Ft_r t_\theta + (2F + RF_R)t_r t_\theta + G_R t_r t_z = g \cdot (t_r(t_\theta, t_z)).$$

Taking $t \cdot t'$ therefore proves the exact identity

$$\frac{1}{2} \frac{d}{dv} |t|^2 = -g \cdot (t_r(t_\theta, t_z)) - \varepsilon k^2 |n_\Phi|^2 |t|^2.$$

For the flux $T = t_r(t_\theta, t_z) = T_N N + T_K K$, the exact production is

$$-g \cdot T = -|g_0|T_N - \Delta g \cdot T, \quad \Delta g = g - g_0.$$

Here g_0 is the leading shear frozen at the representative; it is not identified with the actual shear even there. To keep both sources of its difference, let $V^{(0)}, G^{(0)}$ denote the exact leading chart fields, put $\delta V = V - V^{(0)}$, $\delta G = G - G^{(0)}$, and write

$$g^{(0)} = (\partial_R V^{(0)} - V^{(0)}/R, \partial_R G^{(0)}), \quad g_0 = g^{(0)}(p_0),$$

$$\Delta g = (\partial_R \delta V - \delta V/R, \partial_R \delta G) + g^{(0)}(p) - g^{(0)}(p_0).$$

The leading and actual fields here are the constructed backgrounds. Their proved chart estimates give $|\partial^I \delta V| \leq C_{V,I} \varepsilon^2$ and $|\partial^I \delta G| \leq C_{G,I} \varepsilon^2$. Let $R_- > 0$ be the minimum radius on the compact enlarged chart, L_g bound $D_p g^{(0)}$, and $|p - p_0| \leq C_{\text{box}} S_*^{-3}$, with the connecting segment in that chart. Set

$$C_B = [(C_{V,e_R} + R_-^{-1} C_{V,0})^2 + C_{G,e_R}^2]^{1/2}.$$

The product rule and segment integration give the actual bound

$$|\Delta g| \leq C_B \varepsilon^2 + L_g C_{\text{box}} S_*^{-3} = C_B 2^{-2h\ell} + L_g C_{\text{box}} \ell^{-6}.$$

At p_0 only the segment term is zero. The Borel term remains. For a vector in the prescribed compact stress cone, retain its margin $-T_N \geq c_T |T|$, $c_T > 0$, and $|g_0| \geq g_- > 0$. Then Cauchy-Schwarz proves

$$-g \cdot T \geq [g_- c_T - C_B 2^{-2h\ell} - L_g C_{\text{box}} \ell^{-6}] |T|.$$

One sufficient additional integer threshold is

$$\ell \geq \left\lceil \max \left\{ 1, \left(\frac{4L_g C_{\text{box}}}{g_- c_T} \right)^{1/6}, \frac{\log \max\{1, 4C_B/(g_- c_T)\}}{2h \log 2} \right\} \right\rceil.$$

Each error is then at most $g_- c_T/4$, so the lower bound is $(g_- c_T/2)|T|$, including $T = 0$.

For the actual homogeneous pulse, positivity follows directly from its proved polarization. This requires no assertion that its instantaneous flux lies in the prescribed target cone. Retain

$$t = x(e_r - s_a K_a) + y N_a, \quad s_{\max} = 3u_*/2, \quad \gamma_0 = c_0\sqrt{1+s^2},$$

and choose the constants in the already proved estimates as

$$|n_\Phi - B_s(s, K)| \leq E_n/S_*, \quad B_s \geq B_- > 0, \quad |y/x - \gamma_0| \leq E_\gamma/S_*, \quad c_x P \leq x \leq C_x P.$$

The quantities g_-, g_+, c_-, c_+ denote positive uniform bounds $g_- \leq |g_0| \leq g_+$, $c_- \leq |c_0| \leq c_+$, with $c_0 < 0$. Choose $S_* \geq 2E_n/B_-$. Then $|n_{\Phi, \tan}| \geq B_-/2$. For nonzero vectors a, b , addition and subtraction of $a/|b|$, followed by $||a| - |b|| \leq |a - b|$, gives $|a/|a| - b/|b|| \leq 2|a - b|/|b|$. Therefore

$$|K_a - K| = |N_a - N| \leq \frac{2E_n}{B_- S_*}, \quad |s_a - s| \leq \frac{2(1 + s_{\max})E_n}{B_- S_*}.$$

The last estimate follows by writing $n_{\Phi, r} - s|n_{\Phi, \tan}| = (n_{\Phi, r} - B_s s) + s(B_s - |n_{\Phi, \tan}|)$ and dividing by the actual tangential norm. Define

$$\Gamma_{\max} = c_+ \sqrt{1 + s_{\max}^2}, \quad C_T = \frac{2E_n}{B_-} (1 + 2s_{\max} + \Gamma_{\max}) + E_\gamma.$$

The exact flux and its exact remainder are

$$T = x^2[-sK + \gamma_0 N + e_T], \quad e_T = -(s_a K_a - sK) + (y/x)N_a - \gamma_0 N.$$

Expand the first difference as $(s_a - s)K_a + s(K_a - K)$ and the second as $(y/x - \gamma_0)N_a + \gamma_0(N_a - N)$. The preceding bounds give $|e_T| \leq C_T/S_*$, with all four terms retained. Hence

$$\frac{-g \cdot T}{x^2} = -|g_0|\gamma_0 - g_0 \cdot e_T - \Delta g \cdot (-sK + \gamma_0 N + e_T).$$

The absolute value of the last two terms is at most

$$\mathcal{E}_\ell = g_+ C_T \ell^{-2} + (C_B 2^{-2h\ell} + L_g C_{\text{box}} \ell^{-6})(s_{\max} + \Gamma_{\max} + C_T \ell^{-2}).$$

Set $W = s_{\max} + \Gamma_{\max} + C_T > 0$. In addition to all previous phase and invariant-band thresholds, it suffices to choose

$$\ell \geq \left\lceil \max \left\{ 1, \left(\frac{2E_n}{B_-} \right)^{1/2}, \left(\frac{4g_+ C_T}{g_- c_-} \right)^{1/2}, \left(\frac{8W L_g C_{\text{box}}}{g_- c_-} \right)^{1/6}, \frac{\log \max\{1, 8W C_B/(g_- c_-)\}}{2h \log 2} \right\} \right\rceil.$$

For $\ell \geq 1$, the last factor in $\mathcal{E}_\ell$ is at most W . The three error contributions are then at most $g_- c_-/4, g_- c_-/8, g_- c_-/8$, respectively. Since $-|g_0|\gamma_0 \geq g_- c_-$, this proves

$$\frac{1}{2}g_- c_- c_x^2 P(v)^2 \leq -g \cdot T \leq (g_+ \Gamma_{\max} + \frac{1}{2}g_- c_-) C_x^2 P(v)^2.$$

All constants concern the same selected h , fixed profiles and original compact label family. The displayed thresholds are finite and are added to the previous finite value thresholds; no uniformity as $h \downarrow 0$ is asserted. The exact actual-frame norm is

$$|t|^2 = (1 + s_a^2)x^2 + y^2.$$

The damping $\varepsilon k^2 |n_\Phi|^2 |t|^2$ remains in the energy equation, so positive production is not a claim of positive total energy derivative. Covariance mass uses the proved positive x and the original positive integral of $\psi^2 x^2$; its argument does not omit this damping. The carrier physical wavelength is $\sqrt{Q}/k \asymp Q^{1/2+h/2}$, while the leading physical wave amplitude has power $Q^{-A} \sqrt{\varepsilon} = Q^{-1/2-h/2}$, up to polynomial factors in S_* . Their product has power Q^0 ; this agrees with the retained leading damping $\varepsilon k^2 \asymp 1$.

19 Covariance, signed stress increments, exact curls, and cutoff residuals

This file gives direct proofs for source pp. 81-87, Propositions 7.5-7.6, Lemma 7.7, and Corollary 7.8. The background inputs and pulse estimates are recorded and derived in the preceding files. All averages below are on the extended domain at fixed slow coordinates, before physical evaluation.

19.1 Two columns and their positive coefficients

For real extended vector fields define

$$\mathcal{C}(w) = \langle \langle w_r w_{\tan} \rangle_\theta \rangle_Y, \quad \mathcal{B}(w, v) = \langle \langle w_r v_{\tan} + v_r w_{\tan} \rangle_\theta \rangle_Y, \quad w_{\tan} = (w_\theta, w_z).$$

Expanding the product proves $\mathcal{C}(w + v) = \mathcal{C}(w) + \mathcal{B}(w, v) + \mathcal{C}(v)$ and $D\mathcal{C}(w)[v] = \mathcal{B}(w, v)$. These maps preserve the original component order $(r\theta, rz)$.

For a slow box $\beta = (\ell, a)$, its two signs have disjoint auxiliary rectangles and the same slow cutoff. Write

$$b_\sigma = \chi_g(\xi_g)\psi(v)t_\sigma^h \cos(k\Phi_\sigma), \quad H = [\mathcal{C}(b_+) \mid \mathcal{C}(b_-)].$$

Each b_σ is extended by zero outside its rectangle, using its smooth transverse and temporal cutoffs. The integer $kp \neq 0$ ensures that

$$\frac{1}{2\pi} \int_0^{2\pi} \cos^2(k\Phi_\sigma) d\theta = \frac{1}{2}$$

exactly, whatever the non-angular part of the phase. Haar pullback permits averaging on the band torus. The coordinate map $(\xi_g, \eta_g) \mapsto c_\gamma + \xi_g v_r + \eta_g v_t$ has the retained Jacobian $|\det(v_r, v_t)| = 1 + b_g^2$, and $d\eta_g = c_i dv$. The homogeneous amplitude is independent of ξ_g , since its ODE coefficients depend on slow coordinates and v , and its initial value is fixed with the label. With radial component x , this gives

$$H_\sigma = D_g c_i \int_0^{L_s} \psi^2 x t_{\sigma, \tan}^h dv, \quad h_\sigma = D_g c_i \int_0^{L_s} \psi^2 x^2 dv,$$

$$D_g = \frac{|\det(v_r, v_t)|}{2} \int_{\mathbb{R}} \chi_g(\xi)^2 d\xi > 0.$$

Thus the angular factor $1/2$, the rectangular Jacobian, the time measure $c_i dv$, and the transverse cutoff integral are all present.

The proved bounds $cP \leq x \leq CP$ and $P(v) \asymp \exp(-c(v - L_s/2)^2/L_s)$, with separate upper and lower constants, imply

$$c\sqrt{L_s} \leq \int_0^{L_s} \psi^2 x^2 dv \leq C\sqrt{L_s}.$$

For the lower bound, integrate over $|v - L_s/2| \leq \sqrt{L_s}$, which lies in the region $\psi = 1$ for all sufficiently large L_s ; P there has a fixed positive lower bound. The upper bound follows by extending the Gaussian integral to $\mathbb{R}$. The same substitution $w = (v - L_s/2)/\sqrt{L_s}$ proves

$$\int_0^{L_s} \frac{|v - L_s/2|}{L_s} \psi^2 x^2 dv \leq C.$$

It follows that $h_\sigma \asymp c_i \sqrt{L_s} \asymp S_*^{-1/2}$, and the probability measure proportional to $\psi^2 x^2 dv$ has expectation of $|v - L_s/2|/L_s$ at most $CS_*^{-1/2}$.

Since $t_{\sigma, \tan}^h/x = -s_a K_a + (y/x)N_a$, the previous file proves its uniform difference from $-s(v)K + c_0\sqrt{1+s(v)^2}N$ is $O(S_*^{-1})$. The latter function is Lipschitz in s on the fixed compact range, and $|s(v) - \sigma u_*| = u_*|v - L_s/2|/L_s$. Taking the weighted average therefore gives

$$H_\sigma = h_\sigma(-A_c N - \sigma u_* K + e_\sigma), \quad A_c = -c_0\sqrt{1+u_*^2} > 0, \quad |e_\sigma| \leq CS_*^{-1/2}.$$

Every fixed slow derivative of H_σ is polynomially bounded: differentiate the finite integral using the already proved pulse derivatives, fixed L_s , and fixed label cutoffs. None of these parameter derivatives differentiates a discrete label or moves the integration endpoints.

In coordinates N, K , omit e_σ temporarily and solve the exact two-by-two system:

$$h_+ y_+ = \frac{1}{2}(-T_N/A_c - T_K/u_*), \quad h_- y_- = \frac{1}{2}(-T_N/A_c + T_K/u_*).$$

The cone choice made in the previous file gives a fixed $\eta_c > 0$ such that

$$\frac{|T_K|}{u_*} \leq (1 - \eta_c) \frac{-T_N}{A_c}, \quad T = T_{0,*}.$$

Since N, K are orthonormal and $T_N < 0$, this also gives $-T_N \geq c|T|$. Hence the two displayed reference coefficients are at least $c|T|$ and at most $C|T|$.

The reference column matrix in these coordinates is

$$H_{\text{ref}} = \begin{pmatrix} -A_c h_+ & -A_c h_- \\ -u_* h_+ & u_* h_- \end{pmatrix}, \quad \det H_{\text{ref}} = -2A_c u_* h_+ h_-.$$

Writing $H = (H_{\text{ref}} \text{diag}(h_+^{-1}, h_-^{-1}) + \mathcal{E}) \text{diag}(h_+, h_-)$, the first factor differs from a fixed invertible matrix by $O(S_*^{-1/2})$. For $\|H_0^{-1}\mathcal{E}\| < 1/2$, the exact inverse of $I + H_0^{-1}\mathcal{E}$ is the convergent series $\sum_{n=0}^{\infty} (-H_0^{-1}\mathcal{E})^n$. Indeed multiplication of its finite partial sum through $n = N$ by $I + H_0^{-1}\mathcal{E}$, in either order, gives $I - (-H_0^{-1}\mathcal{E})^{N+1}$, and the last power tends to zero. The sum has norm at most 2, and its difference from I has norm at most $2\|H_0^{-1}\mathcal{E}\|$, by the geometric-series bounds. Thus the exact first-factor inverse is $(I + H_0^{-1}\mathcal{E})^{-1}H_0^{-1}$; the original diagonal $\text{diag}(h_+, h_-)^{-1}$ remains on its left. The coefficient perturbation is then bounded by $CS_*^{-1/2}|T|$ before dividing by h_σ . For a further fixed lower band threshold, it is smaller than half the positive reference coefficient margin. Therefore

$$|\det H| \geq c/S_*, \quad \|H^{-1}\| \leq C\sqrt{S_*}, \quad c\sqrt{S_*}|T_{0,*}| \leq y_\sigma \leq C\sqrt{S_*}|T_{0,*}|, \quad y = H^{-1}T_{0,*}.$$

This proves positivity uniformly up to limiting stress directions at the shell edges. At an edge the actual target is zero; the positive inequalities for y are asserted in the open shell.

The background profile theorem gives $|T_{0,*}| \geq c\zeta$ and $|D^I T_{0,*}| \leq C_I \zeta \delta^{-a_I}$, using the exact chart/profile transition from the first file. Differentiating $HH^{-1} = I$ expresses each fixed derivative of H^{-1} as a finite sum of products of H^{-1} and derivatives of H , hence gives polynomial S_* bounds. Leibniz's rule proves

$$y_\sigma \geq c\sqrt{S_*}\zeta, \quad |D^I y_\sigma| \leq C_I S_*^{b_I} \zeta \delta^{-a_I}.$$

For $a_\sigma = \sqrt{y_\sigma}$, every derivative of positive order is a finite sum

$$c y_\sigma^{1/2-n} \prod_{\nu=1}^n D^{I_\nu} y_\sigma, \quad |I_\nu| \geq 1, \quad \sum I_\nu = I.$$

The numerator contributes ζ^n , while the inverse power contributes $\zeta^{1/2-n}$; the product has exactly the remaining $\sqrt{\zeta}$. The S_* -powers and inverse edge-distance powers are finite at every fixed order. The value bound uses the upper bound for y_σ . Thus

$$|D^I a_\sigma| \leq C_I S_*^{b'_I} \sqrt{\zeta} \delta^{-a'_I},$$

which proves smooth zero extension of every coefficient at both shell edges.

Set $W_0 = \sqrt{\varepsilon} \sum_{\sigma=\pm} a_\sigma b_\sigma$. The two disjoint supports remove the cross products, and the scalar amplitudes are independent of both averaging variables. Consequently

$$\mathcal{C}(W_0) = \varepsilon H(a_+^2, a_-^2)^T = \varepsilon T_{0,*}$$

exactly. The pulse and square-root estimates give the local $\mathcal{W}_{1/2}$ bounds. Multiplying by η_β gives the required slow support.

The physical covariance of the assembled field is obtained with the original factors:

$$Q^{-2A} \varepsilon T_{0,*} = Q^{-2A+h} (Q/q)^{A+1/2} T_0 = Q^{-A+h+1/2} q^{-A-1/2} T_0 = q^{-A-1/2} T_0.$$

The final equality uses $A = 1/2 + h$. Each physical slow point has bounded overlap, and cross-label products vanish. Summing the squared partition $\sum_\beta \eta_\beta^2 = 1$ therefore supplies precisely the leading physical stress, with the sign suitable to cancel the negative stress divergence in Proposition 5.5.

19.2 Signed differential inverse and assembly across bands

Let $\Sigma(R, Z, T)$ be a real auxiliary-independent two-vector in $\mathcal{M}_\alpha$. Keep the previously chosen positive amplitudes fixed. Define

$$d_\Sigma = H^{-1}(\Sigma/\varepsilon), \quad \delta a_\sigma = (d_\Sigma)_\sigma / (2a_\sigma), \quad \mathcal{L}\Sigma = \sqrt{\varepsilon} \sum_{\sigma=\pm} \delta a_\sigma b_\sigma.$$

Here the division is taken only on the open shell before smooth zero extension. Every derivative of $y_\sigma^{-1/2}$ is a finite sum

$$c y_\sigma^{-n-1/2} \prod_{\nu=1}^n D^{I_\nu} y_\sigma,$$

so it has the weight $\zeta^{-1/2}$ with allowed polynomial and edge-distance factors. Since d_Σ has weight $\varepsilon^{\alpha-1}\zeta$, the product defining δa_σ has

$$|D^I \delta a_\sigma| \leq C_I \varepsilon^{\alpha-1} S_*^{b_I} \sqrt{\zeta} \delta^{-a_I}.$$

This proves smooth zero extension and the local $\mathcal{W}_{\alpha-1/2}$ bound for $\mathcal{L}\Sigma$, with the envelope supplied by b_σ . Disjointness and the retained bilinear factor two give

$$\mathcal{B}(W_0, \mathcal{L}\Sigma) = \varepsilon H(2a_+ \delta a_+, 2a_- \delta a_-)^T = \Sigma.$$

This is an exact linear right inverse of $DC(W_0)$, without positivity or smallness assumptions on Σ . It is not an exact nonlinear inverse: the retained identity is

$$\mathcal{C}(W_0 + \mathcal{L}\Sigma) = \varepsilon T_{0,*} + \Sigma + \mathcal{C}(\mathcal{L}\Sigma).$$

The last term lies in $\mathcal{M}_{2\alpha-1}$, as both wave coefficients have exponent $\alpha - 1/2$ and the product has the mean weight ζ .

For a real physical auxiliary-independent stress $\sigma(r, z, t)$ supported in the active shell, whose representatives $Q_\beta^{2A} \sigma$ belong to the stated local $\mathcal{M}_\alpha$ domain, use the source's assembled maps. This domain includes every fixed derivative bound with the weight $\zeta \delta^{-a_I}$, compatibility on overlapping charts, and smooth zero extension at the shell edges. The previously chosen positive amplitudes remain fixed. The following identity holds throughout the active partition domain and extends by zero outside it because the input stress vanishes there:

$$W_{0,\text{phys}}^{\text{as}} = \sum_\beta Q_\beta^{-A} \eta_\beta W_{0,\beta}, \quad \mathcal{L}_{\text{phys}}^{\text{as}} \sigma = \sum_\beta Q_\beta^{-A} \eta_\beta \mathcal{L}_\beta(Q_\beta^{2A} \sigma).$$

The different Q_β are retained. The contribution of box β to the bilinear covariance is

$$Q_\beta^{-2A} \eta_\beta^2 Q_\beta^{2A} \sigma = \eta_\beta^2 \sigma.$$

There are no cross-box terms, so summation gives $\mathcal{B}(W_{0,\text{phys}}^{\text{as}}, \mathcal{L}_{\text{phys}}^{\text{as}} \sigma) = \sigma$. In a reference chart of scale Q , the representatives are

$$W_0^{\text{as}} = Q^A W_{0,\text{phys}}^{\text{as}}, \quad \mathcal{L}^{\text{as}} \Sigma = Q^A \mathcal{L}_{\text{phys}}^{\text{as}}(Q^{-2A} \Sigma).$$

Thus the exact chart identity is $\mathcal{B}(W_0^{\text{as}}, \mathcal{L}^{\text{as}} \Sigma) = \Sigma$. Overlapping Q_β/Q lie between fixed positive bounds, so chart changes preserve all claimed exponents; the bounded covering changes preserve derivatives and averages. Equal label-harmonic contributions are collected before class membership is tested. Auxiliary independence of Σ is essential to taking the amplitudes outside the integral and is explicitly a hypothesis here.

19.3 Curl identity with all cylindrical terms

The normalized cylindrical curl is

$$\text{curl}_* A = (R^{-1} \partial_\theta A_z - D_z A_\theta, D_z A_r - D_r A_z, (D_r + R^{-1}) A_\theta - R^{-1} \partial_\theta A_r).$$

For a smoothly supported $t_m \in \mathcal{W}_\alpha$ with $n_\Phi \cdot t_m = 0$, put

$$C_m = \frac{i n_\Phi \times t_m}{k m |n_\Phi|^2}, \quad A_m = C_m e^{i k m \Phi}.$$

Using the right-handed (e_r, e_θ, e_z) orientation, the exponential differentiation contributes

$$i k m n_\Phi \times C_m = - \frac{n_\Phi \times (n_\Phi \times t_m)}{|n_\Phi|^2} = t_m - \frac{n_\Phi (n_\Phi \cdot t_m)}{|n_\Phi|^2} = t_m.$$

The exact remainder is

$$r_m = (-D_z(C_m)_\theta, D_z(C_m)_r - D_r(C_m)_z, (D_r + R^{-1})(C_m)_\theta).$$

The multiplier $n_\Phi/|n_\Phi|^2$ has all polynomial coefficient bounds. Hence $C_m \in \mathcal{W}_{\alpha+1/2}$, and $r_m \in \mathcal{W}_{\alpha+1/2-\kappa_s}$. The less costly D_z and frame terms satisfy the same weaker common bound. Supports are preserved by local differentiation and smooth extension.

For any smooth normalized vector potential, retain

$$\text{div}_* a = (D_r + R^{-1}) a_r + R^{-1} \partial_\theta a_\theta + D_z a_z.$$

Because $D_r(R^{-1}) = -R^{-2}$, one has $(D_r + R^{-1})(R^{-1}f) = R^{-1}D_rf$. Substitution of the three curl entries gives the six terms

$$R^{-1}D_r\partial_\theta A_z - (D_r + R^{-1})D_zA_\theta + R^{-1}D_z\partial_\theta A_r - R^{-1}\partial_\theta D_rA_z + D_z(D_r + R^{-1})A_\theta - R^{-1}D_z\partial_\theta A_r.$$

They cancel in pairs by the commuting coordinate derivations and $D_z(R^{-1}) = 0$. Thus divergence of the exact curl vanishes. The pullback intertwining proved in the first file transfers this identity exactly to physical space.

For the full amplitude $a_m = t_m + r_m$, its coefficient is angular-independent. Expanding the divergence of $a_me^{ikm\Phi}$ proves

$$ikm n_\Phi \cdot a_m = -((D_r + R^{-1})(a_m)_r + D_z(a_m)_z).$$

The scalar $n_\Phi \cdot a_m = n_\Phi \cdot r_m$ belongs to $\mathcal{W}_{\alpha+1/2-\kappa_s}$, since $n_\Phi \cdot t_m = 0$ and the full phase vector has polynomial coefficient bounds. The displayed left side includes the additional factor ikm , with $k = O(\varepsilon^{-1/2})$ and fixed nonzero harmonic m ; its class is therefore $\mathcal{W}_{\alpha-\kappa_s}$. This is also the class supplied by the amplitude derivative on the right. In particular, the complete amplitude is not asserted to be transverse: its retained longitudinal component is precisely what the divergence identity controls. Physical vector potentials are $Q^{1/2-A}A_*$, because physical curl contributes $Q^{-1/2}$; physical pressures are $Q^{-2A}p_*$. These factors give exactly the prescribed physical velocity and normalized pressure.

Real waves use conjugate harmonic pairs. For real $t \cos(k\Phi)$, the velocity coefficients are $t/2$ for $m = \pm 1$; the factor $i/(km)$ changes by complex conjugation when m changes sign. Thus the potentials and pressures also form real physical fields. No missing factor is introduced by replacing the cosine with exponentials.

19.4 Exact cutoff residual and terminal flatness at each fixed stage

Let the zero-data inverse solve the principal equation for an actual supported source f_m , and set $\hat{t}_m = \psi t_m$, $\hat{\pi}_m = \psi \pi_m$. The principal equation yields exactly

$$\hat{t}_m + \mathcal{K}\hat{t}_m + m^2 d\hat{t}_m + ikmn_\Phi \hat{\pi}_m + f_m = (1 - \psi)f_m + \psi' t_m.$$

The source on the left remains f_m , not ψf_m . This is why the uncanceled $(1 - \psi)f_m$ term is required. The support of either right-hand term lies in $|v - L_s/2| \geq L_s/5$, since $\psi = 1$ on the complementary central region. There the proved Gaussian estimate is $P(v) \leq e^{-cL_s/25} \leq e^{-c'S_*}$. For every fixed coefficient derivative, Leibniz's formula gives the same exponential factor times allowed polynomial powers: derivatives of ψ are supported in the same exterior region, and all derivatives of f_m, t_m already obey the envelope bound. The transverse/slow/shell weights stay unchanged.

When the corresponding exponential and physical factors are differentiated any fixed number of times, only a fixed power Q^{-M} and a polynomial S_*^C are added. This can be verified directly: on band pullbacks, $T_g^i \leq Q^{-1-h}/S_*$, $\Lambda_g^i \leq Q^{-(1+h)\rho_g} S_*^{-\rho_g}$, $k \leq 2Q^{-h/2}$; chart derivatives contribute fixed powers $Q^{-1}, Q^{-D}, Q^{-1/2}$; and the phase itself has only the displayed $p_z Z/\varepsilon$, vH_Φ , $p\theta$, $x_0 R$ terms, with $v = O(S_*)$ and all relevant differentiated coefficients bounded polynomially. The cylindrical frame has $r^{-1} = Q^{-1/2}R^{-1}$, with R bounded away from zero. A fixed product/chain-rule expansion contains finitely many such factors. The exponents M, C may depend on the derivative order, stage, and finite harmonic set, but not on the band.

For every fixed M, C, N , $q \asymp Q = 2^{-\ell}$ gives

$$q^{-N} Q^{-M} S_*^C e^{-c'S_*} \leq C_N \exp(-c'\ell^2 + (M + N)\ell \log 2 + 2C \log \ell) \rightarrow 0.$$

For a direct bound, choose ℓ so that the last two positive terms are at most $c'\ell^2/2$; the remaining Gaussian is bounded and tends to zero. This proves every physical derivative of this fixed-stage cutoff residual is $O(q^N)$ for every N . Homogeneous pulse cutoff tails obey the same argument, with source zero and the retained $\psi' t^h$ term.

At each physical point only a bounded number of slow labels occur, so local summation retains the same bound. On their annular supports, $r^2/(2q) \geq X_a > 0$ implies $q \leq r^2/(2X_a)$. Thus these derivative bounds give arbitrary vanishing order at the singular space-time point $r = z = 0, t = 1$. Every one-sided derivative jet there has value zero; the derivative identities persist at that boundary point by induction and the fundamental theorem of calculus along coordinate segments lying in the time half-space. Outside the annular supports the fields vanish smoothly. At $t = 1, z \neq 0$, however, $q = |z|^{1/D} > 0$, so this estimate does not assert zero values on the whole terminal plane. One-sided smooth jets there are supplied by the parameter-dependent ODE proof. Any continuation across that plane, and the infinite-stage diagonal sum, are the later sections' responsibilities and are not inferred merely from the fixed-stage tail estimate.

19.5 Actual covariance increment with every remainder retained

Let $U = W_0^{\text{as}} + E$ be the actual real divergence-free wave field, with $U \in \mathcal{W}_{1/2}$, $E \in \mathcal{W}_\rho$. Given $\Sigma \in \mathcal{M}_\alpha$ independent of the averaging variables, applying the exact curl construction to $\mathcal{L}^{\text{as}}\Sigma$ gives

$$V = \mathcal{L}^{\text{as}}\Sigma + R, \quad R \in \mathcal{W}_{\alpha-\kappa_s}.$$

The latter exponent is $(\alpha - 1/2) + (1/2 - \kappa_s)$. Then

$$\begin{aligned} \mathcal{C}(U + V) - \mathcal{C}(U) &= \mathcal{B}(U, V) + \mathcal{C}(V) \\ &= \Sigma + \mathcal{B}(E, \mathcal{L}^{\text{as}}\Sigma) + \mathcal{B}(U, R) + \mathcal{C}(V). \end{aligned}$$

The exact bilinear identity at the fixed W_0^{as} is the only cancellation used. Products yield the respective mean exponents

$$\rho + \alpha - \frac{1}{2}, \quad \alpha + \frac{1}{2} - \kappa_s, \quad 2\alpha - 1.$$

For the last exponent, $\kappa_s < 1/2$ and $0 < \varepsilon \leq 1$ imply $R \in \mathcal{W}_{\alpha-1/2}$; hence V is in that class. The radial divergence is the original componentwise operator $D_r + 2/R$ or $D_r + 1/R$, respectively. Its derivative term loses at most κ_s , and its bounded frame terms cause no further loss. Thus all three remainder exponents after radial divergence are their displayed values minus at most κ_s . These are precise local estimates. Their compatibility with the residual-improvement exponents selected in Section 9 must be checked where those exponents are selected.

20 Admissible shear loop, radial modulation, and moment restoration

This file independently proves the operations in Appendix C, pp. 157-165. It retains the original logarithmic coordinate $D_X = X\partial_X$, profile $E > 0, U$, pressure integration constant $\Pi_0(\eta)$, component signs, interval ordering, and all five moments. The existence and numerical parameter margins of the input profile from Appendices A and B are audited by the profiles lane. This file verifies what Appendix C does to that input, including the exact restoration map; it does not claim that its imported input construction has been independently proved here.

20.1 Scalar cone and its four gaps

At $a > 0$, set

$$t_s = -b_s/a, \quad v_s = a(1 + t_s^2), \quad P_c = p_{s,1} + t_s p_{s,2}, \quad J_c = p_{s,2} - t_s p_{s,1}.$$

The relaxed condition is $P_c > 2$, $v_s < U(P_c, J_c)$, with

$$U(P, J) = P + \frac{J^2}{4} - |J| \sqrt{\frac{P-2}{2} + \frac{J^2}{16}}.$$

The admissible condition also requires $v_s > 2$. The quadratic expression

$$G(v) = 2(P - v)^2 - (v - 2)J^2$$

has roots $U(P, J)$ and the same expression with a plus before the square root. For $P > 2$, $U > 2$: writing $w = P - 2 > 0$, the inequality

$$\left(w + \frac{J^2}{4}\right)^2 - J^2 \left(\frac{w}{2} + \frac{J^2}{16}\right) = w^2 > 0$$

proves $w + J^2/4 > |J|\sqrt{w/2 + J^2/16}$. Also $U \leq P$, because the square-root term is at least $J^2/4$, with equality only for $J = 0$. When $J = 0$, $U = P > 2$. Therefore for $v > 2$, admissibility is exactly

$$P - v > 0, \quad 2(P - v)^2 - (v - 2)J^2 > 0.$$

The four continuous smooth gaps on $a > 0$ are a , $v_s - 2$, $P_c - v_s$, and $2(P_c - v_s)^2 - (v_s - 2)J_c^2$. Positivity on a compact parameter set gives one positive lower margin for all four. The root U need only be continuous; no differentiation of its $|J|$ term at zero is used.

20.2 A smooth loop whose average is exactly the original shear

Let $I \times [-1, 1]$ be the fixed compact domain from Appendix C, with relaxed data and admissibility on radial boundary collars. The positive minimum of $P_c(t_s) - 2$ permits a constant

$$0 < d_0 < \frac{1}{2} \min(P_c(t_s) - 2).$$

Here $P_c(t) = p_{s,1} + p_{s,2}t$ and $J_c(t) = p_{s,2} - p_{s,1}t$ for a variable slope t . With normalized θ' -average on $[0, 2\pi]$, define

$$M_e(z) = \frac{1}{2\pi} \int_0^{2\pi} e^{z \sin \theta'} d\theta', \quad t(\theta'; \mu) = t_s + \frac{d_0}{p} \left(\frac{e^{\mu p \sin \theta'}}{M_e(\mu p)} - 1 \right), \quad p = p_{s,2}.$$

For $p = 0$ the quotient is defined by continuity. Its numerator is smooth in p, μ, θ' , zero at $p = 0$, and the identity

$$\frac{G(p)}{p} = \int_0^1 G'(up) du$$

proves joint smoothness, including all parameters and derivatives. Since $M_e(0) = 1, M_e'(0) = 0$, the value there is $t = t_s + d_0 \mu \sin \theta'$.

The definitions give

$$\langle t \rangle_{\theta'} = t_s, \quad P_c(t) = P_c(t_s) + d_0 \left(\frac{e^{\mu p \sin \theta'}}{M_e(\mu p)} - 1 \right) \geq P_c(t_s) - d_0 > 2.$$

The variance is

$$V(\mu, p) = \langle (t - t_s)^2 \rangle = \frac{d_0^2}{p^2} \left(\frac{M_e(2\mu p)}{M_e(\mu p)^2} - 1 \right),$$

with value $d_0^2 \mu^2 / 2$ at $p = 0$. To show strict increase, let $g(z) = \log M_e(z)$. Differentiating under the finite integral gives

$$g''(z) = \langle (\sin \theta' - \langle \sin \theta' \rangle_z)^2 \rangle_z > 0,$$

where the tilted probability density is $e^{z \sin \theta'} / (2\pi M_e(z))$. It is strictly positive on the circle and $\sin \theta'$ is not constant, so the variance is strictly positive. For $p \neq 0, \mu > 0$,

$$\partial_\mu (g(2\mu p) - 2g(\mu p)) = 2p(g'(2\mu p) - g'(\mu p)) > 0.$$

For $p > 0$ both factors on the right are positive; for $p < 0$ both are negative. Thus V strictly increases in $\mu > 0$, including its separately proved formula at $p = 0$.

Here is the required growth bound, without an unproved asymptotic invocation. For $z \geq 1$, write $\theta' = \pi/2 + x$ near the maximum. On a fixed small interval $|x| \leq a < \pi$, there are $c_1, c_2 > 0$ with $1 - c_2 x^2 \leq \cos x \leq 1 - c_1 x^2$. A lower bound integrates over $|x| \leq z^{-1/2} \min(a, 1)$, giving $ce^z z^{-1/2}$. An upper bound on the same interval uses the Gaussian integral and gives $Ce^z z^{-1/2}$. On its complement $\sin \theta' \leq 1 - c_a$, so the contribution is at most $e^{(1-c_a)z}$, also bounded by $Ce^z z^{-1/2}$. For $z < 0$, shift θ' by π to obtain $M_e(z) = M_e(-z)$. Hence

$$ce^{|z|}/\sqrt{|z|} \leq M_e(z) \leq Ce^{|z|}/\sqrt{|z|} \quad (|z| \geq 1).$$

Taking the ratio yields positive constants bounding $M_e(2z)/M_e(z)^2$ above and below by multiples of $\sqrt{|z|}$. Therefore $V(\mu, p) \rightarrow \infty$ for each fixed $p \neq 0$; for $p = 0$ its explicit quadratic formula proves the same.

The compact range of p admits a uniform finite $\mu_{\max}$ such that $V(\mu_{\max}, p) > 3/\min a$. To prove uniformity, for each p_0 select a finite μ_{p_0} giving the strict bound there; continuity gives a neighborhood in p ; a finite subcover gives finitely many such μ , and their maximum works by monotonicity. This step is valid at $p = 0$, so no divergent factor p^{-2} is being used near zero.

All slopes $t(\theta'; \mu)$ for $0 \leq \mu \leq \mu_{\max}$ form a compact smooth family with $P_c(t) > 2$. Continuity of U and the strict inequality $U(P, J) > 2$ established above give a positive minimum of $U - 2$. Choose $0 < \delta_L < 1$ smaller than that minimum, and smaller than the positive minimum of $v_s - 2$ on chosen smaller radial boundary collars.

Choose a smooth $0 \leq \zeta_L \leq 1$ as a function of v_s , equal to one for $v_s \leq 2 + \delta_L/8$ and zero for $v_s \geq 2 + \delta_L/4$. Put

$$v_* = 2 + \delta_L/2, \quad \rho = \zeta_L(v_s)^2(v_* - v_s), \quad v = v_s + \rho,$$

and define $\rho = 0$ where $\zeta_L = 0$. On its possible nonzero support, $v_* - v_s \geq \delta_L/4$, so

$$\sqrt{\rho} = \zeta_L(v_s) \sqrt{v_* - v_s}$$

extends smoothly by zero. Since $v_s > 0$, $0 \leq \rho < v_* < 3$.

There is a unique $0 \leq \mu < \mu_{\max}$ solving $V(\mu, p) = \rho/a$, by the preceding strict monotonicity and endpoint bounds. Its smoothness at $\rho = 0$ needs a distinct verification. Expanding the finite integral in its convergent power series gives $M_e(z) = 1 + z^2/4 + O(z^4)$, and therefore

$$V(\mu, p) = \frac{1}{2} d_0^2 \mu^2 + O(\mu^4 p^2)$$

uniformly for bounded p . Thus the signed square root is $\mu(d_0/\sqrt{2} + O(\mu^2 p^2))$, a smooth odd function of μ near zero with positive derivative $d_0/\sqrt{2}$. Applying the implicit-function theorem to this function and $\sqrt{\rho/a}$, already proved smooth, gives smooth μ there. For $\rho > 0$, strict positivity of $\partial_\mu V$ gives the same conclusion. On the overlap the two local inverses agree by uniqueness.

The three cutoff regimes show $v > 2$ everywhere. Where $\zeta_L \neq 0$, $v \leq v_* < 2 + \delta_L < U(P_c(t), J_c(t))$. Where $\zeta_L = 0$, $\mu = 0, t = t_s, v = v_s > 2$, and the original relaxed condition gives $v < U$. Thus each proposed state lies in the admissible cone.

Define a lifted phase coordinate by

$$\phi(0) = 0, \quad \frac{d\phi}{d\theta'} = \frac{a(1+t^2)}{2\pi v}, \quad (a_L, -b_L) = \frac{v(1, t)}{1+t^2}.$$

Since $\langle t \rangle = t_s$ and $\langle (t - t_s)^2 \rangle = V$,

$$a\langle 1+t^2 \rangle = a(1+t_s^2) + aV = v_s + \rho = v.$$

Therefore $\phi(\theta' + 2\pi) = \phi(\theta') + 1$. Its derivative has a uniform positive lower bound on the compact parameter set; it defines an orientation-preserving circle diffeomorphism with a smooth parameter-dependent inverse. Changing variables proves both original mean components exactly:

$$\int_0^1 a_L d\phi = \frac{a}{2\pi} \int_0^{2\pi} 1 d\theta' = a, \quad \int_0^1 (-b_L) d\phi = \frac{a}{2\pi} \int_0^{2\pi} t d\theta' = at_s = -b_s.$$

Also $-b_L/a_L = t$, $a_L(1+t^2) = v$, so the verified scalar cone inequalities are the desired vector cone gaps. At boundary collars $\zeta_L = 0$, and the reparametrized shear is exactly the original constant-in- ϕ shear. Compactness gives one $\kappa_L > 0$ for all four cone gaps. This completes the complete loop construction, including the mean, boundary, and smoothness claims.

20.3 Exact insertion into the unchanged original radial coordinate

Let $\mathcal{A}, \mathcal{B}$ be the zero-mean periodic antiderivatives

$$\partial_\phi \mathcal{A} = -\frac{1}{2}(a_L - a), \quad \partial_\phi \mathcal{B} = \frac{1}{2}E(b_L - b_s).$$

They exist because the right sides have zero ϕ -mean. One explicit construction for any zero-mean g is

$$\mathcal{I}g(\phi) = \int_0^\phi g(w) dw - \int_0^1 \int_0^s g(w) dw ds.$$

It is periodic, zero mean, smooth in all parameters, and has derivative g . It vanishes identically when $g \equiv 0$. Therefore these antiderivatives vanish on the boundary collars and extend smoothly by zero outside I .

For an integer $N > 0$, keep the source's exact definitions

$$E_N = E \exp(\mathcal{A}(X, \eta, N \log X)/N), \quad U_N = U + \mathcal{B}(X, \eta, N \log X)/N.$$

The modified E_N is positive without approximation. The chain rule in the logarithmic coordinate is $X\partial_X = D_X + N\partial_\phi$ after substitution. Hence

$$\begin{aligned} a_N &= 1 - 2X\partial_X \log E_N \\ &= a - 2D_X \mathcal{A}/N - 2\partial_\phi \mathcal{A} = a_L - 2D_X \mathcal{A}/N, \end{aligned}$$

$$\begin{aligned}
b_N &= \frac{2X\partial_X U_N}{E_N} \\
&= e^{-\mathcal{A}/N} \left(\frac{2D_X U + 2\partial_\phi \mathcal{B}}{E} + \frac{2D_X \mathcal{B}}{NE} \right) \\
&= e^{-\mathcal{A}/N} \left(b_L + \frac{2D_X \mathcal{B}}{NE} \right).
\end{aligned}$$

The axial exponential denominator is retained; replacing it by one would destroy this exact shear formula.

On the fixed compact interval from X_- through the first reserved patch, $X, E, H = \sqrt{2X}E, L$ have positive lower bounds. The phase $N \log X$ is independent of η , so every fixed number m of η -derivatives of $E_N - E, U_N - U, a_N - a_L, b_N - b_L$ is bounded by C_m/N . For the exponential difference, use

$$e^{\mathcal{A}/N} - 1 = \frac{\mathcal{A}}{N} \int_0^1 e^{s\mathcal{A}/N} ds$$

and then differentiate under its fixed finite integral; all factors of $\mathcal{A}$ and its parameter derivatives are uniformly bounded on the periodic compact domain. For radial derivatives, $r \geq 1$ differentiations produce at most N^r from the phase and retain the initial N^{-1} , giving $C_{r,m}N^{r-1}$. Thus no small-radial-derivative assertion is used for the modulation.

20.4 All five moments and the dependence of the integrated stress

Retain exactly

$$\begin{aligned}
M &= \int_0^X U dx, \quad I = \int_0^X \sqrt{2x}E dx, \quad J = \int_0^X U\sqrt{2x}E dx, \\
S &= \int_0^X (U^2 - E^2/2) dx, \quad C_p = \int_0^X E^2/(2x) dx, \quad \Pi = \Pi_0 + C_p.
\end{aligned}$$

On the support of the differences, all scalar weights are bounded, since it is a compact interval with positive left endpoint. For instance

$$U_N^2 - U^2 = (U_N - U)(U_N + U), \quad E_N^2 - E^2 = (E_N - E)(E_N + E).$$

These identities, the product rule, and integration of the uniform C_m/N profile bounds prove every fixed η -derivative of the full prefix moment difference is $O_m(N^{-1})$, uniformly in the endpoint X . The pressure datum is the same, so $\Pi_N - \Pi = C_{p,N} - C_p$ has exactly that bound.

The formulas whose continuity matters are

$$\begin{aligned}
W &= 1 - \frac{2D\eta M + (1 - \eta^2)M_\eta}{X}, \\
Q_s &= -W + \frac{(1 - h)I - D\eta I_\eta - (1 - \eta^2)J_\eta + 2(h - D)\eta J}{XH}, \\
N_s &= -WU + \frac{D(M - \eta M_\eta) + 4h\eta S - (1 - \eta^2)S_\eta}{X} + 4A\eta\Pi - (1 - \eta^2)\Pi_\eta, \\
p_s &= (XQ_s/L, XN_s/(LE)).
\end{aligned}$$

All denominators are bounded away from zero on the fixed interval. Differences of reciprocals are exactly

$$E_N^{-1} - E^{-1} = -(E_N - E)/(E_N E), \quad H_N^{-1} - H^{-1} = -(H_N - H)/(H_N H).$$

Applying the product rule to the displayed formulas proves the m -th parameter derivative bound for $p_{s,N} - p_s$ from the profile and moment bounds through order $m + 1$, with no radial derivative of the differences. This proves the precise assertion needed for the cone stability.

The compact loop has positive four-gap margin. Its values and the original p_s range in a compact subset of $a > 0$; take a fixed small neighborhood still in $a > 0$, on which the four-gap map has a finite Lipschitz constant. The just-proved $O(N^{-1})$ errors in a, b, p_s preserve each gap for sufficiently large finite N . Between I and the first correction patch, $E_N = E, U_N = U$, so the shears themselves equal the originals; the p_s error remains $O(N^{-1})$. The original strict admissible margins on that compact intervening interval are preserved by the same argument.

20.5 Nonlinear moment correction with an exact endpoint map

On the first reserved patch, the original profile is $U = 0, E = K(\eta)X^{-1/2-\lambda}$, where fixed $\lambda > 0$ and positive K have been selected before N . The earlier modulation is zero on this patch. Add two smooth compactly supported U -bumps and three E -bumps with five coefficient functions $c(\eta)$. Each block's bumps are nonnegative, nonzero, and supported on ordered disjoint intervals.

Differentiate the five original moment functionals at zero correction. In row order $(M, J; I, S, C_p)$, the two U -columns have weights 1, $H = \sqrt{2X}E$, and the three E -columns have weights $\sqrt{2X}, -E, E/X$. The off-diagonal blocks are zero because $U = 0$ before the correction. Up to the displayed nonzero factors, the exponent lists are

$$(0, -\lambda), \quad (1/2, -1/2 - \lambda, -3/2 - \lambda).$$

All powers within each block are distinct for the fixed $\lambda > 0$. To establish invertibility directly, a nonzero linear combination of m distinct powers has at most $m - 1$ distinct positive zeros: divide by the least power, use Rolle's theorem on m putative zeros, and apply induction to the derivative, which has $m - 1$ distinct powers and nonzero coefficients wherever the original nonconstant coefficients were nonzero. Thus the evaluation determinant $\det[x_j^{\alpha_i}]$ never vanishes for ordered $0 < x_1 < \dots < x_m$. Its sign is constant on that connected set. Multilinearity and Fubini give

$$\det \left[\int x^{\alpha_i} \beta_j(x) dx \right] = \int \det[x_j^{\alpha_i}] \prod_j \beta_j(x_j) dx_1 \dots dx_m \neq 0.$$

The integrand has a constant nonzero sign on a set of positive measure. This proves each block's rank and hence the full moment Jacobian's rank. Smooth dependence on η , compactness of $[-1, 1]$, and the derivative identity for the inverse give uniform finite bounds at every fixed parameter order. No uniform estimate as $\lambda \rightarrow 0$ is claimed or needed.

The exact moment increment map is $B(\eta)c + Q_\eta(c, c)$, because M, I are linear, J is bilinear in U, E , and S, C_p are quadratic. Let $d_N(\eta)$ be the negative moment discrepancy accumulated before the patch; it obeys $\|d_N\|_{C^m} \leq C_m/N$. On the C^0 ball of radius $r = 2\beta_0\|d_N\|_{C^0}$, where $\beta_0 = \|B^{-1}\|_{C^0 \rightarrow C^0}$, the map

$$\mathcal{T}(c) = B^{-1}(d_N - Q(c, c))$$

maps the ball to itself and has Lipschitz constant at most $2\beta_0\kappa_0r \leq 1/2$, provided $8\beta_0^2\kappa_0\|d_N\|_{C^0} \leq 1$. Thus it has a unique fixed point selected by iteration from zero. The derivative $B + D_cQ(c, c)$ is invertible by the same estimate, giving smooth parameter dependence including one-sided endpoint derivatives. In each preselected finite C^m norm the identical contraction proof works after increasing N , with the relevant multiplication and bilinear norms, and yields C_m/N for the same branch.

The assertion that every other fixed derivative is $O_m(N^{-1})$ can also be justified without demanding simultaneous smallness of infinitely many norms. Differentiate the pointwise equation m times. The highest coefficient derivative is multiplied by the uniformly invertible $B + D_cQ(c, c)$. All other terms contain either a derivative of d_N , $O_m(N^{-1})$, or a lower derivative of c ; induction gives $O_m(N^{-1})$ for them, with the fixed data derivatives included in the constants. The same inverse bound therefore proves the estimate for the highest derivative. Only the finite orders needed for positivity and the cone gaps determine the actual choice of N .

The bump shapes and widths are fixed, so their radial derivatives multiply these small coefficients by fixed constants. The correction is therefore small in the radial derivatives entering a, b_s , in the first parameter derivatives entering p_s , and in all prefix moments. The compact input cone margin on the correction patch persists. Positivity of E there also persists, after one further finite lower bound on N .

At the right endpoint X_{rep} , the five corrected moments equal the original moments exactly as functions of η , while the corrected profiles equal the original profiles for every $X \geq X_{\text{rep}}$. Their derivatives as functions of η also agree. Each difference moment has zero derivative in X there, by the original moment integrands, so it stays zero at every later radius. The common axis pressure datum gives identical Π . The displayed formulas for Q_s, N_s, p_s and the original incompressibility formula

$$V_0 = \frac{X}{L} (2\eta U - 2D\eta M/X - (1 - \eta^2)\partial_\eta(M/X))$$

then prove exact equality of these fields at every later radius. This is the exact morphism from the five-moment endpoint data to exterior pressure, velocity, and stress; it proves preservation of the terminal compensation, both unused reserved correction patches, and the entire exterior. The inner collar is unchanged because both modifications start later and pressure was integrated from the same axis datum.

20.6 Edge direction and the common weight

Appendix C leaves both original collars unchanged. The input edge facts it uses are: at the inner edge, with $y_a = \log(X/X_a)$, the stress is

$$T_0 = e^{-t_1^2/y_a^2} g_a(y_a) B_0(X, \eta), \quad g_a(0) > 0, \quad B_0(X_a, \eta) = F p_{s,r} \neq 0;$$

its limiting direction is a positive multiple of the limiting shear $(a, -b_s) = a(1, t_s)$. At the outer edge, with $y_b = \log(X_b/X)$, the component factorization has common exponential e^{-4/y_b^2} , angular prefactor $y_b^{-3} b_\theta$ with $b_\theta(0, \eta) > 0$, and the direction

$$n = \frac{(b_\theta, y_b^6 b_z)}{\sqrt{b_\theta^2 + y_b^{12} b_z^2}}.$$

The original collar analysis supplies smooth factors and the fixed-order derivative bounds. Their derivation is assigned to the profiles lane; their exact use here is as follows.

The positive factors cancel in $n = T_0/|T_0|$, giving a smooth direction through both endpoints. At the inner edge $n_z - t_s n_\theta = 0$ and $n_\theta + t_s n_z > 0$. At the outer edge $n = (1, 0)$, $t_s = 0$, giving the same two conclusions. In the interior,

$$n_\theta + t_s n_z = \frac{P_c - v_s}{|p_s - (a, -b_s)|} > 0, \quad n_z - t_s n_\theta = \frac{J_c}{|p_s - (a, -b_s)|}.$$

The admissible cone makes

$$2 - (v_s - 2) \frac{(n_z - t_s n_\theta)^2}{(n_\theta + t_s n_z)^2} > 0.$$

Both the denominator and this gap extend continuously to the closed annulus, where the gap has value two at both radial edges. Their positive compact minima permit one $0 < \kappa < 2$ with

$$n_\theta + t_s n_z \geq \kappa, \quad (v_s - 2)(n_z - t_s n_\theta)^2 \leq (2 - \kappa)(n_\theta + t_s n_z)^2.$$

The positive lower bounds on $F, a, v_s - 2$ in the interior and the original endpoint bounds likewise extend uniformly by compactness.

The original stress identity $T_0 = F(p_s - (a, -b_s))$ proves nonvanishing throughout the open shell: if it vanished, $P_c = v_s$, contrary to admissibility. The outside stress remains zero by the exact endpoint restoration and unchanged inner region. Define the original proposed common edge weight

$$\zeta(X) = \exp(-t_1^2/y_a^2 - 4/y_b^2), \quad \delta = \min(1, y_a, y_b),$$

extended by zero outside the open shell. On a sufficiently short inner collar, the second exponential is smooth and bounded above and below by positive constants, while g_a, B_0 have the required nonzero value bounds and fixed derivative bounds. The inner factorization therefore proves $|T_0| \geq c\zeta$ and $|\partial^I T_0| \leq C_I \zeta \delta^{-m_I}$. On a short outer collar the first exponential is similarly a smooth positive factor; $y_b^{-3} b_\theta$ has a lower positive bound and its derivatives have only finite inverse powers of y_b . The supplied outer factor derivative bounds give the same conclusion. On the remaining compact interior both ζ and $|T_0|$ have positive minima and every fixed derivative is bounded, so the same pair of inequalities follows after increasing constants. These three regions cover the annulus. The elementary exponential argument in the first file proves smooth zero extension of all stress derivatives at both boundaries.

All remaining completion claims in C.3 are exact preservation statements: the axis Cartesian formulas remain unchanged on the analytic rectangle; the pressure at infinity and exterior moment values remain unchanged by the proved five-moment map; the exterior heat function is unchanged pointwise; and on $X \geq X_v = X_{\text{end}}$ the original $U = M = 0$ gives $V_0 = 0$ directly by its formula. The third and fourth reserved intervals remain the exact original $U = 0, E = c_{\text{patch}}(1 + \eta^2)^{-1} X^{-1/2-\lambda}$ intervals, with their original order. Fixing this one finite N before the physical scale q , all dyadic bands, and all later correction stages proves the claimed independence of the resulting derivative constants from those later parameters.

21 Independent audit of compactly supported mean corrections

Source: *Finite Time Blowup for Navier-Stokes*, Section 8, printed pp.87-100, released at the source PDF; its URL is recorded in the accompanying provenance ledger. Verified SHA-256: 8c8a94ad9ac824c8b605b9827cadf7beaca48bd10b380de3cfc872a2c37afa81. All statements and proofs in this section were read, and every assigned page was rendered and visually inspected.

This audit finds no contradiction in the internal Section 8 calculations. The proofs below establish the local operator facts and algebra explicitly. The dependency ledger identifies the exact earlier data used; the existence of the actual background and the full iteration are audited elsewhere. In particular, the checks here do not establish the Navier-Stokes endpoint.

21.1 Original coordinates, operators, and spaces

Keep the paper's $A = 1/2 + h$, $D = 1/2 - h$, fixed $h > 0$, $Q = 2^{-\ell}$, $\epsilon = Q^h$, $S_* = \ell^2$, and

$$R = Q^{-1/2}r, \quad Z = Q^{-D}z, \quad T = Q^{-1}(1-t).$$

The positive function $q = q(z, t)$ is independent of radius. The active shell is $X_a < X < X_b$, where $X = QR^2/(2q)$. In a fixed band $q/Q \asymp 1$; hence the radii are bounded above and bounded away from zero. All radial integrals below run from 0 to ∞ , except where their bounds are written otherwise. Auxiliary Haar measure has total mass one on $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$. Angular averaging means componentwise averaging in the cylindrical frame.

The exact constants and vectors are

$$J_g = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad \Lambda_g = 4 - \sqrt{2}, \quad T_g = 4 + \sqrt{2}, \quad b_g = \sqrt{2} - 1,$$

$$v_r = (1, -b_g), \quad v_t = (b_g, 1), \quad \rho_g = \frac{\log \Lambda_g}{\log T_g}, \quad \kappa_s = 10^{-5}, \quad d_r = 2((1+h)\rho_g - h\kappa_s) > 0.$$

The phase map is $Y = v_r r^{d_r} + v_t t \pmod{\mathbb{Z}^2}$. For an independent auxiliary variable Y , differentiation after evaluation is

$$\mathbf{r} = \partial_r + d_r r^{d_r-1} v_r \cdot \partial_Y, \quad \mathbf{t} = \partial_t + v_t \cdot \partial_Y.$$

Both commute with ∂_z , and with each other: the coefficient of the auxiliary radial derivative depends only on r , while the coefficient of the auxiliary temporal derivative is constant. Direct multiplication gives $J_g v_r = \Lambda_g v_r$ and $J_g v_t = T_g v_t$.

For a fixed common covering level i_0 , let $y = J_g^{i_0} Y$, $L = v_r \cdot \partial_y$, $N = v_t \cdot \partial_y$, $M = \Lambda_g^{i_0} Q^{d_r/2}$, and $c = T_g^{i_0} Q^{1+h}$. Then

$$D_r = \partial_R + M d_r R^{d_r-1} L, \quad D_z = \epsilon \partial_Z, \quad t_* = -\epsilon \partial_T + cN, \quad \Delta_0 = D_r^2 + R^{-1} D_r + D_z^2.$$

The integer i_0 , Q , ϵ , M , and c remain fixed under coefficient differentiation. At the band level,

$$i(\ell) = \left\lfloor \log_{T_g} (Q^{-1-h}/S_*) \right\rfloor$$

implies $T_g^{-1} Q^{-1-h}/S_* < T_g^{i(\ell)} \leq Q^{-1-h}/S_*$. Raising to ρ_g and using the displayed definition of d_r proves

$$c \asymp S_*^{-1}, \quad M \asymp \epsilon^{-\kappa_s} S_*^{-\rho_g}, \quad M^{-1} \lesssim \epsilon^{\kappa_s} S_*^{\rho_g}.$$

Bounded changes of covering level multiply these expressions by fixed constants.

The class $\mathcal{M}_\alpha$ consists of angularly independent, compatible torus coefficients supported in the shell with smooth zero extension and, for every fixed multi-index I in (R, Z, T, y_1, y_2) , bounds

$$|\partial^I f| \leq C_{j,I} \epsilon^\alpha S_*^{B_{j,I}} \zeta \delta^{-C'_{j,I}}, \quad \zeta = \exp(-a_a/d_a^2 - a_b/d_b^2),$$

$$d_a = \log(X/X_a), \quad d_b = \log(X_b/X), \quad \delta = \min(1, d_a, d_b), \quad a_a, a_b > 0.$$

The powers may depend on the fixed derivative order and correction stage, but not on the band or point. The class $\mathcal{S}_\alpha$ has the same slow derivative bounds for functions of (Z, T) , without the radial weight. Combining powers into a common larger power is legitimate since $S_* \geq 1$, $\delta \leq 1$. This does not change the source object or its exponent.

Leibniz' rule, $0 < \zeta \leq 1$, and the bounded shell imply $\mathcal{M}_a \mathcal{M}_b \subset \mathcal{M}_{a+b}$. Slow coefficient derivatives preserve the exponent; D_r loses at most κ_s , D_z gains one, the slow part of t_* gains one, and cN preserves it. A slow field has no cN contribution. Each assertion follows term by term from the displayed operators. Multiplying an interior-supported coefficient by a bounded smooth base coefficient preserves its weighted class even when that base coefficient has no global shell weight.

21.2 Exact radial range, inverse, obstruction, and projection

Fix (Z, T) , a covering level, and $e \in \{0, 1, 2\}$. Write

$$\mathcal{D}_e = D_r + e/R.$$

Let $\mathcal{C}$ be the smooth real functions of $R > 0, y \in \mathbb{T}^2$ supported in the closed active shell and smoothly zero across its edges. Define the conjugation

$$(\Gamma H)(R, w) = H(R, w + MR^{d_r} v_r).$$

Its inverse replaces the plus sign by a minus sign. The chain rule gives the exact intertwining identity

$$\partial_R \Gamma(R^e \phi) = \Gamma(R^e \mathcal{D}_e \phi). \quad (\text{MC1})$$

Consequently the full obstruction is the function on the entire torus

$$(K_e f)(w) = \int R^e f(R, w + MR^{d_r} v_r) dR. \quad (\text{MC2})$$

A compactly supported solution of $\mathcal{D}_e \phi = f$ exists exactly when $K_e f = 0$ as a torus function. Necessity follows by integrating (MC1), since $R^e \phi$ vanishes at both radial endpoints. For sufficiency define

$$\phi(R, y) = R^{-e} \int_0^R (R')^e f(R', y + M((R')^{d_r} - R^{d_r}) v_r) dR'. \quad (\text{MC3})$$

It vanishes below the shell. Above the shell it is $R^{-e}(K_e f)(y - MR^{d_r} v_r) = 0$. All derivatives extend by zero, and (MC1) proves the equation. If two compactly supported solutions exist, their difference has $\partial_R \Gamma(R^e \phi) = 0$; the constant is zero below the shell, proving uniqueness.

The obstruction map is onto $C^\infty(\mathbb{T}^2)$: given $F(w)$, choose any smooth interior bump $h(R)$ with integral one and take $f(R, y) = R^{-e} h(R) F(y - MR^{d_r} v_r)$. Substitution in (MC2) gives $K_e f = F$. Thus the strongest exact relationship, at each fixed slow point, is the exact sequence

$$0 \longrightarrow \mathcal{C} \xrightarrow{\mathcal{D}_e} \mathcal{C} \xrightarrow{K_e} C^\infty(\mathbb{T}^2) \longrightarrow 0. \quad (\text{MC4})$$

This identifies all obstructions, not only their Haar averages.

Choose the paper's slow cutoff $\chi_m(X)$, equal to zero near the inner shell edge and one near the outer edge, with derivative supported on a fixed interior interval. For $g = R^e f$, define

$$\begin{aligned} Ig(R, y) &= \int_0^R g(R', y + M((R')^{d_r} - R^{d_r}) v_r) dR', \\ Jg(R, y) &= \int_0^\infty g(R', y + M((R')^{d_r} - R^{d_r}) v_r) dR', \\ T_e f &= R^{-e}(Ig - \chi_m Jg), \quad A_e f = R^{-e}(\partial_R \chi_m) Jg. \end{aligned} \quad (\text{MC5})$$

The conjugation yields $D_r Ig = g$, $D_r Jg = 0$. Since $D_r \chi_m = \partial_R \chi_m$, the product rule yields

$$\mathcal{D}_e T_e = 1 - A_e. \quad (\text{MC6})$$

Moreover $K_e A_e f = K_e f$, because after conjugation the factor $K_e f(w)$ is independent of R and $\int \partial_R \chi_m dR = 1$. Thus $A_e^2 = A_e$, $T_e \mathcal{D}_e = 1$ on $\mathcal{C}$, and T_e is the exact inverse of $\mathcal{D}_e$ on $\ker K_e$. The paper's inverse is an explicitly split inverse with a specified obstruction projector.

Haar averaging (MC2) gives

$$\langle K_e f \rangle = \int R^e \langle f \rangle dR. \quad (\text{MC7})$$

The zero weighted Haar moment is therefore necessary but in general insufficient for exact compact inversion. As an explicit smooth counterexample to that stronger, unclaimed assertion, take a nonzero integer k , the bump h above, and

$$f_M(R, y) = e^{-M^2} R^{-e} h(R) \cos(2\pi k \cdot (y - MR^{d_r} v_r)).$$

Its Haar mean vanishes at every radius, while $K_e f_M = e^{-M^2} \cos(2\pi k \cdot w) \neq 0$. The family has all fixed coefficient derivative bounds: each derivative introduces at most a fixed power of M , absorbed by e^{-M^2} . With the original scale relation for M , these bounds are flat in every power of ϵ . The counterexample does not contradict Lemma 8.2; it proves why its explicitly retained flat remainder cannot be deleted.

21.3 Primitive estimates and the full flatness argument

Set $U = R^{d_r}$ and

$$F(U, Z, T, y) = \frac{g(R, Z, T, y)}{d_r R^{d_r-1}}.$$

Use its smooth zero extension. The change of variables has uniformly bounded derivatives on the shell. Put

$$A_-(U, y) = \int_{-\infty}^0 F(U + u, y + Muv_r) du, \quad A_+(U, y) = \int_0^\infty F(U + u, y + Muv_r) du.$$

Then $Ig = A_-$, $Jg = A_- + A_+$, and $Ig - \chi_m Jg = (1 - \chi_m)A_- - \chi_m A_+$. In these integrals the shift Muv_r does not depend on U, Z, T . Hence every fixed coefficient derivative differentiates only F and the cutoff; it introduces no factor M . All integrations have bounded effective length. Since the source is smoothly zero extended, differentiating moving support creates no boundary term.

For completeness, the precise edge inequality is as follows. For any fixed real $B \geq 0$,

$$\frac{d}{ds} \log(e^{-a/s^2} s^{-B}) = 2a/s^3 - B/s > 0$$

whenever $0 < s < \sqrt{2a/B}$, with no upper restriction needed when $B = 0$. Near the inner edge, only A_- occurs and $0 < d_a(R') \leq d_a(R)$ on its effective segment. The factor associated with d_b stays between positive constants there. This inequality bounds the complete differentiated input weight by $C\zeta(R)\delta(R)^{-B'}$. The outer edge uses only A_+ and the same argument with d_b . On the fixed interior region both radial edge distances are bounded below, so the output weight is bounded below and the unweighted bound suffices. Multiplication by $R^{\pm e}$ and the $R \leftrightarrow U$ chain rule cost fixed constants. This proves $T_e : \mathcal{M}_\alpha \rightarrow \mathcal{M}_\alpha$, including every fixed derivative, shell support, and projection of slow support.

The directional small divisors are controlled without a numerical approximation. For $k = (k_1, k_2) \neq 0$,

$$v_r \cdot k = (k_1 + k_2) - \sqrt{2}k_2,$$

and its product with $(k_1 + k_2) + \sqrt{2}k_2$ is the integer $(k_1 + k_2)^2 - 2k_2^2 \neq 0$. Nonzero follows from the irrationality of $\sqrt{2}$; if $k_2 = 0$, it follows from $k_1 \neq 0$. The conjugate has absolute value at most $C|k|$. Therefore $|v_r \cdot k|^{-1} \leq C(1 + |k|)$. The same calculation for $v_t \cdot k = (k_2 - k_1) + \sqrt{2}k_1$ gives the identical bound.

For $p \geq 1$, the inverse L^{-p} on zero-mean torus functions is the multiplier $(2\pi i v_r \cdot k)^{-p}$ at $k \neq 0$, zero at $k = 0$. For a C^{m+p+3} function, integrate its Fourier coefficient $m + p + 3$ times in a coordinate with $|k_j| \geq |k|/\sqrt{2}$. A derivative of order at most m , followed by the multiplier bound, gives a summand bounded by $C\|H\|_{C^{m+p+3}}(1 + |k|)^{-3}$. The series over $\mathbb{Z}^2$ converges, proving

$$\|L^{-p}H\|_{C^m} \leq C_{m,p}\|H\|_{C^{m+p+3}}. \quad (\text{MC8})$$

It also proves commutation with slow derivatives, smoothness, and reality. There is no assertion of constants uniform in p .

Under $\int R^e \langle f \rangle dR = 0$, the integral of the zero Fourier mode of F is zero. For $F^\circ = F - \langle F \rangle$, differentiation in u gives

$$\frac{d}{du} L^{-1} F^\circ(U + u, y + Muv_r) = \partial_U L^{-1} F^\circ(U + u, y + Muv_r) + M F^\circ(U + u, y + Muv_r).$$

Its full-line integral is zero by smooth compact support. Iterating this identity gives the exact equality

$$Jg = (-M^{-1})^p \int_{\mathbb{R}} \partial_U^p L^{-p} F^\circ(U + u, y + Muv_r) du. \quad (\text{MC9})$$

Each repetition has the same minus sign. All omitted zero-mode integrals, including their slow derivatives, are exactly zero. Fixed-shift differentiation and (MC8) bound any further derivative of (MC9). For an output derivative I , radial input derivatives through $I_R + p$ and torus derivatives through $|I_y| + p + 3$, with the requested slow derivatives, suffice. The coefficient $\partial_R \chi_m$ has fixed interior support. Thus

$$|\partial^I A_e f| \leq C_{j,I,p} \epsilon^{\alpha+p\kappa_s} S_*^{B_{j,I,p}} \zeta \delta^{-B'_{j,I,p}}, \quad \langle A_e f \rangle = 0. \quad (\text{MC10})$$

If f is independent of y , its full characteristic integral is its ordinary weighted radial integral. Under the moment condition the remainder is identically zero, not merely small.

To pass from (MC10) to physical flatness, fix the physical output normalization and derivative order before choosing p . Its graph derivatives require only finitely many coefficient derivatives and cost some finite power q^{-L} times a power of S_* ; L depends on the output order, not on p . The cutoff remainder is interior supported, so no boundary-weight loss remains there. For prescribed $N \geq 0$, choose p with $h(\alpha + p\kappa_s) > N + L$. Since $q \asymp Q$ and $S_* = \ell^2$, the positive excess in the exponent dominates every fixed power of ℓ , proving an $O(q^N)$ bound. This proves every fixed physical derivative is flat. Finite regularity alone would not imply this all-orders conclusion.

21.4 Mean momentum equations derived in the cylindrical frame

Let $e_r = (\cos \theta, \sin \theta, 0)$, $e_\theta = (-\sin \theta, \cos \theta, 0)$, $e_z = (0, 0, 1)$. Then $\partial_\theta e_r = e_\theta$, $\partial_\theta e_\theta = -e_r$. For a divergence-free vector $U = (U_r, U_\theta, U_z)$,

$$(D_r + R^{-1})U_r + R^{-1}\partial_\theta U_\theta + D_z U_z = 0.$$

Expand transport using this identity. For example,

$$(D_r + R^{-1})U_r^2 + R^{-1}\partial_\theta(U_\theta U_r) + D_z(U_z U_r) = U_r D_r U_r + R^{-1}U_\theta \partial_\theta U_r + U_z D_z U_r,$$

because the extra factor multiplying U_r is the displayed divergence. The frame derivative then adds $-U_\theta^2/R$. In the angular component it adds $U_r U_\theta/R$, so its radial flux has weight two. In the axial component the frame contributes nothing. Therefore the transport components are

$$\begin{aligned} (U \cdot \nabla U)_r &= (D_r + R^{-1})U_r^2 + R^{-1}\partial_\theta(U_\theta U_r) + D_z(U_z U_r) - U_\theta^2/R, \\ (U \cdot \nabla U)_\theta &= (D_r + 2R^{-1})(U_r U_\theta) + R^{-1}\partial_\theta U_\theta^2 + D_z(U_z U_\theta), \\ (U \cdot \nabla U)_z &= (D_r + R^{-1})(U_r U_z) + R^{-1}\partial_\theta(U_\theta U_z) + D_z U_z^2. \end{aligned} \tag{MC11}$$

The cylindrical vector Laplacian is

$$\begin{aligned} (\Delta U)_r &= (\Delta_0 + R^{-2}\partial_\theta^2 - R^{-2})U_r - 2R^{-2}\partial_\theta U_\theta, \\ (\Delta U)_\theta &= (\Delta_0 + R^{-2}\partial_\theta^2 - R^{-2})U_\theta + 2R^{-2}\partial_\theta U_r, \\ (\Delta U)_z &= (\Delta_0 + R^{-2}\partial_\theta^2)U_z. \end{aligned}$$

These follow by applying ∂_θ^2 to each component times its frame vector, which produces both the $\mp 2\partial_\theta$ terms and $-U_{r,\theta}$. Their angular averages retain exactly the radial/angular $-R^{-2}$ terms.

Now set $U = (b + \beta, V + v, G + \gamma) + w$, $\langle w \rangle_\theta = 0$, and $W^{ab} = \langle w_a w_b \rangle_\theta$. The identity

$$\langle U_a U_b \rangle_\theta = (U_a - w_a)(U_b - w_b) + W^{ab}$$

keeps the full actual wave covariance. No curl term can be removed from W . Subtract the pure base residual, whose stress contribution is $-(D_r + 2/R)\Sigma_\theta$ in the angular row and $-(D_r + 1/R)\Sigma_z$ in the axial row. Retain its separate flat part. The remaining residual is $(D_r p_m - g_r, E_\theta, E_z)$, with

$$\begin{aligned} E_\theta &= t_* v + (D_r + 2/R)(bv + \beta V + \beta v + W^{r\theta}) \\ &\quad + D_z(Gv + V\gamma + \gamma v + W^{z\theta}) - \epsilon(\Delta_0 - R^{-2})v - (D_r + 2/R)\Sigma_\theta, \\ E_z &= t_* \gamma + (D_r + 1/R)(b\gamma + \beta G + \beta \gamma + W^{rz}) \\ &\quad + D_z(2G\gamma + \gamma^2 + W^{zz} + p_m) - \epsilon\Delta_0 \gamma - (D_r + 1/R)\Sigma_z, \\ g_r &= -t_* \beta - (D_r + 1/R)(2b\beta + \beta^2 + W^{rr}) \\ &\quad - D_z(b\gamma + G\beta + \beta\gamma + W^{zr}) + (2Vv + v^2 + W^{\theta\theta})/R + \epsilon(\Delta_0 - R^{-2})\beta. \end{aligned} \tag{MC12}$$

Every sign follows from (MC11) and subtracting the radial pressure derivative to define g_r . These are precisely the checked mathematical identities in source 8.3.

21.5 Pressure, solenoidal axial realization, and dimensions

Choose the reserved interior bump $\hat{\rho} \in C_c^\infty(I_m)$, $I_m = \{\sqrt{2X} : X \in I_{\text{mean}}\}$, with integral one. In physical variables $\rho_{\text{phys}} = q^{-1/2}\hat{\rho}(r/\sqrt{q})$, and in a chart $\rho = Q^{1/2}\rho_{\text{phys}}$. Changing variable $r = \sqrt{q}x$ proves both physical and chart radial integrals of the bump equal one.

Define

$$P = \int \langle g_r \rangle dR, \quad p_m = T_0(g_r - \rho P).$$

The input has zero radial Haar integral. Equation (MC6) gives exactly

$$D_r p_m - g_r = -\rho P - A_0(g_r - \rho P), \quad \partial_R \langle p_m \rangle = \langle g_r \rangle - \rho P. \quad (\text{MC13})$$

The map $g_r \mapsto p_m$ is linear because the bump is fixed. Radial integration maps $\mathcal{M}_\alpha \rightarrow \mathcal{S}_\alpha$: for each fixed derivative, $\zeta \delta^{-B}$ is bounded and the effective interval has bounded length. The interior bump then maps P back into $\mathcal{M}_\alpha$, and Section 3 proves the pressure estimate and its difference estimate. Its cutoff remainder has zero Haar mean and is flat by (MC10).

For a desired axial increment γ_d with $\int R \langle \gamma_d \rangle dR = 0$, define $\Psi_* = T_1 \gamma_d$ and

$$\Delta \beta = -\epsilon \partial_Z \Psi_*, \quad \Delta \gamma = (D_r + 1/R) \Psi_* = \gamma_d - A_1 \gamma_d. \quad (\text{MC14})$$

Its divergence is

$$(D_r + 1/R) \Delta \beta + D_z \Delta \gamma = -\epsilon (D_r + 1/R) \partial_Z \Psi_* + \epsilon \partial_Z (D_r + 1/R) \Psi_* = 0.$$

This cancellation uses commutation of the differential operators on the already defined potential. It does not commute ∂_Z through the integral operator, whose moving cutoff may depend on Z . The averaged axial increment equals $\langle \gamma_d \rangle$ at each radius, and its weighted integral remains zero. The class bounds are $\Psi_*, \Delta \gamma \in \mathcal{M}_\alpha$, $\Delta \beta \in \mathcal{M}_{\alpha+1}$. If γ_d is torus independent, the axial realization is pointwise exact.

All physical scales can be checked from one change of variable. If $f_* = Q^a f_{\text{phys}}$, then

$$\int R^e f_* dR = Q^{a-(e+1)/2} \int r^e f_{\text{phys}} dr, \quad T_e f_* = Q^{a-1/2} r^{-e} I_{e,\text{phys}}(r^e f_{\text{phys}}). \quad (\text{MC15})$$

Use $u_* = Q^A u_{\text{phys}}$, $g_* = Q^{2A+1/2} g_{\text{phys}}$, $p_* = Q^{2A} p_{\text{phys}}$, $\Psi_* = Q^{A-1/2} \Psi_{\text{phys}}$. In particular, normalizing $-\partial_z \Psi_{\text{phys}}$ yields $-Q^{1/2-D} \partial_Z \Psi_* = -\epsilon \partial_Z \Psi_*$, preserving the minus sign. The pressure primitive scale is Q^{2A} . The resulting moment exponents are

Quantity: Chart value relative to physical value.

$$M_\theta = \int R^2 \langle v \rangle dR: Q^{A-3/2}.$$

$$M_z = \int R \langle \gamma \rangle dR: Q^{A-1}.$$

$$P = \int \langle g_r \rangle dR: Q^{2A}.$$

$$c_\rho = \frac{1}{2} \int R^2 \rho dR: Q^{-1}.$$

$$J_\theta \text{ defined below: } Q^{2A-3/2}.$$

$$J_z \text{ defined below: } Q^{2A-1}.$$

The second term in J_z , involving $R^2 g_r$, has exponent $2A + 1/2 - 3/2 = 2A - 1$, exactly matching its velocity-product term. Thus the fifth moment row has a consistent physical dimension.

21.6 Integrated residuals and both covariance targets

Let bars denote Haar averages. Translation invariance on the torus proves $\overline{D_r f} = \partial_R \bar{f}$, $\overline{t_* f} = -\epsilon \partial_T \bar{f}$, and the corresponding second-derivative statements. For any smoothly supported scalar h ,

$$\int R^e (\partial_R + e/R) h dR = \int \partial_R (R^e h) dR = 0.$$

For the angular radial viscosity, retaining all three terms,

$$\int R^2 h_{RR} dR = 2 \int h dR, \quad \int R h_R dR = - \int h dR, \quad \int (-h) dR = - \int h dR.$$

Their sum is zero. For the axial viscosity, $\int R h_{RR} dR = - \int h_R dR$, which cancels the remaining $\int h_R dR$. The time and axial-viscosity contributions to the two residual moments are respectively

$$-\epsilon \partial_T M_\theta - \epsilon^3 \partial_Z^2 M_\theta, \quad -\epsilon \partial_T M_z - \epsilon^3 \partial_Z^2 M_z.$$

They vanish when the two moments are zero at every slow point. The stress divergences vanish by their own compact support. Finally, compact support and (MC13) give

$$\int R\bar{p}_m dR = -\frac{1}{2} \int R^2 \partial_R \bar{p}_m dR = -\frac{1}{2} \int R^2 \bar{g}_r dR + c_\rho P. \quad (\text{MC16})$$

Set

$$J_\theta = \int R^2 \overline{Gv + V\gamma + \gamma v + W^{z\theta}} dR, \\ J_z = \int R^2 \overline{G\gamma + \gamma^2 + W^{zz}} dR - \frac{1}{2} \int R^2 \bar{g}_r dR, \quad c_\rho = \frac{1}{2} \int R^2 \rho dR.$$

Every surviving term of (MC12) is now accounted for, and the exact identities are

$$\int R^2 \bar{E}_\theta dR = \epsilon \partial_Z J_\theta, \quad \int R \bar{E}_z dR = \epsilon \partial_Z (J_z + c_\rho P). \quad (\text{MC17})$$

The product $c_\rho P$ must stay inside the derivative; c_ρ changes with the moving scale q . These are the complete compatibility moments of the two torus-independent weighted radial equations.

Use the fixed interior bumps σ_θ, σ_z with $\int R^2 \sigma_\theta dR = \int R \sigma_z dR = 1$. Their physical versions have the powers $q^{-3/2} \hat{\sigma}_\theta(r/\sqrt{q})$ and $q^{-1} \hat{\sigma}_z(r/\sqrt{q})$. Define

$$H_\theta = -T_2(\bar{E}_\theta - \sigma_\theta \epsilon \partial_Z J_\theta), \quad H_z = -T_1(\bar{E}_z - \sigma_z \epsilon \partial_Z (J_z + c_\rho P)).$$

Their sources have zero weighted radial integrals by (MC17), and are torus independent, so their A_e remainders vanish exactly. Therefore

$$(D_r + 2/R)H_\theta = -\bar{E}_\theta + \sigma_\theta \epsilon \partial_Z J_\theta, \quad (D_r + 1/R)H_z = -\bar{E}_z + \sigma_z \epsilon \partial_Z (J_z + c_\rho P). \quad (\text{MC18})$$

The minus signs are required because adding $W^{r\theta}$ or W^{rz} contributes a positive divergence in (MC12). Support at a fixed (Z, T) cannot be created when the whole radial source vanishes there. These are exact radial stress inverses with the two explicitly retained bump residuals.

21.7 Fast-time inverse and chart agreement

The same Diophantine proof used for L gives the unique smooth zero-Haar-mean solution of $N\phi = F$, for $\bar{F} = 0$:

$$N^{-1}F = \sum_{k \in \mathbb{Z}^2 \setminus \{0\}} \frac{\hat{F}(k)}{2\pi i v_t \cdot k} e^{2\pi i k \cdot y}, \quad \|N^{-1}F\|_{C^m} \leq C_m \|F\|_{C^{m+4}}. \quad (\text{MC19})$$

All nonzero modes have nonzero divisors. A kernel element therefore has only a constant mode, and the zero-mean condition removes it. The four-derivative count is the $p = 1$ instance of (MC8). This operator is local in the independent variables R, Z, T , hence preserves their closed support and smooth zero extension. It need not preserve support within a proper subset of the torus, which is allowed for $\mathcal{M}_\alpha$.

Define $P_i F(Y) = F(J_g^i Y)$. A mode k pulls back to $(J_g^i)^T k$. This lattice map is injective because $\det J_g = 14 \neq 0$, and hence it preserves the zero Fourier coefficient and Haar average. The exact chain rule and the eigenvector identity prove

$$N_{\text{abs}} P_i = T_g^i P_i N_i, \quad N_{\text{abs}}^{-1} P_i = T_g^{-i} P_i N_i^{-1} \quad (\text{MC20})$$

on zero-mean functions. The inverse identity also follows from uniqueness, so it does not require the covering to be one-to-one.

Define the physical tangential update as $-N_{\text{abs}}^{-1}(E_{\theta, \text{phys}}^\circ, E_{z, \text{phys}}^\circ)$. Its chart representative has multiplier

$$Q^{A-(2A+1/2)} T_g^{-i_0} = Q^{-1-h} T_g^{-i_0} = c^{-1}.$$

Thus $\Delta v = -c^{-1} N^{-1} E_\theta^\circ$, $\gamma_d = -c^{-1} N^{-1} E_z^\circ$. They belong to $\mathcal{M}_\alpha$ when E_θ, E_z do, since $c^{-1} \lesssim S_*$. Both have zero Haar mean at every radius. Realize γ_d by (MC14), and put $a = -A_1 \gamma_d$. Then

$$cN\Delta v = -E_\theta^\circ, \quad cN\Delta \gamma = -E_z^\circ + cNa. \quad (\text{MC21})$$

Translations commute with N , and the cutoff is torus independent, so there is no additional commutator. The remainder cNa is flat by (MC10), with one more requested torus derivative. The slow-time contribution $-\epsilon\partial_T$ is retained and has one extra power of ϵ . The actual velocity is divergence-free and preserves both moments because its angular and axial Haar means vanish pointwise.

Radial chart agreement is equally explicit. Pulling a physical integrand $F(J_g^i Y)$ along the absolute radial shift gives

$$F(J_g^i Y + \Lambda_g^i((r')^{d_r} - r^{d_r})v_r),$$

which is exactly the chart shift after $r = \sqrt{Q}R$. Together with (MC15), this proves the radial operators in different charts define one physical field. Their integer indices are held fixed on each common neighborhood. No derivative of a floor function is taken.

21.8 Explicit five-dimensional moment correction

Preserve the exact reserved-patch base in the paper's variable $x = r/\sqrt{q} \in I_m$:

$$G_q = 0, \quad V_q = a(\eta)x^{-1-2\lambda}, \quad a(\eta) = 2^{1/2+\lambda}c_{\text{patch}}(1+\eta^2)^{-1}, \quad |a(\eta)| \geq a_0 > 0, \quad \lambda > 0. \quad (\text{MC22})$$

This is the specific input base whose construction is assigned to the base audit. The following gives the complete inverse on that specified patch, including a formula beyond just a nonzero determinant.

Choose an interior $x_0 > 0$, a small fixed $d > 0$, and a nonnegative nonzero smooth bump η_0 sufficiently narrow that its three copies

$$\eta_j(x) = e^{-jd}\eta_0(e^{-jd}x), \quad j = 0, 1, 2,$$

have disjoint supports in I_m . Such a choice exists: first choose d so x_0, x_0e^d, x_0e^{2d} remain in the open interval; their finite positive separations then permit a sufficiently small common relative support width. For every real p , define $\mu_p = \int x^p \eta_0(x) dx > 0$. The exact change of variable gives

$$\int x^p \eta_j(x) dx = \mu_p e^{jdp}. \quad (\text{MC23})$$

The required five rows are

$$\begin{aligned} \int R^2 \Delta v &= 0, & \int R \gamma_d &= 0, & \int (2V/R) \Delta v &= -P, \\ \int R^2 (G \Delta v + V \gamma_d) &= -J_\theta, & \int (2RG \gamma_d - RV \Delta v) &= -J_z, \end{aligned} \quad (\text{MC24})$$

with dR understood in each integral. After using physical targets scaled by q with the exact powers in Section 5, the angular powers are $p_0 = 2, p_1 = -2 - 2\lambda, p_2 = -2\lambda$. Set $t_i = e^{dp_i}$. All three are distinct for $\lambda > 0$.

Let $C(s) = u_0 + u_1 s + u_2 s^2$. The angular equations are exactly

$$C(t_0) = 0, \quad C(t_1) = -\frac{P_q}{2a\mu_{p_1}} =: b_1, \quad C(t_2) = \frac{J_{z,q}}{a\mu_{p_2}} =: b_2.$$

Therefore the unique polynomial and all three coefficients are given explicitly by

$$C(s) = b_1 \frac{(s-t_0)(s-t_2)}{(t_1-t_0)(t_1-t_2)} + b_2 \frac{(s-t_0)(s-t_1)}{(t_2-t_0)(t_2-t_1)}. \quad (\text{MC25})$$

Evaluation at the three nodes proves all three equations exactly. Uniqueness follows since a nonzero polynomial of degree at most two cannot have three distinct roots: division by each linear factor would force degree at least three. Equivalently the displayed source matrix has determinant

$$-2a^2 \mu_2 \mu_{-2-2\lambda} \mu_{-2\lambda} (t_1 - t_0)(t_2 - t_0)(t_2 - t_1) \neq 0.$$

For the axial block set $t_b = e^d, t_a = e^{d(1-2\lambda)}$. They are distinct. Its two coefficients are

$$s_1 = -\frac{J_{\theta,q}}{a\mu_{1-2\lambda}(t_a - t_b)}, \quad s_0 = -t_b s_1. \quad (\text{MC26})$$

Then $\mu_1(s_0 + t_b s_1) = 0$, and

$$a\mu_{1-2\lambda}(s_0 + t_a s_1) = -J_{\theta,q}.$$

The determinant is $a\mu_1\mu_{1-2\lambda}(t_a - t_b) \neq 0$. Define the physical corrections, with no change to their powers,

$$\Delta v_{\text{phys}} = q^{-A} \sum_{j=0}^2 u_j \eta_j(r/\sqrt{q}), \quad \gamma_{d,\text{phys}} = q^{-A} \sum_{j=0}^1 s_j \eta_j(r/\sqrt{q}). \quad (\text{MC27})$$

Equations (MC23)-(MC26) prove all five rows (MC24), including the negative $RV\Delta v$ in the last row. The profile-to-chart changes multiply every row and its target by the identical factor from (MC15), so the equations also hold in the fixed Q chart. This is an exact map from the three target functions into the five-dimensional space spanned by the chosen profiles; uniqueness concerns this chosen space, not every possible smooth correction.

All denominators except a are fixed nonzero constants; a^{-1} has bounded slow derivatives for the prescribed smooth base and nonzero lower bound. Equations (MC25)-(MC27), the smooth bounded powers of q/Q , and Leibniz' rule prove $\mathcal{S}_\alpha \rightarrow \mathcal{M}_\alpha$ at every fixed derivative order. On the interior support the weight is bounded below. Constants may deteriorate as $\lambda \downarrow 0$; the construction needs no estimate uniform in that limit. At $\lambda = 0$ the two axial powers coincide, so there is a real degeneracy outside the stated parameter domain.

The axial correction has zero ordinary weighted integral and is torus independent, so (MC14) realizes it exactly: $\Delta\gamma = \gamma_d$. Its potential vanishes below the first bump and above the last bump but may be nonzero between them. Its entire support remains in the reserved interval. The induced radial correction is in $\mathcal{M}_{\alpha+1}$. This explains precisely why local base bounds must hold throughout that interval, rather than only on the union of bump supports.

21.9 Every nonlinear change and its exponent

Keep w , its complete covariance, the base, and the stress fixed. Write $B = \Delta\beta$, $u = \Delta v$, $d = \Delta\gamma = \gamma_d$ only within this calculation; these are the actual fields just constructed, not altered objects. The product differences are

$$\begin{aligned} 2b(\beta + B) + (\beta + B)^2 - (2b\beta + \beta^2) &= 2bB + 2\beta B + B^2, \\ b(\gamma + d) + G(\beta + B) + (\beta + B)(\gamma + d) - (b\gamma + G\beta + \beta\gamma) &= bd + GB + \beta d + \gamma B + Bd, \\ 2V(v + u) + (v + u)^2 - (2Vv + v^2) &= 2Vu + 2vu + u^2. \end{aligned}$$

Substituting these three identities into (MC12) proves the exact radial-source difference

$$\begin{aligned} g_{r,\text{new}} - g_r &= (2V/R)u + R_g, \\ R_g &= -t_* B - (D_r + 1/R)(2bB + 2\beta B + B^2) \\ &\quad - D_z(bd + GB + \beta d + \gamma B + Bd) + (2vu + u^2)/R + \epsilon(\Delta_0 - R^{-2})B. \end{aligned} \quad (\text{MC28})$$

No covariance difference is present because w is unchanged. The finite-dimensional rows cancel only the displayed terms linear in the fixed base. Integrating the radial difference and the expanded flux products gives

$$\begin{aligned} P_{\text{new}} &= \int \langle R_g \rangle dR, \\ (J_\theta)_{\text{new}} &= \int R^2 \langle \gamma u + vd + du \rangle dR, \\ (J_z)_{\text{new}} &= \int R \langle 2\gamma d + d^2 \rangle dR - \frac{1}{2} \int R^2 \langle R_g \rangle dR. \end{aligned} \quad (\text{MC29})$$

To verify the last cancellation, the linear part of the pressure-source moment is

$$-\frac{1}{2} \int R^2 (2V/R)u dR = - \int RVu dR,$$

which combines with $\int 2RGddR$ to form the last row of (MC24). This proves the signs in (MC29) without assuming that new defects vanish.

Take the original hypotheses $v, \gamma \in \mathcal{M}_{0.9}$, $\beta \in \mathcal{M}_{1.9}$, targets in $\mathcal{S}_\alpha$, $\alpha \geq 0.9$. Then $u, d \in \mathcal{M}_\alpha$, $B \in \mathcal{M}_{\alpha+1}$. All increments are torus independent. On their potential support, retain the stated base bounds $b = O(\epsilon S_*^B)$, $V, G = O(S_*^B)$ for every fixed coefficient derivative. The complete term list is:

Term in R_g : Guaranteed exponent.

$$-t_*B = +\epsilon\partial_TB: \alpha + 2.$$

$$-(D_r + 1/R)(2bB): \alpha + 2 - \kappa_s.$$

$$-(D_r + 1/R)(2\beta B): \alpha + 2.9 - \kappa_s.$$

$$-(D_r + 1/R)B^2: 2\alpha + 2 - \kappa_s.$$

$$-D_z(bd): \alpha + 2.$$

$$-D_z(GB): \alpha + 2.$$

$$-D_z(\beta d): \alpha + 2.9.$$

$$-D_z(\gamma B): \alpha + 2.9.$$

$$-D_z(Bd): 2\alpha + 2.$$

$$2vu/R: \alpha + 0.9.$$

$$u^2/R: 2\alpha.$$

$$\epsilon D_r^2 B: \alpha + 2 - 2\kappa_s.$$

$$\epsilon R^{-1} D_r B: \alpha + 2 - \kappa_s.$$

$$\epsilon D_z^2 B: \alpha + 4.$$

$$-\epsilon R^{-2} B: \alpha + 2.$$

Every exponent is at least $\alpha + 0.9 - 2\kappa_s$: the only potentially tight quadratic one is 2α , and $2\alpha - (\alpha + 0.9 - 2\kappa_s) = \alpha - 0.9 + 2\kappa_s \geq 2\kappa_s > 0$. All remaining comparisons follow by subtraction and $\alpha \geq 0.9$. Thus

$$R_g \in \mathcal{M}_{\alpha+0.9-2\kappa_s}, \quad (P_{\text{new}}, (J_\theta)_{\text{new}}, (J_z)_{\text{new}}) \in \mathcal{S}_{\alpha+0.9-2\kappa_s}. \quad (\text{MC30})$$

The two product-only defects in (MC29) actually have bounds $\alpha + 0.9$ and 2α before the R_g moment is added. The table records every transport and viscous term, including those compressed into the source proof's remaining-transport statement.

21.10 Recalculation, endpoint scope, and audit disposition

The exact construction order is forced by the definitions: calculate the full W^{ab} and g_r from the current actual velocity; calculate P and p_m ; calculate E_θ, E_z with that pressure; calculate J_θ, J_z from the same g_r and velocity. Repeat this sequence after every update. In particular, the temporal correction's $a = -A_1\gamma_d$ must occur in all new products. It is insufficient that $\bar{a} = 0$: for example $\bar{a}^2$ can be positive. The exact difference formulas (MC28)-(MC29) apply to the slow five-dimensional update, whose axial remainder is identically zero.

All local operations preserve one-sided slow derivatives on closed positive- q regions because the radial integrals have fixed-shift formulas, Fourier inverse estimates use finitely many derivatives for each requested derivative, and matrix coefficients are smooth functions of the slow data. This statement carries the actual source regularity through those operators; it does not construct missing source regularity. On $q \geq c_0 > 0$, only finitely many bands occur, so no accumulation of covering changes is involved. Near $q = 0$, physical flatness of the radial cutoff remainders was proved in Section 3.

The verified results of this bounded audit are the full radial exact sequence and its cutoff splitting; all derivative losses for the primitive and time inverse; the exact solenoidal velocity realization; the complete integrated mean residual identities; an explicit inverse of the five moment rows; and the exact nonlinear recomputation and stated exponent gain. No Section 8 repair is required by these calculations. This conclusion is limited to these maps and estimates. The actual base input, wave construction, iteration, and passage to an exact solution remain separate portions of the coordinated source audit, listed precisely in `DEPENDENCIES.md`.

22 Residual improvement and the local field (independent reconstruction)

Scope and provenance. The formulas and page assignments in this section were transcribed from the released manuscript, printed pp. 100–116, and then rederived below. Data imported from Sections 5–8 are hypotheses here: the fixed background and primary profiles, the pulse inverse, curl realization, oscillatory covariance, and the temporal and five-equation mean maps. Those earlier constructions are proved in the accompanying bodies at the locations in the dependency ledger. The present text proves their composition, the displayed residual identities, exponent arithmetic, and single-stage summation argument. It does not independently reprove existence of all the imported data.

Put $\varepsilon = Q^h$, $Q = 2^{-\ell}$, $S_* = \ell^2$, and retain the source normalization $R_* = Q^{2A+1/2}R$. Throughout

$$\kappa_s = 10^{-5}, \quad \sigma_j = \frac{1}{5} + \frac{j}{10}, \quad B_j = \frac{1}{2} + \sigma_j, \quad C_j = B_j + \frac{1}{2} = 1 + \sigma_j.$$

The classes $W^\alpha, M^\alpha, S^\alpha$ are exactly those fixed in Section 6.4; polynomial factors in S_* and the radial edge factors $\sqrt{\zeta} \delta^{-M}$ or $\zeta \delta^{-M}$ are retained in every estimate. The angular mean $\langle \cdot \rangle_\theta$ and auxiliary mean $\langle \cdot \rangle_Y$ are different operations. Since $0 < \varepsilon \leq 1$, larger exponents give smaller classes: $W^a \subset W^b$ and $M^a \subset M^b$ for $a \geq b$. Thus the quantities $W^{0.68}$ and $M^{0.9}$ are class bounds; they are not powers of large parameters called W or M .

22.1 Curl identities and the linear residual

For a fixed nonzero harmonic $m \in \mathbb{Z}$, set $k_m = km$, where $k = \lceil \varepsilon^{-1/2} \rceil$, and write $a e^{ik_m \Phi}$. Retain the full phase covector $n_\Phi = \nabla_* \Phi$, without unit normalization. In the fixed label the phase and covector are

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v H_\Phi, \quad H_\Phi = pF + p_z G, \quad F = V/R,$$

$$n_\Phi = (x_0 - v \partial_R H_\Phi, p/R, p_z - \varepsilon v \partial_Z H_\Phi).$$

Here v in the phase is the pulse variable; later v in the mean tuple denotes the azimuthal mean correction, as in the source. The displayed covector follows from $D_r v = D_z v = 0$ and $D_z = \varepsilon \partial_Z$ on auxiliary-independent coefficients. Its fixed positive lower length bound is supplied by the pulse geometry. The imported pulse inverse supplies a transverse amplitude t_m and pressure π_m solving

$$\begin{aligned} L_m(t_m, \pi_m) &:= Q^{1+h} N_{\text{abs}} t_m + K t_m + \varepsilon k_m^2 |n_\Phi|^2 t_m \\ &\quad + i k_m n_\Phi \pi_m = -f_m, \quad n_\Phi \cdot t_m = 0. \end{aligned} \tag{9.1}$$

Its normalized potential coefficient and curl are

$$C_m = \frac{i n_\Phi \times t_m}{k_m |n_\Phi|^2}, \quad \text{curl}_*(C_m e^{ik_m \Phi}) = (t_m + r_m) e^{ik_m \Phi},$$

Indeed, in the retained right-handed (e_r, e_θ, e_z) orientation,

$$i k_m n_\Phi \times C_m = -\frac{n_\Phi \times (n_\Phi \times t_m)}{|n_\Phi|^2} = t_m - \frac{n_\Phi (n_\Phi \cdot t_m)}{|n_\Phi|^2} = t_m.$$

The remaining terms are exactly

$$r_m = (-D_z(C_m)_\theta, D_z(C_m)_r - D_r(C_m)_z, (D_r + R^{-1})(C_m)_\theta).$$

The physical potential is $Q^{1/2-A} C_m e^{ik_m \Phi}$. where the imported curl lemma gives $r_m \in W^{\alpha+1/2-\kappa_s}$ if $t_m \in W^\alpha$. Thus the complete divergence-free amplitude is $a_m = t_m + r_m$; the transport factor in quadratic products is always this complete amplitude.

For an arbitrary amplitude a and pressure coefficient π , direct expansion in cylindrical coordinates gives the remainder after (9.1):

$$L_m^{\text{rem}}(a, \pi) = (-\varepsilon \partial_T + b D_r + G D_z) a + i k_m E_{ik} a \tag{14}$$

$$+ a_r (\partial_R b) e_r + a_z D_z (b, V, G) + \frac{b}{R} a_\theta e_\theta + (D_r \pi, 0, D_z \pi) \tag{15}$$

$$- \varepsilon \{ (D_r^2 + R^{-1} D_r + D_z^2) a - R^{-2} (a_r, a_\theta, 0) \} \tag{16}$$

$$+ 2 i k_m (n_{\Phi,r} D_r + n_{\Phi,z} D_z) a + i k_m (D_r n_{\Phi,r} + R^{-1} n_{\Phi,r} + D_z n_{\Phi,z}) a \tag{17}$$

$$+ 2 i k_m (n_{\Phi,\theta}/R) (-a_\theta, a_r, 0) \}. \tag{9.2}$$

Here $E_{ik} = (t_* + bD_r + (V/R)\partial_\theta + GD_z)\Phi$ is the imported phase-transport defect. The displayed formula follows by differentiating the cylindrical basis: transport of v contributes $(-u_\theta v_\theta, u_\theta v_r, 0)/R$, while the scalar and vector Laplacians give respectively the mixed phase terms and the last angular connection.

Exponent audit. If $f_m \in W^\alpha$, the seven groups in (9.2) gain, in their written order,

$$1 - \kappa_s, \quad \frac{1}{2}, \quad 1, \quad \frac{1}{2} - \kappa_s, \quad 1 - 2\kappa_s, \quad \frac{1}{2} - \kappa_s, \quad \frac{1}{2}.$$

The principal velocity operator applied to the curl correction preserves the same exponent (the fast derivative is already in (9.1), and $\varepsilon k_m^2 = O(1)$). Hence the normalized linear residual belongs to $W^{\alpha+1/2-3\kappa_s}$. Multiplication by a pulse cutoff is done before the curl. The uncanceled $(1 - \psi)f_m$ and ψ'_t terms satisfy, for all fixed Cartesian orders m, N ,

$$|E_{\text{cut}}|_m \leq C_{m,N} Q^{-M} S_*^P e^{-cS_*} \leq C_{m,N,N'} q^{N'}.$$

They are retained in the additive flat remainder and are never reintroduced as forward-pulse sources.

22.2 Nonlinear products and finite-stage source decomposition

If $w \in W^\alpha$ is a wave and a mean field has tangential class M^μ and radial class $M^{\mu+1}$, then every nonzero harmonic of $(w \cdot \nabla_*)v + (v \cdot \nabla_*)w$ lies in $W^{\alpha+\mu-1/2}$. For two curl-generated waves of classes $W^\alpha, W^{\alpha'}$, the nonzero harmonics lie in $W^{\alpha+\alpha'-\kappa_s}$ and the zero harmonic in $M^{\alpha+\alpha'-\kappa_s}$. Indeed, phase differentiation costs $k = O(\varepsilon^{-1/2})$. The complete amplitude is not asserted transverse. Expanding its exact zero divergence gives

$$ik_m(n_\Phi \cdot a_m) = -((D_r + R^{-1})(a_m)_r + D_z(a_m)_z),$$

$$n_\Phi \cdot a_m = n_\Phi \cdot r_m \in W^{\alpha+1/2-\kappa_s}.$$

Thus phase differentiation of an m' harmonic by this wave contributes $ik_{m'}(a_m \cdot n_\Phi)a_{m'}$, which belongs to $W^{\alpha+\alpha'-\kappa_s}$. Equivalently the divergence identity replaces $ik_{m'}n_\Phi \cdot a_m$ by m'/m times the displayed amplitude divergence; m'/m is a fixed constant at each finite stage. Amplitude derivatives cost at most κ_s and cylindrical connections cost none. Since $P_v^2 \leq P_v$ and supports of distinct labels are disjoint, no other loss occurs. Leibniz' rule proves the same statements at every fixed coefficient derivative order.

For a finite stage j , write the physical residual as

$$R(u^{[j]}, p^{[j]}) = G^{[j]} + F^{[j]}, \quad G_*^{[j]} = Q^{2A+1/2}G^{[j]}, \quad F_*^{[j]} = Q^{2A+1/2}F^{[j]}. \quad (9.3)$$

On the common auxiliary torus,

$$G_*^{[j]} - \langle G_*^{[j]} \rangle_\theta = \sum_{\gamma \in \Gamma} \sum_{m \in H_j} f_{\gamma,m}^{[j]} e^{ik_{\gamma m} \Phi_\gamma}, \quad H_j \in \mathbb{Z} \setminus \{0\}, \quad (9.4)$$

with $\text{supp}(f_{\gamma,m}^{[j]}|_{E_\gamma}) \subset \Omega_\gamma$, smooth zero extension to E_γ^+ , and $f_{\gamma,m}^{[j]} \in W^\alpha$. The angular mean belongs to M^μ , with the exact exponents supplied by the finite preceding construction. Radial integration is at fixed (z, t) , so $q(z, t)$ and the local band set do not change; pressure reconstruction and the mean vector potential preserve support and classes. For the stress correction, subtract an interior profile with equal weighted moment. If s is logarithmic distance to an edge and $\zeta \asymp e^{-a/s^2}$, then

$$\int_0^\delta s^{-M} e^{-a/s^2} ds \leq C_{M,a} \delta^{3-M} e^{-a/\delta^2}, \quad \left| \partial_s^k e^{-a/s^2} \right| \leq C_{k,a} s^{-3k} e^{-a/s^2}.$$

For the first inequality substitute $u = a/s^2$, obtaining $\frac{1}{2}a^{(1-M)/2} \int_{a/\delta^2}^\infty u^{(M-3)/2} e^{-u} du$. With $u = u_0 + y$ and $u_0 \geq 1$, a nonpositive power is bounded by $u_0^{(M-3)/2}$, while a positive power is bounded by this factor times $(1+y)^{(M-3)/2}$. Integration in y gives the claimed bound. For the second, induction expresses the derivative as e^{-a/s^2} times a polynomial in s^{-1} of degree at most $3k$. This also proves that every fixed inverse-distance power times either edge factor tends to zero with all derivatives. Thus the correction remains in M^α with only polynomial S_* losses. The amplitude division is exact:

$$d\Sigma = H^{-1}(\Sigma/\varepsilon), \quad \delta a_\sigma = \frac{(d\Sigma)_\sigma}{2a_\sigma}, \quad |D_I \delta a_\sigma| \lesssim \varepsilon^{\alpha-1} S_*^{P_I} \sqrt{\zeta} \delta^{-M_I},$$

and $\sqrt{\varepsilon} \delta a_\sigma b_\sigma \in W^{\alpha-1/2}$ with the retained factor P_v . This proves the pulse inverse hypotheses at every finite stage without shrinking the common threshold.

The flat term satisfies, for every fixed j, m, N ,

$$|F^{[j]}|_m \leq C_{j,m,N} q^N. \quad (9.5)$$

For an increment (w, π) the exact residual identity is

$$R(u + w, p + \pi) = R(u, p) + \partial_t w - \Delta w + \nabla \pi + (u \cdot \nabla)w + (w \cdot \nabla)u + (w \cdot \nabla)w. \quad (9.5a)$$

This proves inductively that all products use complete cutoff curls and that new cutoff tails are added to F . The exact mean identities, with g_r, E_θ, E_z defined by the imported formulas (8.3), are

$$\begin{aligned} W_{ab} &= \langle w_a w_b \rangle_\theta, \quad P = \int \langle g_r \rangle_Y dR, \quad p_m = \mathcal{T}^0(g_r - \rho P), \\ \langle G_* + F_* - E_{B,*} \rangle_\theta &= (D_r p_m - g_r, E_\theta, E_z). \end{aligned} \quad (9.6)$$

22.3 Finite correction state and initialization

Write the normalized finite state as

$$u_* = (b + \beta, V + v, G + \gamma) + w, \quad p_* = p_{B,*} + p_m + p_w, \quad \langle w \rangle_\theta = \langle p_w \rangle_\theta = 0. \quad (9.7)$$

At stage j , the nonzero wave residual, tangential mean residual, and defects obey

$$\begin{aligned} G_{\text{wave}}^{[j]} &\in W^{B_j}, \quad E_\theta, E_z \in M^{C_j}, \quad (P, J_\theta, J_z) \in S^{C_j}, \\ B_j &= \frac{1}{2} + \sigma_j, \quad C_j = 1 + \sigma_j. \end{aligned} \quad (9.8)$$

The cumulative fields satisfy the deliberately retained budgets

$$w \in W^{1/2}, \quad w - w_0^{\text{tan}} \in W^{0.68}, \quad v, \gamma, p_m \in M^{0.9}, \quad \beta \in M^{1.9}, \quad (9.9)$$

and the moments remain exact:

$$\int_0^\infty R^2 \langle v \rangle_Y dR = 0, \quad \int_0^\infty R \langle \gamma \rangle_Y dR = 0. \quad (9.10)$$

The primary transverse amplitudes have class $W^{1/2}$; their linear residual has $1 - 3\kappa_s$, and wave nonlinearities $1 - \kappa_s$. Before mean updates, the exact balances and defects are $M^{1-\kappa_s}$ and $S^{1-\kappa_s}$. The leading auxiliary covariance cancels exactly because the squared slow partition functions sum to one. Writing $\Sigma_a^{(0)}$ for this tensor and Σ_a for the full tensor,

$$\langle E_\theta \rangle_Y = (\partial_R + 2/R)(\langle W_{r\theta} \rangle_Y - \Sigma_\theta^{(0)}) - (\partial_R + 2/R)(\Sigma_\theta - \Sigma_\theta^{(0)}) + \varepsilon \partial_Z \langle W_{z\theta} \rangle_Y, \quad (18)$$

$$\langle E_z \rangle_Y = (\partial_R + 1/R)(\langle W_{rz} \rangle_Y - \Sigma_z^{(0)}) - (\partial_R + 1/R)(\Sigma_z - \Sigma_z^{(0)}) + \varepsilon \partial_Z \langle W_{zz} + p_m \rangle_Y. \quad (19)$$

Cross terms contain a curl correction and are $M^{3/2-\kappa_s}$; axial and higher covariance terms are at least $M^{2-\kappa_s}$. Hence the averaged balances are $M^{1.49}$. The temporal inverse at order $H_0 = 1 - \kappa_s$ gives tangential increment M^{H_0} and radial increment M^{H_0+1} . After pressure reconstruction the new tangential mean residual terms have order at least $H_0 + 1 - 2\kappa_s$, while their unchanged auxiliary averages retain order 1.49. Wave-mean interaction has wave order at least H_0 ; defects remain S^{H_0} . The five-equation map cancels the three linear defect contributions, leaving defects above $1.8 - \kappa_s$. Since $1 - \kappa_s > 0.68$ and $H_0 > 0.9$, this is stage 0: $B_0 = 0.7$, $C_0 = 1.2$, with (9.9)–(9.10).

22.4 One correction cycle

Assume a state at stage j and abbreviate $B = B_j$, $C = C_j$. Recompute g_r from complete fields and set $P = \int \langle g_r \rangle_Y dR$, $p_m = \mathcal{T}^0(g_r - \rho P)$ after each substep.

(i) Supported nonzero harmonics. For each coefficient in (9.4), solve the pulse equation with zero data on the entrance path and set

$$\Delta w_m = \text{curl}_* \left(\frac{in_\Phi \times (\psi t_m)}{k_m |n_\Phi|^2} e^{ik_m \Phi} \right), \quad \Delta p_{w,m} = \psi \pi_m e^{ik_m \Phi}.$$

The increment is W^B , its pressure is $W^{B+1/2}$, and the four resulting nonzero-harmonic errors have exponents

$$B + \frac{1}{2} - 3\kappa_s, \quad B + \frac{1}{2} - \kappa_s, \quad 2B - \kappa_s, \quad B + 0.4.$$

The last $B + 0.4$ estimate is a nonzero-harmonic (wave) estimate. Separately, the mean covariance changes have exponent at least $B + 1/2 = C$; a radial divergence costs κ_s . Thus $E_\theta, E_z, \Delta g_r, \Delta p_m \in M^{C-\kappa_s}$ and $(P, J_\theta, J_z) \in S^{C-\kappa_s}$. The exact integrated identities improve the weighted moments by one:

$$\int R^2 \langle E_\theta \rangle_Y dR, \quad \int R \langle E_z \rangle_Y dR \in S^{C+1-\kappa_s}. \quad (9.12)$$

Precisely, compact radial support and the two moment constraints give

$$\int R^2 \langle E_\theta \rangle_Y dR = \varepsilon \partial_Z J_\theta, \quad \int R \langle E_z \rangle_Y dR = \varepsilon \partial_Z (J_z + c_\rho P), \quad c_\rho = \frac{1}{2} \int R^2 \rho dR.$$

Multiplying a radial flux $(\partial_R + e/R)h$ by R^e turns its integral into a zero boundary term. For the angular viscosity the integrals of $R^2 h_{RR}$, Rh_R , and $-h$ are respectively $2 \int h$, $-\int h$, and $-\int h$; axial radial viscosity cancels in the same way. Time and axial derivatives of the two zero moments vanish. Finally $\int R \langle p_m \rangle_Y dR = -\frac{1}{2} \int R^2 \langle g_r \rangle_Y dR + c_\rho P$. These calculations leave exactly the two displayed axial flux derivatives and account for the $c_\rho P$ term without removing it from its derivative. The $\varepsilon \partial_Z$ supplies the claimed gain of one.

(ii) Auxiliary-averaged stress correction. In physical variables let $F_2 = Q^{-2A-1/2} \langle E_\theta \rangle_Y$, $F_1 = Q^{-2A-1/2} \langle E_z \rangle_Y$. Choose fixed profiles $\hat{b}_e$ with $\int x^e \hat{b}_e(x) dx = 1$, set $b_e(r) = q^{-(e+1)/2} \hat{b}_e(r/\sqrt{q})$, and define

$$M_e = \int_0^\infty r^e F_e(r) dr, \quad \sigma_e(r) = -r^{-e} \int_0^r (r')^e (F_e(r') - b_e(r') M_e) dr'.$$

The equal-moment subtraction implies σ_e vanishes outside the annulus and $(\partial_r + e/r)\sigma_e = -F_e + b_e M_e$. Put $\Sigma = Q^{2A}(\sigma_2, \sigma_1)$. The edge estimates above give $\Sigma \in M^{C-\kappa_s}$. Applying the physical signed amplitude map divides by the fixed positive $2\sqrt{y_\sigma}$ and realizes the prescribed covariance with w_0^{tan} ; no square root of a perturbed covariance is taken.

Write the signed increment as $s = t_s + r_s$. Its exact covariance change is

$$\begin{aligned} B(a, b) &= \langle \langle a \otimes b + b \otimes a \rangle_\theta \rangle_Y, \quad \langle \Delta W \rangle_Y = \langle \langle (w + s) \otimes (w + s) - w \otimes w \rangle_\theta \rangle_Y. \\ \langle \Delta W \rangle_Y &= B(w_0^{\text{tan}}, t_s) + B(w - w_0^{\text{tan}}, t_s) \\ &\quad + B(w, r_s) + \langle \langle s \otimes s \rangle_\theta \rangle_Y. \end{aligned} \quad (9.13)$$

This is the distributive law with $w = w_0^{\text{tan}} + (w - w_0^{\text{tan}})$ and $s = t_s + r_s$; in particular the complete self-interaction is retained. The four new wave errors from this signed increment are, respectively,

$$B + \frac{1}{2} - 4\kappa_s, \quad B + 0.4 - \kappa_s, \quad B + \frac{1}{2} - 2\kappa_s, \quad 2B - 3\kappa_s.$$

The first term has radial-tangential component Σ and cancels the target average. The remaining terms before divergence have exponents $C + 0.18 - \kappa_s$, $C + 1/2 - 2\kappa_s$, and $C + \sigma_j - 2\kappa_s$; radial divergence costs κ_s . Together with (9.12), pressure reconstruction yields

$$\langle E_\theta \rangle_Y, \langle E_z \rangle_Y \in M^{C+0.17}, \quad E_\theta, E_z \in M^H, \quad (P, J_\theta, J_z) \in S^H, \quad H = C - 2\kappa_s. \quad (9.14)$$

(iii) Temporal removal of auxiliary oscillation. The imported radial mean inverse is the split operator for the full torus obstruction $K_e f(w) = \int R^e f(R, w + MR^{d_r} v_r) dR$: $D_e \mathcal{T}_e = 1 - \mathcal{A}_e$, $K_e \mathcal{A}_e = K_e$, and $\mathcal{A}_e^2 = \mathcal{A}_e$. The vanishing Haar moment alone is not used as an inversion hypothesis; the projector contribution is retained as the flat remainder. The temporal inverse N^{-1} acts on the zero auxiliary average

subspace. The radial inverse is used with its projector term; zero auxiliary average does not assert $K_e f = 0$. Set $E_a^\circ = E_a - \langle E_a \rangle_Y$ and use the zero-average inverse:

$$\Delta v = -c_{i0}^{-1} N_{i0}^{-1} E_\theta^\circ, \quad \gamma_d = -c_{i0}^{-1} N_{i0}^{-1} E_z^\circ, \quad \Psi_* = \mathcal{T}^1 \gamma_d, \quad \Delta \beta = -\varepsilon \partial_Z \Psi_*, \quad \Delta \gamma = (D_r + R^{-1}) \Psi_*.$$

The actual $\Delta \gamma$ includes the flat reconstruction remainder $F_{\text{ax}} = c_{i0} N_{i0} (\Delta \gamma - \gamma_d)$. Fast-time identities are $c_{i0} N_{i0} \Delta v = -E_\theta^\circ$ and $c_{i0} N_{i0} \Delta \gamma = -E_z^\circ + F_{\text{ax}}$. Subtracting old from new balances gives exactly

$$E_{\theta, \text{new}} - \langle E_\theta \rangle_Y = -\varepsilon \partial_T \Delta v - \varepsilon (\Delta_0 - R^{-2}) \Delta v + (D_r + 2/R)((b + \beta) \Delta v + (V + v) \Delta \beta + \Delta \beta \Delta v) \quad (20)$$

$$+ D_z((G + \gamma) \Delta v + (V + v) \Delta \gamma + \Delta \gamma \Delta v), \quad (21)$$

$$E_{z, \text{new}} - \langle E_z \rangle_Y = -\varepsilon \partial_T \Delta \gamma - \varepsilon \Delta_0 \Delta \gamma + F_{\text{ax}} + (D_r + R^{-1})((b + \beta) \Delta \gamma + (G + \gamma) \Delta \beta + \Delta \beta \Delta \gamma) \quad (22)$$

$$+ D_z(2(G + \gamma) \Delta \gamma + (\Delta \gamma)^2 + \Delta p_m). \quad (23)$$

Slow time, radial flux, axial flux, and viscosity have gains respectively $H+1, H+1-\kappa_s, H+1, H+1-2\kappa_s$. Thus, after placing F_{ax} in the flat remainder, $E_\theta, E_z \in M^{\min(C+0.17, H+1-2\kappa_s)}$ and the wave change is W^H .

(iv) Compatibility defects. Apply the five-equation map to (P, J_θ, J_z) at order H . The azimuthal and prescribed axial increments are M^H , their induced radial increment is M^{H+1} , and the first two rows preserve (9.10). The term $2V\Delta v/R$ and its two moment contributions are exactly the linear terms canceled by the last three rows. Every remaining term has exponent at least $H+0.9-2\kappa_s$, hence

$$(P, J_\theta, J_z) \in S^{H+0.9-2\kappa_s} = S^{C+0.9-4\kappa_s}.$$

Here is the complete calculation behind the defect estimate. Write $u = \Delta v$, $d = \Delta \gamma = \gamma_d$, and $B_r = \Delta \beta$ in this paragraph only. The slow moment correction has zero axial projector remainder, because its prescribed axial mean satisfies the exact weighted moment constraint. Since the wave is unchanged, subtraction gives

$$g_{r, \text{new}} - g_r = (2V/R)u + R_g,$$

$$\begin{aligned} R_g = & -t_* B_r - (D_r + R^{-1})(2bB_r + 2\beta B_r + B_r^2) \\ & - D_z(bd + GB_r + \beta d + \gamma B_r + B_r d) \\ & + R^{-1}(2vu + u^2) + \varepsilon(\Delta_0 - R^{-2})B_r. \end{aligned}$$

The last three rows of the fixed linear system imply exactly

$$\begin{aligned} P_{\text{new}} &= \int \langle R_g \rangle_Y dR, \quad J_{\theta, \text{new}} = \int R^2 \langle \gamma u + vd + du \rangle_Y dR, \\ J_{z, \text{new}} &= \int R \langle 2\gamma d + d^2 \rangle_Y dR - \frac{1}{2} \int R^2 \langle R_g \rangle_Y dR. \end{aligned}$$

For $u, d \in M^H$ and $B_r \in M^{H+1}$ the termwise exponents are

term	exponent
$-t_* B_r$	$H+2$
$(D_r + R^{-1})bB_r$	$H+2-\kappa_s$
$(D_r + R^{-1})\beta B_r$	$H+2.9-\kappa_s$
$(D_r + R^{-1})B_r^2$	$2H+2-\kappa_s$
$D_z b d, D_z G B_r$	$H+2$
$D_z \beta d, D_z \gamma B_r$	$H+2.9$
$D_z B_r d$	$2H+2$
$2vu/R, u^2/R$	$H+0.9, 2H$
$\varepsilon D_r^2 B_r$	$H+2-2\kappa_s$
$\varepsilon R^{-1} D_r B_r$	$H+2-\kappa_s$
$\varepsilon D_z^2 B_r, -\varepsilon R^{-2} B_r$	$H+4, H+2$

Every entry is at least $H+0.9-2\kappa_s$, since $H = C - 2\kappa_s \geq 1.19998 > 0.9$. The two product-only defect expressions have exponents at least $\min(H+0.9, 2H)$ before the R_g moment. This establishes the displayed defect order with every retained term. The closure inequalities are

$$\min\{\frac{1}{2} - 3\kappa_s, \frac{1}{2} - \kappa_s, B - \kappa_s, 0.4\} \geq 0.4, \quad \min\{\frac{1}{2} - 4\kappa_s, 0.4 - \kappa_s, \frac{1}{2} - 2\kappa_s, B - 3\kappa_s\} \geq 0.4 - \kappa_s,$$

$$H - B = \frac{1}{2} - 2\kappa_s > 0.1, \quad \min\{0.17, 1 - 4\kappa_s\} = 0.17 > 0.1, \quad 0.9 - 4\kappa_s > 0.1.$$

The numerical margins are exact decimal rationals: $0.18 - 2\kappa_s = 0.17998 > 0.17$, $\sigma_j - 3\kappa_s \geq 0.19997 > 0.17$, $0.4 - \kappa_s = 0.39999$, $H - B = 0.49998$, $1 - 4\kappa_s = 0.99996$, and $0.9 - 4\kappa_s = 0.89996$. All exceed the needed 0.1 gain in the relevant comparison. Since $B \geq 0.7$, the next state has $B_{j+1} = B_j + 0.1$ and $C_{j+1} = C_j + 0.1$. The smallest signed increment is $B - \kappa_s \geq 0.69999 > 0.68$; mean and radial increments retain 0.9 and 1.9. This proves Proposition 9.6 by induction, with pressure recomputed after every substep and all retained flat terms in F .

22.5 Common domain and Cartesian derivative budgets

Choose one $q_{\text{big}} \leq q_*$ on which the fixed background, positive primary amplitudes, pulse propagators, and mean correction profiles are defined. The propagator for every later harmonic is the fixed fundamental propagator times a forward damping factor of modulus at most one, so no stage-dependent threshold is needed. Signed updates divide by the original $2\sqrt{y_\sigma}$; the five-equation map, pressure map, temporal inverse, and modified radial integrals are fixed linear maps. Thus all finite states exist on $0 < q < q_{\text{big}}$.

For a fixed stage, let $D_{v,a}(m)$ count source derivatives needed at vertex v of the finite directed acyclic graph of sums, products, derivatives, and inverses:

$$D_{v,a}(m) = \max_{w \rightarrow v} D_{w,a}(m + d_{v,w}), \quad D_{b,a}(m) = m \ (b = a), \quad D_{b,a}(m) = 0 \ (b \neq a).$$

The graph is finite at each stage. Extra derivatives of $N^{-1}D_r f$ do not repeat the explicit radial loss; they only increase $D_{v,a}(m)$ and polynomial S_* powers.

On perturbation supports $r \asymp \sqrt{Q}$, $q \asymp Q$. In a fixed chart the physical derivative losses are

$$\partial_r : \frac{1}{2} + h\kappa_s, \quad D_z : \frac{1}{2} - h, \quad \partial_t : 1 + h, \quad \text{frame derivative} : \frac{1}{2}. \quad (9.19)$$

The phase obeys $|\nabla_x^a \partial_t^b \Phi_\gamma| \lesssim Q^{-a/2-b(1+h)} S_*^{P_{a,b}}$, while $k \leq 2Q^{-h/2}$ gives, with $s_x = 1/2 + h/2$ and $s_t = 1 + 3h/2$,

$$|\nabla_x^a \partial_t^b e^{ikm\Phi}| \lesssim Q^{-as_x - bs_t} S_*^{P_{a,b,m}}. \quad (9.19a)$$

These dominate (9.19), including chart transitions. For every derivative order m , a sufficient common loss is

$$\ell_m = 2A + (m+1)(1 + 3h/2).$$

Therefore, with $g_j = hj/10$, each retained stage increment obeys

$$|Z_j|_m + |\Delta u_j|_m \leq C_{j,m} q^{g_j - \ell_m} (1 + |\log q|)^{P_{j,m}}, \quad (9.17)$$

$$Z_j = (A_j, B_j, p_j), \quad \Delta u_j = \text{curl } A_j + B_j.$$

The finite sums satisfy

$$|(u^{[j]}, p^{[j]})|_m \leq C_{j,m} q^{-K_m} (1 + |\log q|)^{P_{j,m}},$$

and their residuals

$$|R(u^{[j]}, p^{[j]})|_m \leq C_{j,m} q^{h\sigma_j - K_m} (1 + |\log q|)^{P_{j,m}} + E_{j,m}, \quad |E_{j,m}| \leq C_{j,m,N} q^N. \quad (9.18)$$

The fixed physical rescalings Q^{-A} , Q^{-2A} , $Q^{1/2-A}$ are bounded by Q^{-2A} ; one extra spatial derivative accounts for recovering velocity from a potential. On overlaps, prescribed torus coordinate changes and smooth zero extensions prove the same estimates across support and band boundaries.

22.6 Realization by a diagonal single-stage sum

Let $U_0 = (u^{[0]}, p^{[0]})$ and, for $j \geq 1$, let A_j^w be the physical wave-potential increment, Ψ_j its azimuthal mean potential, and b_j its direct azimuthal velocity. Set

$$A_j = A_j^w + \Psi_j e_\theta, \quad B_j = b_j e_\theta, \quad p_j = p^{[j]} - p^{[j-1]}, \quad Z_j = (A_j, B_j, p_j),$$

so $\Delta u_j = \text{curl } A_j + B_j$. In a fixed chart the pressure increment is the exact telescoping identity

$$Q^{2A} p_j = p_w^{[j]} - p_w^{[j-1]} + \mathcal{T}^0 \left(g_r^{[j]} - g_r^{[j-1]} - \rho(P^{[j]} - P^{[j-1]}) \right). \quad (9.16)$$

Choose a fixed smooth χ with $\chi(s) = 1$ for $s \leq 1/2$, $\chi(s) = 0$ for $s \geq 1$, and $0 \leq \chi \leq 1$. We now construct, rather than assume, a sequence $a_j \uparrow \infty$ which gives the claimed residual bounds. It will define

$$\begin{aligned} A &= A_0 + \sum_{j \geq 1} \chi(a_j q) A_j, \\ B e_\theta &= B_0 e_\theta + \sum_{j \geq 1} \chi(a_j q) B_j, \\ p_{\text{loc}} &= p_0 + \sum_{j \geq 1} \chi(a_j q) p_j, \\ u_{\text{loc}} &= \text{curl } A + B e_\theta. \end{aligned} \tag{9.21}$$

For a multi-index β in physical space and time, the similarity map gives $|\partial^\beta q| \leq C_\beta q^{1-|\beta|}$ on $0 < q < q_{\text{big}} < 1$; the sharper axial loss is $D < 1$. The multivariate chain rule expresses each derivative of a cutoff as a sum of

$$a_j^k \chi^{(k)}(a_j q) \prod_{\nu=1}^k \partial^{\beta_\nu} q, \quad \sum_{\nu} \beta_\nu = \beta, \quad |\beta_\nu| \geq 1.$$

For $k \geq 1$, its support has $1/2 \leq a_j q \leq 1$, so pairing a_j^k with the q^k from the product proves $|\partial^\beta \chi(a_j q)| \leq C_\beta q^{-|\beta|}$, independently of a_j . The undifferentiated cutoff is bounded by one.

Define the actual cutoff increment

$$V_j = \text{curl}(\chi(a_j q) A_j) + \chi(a_j q) B_j, \quad \Pi_j = \chi(a_j q) p_j.$$

The exact product rule retains

$$V_j = \chi(a_j q) \Delta u_j + a_j \chi'(a_j q) \nabla q \times A_j.$$

Thus (9.17) and Leibniz's rule give constants $\widehat{C}_{j,m}, \widehat{P}_{j,m}$ independent of a_j such that

$$|(V_j, \Pi_j)|_m \leq \widehat{C}_{j,m} q^{g_j - L_m} (1 + |\log q|)^{\widehat{P}_{j,m}}, \quad L_m = \ell_{m+1} + m + 1,$$

on their support $q \leq a_j^{-1}$. This deliberately generous stage-independent loss includes every derivative of the cutoff and the additional curl.

Recursively choose $a_j \geq \max(2a_{j-1}, 2/q_{\text{big}}, j)$ sufficiently large that the finitely many inequalities

$$\widehat{C}_{j,m} (1 + |\log q|)^{\widehat{P}_{j,m}} q^{g_j/2} \leq 2^{-j} \quad (0 < q \leq a_j^{-1}, \quad 0 \leq m \leq j)$$

hold. Such a choice exists because $g_j = hj/10 > 0$ and $q^\delta (1 + |\log q|)^P \rightarrow 0$ for every $\delta > 0$ and finite P . Consequently $|(V_j, \Pi_j)|_m \leq 2^{-j} q^{g_j/2 - L_m}$ for $m \leq j$. On a compact subset of the local domain, q has a positive lower bound, so only finitely many cutoff terms survive. This proves smoothness of (9.21).

Each representative has smooth zero extension and vanishes near the axis for fixed $t < 1$. Each $B_j = b_j e_\theta$ is an axisymmetric angular vector, so in physical cylindrical coordinates $\text{div}(\chi(a_j q) B_j) = r^{-1} \partial_\theta(\chi(a_j q) b_j) = 0$. Divergence of each curl vanishes. Thus $\text{div } u_{\text{loc}} = 0$, including all cutoff transitions.

Fix a finite $J \geq \max(1, m)$ and restrict to $q < (2a_J)^{-1}$. Then all the first J cutoffs equal one, and for $e_J = (u_{\text{loc}}, p_{\text{loc}}) - (u^{[J]}, p^{[J]})$,

$$|e_J|_m \leq \sum_{j > J} 2^{-j} q^{g_j/2 - L_m} \leq 2^{-J} q^{g_{J+1}/2 - L_m}. \tag{9.23}$$

This is a comparison with one fixed stage. It does not estimate an infinite sum of stage residuals.

Write $e_J = (v_J, \pi_J)$. The residual difference is exactly

$$R(u^{[J]} + v_J, p^{[J]} + \pi_J) - R(u^{[J]}, p^{[J]}) = \partial_t v_J - \Delta v_J + \nabla \pi_J + (u^{[J]} \cdot \nabla) v_J + (v_J \cdot \nabla) u^{[J]} + (v_J \cdot \nabla) v_J.$$

For the Cartesian jet norm, Leibniz's rule bounds its order m by $C_m |e_J|_{m+2} (1 + |u^{[J]}|_{m+2} + |e_J|_{m+2})$. Given m, N , choose one $J \geq m + 2$ so large that

$$g_{J+1}/2 - L_{m+2} > N + K_{m+2} + 2, \quad h\sigma_J - K_m > N + 1.$$

This is possible since both g_j and σ_j tend to infinity. Keep this J fixed. The finite-stage logarithm is $O(q^{-1})$, with a constant depending on J, m , so (9.23) and the finite-stage growth bound imply that the

residual difference is $O(q^N)$. Equation (9.18) gives the same estimate for the fixed residual, including $E_{J,m}$ by its fixed-stage flatness. Adding these two estimates proves for all fixed m, N

$$|R(u_{\text{loc}}, p_{\text{loc}})|_m = O(q^N) \quad (q \downarrow 0). \quad (9.20)$$

No bound uniform in J on the flat remainder constants was assumed or used. On compact sets $c \leq q \leq c'$ only finitely many stages and bands are active. The imported endpoint bounds apply to each of these finite coefficients with the discrete label choices held fixed. Since $L = 1 - 2h\eta^2 \geq 1 - 2h > 0$, coordinate conversion preserves bounds through every fixed order. Bounding one additional time derivative and integrating it gives uniform one-sided limits of each derivative at $t = 1$. Integrating spatial derivatives on compact coordinate segments shows that the limiting jets are compatible with spatial differentiation; integrating in time gives compatibility of successive time derivatives. Thus the finite sum has the asserted endpoint regularity away from the singular point. Outside a fixed X_{ext} all annular corrections vanish; the remaining field is the exact exterior heat profile with centrifugal pressure. Its precise formula is

$$K(r, \tau) = r^{-1-2h} H_{\text{ext}}(\tau/r^2), \quad H_{\text{ext}}(s) = 2^A c_\infty H(4s),$$

$$p_{\text{ext}}(r, \tau) = - \int_r^\infty K(\rho, \tau)^2 \frac{d\rho}{\rho}.$$

Indeed the imported leading field is $c_\infty(r^2/2)^{-A} H(4\tau/r^2)$ and $2A = 1 + 2h$, which gives this exact prefactor. The integral defining pressure converges because $K = O(r^{-1-2h})$ at infinity. It has derivative $\partial_r p_{\text{ext}} = K^2/r$. Consequently the radial component of the momentum residual is $-K^2/r + \partial_r p_{\text{ext}} = 0$; the axial component is zero, and the angular component is zero by the imported exact heat equation $\partial_t K = (\partial_r^2 + r^{-1}\partial_r - r^{-2})K$.

The exterior profile integral is

$$H(Z) = \frac{1}{\Gamma(1+h)} \int_0^\infty e^{-v} v^h (1+Zv)^{-h} dv, \quad Z \geq 0.$$

Its m th derivative is the same integral with factor $(-1)^m (h)_m v^m (1+Zv)^{-m}$. The absolute integrand is bounded by $e^{-v} v^{h+m} (h)_m / \Gamma(1+h)$, an integrable function independent of Z . Differentiation under the integral and the Gamma recurrence therefore prove

$$\sup_{s \geq 0} |H_{\text{ext}}^{(m)}(s)| \leq 2^A c_\infty 4^m (h)_m (1+h)_m.$$

Here $(a)_0 = 1$ and $(a)_m = \prod_{\nu=0}^{m-1} (a + \nu)$. Thus the exterior derivative estimate is uniform on the whole nonnegative half-line, including its endpoint.

Near the axis the base streamfunction is r^2 times a smooth function and the base angular field is r times a smooth scalar, so (9.21) has a smooth Cartesian representative. At $(X, \eta) = (X_{\text{in}}, 0)$, all annular corrections vanish and the leading profile yields, with $q = \tau = 1 - t$,

$$u_{\theta, \text{loc}}(\sqrt{2X_{\text{in}}\tau}, 0, 0, 1 - \tau) = \tau^{-A} (e_0 + O(\tau^{2h})), \quad e_0 > 0. \quad (9.22)$$

These endpoint, exterior, and inner statements are consequences of the imported profile, base, oscillation and mean results plus the finite-stage proof above; they are recorded here without a theorem-complete claim.

23 The terminal force, the comparison argument, and the two official domains

This chapter reconstructs the final argument of the released manuscript, Section 10, pp. 117–126. It uses the actual local fields constructed in its preceding sections. The complete independent review of those constructions is recorded separately; this chapter is not a declaration that the entire manuscript has already passed that review. The maps, cutoff terms, limiting arguments and comparison estimates used here are written out explicitly.

23.1 The local representatives and the localization map

The retained parameters are

$$0 < h < \frac{1}{100}, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad \tau = 1 - t = q(1 - \eta^2), \quad z = q^D \eta, \quad X = \frac{r^2}{2q}.$$

Here $r = (x_1^2 + x_2^2)^{1/2}$, $q > 0$, $|\eta| < 1$ before the terminal time. The constant $q_* > 0$ is the fixed local-domain threshold from Theorem 3.1 (chosen there uniformly for the finite correction stages). The local domain is $\Omega_* = \{\tau > 0, q < q_*\}$. Write the actual local solution representatives as

$$u_{\text{loc}} = \text{curl } \mathcal{A} + B e_\theta, \quad p_{\text{loc}}.$$

The separate scalar B is independent of θ . This notation $\mathcal{A}$ for the vector potential avoids confusing it with the retained scalar exponent A .

The background contribution to the potential is determined by the actual axial coefficients U_n :

$$S_n = q^{1-A+2nh} \int_0^X U_n(X', \eta) dX', \quad S = S_0 + \sum_{n \geq 1} \chi(c_n q) S_n, \quad \mathcal{A}_{\text{base}} = \frac{S}{r} e_\theta.$$

Each full radial integral of U_n is zero. Consequently $S_n = 0$ beyond the common axial support, with its cutoff retained in S . Near the axis $S = r^2 a(r^2, z, t)$ and hence

$$\mathcal{A}_{\text{base}} = a(r^2, z, t)(-x_2, x_1, 0).$$

This is a smooth Cartesian vector field; no division by the coordinate r remains in this representative. The wave potentials are supported in the active annulus. A mean axial increment $\gamma_{d,\text{phys}}$ has potential coefficient

$$\Psi = r^{-1} I_c(r \gamma_{d,\text{phys}}), \quad I_c g = I g - \chi_m J g, \quad I g(r) = \int_0^r g(s) ds, \quad J g = \int_0^\infty g(s) ds.$$

Beyond the source and transition of χ_m , $I g = J g$ and $\chi_m = 1$; therefore this potential vanishes there as well. These facts give the actual representative $\mathcal{A} = 0$ for $X \geq X_{\text{ext}}$. All annular terms vanish near the axis for each fixed $t < 1$.

The coordinates give an elementary useful estimate without an asymptotic replacement. If $1 - \eta^2 \geq 1/2$, then $q \leq 2\tau$. Otherwise $|z| = q^D |\eta| > q^D / \sqrt{2}$, so $q < (\sqrt{2}|z|)^{1/D}$. Thus, for a fixed $C_0 > 0$,

$$q \leq C_0(\tau + |z|^{1/D}).$$

Choose $0 < \tau_0 < 1/2$ and $z_0 > 0$ such that $C_0(\tau_0 + z_0^{1/D}) < q_*/2$. Choose a smooth axisymmetric χ_x , smooth in (r^2, z) , equal to one for $r \leq r_0/2$, $|z| \leq z_0/2$, with compact support in $\{r < r_0, |z| < z_0\}$. Choose χ_t zero for $1 - t \geq \tau_0$ and one for $0 \leq 1 - t \leq \tau_0/2$. Put $c = \chi_x \chi_t$, and define, before time one,

$$u = \text{curl}(c\mathcal{A}) + cB e_\theta, \quad p = cp_{\text{loc}}. \quad (\text{E.1})$$

These expressions are extended by zero off the cutoff support. The support is contained in the fixed compact set $K = \text{supp } \chi_x$, and its part in the local domain has the strict margin $q < q_*/2$. Smoothness of the cutoffs, the Cartesian axis formulas and this margin prove smoothness of the extended fields.

The exact product formula is

$$u = cu_{\text{loc}} + \nabla c \times \mathcal{A}. \quad (\text{E.2})$$

Incompressibility follows directly:

$$\text{div } u = \text{div } \text{curl}(c\mathcal{A}) + r^{-1} \partial_\theta(cB) = 0.$$

The axis identity extends by Cartesian smoothness. The extra term in (E.2) is retained throughout; localization has not been asserted to commute with curl. The time cutoff gives $u = p = 0$ on an interval after time zero.

Define the full residual

$$f = \partial_t u + (u \cdot \nabla) u - \Delta u + \nabla p. \quad (\text{E.3})$$

This defines the required force before time one and includes every derivative and quadratic product of the cutoffs in (E.1). Taking the divergence of this exact identity gives

$$-\Delta p = \sum_{i,j=1}^3 \partial_i \partial_j (u_i u_j) - \text{div } f.$$

There is no divergence-free restriction on the prescribed force in this identity or in the official forced alternatives.

23.2 All terminal derivatives, including the exterior pressure

The local construction supplies bounds on every derivative of $\mathcal{A}, Be_\theta, p_{\text{loc}}$ on compact sets on which q is bounded away from zero and q_* . These are bounds for the actual representatives on the closed parameter interval $-1 \leq \eta \leq 1$. For one such set K_0 , a component G , a spatial multi-index α and an integer $j \geq 0$, the identity

$$\partial_x^\alpha \partial_t^j G(x, t') - \partial_x^\alpha \partial_t^j G(x, t) = \int_t^{t'} \partial_x^\alpha \partial_s^{j+1} G(x, s) ds$$

shows that this derivative is uniformly Cauchy on K_0 as $t, t' \uparrow 1$. Multiplication by the fixed cutoffs and the finite differential polynomial (E.3) preserves this property.

For positive radius and sufficiently small q , the representative is the exact heat exterior. With $s = r^2/2$, its azimuthal velocity is

$$K_{\text{heat}}(r, \tau) = c_\infty s^{-A} H(2\tau/s) = r^{-1-2h} H_{\text{ext}}(\tau/r^2), \quad H_{\text{ext}}(Z) = 2^A c_\infty H(4Z). \quad (\text{E.4})$$

The integral defining H in the exterior construction can be differentiated j times using an integrable majorant proportional to $e^{-v} v^{h+j}$, uniformly for its nonnegative argument. Thus

$$|\partial_\tau^j K_{\text{heat}}(r, \tau)| \leq C_j r^{-1-2h-2j}, \quad \left| \partial_\tau^j \frac{K_{\text{heat}}(\rho, \tau)^2}{\rho} \right| \leq C_j \rho^{-3-4h-2j}. \quad (\text{E.5})$$

The second bound also follows by writing the complete Leibniz sum $\sum_{a=0}^j \binom{j}{a} (\partial_\tau^a K_{\text{heat}})(\partial_\tau^{j-a} K_{\text{heat}})/\rho$; each summand has exactly the same displayed radial power.

The pressure representative is fixed by radial infinity:

$$p_{\text{loc}}(r, \tau) = - \int_r^\infty \frac{K_{\text{heat}}(\rho, \tau)^2}{\rho} d\rho. \quad (\text{E.6})$$

On $r \in [r_-, r_+]$ with $r_- > 0$, the majorant in (E.5) is integrable over $[r_-, \infty)$. It justifies all time differentiations in (E.6), and dominated convergence gives their limits at $\tau = 0$ uniformly on this radial interval. Differentiation of the lower endpoint gives the radial derivatives; differentiating (E.4) gives the velocity's radial derivatives. The sign $\partial_t = -\partial_\tau$ is retained. Cartesian derivatives follow from these radial derivatives with bounded coefficients on a positive-radius compact set. Since $\mathcal{A} = 0$ here, the localized expressions involve only these fields and the fixed cutoffs. The heat equation and $\partial_\tau p_{\text{loc}} = K_{\text{heat}}^2/r$ give zero uncut residual.

These two arguments cover the terminal spatial slice away from the origin. At $z \neq 0$, $q \geq |z|^{1/D} > 0$. At $z = 0$ and $x \neq 0$, the radius is positive and the heat exterior applies in a neighborhood. There are therefore limits for all mixed derivatives of (E.3), uniform on each compact spatial set excluding the origin.

Near the origin both cutoffs are one. For $X \leq X_{\text{ext}}$ the constructed local residual has every derivative of order $O(q^N)$ for every integer $N \geq 0$, and beyond X_{ext} it is exactly zero. The coordinate bound above therefore gives

$$|\partial_x^\alpha \partial_t^j f(x, t)| \leq C_{\alpha, j, N} (\tau + |z|^{1/D})^N \quad (\text{E.7})$$

on a fixed neighborhood of $(0, 1)$. For any tolerance choose first a small ball and then a late time making (E.7) smaller than that tolerance. On the remaining compact part of K , the previous uniform convergence applies using a finite cover. All derivatives vanish off K . They are consequently uniformly Cauchy on all of $\mathbb{R}^3$.

Let $F_j(x) = \lim_{t \uparrow 1} \partial_t^j f(x, t)$. To identify spatial derivatives, pass to the uniform limit in

$$\partial_x^\alpha \partial_t^j f(x + se_i, t) - \partial_x^\alpha \partial_t^j f(x, t) = \int_0^s \partial_{x_i} \partial_x^\alpha \partial_t^j f(x + ve_i, t) dv.$$

Induction on $|\alpha|$ shows that all the derivative limits are $\partial_x^\alpha F_j$. Thus $F_j \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, $\text{supp } F_j \subset K$, and all its spatial derivatives vanish at the origin. Passing to the limit in the time integral gives the exact compatibility identity

$$\partial_x^\alpha \partial_t^j f(x, t) = \partial_x^\alpha F_j(x) - \int_t^1 \partial_x^\alpha \partial_s^{j+1} f(x, s) ds. \quad (\text{E.8})$$

23.3 A concrete smooth extension of the prescribed force

Fix $\chi_0 \in C_c^\infty(\mathbb{R})$ equal to one near zero and zero on $[1, \infty)$. For $\sigma = t - 1 \geq 0$ define

$$f(x, 1 + \sigma) = \sum_{j=0}^{\infty} \chi_0(b_j \sigma) \frac{\sigma^j}{j!} F_j(x). \quad (\text{E.9})$$

The positive integers b_j will be specified by finite inequalities. For a temporal derivative of order m , the complete product rule is

$$\partial_\sigma^m \left(\chi_0(b\sigma) \frac{\sigma^j}{j!} \right) = \sum_{\substack{0 \leq a \leq m \\ m-a \leq j}} \binom{m}{a} b^a \chi_0^{(a)}(b\sigma) \frac{\sigma^{j-m+a}}{(j-m+a)!}.$$

On $\sigma \geq 0$ every term is supported in $0 \leq \sigma \leq b^{-1}$. Its supremum is bounded by its fixed coefficient times b^{m-j} . For example one can use the finite explicit constant

$$C_{j,m} = \sum_{\substack{0 \leq a \leq m \\ m-a \leq j}} \binom{m}{a} \frac{\|\chi_0^{(a)}\|_\infty}{(j-m+a)!}.$$

Thus, with the usual maximum norm over derivatives up to order a_0 ,

$$\left\| \partial_x^\alpha \partial_\sigma^m \left(\chi_0(b\sigma) \frac{\sigma^j}{j!} F_j \right) \right\|_\infty \leq C_{j,m} b^{m-j} \|F_j\|_{C^{|\alpha|}}. \quad (\text{E.10})$$

Take $b_0 = 1$. For $j \geq 1$ take the least integer $b > b_{j-1}$ for which

$$C_{j,m} b^{m-j} \|F_j\|_{C^{a_0}} \leq 2^{-j} \quad \text{for every } a_0, m \geq 0, \quad a_0 + m \leq \lfloor j/2 \rfloor. \quad (\text{E.11})$$

There are finitely many constraints, and each exponent $m-j$ is strictly negative. Therefore the set defining this least integer is nonempty. This specifies the sequence without a uniform bound on the size of the jets F_j .

For fixed α, m , all sufficiently late summands in (E.9) satisfy the summable bound 2^{-j} . The series of each differentiated order converges uniformly. Repeated integration of this uniform series shows that it is the corresponding derivative of (E.9). At $\sigma = 0$ the cutoff in each fixed summand is constant on a neighborhood, so its m th derivative contributes only when $j = m$, with value F_m . The mixed derivatives therefore match (E.8) exactly, proving smooth gluing across $t = 1$. The same uniform convergence proves smoothness at the spatial support boundary and at $t = 2$. The original field vanishes near $t = 0$. Hence

$$f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3). \quad (\text{E.12})$$

If $K \subset B(0, R_0)$ and $M_{\alpha,m} = \|\partial_x^\alpha \partial_t^m f\|_\infty$, its actual support implies, for every integer $k \geq 0$,

$$|\partial_x^\alpha \partial_t^m f(x, t)| \leq M_{\alpha,m} (3 + R_0)^k (1 + |x| + t)^{-k}, \quad t \geq 0. \quad (\text{E.13})$$

On the support $1 + |x| + t \leq 3 + R_0$; off the support the left side is zero. This proves all the original whole-space force decay conditions.

23.4 The energy estimate retains the dissipation

On every $[0, T]$ with $T < 1$, the fields in (E.1) are smooth with fixed compact support. Integration by parts and incompressibility give

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_2^2 + \|\nabla u(t)\|_2^2 = \langle f(t), u(t) \rangle. \quad (\text{E.14})$$

The transport integral is the integral of $\text{div}(u|u|^2/2)$ and the pressure integral is the integral of $\text{div}(pu)$, so both vanish. Put $F(t) = \int_0^t \|f(s)\|_2 ds$. For $\delta > 0$ division of (E.14) by $(\|u\|_2^2 + \delta^2)^{1/2}$ yields

$$\frac{d}{dt} (\|u\|_2^2 + \delta^2)^{1/2} \leq \|f(t)\|_2.$$

After integration from zero and $\delta \downarrow 0$, $\|u(t)\|_2 \leq F(t)$. Returning to the original equality, without discarding its dissipation, gives

$$\|u(t)\|_2^2 + 2 \int_0^t \|\nabla u(s)\|_2^2 ds \leq 2 \int_0^t F'(s) F(s) ds = F(t)^2. \quad (\text{E.15})$$

The smooth compact force has $F(1) < \infty$. This proves a uniform kinetic-energy bound and finite total dissipation on $[0, 1)$ by monotone convergence.

23.5 Comparison without a pressure condition at infinity

Fix $T < 1$. Let (v, P) be any smooth solution with the same force and zero datum, with $\sup_{0 \leq t \leq T} \|v(t)\|_2 < \infty$. Set

$$w = v - u, \quad \pi = P - p, \quad g_{ij} = v_i v_j - u_i u_j = w_i w_j + w_i u_j + u_i w_j.$$

Then w is uniformly in L^2 and g uniformly in L^1 . The difference equation, with the tensor convention $(\operatorname{div} g)_i = \sum_j \partial_j g_{ij}$, is

$$\partial_t w + (v \cdot \nabla) w + (w \cdot \nabla) u = \Delta w - \nabla \pi, \quad \operatorname{div} w = 0. \quad (\text{E.16})$$

Its conservative version has $\operatorname{div} g$ as the transport term.

Use the Fourier convention with derivative multiplier $i\xi_i$ and Riesz multiplier $i\xi_i/|\xi|$. Define

$$\pi_* = \sum_{i,j} R_i R_j g_{ij}. \quad (\text{E.17})$$

For $s > 3/2$, boundedness of the Fourier transform of each L^1 component gives

$$\|\pi_*\|_{H^{-s}}^2 \leq C_s \left(\sum_{i,j} \|g_{ij}\|_1 \right)^2 \int_{\mathbb{R}^3} (1 + |\xi|^2)^{-s} d\xi < \infty.$$

The value of the multiplier at the origin does not affect this distribution. Its sign gives $\Delta \pi_* = -\sum_{i,j} \partial_i \partial_j g_{ij}$, the same Poisson equation as π .

The pressure gradient itself is fixed by the equation. For $a \in C_c^\infty(0, T)$ its time integral satisfies

$$\int_0^T a \nabla \pi dt = \Delta \int_0^T a w dt + \int_0^T a' w dt - \operatorname{div} \int_0^T a g dt.$$

These terms belong respectively to H^{-2} , L^2 and H^{-3} . The last inclusion follows from $L^1 \subset H^{-2}$ by the preceding Fourier bound and one derivative. Also $\pi_* \in L_t^\infty H_x^{-2}$. Consequently

$$H_a = \int_0^T a (\nabla \pi - \nabla \pi_*) dt \in H^{-3}, \quad \Delta H_a = 0.$$

The Fourier transform of H_a is a function with finite weighted L^2 norm. The equation $|\xi|^2 \widehat{H}_a = 0$ makes that function zero off the singleton $\{0\}$; a function supported on a set of Lebesgue measure zero is zero almost everywhere. Thus $H_a = 0$. Testing in space and time yields $\nabla \pi = \nabla \pi_*$ in the interior. No spatial growth hypothesis on P has been inserted.

Choose $0 \leq \phi \leq 1$ smooth and compactly supported, equal to one on the unit ball. For $R \geq 1$ put

$$\phi_R(x) = \phi(x/R), \quad \chi_R = \phi_R^8, \quad E_R = \int \chi_R |w|^2, \quad A_R = \left(\int \chi_R |\nabla w|^2 \right)^{1/2}, \quad B_R = \|\phi_R^4 w\|_6.$$

The ordinary three-dimensional homogeneous Sobolev inequality and the full product rule give

$$B_R \leq C \|\nabla(\phi_R^4 w)\|_2 \leq C(A_R + R^{-1} \|w\|_2). \quad (\text{E.18})$$

Using the classical L^p boundedness of Riesz transforms for $1 < p < \infty$, write exactly

$$\phi_R^4 \pi_* = \sum_{i,j} R_i R_j (\phi_R^4 g_{ij}) + \sum_{i,j} [\phi_R^4, R_i R_j] g_{ij}. \quad (\text{E.19})$$

The first sum has $L^{3/2}$ norm at most $C_T(B_R + 1)$, since

$$\|\phi_R^4 w_i w_j\|_{3/2} \leq B_R \|w\|_2, \quad \|\phi_R^4 w_i u_j\|_{3/2} \leq \|w\|_2 \|u\|_6,$$

and the same estimate holds for $u_i w_j$.

The nonlocal kernel of $R_i R_j$ is bounded by $C|x - y|^{-3}$. Its possible local identity term cancels in the commutator. The Lipschitz and supremum bounds on ϕ_R^4 therefore majorize its kernel by

$$K_R(x - y) = C|x - y|^{-3} \min\{|x - y|/R, 1\}.$$

The radial integral is

$$\|K_R\|_{4/3}^{4/3} \leq C \left(R^{-4/3} \int_0^R r^{-2/3} dr + \int_R^\infty r^{-2} dr \right) \leq CR^{-1}. \quad (\text{E.20})$$

Convolution of this kernel with $g \in L^1$ bounds the second sum of (E.19) in $L^{4/3}$ by $C_T R^{-3/4}$. One may obtain (E.19) first for compact smooth cutoffs of g , then pass to the limit: g converges in L^1 , its Riesz transform in H^{-s} , and the commutator in $L^{4/3}$ by (E.20). On every compact set the first term is in $L^{3/2}$ because w is smooth there. Thus π_* is locally integrable and may be used in the following pressure pairing.

Testing the equality of pressure gradients against the compact field $a(t)\chi_R w$ and using $\operatorname{div} w = 0$ gives, for almost every t ,

$$\int \pi w \cdot \nabla \chi_R = \int \pi_* w \cdot \nabla \chi_R.$$

Since $|\nabla \chi_R| \leq CR^{-1}\phi_R^7$, (E.19) pairs with $CR^{-1}\phi_R^3|w|$. The exact weighted interpolations are

$$\|\phi_R^3 w\|_3 \leq \|\phi_R^2 w\|_3 \leq B_R^{1/2} \|w\|_2^{1/2}, \quad \|\phi_R^3 w\|_4 \leq B_R^{3/4} \|w\|_2^{1/4}.$$

For the first use $\phi_R^2|w| = (\phi_R^4|w|)^{1/2}|w|^{1/2}$; for the second use powers $3/4$ and $1/4$. Hölder then proves

$$\left| \int \pi w \cdot \nabla \chi_R \right| \leq C_T R^{-1} \left((B_R + 1) B_R^{1/2} + R^{-3/4} B_R^{3/4} \right). \quad (\text{E.21})$$

All integrations in the difference energy identity are compact:

$$\begin{aligned} \frac{1}{2} E'_R + A_R^2 = & - \int \chi_R (w \cdot \nabla) u \cdot w + \frac{1}{2} \int |w|^2 \Delta \chi_R \\ & + \frac{1}{2} \int |w|^2 v \cdot \nabla \chi_R + \int \pi w \cdot \nabla \chi_R. \end{aligned} \quad (\text{E.22})$$

For R sufficiently large, $\chi_R = 1$ on a neighborhood of the fixed support of u , hence $v = w$ on $\operatorname{supp} \nabla \chi_R$. The transport term is at most

$$CR^{-1} \int \phi_R^6 |w|^3 \leq C_T R^{-1} B_R^{3/2},$$

where $0 \leq \phi_R \leq 1$ and the preceding interpolation were used. The Laplacian term is bounded by $C_T R^{-2}$. The remaining non-flux term is at most $\|\nabla u\|_\infty E_R$. Substitute (E.18) into (E.21). Every resulting positive power of A_R is at most $3/2 < 2$. For example the scalar Young inequality gives $R^{-1} A_R^{3/2} \leq \delta A_R^2 + C_\delta R^{-4}$; the smaller powers are absorbed by the same method. Choosing the finite sum of dissipation coefficients below $1/2$ proves

$$\frac{1}{2} E'_R + \frac{1}{2} A_R^2 \leq \|\nabla u\|_\infty E_R + \frac{C_T}{R}.$$

Here $\|\nabla u\|_\infty$ is bounded on $[0, T]$, and $E_R(0) = 0$. Integrating the differential inequality gives explicitly

$$E_R(t) \leq \frac{2C_T}{R} \int_0^t \exp \left(2 \int_s^t \|\nabla u(a)\|_\infty da \right) ds \leq \frac{C'_T}{R}.$$

Each fixed ball has $\chi_R = 1$ for all sufficiently large R . Letting $R \rightarrow \infty$ shows its integral of $|w|^2$ is zero. Smoothness gives $w = 0$ everywhere on $[0, T]$.

23.6 The actual path of velocity growth

The local inner asymptotic retained from the constructed profiles is

$$x_\tau = (\sqrt{2X_{\text{in}}\tau}, 0, 0), \quad u_\theta(x_\tau, 1 - \tau) = \tau^{-A}(e_0 + O(\tau^{2h})), \quad e_0 > 0. \quad (\text{E.23})$$

Along this path $z = 0$, $q = \tau$, $X = X_{\text{in}}$; the annular corrections vanish, and both localization cutoffs are one for small τ . The cutoff sum of all positive-order background terms is included in the displayed error. Since $A > 0$ and $h > 0$, the bracket is at least $e_0/2$ for sufficiently small τ and the velocity diverges. Its spatial points converge to the origin.

Every global smooth competitor satisfying the official uniform kinetic-energy condition agrees with u on each $[0, T]$, $T < 1$, by the comparison calculation. Such a competitor is bounded on a compact neighborhood of $(0, 1)$ by continuity, contradicting (E.23). This conclusion uses the actual earlier profile and residual construction; it does not follow from a singular expression alone. The prescribed force is nonzero, since $f = 0$ would make (E.15) imply $u = 0$.

23.7 Exact viscosity and parabolic maps

For the residual convention $\mathcal{R}_\nu(u, p, f) = \partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p - f$, define the map from viscosity one to viscosity $\nu > 0$:

$$\mathcal{A}_\nu(u, p, f)(x, t) = (\sqrt{\nu} u(x/\sqrt{\nu}, t), \nu p(x/\sqrt{\nu}, t), \sqrt{\nu} f(x/\sqrt{\nu}, t)). \quad (\text{E.24})$$

With $y = x/\sqrt{\nu}$, its time, convection, viscous and pressure terms are respectively $\sqrt{\nu} \partial_t u$, $\sqrt{\nu} (u \cdot \nabla_y)u$, $-\sqrt{\nu} \Delta_y u$, $\sqrt{\nu} \nabla_y p$. Therefore

$$\mathcal{R}_\nu(\mathcal{A}_\nu(u, p, f))(x, t) = \sqrt{\nu} \mathcal{R}_1(u, p, f)(y, t), \quad \operatorname{div}_x u^\nu = \operatorname{div}_y u.$$

The map is invertible on its image: $(v, P, F) \mapsto (\nu^{-1/2} v(\sqrt{\nu} y, t), \nu^{-1} P(\sqrt{\nu} y, t), \nu^{-1/2} F(\sqrt{\nu} y, t))$. It preserves time and zero initial data, sends K to $\sqrt{\nu} K$, and obeys, for every α, m ,

$$\partial_x^\alpha \partial_t^m f^\nu = \nu^{(1-|\alpha|)/2} (\partial_y^\alpha \partial_t^m f)(y, t).$$

The volume Jacobian is $\nu^{3/2}$, so

$$\|u^\nu(t)\|_2^2 = \nu^{5/2} \|u(t)\|_2^2, \quad \nu \int_0^1 \|\nabla u^\nu(t)\|_2^2 dt = \nu^{5/2} \int_0^1 \|\nabla u(t)\|_2^2 dt. \quad (\text{E.25})$$

The same inverse map sends any bounded-energy competitor to the already excluded viscosity-one competitor.

At a fixed viscosity define the parabolic map

$$\mathcal{S}_\lambda(v, P, F)(x, t) = (\lambda v(\lambda x, \lambda^2 t), \lambda^2 P(\lambda x, \lambda^2 t), \lambda^3 F(\lambda x, \lambda^2 t)), \quad \lambda > 0. \quad (\text{E.26})$$

Its momentum residual is λ^3 times the original residual at $(\lambda x, \lambda^2 t)$, and divergence is λ^2 times the original divergence. Its inverse is $\mathcal{S}_{1/\lambda}$. The instantaneous energy factor is λ^{-1} .

The formalization uses a different viscosity adapter,

$$\mathcal{T}_\nu(u, p, f)(x, t) = (\nu u(x, \nu t), \nu^2 p(x, \nu t), \nu^2 f(x, \nu t)). \quad (\text{E.27})$$

Every momentum term at viscosity ν has factor ν^2 times its viscosity-one counterpart. Its singular time is $1/\nu$, its energy factor is ν^2 , and it preserves the unit spatial period. These two actual adapters are related by the exact identity on the entire velocity–pressure–force triple

$$\mathcal{T}_\nu = \mathcal{S}_{\sqrt{\nu}} \circ \mathcal{A}_\nu. \quad (\text{E.28})$$

Indeed the arguments become $(\sqrt{\nu} x / \sqrt{\nu}, (\sqrt{\nu})^2 t) = (x, \nu t)$, and the three prefactors become ν, ν^2, ν^2 . The energy factors compose as $\nu^{-1/2} \nu^{5/2} = \nu^2$. For a unit-periodic input, $\mathcal{A}_\nu$ gives period $\sqrt{\nu}$, and $\mathcal{S}_{\sqrt{\nu}}$ returns it to period one. Thus their different intermediate periods and singular times are accounted for by a proved morphism. This identity concerns the two adapters; it does not assert without inspection that their input representatives in the paper and Lean source are literally identical.

23.8 The unit-periodic construction, with periodic pressure

Fix $\nu > 0$ and use the fields at that viscosity just constructed. Choose $\lambda > 1$ with $\lambda^{-1} \sqrt{\nu} K \in Q_0 = (-1/2, 1/2)^3$, and let $t_0 = 1 - \lambda^{-2}$. For $t \geq t_0$ put

$$\tilde{u}(x, t) = \lambda u^\nu(\lambda x, \lambda^2(t - t_0)), \quad \tilde{p}(x, t) = \lambda^2 p^\nu(\lambda x, \lambda^2(t - t_0)), \quad \tilde{f}(x, t) = \lambda^3 f^\nu(\lambda x, \lambda^2(t - t_0)). \quad (\text{E.29})$$

Use the first two formulas before time one and the force formula for all subsequent time; extend all three by zero for $t < t_0$. The original fields vanish on an initial time interval, so this extension is smooth. All momentum terms scale by λ^3 and the viscosity stays ν . Original time one is reached at time one because $\lambda^2(1 - t_0) = 1$.

Define on $\mathbb{R}^3 / \mathbb{Z}^3$

$$U(x, t) = \sum_{k \in \mathbb{Z}^3} \tilde{u}(x + k, t), \quad P_{\text{per}}(x, t) = \sum_{k \in \mathbb{Z}^3} \tilde{p}(x + k, t), \quad F_{\text{per}}(x, t) = \sum_{k \in \mathbb{Z}^3} \tilde{f}(x + k, t). \quad (\text{E.30})$$

The supports of distinct translates have a positive gap, since the unshifted compact support lies strictly inside Q_0 . The sums are locally finite and at each point at most one translate is nonzero. The support of each derivative remains in the same closed support. Consequently the full convection term is

$$(U \cdot \nabla)U = \sum_{k \in \mathbb{Z}^3} (\tilde{u}(x+k, t) \cdot \nabla) \tilde{u}(x+k, t).$$

All cross terms vanish by disjointness, not by linearity of the equation. Termwise differentiation now proves

$$\partial_t U + (U \cdot \nabla)U - \nu \Delta U + \nabla P_{\text{per}} = F_{\text{per}}, \quad \text{div } U = 0, \quad U(\cdot, 0) = 0.$$

Integer translation reindexes each sum, proving unit periodicity of velocity, pressure and force separately. The force has time support in $[t_0, 1+\lambda^{-2}]$, and hence every prescribed periodic force derivative has arbitrary polynomial time decay.

The precise path of growth is

$$\tilde{x}_\tau = \lambda^{-1} \sqrt{\nu} x_\tau, \quad \tilde{t}_\tau = 1 - \lambda^{-2} \tau, \quad U_\theta(\tilde{x}_\tau, \tilde{t}_\tau) = \lambda \sqrt{\nu} \tau^{-A} (e_0 + O(\tau^{2h})).$$

For small τ this lies inside Q_0 and only its zeroth translate contributes. The value diverges as before. Two smooth periodic solutions with the same data have difference w satisfying, on each $[0, T]$, $T < 1$,

$$\frac{1}{2} \frac{d}{dt} \|w\|_{L^2(\mathbb{T}^3)}^2 + \nu \|\nabla w\|_{L^2(\mathbb{T}^3)}^2 \leq \|\nabla U\|_\infty \|w\|_{L^2(\mathbb{T}^3)}^2.$$

Periodic integration removes the transport divergence and pressure gradient; the remaining quadratic term has the displayed bound. The zero datum and integration of this inequality force $w = 0$. A global smooth periodic competitor therefore contradicts the same growth path. Both the original unit periods and the pressure periodicity required by the official correction have been preserved.

24 The exact constructed objects and the order of the argument

Retain the original right-handed cylindrical coordinates

$$x = (r \cos \theta, r \sin \theta, z), \quad s = r^2/2, \quad \tau = 1 - t = q(1 - \eta^2), \quad z = q^D \eta, \quad X = s/q,$$

where

$$0 < h < 1/100, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad d = 1 - \eta^2, \quad L = 1 - 2h\eta^2, \quad -1 \leq \eta \leq 1.$$

The physical chart for $t < 1$ has $|\eta| < 1$; the profile endpoints have one-sided derivatives. The coordinate and Cartesian extension proofs are in Section 31. No radial dilation below changes these coordinates. The fields are

$$u_\theta = q^{-A} E, \quad u_z = q^{-A} U, \quad ru_r = V_0, \quad p = q^{-2A} \Pi, \quad E = \sqrt{2X} F = \sqrt{2X} \phi / C, \quad H = \sqrt{2X} E.$$

Every integral and component order in this document is the original one:

$$\begin{aligned} M &= \int_0^X U \, dx, & I &= \int_0^X H \, dx, & J &= \int_0^X UH \, dx, \\ S &= \int_0^X (U^2 - E^2/2) \, dx, & C_p &= \int_0^X E^2/(2x) \, dx, & \Pi &= \Pi_0 + C_p. \end{aligned}$$

The two tangential components are ordered $(r\theta, rz)$. In particular,

$$\begin{aligned} V_0 &= (2\eta XU - 2D\eta M - dM_\eta)/L, \\ W &= 1 - (2D\eta M + dM_\eta)/X, \\ Q_s &= -W + \frac{(1-h)I - D\eta I_\eta - dJ_\eta + 2(h-D)\eta J}{XH}, \\ N_s &= -WU + \frac{D(M - \eta M_\eta) + 4h\eta S - dS_\eta}{X} + 4A\eta \Pi - d\Pi_\eta, \\ p_s &= (XQ_s/L, XN_s/(LE)), \quad a = 1 - 2X\partial_X \log E, \quad b_s = 2XU_X/E, \\ T_0 &= F\{p_s - (a, -b_s)\}. \end{aligned} \tag{PA.1}$$

At the regular axis the formulas are extended by the proved radial primitive identities, rather than evaluated as indeterminate quotients.

The construction uses the following directed chain:

$$\begin{aligned}
& \text{outer schedule and moment adjustments} \longrightarrow \Pi_0 \text{ and the outer family,} \\
& \Pi_0 \longrightarrow \text{analytic axis and inner continuation} \longrightarrow \text{five-row join,} \\
& \text{regular join and total moments} \longrightarrow \text{backward stress and endpoint directions,} \\
& \text{joined relaxed shear and admissible collars} \longrightarrow \text{variance loop,} \\
& \text{variance loop} \longrightarrow \text{finite-frequency field and five-row repair.}
\end{aligned} \tag{PA.2}$$

The outer schedule does not use a regular axis field. Its temporary inner branch $U_o = 4\eta$, $E_o = P_* f(X/X_R)^{1/10}$, $f = (1 + \eta^2)^{-1}$, has convergent five prefix moments. The later backward-stress calculation is applied to the regular joined field actually constructed below. Thus it is a later use of the schedule, and supplies no premise needed to construct that schedule.

25 The outer pressure is the actual axis datum

The outer schedule in Section 33 first fixes

$$M_d > 0, \quad T_d = e^{M_d} + 10, \quad P_* > e^{T_d}, \quad \lambda > 0, \quad 0 < h < \min\{1/100, \lambda, e^{-T_d}\}. \tag{PA.3}$$

Its successive slope changes, finite pulse, affine two-bump axial solve, pressure-neutral angular solve, amplitude root, and finite tail waiting length are proved there. In particular no stage length depends on η or on X_R . After omitting only the angular edit of zero total pressure increment, write its swirl exactly as

$$E_{\text{sched}}(y, \eta) = c(y)f(\eta)^{\vartheta(y)}, \quad y = \log(X/X_R), \quad 0 \leq \vartheta \leq 1.$$

Here $\vartheta = 1$ and $c(y) = P_* e^{y/10}$ for $y \leq 0$; in the final tail $\vartheta = 0$ and $c(y)$ decays with exponent $-A$. Define, before selecting X_R ,

$$\Pi_0(\eta) = -\frac{1}{2} \int_{-\infty}^{\infty} c(y)^2 f(\eta)^{2\vartheta(y)} dy. \tag{PA.4}$$

For $z = x + iu$, $|u| < 1/4$, the real part of $1 + z^2$ is $1 + x^2 - u^2 > 15/16$. The analytic principal logarithm of $1 + z^2$ therefore defines $f(z)^{2\vartheta} = \exp[-2\vartheta \log(1 + z^2)]$ throughout this strip. On each compact subset it is bounded uniformly for $0 \leq \vartheta \leq 1$. The integrable bounds $Ce^{y/5}$ at $-\infty$ and $Ce^{-(1+2h)y}$ at $+\infty$, and the finite intervening length, give uniform convergence on compact complex subsets. Integrating on rectangles and applying Fubini and Cauchy's theorem proves holomorphy of (PA.4); alternatively, uniform convergence of truncated holomorphic integrals proves the same statement. The integrand is real positive for real η and even. As $f' = -2\eta f^2$, dominated differentiation gives the stronger exact bounds

$$\Pi_0 \leq -\frac{5}{2} P_*^2 f^2, \quad \eta \Pi'_0 = 2\eta^2 f \int \vartheta c^2 f^{2\vartheta} dy \geq 10 P_*^2 \eta^2 f^3 > 0 \quad (\eta \neq 0). \tag{PA.5}$$

The reference half-line contributes $\int_{-\infty}^0 e^{y/5} dy = 5$; all other contributions to the derivative have the same sign.

The scale independence and its survival under the actual modifications are exact. The substitution $X = X_R e^y$ gives $dX/(2X) = dy/2$, so (PA.4) contains no X_R . The two angular bumps restore their total $\int E^2 dy$ to its pre-edit value. The heat replacement and its earlier three-row correction restore together the total increments in C_p, S, I (with I understood as its convergent subtracted total). Their equations are proved in Section 33; M, J are unchanged because $U = 0$ on both supports. Thus the complete prepared outer profile has

$$\int_0^{\infty} E_o^2/(2X) dX = -\Pi_0 \tag{PA.6}$$

for every sufficiently large X_R , with this very same function Π_0 . The pressure used in the axis equation is (PA.4), not a later approximation to it.

For completeness, positivity at the turning point of the axis equation has an explicit bound. Choose $0 < j_0 \leq 1/20$, put

$$U_* = 4\eta + j_0, \quad H_* = D\eta + dU_*, \quad Z_* = -A(1 - 2\eta U_*)U_* - H_* U'_* - d\Pi'_0 + 4A\eta \Pi_0.$$

The function $H_*/d = D\eta/(1 - \eta^2) + 4\eta + j_0$ has derivative $D(1 + \eta^2)/(1 - \eta^2)^2 + 4 > 0$, with endpoint limits of opposite infinite sign. Its unique zero is $\eta_0 = -s$, where

$$0 < s \leq 1/80, \quad j_0/5 < s < j_0/4, \quad d_0 = 1 - s^2 \geq 6399/6400, \quad U_*(\eta_0) = Ds/d_0.$$

The first bound follows from $s = j_0/(4 + D/d_0)$; it then gives $D/d_0 \leq 3200/6399 < 1$, proving the strict lower bound. At this zero, $-H_*U'_* = 0$ and $-d_0\Pi'_0(\eta_0) > 0$. Using $f^2 \geq 1/4$ and (PA.5), the positive pressure term is at least $Aj_0P_*^2/2$. On the other hand,

$$U_*(\eta_0)/j_0 \leq 800/6399, \quad 2sU_*(\eta_0) \leq 1/6399,$$

so its only negative term has absolute value at most

$$Aj_0 \frac{5120000}{40947201} < Aj_0/7.$$

Consequently

$$Z_*(\eta_0) > Aj_0(P_*^2/2 - 1/7) > 0. \quad (\text{PA.7})$$

This proves the exact hypothesis of the axis construction for the already selected $P_* > 1$, without a further dependence of P_* on j_0 or X_R . Choose a neighborhood V of η_0 with $Z_* > Z_*(\eta_0)/2$ and $0 < \delta_* < Z_*(\eta_0)/4$. On the compact set $K = \{|Z_*| \leq \delta_*\}$, the continuous function $|H_*|$ has a positive minimum m when K is nonempty. Choose $0 < \sigma_*^2 < m^2/99$; for empty K take any sufficiently small positive σ_* . Then

$$\chi = H_*^2/(H_*^2 + \sigma_*^2) > .99 \quad \text{on } K. \quad (\text{PA.8})$$

After a small real and complex enlargement, L and $H_*^2 + \sigma_*^2$ remain nonzero and Π_0 remains holomorphic. These are all the analytic datum hypotheses used by the coefficient-space contraction in Section 36. Denote the selected simply connected complex neighborhood, with compact closure inside that nonvanishing domain, by Ω .

26 One finite choice of radius, amplitude, and joining widths

The outer schedule estimates use finite constants formed from fixed smooth steps, finite intervals, and their displayed convolution integrals. The outer proof first chooses M_d and P_* so the axial-decrease cone gaps are strictly positive, then chooses λ so that

$$C_{\text{pre}}\sqrt{\lambda}(1 + \log(1/\lambda)) \quad \text{and} \quad C_{\text{pre}}\lambda(1 + \log(1/\lambda))$$

are smaller than the fixed strict allowances in its pulse and power-law estimates. Those allowances are positive: the two pulse quadratic upper bounds are $2.41^2/8 < .74$ and $4.82^2/14 < 1.68$, and the amplitude-root endpoint gaps exceed .047 and .038 before the small error. Choosing h subsequently small relative to both λ and e^{-T_d} gives (PA.3) and the remaining positive outer gaps. These assertions use the explicit estimates proved in the outer component, not an assumed cone theorem.

Here are precise inequalities for the later choices. All constants in them are finite norms, minima, or Lipschitz constants of the displayed maps on specified compact sets. Fix the finite derivative orders required for the joining inequalities, including one extra η derivative for (PA.1). On $R = [e^{-8}, e^{-5}] \times [-1, 1]$, the ideal pair is

$$U_{\text{id}} = 4\eta, \quad E_{\text{id}} = P_*f x^{1/10}.$$

Its angular source is exactly

$$S_{q,\text{id}} = \frac{9}{5} - h + \frac{16}{5}h\eta^2 + \frac{2\eta^2(D + 4d)}{1 + \eta^2} > 0,$$

and the positive regular weighted primitive gives $Q_{\min} = \min_R Q_{s,\text{id}} > 0$. Put $e_* = \min_R E_{\text{id}} > 0$ and $w_* = 1 + \max_R |N_{s,\text{id}}/(E_{\text{id}}Q_{s,\text{id}})|$. Choose a positive tolerance ε_m in the normalized field/moment map and fixed restoration/repair maps sufficiently small to imply

$$Q_s \geq Q_{\min}/2, \quad E \geq e_*/2, \quad .7 \leq a \leq .9, \quad |N_s/(EQ_s)| \leq w_*, \quad |b_s| \leq .1/(1 + w_*). \quad (\text{PA.9})$$

Such a positive tolerance follows directly from the rational formulas (PA.1), the fixed bump Jacobian inverse and quadratic contraction proved below. The denominators have positive minima at the ideal pair. In particular the tolerance contains no X_R .

Select j_0 and then Λ so the finite bounds $C_k j_0 + C_k^{\text{nat}}/\Lambda$ are less than one quarter of the required joining-error allowance for each of the selected orders. In the radial variable $Y = \Lambda X$, write $\phi = \phi_* \Phi$. Also impose all the finitely many axis-contraction and endpoint bounds proved in the axis component:

$$\Lambda \geq \max\{1, 2M, 2K\}, \quad \Phi \geq c_0 > 0, \quad S_q \geq 2.5 + .95\Lambda L\chi,$$

and its two exit alternatives. Here M, K are the actual radius-one-ball norm and Lipschitz bounds for the rescaled remainder operators, which are uniform in the later amplitude after its complex bound is imposed. Choose the amplitude-independent bounds B_k and then

$$T_{\text{sh}} \geq 20\|\sigma'\|_\infty(B_0 + \|\log f\|_\infty), \quad X_{\text{sep}} = 110e^{T_{\text{sh}}}. \quad (\text{PA.10})$$

The complete proof of their independence from subsequent C and smaller joining widths is in Section 37.

Now choose C , once, to exceed all the following finite bounds:

$$\begin{aligned} C > 1, \quad C \geq \sup_{\bar{\Omega}} |\phi_*|, \quad C \geq C_0, \quad C \geq (R_*/110)^{1/10}/P_*, \\ (CP_*)^{10} > e^{T_{\text{sh}}+8}, \quad X_R := 110(CP_*)^{10} > 14e^8/(3Q_{\min}). \end{aligned} \quad (\text{PA.11})$$

The threshold R_* includes the prepared outer cone, heat compensation, and terminal $-ZH'/\mathcal{H} < h/4$ requirements, all depending only on the fixed outer data. The threshold C_0 includes the uniform exit and reference inequalities $p_{1,r} + p_{2,r}^2/p_{1,r} > 2 + c$. Also require, for the finite selected orders, the vanishing early normalized moment errors and ideal early moments to use less than one quarter of the remaining matching allowance. This is possible because

$$\begin{aligned} x_{\text{sep}} &= e^{T_{\text{sh}}}/(CP_*)^{10} \rightarrow 0, \\ \|\widehat{M}\|_k + \|\widehat{S}\|_k &\leq K_k x_{\text{sep}}, \quad \|\widehat{I}\|_k + \|\widehat{J}\|_k \leq K_k C^{-1} x_{\text{sep}}^{3/2}, \quad \|C_p\|_k \leq K_k C^{-2}, \end{aligned} \quad (\text{PA.12})$$

with constants already fixed; their exact integral proof is included below. The complex supremum in (PA.11) gives uniform η derivatives by Cauchy's estimate; a real supremum alone would not do so.

After fixing this C , first choose $\kappa_0 > 0$ so

$$\kappa_0 < 1/2, \quad \kappa_0 V_{\max} < 1/4, \quad \kappa_0 \sup p_{1,r} < .8,$$

and so the $O(\kappa_0)$ value and moment errors fit their positive allowances. The number $V_{\max}$ is a common bound on the family $0 < t_1 \leq t_*$, with t_* fixed before κ_0 . Now choose $0 < t_1 \leq t_*$ and the two final widths, of sum $\omega_{\text{fin}} > 0$, sufficiently small that the remaining $C_k^{\text{join}}(\Lambda)(t_1 + \kappa_0 + \omega_{\text{fin}})$ errors fit the matching allowance and all the activation comparison gaps. The full derivation has no factor $1/\kappa_0$ in the activation slope. It proves the exact $e^{-t_1^2/y_a^2}$ factor before any division by it. Thus none of these choices forces a new earlier choice of T_{sh} or C .

The relaxed-cone verification on the matching patch is quantitative. From (PA.9),

$$v_s = a + b_s^2/a \leq .9 + .01/.7 < 1, \quad G = Q_s - b_s N_s/(aE) \geq 6Q_s/7 \geq 3Q_{\min}/7.$$

Since $L \leq 1$ and $x \geq e^{-8}$, (PA.11) gives

$$P_c = X_R x G/L > 2. \quad (\text{PA.13})$$

The smaller root $U(P_c, J_c)$ of the cone polynomial is strictly greater than 2, as proved in the core and loop components. Thus (PA.13) proves the strict relaxed cone throughout the actual restoration and correction, with the single previously fixed tolerance.

27 Every row of both five-moment repairs and the exterior map

Write $u = \delta U$, $e = \delta E$. At any fixed reference pair the complete five increment integrands, in order (M, I, J, S, C_p) , are

$$\begin{pmatrix} u \\ \sqrt{2X}e \\ Hu + U\sqrt{2X}e + \sqrt{2X}ue \\ 2Uu + u^2 - Ee - e^2/2 \\ Ee/X + e^2/(2X) \end{pmatrix}. \quad (\text{PA.14})$$

This follows by multiplying out the two quadratic and one bilinear original integrands. In particular the C_p row includes its factor $1/X$ and the S row its negative angular quadratic term. Under $X = X_R x$, the exact moment transformation is

$$(M, I, J, S, C_p) = (X_R \widehat{M}, X_R^{3/2} \widehat{I}, X_R^{3/2} \widehat{J}, X_R \widehat{S}, \widehat{C}_p), \quad H = X_R^{1/2} \widehat{H}. \quad (\text{PA.15})$$

Substitution in (PA.1) cancels X_R in Q_s, N_s ; it retains the factor X_R in p_s .

For the inner join, on $e^{-6} < x < e^{-5}$ the base pair is $U = 4\eta$, $E = P_* f x^{1/10}$. Take two nonnegative ordered disjoint bumps for u and three for e . The invertible row changes $J \mapsto J - 4\eta I$, $S \mapsto S - 8\eta M$ use their displayed base values, while retaining all quadratic terms in (PA.14). The derivative blocks are

$$(1, \sqrt{2} P_* f x^{3/5}), \quad (\sqrt{2} x^{1/2}, -P_* f x^{1/10}, P_* f x^{-9/10}). \quad (\text{PA.16})$$

For the later shear repair on a compact subinterval of I_1 , the base pair is $U = 0$, $E = K(\eta) X^{-1/2-\lambda}$ with $K > 0$. In order $(M, J; I, S, C_p)$, the blocks are exactly

$$(1, \sqrt{2} K X^{-\lambda}), \quad (\sqrt{2} X, -K X^{-1/2-\lambda}, K X^{-3/2-\lambda}). \quad (\text{PA.17})$$

The powers in (PA.16) and (PA.17) are distinct within each block; (PA.17) uses the already fixed $\lambda > 0$. Their matrices are invertible. Indeed a nonzero combination of m distinct powers has at most $m-1$ positive zeros: divide by the lowest power and apply Rolle's theorem and induction to its derivative. The evaluation determinant consequently has constant nonzero sign on the connected set of ordered positive evaluation points. Multilinearity expresses the bump-moment determinant as its integral against the nonnegative product of the bumps. This product is positive on a set of positive measure, so the integral is nonzero. Compactness in η and the adjugate identity bound every fixed derivative of the inverse.

The exact increment map is $B(\eta)c + Q_\eta(c, c)$. Let ν_j denote the C^j multiplication norm of B^{-1} and q_j the bilinear norm of Q . For each of the finitely selected orders j , impose

$$8\nu_j^2 q_j \|d\|_{C^j} \leq 1. \quad (\text{PA.18})$$

The map $c \mapsto B^{-1}(d - Q(c, c))$ is a contraction on the radius $2\nu_j \|d\|_{C^j}$ ball: its Lipschitz constant is at most $4\nu_j^2 q_j \|d\|_{C^j} \leq 1/2$, and its image norm is at most three quarters of that radius. Starting from zero in C^0 and C^j selects the same solution. The pointwise Jacobian differs from B by a perturbation whose B^{-1} -norm is at most $1/2$; hence it is invertible. The implicit function theorem then gives smoothness at every remaining fixed order, with one-sided endpoint derivatives. These statements prove the complete nonlinear repair maps, rather than replacing them by their linearizations.

The join takes place at $X_h = X_R e^{-5}$. Its two field definitions agree with the identical ideal pair on a neighborhood, so every radial jet agrees. The first repair sets all five moments equal there as functions of η . For every $X \geq X_h$, integrating (PA.14) from X_h gives zero, since $u = e = 0$ there. Hence every moment and every η derivative agrees for all such X . The common Π_0 gives identical Π . Formula (PA.1) then gives identical V_0, W, Q_s, N_s, p_s ; equality of field jets gives identical a, b_s, F, T_0 . This proves the exact map from the five matching values to the entire outer field and stress. The identical argument applies after the second repair.

In particular the actual regular joined pair has

$$M(\infty) = J(\infty) = S(\infty) = 0, \quad \int_0^\infty (H - H_{\text{pow}}) dX = 0, \quad H_{\text{pow}} = \sqrt{2X} c_\infty X^{-A}, \quad (\text{PA.19})$$

because its difference from the prepared outer pair is supported before X_h , and its I difference at X_h is zero. The difference $H - H_{\text{pow}}$ is integrable at zero since $h < 1$, and at infinity since the heat difference is $O(X^{-1-h})$. Its pressure is exactly

$$\Pi = \Pi_0 + C_p = C_p - C_p(\infty) = - \int_X^\infty E(x, \eta)^2 / (2x) dx. \quad (\text{PA.20})$$

28 Applying the outer endpoint calculation to the actual join

The joined field is Cartesian smooth at the axis by the axis component, agrees pointwise with the heat-corrected outer profile for $X \geq X_h < X_{\text{tail}}$, and has (PA.19)–(PA.20). We now prove precisely the backward-stress input formerly listed separately from the axis construction.

Put $K_{\text{pow}} = c_\infty(r^2/2)^{-A}$. The exact substitution $r = \sqrt{2qX}$, with $r dr = q dX$, gives

$$\int_0^\infty r^2(u_\theta - K_{\text{pow}}) dr = q^{3/2-A} \int_0^\infty (H - H_{\text{pow}}) dX = 0. \quad (\text{PA.SUB})$$

The heat expansion gives $u_\theta - K_{\text{pow}} = O_h(\tau r^{-3-2h})$, $\partial_t u_\theta = O_h(r^{-3-2h})$, and $r^2 \partial_r u_\theta - r u_\theta = O_h(r^{-2h})$ in the exterior. Their weighted tails have integrable majorant $C r^{-1-2h}$ for fixed $h > 0$. At zero the subtracted weight is r^{1-2h} and the field is regular. Dominated differentiation therefore gives zero integrated angular time term. The axial angular-momentum flux has integral $q^{3/2-2A} J(\infty) = 0$. The radial transport boundary vanishes at zero by regularity and at infinity because $U = M = V_0 = 0$. Its viscosity boundary vanishes by the displayed heat estimate. Thus $\int_0^\infty r^2 \mathcal{R}_\theta^{(0)} dr = 0$.

For the axial component $\int r u_z dr = q^{1-A} M(\infty) = 0$. The canonical pressure obeys, by absolute convergence and Fubini,

$$\int_0^\infty r p(r) dr = -\frac{1}{2} \int_0^\infty r u_\theta(r)^2 dr.$$

The integrated axial momentum flux is consequently $q^{1-2A} S(\infty) = 0$. Its transport and viscosity radial boundaries vanish, and the pressure boundary $r^2 p \rightarrow 0$ follows from $p = O(r^{-2-4h})$. Hence $\int_0^\infty r \mathcal{R}_z^{(0)} dr = 0$. Here $\mathcal{R}_j^{(0)} = \mathcal{R}(u, p) \cdot e_j + \partial_{zz} u_j$, with viscosity one, exactly as in the core calculation. The leading residuals exclude only the axial viscosity term in this identity. Their regular forward primitives are therefore exactly

$$T_\theta = r^{-2} \int_r^\infty r'^2 \mathcal{R}_\theta^{(0)}(r') dr', \quad T_z = r^{-1} \int_r^\infty r' \mathcal{R}_z^{(0)}(r') dr', \quad T = q^{-A-1/2} T_0. \quad (\text{PA.21})$$

All original hypotheses of the outer endpoint calculation have now been proved for this specific joined field.

Section 38 derives (PA.21) directly once more, evaluates the heat-tail residual including its pressure term and both integration signs, and proves

$$b_s = 0, \quad 2 + h < a \leq 2 + 2h, \quad T_\theta > 0, \quad 2 - (a - 2)(T_z/T_\theta)^2 \geq \kappa_o > 0 \quad (\text{PA.22})$$

on $e^{1/2} X_{\text{tail}} \leq X < X_b = e^3 X_{\text{tail}}$. Its exact endpoint factors, with $y_b = \log(X_b/X)$, are

$$T_{0,\theta} = e^{-4/y_b^2} y_b^{-3} b_\theta(y_b, \eta), \quad T_{0,z} = e^{-4/y_b^2} y_b^3 b_z(y_b, \eta), \quad (\text{PA.23})$$

with smooth factors and

$$b_\theta(0, \eta) = \frac{16\rho_o E_{\text{pow}}(X_b)}{\sqrt{2X_b}} \mathcal{H}(2d/X_b) e > 0.$$

This coefficient keeps the cutoff, heat, radial, and factor 16 constants. The stress is zero for $X \geq X_b$ by the exact radial heat equation and (PA.21). The inner continuation component proves, for $y_a = \log(X/X_a)$,

$$T_0 = e^{-t_1^2/y_a^2} g_a(y_a) B_a(y_a, \eta), \quad g_a(0) = (1 - \kappa_o) e > 0, \quad B_a(0, \eta) = F(X_a, \eta)(a, -b_s)(X_a, \eta) \neq 0. \quad (\text{PA.24})$$

Its stress is zero through $X_a = 4/\Lambda$, and its exit has $v_s > 2 + .2$. Thus the actual joined profile supplies the two admissible boundary collars needed for the next construction.

29 A single finite loop insertion and restoration

Choose X_- strictly inside the first admissible collar, and $X_{\text{good}} < X_+ < \inf I_1$. The rectangle $I = [X_-, X_+] \times [-1, 1]$ contains all potentially nonadmissible points and has admissible radial boundary collars. Shrink X_{an} so $X_a < X_{\text{an}} < X_-$. Choose $[Y_0, Y_1] \subset I_1$ with $X_+ < Y_0 < Y_1$. The exact relaxed data $a > 0$, $P_c > 2$, $v_s < U(P_c, J_c)$ follow from the preceding inner/outer construction on all of I . The required compact strict inequalities are actual positive continuous minima. The admissible condition holds throughout $[X_+, Y_1]$.

The component proof in Section 39 constructs the smooth variance loop from these very data. It proves its $p_{s,2} = 0$ extension, strict variance monotonicity, finite uniform variance range, smooth inverse at zero variance, and exact mean after the positive circle reparameterization. The output satisfies, in phase $\varphi \in \mathbb{R}/\mathbb{Z}$,

$$\int_0^1 (a_L, -b_L) d\varphi = (a, -b_s),$$

with four strictly positive cone gaps on the compact loop family, and equals the original shear on the two boundary collars. The positive compact derivative of the phase map gives a smooth inverse on the same parameter rectangle.

The zero-mean periodic primitives $\mathcal{A}, \mathcal{B}$ have

$$\partial_\varphi \mathcal{A} = -(a_L - a)/2, \quad \partial_\varphi \mathcal{B} = E(b_L - b_s)/2.$$

They are zero on those collars. Set

$$E_N = E \exp[\mathcal{A}(X, \eta, N \log X)/N], \quad U_N = U + \mathcal{B}(X, \eta, N \log X)/N.$$

The exact chain rule, with D_X holding phase fixed, is

$$a_N = a_L - 2D_X \mathcal{A}/N, \quad b_N = e^{-\mathcal{A}/N} \{b_L + 2D_X \mathcal{B}/(NE)\}. \quad (\text{PA.25})$$

For each fixed number of η derivatives, field differences, shear errors relative to the loop, and all prefix-moment differences are $O(N^{-1})$ on the fixed interval $[X_-, Y_1]$. This follows directly from (PA.14) and the identity $e^v - 1 = v \int_0^1 e^{sv} ds$. The phase has no η dependence. Formula (PA.1) then bounds $p_{s,N} - p_s$ by C/N , using one extra η derivative and no radial derivative of the field differences. Actual high radial derivatives can grow as N^{r-1} ; N is fixed before they are used.

Here is one simultaneous finite bound for N . Let μ_L be the minimum of the four cone gaps on the loop and μ_R their minimum on the original compact segment $[X_+, Y_1]$. Choose fixed neighborhoods of these states in $a > 0$, and let L_Ψ be a common Lipschitz constant for

$$\Psi(a, b, p) = (a, v - 2, c - v, 2(c - v)^2 - (v - 2)j^2), \quad v = a + b^2/a, \quad c = p_1 - bp_2/a, \quad j = p_2 + bp_1/a.$$

Let B_{mod}/N bound the modulation errors in its three inputs, and let B_{rep}/N bound the repair's changes of those inputs, including all its prefix moments. Let D_j/N bound the defect for the repair (PA.17), $j = 0, 2$, and let B_E/N bound its change in E . These constants are fixed by the loop, bump shapes, and all the parameters chosen above. Take an integer N satisfying

$$\begin{aligned} N &> 1, \quad N > 2L_\Psi B_{\text{mod}}/\min(\mu_L, \mu_R), \\ N &> 4L_\Psi B_{\text{rep}}/\mu_R, \quad N > 8 \max_{j=0,2} \nu_j^2 q_j D_j, \\ N &> 2B_E / \min_{[Y_0, Y_1] \times [-1, 1]} E. \end{aligned} \quad (\text{PA.26})$$

Increase it finitely, when necessary, to put the input changes inside the chosen neighborhoods. The first bound preserves half the loop and intervening margins. The C^2 contraction (PA.18) gives the exact repair with $\|c\|_{C^2} \leq 2\nu_2 D_2/N$; its fixed bump shapes give the bound defining B_{rep} . The next bound preserves at least $\mu_R/4$ on the repair. The last preserves $E > 0$. Equation (PA.26) imposes only finitely many inequalities. Every other fixed parameter derivative is finite by implicit differentiation of the same coefficient equation, without reselecting N .

At Y_1 all five moments are restored. The exterior map proved after (PA.18) therefore proves exact equality of $\Pi, V_0, Q_s, N_s, p_s, a, b_s, T_0$ beyond Y_1 . The inner analytic rectangle is unchanged because both perturbations start later and use the same axis pressure. The heat compensation in I_2 , and the unused I_3, I_4 , consequently remain exactly their original profiles.

30 Closed leading-profile conclusion

Leading-profile assembly theorem. For the particular h selected in (PA.3) with the finite smallness choices in Section 26, the preceding construction produces the profiles and every property of [NS, Theorem 4.6]. This chosen h lies in the original range $0 < h < 1/100$. The original coordinates and component signs are retained. All leading-profile constants are independent of physical q , later dyadic bands, and later correction stages.

Proof. Cartesian smoothness, positivity of ϕ and E for $X > 0$, and the common complex parameter neighborhood for all fixed radial derivatives through $[0, X_{\text{an}}]$ follow from the explicit analytic axis and continuation construction, unchanged by both later supports. The average identity $M/X = \int_0^1 U(vX, \eta) dv$ supplies the smooth extension of V_0/X . Equation (PA.20) proves the requested canonical pressure. Differentiating it gives $\Pi_X = E^2/(2X) = F^2$, including at $X = 0$. The exact tangential divergence identities

in the core component, retaining radial viscosity, hold for every constructed profile; through X_a its two stress-free equations were solved by contraction. Equations (PA.21)–(PA.24) and the final restoration prove zero stress outside $[X_a, X_b]$. On its interior the admissible cone is now strict. Since $T_0 = 0$ would imply $P_c = v_s$, it is nonzero there.

The unit direction is smooth on the open annulus. At the inner edge (PA.24) gives $n = B_a/|B_a|$, with direction parallel to $(a, -b_s)$; at the outer edge (PA.23) gives

$$n = \frac{(b_\theta, y_b^6 b_z)}{(b_\theta^2 + y_b^{12} b_z^2)^{1/2}},$$

which is smooth through the edge and equals $(1, 0)$ there. Put $A_n = n_\theta + t_s n_z$ and $B_n = n_z - t_s n_\theta$, where $t_s = -b_s/a$. In the interior, the exact scalar products in (PA.1) give

$$A_n = (P_c - v_s)/|p_s - (a, -b_s)| > 0, \quad B_n = J_c/|p_s - (a, -b_s)|,$$

so $G_n = 2 - (v_s - 2)B_n^2/A_n^2 > 0$. At both edges $B_n = 0$, $A_n > 0$, and $G_n = 2$. Compactness therefore gives positive minima of A_n, G_n . The choice $\kappa = \min\{1, \min A_n, \min G_n\}$ is in $(0, 2)$ and gives exactly (4.26). Positivity of $F, a, v_s - 2$ has positive compact minima, using the inner $v_s > 2 + .2$ bound, outer $a > 2 + h$ bound, and the finite-loop gaps. Thus the lower bounds extend over the closed annulus.

Define, with zero extension outside the open annulus,

$$\zeta(X) = \exp(-t_1^2/y_a^2 - 4/y_b^2), \quad \delta = \min\{1, y_a, y_b\}.$$

Differentiating either exponential introduces only finitely many inverse powers of its logarithmic distance. The exponential dominates each such power, so the extension is smooth with every radial derivative zero at the edges. On a fixed small inner collar (PA.24) gives $|T_0| \geq c\zeta$, $|\partial^\alpha T_0| \leq C_\alpha \zeta \delta^{-m_\alpha}$. The same holds on a fixed outer collar by (PA.23); its y_b^{-3} factor is bounded below there, and all its derivatives cost finite inverse powers. On the intervening compact interior, $|T_0|$ and ζ have positive minima and all fixed derivatives are bounded. Taking the minimum of the three lower constants gives one $c > 0$ independent of α , and increasing the upper constants proves (4.27) on the whole open annulus.

The second repair restores (PA.19)–(PA.20). In the unchanged exterior the field is

$$U = V_0 = 0, \quad E = c_\infty X^{-A} \mathcal{H}(2d/X), \quad \mathcal{H}(Z) = \Gamma(1+h)^{-1} \int_0^\infty e^{-v} v^h (1+Zv)^{-h} dv.$$

The core and outer calculations prove this positive heat factor, all its one-sided derivatives at zero, and the exact radial heat equation for $q^{-A}E = c_\infty s^{-A} \mathcal{H}(2\tau/s)$. Above the original X_v , $U = M = J = 0$, so (PA.1) gives $V_0 = 0$ on the entire outer collar as well. The exact exterior map preserves $I_{\text{pos}} = I_3$ and $I_{\text{mean}} = I_4$, with $\sup I_3 < \inf I_4 < X_v$ and $U = 0$, $E = c_{\text{patch}}(1+\eta^2)^{-1} X^{-1/2-\lambda}$ there. All radii and widths, the circle loop, its bumps, and the single integer N were selected in (PA.3)–(PA.26) before physical q or any later derivative stage. Every resulting fixed derivative is a finite derivative of these fixed smooth functions. This proves the final independence assertion and all six parts. $\square$

The remaining sections supply the complete independently written component calculations used above. They are embedded in this file; compilation and reading require no external TeX inputs. The proof order within the main construction distinguishes the outer schedule from its later use in the backward-stress calculation. References to [NS] below identify equations in the fixed source, not additional unproved inputs.

31 Exact leading-field maps and radial primitives

31.1 1. Coordinates and their exact Jacobian

Take right-handed cylindrical coordinates

$$x = (r \cos \theta, r \sin \theta, z), \quad e_\theta = (-\sin \theta, \cos \theta, 0), \quad \theta \in \mathbb{R}/(2\pi\mathbb{Z}), \quad s = r^2/2.$$

Put

$$\begin{aligned} \tau &= 1 - t > 0, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad 0 < h < \frac{1}{2}, \\ z &= q^D \eta, \quad \tau = q(1 - \eta^2), \quad X = s/q, \\ d &= 1 - \eta^2, \quad L = 1 - 2h\eta^2. \end{aligned}$$

For fixed $\tau > 0, z \in \mathbb{R}$, eliminating η gives $g(q) = q - z^2 q^{2h} = \tau$. At $q_* = |z|^{1/D}$, $g(q_*) = 0$; for $z = 0$ take $q_* = 0$. For $q > q_*$, $z^2 q^{2h-1} < 1$, and

$$g'(q) = 1 - 2hz^2 q^{2h-1} \geq 1 - 2h > 0.$$

Moreover $g(q) \rightarrow \infty$. Thus exactly one $q > q_*$ exists, $|\eta| < 1$, and the implicit function theorem makes q, η smooth for $t < 1$. The values $\eta = \pm 1$ used for profiles are one-sided parameter endpoints; they are not additional points with $\tau > 0$.

Differentiation at fixed physical coordinates gives

$$q_t = -L^{-1}, \quad \eta_t = \frac{D\eta}{qL}, \quad X_t = \frac{X}{qL}, \quad q_z = \frac{2\eta q^{1-D}}{L}, \quad \eta_z = \frac{d}{q^D L}, \quad X_z = -\frac{2\eta X}{q^D L}.$$

The formula for η_z uses the exact equality $L - 2D\eta^2 = d$. Consequently, for every real b and smooth profile f ,

$$\begin{aligned} \partial_t(q^b f) &= q^{b-1} L^{-1}(-bf + D\eta f_\eta + X f_X), \\ \partial_z(q^b f) &= q^{b-D} L^{-1}(2b\eta f + df_\eta - 2\eta X f_X). \end{aligned}$$

No constant or power of q is discarded in these maps.

31.2 2. Cartesian axis extension and incompressibility

Let $C > 1$, $F = \phi/C$, $E = \sqrt{2X}F$, and prescribe

$$u_\theta = q^{-A}E, \quad u_z = q^{-A}U, \quad ru_r = V_0 = Xv_0, \quad p = q^{-2A}\Pi.$$

For F, U, v_0, Π smooth on $[0, R] \times [-1, 1]$, the Cartesian components are exactly

$$u_1 = \frac{v_0}{2q}x_1 - q^{-A-1/2}Fx_2, \quad u_2 = \frac{v_0}{2q}x_2 + q^{-A-1/2}Fx_1, \quad u_3 = q^{-A}U.$$

Since $X = (x_1^2 + x_2^2)/(2q)$, these formulas prove smoothness across $r = 0$ for each $t < 1$. Smoothness is imposed on F , not on E as a function of X . This explicitly proves the map from scalar profile regularity to Cartesian velocity regularity.

Incompressibility is $\partial_s(ru_r) + \partial_z u_z = 0$. Because $A + D = 1$, its exact scalar equation is

$$(V_0)_X = L^{-1}(2A\eta U - dU_\eta + 2\eta XU_X).$$

Integrate with $V_0(0, \eta) = 0$. Writing $M(X, \eta) = \int_0^X U(x, \eta) dx$, the result is

$$V_0 = \frac{2\eta XU - 2D\eta M - dM_\eta}{L}.$$

The extension of M/X at zero is smooth because $M/X = \int_0^1 U(vX, \eta) dv$; differentiating this integral proves every mixed derivative assertion. Thus the formula also proves smoothness of V_0/X .

The radial derivative of pressure and centrifugal term have the common factor $q^{-2A-1/2}$, with respective coefficients $\sqrt{2X}\Pi_X$ and $E^2/\sqrt{2X}$. Their exact equality is

$$\Pi_X = \frac{E^2}{2X} = F^2.$$

The other radial terms are still present in the full Navier-Stokes residual. Radial time differentiation, both material-transport contributions and radial viscosity have scale $q^{-3/2} = q^{2h}q^{-2A-1/2}$. Axial viscosity of the radial velocity has scale $q^{-1/2-2D} = q^{-3/2+2h} = q^{4h}q^{-2A-1/2}$. Axial viscosity has scale $q^{-A-2D} = q^{2h}q^{-A-1}$ in the tangential equations. These equalities identify every remaining type of term without setting any of them to zero.

31.3 3. Exact transport and stress integration

All residual identities here have viscosity one, as defined on source p. 6. Define

$$H = \sqrt{2X}E = 2XF, \quad l = X\partial_X \log H, \quad W = 1 - \frac{2D\eta M + dM_\eta}{X}, \quad H_c = D\eta + dU.$$

Applying the preceding derivative maps and $u_r = V_0/r$ gives, for arbitrary b, f ,

$$(\partial_t + u_r\partial_r + u_z\partial_z)(q^b f) = \frac{q^{b-1}}{L} \{WXf_X + H_c f_\eta - b(1 - 2\eta U)f\}.$$

Indeed the coefficient of Xf_X before substituting V_0 is $1 + LV_0/X - 2\eta U$, exactly W . Angular momentum is $ru_\theta = q^{-h}H$. Hence the angular material derivative equals $-q^{-h-1}HS_q/L$, where

$$S_q = -Wl - h(1 - 2\eta U) - H_c(\log E)_\eta.$$

For the axial velocity, add $\partial_z p$. Since $-2A - D = -A - 1$, the result is $-q^{-A-1}S_n/L$, with

$$S_n = -WXU_X - A(1 - 2\eta U)U - H_c U_\eta - d\Pi_\eta + 4A\eta\Pi + 2\eta X\Pi_X.$$

Let $D_X = X\partial_X$. Solve

$$D_X Q_s + (1 + l)Q_s = S_q, \quad D_X N_s + N_s = S_n.$$

Their integrating factors are XH and X . For $\phi > 0$ smooth at the axis, $H = 2X\phi/C$, and any nonzero integration constant produces an X^{-2} singularity in Q_s or an X^{-1} singularity in N_s . Therefore the unique regular choices are

$$Q_s = \frac{\int_0^X H S_q dx}{XH}, \quad N_s = \frac{1}{X} \int_0^X S_n dx.$$

In particular,

$$Q_s = \frac{1}{F(X, \eta)} \int_0^1 v F(vX, \eta) S_q(vX, \eta) dv, \quad N_s = \int_0^1 S_n(vX, \eta) dv,$$

which prove the scalar smooth extensions and give $Q_s(0) = S_q(0)/2$, $N_s(0) = S_n(0)$.

Define

$$p_s = \left(\frac{XQ_s}{L}, \frac{XN_s}{LE} \right), \quad a = 1 - 2D_X \log E = 2 - 2l, \\ b_s = \frac{2D_X U}{E}, \quad s_{\text{shear}} = (a, -b_s), \quad T_0 = F(p_s - s_{\text{shear}}).$$

Here s_{shear} is only a typographical disambiguation from the unchanged coordinate $s = r^2/2$; it is exactly the manuscript's vector s . The stress components remain ordered $(r\theta, rz)$.

For any profile G , at fixed t, z ,

$$(\partial_r + k/r)(q^{-A-1/2}G) = q^{-A-1} \left(\sqrt{2X} G_X + \frac{k}{\sqrt{2X}} G \right).$$

Substituting $G = FXQ_s/L = HQ_s/(2L)$, $k = 2$, produces $q^{-A-1}ES_q/L$. Substituting $G = FXN_s/(LE) = \sqrt{X/2}N_s/L$, $k = 1$, produces $q^{-A-1}S_n/L$. The remaining stress terms are exactly

$$\partial_r u_\theta - u_\theta/r = -q^{-A-1/2}Fa, \quad \partial_r u_z = q^{-A-1/2}Fb_s.$$

The differential identities

$$(\partial_r + 2/r)(\partial_r u_\theta - u_\theta/r) = \partial_{rr} u_\theta + r^{-1}\partial_r u_\theta - r^{-2}u_\theta, \\ (\partial_r + 1/r)\partial_r u_z = \partial_{rr} u_z + r^{-1}\partial_r u_z$$

now prove, including both signs,

$$R_\theta^{(0)} = -(\partial_r + 2/r)(q^{-A-1/2}T_{0,\theta}), \quad R_z^{(0)} = -(\partial_r + 1/r)(q^{-A-1/2}T_{0,z}),$$

where $R_j^{(0)} = \mathcal{R}(u, p) \cdot e_j + \partial_{zz} u_j$. Thus only axial viscosity has been removed from these exact identities.

Zero stress is equivalent to $Q_s = -2L\phi_X/\phi$ and $N_s = -2LU_X$. Applying the two radial operators and canceling the two opposite quadratic logarithmic-derivative terms gives exactly

$$-2L \frac{X\phi_{XX} + 2\phi_X}{\phi} = S_q, \quad -2L(XU_{XX} + U_X) = S_n.$$

Conversely, these equations and regularity imply the same Q_s, N_s , by uniqueness of the regular radial primitives, hence zero stress. The first displayed fraction was checked directly against PDF p. 27.

31.4 4. The five moments and their exact matching map

Retain all five quantities

$$\begin{aligned} M &= \int_0^X U \, dx, \quad I = \int_0^X H \, dx, \quad J = \int_0^X UH \, dx, \\ S &= \int_0^X (U^2 - E^2/2) \, dx, \quad C_p = \int_0^X \frac{E^2}{2x} \, dx, \quad \Pi = \Pi_{\text{ax}} + C_p. \end{aligned}$$

Since $(XW)_X = 1 - 2D\eta U - dU_\eta$, integration by parts in $-WXH_X$ gives $-XWH$ and the remaining integrand $(1-h)H - D\eta H_\eta - d(UH)_\eta + 2(h-D)\eta UH$. Hence

$$Q_s = -W + \frac{(1-h)I - D\eta I_\eta - dJ_\eta + 2(h-D)\eta J}{XH}.$$

For the axial primitive, integrate $-WXU_X$ in the same way. Its velocity terms become $D(U - \eta U_\eta) + 4h\eta U^2 - d(U^2)_\eta$. Pressure contributes

$$X(4A\eta\Pi - d\Pi_\eta) - 2h\eta \int_0^X E^2 \, dx + \frac{d}{2}\partial_\eta \int_0^X E^2 \, dx,$$

because $\int_0^X \Pi \, dx = X\Pi - \frac{1}{2} \int_0^X E^2 \, dx$. Thus

$$N_s = -WU + \frac{D(M - \eta M_\eta) + 4h\eta S - dS_\eta}{X} + 4A\eta\Pi - d\Pi_\eta.$$

The independent exact check differentiates both right-hand primitives, retains $\Pi_X = H^2/(4X^2)$, and obtains precisely HS_q and S_n .

If two pairs (U_i, E_i) agree for $X \geq X_h > 0$ and their five moment values at X_h agree as functions of η , all five moment differences are identically zero beyond X_h : their radial derivatives are zero and their initial values are zero. All parameter derivatives also agree. With the same Π_{ax} , the explicit formulas then give identical Π, V_0, Q_s, N_s, p_s . Agreement of profiles on an interval gives agreement of radial derivatives, hence of a, b_s, T_0 . This proves the complete morphism from matching data to the exterior field and stress; pressure matching alone would not suffice.

On a compact X -rectangle bounded away from zero and with $E_i \geq e_{\min} > 0$, the displayed formulas are compositions of addition, multiplication and reciprocal on denominators X, L, E, H bounded away from zero. For example $H_2^{-1} - H_1^{-1} = -(H_2 - H_1)/(H_1 H_2)$. The product rule through k parameter derivatives bounds the output difference in (Q_s, N_s, p_s) by the input differences in (U, E, M, I, J, S, C_p) through order $k + 1$. This uses no radial derivative of the difference. The one extra parameter derivative is necessary because W contains M_η and the primitive formulas contain moment derivatives.

Under the exact coordinate substitution $X = \rho x$, use

$$\begin{aligned} \widehat{U}(x, \eta) &= U(\rho x, \eta), \quad \widehat{E}(x, \eta) = E(\rho x, \eta), \quad H = \rho^{1/2} \widehat{H}, \\ (M, I, J, S, C_p) &= (\rho \widehat{M}, \rho^{3/2} \widehat{I}, \rho^{3/2} \widehat{J}, \rho \widehat{S}, \widehat{C}_p), \quad \widehat{Q}_s = Q_s, \quad \widehat{N}_s = N_s, \quad \widehat{p}_s = \rho^{-1} p_s. \end{aligned}$$

Direct substitution proves that precisely the same rational formulas hold on the fixed x -rectangle. This is an invertible change of variables retaining all factors, not a replacement of the original moment data.

31.5 5. Cone equivalence and finite-dimensional solve

For $a > 0$, define

$$t_s = -b_s/a, \quad v_s = a(1 + t_s^2), \quad P_c = p_{s,1} + t_s p_{s,2}, \quad J_c = p_{s,2} - t_s p_{s,1}.$$

The roots in v of $2(P_c - v)^2 - (v - 2)J_c^2$ are

$$v_\pm = P_c + J_c^2/4 \pm |J_c| \sqrt{(P_c - 2)/2 + J_c^2/16}.$$

For $P_c > 2$, evaluate the polynomial at 2 and P_c : it is positive at 2, nonpositive at P_c , and has its vertex at or above P_c . Consequently $2 < v_- \leq P_c$. This includes $J_c = 0$, where the two roots are P_c . Therefore for $v_s > 2$, the admissible inequality $v_s < v_-$ is exactly

$$P_c > v_s, \quad (v_s - 2)J_c^2 < 2(P_c - v_s)^2.$$

In particular $v_s \leq 2$, $P_c > 2$ always implies the strict relaxed inequality $v_s < v_-$.

For $p_{s,2} = wp_{s,1}$, put $c = 1 - b_s w/a$, $j = w + b_s/a$, $v = a + b_s^2/a$. Direct expansion gives

$$(v-2)j^2 - 2c^2 = (1 + b_s^2/a^2)((a-2)w^2 + 2b_s w + b_s^2/a - 2).$$

On the compact set specified in Lemma 4.5, choose actual minima $c_K > 0$, $\gamma_K = \min(2c^2 - (v-2)j^2) > 0$, and $B_K \geq \max\{1, v, cv\}$. For

$$p_{s,1} \geq P_K = \max\{(B_K + 2)/c_K, 8B_K/\gamma_K\},$$

one has $P_c > \max\{2, v\}$ and

$$\frac{2(P_c - v)^2 - (v-2)J_c^2}{p_{s,1}^2} \geq \gamma_K - 4B_K/p_{s,1} \geq \gamma_K/2.$$

These are exactly the source constants, with the endpoint $v = 2$ covered by the preceding root calculation.

Stress coordinates satisfy, by direct scalar products,

$$T_{0,\theta} + t_s T_{0,z} = F(P_c - v_s), \quad T_{0,z} - t_s T_{0,\theta} = FJ_c.$$

Thus a positive margin for the two stress inequalities proves nonvanishing and controls direction independently of magnitude.

For completeness, the moment matrix invertibility used later is proved as follows. Distinct real exponents α_i form a Chebyshev family on $(0, \infty)$. For m exponents, divide a nonzero combination by its smallest power and differentiate. If the original combination had m distinct positive zeros, Rolle's theorem would give $m-1$ zeros of a combination of $m-1$ distinct powers, contradicting the induction starting with one nonzero power. Therefore the determinant $\det[x_j^{\alpha_i}]$ never vanishes for ordered $x_1 < \dots < x_m$, and its sign is constant on that connected set. Multilinearity expresses the bump moment determinant as the integral of this determinant against $\prod_j \beta_j(x_j) \geq 0$. Ordered disjoint nontrivial supports make the integral nonzero. Compact smooth parameter families then have bounded inverse matrices and bounded fixed derivatives by the adjugate formula.

For the exact quadratic moment equation $Bc + Q(c, c) = d$, multiplication by B^{-1} has norm ν_k , and Q has bilinear norm q_k on C^k . On the ball of radius $r_k = 2\nu_k \|d\|_{C^k}$, the map $c \mapsto B^{-1}(d - Q(c, c))$ has image norm at most $r_k/2 + \nu_k q_k r_k^2$ and Lipschitz constant at most $2\nu_k q_k r_k$. The source condition $8\nu_k^2 q_k \|d\|_{C^k} \leq 1$ gives invariance and Lipschitz constant at most $1/2$. Iteration from zero in C^0 and C^k gives the same fixed point. Its pointwise derivative matrix is $B + Q(c, \cdot) + Q(\cdot, c)$; the corresponding B^{-1} -perturbation has norm at most $1/2$, proving invertibility. The finite-dimensional implicit function theorem therefore gives all higher fixed parameter derivatives without imposing infinitely many smallness conditions.

31.6 6. Direct heat-exterior verification

The function named $\mathcal{H}$ here is exactly the source's $H(Z)$ in (4.29), distinguished typographically from angular momentum $H(X, \eta)$:

$$\mathcal{H}(Z) = \frac{1}{\Gamma(1+h)} \int_0^\infty e^{-v} v^h (1+Zv)^{-h} dv, \quad Z \geq 0.$$

For each k , its k -th derivative is the integral with factor $(-1)^k (h)_k v^k (1+Zv)^{-h-k}$. The integrable majorant is $(h)_k e^{-v} v^{h+k}$, proving smooth one-sided derivatives at $Z = 0$ and positivity. Differentiating $e^{-v} v^{h+1} (1+Zv)^{-h-1}$ with respect to v shows

$$Z^2 \mathcal{H}'' + ((2+2h)Z + 1) \mathcal{H}' + h(1+h) \mathcal{H} = 0.$$

In fact the numerator of the ODE integrand after taking the common denominator $(1+Zv)^{h+2}$ is $h(h+1-v-Zv^2)$, exactly h times that derivative numerator; both boundary values vanish. This is a full integration-by-parts proof.

For $K = c_\infty s^{-A} \mathcal{H}(2\tau/s)$, let $Z = 2\tau/s$. Then

$$\partial_t K = -2c_\infty s^{-A-1} \mathcal{H}', \quad \partial_s K = c_\infty s^{-A-1} (-A\mathcal{H} - Z\mathcal{H}'),$$

$$(\partial_{rr} + r^{-1} \partial_r - r^{-2}) K = c_\infty s^{-A-1} \{(2A^2 - \frac{1}{2})\mathcal{H} + (4A+2)Z\mathcal{H}' + 2Z^2 \mathcal{H}''\}.$$

Using $2A^2 - 1/2 = 2h(1+h)$ and the preceding ODE proves the exact radial heat equation. Also

$$A\mathcal{H} + Z\mathcal{H}' = \frac{1}{\Gamma(1+h)} \int_0^\infty e^{-v} v^h \frac{A + Zv/2}{(1+Zv)^{h+1}} dv > 0,$$

so $K_r = rK_s < 0$ for $r > 0$. Finally $q^{-A} c_\infty X^{-A} \mathcal{H}(2d/X) = c_\infty s^{-A} \mathcal{H}(2\tau/s)$ exactly; the exterior has no residual hidden by a coordinate substitution.

32 Power moment maps and exact corrections

32.1 An invertible moment map and its quadratic inverse

For distinct real exponents $\alpha_1, \dots, \alpha_m$, a nonzero sum $\sum c_i x^{\alpha_i}$ has at most $m - 1$ positive zeros. To prove this by induction, order the exponents and divide by the smallest power. If the result had m zeros, its derivative would have $m - 1$ zeros by Rolle's theorem. That derivative is either zero, in which case the original result is constant, or is a nonzero combination of $m - 1$ distinct powers, which has at most $m - 2$ zeros by induction. Both alternatives exclude m zeros.

For ordered disjoint intervals $I_1 < \dots < I_m$ and nonnegative nonzero $\beta_j \in C_c^\infty(I_j)$, the determinant of $(x_j^{\alpha_i})$ is nonzero at every $x_1 < \dots < x_m$, and its continuous sign is fixed on that connected set. Multilinearity and Fubini give

$$\det \left(\int x^{\alpha_i} \beta_j(x) dx \right) = \int_{I_1 \times \dots \times I_m} \det(x_j^{\alpha_i}) \prod_j \beta_j(x_j) dx_1 \dots dx_m \neq 0.$$

Smooth families on compact parameter sets have bounded inverse derivatives, by the determinant formula and $\partial B^{-1} = -B^{-1}(\partial B)B^{-1}$. Under $x = e^y$, this is also the exponential-weight statement in [NS, Lemma A.1]. For two identical translated logarithmic bumps at centers separated by $\Delta > 0$, after dividing row i by $e^{s_i c_1}$ the determinant is $b(s_1)b(s_2)(e^{s_2 \Delta} - e^{s_1 \Delta})$, where $b(s) = \int e^{st} \beta(t) dt > 0$. On a fixed compact slope interval, the mean value theorem gives an inverse bound $O(|s_1 - s_2|^{-1})$. In particular the near-coincident weights in this construction cost exactly an allowed $O(\lambda^{-1})$ upper bound, not a constant independent of λ .

For completeness the finite nonlinear solve can be made fully quantitative. Let $F_\eta(c) = B(\eta)c + Q_\eta(c, c)$, with Q bilinear. In a chosen $C^k([-1, 1])$ norm let multiplication by B^{-1} have norm β_k and the bilinear map have norm κ_k . If $8\beta_k^2 \kappa_k \|d\|_{C^k} \leq 1$, the map $\Phi(c) = B^{-1}(d - Q(c, c))$ preserves the ball of radius $r = 2\beta_k \|d\|_{C^k}$, since

$$\|\Phi(c)\| \leq \beta_k \|d\| + \beta_k \kappa_k r^2 \leq r/2 + r/4 < r.$$

Its Lipschitz constant is at most $2\beta_k \kappa_k r \leq 1/2$. Iteration from zero therefore converges to the unique solution in this ball. At $k = 0$, $\|B^{-1} D_c Q(c, c)\| \leq 1/2$, so the Jacobian is invertible. Pointwise implicit differentiation proves smoothness. Solving in any fixed C^k ball gives the same branch because the iterates are identical. This proves [NS, Lemma A.2] and also specifies the branch selected in every use.

The five original leading moments are

$$\begin{aligned} M &= \int_0^X U dx, & I &= \int_0^X \sqrt{2x} E dx, & J &= \int_0^X U \sqrt{2x} E dx, \\ S &= \int_0^X (U^2 - E^2/2) dx, & C_p &= \int_0^X E^2/(2x) dx, & \Pi &= \Pi_{\text{ax}} + C_p. \end{aligned}$$

The substitution $X = \rho x$ is an exact map: the factors of these five moments are $(\rho, \rho^{3/2}, \rho^{3/2}, \rho, 1)$, with $H = \rho^{1/2} \sqrt{2x} E$. No factor is discarded. At $U = u_c(\eta)$ and $E = e_* f_*(\eta) x^\alpha$, take the differential of the rows $J - u_c I$ and $S - 2u_c M$, where u_c in the row operation stays at its original value. These differentials are $\int \sqrt{2x} E \delta U dx$ and $-\int E \delta E dx$. Together with M, I, C_p , they give the two blocks of weights

$$(1, x^{\alpha+1/2}), \quad (x^{1/2}, x^\alpha, x^{\alpha-1}),$$

with the factors $e_* f_*$ and $\sqrt{2}$ retained in their row multipliers. The exponents are distinct precisely when $\alpha \notin \{-1/2, 1/2, 3/2\}$. The functionals are degree at most two, so their unlisted remainder is exactly a quadratic bilinear map. This derives [NS, Corollary A.3].

33 The complete outer schedule and its exact moment closure

33.1 Definitions of all stages and the adjustable coefficients

Use the fixed step

$$\sigma(y) = \frac{e^{-1/y^2}}{e^{-1/y^2} + e^{-1/(1-y)^2}} \quad (0 < y < 1), \quad \sigma = 0 \quad (y \leq 0), \quad \sigma = 1 \quad (y \geq 1).$$

The schedule chooses, in order, M_d large, $T_d = e^{M_d} + 10$, $P_* > e^{T_d}$, $\lambda > 0$ small depending on these choices, and $h > 0$ sufficiently small relative to both λ and e^{-T_d} . Only after these choices is X_R made large. All estimates denoted C_{pre} below depend on the first three choices and the fixed steps, not on the later X_R or amplitude factor C . Smallness requirements use finitely many derivatives, and parameters can be chosen to meet all of them.

Write $x = X/X_R$ and, initially, $y = \log x$. The reference fields for $y \leq 0$ are $U = 4\eta$, $E = P_* f e^{y/10}$, $f = (1 + \eta^2)^{-1}$, $J_0 = \log(1 + \eta^2)$. In each next stage y is reset to zero at that stage's left endpoint, and $l = X \partial_X \log(\sqrt{2X} E)$. The fields are fully specified by the following data, with values at endpoints matched by integration of $\partial_y \log E = l - 1/2$:

1. On a unit interval, $U = 4\eta$, $l = (3/5)(1 - \sigma(y))$. On the following interval $0 \leq y \leq T_d$, set $l = 0$, $U = k(y)\eta$, $k = 4[1 - \sigma(\log(1+y)/M_d)]$. Thus k vanishes for its last eleven units, and $|k'| \leq e_d/(1+y)$ with $e_d = 4\|\sigma'\|_\infty/M_d$. The swirl on this interval is $E = P_1 f e^{-y/2}$, where $P_1 \asymp P_*$.
2. On a unit interval set $U = 0$, $l = -\lambda\sigma(y)$, and then hold $l = -\lambda$ for $T_w = 60 \log(1/\lambda)$. Reserve $(T_w - 25, T_w - 20)$, $(T_w - 20, T_w - 15)$, $(T_w - 14, T_w - 9)$, and $(T_w - 8, T_w - 3)$ for the four later corrections.
3. At $X = X_p$ start a pulse with $0 \leq y \leq 13/\lambda$, $E = e_b f e^{-(1/2+\lambda)y}$, $U = E R_b$, $\xi_b = \lambda y$. Define $\varphi_b(\xi) = \int_0^\xi \sigma(v/.02) dv$ for $\xi \geq 0$, extended by zero to negative arguments, and $R_0(\xi) = \varphi_b(\xi)[1 - \sigma(\xi - 10)]$. Set $R_b = \text{Amp}(\eta) R_0(\lambda y) + c_1 \beta_1(y) + c_2 \beta_2(y)$, where the two identical width .3 bumps have centers $13/\lambda - 3$ and $13/\lambda - 1$. Choose c_1, c_2 to make $M = J = 0$ at the pulse end.
4. From the pulse endpoint $E = e_{\text{end}} f$, set $U = 0$ permanently and

$$\log E = \log e_{\text{end}} - (1/2 + \lambda)y - \vartheta_f(y) J_0 - (1 - \vartheta_f(y)) \log 2, \quad \vartheta_f = 1 - \sigma(y/T_f).$$

A fixed sufficiently large T_f ensures $-\lambda - .1 \leq l \leq -\lambda$. Next retain the now η independent exponential for $30 \log(1/\lambda)$. Two relative swirl bumps of width .3, three and one units before its end, set $I = XH/(1 - \lambda)$ there and make their total integral $\int (E^2 - E_{\text{unedited}}^2) dy$ zero.

5. Starting at X_{rel} , vary l from $-\lambda$ to -1 in a unit interval, hold $l = -1$ for $4 \log(1/h)$, and vary from -1 to $-h$ in a unit interval, all using σ . The terminal multiplier is $f_o(y) = 1 - \rho_o \psi_o(y)$ on $0 \leq y \leq 3$, $\psi_o = 1 - \sigma((y - 1)/2)$, $\rho_o = c_o h > 0$ with c_o fixed so that $0 \leq f'_o/f_o < h/4$. Define

$$Q_p = \int_0^3 \exp\left(\int_0^v (1 + l(s)) ds\right) \frac{f'_o(v)}{f_o(v)} dv, \quad l = -h + f'_o/f_o$$

on this terminal interval. Hold $l = -h$ before it until Q_s reaches Q_p . Then apply this terminal interval, and retain $l = -h$ thereafter. On and after the terminal interval the field is exactly $E_{\text{pow}} f_o$, $E_{\text{pow}} = c_\infty X^{-A}$, with c_∞ independent of η .

All stage endpoints divided by X_R are constants independent of η . The following calculations prove that the apparently adjustable stages have the stated finite solutions, and determine the pulse amplitude.

33.2 Pulse corrections, angular correction, and amplitude root

During the long interval T_w , the quantities XE and XHE grow at rates $1/2 - \lambda$ and $1/2 - 2\lambda$, respectively, whereas XE^2 decays at rate 2λ . The preceding M, J are constant once $U = 0$. Since $e^{-T_w/2} = \lambda^{30}$ and $\lambda T_w = 60 \log(1/\lambda) \rightarrow 0$, it follows, with one η derivative as well, that

$$e_b \leq C_{\text{pre}} \lambda^{30}, \quad \left\| \frac{M(X_p)}{X_p e_b f} \right\|_{C^1} + \left\| \frac{J(X_p)}{X_p H(X_p) e_b f} \right\|_{C^1} \leq C_{\text{pre}} \lambda^{29},$$

$$\left\| \frac{S(X_p)}{X_p e_b^2 f^2} \right\|_{C^1} \leq C_{\text{pre}} (1 + \log(1/\lambda)).$$

For the last inequality, express S as an integral of $X(U^2 - E^2/2)$ in logarithmic coordinates. The finite preceding stages contribute a fixed constant in this scale and the long $U = 0$ stage contributes its length times a bounded factor $e^{-2\lambda y}$. These statements prove (A.14) with all scale factors retained.

The exact two equations for the pulse correction are

$$\begin{pmatrix} \int e^{s_1 y} \beta_1 dy & \int e^{s_1 y} \beta_2 dy \\ \int e^{s_2 y} \beta_1 dy & \int e^{s_2 y} \beta_2 dy \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = - \begin{pmatrix} M(X_p)/(X_p e_b f) + \text{Amp} \int e^{s_1 y} R_0(\lambda y) dy \\ J(X_p)/(X_p H(X_p) e_b f) + \text{Amp} \int e^{s_2 y} R_0(\lambda y) dy \end{pmatrix},$$

where $s_1 = 1/2 - \lambda$, $s_2 = 1/2 - 2\lambda$. Divide each row by its exponential value at the first bump center. The main pulse is zero after $11/\lambda$, leaving a gap at least $2/\lambda - 3$; both slopes exceed .4 for small λ . Its normalized moments are at most a polynomial in λ^{-1} times $(1 + |\text{Amp}|)e^{-.4(2/\lambda-3)}$. The prior moments have an even larger exponential gap. The matrix inverse costs $O(\lambda^{-1})$. Reducing the positive exponential constant absorbs every such fixed polynomial. Hence the exact affine solution $c = c^{(0)} + \text{Amp} c^{(1)}$ satisfies

$$\|c^{(0)}\|_{C^1} + \|c^{(1)}\|_{C^1} \leq C_{\text{pre}} e^{-c/\lambda}.$$

The same argument applies to any fixed derivative, with additional powers of λ^{-1} for ξ_b derivatives. This proves exact $M = J = 0$ and (A.15).

For angular momentum put $r_I = I/(XH)$. Since $I_X = H$, $r'_I + (1+l)r_I = 1$. During the uniform interval following interpolation, $r_I - (1-\lambda)^{-1}$ decays at rate $1-\lambda$. At the first swirl bump center its discrepancy is $O_{C^1}(C_{\text{pre}}\lambda^{28})$; the exponent 28 is a valid relaxed bound for $e^{-(1-\lambda)(30\log(1/\lambda)-3)}$. For the relative edit $E \mapsto E(1 + c_1\beta_1 + c_2\beta_2)$, the linear angular row has logarithmic weight $e^{(1-\lambda)y}$ and the linear pressure row has weight $2e^{(-1-2\lambda)y}$. Their slope separation tends to 2, so their normalized inverse is bounded. The pressure row alone has its exact quadratic remainder. The contraction proved above yields coefficients $O_{C^1}(C_{\text{pre}}\lambda^{28})$, with both exact targets met. All fixed radial derivatives of these bumps are of the same size, so they preserve $E > 0$ and $l \leq -\lambda/2$.

After the correction, $I = XH/(1-\lambda)$ and $U = M = J = 0$ at the start of the exterior transition. The exact radial identities of (4.16) give

$$Q_s = -1 + (1-h)I/(XH) = (\lambda-h)/(1-\lambda).$$

Subsequently $Q'_s + (1+l)Q_s = -l-h$. During the first transition the nonnegative source and bounded integrating factor retain a positive $c\lambda$ lower bound. The $l = -1$ hold gives $Q'_s = 1-h$, so its end value is comparable to $\log(1/h)$. The next unit transition changes this by bounded factors. On the other hand $Q_p \asymp \rho_o \asymp h$: f_o is bounded above and below, the interval has fixed length, and $\int f'_o = \rho_o$. The intervening $l = -h$ solution is $Q_{\text{in}}e^{-(1-h)y}$; the exact required length is $(1-h)^{-1}\log(Q_{\text{in}}/Q_p) > 0$ for sufficiently small h . On the terminal interval the resulting solution is

$$Q_s(y) = \int_y^3 \exp\left(\int_y^v (1+l(s)) ds\right) \frac{f'_o(v)}{f_o(v)} dv.$$

Differentiating this expression proves the equation and its zero right endpoint. Thus $Q_s = 0$ afterward and $I = XH_{\text{pow}}/(1-h)$. Since $\int_0^X H_{\text{pow}} dx = XH_{\text{pow}}/(1-h)$ for $h < 1$, this is exactly $\int_0^\infty (H - H_{\text{pow}}) dX = 0$. The subtraction is integrable at zero, and is identically zero at large X at this pre-heat stage.

For the other moment, note the exact identities on the $l = -1$ hold:

$$\frac{(XE^2)_{\text{end}}}{(XE^2)_{\text{start}}} = h^8, \quad \frac{E_{\text{end}}}{E_{\text{start}}} = h^6.$$

They follow by integrating $(\log(XE^2))' = 2l = -2$ and $(\log E)' = -3/2$ over $4\log(1/h)$. The first transition and hold have total $\int XE^2 dy \leq CX_{\text{rel}}e_{\text{rel}}^2$. After the hold $l \leq -3h/4$, so the remaining integral is at most $CX_{\text{rel}}e_{\text{rel}}^2 h^8 \int_0^\infty e^{-3hy/2} dy \leq CX_{\text{rel}}e_{\text{rel}}^2 h^7$. This proves the h uniform bound (A.18). Interpolation and its next hold cost at most $C(1+\log(1/\lambda))X_{\text{end}}e_{\text{end}}^2$; their first parameter derivatives satisfy the same bounds because f^ϑ and its derivatives are bounded and the exterior is parameter independent.

There is also an exact algebraic specification of the amplitude root. Substitute the affine c above and write $R_b = R^{(0)} + \text{Amp} R^{(1)}$. In the full $S(\infty)$, every term outside $\int_{\text{pulse}} E^2 R_b^2 dX$ is independent of Amp . Consequently this moment is an explicitly defined quadratic polynomial $a(\eta)\text{Amp}^2 + b(\eta)\text{Amp} + c(\eta)$, where, without dropping any end bumps,

$$\begin{aligned} a &= \int_{\text{pulse}} E^2 (R^{(1)})^2 dX > 0, & b &= 2 \int_{\text{pulse}} E^2 R^{(0)} R^{(1)} dX, \\ c &= S_{\text{no axial pulse}}(\infty) + \int_{\text{pulse}} E^2 (R^{(0)})^2 dX. \end{aligned}$$

Changing variables $\xi_b = \lambda y$ in this polynomial gives

$$\frac{\lambda S(\infty)}{X_p e_b^2 f^2} = \text{Amp}^2 K_b - \frac{1 - e^{-26}}{4} + \mathcal{E}(\text{Amp}, \eta), \quad K_b = \int_0^{13} e^{-2\xi} R_0(\xi)^2 d\xi,$$

$$\|\mathcal{E}\|_{C^1([.9,1.2] \times [-1,1])} \leq C_{\text{pre}} \lambda (1 + \log(1/\lambda)).$$

Indeed the prior S and the post-pulse negative $E^2/2$ integral have precisely the bounds just proved after multiplication by λ , whereas the affine end bumps are exponentially small; all cross terms are retained in $\mathcal{E}$. On $.02 \leq \xi \leq 9$, $R_0 \geq \xi - .02$, and globally $0 \leq R_0 \leq \xi$. Therefore

$$\frac{e^{-.04}}{4} [1 - e^{-17.96}(1 + 17.96 + 2(8.98)^2)] \leq K_b \leq \frac{1}{4},$$

whose lower bound exceeds .20. At $\text{Amp} = .9$ the principal value is below $-.047$; at 1.2 it exceeds .038; its derivative is at least .36. For sufficiently small λ the exact polynomial has a unique root on this interval. It is the positive branch $(-b + \sqrt{b^2 - 4ac})/(2a)$, with its nonzero derivative already proved by the bracket estimate. Implicit differentiation gives $|\partial_\eta \text{Amp}| \leq C_{\text{pre}} \lambda (1 + \log(1/\lambda))$. This constructs the smooth amplitude and proves both required moments, including all terms in (A.19)–(A.20).

34 Pressure datum and the cone margins on every outer stage

34.1 Exact pressure and its parameter analyticity

Omit only the swirl bumps whose total $\int E^2 dy$ was just proved to be zero. The remaining scheduled field has the form $E = c(y)f(\eta)^{\vartheta(y)}$, $0 \leq \vartheta \leq 1$, with c, ϑ and all interval lengths independent of η and X_R . On one simply connected complex neighborhood of $[-1, 1]$ avoiding $\pm i$, choose the logarithm of f . The holomorphic functions $\exp(\vartheta \log f)$ are uniformly bounded for $0 \leq \vartheta \leq 1$. The squared integrand is bounded by $Ce^{y/5}$ as $y \rightarrow -\infty$ and $Ce^{-(1+2h)y}$ as $y \rightarrow \infty$. Uniform convergence of the integral on compact complex subsets therefore defines a holomorphic function

$$\Pi_{\text{ax}}(\eta) = -\frac{1}{2} \int_{-\infty}^{\infty} E_{\text{sched}}(y, \eta)^2 dy.$$

It is even, independent of X_R , and its derivative is an integral of $\vartheta J'_0 c^2 f^{2\vartheta}$. This has the sign of η and is strict for $\eta \neq 0$ because the reference interval has $\vartheta = 1$. The reference interval contributes exactly $-\frac{1}{2} P_*^2 f^2 \int_{-\infty}^0 e^{y/5} dy = -\frac{5}{2} P_*^2 f^2$. Restoring the pressure-neutral bumps leaves the total unchanged. Hence the actual pressure is exactly

$$\Pi(X, \eta) = -\frac{1}{2} \int_{\log(X/X_R)}^{\infty} E(v, \eta)^2 dv.$$

This proves all assertions of [NS, Lemma A.5]. It also proves the precise preservation map: an edit of zero total $\int \delta(E^2)/(2X) dX$ keeps forward pressure unchanged before and after its support; it need not keep pressure unchanged inside that support.

34.2 The exact quantities used for the cone test

Retain

$$\begin{aligned} W &= 1 - 2D\eta M/X - dM_\eta/X, \quad H_c = D\eta + dU, \quad a = 2 - 2l, \quad b_s = 2XU_X/E, \\ Q_s &= -W + \frac{(1-h)I - D\eta I_\eta - dJ_\eta + 2(h-D)\eta J}{XH}, \\ N_s &= -WU + \frac{D(M - \eta M_\eta) + 4h\eta S - dS_\eta}{X} + 4A\eta\Pi - d\Pi_\eta. \end{aligned}$$

These formulas follow by integrating the source equations

$$Q'_s + (1+l)Q_s = -Wl - h(1-2\eta U) - H_c(\log E)_\eta,$$

$$N'_s + N_s = -WU' - A(1-2\eta U)U - H_c U_\eta - d\Pi_\eta + 4A\eta\Pi + 2\eta X\Pi_X.$$

For example $(XW)_X = 1 - 2D\eta U - dU_\eta$; integration by parts of $-WXH_X$ produces $-XWH$ and the displayed I, J terms. In the pressure terms use $\int_0^X \Pi, dx = X\Pi - \int_0^X E^2/2, dx$. These operations produce the exact S and M terms above, and establish the map from the five original moments to stress coefficients in [NS, (4.16)].

Set $w = N_s/(EQ_s)$ where $Q_s > 0$, and $p_{s,1} = XQ_s/L$, $p_{s,2} = wp_{s,1}$. To explain precisely the sufficient test used below, put

$$c = 1 - b_s w/a, \quad j = w + b_s/a, \quad v_s = a + b_s^2/a, \quad P_c = p_{s,1}c, \quad J_c = p_{s,1}j.$$

Direct expansion gives

$$2c^2 - (v_s - 2)j^2 = \left(1 + \frac{b_s^2}{a^2}\right) \left[2 - 2b_s w - \frac{b_s^2}{a} - (a - 2)w^2\right].$$

Thus on a compact set with $a > 0$, $a - b_s w > 0$, and the bracket positive, there are constants $c_* > 0$, $G_* > 0$, and $B < \infty$ bounding these expressions away from zero and bounding v_s, cv_s . For $p_{s,1} \geq \max((B + 2)/c_*, 8B/G_*)$, $P_c > \max(2, v_s)$ and

$$\frac{2(P_c - v_s)^2 - (v_s - 2)J_c^2}{p_{s,1}^2} \geq G_* - 4B/p_{s,1} \geq G_*/2.$$

For $v_s > 2$ these are exactly the admissible cone inequalities; for $v_s \leq 2$ they imply the relaxed cone. Equivalence follows by solving the quadratic $2(P_c - v)^2 - (v - 2)J_c^2 = 0$, whose smaller root is the function in (4.21), between 2 and P_c . This verifies the finite threshold used in Lemma 4.5.

34.3 Reference and axial-decrease stages

On the ideal reference interval $U = 4\eta$, $W = 1 - 4L$, $l = 3/5$, so

$$Q_s = \frac{(3/5)(4L - 1) - h(1 - 8\eta^2) + (D + 4d)\eta J'_0}{8/5} > c > 0$$

for the chosen small h . Here $a = 4/5$, $b_s = 0$. Its N_s geometric source is $O(|\eta|)$ and its pressure source is $O(P_*^2|\eta|)$; these estimates follow by substituting the constant U and the bounded backward pressure integral above. The vanishing at zero follows from even pressure and odd U , and one parameter derivative removes the $|\eta|$ factor. The $N'_s + N_s$ convolution has finite bounds on each prescribed reference interval $x \geq x_- > 0$.

During the first unit transition, $b_s = 0$, $.8 \leq a \leq 2$, and the angular source is at least $-h$ since $W < 0$, $l \geq 0$, and $H_c J'_0 \geq 0$. The initial positive lower bound and variation of constants retain $Q_s > c$ for small h . With $b_s = 0$ and $a \leq 2$, the second sufficient cone bracket is automatically at least 2.

During axial decrease, $l = 0$, $a = 2$, $b_s = 2k'\eta/E$. The angular source is exactly $-h(1 - 2k\eta^2) + (D + dk)\eta J'_0$. Since $\eta J'_0 = 2\eta^2/(1 + \eta^2) \asymp \eta^2$, convolution with e^{-y} and $h \ll e^{-T_d}$ gives $Q_s \geq c(\eta^2 + e^{-y})$ for $0 \leq y \leq T_d$. The geometric axial source is $O(|\eta|)$. The pressure source is $O(E^2|\eta|)$ because subsequent scheduled swirl decays at least $e^{-\Delta y/2}$ and $|\partial_\eta \log E| \leq |J'_0|$; the later pressure-neutral angular bumps may be omitted from the backward integral. Solving the N_s equation yields

$$|N_s|/E^2 \leq C|\eta|(e^{T_d}/P_*^2 + 1 + y) \leq C|\eta|(1 + y).$$

Consequently $|b_s w| \leq C|k'|\eta^2(1 + y)/(\eta^2 + e^{-y}) \leq Ce_d$ and $b_s^2 \leq Ce_d^2 e^{T_d}/P_*^2$. Choose M_d so Ce_d is small, then $P_* > e^{T_d}$ sufficiently large. Both required strict margins follow, uniformly in η and this whole stage.

34.4 Intermediate power law and the complete pulse

Write $M/X = \bar{k}\eta$ during axial decrease. Then $\bar{k}' = k - \bar{k}$. Because $0 \leq k \leq 4$ and its final eleven units are zero, $L\bar{k} < 1/2$ at the next transition. When $U = 0$ this gives $W \geq 1/2$. Write $c_\eta = D\eta J'_0 \asymp \eta^2$. The angular source is $(-l)W - h + c_\eta$. During the unit transition $0 \geq l \geq -\lambda$, the prior positive lower bound persists; w has a finite bound depending only on earlier choices. Thus $\sqrt{\lambda}|w|$ is small. The cone is admissible as soon as $l < 0$, and relaxed at its initial endpoint.

On the constant-slope interval, $\bar{k}(y) = \bar{k}(0)e^{-y}$ and

$$Q'_s + (1 - \lambda)Q_s = \lambda - h + c_\eta - \lambda L\bar{k}(0)e^{-y}.$$

The right side is at least $\lambda/4 + c_\eta$ for small h/λ . Its explicit convolution therefore gives

$$Q_s \geq c_{\text{pre}}(\eta^2 + \lambda + e^{-(1-\lambda)y}), \quad Q_s - \frac{\lambda - h + c_\eta}{1 - \lambda} = O_{C^1}(C_{\text{pre}}(1 + y)e^{-(1-\lambda)y}).$$

There is no geometric N_s source when $U = 0$; the pressure source is $O(E^2|\eta|)$. Its convolution gives $|N_s/E| \leq C_{\text{pre}}|\eta|(1 + y)e^{-(1/2-\lambda)y}$. Using $|\eta|/(\eta^2 + e^{-(1-\lambda)y}) \leq \frac{1}{2}e^{(1-\lambda)y/2}$ proves

$$\sqrt{\lambda}|w| \leq C_{\text{pre}}\sqrt{\lambda}(1 + y)e^{\lambda y/2} = o(1) \quad (0 \leq y \leq 60 \log(1/\lambda)).$$

Since $a = 2 + 2\lambda$, $b_s = 0$, this supplies the strict cone bracket.

During the pulse set $m = (M/X)/E$ and $\beta = 1/2 - \lambda$. The exact identity $m' + \beta m = R_b$ follows by differentiating M/X and E . With the pulse extended flatly to the left, write its convolution as $\int_0^\infty e^{-\beta v} R_b(\lambda(y-v)) dv$ plus $m(0)e^{-\beta y}$. Taylor's formula under this integral gives

$$m = R_b/\beta - \lambda \partial_{\xi_b} R_b/\beta^2 + O(\lambda^2) + m(0)e^{-\beta y},$$

because the integral of $v^2 e^{-\beta v}$ is finite uniformly for $\beta > .4$. The end bumps have exponentially small fixed ξ_b derivatives. The same estimate holds after one η derivative.

Since $E \leq C_{\text{pre}} \lambda^{30}$, all terms from $W = 1 - 2D\eta Em - d(Em)_\eta$ and from dU in the angular source have size $O(C_{\text{pre}} E)$, including one parameter derivative. The initial constant-slope estimate and exponential convolution give

$$Q_s = \frac{\lambda - h + c_\eta}{1 - \lambda} + O(C_{\text{pre}} \lambda^{28}).$$

The backward pressure and its first derivative are $O(E^2)$. For S , integration over $0 \leq \lambda y \leq 13$ bounds $S/(XE^2)$ by $C_{\text{pre}} e^{26}/\lambda$ in C^1 . The numerator accumulates at most a fixed integral in ξ_b divided by λ and the denominator loses at most e^{-26} ; this explains the constant e^{26}/λ , not $e^{26/\lambda}$. Substituting $M/X = Em$ in the exact moment formula for N_s gives

$$N_s/E = -R_b + (D + c_\eta)m - D\eta m_\eta + O(C_{\text{pre}} \lambda^{27}).$$

For example $D(M - \eta M_\eta)/(XE) = D(m - \eta m_\eta + \eta J'_0 m)$, giving exactly $(D + c_\eta)m$. The S and pressure remainders are at most CEe^{26}/λ in this quotient, which is smaller than the stated bound for sufficiently small λ . The difference $W - 1$ similarly costs $O(E)$.

The amplitude derivative and the initial-moment bounds show $|m_\eta| \leq C_{\text{pre}} \lambda(1 + \log(1/\lambda))$. Since $|\eta|/(\lambda + c_\eta) \leq C/\sqrt{\lambda}$, the quotient of ηm_η by Q_s is $o(1)$. The preceding formulas therefore imply, uniformly throughout the pulse,

$$w = 2R_b - C_d \partial_{\xi_b} R_b + o(1), \quad C_d = \frac{\lambda(D + c_\eta)(1 - \lambda)}{\beta^2(\lambda - h + c_\eta)} \leq 2 + o(1).$$

The coefficient of R_b before this last replacement is exactly $(1 - \lambda)/\beta$, since $D + c_\eta - \beta = \lambda - h + c_\eta$. The quotient defining C_d is maximal at $c_\eta = 0$, because its derivative in c_η has sign $\lambda - h - D < 0$; $h/\lambda \rightarrow 0$ gives the bound. Also $b_s = -(1 + 2\lambda)R_b + 2\lambda \partial_{\xi_b} R_b = -R_b + O(\lambda)$. Apart from exponentially small end bumps, $R_b \geq 0$ and $\partial_{\xi_b} R_b \leq 1.2$, since the declining cutoff has a nonpositive contribution. The exact upper bounds for the two leading quadratics are

$$\sup_{R \geq 0} (-2R^2 + 2.41R) = 2.41^2/8 < .74,$$

$$\sup_{R \geq 0} (-3.5R^2 + 4.82R) = 4.82^2/14 < 1.68.$$

The uniformly $o(1)$ remainders can be made smaller than the strict numerical gaps. Hence $a - b_s w > 1$ and $2b_s w + b_s^2/a + (a - 2)w^2 < 2$ on the complete pulse, including its descending and correction parts. Here $v_s \geq a = 2 + 2\lambda > 2$.

34.5 Interpolation and exterior transitions

After the pulse $U = M = J = 0$, so $W = 1$. The angular source during interpolation is $-l - h + \vartheta_f c_\eta$, giving $Q_s \geq c\lambda$. The tiny angular correction preserves this bound. Since the complete amplitude solve gives $S(\infty) = 0$,

$$S(X) = \frac{1}{2} \int_X^\infty E(X')^2 dX'.$$

Before exterior release, $l \leq -\lambda/2$ and the later release has the h uniform integral bound proved above. Therefore $(|S| + |S_\eta|)/X \leq CE^2/\lambda$ and $|\Pi| + |\Pi_\eta| \leq CE^2$. The exact moment formula yields $|w| \leq CE/\lambda^2 \ll 1$, because pulse end reduces E exponentially in $1/\lambda$. Here $b_s = 0$ and $2 < a \leq 2 + 2\lambda + .2 + o(1)$, so both cone margins remain strict.

On the parameter-independent release and tail, the same backward integrals give the exact expression

$$N_s = 2\eta \int_y^\infty (he^{v-y} - A)E(v)^2 dv.$$

Since $l \leq -3h/4$, $E(v)^2 \leq E(y)^2 e^{-(1+3h/2)(v-y)}$. Thus the he^{v-y} integral is at most $\frac{2}{3}E(y)^2$ and the other at most $E(y)^2/(1+3h/2)$. In particular $|N_s| \leq CE^2$ with a constant independent of small h . Before the steep hold, $Q_s \geq c\lambda$ and the amplitude is already small relative to λ . During the hold it remains above this bound. After the hold, $E \leq Ce_{\text{rel}}h^6$, while $Q_s \geq ch$ up to terminal coordinate $y = 1/2$. The latter follows from its explicit positive terminal integral and its intervening monotone decay to $Q_p \asymp h$. Consequently $|w| \leq CE/Q_s \ll 1$, even if $h \ll e_{\text{rel}}$. Here $b_s = 0$, $2 < a \leq 4$, so the sufficient cone test has uniform strict slack.

All these intervals are compact once the sequential parameters are fixed. Under the exact moment scaling $X = X_R x$, every X_R factor cancels in Q_s, N_s , while $p_{s,1} = X_R x Q_s / L$ grows linearly with X_R . Thus one finite increase of X_R meets the threshold proved above on the whole designated interval. This completes the local reconstruction of [NS, Proposition A.4], including each part of its cone verification.

35 The heat factor, its exact equation, and the compensated replacement

35.1 A positive solution of the radial swirl heat equation

Define for $Z \geq 0$,

$$\mathcal{H}(Z) = \frac{1}{\Gamma(1+h)} \int_0^\infty e^{-v} v^h (1+Zv)^{-h} dv.$$

Dominated differentiation gives for every nonnegative integer m

$$\mathcal{H}^{(m)}(Z) = \frac{(-1)^m (h)_m}{\Gamma(1+h)} \int_0^\infty e^{-v} v^{h+m} (1+Zv)^{-h-m} dv, \quad \mathcal{H}^{(m)}(0) = (-1)^m (h)_m (1+h)_m.$$

For fixed m the integrand after deleting the last factor is integrable uniformly for $Z \geq 0$. Hence $\mathcal{H}$ is positive, smooth up to zero from the right, and has bounded fixed derivatives on every bounded nonnegative interval (in fact on the entire half-line for absolute derivatives). For $0 < h \leq h_0 < 1/2$ and fixed $m \geq 1$, the last factorial product is at most $C_m h$. Taylor's formula therefore proves

$$\mathcal{H}(Z) = 1 - h(1+h)Z + O_h(Z^2), \quad |\mathcal{H}(Z) - 1| + |Z\mathcal{H}'(Z)| \leq ChZ$$

on a fixed bounded nonnegative interval.

The integral solves the required ODE by an exact polynomial calculation. Multiply $Z^2\mathcal{H}'' + (1+2(1+h)Z)\mathcal{H}' + h(1+h)\mathcal{H}$ by $\Gamma(1+h)/h$, and use common integrand $e^{-v}v^h(1+Zv)^{-h-2}$. Its remaining polynomial is

$$(1+h)Z^2v^2 - (1+2(1+h)Z)v(1+Zv) + (1+h)(1+Zv)^2 = (1+h) - v - Zv^2.$$

This is precisely the polynomial from $\partial_v[e^{-v}v^{1+h}(1+Zv)^{-1-h}]$ in the same common factor. The boundary values at zero and infinity are both zero. Thus

$$Z^2\mathcal{H}'' + (1+2(1+h)Z)\mathcal{H}' + h(1+h)\mathcal{H} = 0. \quad (\text{PA.HEAT})$$

Set $K(r, t) = c_\infty s^{-A} \mathcal{H}(2\tau/s)$, $s = r^2/2$, $Z = 2\tau/s$. Direct differentiation gives

$$K_t = -2c_\infty s^{-A-1} \mathcal{H}',$$

$$(\partial_{rr} + r^{-1}\partial_r - r^{-2})K = 2c_\infty s^{-A-1} \{Z^2\mathcal{H}'' + (2A+1)Z\mathcal{H}' + (A^2 - 1/4)\mathcal{H}\}.$$

Since $2A+1 = 2(1+h)$ and $A^2 - 1/4 = h(1+h)$, (PA.HEAT) proves the heat equation, with its swirl term $-r^{-2}K$.

Normalize the positive integrand defining $\mathcal{H}$ to a probability density. Then $-Z\mathcal{H}'/\mathcal{H}$ is its average of $hZv/(1+Zv)$, so $0 \leq -Z\mathcal{H}'/\mathcal{H} < h$. It follows that $rK_r/(2K) = -A - Z\mathcal{H}'/\mathcal{H} < -1/2$, in particular $K_r < 0$. At $Z = 0$ this is still strict because $h > 0$. Finally $Z = 2(1-\eta^2)/X$ gives only factors $-4\eta/X$ and $-4/X$ when differentiated in η , never a denominator $1-\eta^2$. Consequently all fixed mixed X, η derivatives have one-sided limits at $\eta = \pm 1$, and for each fixed logarithmic radial and parameter derivative,

$$\frac{E_{\text{heat}}}{E_{\text{pow}}} = \mathcal{H}(2d/X) = 1 - \frac{2h(1+h)d}{X} + O_{h,j,m}(X^{-2}).$$

This proves [NS, Lemma A.6] on its stated nonnegative domain. No analytic continuation through negative Z is used or required.

35.2 The three heat discrepancies and their exact cancellation

Let $y = \log(X/X_{\text{tail}})$, $X_K = e^2 X_{\text{tail}}$, $X_b = e^3 X_{\text{tail}}$, and choose $\chi_K = 0$ for $y \leq .2$, $\chi_K = 1$ for $y \geq .5$. Replace the reference swirl E_{cl} by

$$E_{\text{cl}}[1 + \chi_K(y)(\mathcal{H}(2d/X) - 1)].$$

For $x = X/X_K \geq 1$ and its reference amplitude e_K , the preceding derivative bounds give exactly

$$|(X\partial_X)^j \partial_\eta^m (E - E_{\text{cl}})| \leq C_{j,m} e_K X_K^{-1} x^{-A-1}.$$

The profile is comparable to $e_K x^{-A}$ with constants depending on the fixed schedule. Thus the pressure, S , and subtracted angular discrepancies obey, after m parameter derivatives,

$$\begin{aligned} |\Delta C_p|_m &\leq C_m \frac{e_K^2}{X_K} \int_1^\infty x^{-2A-2} dx, \quad |\Delta S|_m \leq C_m e_K^2 \int_1^\infty x^{-2A-1} dx, \\ \left| \Delta \int (H - H_{\text{pow}}) dX \right|_m &\leq C_m e_K X_K^{1/2} \int_1^\infty x^{-A-1/2} dx = C_m e_K X_K^{1/2} / h. \end{aligned}$$

The powers arise respectively from $\delta(E^2)dX/(2X)$, $-\delta(E^2)dX/2$, and $\sqrt{2X}\delta E, dX$. The final $1/h$ is retained: h has already been fixed before X_R grows.

At the earlier second reserved patch write $x_* = X/X_*$, $E = e_* f x_*^{-1/2-\lambda}$, $U = 0$, and add $\delta E = e_* \sum_{i=1}^3 c_i(\eta) \beta_i(x_*)$. Divide the three rows by e_*^2 , $X_* e_*^2$, and $X_*^{3/2} e_*$. The exact linear weights are then

$$f x_*^{-3/2-\lambda}, \quad -f x_*^{-1/2-\lambda}, \quad \sqrt{2} x_*^{1/2}.$$

The first two rows have exact quadratic terms and the third is linear. The moment determinant is nonzero by the moment-map calculation above; its inverse is uniform in η . Since X_*/X_K and e_*/e_K are fixed, all three normalized discrepancies are $O_m(X_K^{-1})$. The contraction gives the exact correction with $\|c\|_{C^m} \leq C_m X_K^{-1}$ for any prescribed finite m , after choosing X_R large. More generally its higher fixed derivative bounds follow by implicit differentiation with the same small forcing: the Jacobian stays invertible and every differentiated equation has only lower derivative products and an $O_m(X_K^{-1})$ inhomogeneity. This justifies the stated O_m estimates without selecting infinitely many smallness thresholds at once.

Both edits have $U = 0$, so M, J are unchanged exactly. All three other total moments are restored by this solve. The total pressure integral is therefore unchanged, so the analytic axis pressure and every profile before the patch remain unchanged. On the compact logarithmic interval up to $y = .5$, the exact moment formulas transfer the $O(X_K^{-1})$ changes of profiles and moments to Q_s, N_s , costing one more η derivative. Indeed substituting $I = X_R^{3/2} \hat{I}$, $J = X_R^{3/2} \hat{J}$, $M = X_R \hat{M}$, $S = X_R \hat{S}$, $H = X_R^{1/2} \hat{H}$ cancels every X_R in those formulas. Positive lower bounds for E, Q_s and the already strict cone margins persist by continuity, uniformly on this fixed compact interval. This proves [NS, Proposition A.7] and its preservation claims; the remaining endpoint is treated next rather than being inferred from a compact-interior estimate.

36 Analytic axis construction and endpoint inequalities

36.1 1. Axis data and the complementary parameter regions

Keep $A = 1/2 + h$, $D = 1/2 - h$, $d = 1 - \eta^2$, $L = 1 - 2h\eta^2$, and the already fixed $0 < h \leq 10^{-2}$. The datum constructed in (PA.4) is holomorphic near $[-1, 1]$, real and even on that interval, and has

$$\Pi_0 \leq -\frac{5}{2} P_*^2 (1 + \eta^2)^{-2}, \quad \eta \Pi'_0(\eta) > 0 \quad (\eta \neq 0).$$

Choose $0 < j_0 \leq 0.05$ and set exactly

$$\begin{aligned} U_* &= 4\eta + j_0, \quad H_* = D\eta + dU_*, \quad W_* = 1 - dU'_* - 2D\eta U_*, \\ Z_* &= -A(1 - 2\eta U_*)U_* - H_* U'_* - d\Pi'_0 + 4A\eta \Pi_0. \end{aligned}$$

For $-1 < \eta < 1$,

$$\frac{H_*}{d} = \frac{D\eta}{1 - \eta^2} + 4\eta + j_0, \quad \partial_\eta(H_*/d) = D \frac{1 + \eta^2}{(1 - \eta^2)^2} + 4 > 0.$$

Its endpoint limits are $-\infty, +\infty$, so it has exactly one zero η_0 . Its value at zero is $j_0 > 0$, hence $\eta_0 < 0$. The root equation yields

$$|\eta_0| = \frac{j_0}{4 + D/(1 - \eta_0^2)}.$$

First this is at most $j_0/4 \leq 1/80$. Substitution back gives the useful explicit bounds $j_0/5 < |\eta_0| < j_0/4$. Further,

$$-W_* = 3 - 8h\eta^2 + (1 - 2h)j_0\eta \geq 3 - 0.08 - 0.05 = 2.87 > 2.8.$$

At the zero, $H_*U'_* = 0$ and $-d\Pi'_0 \geq 0$. The term $4A\eta_0\Pi_0$ is positive. Using $(1 + \eta_0^2)^{-2} \geq 1/4$, it is at least $(A/2)j_0P_*^2$. The remaining negative term has magnitude bounded by a fixed multiple of j_0 , since $U_* = -D\eta_0/d$ at the zero. Therefore the already large fixed P_* gives $Z_*(\eta_0) > 0$, quantitatively at least a positive constant times $j_0P_*^2$.

Continuity gives an open neighborhood V of η_0 on which Z_* is bounded below by a positive number. Choose $\delta_* > 0$ smaller than half that lower bound. The compact set $K_\delta = \{\eta : |Z_*(\eta)| \leq \delta_*\}$ avoids V , hence $|H_*| \geq m > 0$ there (if this set is empty, the following restriction is vacuous). Choose $\sigma_* > 0$ with $\sigma_*^2 < m^2/99$. Then

$$\chi = \frac{H_*^2}{H_*^2 + \sigma_*^2} > 0.99 \quad \text{on } K_\delta.$$

Thus $\chi \leq 0.99$ implies $|Z_*| > \delta_*$, exactly the later endpoint alternative. Strict margins allow the same choices on a slightly larger compact interval $I \supset [-1, 1]$. There is a bounded simply connected complex neighborhood Ω of I on whose closure L and $H_*^2 + \sigma_*^2$ are nonzero and Π_0 is holomorphic: take a sufficiently thin neighborhood of the compact real interval, avoiding the closed zero sets.

Define

$$\zeta_* = -\frac{LH_*}{H_*^2 + \sigma_*^2}, \quad \xi_0 = \Lambda\zeta_*, \quad \phi_*(\eta) = \exp\left(\Lambda \int_0^\eta \zeta_*(w) dw\right).$$

The primitive is single-valued on Ω , and $\phi_* > 0$ on I . The exact identity needed below is $H_*\zeta_*/L = -\chi$. After choosing Λ , impose

$$C \geq \sup_{\eta \in \Omega} |\phi_*(\eta)|.$$

This supremum is finite after choosing Ω with compact closure inside the holomorphic domain. Then $g = \phi_*/C$ has a complex bound independent of both later large parameters. A real-interval bound alone would not establish the required parameter derivative bounds.

36.2 2. The coefficient space, with all derivative losses accounted for

Set $Y = \Lambda X$. For coefficient sequences $F = \sum_{\alpha \geq 0} F_\alpha(\eta)Y^\alpha$, use precisely

$$a_{\alpha\beta} = \frac{20^{-\alpha}\rho^{-\beta}\beta!(\alpha+\beta)}{(\alpha+1)^2(\beta+1)^2}, \quad \|F\|_\rho = \sup_{\alpha, \beta \geq 0} \sup_{\eta \in I} \frac{|\partial_\eta^\beta F_\alpha(\eta)|}{a_{\alpha\beta}}.$$

Completeness follows coefficient by coefficient: a norm-Cauchy sequence converges uniformly for every derivative, and the fundamental theorem of calculus identifies each derivative limit with the derivative of its predecessor. The supremum bound passes to the limit, so this is a Banach space.

Let $C_{\text{sq}} = 8 \sum_{n \geq 1} n^{-2}$. For $0 \leq i \leq N/2$, $(N+1)^2/(N-i+1)^2 \leq 4$, while the other half has the analogous bound after $i \mapsto N-i$. Therefore

$$\sum_{i=0}^N \frac{(N+1)^2}{(i+1)^2(N-i+1)^2} \leq C_{\text{sq}}.$$

In a coefficient product and its β -th derivative, the Leibniz factor $\binom{\beta}{k}$ cancels the factorials in the two input weights. The remaining binomial quotient is at most one because

$$\binom{i+k}{k} \binom{\alpha-i+\beta-k}{\beta-k} \leq \binom{\alpha+\beta}{\beta}.$$

It counts subsets selecting specified numbers from the two fixed blocks. Applying the preceding convolution bound once in the radial and once in the parameter index proves $\|FG\|_\rho \leq C_{\text{sq}}^2 \|F\|_\rho \|G\|_\rho$.

Define $\mathcal{I}F(Y) = \int_0^Y F(v) dv$, $\mathcal{A}F = Y^{-1}\mathcal{I}F$, and let $\mathcal{J}_\nu F$, $\nu = 1, 2$, be the regular solution with zero value at $Y = 0$ of $Y G_{YY} + \nu G_Y = F$. Comparing coefficients gives

$$(\mathcal{J}_\nu F)_{\alpha+1} = \frac{F_\alpha}{(\alpha+1)(\alpha+\nu)}, \quad (\mathcal{J}_\nu F)_0 = 0.$$

The exact weight ratios are

$$\frac{a_{\alpha\beta}}{a_{\alpha+1,\beta}} = 20 \frac{(\alpha+2)^2}{(\alpha+1)^2} \frac{\alpha+1}{\alpha+\beta+1} \leq 80,$$

$$\frac{a_{i,k+1}}{a_{i+1,k}} = \frac{20}{\rho} (i+1) \frac{(i+2)^2}{(i+1)^2} \frac{(k+1)^2}{(k+2)^2} \leq \frac{80}{\rho} (i+1).$$

These prove boundedness of multiplication by Y , $\mathcal{I}$, $\mathcal{J}_\nu$, averaging (which divides coefficient α by $\alpha+1$), and $\partial_\eta \mathcal{I}$. The last assertion uses its integration divisor $i+1$ to cancel the last factor in the second ratio.

For $\mathcal{J}_\nu[(\partial_\eta F)(Y \partial_Y G)]$, consider output degree $N = \alpha+1$, split the input radial degrees as i and $\alpha-i$, and allocate the added degree to the differentiated F -factor. Put $I_1 = i+1$, $J_1 = \alpha-i$. The weight comparison and radial inverse give, after dividing by the output weight, at most

$$\frac{80}{\rho} \|F\|_\rho \|G\|_\rho \sum_{i=0}^{\alpha} \sum_{k=0}^{\beta} \frac{I_1 J_1}{N(\alpha+\nu)} \frac{(N+1)^2}{(I_1+1)^2 (J_1+1)^2} \frac{(\beta+1)^2}{(k+1)^2 (\beta-k+1)^2}.$$

The first fraction is at most one, including the zero endpoints; the two sums are bounded by C_{sq} . If the logarithmic derivative is absent, replace J_1 by one, still giving a fraction at most one. If F is averaged, its divisor $i+1$ cancels I_1 . If the parameter derivative is absent, use the first weight ratio instead. Inserting further undifferentiated factors adds the same convolution bounds with unchanged degree allocation. This proves every nonlinear operator estimate invoked below, including the case where both differentiated factors are copies of the same unknown.

A holomorphic coefficient $b(\eta)$ bounded on a larger neighborhood belongs as degree zero to this space: Cauchy's estimate gives $|b^{(\beta)}| \leq M\beta! r^{-\beta}$, and choosing $\rho < r$ bounds $(\beta+1)^2(\rho/r)^\beta$. This applies uniformly to g , since its complex supremum is at most one.

36.3 3. Exact rescaling and contraction

Write the original fields as

$$\phi = \phi_* \Phi, \quad U = U_* + \Lambda^{-1}u, \quad \Pi = \Pi_0 + \Lambda^{-1}p, \quad p = \mathcal{I}(g^2 \Phi^2), \quad \Phi(0) = 1, \quad u(0) = 0.$$

Thus $\Pi_X = g^2 \Phi^2 = \phi^2 / C^2$ exactly. Set

$$B = -2D\eta Au - d\partial_\eta Au, \quad W = W_* + \Lambda^{-1}B, \quad H_c = H_* + \Lambda^{-1}du.$$

Substitute these into the two stress-free equations in `core_exact.md`. The unchanged equations are

$$2(Y\Phi_{YY} + 2\Phi_Y) = -\chi\Phi + \Lambda^{-1}R_1, \quad 2(Yu_{YY} + u_Y) = -Z_*/L + \Lambda^{-1}R_2,$$

where

$$LR_1 = [W + h(1 - 2\eta U) + du\zeta_*]\Phi + WY\Phi_Y + H_c\Phi_\eta,$$

$$LR_2 = [A(1 - 4\eta U_*) + dU_*']u - 2A\eta\Lambda^{-1}u^2 + WYu_Y + H_*u_\eta + d\Lambda^{-1}uu_\eta - 4A\eta p + d\partial_\eta p - 2\eta Y p_Y.$$

Both identities have been checked as exact rational expressions by `exact_checks.py`, before any estimate. In particular the pressure terms retain all three signs and coefficients. The angular $-\chi\Phi$ comes from $H_*\zeta_*/L = -\chi$; this term is order one and must be inverted, not treated as small.

The only products with both an unintegrated parameter derivative and a logarithmic radial derivative are $(\partial_\eta Au)Y\Phi_Y$ and $(\partial_\eta Au)Yu_Y$. They have the precise structure proved above. The term $\partial_\eta p = \partial_\eta \mathcal{I}(g^2 \Phi^2)$ is bounded, and $Yp_Y = Yg^2 \Phi^2$. Every other monomial is controlled by multiplication and a radial inverse, with at most one of the two derivative types. Subtracting two monomials by replacing their factors one at a time proves the corresponding local Lipschitz estimates. All constants are finite on a fixed ball, uniformly for $\Lambda \geq 1$ and C satisfying the complex bound.

Let M_χ be the bounded multiplier norm of χ , and set $T = \mathcal{J}_2\chi/2$, multiplication preceding integration. A sequence vanishing below degree b obeys

$$\|\mathcal{J}_2 F\|_\rho \leq \frac{80}{(b+1)(b+2)} \|F\|_\rho.$$

Multiplication by χ preserves that vanishing order and $\mathcal{J}_2$ increases it by one. Therefore

$$\|T^k\| \leq \frac{(40M_\chi)^k}{k!(k+1)!}.$$

The absolutely convergent series $\sum_{k \geq 0} (-T)^k$ is a two-sided inverse of $1 + T$: multiply finite partial sums by $1 + T$ and use $\|T^{k+1}\| \rightarrow 0$. No smallness of χ is required.

The center of the contraction is

$$\Phi_0 = f_0(Y\chi), \quad u_0 = -\frac{YZ_*}{2L}, \quad f_0(z) = \sum_{\alpha \geq 0} \frac{(-z/2)^\alpha}{\alpha!(\alpha+1)!}.$$

Its recurrence verifies $2(zf_0'' + 2f_0') = -f_0$, $f_0(0) = 1$, so $\Phi_0 = (1 + T)^{-1}1$. On the closed radius-one ball about (Φ_0, u_0) , let M bound the pair $((1 + T)^{-1}\mathcal{J}_2R_1/2, \mathcal{J}_1R_2/2)$, and let K be its Lipschitz constant. These are the finite constants supplied by the proved estimates. Choosing $\Lambda \geq \max\{1, 2M, 2K\}$ makes the map

$$(\Phi, u) \mapsto \left(\Phi_0 + \frac{(1 + T)^{-1}\mathcal{J}_2R_1}{2\Lambda}, u_0 + \frac{\mathcal{J}_1R_2}{2\Lambda} \right)$$

invariant and a contraction with factor at most $1/2$. Its unique fixed point has norm distance at most M/Λ . This explicitly supplies the existential large-parameter threshold from finite operator constants.

To recover analytic functions, for $R < 20$ use

$$\sum_{\alpha \geq 0} \binom{\alpha + \beta}{\beta} (R/20)^\alpha = (1 - R/20)^{-\beta-1}.$$

The coefficient norm therefore bounds parameter derivatives on $|Y| \leq R$ by a constant times $\beta![\rho(1 - R/20)]^{-\beta}$. It gives analytic continuation in a common positive parameter radius. Choose $4.1 < R_1 < R_2 < 20$; Cauchy's formula on the larger radial disk and a slightly smaller parameter neighborhood gives, for every fixed r, s ,

$$\sup_{[0, 4.1] \times I} |\partial_Y^r \partial_\eta^s (\Phi - f_0(Y\chi), u + YZ_*/(2L))| \leq C_{r,s}/\Lambda.$$

The same parameter domain serves every fixed radial derivative because those derivatives use radial Cauchy estimates on the fixed larger disk. For original radial derivatives the factor is exactly $\partial_X^r = \Lambda^r \partial_Y^r$, giving $C_{r,s}\Lambda^{r-1}$. Real coefficients and uniqueness imply real profiles on the real rectangle.

36.4 4. Positivity and endpoint alternative with the original constants

For $0 \leq z \leq 4.1$, put $t = z/2 \leq 2.05$. The magnitude ratio of consecutive terms of f_0 is $t/[(n+1)(n+2)]$, so the alternating tail after the cubic has decreasing magnitudes. Consequently

$$f_0(z) \geq 1 - t/2 + t^2/12 - t^3/144 \geq 305719/1152000 > 0.265.$$

The cubic derivative is $-1/2 + t/6 - t^2/48 < 0$: its discriminant is negative and its leading coefficient is negative. Its minimum is thus the displayed exact endpoint value. Grouping the alternating series after the constant term gives $f_0 \leq 1$. The proved $O(\Lambda^{-1})$ bound gives $\Phi \geq c_0 > 0$ when Λ is sufficiently large. In particular division by ϕ in the original equation is justified. Uniform complex derivative bounds then give a smaller common complex domain with no zeros; derivatives of $\log \Phi$ are bounded there.

Differentiate $\chi = H_*^2/(H_*^2 + \sigma_*^2)$:

$$|H_*\chi'| = 2|H_*'| \frac{H_*^2\sigma_*^2}{(H_*^2 + \sigma_*^2)^2} \leq 2\|H_*'\|_\infty \chi.$$

Since f_0 is positive and has bounded fixed derivatives on the stated compact range,

$$|D_X \log \Phi| + |H_*\partial_\eta \log \Phi| \leq C\chi + C/\Lambda.$$

Here $D_X = Y\partial_Y$ and no division by χ is used. Also

$$-H_c\xi_0 = \Lambda L\chi - du\zeta_*, \quad |du\zeta_*| \leq C_{\sigma_*}\sqrt{\chi},$$

because $|H_*|/(H_*^2 + \sigma_*^2) \leq \sigma_*^{-1}\sqrt{\chi}$. Thus the full source has the lower bound

$$S_q \geq \Lambda L\chi - W_* - 10h - C\chi - C\sigma_*\sqrt{\chi} - C/\Lambda.$$

For any fixed small $\varepsilon > 0$, Young's inequality in the exact form $K\sqrt{\chi} \leq \varepsilon\Lambda L\chi + K^2/(4\varepsilon\Lambda L)$ absorbs the square-root term. Increase Λ so the linear $C\chi$ term uses the remaining part of a total $0.05\Lambda L\chi$ allowance. Since $-W_* > 2.8$, $10h \leq 0.1$, and the remaining constant error can be made at most 0.2, this gives

$$S_q \geq 2.5 + 0.95\Lambda L\chi.$$

The integral formula

$$Q_s(X, \eta) = \frac{\int_0^X x\Phi(\Lambda x, \eta)S_q(x, \eta) dx}{X^2\Phi(\Lambda X, \eta)}$$

proves $p_1/X = Q_s/L \geq c_1 > 0$. Stress freedom gives the exact identities

$$p_1 = \frac{XQ_s}{L} = a = -2Y\Phi_Y/\Phi, \quad n_s = N_s/L = -2U_X = -2u_Y = Z_*/L + O(\Lambda^{-1}), \quad p_2 = Xn_s/E.$$

All fixed parameter derivatives of $\Phi, \log \Phi, u, p_1, n_s$ are bounded uniformly in subsequent C , with Λ fixed; the radial scaling factor has already been accounted for.

At $X_0 = 4/\Lambda$, first take $\chi > 0.99$, so $t = 2\chi \in (1.98, 2]$. The series for $f_0(z) + zf'_0(z)$ has terms $(-t)^n/(n!)^2$. Its alternating remainder beyond degree four is negative, giving

$$f_0 + zf'_0 \leq 1 - t + t^2/4 - t^3/36 + t^4/576.$$

The derivative is $-1 + t/2 - t^2/12 + t^3/144$; on $[1.98, 2]$ its first two terms are nonpositive and its last two have sum $-t^2(12 - t)/144 < 0$. Thus the polynomial decreases, and its value at 1.98 is exactly $-75535511/400000000 < -0.188 < -0.18$. Since $f_0 \leq 1$,

$$-2zf'_0/f_0 = 2 - 2(f_0 + zf'_0)/f_0 > 2.36.$$

Taking Λ large gives $p_1 > 2.3$.

On $\chi \leq 0.99$, $|Z_*| > \delta_*$. Since $L \leq 1$ on $[-1, 1]$, the preceding error estimate gives $|n_s| \geq \delta_*/2$. At the same endpoint, exactly

$$|p_2| = C\sqrt{2/\Lambda} \frac{|n_s|}{\phi_*\Phi}.$$

For fixed Λ , $\phi_*\Phi$ has a finite positive upper bound independent of subsequent C ; p_1 has a finite upper bound independent of C . Thus choosing C large makes $p_2^2/p_1 > 2.3$ uniformly on this complementary compact set. The two alternatives yield

$$p_1 + p_2^2/p_1 > 2 + 0.2 \quad \text{at } X_0 = 4/\Lambda,$$

retaining the source's allowed $c_{\text{ex}} = 0.2$. The proof uses distinct regions only after constructing their exact covering through χ and Z_* ; no missing case at $H_* = 0$ remains.

37 Continuation, exact moment repair, and assembly of the annular profile

This file gives the independent continuation derivation for source pp. 150-157 and checks the assembly on pp. 36-45. Original source parameters remain $0 < h < 1/100$, $X_0 = X_a = 4/\Lambda$, $X_b^{\text{ref}} = 100$, $X_i = 110$. The notation X_b^{ref} distinguishes the temporary number 100 in Appendix B from the final outer annular edge $X_b = e^3 X_{\text{tail}}$ in Section 4; neither numerical value or role is changed. The core and axis sections above provide all field definitions and identities.

37.1 1. The reference continuation and its estimates

The smooth step is exactly

$$\sigma(y) = \frac{e^{-1/y^2}}{e^{-1/y^2} + e^{-1/(1-y)^2}} \quad (0 < y < 1), \quad \sigma = 0 \ (y \leq 0), \quad \sigma = 1 \ (y \geq 1).$$

It is increasing: the ratio of its second denominator term to its first has logarithmic derivative $-2/(1-y)^3 - 2/y^3 < 0$. It is smooth and flat at both endpoints by the corresponding exponential formulas. Choose fixed $\bar{t} > 0$ with $4e^{2\bar{t}} < 4.1$. Set $y = \log(X/X_0)$. For $0 < t_1 \leq \bar{t}$, retain the analytic axis profile through $y = t_1$, then prescribe on $(t_1, 2t_1)$

$$\partial_y \log \phi_r = (1 - \sigma((y - t_1)/t_1))D_X \log \phi_{\text{nat}}, \quad \partial_y U_r = (1 - \sigma((y - t_1)/t_1))D_X U_{\text{nat}}.$$

After $2t_1$ keep ϕ_r, U_r constant in y . All endpoint jets match because the step is flat. Pressure and the integrated coefficients continue to be computed from the axis by the original integral definitions.

The axis approximation bounds $U_{\text{nat}} - U_*$ and its fixed parameter derivatives by $O(\Lambda^{-1})$; its cutoff variation integrates over an interval of length t_1 and is $O(t_1/\Lambda)$. Thus $U_r - U_*$ and its radial average remain small uniformly through X_i : explicitly

$$\|\mathcal{A}_X U_r - U_*\|_{C_\eta^k} \leq \sup_{0 \leq X \leq X_i} \|U_r - U_*\|_{C_\eta^k}.$$

No factor equal to the long logarithmic interval appears. Likewise integrate the bounded natural logarithmic slope to bound $\log \phi_r - \log \phi_*$. Its leading parameter derivative remains ξ_0 , with the $O(\chi) + O(\Lambda^{-1})$ weighted-gradient errors proved for the natural solution and an additional error tending to zero with t_1 after Λ is fixed. In particular $CE_r = \sqrt{2X}\phi_r$, its reciprocal on $[X_0, X_i]$, and all their fixed parameter derivatives have finite bounds independent of subsequent large C and sufficiently small t_1 .

Pressure satisfies the exact identity and estimate

$$\Pi_r - \Pi_0 = C^{-2} \int_0^X \phi_r(x, \eta)^2 dx, \quad \sup_{X \leq X_i} \|\Pi_r - \Pi_0\|_{C_\eta^k} \leq K_k C^{-2}.$$

The axis factor makes this integral finite at zero. The natural shear has $a = p_1 > 0$; cutting its nonpositive $D_X \log \phi$ to zero proves $l_r \leq 1$. In the angular source the unchanged large term is $-H_* \xi_0 = \Lambda L \chi$. Replacing H_* by $H_{c,r}$ costs $K_{\sigma_*} \sqrt{\chi} + o_{t_1}(1)$; the other weighted gradients cost $K\chi + K/\Lambda$. Young's inequality as in `axis_exact.md`, now allocating a total $0.06\Lambda L \chi$, and the strict $-W_* > 2.8$ margin give

$$S_{q,r} \geq 0.94\Lambda L \chi + 2.4.$$

Here one first enlarges Λ , then C , and then decreases t_1 ; all residual constant errors have a strictly positive allowance. The axial source is

$$S_{n,r} = Z_* + O(\Lambda^{-1}) + o_{t_1}(1) + O(C^{-2})$$

uniformly on the finite interval. Indeed all changes of U and of its parameter derivative are small, the logarithmic radial derivative is $O(\Lambda^{-1})$ on the cutoff and zero later, and pressure is controlled by the preceding exact integral.

The defining primitive equations imply

$$D_X p_{1,r} = X S_{q,r}/L - l_r p_{1,r}, \quad D_X n_{s,r} + n_{s,r} = S_{n,r}/L.$$

Positive regular initial data and $l_r \leq 1$ imply $p_{1,r} > 0$. Multiplying the first inequality by the integrating factor e^y , keeping the two positive source terms separately, gives

$$p_{1,r}(y) \geq p_{1,r}(0)e^{-y} + 1.88\chi(e^y - e^{-y}),$$

$$p_{1,r}(X) \geq p_{1,r}(X_0)X_0/X + 1.2(X - X_0^2/X).$$

For the first, $X = X_0 e^y = (4/\Lambda)e^y$ supplies the exact factor $0.94\Lambda X_0/2 = 1.88$. For the second, use $L \leq 1$, integrate $2.4X$ against the same factor, and convert back to X . On $\chi > 0.99$, the first lower bound remains above 2.3: replace its initial value by 2.3; its derivative is positive for $y \geq 0$, since $1.88\chi > 2.3/2$. On $\chi \leq 0.99$, $|Z_*| > \delta_*$, and the exact primitive for $n_{s,r}$ retains its sign and a fixed positive lower bound when the displayed errors are small. Because $p_{2,r} = X n_{s,r}/E_r$, increasing C then makes $p_{2,r}^2/p_{1,r}$ large uniformly. This proves

$$v_r := p_{1,r} + p_{2,r}^2/p_{1,r} > 2 + c$$

on the entire reference interval, for a fixed $c > 0$. The second comparison yields $p_{1,r} > 3$ on $[100, 110]$, since $X_0 \leq 4$. Upper bounds for $p_{1,r}, n_{s,r}$ and their fixed parameter derivatives follow directly from their integral representations and the finite coefficient bounds. They are independent of subsequent C and of the small reference widths.

37.2 2. Activation of stress, with division by the flat factor proved

Fix a sufficiently large finite C . Choose $t_* > 0$ so the preceding bounds hold for every $0 < t_1 \leq t_*$. On this family, v_r has a finite common upper bound $V_{\max}$; it may depend on C , which has already been fixed. Choose $0 < \kappa_0 < 1/2$ later small enough relative to this bound. On $0 < y < t_1$ put

$$e_a = (1 - \kappa_0)\sigma(y/t_1), \quad \kappa = 1 - e_a, \quad a = \kappa p_{1,r}, \quad D_X U = -\kappa X n_{s,r}/2, \quad \partial_y \log \phi = -\kappa p_{1,r}/2.$$

The natural reference is stress-free here. Therefore the exact differences obey

$$\partial_y(\log \phi - \log \phi_r) = e_a p_{1,r}/2, \quad \partial_y(U - U_r) = e_a X n_{s,r}/2.$$

For a fixed parameter derivative, the integral from zero is bounded by a constant times $y e_a(y)$, since e_a is increasing. The constants are uniform in smaller t_1, κ_0 after Λ, C are fixed. The five moment differences are integrals of these value differences, with fixed smooth weights on the activation interval; their parameter derivatives have the same bound. The proved moment map in `core_exact.md`, using one further parameter derivative, then gives

$$p_s - p_{s,r} = O(y e_a), \quad E_r/E = 1 + O(y e_a).$$

The actual shear vector is exactly

$$s_{\text{shear}} = \kappa(p_{1,r}, p_{2,r} E_r/E).$$

Because κ cancels before forming its slope, there is no inverse power of κ_0 in

$$t_s = (p_{2,r}/p_{1,r})(E_r/E), \quad v_s = \kappa v_r + O(y e_a), \quad P_c = v_r + O(y e_a), \quad J_c = O(y e_a).$$

Consequently $P_c - v_s = e_a v_r + O(y e_a) \geq c e_a$ for small fixed t_1 , and $P_c > 2 + c'$. The ratio $(v_s - 2)_+ J_c^2 / (P_c - v_s)^2$ is at most a fixed constant times y^2 , hence below 2 for small t_1 . The exact cone equivalence proves admissibility wherever $v_s > 2$ and the relaxed condition everywhere else. Since $v_s(0) = v_r(0) > 2 + c$, continuity on the compact parameter interval supplies a first collar of positive width with $v_s > 2 + c/2$.

The stronger endpoint assertion requires more than an $O(y e_a)$ estimate. Here is the exact flat-integral calculation. For $c > 0$ and smooth b , substitute $u = y/\sqrt{1 + y^2 v}$ to obtain

$$\int_0^y e^{-c/u^2} u^{-j} b(u, \eta) du = \frac{1}{2} e^{-c/y^2} y^{3-j} \int_0^\infty e^{-c v} (1 + y^2 v)^{(j-3)/2} b\left(\frac{y}{\sqrt{1 + y^2 v}}, \eta\right) dv.$$

For each fixed derivative the final integrand is bounded by $e^{-c v}$ times a polynomial in v , uniformly on the compact parameter set; denominators $1 + y^2 v$ are at least one. Differentiation under the integral proves a smooth coefficient through $y = 0$, with limiting value $b(0, \eta)/c$. This proves the needed result from Appendix A.9 directly.

Near zero, the specific step gives $e_a = e^{-t_1^2/y^2} g_a(y)$, where

$$g_a(y) = \frac{1 - \kappa_0}{e^{-t_1^2/y^2} + e^{-1/(1-y/t_1)^2}}$$

extends smoothly with $g_a(0) = (1 - \kappa_0)e > 0$. Taking $j = 0$ in the flat-integral formula proves

$$\log \phi - \log \phi_r, \quad U - U_r \in e_a y^3 C^\infty.$$

This class is preserved by smooth compositions, products with smooth functions, and division by nonvanishing fields: for example $(e^{e_a y^3 b} - 1)/(e_a y^3) = b \int_0^1 e^{v e_a y^3 b} dv$ is smooth. Moment and pressure differences acquire a second integration and belong to $e_a y^6 C^\infty$. Substitution into the exact moment formulas therefore shows

$$T_0 = e_a B_0(y, \eta), \quad B_0 \in C^\infty, \quad B_0(0, \eta) = F(X_0, \eta) p_{s,r}(X_0, \eta) \neq 0.$$

At the endpoint the latter vector is a positive multiple of the limiting shear. Its component along $(1, t_s)$ is positive, and that along $(-t_s, 1)$ is zero. Both cone-direction margins consequently stay positive on a smaller collar. Every derivative of the exponential adds only finitely many powers of y^{-1} , so for every fixed mixed derivative

$$|T_0| \geq c e^{-t_1^2/y^2}, \quad |\partial^\alpha T_0| \leq C_\alpha e^{-t_1^2/y^2} y^{-N_\alpha}.$$

Since $X_0 > 0$, converting y -derivatives to X -derivatives changes only fixed smooth coefficients. Thus the vector direction is smooth through the zero-stress edge, not merely convergent along individual paths.

37.3 3. Continuation to 110 and preservation of parameter analyticity

After activation retain the same shear prescription with $\kappa = \kappa_0$ through the reference cutoff and then through $X = 100$. The natural reference slopes are nonzero only in its initial interval of length $2t_1$. Integrating the difference on that interval gives $O(t_1)$. Beyond it, the reference slopes vanish, and the new slopes have size at most κ_0 times the bounded functions $p_{1,r}/2$, $Xn_{s,r}/2$. Integration on the fixed interval gives $O(\kappa_0)$. Thus field values, logarithms and the required parameter derivatives differ from reference by $O(t_1 + \kappa_0)$. Their integrated coefficients obey the same estimate with finite constants depending on the already fixed C, Λ .

Choose κ_0 small enough that $\kappa_0 V_{\max} < 1/4$ and that its comparison errors use at most one quarter of the minimum $v_r - 2$. Then choose t_1 small for the other error quarters and for the activation inequalities. The same exact slope formula yields $P_c > 2 + c/2$, $v_s < 1$. The relaxed cone therefore holds regardless of J_c . This order works for the entire family of reference widths, avoiding a circular choice of κ_0 from a width that itself depends on κ_0 .

Starting at $X = 100$, multiply the axial prescription by a smooth β from one to zero on a short logarithmic interval. The integrated coefficients change by arbitrarily small errors when the interval is short, while

$$P_c = p_{1,r} + \beta p_{2,r}^2 / p_{1,r} + o(1) > 2, \quad v_s < 1,$$

using $p_{1,r} > 3$ there. Then hold $D_X U = 0$ and interpolate a convexly from $\kappa_0 p_{1,r}$ to 0.8, after arranging $\kappa_0 \sup p_{1,r} < 0.8$. During this second interval $v_s = a \leq 0.8$, $P_c = p_1 > 2$ by continuity and its strictly positive initial margin. Both intervals can fit strictly between 100 and 110.

On the final interval, $a = 0.8$, $l = 0.6$, $D_X U = 0$. The unchanged leading gradient and the already small field changes give $S_q > 1$: its constant source contribution is $0.6(-W_*) > 1.68$, while the large ΛL_X term absorbs the previously identified weighted errors. At a hypothetical first downward crossing $p_1 = 2$,

$$D_X p_1 = X S_q / L - 0.6 p_1 > X - 1.2 > 0,$$

a contradiction. Hence $p_1 > 2$ persists through $X_i = 110$.

On a small first collar, all reference coefficients arise from the analytic axis solution, its derivatives and forward integrals. The only cutoffs depend on y , not on η . Choose a compact complex neighborhood on which the natural Φ , L , and the relevant denominators do not vanish. Integration in the real radial variable preserves holomorphy with uniform bounds on that neighborhood. Exponentiating the prescribed logarithm preserves nonvanishing. Radial differentiation changes the cutoff bounds but does not change the holomorphic parameter domain. A smaller fixed $t_c \in (0, t_1)$, with $4e^{t_c} < 4.1$, therefore gives the common parameter domain for $F, U, V_0/X, \Pi$ and every fixed radial derivative throughout $[0, X_0 e^{t_c}]$. Subsequent changes have later support, so their forward primitives leave this rectangle exactly unchanged.

37.4 4. Transition length fixed before amplitude

Let $\ell_i = \log(CE(X_i, \eta))$, $G_i = U(X_i, \eta)$, and $L_0 = \log(X_i/X_0)$. Write M_k, N_k for the reference bounds on $p_{1,r}, n_{s,r}$, which are independent of subsequent C . Before the final interpolation the logarithmic ϕ slope is $-\kappa p_{1,r}/2$; during it the slope is a convex interpolation with -0.4 ; after it the slope is exactly -0.4 . Hence

$$\|\log \phi(X_i)\|_{C_\eta^k} \leq \|\log \phi_* + \log \Phi(4, \cdot)\|_{C_\eta^k} + \frac{1}{2}(M_k + 0.8)L_0.$$

Adding $\frac{1}{2} \log(2X_i)$ proves $\|\ell_i\|_{C_\eta^k} \leq B_k$, where B_k is independent of subsequent C and smaller transition widths. The axial change on activation is at most $X_0 e^{\bar{t}} N_k t_1 / 2$; the remaining small-shear part contributes at most $X_i N_k \kappa_0 / 2$. The natural-axis difference is $O(\Lambda^{-1})$ with a constant independent of Λ . Therefore

$$\|G_i - U_*\|_{C_\eta^k} \leq C_k^{\text{nat}} / \Lambda + C_k^{\text{join}}(\Lambda)(t_1 + \kappa_0 + \omega_{\text{fin}}),$$

where ω_{fin} is the sum of the two final logarithmic widths. No inverse cutoff width occurs in these integrals.

With $f = (1 + \eta^2)^{-1}$, choose a finite positive T_{sh} satisfying exactly the source lower bound

$$T_{\text{sh}} \geq 20 \|\sigma'\|_\infty (B_0 + \|\log f\|_\infty).$$

For $y_i = \log(X/X_i)$, prescribe

$$\log E = -\log C + y_i/10 + (1 - \sigma(y_i/T_{\text{sh}}))\ell_i + \sigma(y_i/T_{\text{sh}})\log f, \quad U = G_i.$$

At the left endpoint this matches $E(X_i)$ and its radial jet, because both original and new profiles have $D_X \log E = 0.1$ there and the step is flat. At the right endpoint it matches $C^{-1}f(X/X_i)^{1/10}$ with its entire radial jet. Direct differentiation gives

$$l = 0.6 + T_{\text{sh}}^{-1} \sigma'(y_i/T_{\text{sh}})(\log f - \ell_i) \in [0.55, 0.65], \quad a \in [0.7, 0.9], \quad b_s = 0.$$

The old parameter gradient satisfies $-H_c \ell'_i \geq \Lambda L \chi - K \chi - K_{\sigma_*} \sqrt{\chi} - K/\Lambda - o(1)$. The new one is

$$-H_c(\log f)' = \frac{2\eta H_c}{1 + \eta^2} \geq -K j_0^2 - K/\Lambda - o(1).$$

For the latter bound, the exact identity $\eta H_* = (D + 4d)\eta^2 + dj_0\eta$ and completion of the square bound its negative part by $K j_0^2$; the $U - U_*$ error is already controlled. These gradients are combined with nonnegative coefficients summing to one. The old large positive term absorbs its correspondingly weighted errors, and the new negative part is small. Since $-W$ is close to $-W_* > 2.8$ and $l \geq 0.55$, the constant contribution exceeds 1.54 before the small h, j_0, Λ^{-1} errors. Thus $S_q > 1$. The same crossing argument at $p_1 = 2$, now $l \leq 0.65$, gives $D_X p_1 > X - 1.3 > 0$. Since $b_s = 0$, this proves $P_c = p_1 > 2$, $v_s = a < 1$, hence the relaxed cone, along the full transition and the following ideal-angular interval.

37.5 5. The five matching equations, including all scaling factors

Set

$$X_R = 110(CP_*)^{10}, \quad x = X/X_R, \quad X_{\text{sep}} = 110e^{T_{\text{sh}}}, \quad x_{\text{sep}} = \frac{e^{T_{\text{sh}}}}{(CP_*)^{10}}.$$

Here X_{sep} is fixed before C . The ideal pair is $U_0 = 4\eta$, $E_0 = P_* f x^{1/10}$. The exact normalized moments are

$$\begin{aligned} \widehat{M}_0 &= 4\eta x, \quad \widehat{I}_0 = \frac{5\sqrt{2}}{8} P_* f x^{8/5}, \quad \widehat{J}_0 = 4\eta \widehat{I}_0, \\ \widehat{S}_0 &= 16\eta^2 x - \frac{5}{12} P_*^2 f^2 x^{6/5}, \quad C_{p,0} = \frac{5}{2} P_*^2 f^2 x^{1/5}. \end{aligned}$$

These follow by direct integration, retaining the factors

$$(M, I, J, S, C_p) = (X_R \widehat{M}, X_R^{3/2} \widehat{I}, X_R^{3/2} \widehat{J}, X_R \widehat{S}, C_p).$$

On $[0, X_{\text{sep}}]$, fixed parameter derivatives of U and CE are bounded independently of later C ; near zero $CE = O(\sqrt{X})$. Thus

$$\|\widehat{M}\|_{C^k} + \|\widehat{S}\|_{C^k} \leq K_k x_{\text{sep}}, \quad \|\widehat{I}\|_{C^k} + \|\widehat{J}\|_{C^k} \leq K_k C^{-1} x_{\text{sep}}^{3/2}, \quad \|C_p\|_{C^k} \leq K_k C^{-2}$$

at $x = x_{\text{sep}}$. For pressure the proof uses the physical integral $C^{-2} \int_0^{X_{\text{sep}}} (CE)^2/(2X) dX$, finite because $CE = O(\sqrt{X})$. All constants are fixed before increasing C . The ideal moments also tend to zero with the displayed exact powers. Their pressure increment is $O(C^{-2})$, because $x_{\text{sep}}^{1/5} = e^{T_{\text{sh}}/5} (CP_*)^{-2}$.

Beyond X_{sep} , the angular profile is exactly ideal, since $C^{-1}f(X/X_i)^{1/10} = P_* f x^{1/10}$. Choose C so $x_{\text{sep}} < e^{-8}$. Restore G_i to 4η by a fixed smooth step on $-8 < \log x < -7$, retaining E_0 . The size of this change, and its fixed logarithmic radial and parameter derivatives, is bounded by fixed constants times $\|G_i - 4\eta\|$. The moment errors accumulated to the correction interval are bounded by

$$K_k \|G_i - 4\eta\|_{C_\eta^{k+1}} + o_C(1).$$

Every relevant radial weight is integrable at zero: their actual exponents are $0, 3/5, 1/5$ and $-4/5$ for the pressure integrand, all greater than -1 . No divergent contribution is discarded.

On $-6 < \log x < -5$, choose two fixed ordered nonnegative bumps for U , three for E . At the ideal pair make the invertible row operations $J \mapsto J - 4\eta I$, $S \mapsto S - 8\eta M$, with 4η held at its base value. The linearized axial block has weights $1, f x^{3/5}$; the azimuthal block has weights $x^{1/2}, f x^{1/10}, f x^{-9/10}$, retaining fixed nonzero factors $P_*, \sqrt{2}$ in the matrix. The exponents in each block are distinct. The Chebyshev determinant proof and quadratic contraction in `core_exact.md` therefore solve all five moment equations exactly, with small smooth coefficients. The row operations do not impose that the corrected U remain constant; the full nonlinear differences still include $\delta U \delta E$, $(\delta U)^2$, and $(\delta E)^2$.

To verify one tolerance can precede C , put $R = [e^{-8}, e^{-5}]$, $Q_{\min} = \min Q_{s,0} > 0$, $e_* = \min E_0 > 0$, and $w_* = 1 + \max |N_{s,0}/(E_0 Q_{s,0})|$ on $R \times [-1, 1]$. The ideal source is positive directly: for this pair

$$S_{q,0} = \frac{9}{5} - h + \frac{16}{5}h\eta^2 + \frac{2\eta^2(D+4d)}{1+\eta^2} > 0,$$

as $W = -3 + 8h\eta^2$, $l = 3/5$, $H_c = (D+4d)\eta$. Its regular weighted primitive gives $Q_{s,0} > 0$. The compact minima thus exist without importing the positivity of a separate abbreviated formula. The exact moment map has no X_R in Q_s, N_s , so a fixed small tolerance in the fields and normalized moments through one extra parameter derivative ensures

$$Q_s \geq Q_{\min}/2, \quad E \geq e_*/2, \quad |a - 0.8| \leq 0.1, \quad |N_s/(EQ_s)| \leq w_*, \quad |b_s| \leq 0.1/(1+w_*).$$

Then, retaining the source numbers,

$$v_s \leq 0.9 + 0.01/0.7 < 1, \quad G := Q_s - b_s N_s/(aE) \geq (1 - 1/7)Q_s \geq 3Q_{\min}/7, \quad P_c = X_R x G/L.$$

Choosing X_R large now gives $P_c > 2$ without shrinking the tolerance. First choose j_0 , then Λ , so $j_0 + C_k^{\text{mat}}/\Lambda$ fits the tolerance. Next fix the B_k and T_{sh} , then enlarge C for the vanishing moment contributions. Finally decrease $\kappa_0, t_1, \omega_{\text{fin}}$ for the remaining G_i error. These choices prove the exact order

$$(M_d, T_d, P_*, \lambda, h) \rightarrow \text{moment tolerance}, j_0 \rightarrow \delta_*, \sigma_*, \Lambda \rightarrow (B_k), T_{\text{sh}} \rightarrow C, X_R \rightarrow \kappa_0, t_1, \text{final widths}.$$

Only finitely many parameter derivative orders need to be small. Smooth implicit differentiation supplies all remaining fixed orders after the choices. At $X_h = X_R e^{-5}$, all five moments and both fields agree with the ideal pair, with the same axis pressure Π_0 , so the exact exterior-preservation map from `core_exact.md` applies.

38 Exterior stress, exact flat factors, and the order-one edge estimate

38.1 Convergent moment subtraction and backward stress formulas

Apply this calculation to the actual regular joined field constructed above. The five-row join has proved all the required total moments. In particular, $M(\infty) = J(\infty) = S(\infty) = 0$ and the compensated subtracted angular moment gives

$$I(X) = \frac{XH_{\text{pow}}(X)}{1-h} - \int_X^\infty (H - H_{\text{pow}})dx, \quad S(X) = \frac{1}{2} \int_X^\infty E^2 dx.$$

The exact pressure is the backward integral already established. These identities, substituted into the moment formulas, determine the same forward stress everywhere in the tail.

To prove the backward physical stress representation rather than assume its boundary constants vanish, put $K_{\text{pow}} = c_\infty(r^2/2)^{-A}$. For $r \rightarrow \infty$, uniformly on bounded $\tau \geq 0$ intervals, the heat expansion gives

$$K - K_{\text{pow}} = O_h(\tau r^{-3-2h}), \quad K_t = O_h(r^{-3-2h}), \quad r^2 K_r - rK = O_h(r^{-2h}).$$

Thus the weighted difference and its time derivative are dominated at infinity by Cr^{-1-2h} , whose integral from R_* is $R_*^{-2h}/(2h)$. At zero the subtracted pure power has weight r^{1-2h} , integrable for $h < 1$, while the regular field is integrable by axis smoothness. The substitution $r = \sqrt{q}R$, $dX = R dR$ proves exactly

$$\int_0^\infty r^2(u_\theta - K_{\text{pow}})dr = q^{3/2-A} \int_0^\infty (H - H_{\text{pow}})dX = 0.$$

Time differentiation is justified by the displayed integrable majorant on compact physical parameter neighborhoods, proving (PA.SUB) and the zero integrated time term. The angular transport flux has zero axial integral because it is $q^{3/2-2A}J(\infty) = 0$; its radial boundary vanishes by regularity at zero and $U = M = V_0 = 0$ in the exterior. The radial viscous boundary is zero by the r^{-2h} estimate above.

For the axial component $\int r u_z dr = q^{1-A}M(\infty) = 0$. Fubini with the absolutely convergent positive pressure integrand yields

$$\int_0^\infty r p(r) dr = - \int_0^\infty \frac{u_\theta(r')^2}{r'} \left(\int_0^{r'} r dr \right) dr' = -\frac{1}{2} \int_0^\infty r' u_\theta(r')^2 dr'.$$

The pressure has decay $O(r^{-2-4h})$, hence $r^2 p \rightarrow 0$, and its contribution to the integrated axial flux is exactly $q^{1-2A} S(\infty) = 0$. All axial radial transport and viscosity boundaries vanish as well. Therefore both leading residuals have zero weighted totals, so the stresses defined with a minus sign by integration from the axis are exactly

$$T_\theta(r) = r^{-2} \int_r^\infty r'^2 \mathcal{R}_\theta^{(0)}(r') dr', \quad T_z(r) = r^{-1} \int_r^\infty r' \mathcal{R}_z^{(0)}(r') dr'. \quad (\text{PA.BACK})$$

For $X \geq X_b$ the field is $(u_r, u_\theta, u_z) = (0, K, 0)$ and pressure is its z independent backward radial integral. The heat equation makes both leading residuals zero there, proving zero exterior stress.

38.2 An exact flat integral map

For $c > 0$, integer $j \geq 0$, and a smooth $b(u, \eta)$ on a closed compact parameter rectangle, substitute $u = \delta(1 + \delta^2 v)^{-1/2}$. Then $du = -\frac{1}{2} \delta^3 (1 + \delta^2 v)^{-3/2} dv$ and $c/u^2 = c/\delta^2 + cv$. Hence

$$\int_0^\delta e^{-c/u^2} u^{-j} b(u, \eta) du = \frac{1}{2} e^{-c/\delta^2} \delta^{3-j} B(\delta, \eta), \quad (\text{PA.FLAT})$$

where the coefficient is explicitly

$$B(\delta, \eta) = \int_0^\infty e^{-cv} (1 + \delta^2 v)^{(j-3)/2} b(\delta/\sqrt{1 + \delta^2 v}, \eta) dv.$$

Every fixed derivative of the integrand is bounded by e^{-cv} times a polynomial in v , uniformly in the compact parameters and $0 \leq \delta \leq \delta_0$. This follows directly from differentiating $\delta(1 + \delta^2 v)^{-1/2}$; its denominators are at least one and all its numerators are polynomials in δ, v . Dominated differentiation proves smoothness of B with all fixed derivatives bounded, and $B(0, \eta) = b(0, \eta)/c$. This proves the exact map in [NS, Lemma A.9], including the factor $1/2$.

38.3 Positive angular stress and the limiting direction

For terminal $y \geq .5$, the heat cutoff is one, so $u_\theta = Kf$, $f = f_o(y)$, and $u_r = u_z = 0$. At fixed z, t , $y_r = 2/r$, while $y_t = (qL)^{-1}$, $y_z = -2\eta/(q^D L)$. The leading angular residual, which omits axial viscosity, is exactly

$$\mathcal{R}_\theta^{(0)} = Kf'/(qL) - K(f_{rr} + r^{-1}f_r) - 2K_r f_r.$$

Differentiate the physical pressure $p = -\int_r^\infty K(r')^2 f(r')^2 dr'/r'$. Since $K_z = 0$ and $dr'/r' = dy'/2$, this gives the sign and factor

$$p_z(r) = \frac{2\eta}{q^D L} \int_y^3 K(y')^2 f(y') f'(y') dy'.$$

Inserting the angular residual into (PA.BACK) and integrating its viscous terms by parts gives

$$T_\theta = Kf_r + \frac{1}{r^2} \int_r^\infty (r'K - r'^2 K_{r'}) f_{r'} dr' + \frac{1}{qLr^2} \int_r^\infty r'^2 K f' dr'. \quad (\text{PA.POS})$$

To check the integration sign, the term $-\int r'^2 K f_{r'} dr'$ contributes

$$r^2 K f_r + \int (2r'K + r'^2 K_{r'}) f_{r'} dr'.$$

The two remaining viscous terms leave exactly $r'K - r'^2 K_{r'}$ in this integral. Every term in (PA.POS) is nonnegative since $K > 0$, $K_r < 0$, $f' \geq 0$. The remaining interval has logarithmic length at most 2.5; radii and heat factors there have uniformly bounded ratios. Since $\int_y^3 f' dv = \rho_o \psi_o(y)$, its last term implies

$$T_\theta \geq c \frac{rK}{qL} \rho_o \psi_o(y) > 0 \quad (y < 3).$$

The pressure formula and its next radial integral similarly give

$$|p_z| \leq Cq^{-D} K^2 \rho_o \psi_o, \quad |T_z| \leq C r q^{-D} K^2 \rho_o \psi_o, \quad |T_z/T_\theta| \leq C q^{1-D} K = C E_{\text{pow}} \mathcal{H}(2d/X).$$

The final equality uses $1 - D = A$. The h^6 amplitude reduction makes this ratio small. Furthermore, since $\partial_y Z = -Z$,

$$a = 2 + 2h + 2Z\mathcal{H}'/\mathcal{H} - 2f'/f.$$

For sufficiently large X_R , $-Z\mathcal{H}'/\mathcal{H} < h/4$ uniformly in η ; also $f'/f < h/4$ by construction. Hence $2 + h < a \leq 2 + 2h$ and $b_s = 0$. The exact stress identities are $P_c - a = T_{0,\theta}/F > 0$, $J_c = T_{0,z}/F$ with $F = E/\sqrt{2X} > 0$. Thus the cone is exactly $(a - 2)(T_z/T_\theta)^2 < 2$, with uniform strict slack throughout the terminal interval. This also covers the plateau $.5 \leq y \leq 1$: the positive backward time integral persists there even though $f'(y) = 0$.

Near $\delta = 3 - y = 0$, the specified step has the exact factorization

$$\psi_o = e^{-4/\delta^2} g(\delta), \quad g(\delta) = \frac{1}{e^{-(1-\delta/2)^{-2}} + e^{-4/\delta^2}}, \quad g(0) = e,$$

$$f' = \rho_o e^{-4/\delta^2} \delta^{-3} [8g(\delta) + \delta^3 g'(\delta)].$$

The normalized boundary term in (PA.POS) is

$$q^{A+1/2} K f_r = \frac{2E_{\text{pow}}(X) \mathcal{H}(2d/X)}{\sqrt{2X}} f'.$$

The two remaining angular integrals contain f' , so (PA.FLAT) with $c = 4, j = 3$ makes each e^{-4/δ^2} times a smooth coefficient. Relative to the boundary factor $e^{-4/\delta^2} \delta^{-3}$ they vanish to at least order δ^3 . Therefore

$$T_{0,\theta} = e^{-4/\delta^2} \delta^{-3} b_\theta(\delta, \eta), \quad b_\theta(0, \eta) = \frac{16\rho_o E_{\text{pow}}(X_b)}{\sqrt{2X_b}} \mathcal{H}(2d/X_b) g(0) > 0.$$

The heat factor has a positive minimum on $[0, 2/X_b]$, so b_θ has a positive lower bound on a sufficiently short closed collar uniformly for $|\eta| \leq 1$. The pressure integral contains one f' factor; the same substitution gives $p_z = q^{-D-2A} e^{-4/\delta^2}$ times a smooth coefficient. Its further radial integration for T_z uses (PA.FLAT) with $j = 0$, giving

$$T_{0,z} = e^{-4/\delta^2} \delta^3 b_z(\delta, \eta).$$

The powers of q cancel exactly: $q^{A+1/2} q^{1/2} q^{-D-2A} = q^{1-D-A} = 1$. The remaining coefficients depend only on $X, \eta, \mathcal{H}(2d/X), L^{-1}$, with their already proved closed-parameter smoothness. Consequently

$$T_{0,z}/T_{0,\theta} = \delta^6 b_z/b_\theta \longrightarrow 0$$

smoothly, and the stress direction extends to $(1, 0)$ at the outer edge. Every fixed profile derivative costs only a finite inverse power of δ , and $|T_0| \geq c e^{-4/\delta^2} \delta^{-3}$ there. This proves (A.48)–(A.51) including the precise leading coefficient and the independence from the physical scale q .

39 Admissible shear loop, radial modulation, and moment restoration

39.1 Scalar cone and its four gaps

At $a > 0$, set

$$t_s = -b_s/a, \quad v_s = a(1 + t_s^2), \quad P_c = p_{s,1} + t_s p_{s,2}, \quad J_c = p_{s,2} - t_s p_{s,1}.$$

The relaxed condition is $P_c > 2, v_s < U(P_c, J_c)$, with

$$U(P, J) = P + \frac{J^2}{4} - |J| \sqrt{\frac{P-2}{2} + \frac{J^2}{16}}.$$

The admissible condition also requires $v_s > 2$. The quadratic expression

$$G(v) = 2(P - v)^2 - (v - 2)J^2$$

has roots $U(P, J)$ and the same expression with a plus before the square root. For $P > 2, U > 2$: writing $w = P - 2 > 0$, the inequality

$$\left(w + \frac{J^2}{4}\right)^2 - J^2 \left(\frac{w}{2} + \frac{J^2}{16}\right) = w^2 > 0$$

proves $w + J^2/4 > |J|\sqrt{w/2 + J^2/16}$. Also $U \leq P$, because the square-root term is at least $J^2/4$, with equality only for $J = 0$. When $J = 0$, $U = P > 2$. Therefore for $v > 2$, admissibility is exactly

$$P - v > 0, \quad 2(P - v)^2 - (v - 2)J^2 > 0.$$

The four continuous smooth gaps on $a > 0$ are a , $v_s - 2$, $P_c - v_s$, and $2(P_c - v_s)^2 - (v_s - 2)J_c^2$. Positivity on a compact parameter set gives one positive lower margin for all four. The root U need only be continuous; no differentiation of its $|J|$ term at zero is used.

39.2 A smooth loop whose average is exactly the original shear

Use the actual joined data on the compact rectangle $I \times [-1, 1]$ chosen in Section 29; the preceding construction proves its relaxed condition and admissibility on both radial boundary collars. The positive minimum of $P_c(t_s) - 2$ permits a constant

$$0 < d_0 < \frac{1}{2} \min(P_c(t_s) - 2).$$

Here $P_c(t) = p_{s,1} + p_{s,2}t$ and $J_c(t) = p_{s,2} - p_{s,1}t$ for a variable slope t . With normalized θ' -average on $[0, 2\pi]$, define

$$M_e(z) = \frac{1}{2\pi} \int_0^{2\pi} e^{z \sin \theta'} d\theta', \quad t(\theta'; \mu) = t_s + \frac{d_0}{p} \left(\frac{e^{\mu p \sin \theta'}}{M_e(\mu p)} - 1 \right), \quad p = p_{s,2}.$$

For $p = 0$ the quotient is defined by continuity. Its numerator is smooth in p, μ, θ' , zero at $p = 0$, and the identity

$$\frac{G(p)}{p} = \int_0^1 G'(up) du$$

proves joint smoothness, including all parameters and derivatives. Since $M_e(0) = 1, M_e'(0) = 0$, the value there is $t = t_s + d_0 \mu \sin \theta'$.

The definitions give

$$\langle t \rangle_{\theta'} = t_s, \quad P_c(t) = P_c(t_s) + d_0 \left(\frac{e^{\mu p \sin \theta'}}{M_e(\mu p)} - 1 \right) \geq P_c(t_s) - d_0 > 2.$$

The variance is

$$V(\mu, p) = \langle (t - t_s)^2 \rangle = \frac{d_0^2}{p^2} \left(\frac{M_e(2\mu p)}{M_e(\mu p)^2} - 1 \right),$$

with value $d_0^2 \mu^2 / 2$ at $p = 0$. To show strict increase, let $g(z) = \log M_e(z)$. Differentiating under the finite integral gives

$$g''(z) = \langle (\sin \theta' - \langle \sin \theta' \rangle_z)^2 \rangle_z > 0,$$

where the tilted probability density is $e^{z \sin \theta'} / (2\pi M_e(z))$. It is strictly positive on the circle and $\sin \theta'$ is not constant, so the variance is strictly positive. For $p \neq 0, \mu > 0$,

$$\partial_\mu(g(2\mu p) - 2g(\mu p)) = 2p(g'(2\mu p) - g'(\mu p)) > 0.$$

For $p > 0$ both factors on the right are positive; for $p < 0$ both are negative. Thus V strictly increases in $\mu > 0$, including its separately proved formula at $p = 0$.

Here is the required growth bound, without an unproved asymptotic invocation. For $z \geq 1$, write $\theta' = \pi/2 + x$ near the maximum. On a fixed small interval $|x| \leq a < \pi$, there are $c_1, c_2 > 0$ with $1 - c_2 x^2 \leq \cos x \leq 1 - c_1 x^2$. A lower bound integrates over $|x| \leq z^{-1/2} \min(a, 1)$, giving $ce^{zz^{-1/2}}$. An upper bound on the same interval uses the Gaussian integral and gives $Ce^{zz^{-1/2}}$. On its complement $\sin \theta' \leq 1 - c_a$, so the contribution is at most $e^{(1-c_a)z}$, also bounded by $Ce^{zz^{-1/2}}$. For $z < 0$, shift θ' by π to obtain $M_e(z) = M_e(-z)$. Hence

$$ce^{|z|}/\sqrt{|z|} \leq M_e(z) \leq Ce^{|z|}/\sqrt{|z|} \quad (|z| \geq 1).$$

Taking the ratio yields positive constants bounding $M_e(2z)/M_e(z)^2$ above and below by multiples of $\sqrt{|z|}$. Therefore $V(\mu, p) \rightarrow \infty$ for each fixed $p \neq 0$; for $p = 0$ its explicit quadratic formula proves the same.

The compact range of p admits a uniform finite $\mu_{\max}$ such that $V(\mu_{\max}, p) > 3/\min a$. To prove uniformity, for each p_0 select a finite μ_{p_0} giving the strict bound there; continuity gives a neighborhood in

p ; a finite subcover gives finitely many such μ , and their maximum works by monotonicity. This step is valid at $p = 0$, so no divergent factor p^{-2} is being used near zero.

All slopes $t(\theta'; \mu)$ for $0 \leq \mu \leq \mu_{\max}$ form a compact smooth family with $P_c(t) > 2$. Continuity of U and the strict inequality $U(P, J) > 2$ established above give a positive minimum of $U - 2$. Choose $0 < \delta_L < 1$ smaller than that minimum, and smaller than the positive minimum of $v_s - 2$ on chosen smaller radial boundary collars.

Choose a smooth $0 \leq \zeta_L \leq 1$ as a function of v_s , equal to one for $v_s \leq 2 + \delta_L/8$ and zero for $v_s \geq 2 + \delta_L/4$. Put

$$v_* = 2 + \delta_L/2, \quad \rho = \zeta_L(v_s)^2(v_* - v_s), \quad v = v_s + \rho,$$

and define $\rho = 0$ where $\zeta_L = 0$. On its possible nonzero support, $v_* - v_s \geq \delta_L/4$, so

$$\sqrt{\rho} = \zeta_L(v_s)\sqrt{v_* - v_s}$$

extends smoothly by zero. Since $v_s > 0$, $0 \leq \rho < v_* < 3$.

There is a unique $0 \leq \mu < \mu_{\max}$ solving $V(\mu, p) = \rho/a$, by the preceding strict monotonicity and endpoint bounds. Its smoothness at $\rho = 0$ needs a distinct verification. Expanding the finite integral in its convergent power series gives $M_e(z) = 1 + z^2/4 + O(z^4)$, and therefore

$$V(\mu, p) = \frac{1}{2}d_0^2\mu^2 + O(\mu^4p^2)$$

uniformly for bounded p . Thus the signed square root is $\mu(d_0/\sqrt{2} + O(\mu^2p^2))$, a smooth odd function of μ near zero with positive derivative $d_0/\sqrt{2}$. Applying the implicit-function theorem to this function and $\sqrt{\rho/a}$, already proved smooth, gives smooth μ there. For $\rho > 0$, strict positivity of $\partial_\mu V$ gives the same conclusion. On the overlap the two local inverses agree by uniqueness.

The three cutoff regimes show $v > 2$ everywhere. Where $\zeta_L \neq 0$, $v \leq v_* < 2 + \delta_L < U(P_c(t), J_c(t))$. Where $\zeta_L = 0$, $\mu = 0, t = t_s, v = v_s > 2$, and the original relaxed condition gives $v < U$. Thus each proposed state lies in the admissible cone.

Define a lifted phase coordinate by

$$\phi(0) = 0, \quad \frac{d\phi}{d\theta'} = \frac{a(1+t^2)}{2\pi v}, \quad (a_L, -b_L) = \frac{v(1, t)}{1+t^2}.$$

Since $\langle t \rangle = t_s$ and $\langle (t - t_s)^2 \rangle = V$,

$$a\langle 1+t^2 \rangle = a(1+t_s^2) + aV = v_s + \rho = v.$$

Therefore $\phi(\theta' + 2\pi) = \phi(\theta') + 1$. Its derivative has a uniform positive lower bound on the compact parameter set; it defines an orientation-preserving circle diffeomorphism with a smooth parameter-dependent inverse. Changing variables proves both original mean components exactly:

$$\int_0^1 a_L d\phi = \frac{a}{2\pi} \int_0^{2\pi} 1 d\theta' = a, \quad \int_0^1 (-b_L) d\phi = \frac{a}{2\pi} \int_0^{2\pi} t d\theta' = at_s = -b_s.$$

Also $-b_L/a_L = t$, $a_L(1+t^2) = v$, so the verified scalar cone inequalities are the desired vector cone gaps. At boundary collars $\zeta_L = 0$, and the reparametrized shear is exactly the original constant-in- ϕ shear. Compactness gives one $\kappa_L > 0$ for all four cone gaps. This completes the complete loop construction, including the mean, boundary, and smoothness claims.

39.3 Exact signed square root at zero variance

The following identity also proves the preceding local smoothness without an asymptotic remainder. Define

$$R_e(z) = M_e(2z)/M_e(z)^2, \quad H_e(z) = (R_e(z) - 1)/z^2 \quad (z \neq 0), \quad H_e(0) = 1/2.$$

The convergent power series of the finite circle integral gives an even real analytic M_e , with $M_e(z) = 1 + z^2/4 + O(z^4)$. Its positive real values give an even real analytic R_e , and multiplication of the power series gives $R_e(z) = 1 + z^2/2 + O(z^4)$. Thus the quotient H_e extends real analytically and evenly through zero. For $z \neq 0$, Cauchy-Schwarz gives the strict inequality

$$\langle e^{z \sin \theta'} \rangle^2 < \langle e^{2z \sin \theta'} \rangle \langle 1 \rangle = M_e(2z).$$

Equality would require $e^{z \sin \theta'}$ to be constant almost everywhere; continuity and the positive uniform measure make that impossible for $z \neq 0$. Consequently $H_e(z) > 0$ on the entire real line. Exactly,

$$V(\mu, p) = d_0^2 \mu^2 H_e(\mu p), \quad F_e(\mu, p) = d_0 \mu \sqrt{H_e(\mu p)}.$$

The latter function is a globally smooth signed square root. At zero its μ derivative is $d_0/\sqrt{2}$ for every p . For $\mu > 0$ it is positive, and $2F_e \partial_\mu F_e = \partial_\mu V > 0$, so the derivative stays positive. The equation $F_e(\mu, p) = \sqrt{\rho}/\sqrt{a}$ has a smooth implicit solution, including points where $p = \rho = 0$, because $\sqrt{\rho} = \zeta_L(v_s) \sqrt{v_* - v_s}$ is the smooth zero extension already proved above. The strict variance monotonicity identifies all local solutions with the same globally defined μ . This proves every required parameter derivative at zero variance as well as at $p = 0$.

39.4 Exact insertion into the unchanged original radial coordinate

Let $\mathcal{A}, \mathcal{B}$ be the zero-mean periodic antiderivatives

$$\partial_\phi \mathcal{A} = -\frac{1}{2}(a_L - a), \quad \partial_\phi \mathcal{B} = \frac{1}{2}E(b_L - b_s).$$

They exist because the right sides have zero ϕ -mean. One explicit construction for any zero-mean g is

$$\mathcal{I}g(\phi) = \int_0^\phi g(w) dw - \int_0^1 \int_0^s g(w) dw ds.$$

It is periodic, zero mean, smooth in all parameters, and has derivative g . It vanishes identically when $g \equiv 0$. Therefore these antiderivatives vanish on the boundary collars and extend smoothly by zero outside I .

For an integer $N > 0$, keep the source's exact definitions

$$E_N = E \exp(\mathcal{A}(X, \eta, N \log X)/N), \quad U_N = U + \mathcal{B}(X, \eta, N \log X)/N.$$

The modified E_N is positive without approximation. The chain rule in the logarithmic coordinate is $X \partial_X = D_X + N \partial_\phi$ after substitution. Hence

$$\begin{aligned} a_N &= 1 - 2X \partial_X \log E_N \\ &= a - 2D_X \mathcal{A}/N - 2\partial_\phi \mathcal{A} = a_L - 2D_X \mathcal{A}/N, \\ b_N &= \frac{2X \partial_X U_N}{E_N} \\ &= e^{-\mathcal{A}/N} \left(\frac{2D_X U + 2\partial_\phi \mathcal{B}}{E} + \frac{2D_X \mathcal{B}}{NE} \right) \\ &= e^{-\mathcal{A}/N} \left(b_L + \frac{2D_X \mathcal{B}}{NE} \right). \end{aligned}$$

The axial exponential denominator is retained; replacing it by one would destroy this exact shear formula.

On the fixed compact interval from X_- through the first reserved patch, $X, E, H = \sqrt{2X}E, L$ have positive lower bounds. The phase $N \log X$ is independent of η , so every fixed number m of η -derivatives of $E_N - E, U_N - U, a_N - a_L, b_N - b_L$ is bounded by $C_{r,m}/N$. For the exponential difference, use

$$e^{\mathcal{A}/N} - 1 = \frac{\mathcal{A}}{N} \int_0^1 e^{s\mathcal{A}/N} ds$$

and then differentiate under its fixed finite integral; all factors of $\mathcal{A}$ and its parameter derivatives are uniformly bounded on the periodic compact domain. For radial derivatives, $r \geq 1$ differentiations produce at most N^r from the phase and retain the initial N^{-1} , giving $C_{r,m}N^{r-1}$. Thus no small-radial-derivative assertion is used for the modulation.

39.5 All five moments and the dependence of the integrated stress

Retain exactly

$$M = \int_0^X U dx, \quad I = \int_0^X \sqrt{2x} E dx, \quad J = \int_0^X U \sqrt{2x} E dx,$$

$$S = \int_0^X (U^2 - E^2/2) dx, \quad C_p = \int_0^X E^2/(2x) dx, \quad \Pi = \Pi_0 + C_p.$$

On the support of the differences, all scalar weights are bounded, since it is a compact interval with positive left endpoint. For instance

$$U_N^2 - U^2 = (U_N - U)(U_N + U), \quad E_N^2 - E^2 = (E_N - E)(E_N + E).$$

These identities, the product rule, and integration of the uniform C_m/N profile bounds prove every fixed η -derivative of the full prefix moment difference is $O_m(N^{-1})$, uniformly in the endpoint X . The pressure datum is the same, so $\Pi_N - \Pi = C_{p,N} - C_p$ has exactly that bound.

The formulas whose continuity matters are

$$\begin{aligned} W &= 1 - \frac{2D\eta M + (1 - \eta^2)M_\eta}{X}, \\ Q_s &= -W + \frac{(1 - h)I - D\eta I_\eta - (1 - \eta^2)J_\eta + 2(h - D)\eta J}{XH}, \\ N_s &= -WU + \frac{D(M - \eta M_\eta) + 4h\eta S - (1 - \eta^2)S_\eta}{X} + 4A\eta\Pi - (1 - \eta^2)\Pi_\eta, \\ p_s &= (XQ_s/L, XN_s/(LE)). \end{aligned}$$

All denominators are bounded away from zero on the fixed interval. Differences of reciprocals are exactly

$$E_N^{-1} - E^{-1} = -(E_N - E)/(E_N E), \quad H_N^{-1} - H^{-1} = -(H_N - H)/(H_N H).$$

Applying the product rule to the displayed formulas proves the m -th parameter derivative bound for $p_{s,N} - p_s$ from the profile and moment bounds through order $m + 1$, with no radial derivative of the differences. This proves the precise assertion needed for the cone stability.

The compact loop has positive four-gap margin. Its values and the original p_s range in a compact subset of $a > 0$; take a fixed small neighborhood still in $a > 0$, on which the four-gap map has a finite Lipschitz constant. The just-proved $O(N^{-1})$ errors in a, b, p_s preserve each gap for sufficiently large finite N . Between I and the first correction patch, $E_N = E, U_N = U$, so the shears themselves equal the originals; the p_s error remains $O(N^{-1})$. The original strict admissible margins on that compact intervening interval are preserved by the same argument.

39.6 Nonlinear moment correction with an exact endpoint map

On the first reserved patch, the original profile is $U = 0, E = K(\eta)X^{-1/2-\lambda}$, where fixed $\lambda > 0$ and positive K have been selected before N . The earlier modulation is zero on this patch. Add two smooth compactly supported U -bumps and three E -bumps with five coefficient functions $c(\eta)$. Each block's bumps are nonnegative, nonzero, and supported on ordered disjoint intervals.

Differentiate the five original moment functionals at zero correction. In row order $(M, J; I, S, C_p)$, the two U -columns have weights $1, H = \sqrt{2X}E$, and the three E -columns have weights $\sqrt{2X}, -E, E/X$. The off-diagonal blocks are zero because $U = 0$ before the correction. Up to the displayed nonzero factors, the exponent lists are

$$(0, -\lambda), \quad (1/2, -1/2 - \lambda, -3/2 - \lambda).$$

All powers within each block are distinct for the fixed $\lambda > 0$. To establish invertibility directly, a nonzero linear combination of m distinct powers has at most $m - 1$ distinct positive zeros: divide by the least power, use Rolle's theorem on m putative zeros, and apply induction to the derivative, which has $m - 1$ distinct powers and nonzero coefficients wherever the original nonconstant coefficients were nonzero. Thus the evaluation determinant $\det[x_j^{\alpha_i}]$ never vanishes for ordered $0 < x_1 < \dots < x_m$. Its sign is constant on that connected set. Multilinearity and Fubini give

$$\det \left[\int x^{\alpha_i} \beta_j(x) dx \right] = \int \det[x_j^{\alpha_i}] \prod_j \beta_j(x_j) dx_1 \dots dx_m \neq 0.$$

The integrand has a constant nonzero sign on a set of positive measure. This proves each block's rank and hence the full moment Jacobian's rank. Smooth dependence on η , compactness of $[-1, 1]$, and the derivative identity for the inverse give uniform finite bounds at every fixed parameter order. No uniform estimate as $\lambda \rightarrow 0$ is claimed or needed.

The exact moment increment map is $B(\eta)c + Q_\eta(c, c)$, because M, I are linear, J is bilinear in U, E , and S, C_p are quadratic. Let $d_N(\eta)$ be the negative moment discrepancy accumulated before the patch; it obeys $\|d_N\|_{C^m} \leq C_m/N$. On the C^0 ball of radius $r = 2\beta_0\|d_N\|_{C^0}$, where $\beta_0 = \|B^{-1}\|_{C^0 \rightarrow C^0}$, the map

$$\mathcal{T}(c) = B^{-1}(d_N - Q(c, c))$$

maps the ball to itself and has Lipschitz constant at most $2\beta_0\kappa_0r \leq 1/2$, provided $8\beta_0^2\kappa_0\|d_N\|_{C^0} \leq 1$. Thus it has a unique fixed point selected by iteration from zero. The derivative $B + D_cQ(c, c)$ is invertible by the same estimate, giving smooth parameter dependence including one-sided endpoint derivatives. In each preselected finite C^m norm the identical contraction proof works after increasing N , with the relevant multiplication and bilinear norms, and yields C_m/N for the same branch.

The assertion that every other fixed derivative is $O_m(N^{-1})$ can also be justified without demanding simultaneous smallness of infinitely many norms. Differentiate the pointwise equation m times. The highest coefficient derivative is multiplied by the uniformly invertible $B + D_cQ(c, c)$. All other terms contain either a derivative of d_N , $O_m(N^{-1})$, or a lower derivative of c ; induction gives $O_m(N^{-1})$ for them, with the fixed data derivatives included in the constants. The same inverse bound therefore proves the estimate for the highest derivative. Only the finite orders needed for positivity and the cone gaps determine the actual choice of N .

The bump shapes and widths are fixed, so their radial derivatives multiply these small coefficients by fixed constants. The correction is therefore small in the radial derivatives entering a, b_s , in the first parameter derivatives entering p_s , and in all prefix moments. The compact input cone margin on the correction patch persists. Positivity of E there also persists, after one further finite lower bound on N .

At the right endpoint X_{rep} , the five corrected moments equal the original moments exactly as functions of η , while the corrected profiles equal the original profiles for every $X \geq X_{\text{rep}}$. Their derivatives as functions of η also agree. Each difference moment has zero derivative in X there, by the original moment integrands, so it stays zero at every later radius. The common axis pressure datum gives identical II. The displayed formulas for Q_s, N_s, p_s and the original incompressibility formula

$$V_0 = \frac{X}{L} (2\eta U - 2D\eta M/X - (1 - \eta^2)\partial_\eta(M/X))$$

then prove exact equality of these fields at every later radius. This is the exact morphism from the five-moment endpoint data to exterior pressure, velocity, and stress; it proves preservation of the terminal compensation, both unused reserved correction patches, and the entire exterior. The inner collar is unchanged because both modifications start later and pressure was integrated from the same axis datum.

39.7 Edge direction and the common weight

Appendix C leaves both original collars unchanged. The input edge facts it uses are: at the inner edge, with $y_a = \log(X/X_a)$, the stress is

$$T_0 = e^{-t_1^2/y_a^2} g_a(y_a) B_0(X, \eta), \quad g_a(0) > 0, \quad B_0(X_a, \eta) = Fp_{s,r} \neq 0;$$

its limiting direction is a positive multiple of the limiting shear $(a, -b_s) = a(1, t_s)$. At the outer edge, with $y_b = \log(X_b/X)$, the component factorization has common exponential e^{-4/y_b^2} , angular prefactor $y_b^{-3}b_\theta$ with $b_\theta(0, \eta) > 0$, and the direction

$$n = \frac{(b_\theta, y_b^6 b_z)}{\sqrt{b_\theta^2 + y_b^{12} b_z^2}}.$$

The inner continuation and outer endpoint calculations above prove these smooth factors and all fixed-order derivative bounds for the constructed field. Their exact use is as follows.

The positive factors cancel in $n = T_0/|T_0|$, giving a smooth direction through both endpoints. At the inner edge $n_z - t_s n_\theta = 0$ and $n_\theta + t_s n_z > 0$. At the outer edge $n = (1, 0)$, $t_s = 0$, giving the same two conclusions. In the interior,

$$n_\theta + t_s n_z = \frac{P_c - v_s}{|p_s - (a, -b_s)|} > 0, \quad n_z - t_s n_\theta = \frac{J_c}{|p_s - (a, -b_s)|}.$$

The admissible cone makes

$$2 - (v_s - 2) \frac{(n_z - t_s n_\theta)^2}{(n_\theta + t_s n_z)^2} > 0.$$

Both the denominator and this gap extend continuously to the closed annulus, where the gap has value two at both radial edges. Their positive compact minima permit one $0 < \kappa < 2$ with

$$n_\theta + t_s n_z \geq \kappa, \quad (v_s - 2)(n_z - t_s n_\theta)^2 \leq (2 - \kappa)(n_\theta + t_s n_z)^2.$$

The positive lower bounds on $F, a, v_s - 2$ in the interior and the original endpoint bounds likewise extend uniformly by compactness.

The original stress identity $T_0 = F(p_s - (a, -b_s))$ proves nonvanishing throughout the open shell: if it vanished, $P_c = v_s$, contrary to admissibility. The outside stress remains zero by the exact endpoint restoration and unchanged inner region. Define the original proposed common edge weight

$$\zeta(X) = \exp(-t_1^2/y_a^2 - 4/y_b^2), \quad \delta = \min(1, y_a, y_b),$$

extended by zero outside the open shell. On a sufficiently short inner collar, the second exponential is smooth and bounded above and below by positive constants, while g_a, B_0 have the required nonzero value bounds and fixed derivative bounds. The inner factorization therefore proves $|T_0| \geq c\zeta$ and $|\partial^I T_0| \leq C_I \zeta \delta^{-m_I}$. On a short outer collar the first exponential is similarly a smooth positive factor; $y_b^{-3} b_\theta$ has a lower positive bound and its derivatives have only finite inverse powers of y_b . The supplied outer factor derivative bounds give the same conclusion. On the remaining compact interior both ζ and $|T_0|$ have positive minima and every fixed derivative is bounded, so the same pair of inequalities follows after increasing constants. These three regions cover the annulus. The exponential derivative calculation in the closed theorem proof proves smooth zero extension of all stress derivatives at both boundaries.

All remaining completion claims in C.3 are exact preservation statements: the axis Cartesian formulas remain unchanged on the analytic rectangle; the pressure at infinity and exterior moment values remain unchanged by the proved five-moment map; the exterior heat function is unchanged pointwise; and on $X \geq X_v = X_{\text{end}}$ the original $U = M = 0$ gives $V_0 = 0$ directly by its formula. The third and fourth reserved intervals remain the exact original $U = 0, E = c_{\text{patch}}(1 + \eta^2)^{-1} X^{-1/2-\lambda}$ intervals, with their original order. Fixing this one finite N before the physical scale q , all dyadic bands, and all later correction stages proves the claimed independence of the resulting derivative constants from those later parameters.

40 Exact coordinate, evaluation, and covering maps

This section audits the coordinate and auxiliary maps on their original domains. For the physical and profile coordinate maps retain the base construction's range $0 < h < 1/2$. Its constants are

$$A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad d = 1 - \eta^2, \quad L = 1 - 2h\eta^2.$$

For the auxiliary and oscillatory construction below retain its narrower original range $0 < h < 1/100$. The letter t denotes physical time. The symbol $\mathbf{t}$ below is an extended differential operator, not a second time coordinate. No statement at the excluded point $(z, t) = (0, 1)$ is inferred from the maps.

40.1 The global $\mathbf{q}$ map and its boundary

Define

$$\Psi : (0, \infty) \times (-1, 1) \longrightarrow \mathbb{R} \times (-\infty, 1), \quad \Psi(q, \eta) = (z, t) = (q^D \eta, 1 - q(1 - \eta^2)).$$

This is a smooth bijection with smooth inverse. Indeed, for given z and $\tau = 1 - t > 0$, put $q_0 = |z|^{1/D}$. On $q > q_0$ the function $f_z(q) = q - z^2 q^{2h}$ has derivative

$$f'_z(q) = 1 - 2hz^2 q^{2h-1} \geq 1 - 2h > 0.$$

It tends to zero as $q \downarrow q_0$ (also when $z = 0$), and tends to infinity as $q \rightarrow \infty$, since $2h < 1$. The intermediate value theorem and strict monotonicity give exactly one root $f_z(q) = \tau$. Setting $\eta = zq^{-D}$ gives $|\eta| < 1$ and both displayed equations for Ψ . Conversely, any preimage must give that root and that value of η . The exact derivative matrix and determinant are

$$\frac{\partial(z, t)}{\partial(q, \eta)} = \begin{pmatrix} Dq^{D-1}\eta & q^D \\ -d & 2q\eta \end{pmatrix}, \quad \det \frac{\partial(z, t)}{\partial(q, \eta)} = q^D L > 0.$$

Inverting this matrix proves smoothness of the inverse by the inverse function theorem, and proves, with the other physical coordinate fixed,

$$q_t = -\frac{1}{L}, \quad \eta_t = \frac{D\eta}{qL}, \quad q_z = \frac{2\eta q^{1-D}}{L}, \quad \eta_z = \frac{d}{q^D L}.$$

The same map on $(0, \infty) \times [-1, 1]$ is a bijection onto

$$\{(z, t) : t < 1\} \cup \{(z, 1) : z \neq 0\}.$$

For $t = 1$ the root is $q = |z|^{1/D} > 0$ and $\eta = \text{sign } z$. Since the determinant remains positive there, the inverse has a smooth extension to a neighborhood of every such boundary point and hence all one-sided derivatives. There is no point of this domain above $(0, 1)$.

Retain $s = \frac{1}{2}r^2$ and $X = s/q$ for $r \geq 0$. For $r > 0$,

$$\partial_r X = \frac{r}{q}, \quad X_t = \frac{X}{qL}, \quad X_z = -\frac{2\eta X}{q^D L}.$$

The inverse radial relation is $r = \sqrt{2qX}$, so the original physical radius is preserved. For every $b \in \mathbb{R}$ and smooth $f(X, \eta)$, the ordinary product and chain rules, with no change of q, X, η , give

$$\begin{aligned} \partial_t(q^b f) &= q^{b-1} T_b f, & T_b f &= L^{-1}(-bf + D\eta f_\eta + X f_X), \\ \partial_z(q^b f) &= q^{b-D} Z_b f, & Z_b f &= L^{-1}(2b\eta f + df_\eta - 2\eta X f_X). \end{aligned}$$

Consequently $\partial_{zz}(q^b f) = q^{b-2D} Z_{b-D} Z_b f$ with that exact order and shifted subscript. These formulas contain no division by d and remain valid at $\eta = \pm 1$ through the boundary extension.

40.2 Chart-to-profile maps with fixed Q

For a fixed integer ℓ , set

$$Q = 2^{-\ell}, \quad \varepsilon = Q^h, \quad S_* = \ell^2, \quad R = \frac{r}{\sqrt{Q}}, \quad Z = \frac{z}{Q^D}, \quad T = \frac{1-t}{Q}.$$

Use $s_Q = q/Q$ in the next formulas; it is different from $s = \frac{1}{2}r^2$. The exact transition maps are

$$Z = s_Q^D \eta, \quad T = s_Q(1 - \eta^2), \quad X = \frac{R^2}{2s_Q}, \quad R_{\text{profile}} = \frac{r}{\sqrt{q}} = \frac{R}{\sqrt{s_Q}}.$$

In particular $T = s_Q - Z^2 s_Q^{2h}$, whose derivative in s_Q at fixed Z is L . Direct inversion, or the physical formulas above, gives

$$\begin{aligned} (s_Q)_T &= \frac{1}{L}, & (s_Q)_Z &= \frac{2\eta s_Q^{1-D}}{L}, & \eta_T &= -\frac{D\eta}{s_Q L}, & \eta_Z &= \frac{d}{s_Q^D L}, \\ X_R &= \frac{R}{s_Q}, & X_T &= -\frac{X}{s_Q L}, & X_Z &= -\frac{2\eta X}{s_Q^D L}. \end{aligned}$$

The inverse map from (X, s_Q, η) to (R, Z, T) is explicitly

$$(R, Z, T) = (\sqrt{2Xs_Q}, s_Q^D \eta, s_Q d).$$

For any fixed $0 < X_a < X_b < \infty$, all fixed derivatives of this map and its inverse are bounded on

$$X_a \leq X \leq X_b, \quad \frac{1}{2} \leq s_Q \leq 2, \quad -1 \leq \eta \leq 1.$$

To justify the assertion at every derivative order, the first derivative formulas have denominators consisting only of positive powers of s_Q, L , and R for the inverse radial relation. On this set these are bounded below by $1/2, 1 - 2h$, and $\sqrt{X_a}$, respectively. Differentiating any finite product of these factors again produces a finite sum with the same types of denominators and bounded numerator factors. Induction on the derivative order proves the assertion. This establishes the exact chart/profile derivative morphism including all edge parameters; it never identifies R with R_{profile} .

40.3 Physical evaluation as a differential algebra map

From this subsection onward take $0 < h < 1/100$, as in the original oscillatory construction. Let $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ with Haar measure of mass one, let $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$, and let $r > 0$, $z \in \mathbb{R}$, $t < 1$ or a retained boundary point with $q > 0$. Set

$$J_g = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad b_g = \sqrt{2} - 1, \quad v_r = (1, -b_g), \quad v_t = (b_g, 1),$$

$$\Lambda_g = 4 - \sqrt{2}, \quad T_g = 4 + \sqrt{2}, \quad \rho_g = \frac{\log \Lambda_g}{\log T_g}, \quad \kappa_s = 10^{-5}, \quad d_r = 2((1+h)\rho_g - h\kappa_s).$$

Matrix multiplication gives $J_g v_r = \Lambda_g v_r$ and $J_g v_t = T_g v_t$. Also $v_r \cdot v_t = 0$, $|v_r|^2 = |v_t|^2 = 1 + b_g^2$, and $\det(v_r, v_t) = 1 + b_g^2 > 0$. Since $\Lambda_g > 2$ and $T_g < 6$, $\rho_g > \log 2 / \log 6 > 1/3 > \kappa_s$, so $d_r > 0$.

For a smooth extended scalar $F(r, \theta, z, t, Y)$ define

$$\gamma(r, t) = v_r r^{d_r} + v_t t \pmod{\mathbb{Z}^2}, \quad (\mathcal{E}F)(r, \theta, z, t) = F(r, \theta, z, t, \gamma(r, t)).$$

This map preserves addition, multiplication, and constants. Its intertwining identities are exactly

$$\partial_t \mathcal{E} = \mathcal{E} \mathfrak{t}, \quad \partial_r \mathcal{E} = \mathcal{E} r, \quad \partial_z \mathcal{E} = \mathcal{E} \partial_z, \quad \partial_\theta \mathcal{E} = \mathcal{E} \partial_\theta,$$

where

$$\mathfrak{t} = \partial_t + v_t \cdot \partial_Y, \quad r = \partial_r + d_r r^{d_r-1} v_r \cdot \partial_Y.$$

For example, differentiating $F(r, \theta, z, t, \gamma)$ in r produces $F_r + (d_r r^{d_r-1} v_r) \cdot F_Y$, evaluated at $Y = \gamma$; the other three identities follow by the same explicit chain rule. All four unscaled coordinate derivations on the right commute: every possible coefficient derivative in a commutator is zero, except for radial differentiation of $d_r r^{d_r-1}$, which never appears in a commutator with a different operator. Iterating the identities proves them for every ordered composition of the coordinate derivatives.

The use of r^{d_r} is restricted to $r > 0$. If an extended field is supported in $X_a \leq r^2/(2q) \leq X_b$ with $X_a > 0$, its evaluation vanishes in a neighborhood of $r = 0$ at every fixed (z, t) with $q > 0$. More precisely, on a neighborhood where $q \geq q_* > 0$ it vanishes for $r < \sqrt{2X_a q_*}$. Its zero extension in Cartesian coordinates is therefore smooth near that axis neighborhood without differentiating r^{d_r} at zero.

40.4 Integer coverings, differential intertwining, and Haar measure

For each integer $i \geq 0$, define

$$\pi_i : \mathbb{T}^2 \longrightarrow \mathbb{T}^2, \quad \pi_i([Y]) = [J_g^i Y], \quad P_i f = f \circ \pi_i.$$

Integer entries make the map well-defined. A lift y of any target point has preimage $[J_g^{-i} y]$, so the map is surjective. For $i = 1$ the integer column swap followed by the row operation $\text{row}_2 \mapsto \text{row}_2 - 5\text{row}_1$ and the column operation $\text{col}_2 \mapsto \text{col}_2 - 3\text{col}_1$ reduces J_g to $\text{diag}(1, -14)$. All these operations are invertible integer torus maps. The diagonal map has fourteen kernel points, so $|\ker \pi_1| = 14$. The map $\pi_1 : \ker \pi_{i+1} \rightarrow \ker \pi_i$ is surjective: lift each point of $\ker \pi_i$ through the surjective π_1 . Every fiber is a coset of $\ker \pi_1$ and has fourteen points. Induction gives $|\ker \pi_i| = 14^i$.

Write $L_i = v_r \cdot \partial_{Y_i}$ and $N_i = v_t \cdot \partial_{Y_i}$. The chain rule and the two eigenvector identities prove

$$(v_r \cdot \partial_Y) P_i f = \Lambda_g^i P_i L_i f, \quad (v_t \cdot \partial_Y) P_i f = T_g^i P_i N_i f.$$

For $n \in \mathbb{Z}^2$, the character $e^{2\pi i n \cdot Y_i}$ pulls back to the character of frequency $(J_g^i)^T n$. Since J_g^i is invertible over $\mathbb{R}$, this frequency is zero precisely for $n = 0$. Integrating the Fourier series of a smooth function, which converges absolutely, thus gives

$$\int_{\mathbb{T}^2} P_i f dY = \int_{\mathbb{T}^2} f dY_i.$$

This proves preservation of the normalized Haar measure exactly.

The image of P_i consists precisely of the smooth functions invariant under translations by $\ker \pi_i$. Necessity follows from $\pi_i(Y + k) = \pi_i(Y)$ for k in that kernel. For sufficiency define $f(y) = F(Y)$ for any preimage Y of y ; invariance makes it independent of the preimage, and local inverses of the nonsingular linear cover prove smoothness. In Fourier coordinates the same condition is that the support is contained

in $(J_g^i)^T \mathbb{Z}^2$. To prove the nontrivial direction of this last equivalence, invariance implies that any nonzero Fourier coefficient at m satisfies $m \cdot J_g^{-i} k \in \mathbb{Z}$ for every $k \in \mathbb{Z}^2$, since $[J_g^{-i} k] \in \ker \pi_i$. Thus $(J_g^{-i})^T m \in \mathbb{Z}^2$, which is exactly the asserted sublattice condition.

For completeness, the directional inverses used with these maps are unambiguous on the zero-mean subspace. If $n \neq 0$ then

$$\begin{aligned} (v_r \cdot n)((n_1 + n_2) + \sqrt{2}n_2) &= (n_1 + n_2)^2 - 2n_2^2, \\ (v_t \cdot n)((n_2 - n_1) - \sqrt{2}n_1) &= (n_2 - n_1)^2 - 2n_1^2. \end{aligned}$$

Each right side is a nonzero integer: a zero would force the indicated integer ratio to equal $\sqrt{2}$, or force both coordinates to vanish. The conjugate factors have absolute value at most $C_g|n|$, $C_g = \sqrt{4 + 2\sqrt{2}}$, by the Euclidean Cauchy–Schwarz inequality. Hence $|v_r \cdot n|, |v_t \cdot n| \geq (C_g|n|)^{-1}$. The inverse multiplier is exactly $(2\pi i(v \cdot n))^{-1}$ at $n \neq 0$ and zero at $n = 0$. It grows at most linearly in $|n|$, so preserves rapid Fourier decay and gives $\|(v \cdot \partial_Y)^{-1} f\|_{H^s} \leq C_s \|f\|_{H^{s+1}}$. Uniqueness follows because any smooth zero-mean function in the kernel has every Fourier coefficient zero. The exact intertwining for the inverses is

$$L_{\text{abs}}^{-1} P_i = \Lambda_g^{-i} P_i L_i^{-1}, \quad N_{\text{abs}}^{-1} P_i = T_g^{-i} P_i N_i^{-1}.$$

Translation, averaging, and these inverses preserve the Fourier sublattice, hence preserve descent. Integration in r at fixed z, t, Y also preserves descent whenever the integrand and integration limits are defined and the latter are independent of Y ; deck translation commutes pointwise with that integral.

40.5 The original scaled operators and their threshold

For $\ell \geq 4$, $2^\ell \geq \ell^2$: it holds at four, and its induction step follows from $2\ell^2 \geq (\ell + 1)^2$ for $\ell \geq 4$. Therefore $Q^{-1-h}/S_* > 1$ and the integer

$$i(\ell) = \left\lfloor \log_{T_g} \frac{Q^{-1-h}}{S_*} \right\rfloor$$

is nonnegative. The floor inequality is

$$T_g^{-1} Q^{-1-h} / S_* < T_g^{i(\ell)} \leq Q^{-1-h} / S_*.$$

Combining the fixed- Q coordinate map with the cover intertwining gives, on the band representative,

$$\begin{aligned} \mathbf{t}_* &= Q^{1+h} \mathbf{t} = -\varepsilon \partial_T + c_i N_i, & c_i &= T_g^i Q^{1+h}, \\ D_r &= \sqrt{Q} \mathbf{r} = \partial_R + M_i d_r R^{d_r-1} L_i, & M_i &= \Lambda_g^i Q^{d_r/2}, \\ D_z &= \sqrt{Q} \partial_z = \varepsilon \partial_Z, & D_\theta &= R^{-1} \partial_\theta. \end{aligned}$$

Here the sign of the time term is fixed by $t = 1 - QT$. The constants obey

$$T_g^{-1} S_*^{-1} < c_i \leq S_*^{-1}, \quad T_g^{-\rho_g} \varepsilon^{-\kappa_s} S_*^{-\rho_g} < M_i \leq \varepsilon^{-\kappa_s} S_*^{-\rho_g},$$

because $\Lambda_g^i = (T_g^i)^{\rho_g}$ and $d_r/2 = (1 + h)\rho_g - h\kappa_s$. Thus each application of D_r to a coefficient incurs at most the retained factor $\varepsilon^{-\kappa_s}$, with the power $S_*^{-\rho_g}$ still present in its fast part. Every derivative of $d_r R^{d_r-1}$ is bounded on the fixed positive chart annulus. Applying D_z incurs the factor ε , and the two parts of $\mathbf{t}_*$ are exactly the displayed slow and fast parts.

The operators $D_r, D_z, \mathbf{t}_*, \partial_\theta$ commute pairwise. The orthonormal angular derivative has instead the exact frame commutator

$$[D_r, D_\theta] = -R^{-1} D_\theta.$$

Indeed, $D_r(R^{-1}) = -R^{-2}$ and $[D_r, \partial_\theta] = 0$. Accordingly, the cylindrical formulas retain their frame terms. For an extended vector in component order (r, θ, z) they are

$$\begin{aligned} \text{div}_* u &= (D_r + R^{-1})u_r + D_\theta u_\theta + D_z u_z, \\ \text{curl}_* u &= (D_\theta u_z - D_z u_\theta, D_z u_r - D_r u_z, (D_r + R^{-1})u_\theta - D_\theta u_r). \end{aligned}$$

These follow by inserting the evaluation intertwining into the usual cylindrical formulas, or directly differentiating the basis vectors $\partial_\theta e_r = e_\theta$, $\partial_\theta e_\theta = -e_r$. The scalar spatial Laplacian is $D_r^2 + R^{-1} D_r + D_z^2 + R^{-2} \partial_\theta^2$. For physical fields $u_{\text{phys}} = Q^{-A} \mathcal{E} u_*$ and $p_{\text{phys}} = Q^{-2A} \mathcal{E} p_*$, multiplying the physical equation by $Q^{2A+1/2}$ gives the original time coefficient Q^{1+h} , first-spatial-derivative coefficient $\sqrt{Q}$, and viscosity-one coefficient $Q^h = \varepsilon$. These are the identities $2A + 1/2 - A = 1 + h$ and $2A + 1/2 - A - 1 = h$; none of $D_r, D_z, \mathbf{t}_*$ is replaced by its ordinary slow-coordinate part.

40.6 Common torus, local band range, and exact pulse path

Fix a physical point with $q = q_0 > 0$ and restrict to a neighborhood U such that $q(U) \subset (q_0/2, 2q_0)$. For bands with $q/Q_\ell \in [1/2, 2]$ somewhere on U , one has $Q_\ell \in (q_0/4, 4q_0)$. Hence only finitely many occur and the largest difference of their indices is at most four. This restriction on U is necessary for a uniform bound; finiteness of an arbitrary band set alone would not provide it.

Choose an integer $\ell_0 > 4$ below all active indices. On $x \geq \ell_0$ put

$$f(x) = \frac{(1+h)x \log 2 - 2 \log x}{\log T_g}.$$

For $|\ell - \ell'| \leq 4$, the mean value theorem and $||a| - |b|| \leq |a - b| + 1$ give

$$|i(\ell) - i(\ell')| \leq 1 + \frac{4(1+h) \log 2 + 8/\ell_0}{\log T_g}.$$

Take the ceiling of this expression as $\Delta_{\max}$. Define $i_0 = \min_{\ell \in \mathcal{L}(U)} i(\ell)$ and the common auxiliary coordinate $H = J_g^{i_0} Y$. For a particular band set $\Delta = i - i_0$. Its representative is pulled back through

$$P_\Delta : f(Y_i) \mapsto f(J_g^\Delta H), \quad 0 \leq \Delta \leq \Delta_{\max}.$$

The equality $P_{i_0} P_\Delta = P_i$ proves compatibility with the absolute torus. On an overlap of two neighborhoods, choose the lesser of their two i_0 values and pull both representatives to that torus; their pullbacks agree whenever they represent the same absolute field. Injectivity of pullback by a surjective covering proves this agreement as an equality of functions, not merely of averages. Because the maps are independent of the slow coordinates, slow derivatives preserve it. For the same fixed Q , the exact common coefficients are

$$c_{i_0} = T_g^{-\Delta} c_i, \quad M_{i_0} = \Lambda_g^{-\Delta} M_i.$$

Thus the common and band formulas give the same absolute scaled operators. Factors coming from J_g^Δ are uniformly bounded because Δ lies in the displayed finite set.

Consider an injective rectangle in the band torus with lift

$$Y_i = c_\gamma + k + \xi_g v_r + \eta_g v_t, \quad |\xi_g| < r_0, \quad |\eta_g| < r_0.$$

Its full preimage consists exactly of the copies

$$H = J_g^{-\Delta} (c_\gamma + k + \xi_g v_r + \eta_g v_t) \pmod{\mathbb{Z}^2}, \quad [k] \in \mathbb{Z}^2 / J_g^\Delta \mathbb{Z}^2.$$

Each copy maps diffeomorphically onto the rectangle. Any preimage gives one such integer k by lifting its image. If two copies meet, the corresponding band points agree; injectivity of the band rectangle forces their ξ_g, η_g coordinates to agree, and then the two k values differ by $J_g^\Delta \mathbb{Z}^2$. This proves completeness and pairwise disjointness of the copies. Their number is 14^Δ by the proved kernel count.

Set $v = (\eta_g + r_0)/c_i$ and $L_s = 2r_0/c_i$. With $\lambda_t(w) = v_t \cdot w / (1 + b_g^2)$ one has $\lambda_t(v_t) = 1$, $\lambda_t(v_r) = 0$, and $\lambda_t J_g^\Delta = T_g^\Delta \lambda_t$ because J_g is symmetric and v_t is its eigenvector. In the band rectangle, $L_i \eta_g = N_i \xi_g = 0$ and $L_i \xi_g = N_i \eta_g = 1$. Consequently

$$D_r v = D_z v = 0, \quad \mathbf{t}_* v = 1.$$

For a fixed lift of a common rectangle, the path at fixed transverse and slow coordinates from pulse value $v(H)$ to $w \in [0, L_s]$ is exactly

$$H_w = H + T_g^{-\Delta} c_i (w - v(H)) v_t.$$

Its image in the band rectangle changes by $c_i (w - v(H)) v_t$, and hence has pulse value w and unchanged ξ_g . Its derivatives are

$$D_H v = \frac{T_g^\Delta}{c_i} \lambda_t, \quad D_H H_w = I - v_t \lambda_t, \quad \partial_w H_w = T_g^{-\Delta} c_i v_t.$$

All derivatives of order at least two in H, w vanish. The first displayed derivative is $O(S_*)$ and $D_H H_w$ is bounded independently of the band. Changing a lift or moving by a deck translation just changes the integer k ; the formula for the path changes by the same deck translation. Thus an initial-value problem whose coefficients and initial datum are pullbacks of the same band data has equal solutions on deck-related copies: subtract their pulled-back equations to obtain the same initial-value problem with zero difference datum, whose uniqueness gives zero difference. This is the exact descent statement for that solution. The path itself takes values at individual auxiliary points and performs no averaging.

40.7 Separation for all products used in covariance assembly

The slow labels are boxes $\beta = (\ell, a)$, with the two signs retained as distinct labels $\gamma = (\ell, a, \sigma)$. The squared partitions are $\sum_{\ell} \chi_{\ell}(q)^2 = 1$ and, in each band chart, $\sum_a \chi_{\ell,a}^2 = 1$. Define $\eta_{\beta} = \chi_{\ell}(q) \chi_{\ell,a}$. Then

$$\sum_{\beta=(\ell,a)} \eta_{\beta}^2 = 1$$

on their active domain; the signs do not duplicate this sum. Products of these nonnegative summands give the identity directly. Choose a representative in the intersection of the active shell and every selected grid support. The mesh in the three slow chart coordinates is $S_*^{-3} = \ell^{-6}$, and each grid support and its fixed enlarged version are contained in a box with side length at most a fixed constant times that mesh.

Join two distinct labels whenever their enlarged slow supports meet and their band indices differ by at most four, and join the two signs of each box. This countable graph has a finite upper bound d_g on its vertex degree. Here is the geometric count: for bands differing by at most four, each chart coordinate changes by a diagonal factor bounded above and below by fixed powers of 2^4 . Their mesh ratio is $(\ell/\ell')^6$, also bounded above and below since $\ell, \ell' \geq \ell_0 > 4$ and $|\ell - \ell'| \leq 4$. Thus the image of one enlarged box meets at most a fixed number of grid boxes in each of three coordinates, and hence at most the product of those three numbers. There are at most nine band indices and two signs to count. These bounds prove the finite degree assertion. Enumerate the labels and at each step choose the least unused color among $1, \dots, d_g + 1$ on its already colored neighbors. There are at most d_g such neighbors, so this yields a proper coloring of all labels.

For these finitely many colors choose centers $c_{\mu} \in \mathbb{T}^2$ such that

$$c_{\mu} \neq J_g^{\Delta} c_{\nu} \pmod{\mathbb{Z}^2}, \quad 0 \leq \Delta \leq \Delta_{\max},$$

apart from the unavoidable case $\Delta = 0, \mu = \nu$. This choice can be made with rational coordinates. To prove existence, regard the centers as a point in the finite product of tori. For distinct colors each forbidden equality is a closed set with empty interior: holding the other center fixed, arbitrary small changes of c_{μ} leave that equality. For equal colors and $\Delta > 0$, the matrix $J_g^{\Delta} - I$ is nonsingular, since its eigenvalues are $\Lambda_g^{\Delta} - 1, T_g^{\Delta} - 1 > 0$. Its torus kernel is finite: preimages can be represented by integer vectors in the bounded image of a unit fundamental square, and there are only finitely many such integer vectors. Thus this forbidden set also has empty interior. The complement of each of the finitely many closed forbidden sets is open and dense. Repeatedly intersecting any nonempty open set with these complements shows that their full intersection is nonempty and open. Rational tuples are dense, so one lies in that intersection. For this fixed choice the finitely many forbidden differences have a positive minimum distance d_c from $\mathbb{Z}^2$.

Let

$$C_J = \max_{0 \leq \Delta \leq \Delta_{\max}} \|J_g^{\Delta}\|, \quad C_v = |v_r| + |v_t|.$$

Choose $r_0 > 0$ so small that

$$4r_0 C_v < 1, \quad 2r_0 C_v (C_J + 1) < d_c.$$

The closed enlarged rectangles with $|\xi_g|, |\eta_g| \leq 2r_0$ are injective, because the difference between any two of their lifts has length at most $4r_0 C_v < 1$ and so cannot be a nonzero integer vector. Suppose the absolute preimages of two such rectangles for joined labels, at levels i and $i + \Delta$, have a common point. Writing the two rectangle errors as e_{ν}, e_{μ} , each of length at most $2r_0 C_v$, gives

$$c_{\mu} - J_g^{\Delta} c_{\nu} = J_g^{\Delta} e_{\nu} - e_{\mu} \pmod{\mathbb{Z}^2}.$$

The left side is one of the forbidden differences: at $\Delta = 0$ joined vertices have distinct colors, and at $\Delta > 0$ all color pairs were excluded. Its distance from the lattice is at least d_c . The right side has distance at most $2r_0 C_v (C_J + 1) < d_c$, a contradiction. Hence the closed absolute preimages are disjoint for every pair that can interact at a slow point. Passing to a common torus preserves their disjointness: any intersection there would lift by its surjective map from the absolute torus to an intersection already excluded.

For actual smooth coefficients supported in these products of slow supports and auxiliary rectangles, every derivative has support contained in the original closed support. Outside that support the function vanishes on a neighborhood, and therefore all derivatives vanish there, which proves this support assertion. Consequently every product of coefficients or of any of their derivatives from distinct active labels is identically zero on the extended domain. Since $\mathcal{E}$ preserves products, the same vanishing holds after physical evaluation. This proves the cross-label cancellations used in covariance and differential covariance assembly, including the two opposite signs of a single box and their support boundaries.

40.8 Averaging, evaluation, and the preserved covariance target

Let $\mathcal{P}F = \int_{\mathbb{T}^2} F dY$ and let $\mathcal{J}$ embed an auxiliary-independent function as a constant function of Y . Then

$$\mathcal{P}\mathcal{J} = I, \quad \mathcal{E}\mathcal{J} = I, \quad \mathcal{E}F = \mathcal{P}F + \mathcal{E}(F - \mathcal{J}\mathcal{P}F).$$

The cover identities prove agreement of absolute, common, and band averages. Integrating a directional derivative of a periodic smooth function gives zero, so

$$\mathcal{P}rF = \partial_r \mathcal{P}F, \quad \mathcal{P}tF = \partial_t \mathcal{P}F,$$

and the analogous z, θ identities hold. This gives an exact differential map to the auxiliary mean.

There is no operation on the evaluated field alone that recovers this mean for all extended fields. For any $n \in \mathbb{Z}^2 \setminus \{0\}$, take

$$F(r, \theta, z, t, Y) = e^{2\pi i n \cdot Y} - e^{2\pi i n \cdot \gamma(r, t)}.$$

It is smooth on the specified $r > 0$ domain, $\mathcal{E}F = 0$, and $\mathcal{P}F = -e^{2\pi i n \cdot \gamma(r, t)} \neq 0$. Any proposed recovery map would therefore assign to the zero evaluated field both zero and this nonzero function, which is impossible. Real and imaginary parts give real counterexamples. In particular an auxiliary covariance identity refers to the extended construction; it is not an identity obtained by averaging the physically evaluated field over a missing auxiliary variable.

Finally, for a real e , an integrable coefficient with $f_* = Q^{a_f} f_{\text{phys}}$, and corresponding radial integration intervals $I_r = \sqrt{Q}I_R$, the substitution $r = \sqrt{Q}R$ proves

$$\int_{I_R} R^e \langle f_* \rangle_Y dR = Q^{a_f - (e+1)/2} \int_{I_r} r^e \langle f_{\text{phys}} \rangle_Y dr.$$

This includes both the original power weight and the measure. Agreement of the three auxiliary averages follows from the proved cover morphisms. The identity makes no claim at a nonintegrable endpoint or at $q = 0$.

41 Phase, constrained evolution, and every fixed derivative

This section derives the pulse equations and estimates of source pp. 73–81. The exact auxiliary maps and operators were proved above. The actual-base expansion proved in the initial-state section below gives, on the fixed enlarged annulus, the chart radial bound $b = O(\varepsilon)$ and the tangential differences $V - V_0, G - G_0 = O(\varepsilon^2)$, at every fixed slow derivative. Here those exact chart data enter the pulse system, whose phase, constrained evolution, and derivative estimates are calculated explicitly.

41.1 Cone coordinates and the phase gradient

Write $F = V/R$, $g = (RF_R, G_R)$ in component order (θ, z) . At the fixed representative of a slow box, the leading shear is

$$g_0 = F_0(-a, b_s), \quad N = g_0/|g_0|, \quad K = (-N_z, N_\theta).$$

Let $t_s = -b_s/a$ and $v_s = a(1 + t_s^2)$. Since $a, F_0 > 0$,

$$N = -\frac{(1, t_s)}{\sqrt{1 + t_s^2}}, \quad K = \frac{(t_s, -1)}{\sqrt{1 + t_s^2}}, \quad |g_0| = aF_0\sqrt{1 + t_s^2}.$$

Retain the source's

$$\lambda_0^2 = -2F_0N_\theta(2F_0N_\theta + |g_0|), \quad c_0 = \lambda_0/(2F_0N_\theta).$$

Substitution gives

$$\lambda_0^2 = 2F_0^2 \left(a - \frac{2}{1 + t_s^2} \right) = 2aF_0^2(1 - 2/v_s) > 0, \quad c_0^2 = (v_s - 2)/2, \quad c_0 < 0.$$

The positive lower bounds on $F_0, a, v_s - 2$ and compact upper bounds give positive lower bounds on $\lambda_0, |c_0|$, and finite upper bounds.

At the same, unfrozen profile point, $T_0 = F(p_s - (a, -b_s))$, $P_c = p_{s,1} + t_s p_{s,2}$, $J_c = p_{s,2} - t_s p_{s,1}$. Consequently

$$T_0 \cdot N = -\frac{F(P_c - v_s)}{\sqrt{1 + t_s^2}}, \quad T_0 \cdot K = -\frac{FJ_c}{\sqrt{1 + t_s^2}}.$$

Multiplication of T_0 by the positive chart factor $(Q/q)^{A+1/2}$ changes neither sign nor quotient. Thus the two stress cone inequalities are exactly $T_{0,*} \cdot N < 0$ and

$$\left| c_0 \frac{T_{0,*} \cdot K}{T_{0,*} \cdot N} \right| < 1.$$

The weighted direction at a vanishing edge is used, not a division of zero values. Its smooth extension and strict compact margin give a supremum $\mu < 1$. Choosing $u_* > 0$ with $u_*/\sqrt{1+u_*^2} > \mu$ is possible because that function increases to one. All quantities now frozen at representatives are kept fixed under differentiation. Uniform continuity in the compact chart domain makes the same strict cone margin valid on every enlarged box of diameter CS_*^{-3} , after one lower bound on ℓ .

Retain $k = \lceil \varepsilon^{-1/2} \rceil$, so for $0 < \varepsilon \leq 1$,

$$1 \leq \varepsilon k^2 \leq (1 + \sqrt{\varepsilon})^2 \leq 4, \quad k^{-1} \leq \sqrt{\varepsilon}.$$

Define

$$B_s^2 = \frac{\lambda_0}{\varepsilon k^2 (1 + u_*^2)^{3/2}}, \quad s(v) = \sigma(u_*/2 + u_*v/L_s), \quad x_0 = \sigma B_s u_*/2.$$

The B_s have fixed positive lower and upper bounds. Set

$$(\tilde{p}/R_0, p_z) = B_s \left(K - \frac{\sigma u_* g_0}{L_s |g_0|^2} \right).$$

Choose kp as the specified nearest nonzero integer to $k\tilde{p}$. The error $|p - \tilde{p}| \leq k^{-1}$ suffices, including when the nearest unrestricted integer would have been zero. The integer is constant with the label. Put

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v H_\Phi, \quad H_\Phi = pF + p_z G.$$

Because $D_r v = D_z v = 0$ and the base is auxiliary-independent,

$$n_\Phi = \nabla_* \Phi = (x_0 - v(H_\Phi)_R, p/R, p_z - \varepsilon v(H_\Phi)_Z).$$

At the representative, before rounding, $(\tilde{p}/R_0, p_z) \cdot g_0 = -\sigma B_s u_*/L_s$. Variation across the slow box changes this by $O(S_*^{-3})$; the background change contributes $O(\varepsilon^2)$; rounding contributes $O(k^{-1})$. Hence

$$|(H_\Phi)_R + \sigma B_s u_*/L_s| \leq C(S_*^{-3} + \varepsilon^2 + k^{-1}).$$

Multiplication by $v \leq L_s = O(S_*)$ gives the radial comparison with $B_s s(v)$. The tangential comparison has contributions $O(S_*^{-1})$ from the deliberate tilt, $O(S_*^{-3} + \varepsilon^2 + k^{-1})$ from freezing and rounding, and $O(\varepsilon S_*)$ from the axial phase term. A single sufficiently large ℓ_* satisfies

$$S_*^2(\varepsilon + \varepsilon^2 + k^{-1}) \leq 1$$

for every $\ell \geq \ell_*$, since $S_* = \ell^2$ and $\varepsilon = 2^{-h\ell}$. Therefore

$$|n_\Phi - B_s(s(v), K)| \leq C/S_*.$$

Differentiation in v gives $n'_\Phi = (-(H_\Phi)_R, 0, -\varepsilon(H_\Phi)_Z)$, also $O(S_*^{-1})$. Each fixed number of slow derivatives of these coefficients is bounded by a polynomial in S_* : the only growing factors in these formulas are v , ε^{-1} in the phase itself, and the fixed chart derivatives. The ε^{-1} term of Φ disappears in n_Φ ; it is not discarded from the full exponential when physical derivatives are later taken.

The exact transport defect follows term by term:

$$\mathbf{t}_* \Phi = -H_\Phi + \varepsilon v(H_\Phi)_T, \quad F\partial_\theta \Phi + GD_z \Phi = H_\Phi - \varepsilon vG(H_\Phi)_Z.$$

Thus

$$E_{ik} := (\mathbf{t}_* + bD_r + F\partial_\theta + GD_z)\Phi = \varepsilon v((H_\Phi)_T - G(H_\Phi)_Z) + bn_{\Phi,r}.$$

The background satisfies $D^I b = O_I(\varepsilon)$, while every fixed coefficient derivative of $n_{\Phi,r}$ is polynomially bounded in S_* . The exact product rule therefore gives

$$D^I(bn_{\Phi,r}) = \sum_{J \leq I} \binom{I}{J} (D^J b)(D^{I-J} n_{\Phi,r}) = O_I(\varepsilon S_*^{d_I}).$$

For example, at fixed pulse coordinate, $\partial_R n_{\Phi,r} = -v(H_\Phi)_{RR}$, whose bound can contain a factor S_* . Auxiliary coefficient derivatives of v are $O(S_*)$, and derivatives of the fixed affine path of order two or greater vanish. Applying the same finite product rule to $\varepsilon v((H_\Phi)_T - G(H_\Phi)_Z)$ proves $D^I E_{ik} = O_I(\varepsilon S_*^{b_I})$ for every fixed I , on the same domain. This introduces no smallness condition on the higher derivatives.

41.2 Exact pressure elimination and plane isomorphism

Set

$$\mathcal{K} = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad d = \varepsilon k^2 |n_\Phi|^2, \quad \mathcal{A}_\Phi = -\mathcal{K} + \frac{n_\Phi(n_\Phi^T \mathcal{K} - (n'_\Phi)^T)}{|n_\Phi|^2}.$$

For the original equation

$$t'_m + \mathcal{K}t_m + m^2 dt_m + ikmn_\Phi \pi_m = -f_m, \quad n_\Phi \cdot t_m = 0,$$

differentiate its constraint to get $n_\Phi \cdot t'_m = -n'_\Phi \cdot t_m$. Taking its normal component then forces

$$\pi_m = \frac{i}{km} \frac{n_\Phi \cdot \mathcal{K}t_m - n'_\Phi \cdot t_m + n_\Phi \cdot f_m}{|n_\Phi|^2}.$$

The factor has sign $+i/(km)$: multiplying it by $ikmn_\Phi$ gives the negative of the required normal vector. Substituting gives the projected equation

$$t'_m = \mathcal{A}_\Phi t_m - m^2 dt_m - \text{proj}_{n_\Phi^\perp} f_m.$$

Conversely this projected equation and the displayed pressure imply the original equation provided the constraint holds. For any solution of the projected equation,

$$(n_\Phi \cdot t_m)' = -m^2 d(n_\Phi \cdot t_m).$$

An initially zero constraint remains zero, proving equivalence, existence, and uniqueness once the finite-dimensional initial-value equation is solved.

Since $n_{\Phi, \tan}$ stays close to $B_s K$, its norm has a uniform positive lower bound. The nonzero p/R also prevents it from being literally zero. Define

$$K_a = n_{\Phi, \tan} / |n_{\Phi, \tan}|, \quad N_a = ((K_a)_z, -(K_a)_\theta), \quad s_a = n_{\Phi, r} / |n_{\Phi, \tan}|.$$

Then $N_a \perp K_a$, $|N_a| = |K_a| = 1$, and

$$U = [e_r - s_a K_a, N_a], \quad M = \begin{pmatrix} 1 & 1 \\ c_0 \sqrt{1+s^2} & -c_0 \sqrt{1+s^2} \end{pmatrix}, \quad B = UM$$

maps $\mathbb{C}^2$ isomorphically onto the actual complex plane $n_\Phi^\perp$. The exact left inverse is

$$B^\ell = M^{-1} \begin{pmatrix} e_r^T \\ N_a^T \end{pmatrix}$$

because the last displayed row matrix times U is I_2 . The determinant of M is $-2c_0 \sqrt{1+s^2}$, bounded away from zero. All factors and their inverses are uniformly bounded in value and polynomially bounded in each fixed coefficient derivative.

For clarity the ambient map also has an exact formula. Regard K_a as the three-vector with zero radial component, and let $W_a = (e_r^T; N_a^T)$. Direct multiplication gives $W_a U = I_2$. For any $w = w_r e_r + w_K K_a + w_N N_a \in \mathbb{C}^3$,

$$BB^\ell w = UW_a w = w_r(e_r - s_a K_a) + w_N N_a = w - K_a(w_K + s_a w_r).$$

Since $n_\Phi = |n_{\Phi, \tan}|(s_a e_r + K_a)$, this proves

$$B^\ell B = I_2, \quad BB^\ell = I_3 - \frac{K_a n_\Phi^T}{|n_{\Phi, \tan}|}.$$

Thus BB^ℓ is the projection onto $n_\Phi^\perp$ along $\text{span}_\mathbb{C}\{K_a\}$. Its restriction to $n_\Phi^\perp$ is the identity; its action on K_a is zero. In particular the inverse assertion is not made on all of $\mathbb{C}^3$.

The moving-plane differential equation has the exact coordinate form

$$z'_m = B^\ell(\mathcal{A}_\Phi B - B')z_m - m^2 dz_m - B^\ell \text{proj}_{n_\Phi^\perp} f_m.$$

For its converse, differentiate $n_\Phi^T B = 0$ to obtain $n_\Phi^T B' = -(n'_\Phi)^T B$. The definition of $\mathcal{A}_\Phi$ gives $n_\Phi^T \mathcal{A}_\Phi B = -(n'_\Phi)^T B$. Therefore each column of $\mathcal{A}_\Phi B - B'$ lies in $n_\Phi^\perp$. Both Bz_m and the projected force lie there as well. Multiplying the coordinate equation by B and using $BB^\ell = I$ on these three terms yields

$$Bz'_m = (\mathcal{A}_\Phi B - B')z_m - m^2 dBz_m - \text{proj}_{n_\Phi^\perp} f_m.$$

Adding $B'z_m$ proves the projected equation for $t_m = Bz_m$, and $n_\Phi^T t_m = 0$ holds by construction. The displayed pressure then recovers the full equation with its actual normal forcing. Conversely, a constrained solution has $z_m = B^\ell t_m$ and $t_m = Bz_m$. Differentiating the latter identity and multiplying the projected equation by B^ℓ gives exactly the coordinate equation above. Both inverse compositions, including their initial data and the unique pressure, are therefore proved.

For the reference normal $n_{\text{ref}} = B_s(s, K)$, use $e = e_r - sK$ and N as an orthogonal plane basis, of lengths $\sqrt{1+s^2}$ and 1. The actual source matrix $\mathcal{K}_0$ gives

$$\mathcal{K}_0 e = (2F_0 s K_\theta, 2F_0 e_\theta + g_0), \quad \mathcal{K}_0 N = (-2F_0 N_\theta, 0).$$

Consequently

$$\begin{aligned} e \cdot (-\mathcal{K}_0 e) &= sK \cdot g_0 = 0, & e \cdot (-\mathcal{K}_0 N) &= 2F_0 N_\theta, \\ N \cdot (-\mathcal{K}_0 e) &= -(2F_0 N_\theta + |g_0|), & N \cdot (-\mathcal{K}_0 N) &= 0. \end{aligned}$$

Orthogonal projection onto $n_{\text{ref}}^\perp$ leaves these scalar products unchanged. Dividing the first coordinate by $1+s^2$ produces

$$\mathcal{D} = \begin{pmatrix} 0 & 2F_0 N_\theta / (1+s^2) \\ -(2F_0 N_\theta + |g_0|) & 0 \end{pmatrix}.$$

Its two eigenvectors are exactly the columns of M , with eigenvalues $\lambda = \lambda_0 / \sqrt{1+s^2}$ and $-\lambda$, since $c_0 = \lambda_0 / (2F_0 N_\theta)$ and $\lambda_0^2 = -2F_0 N_\theta (2F_0 N_\theta + |g_0|)$. This verifies the orientation and signs of the growing coordinate.

Replacing the background and normal by the actual ones changes the algebraic projected matrix by $O(S_*^{-1})$. This follows from the explicit rational expression for the projection, the uniform positive denominators, the value bounds already proved, and the mean-value formula on their common compact range. Also $K_a - K, s_a - s = O(S_*^{-1})$, while their v -derivatives and s' are $O(S_*^{-1})$. Hence $B' = O(S_*^{-1})$ and the normal-motion term in $\mathcal{A}_\Phi$ is $O(S_*^{-1})$. Therefore

$$B^\ell (\mathcal{A}_\Phi B - B') = \text{diag}(\lambda, -\lambda) + E, \quad |E| \leq C/S_*.$$

Every fixed derivative of E is polynomially bounded; no derivative-smallness claim is needed. The damping comparison is

$$d_{\text{ref}} = \varepsilon k^2 B_s^2 (1+s^2), \quad |d - d_{\text{ref}}| \leq C/S_*,$$

because εk^2 is bounded and $|n_\Phi|^2 - B_s^2 (1+s^2)$ is $O(S_*^{-1})$.

41.3 Envelope and uniform forward inverse

Let

$$a_{\text{net}}(y) = \frac{\lambda_0}{\sqrt{1+y^2}} - \frac{\lambda_0(1+y^2)}{(1+u_*^2)^{3/2}}, \quad y = |s(v)| = u_*/2 + u_* v / L_s.$$

It is zero at $v = L_s/2$, and direct differentiation gives

$$\frac{d}{dv} a_{\text{net}}(|s(v)|) = -\frac{u_* \lambda_0 |s(v)|}{L_s} \left((1+s(v)^2)^{-3/2} + 2(1+u_*^2)^{-3/2} \right).$$

Since $u_*/2 \leq |s| \leq 3u_*/2$, its value lies between $-C/L_s$ and $-c/L_s$, for fixed positive c, C . Define $P(v) = \exp(\int_{L_s/2}^v a_{\text{net}}(|s(w)|) dw)$. Integrating the derivative inequality on either side of the midpoint, keeping the direction of each integral, gives

$$-C_1(v - L_s/2)^2 / L_s \leq \log P(v) \leq -c_1(v - L_s/2)^2 / L_s.$$

This proves the two Gaussian bounds and $0 < P \leq 1$ with the source's integral normalization.

Writing $t_m = Bz_m$ gives

$$z'_m = A_m z_m + g_m, \quad A_m = \text{diag}(\lambda, -\lambda) + E - m^2 dI_2, \quad g_m = -B^\ell \text{proj}_{n_\Phi^\perp} f_m.$$

For the complex Euclidean norm, taking the real part of $z^* A_1 z$ bounds the upper right derivative of $|z|$ by $(\lambda - d_{\text{ref}} + C/S_*)|z|$. Integration gives

$$\|V_1(v, w)\| \leq \exp(C(v-w)/S_*) P(v)/P(w) \leq C' P(v)/P(w).$$

The factor C' is fixed because $0 \leq v - w \leq L_s \asymp S_*$. Since the difference of A_m from A_1 is the scalar matrix $-(m^2 - 1)dI_2$, differentiation directly verifies

$$V_m(v, w) = \exp\left(-(m^2 - 1) \int_w^v d(a) da\right) V_1(v, w).$$

For every nonzero integer m , $m^2 - 1 \geq 0$ and $d > 0$, so the same value bound is valid for all such harmonics. This is the exact mechanism preventing a new small-domain requirement when the finite harmonic set increases.

If the source obeys

$$|D^I f_m| \leq C_I \varepsilon^\alpha S_*^{b_I} \sqrt{\zeta} \delta^{-a_I} P(v),$$

the zero-data Duhamel integral gives

$$z_m(v) = \int_0^v V_m(v, w) g_m(w) dw, \quad |z_m(v)| \leq C \varepsilon^\alpha S_*^{b+1} \sqrt{\zeta} \delta^{-a} P(v).$$

The source's $P(w)$ cancels the propagator's denominator exactly. The shell weights are constant along the integration path because X is constant there.

For completeness, parameter differentiation can be performed without differentiating a possibly exponentially large propagator in isolation. Hold v fixed and use slow variables and the initial transverse coordinate as parameters. They commute with ∂_v ; the zero initial data and all their parameter derivatives vanish. For each multi-index I ,

$$(\partial^I z_m)' = A_m \partial^I z_m + \partial^I g_m + \sum_{0 < J \leq I} \binom{I}{J} (\partial^J A_m) \partial^{I-J} z_m.$$

At fixed finite harmonic bound M , each $\partial^J A_m$ is bounded by $C_{J,M} S_*^{c_J}$. Induction on $|I|$, using the already proved propagator bound, shows that every source term on the right has the same $\varepsilon^\alpha \sqrt{\zeta} P$ weight with a larger inverse-distance exponent and polynomial degree. A further time integral contributes only L_s . One may choose the degree at step I to exceed

$$1 + \max\{b_I, \max_{0 < J \leq I} (c_J + b'_{I-J})\}.$$

Pulse derivatives follow from the differential equation, and mixed pulse/parameter derivatives follow by repeated differentiation of that equation. Returning to the current H uses $D_H v = O(S_*)$, $D_H H_w = I - v_t \lambda_t$, with higher affine derivatives zero, and therefore introduces only additional polynomial powers. Multiplying by B yields the stated bound for t_m . The explicit pressure has the extra factor $(km)^{-1} = O(\varepsilon^{1/2})$; its other factors have the proved derivative bounds. Thus the exponent for pressure is $\alpha + 1/2$.

The support argument also follows from this equation. If a fixed slow/transverse point is outside the source's closed support, the entire source path is zero; uniqueness gives a zero solution. At a boundary all source jets vanish by the prescribed smooth zero extension, so the differentiated equations inductively give zero solution jets. Across a shell edge the retained $\sqrt{\zeta} \delta^{-a_I}$ bounds give the same conclusion. Deck translations preserve the path and the prescribed data, giving common-torus descent by uniqueness. Smooth extension to the rectangle enlargement follows from smooth finite-dimensional ODE dependence on its coefficients and data; no temporal zero extension is asserted before cutoff. On compact sets with $q > 0$, one-sided background/source derivatives at $\tau = 0$ pass through the same parameter equations and pressure formula.

Only the lower bounds for denominators and finitely many value comparisons determine q_* . The induction above changes constants and polynomial degrees for each fixed M, I ; it does not impose further smallness inequalities. This proves the quantifier statement of Corollary 7.3.

41.4 The real homogeneous pulse and energy transfer

For $m = 1, f = 0$, specify $z_+(0) = P(0) > 0, z_-(0) = 0$. While $z_+ > 0$, the ratio $r = z_-/z_+$ obeys exactly

$$r' = E_{21} + (-2\lambda + E_{22} - E_{11})r - E_{12}r^2.$$

Let $c_\lambda = \inf \lambda > 0$. On $r = \pm K_b/S_*$, the outward derivative is at most

$$\frac{-2c_\lambda K_b + C(1 + K_b/S_* + K_b^2/S_*^2)}{S_*}.$$

Choose fixed $K_b > C/c_\lambda$, then enlarge the fixed band threshold to make the numerator negative. The closed interval $[-K_b/S_*, K_b/S_*]$ is invariant until any putative first zero of z_+ . On this interval,

$$(\log(z_+/P))' = -(d - d_{\text{ref}}) + E_{11} + E_{12}r = O(S_*^{-1}).$$

Since $z_+(0) = P(0)$ and L_s/S_* is bounded, integration proves $cP \leq z_+ \leq CP$, ruling out that putative zero. For all v ,

$$x = z_+ + z_- = z_+(1 + r), \quad y = c_0 \sqrt{1 + s^2}(z_+ - z_-),$$

so

$$cP \leq x \leq CP, \quad y/x = c_0 \sqrt{1 + s^2} + O(S_*^{-1}).$$

The same differentiated equation used for the inverse proves every fixed derivative bound, with the nonzero initial value retained: it is fixed with the label, so all slow/transverse derivatives of that datum vanish. The value solution contributes $\|V_1(v, 0)\|P(0) \leq CP(v)$. Pulse derivatives are obtained from the equation. Pressure again has the extra $\varepsilon^{1/2}$.

For the real homogeneous amplitude t , $n_\Phi \cdot t = 0$ makes the normal component of $\mathcal{A}_\Phi t$ perpendicular to t . Direct multiplication, keeping both cylindrical $2F$ terms, gives

$$t \cdot \mathcal{K}t = -2Ft_r t_\theta + (2F + RF_R)t_r t_\theta + G_R t_r t_z = g \cdot (t_r(t_\theta, t_z)).$$

Taking $t \cdot t'$ therefore proves the exact identity

$$\frac{1}{2} \frac{d}{dv} |t|^2 = -g \cdot (t_r(t_\theta, t_z)) - \varepsilon k^2 |n_\Phi|^2 |t|^2.$$

For the flux $T = t_r(t_\theta, t_z) = T_N N + T_K K$, the exact production is

$$-g \cdot T = -|g_0|T_N - \Delta g \cdot T, \quad \Delta g = g - g_0.$$

Here g_0 is the leading shear frozen at the representative; it is not identified with the actual shear even there. To keep both sources of its difference, let $V^{(0)}, G^{(0)}$ denote the exact leading chart fields, put $\delta V = V - V^{(0)}$, $\delta G = G - G^{(0)}$, and write

$$g^{(0)} = (\partial_R V^{(0)} - V^{(0)}/R, \partial_R G^{(0)}), \quad g_0 = g^{(0)}(p_0),$$

$$\Delta g = (\partial_R \delta V - \delta V/R, \partial_R \delta G) + g^{(0)}(p) - g^{(0)}(p_0).$$

The leading and actual fields here are the constructed backgrounds. Their proved chart estimates give $|\partial^I \delta V| \leq C_{V,I} \varepsilon^2$ and $|\partial^I \delta G| \leq C_{G,I} \varepsilon^2$. Let $R_- > 0$ be the minimum radius on the compact enlarged chart, L_g bound $D_p g^{(0)}$, and $|p - p_0| \leq C_{\text{box}} S_*^{-3}$, with the connecting segment in that chart. Set

$$C_B = [(C_{V,e_R} + R_-^{-1} C_{V,0})^2 + C_{G,e_R}^2]^{1/2}.$$

The product rule and segment integration give the actual bound

$$|\Delta g| \leq C_B \varepsilon^2 + L_g C_{\text{box}} S_*^{-3} = C_B 2^{-2h\ell} + L_g C_{\text{box}} \ell^{-6}.$$

At p_0 only the segment term is zero. The Borel term remains. For a vector in the prescribed compact stress cone, retain its margin $-T_N \geq c_T |T|$, $c_T > 0$, and $|g_0| \geq g_- > 0$. Then Cauchy-Schwarz proves

$$-g \cdot T \geq [g_- c_T - C_B 2^{-2h\ell} - L_g C_{\text{box}} \ell^{-6}] |T|.$$

One sufficient additional integer threshold is

$$\ell \geq \left\lceil \max \left\{ 1, \left(\frac{4L_g C_{\text{box}}}{g_- c_T} \right)^{1/6}, \frac{\log \max\{1, 4C_B/(g_- c_T)\}}{2h \log 2} \right\} \right\rceil.$$

Each error is then at most $g_- c_T/4$, so the lower bound is $(g_- c_T/2)|T|$, including $T = 0$.

For the actual homogeneous pulse, positivity follows directly from its proved polarization. This requires no assertion that its instantaneous flux lies in the prescribed target cone. Retain

$$t = x(e_r - s_a K_a) + y N_a, \quad s_{\text{max}} = 3u_*/2, \quad \gamma_0 = c_0 \sqrt{1 + s^2},$$

and choose the constants in the already proved estimates as

$$|n_\Phi - B_s(s, K)| \leq E_n/S_*, \quad B_s \geq B_- > 0, \quad |y/x - \gamma_0| \leq E_\gamma/S_*, \quad c_x P \leq x \leq C_x P.$$

The quantities g_-, g_+, c_-, c_+ denote positive uniform bounds $g_- \leq |g_0| \leq g_+$, $c_- \leq |c_0| \leq c_+$, with $c_0 < 0$. Choose $S_* \geq 2E_n/B_-$. Then $|n_{\Phi, \tan}| \geq B_-/2$. For nonzero vectors a, b , addition and subtraction of $a/|b|$, followed by $||a| - |b|| \leq |a - b|$, gives $|a/|a| - b/|b|| \leq 2|a - b|/|b|$. Therefore

$$|K_a - K| = |N_a - N| \leq \frac{2E_n}{B_- S_*}, \quad |s_a - s| \leq \frac{2(1 + s_{\max})E_n}{B_- S_*}.$$

The last estimate follows by writing $n_{\Phi, r} - s|n_{\Phi, \tan}| = (n_{\Phi, r} - B_s s) + s(B_s - |n_{\Phi, \tan}|)$ and dividing by the actual tangential norm. Define

$$\Gamma_{\max} = c_+ \sqrt{1 + s_{\max}^2}, \quad C_T = \frac{2E_n}{B_-} (1 + 2s_{\max} + \Gamma_{\max}) + E_\gamma.$$

The exact flux and its exact remainder are

$$T = x^2[-sK + \gamma_0 N + e_T], \quad e_T = -(s_a K_a - sK) + (y/x)N_a - \gamma_0 N.$$

Expand the first difference as $(s_a - s)K_a + s(K_a - K)$ and the second as $(y/x - \gamma_0)N_a + \gamma_0(N_a - N)$. The preceding bounds give $|e_T| \leq C_T/S_*$, with all four terms retained. Hence

$$\frac{-g \cdot T}{x^2} = -|g_0|\gamma_0 - g_0 \cdot e_T - \Delta g \cdot (-sK + \gamma_0 N + e_T).$$

The absolute value of the last two terms is at most

$$\mathcal{E}_\ell = g_+ C_T \ell^{-2} + (C_B 2^{-2h\ell} + L_g C_{\text{box}} \ell^{-6})(s_{\max} + \Gamma_{\max} + C_T \ell^{-2}).$$

Set $W = s_{\max} + \Gamma_{\max} + C_T > 0$. In addition to all previous phase and invariant-band thresholds, it suffices to choose

$$\ell \geq \left\lceil \max \left\{ 1, \left(\frac{2E_n}{B_-} \right)^{1/2}, \left(\frac{4g_+ C_T}{g_- c_-} \right)^{1/2}, \left(\frac{8W L_g C_{\text{box}}}{g_- c_-} \right)^{1/6}, \frac{\log \max\{1, 8W C_B/(g_- c_-)\}}{2h \log 2} \right\} \right\rceil.$$

For $\ell \geq 1$, the last factor in $\mathcal{E}_\ell$ is at most W . The three error contributions are then at most $g_- c_-/4, g_- c_-/8, g_- c_-/8$, respectively. Since $-|g_0|\gamma_0 \geq g_- c_-$, this proves

$$\frac{1}{2} g_- c_- c_x^2 P(v)^2 \leq -g \cdot T \leq (g_+ \Gamma_{\max} + \frac{1}{2} g_- c_-) C_x^2 P(v)^2.$$

All constants concern the same selected h , fixed profiles and original compact label family. The displayed thresholds are finite and are added to the previous finite value thresholds; no uniformity as $h \downarrow 0$ is asserted. The exact actual-frame norm is

$$|t|^2 = (1 + s_a^2)x^2 + y^2.$$

The damping $\varepsilon k^2 |n_\Phi|^2 |t|^2$ remains in the energy equation, so positive production is not a claim of positive total energy derivative. Covariance mass uses the proved positive x and the original positive integral of $\psi^2 x^2$; its argument does not omit this damping. The carrier physical wavelength is $\sqrt{Q}/k \asymp Q^{1/2+h/2}$, while the leading physical wave amplitude has power $Q^{-A} \sqrt{\varepsilon} = Q^{-1/2-h/2}$, up to polynomial factors in S_* . Their product has power Q^0 ; this agrees with the retained leading damping $\varepsilon k^2 \asymp 1$.

42 The complete radial obstruction and the five-equation correction

This fragment proves the radial and finite-dimensional maps used in the initial mean correction. The input fields remain the actual fields: a nonzero flat field is retained in the field and its products. In particular, the proof does not identify Haar cancellation with exact characteristic cancellation. The reference formulas checked here are `mean_corrections/mean_corrections_body.tex`, (MC1)–(MC10), (MC13)–(MC15), and (MC22)–(MC30), together with the reserved-support construction in `base_heat/derivation.tex`.

42.1 Coordinates, domains, and fixed parameters

Fix the original parameters

$$0 < h < \frac{1}{100}, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad \kappa_s = 10^{-5}, \quad Q = 2^{-\ell}, \quad \epsilon = Q^h, \quad S_* = \ell^2,$$

where the integer band index satisfies $\ell \geq 1$. The physical variables are $r > 0, z \in \mathbb{R}, t < 1$. Write $\tau = 1 - t$. The equations

$$z = q^D \eta, \quad \tau = q(1 - \eta^2), \quad X = \frac{r^2}{2q}, \quad L_q = 1 - 2h\eta^2 \quad (\text{RF1})$$

specify $q > 0, |\eta| < 1$ uniquely: the derivative of $q - z^2 q^{2h}$, on $q > |z|^{1/D}$, equals $1 - 2hz^2 q^{2h-1} \geq 1 - 2h > 0$; the function starts at zero and tends to infinity. Implicit differentiation gives

$$q_t = -L_q^{-1}, \quad q_z = 2\eta q^{1-D}/L_q, \quad \eta_t = D\eta/(qL_q), \quad \eta_z = (1 - \eta^2)/(q^D L_q). \quad (\text{RF2})$$

These formulas also give one-sided derivatives at $|\eta| = 1$ when $q > 0$. There is no division by $1 - \eta^2$.

In a fixed band put

$$R = Q^{-1/2}r, \quad Z = Q^{-D}z, \quad T = Q^{-1}\tau, \quad s = q/Q, \quad X = R^2/(2s).$$

The source band has $s_- = 1/2 \leq s \leq s_+ = 2$; we retain the symbols s_-, s_+ when writing its fixed endpoint bounds. They give a compact range for $Z = s^D \eta, T = s(1 - \eta^2)$, including its one-sided boundary. Equations (RF2) give

$$\partial_Z s = 2\eta s^{1-D}/L_q, \quad \partial_T s = L_q^{-1}, \quad \partial_Z \eta = (1 - \eta^2)/(s^D L_q), \quad \partial_T \eta = -D\eta/(sL_q). \quad (\text{RF3})$$

All repeated derivatives are bounded on that compact domain, because every differentiated denominator is a power of s or L_q , with positive lower bounds. This proves the needed slow-coordinate bounds. The active shell is

$$X_a < X < X_b, \quad 0 < X_a < X_b < \infty, \quad R_a(Z, T) = \sqrt{2sX_a}, \quad R_b(Z, T) = \sqrt{2sX_b}. \quad (\text{RF4})$$

Every field supported in its closed shell is extended smoothly by zero across both edges. The resulting uniform radial interval stays away from $R = 0$.

Retain the integer matrix and its exact eigenvectors:

$$J_g = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad \Lambda_g = 4 - \sqrt{2}, \quad T_g = 4 + \sqrt{2}, \quad v_r = (1, 1 - \sqrt{2}), \quad v_t = (\sqrt{2} - 1, 1),$$

$$\rho_g = \frac{\log \Lambda_g}{\log T_g}, \quad d_r = 2((1 + h)\rho_g - h\kappa_s) > 0.$$

The positivity follows, for example, from $\Lambda_g > 2, T_g < 8$, so $\rho_g > 1/3$, while $0 < h < 1/2$. Direct matrix multiplication proves $J_g v_r = \Lambda_g v_r$ and $J_g v_t = T_g v_t$. At a fixed common covering level i_0 ,

$$y = J_g^{i_0} Y \in \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2, \quad M = \Lambda_g^{i_0} Q^{d_r/2}, \quad c = T_g^{i_0} Q^{1+h}, \quad L = v_r \cdot \partial_y, \quad N = v_t \cdot \partial_y.$$

The exact normalized differential operators are

$$D_r = \partial_R + M d_r R^{d_r-1} L, \quad D_z = \epsilon \partial_Z, \quad t_* = -\epsilon \partial_T + c N, \quad \Delta_0 = D_r^2 + R^{-1} D_r + D_z^2. \quad (\text{RF5})$$

Here Q, i_0, ϵ, M, c are fixed under all coefficient differentiation. The phase in physical variables is $Y = v_r r^{d_r} + v_t t$; the chain rule gives exactly (RF5). The operators D_r, D_z, t_* commute pairwise, since the only nonconstant coefficient in them depends just on R .

For $i(\ell) = \lfloor \log_{T_g}(Q^{-1-h}/S_*) \rfloor$,

$$T_g^{-1} S_*^{-1} < T_g^{i(\ell)} Q^{1+h} \leq S_*^{-1}, \quad T_g^{-\rho_g} \epsilon^{-\kappa_s} S_*^{-\rho_g} < \Lambda_g^{i(\ell)} Q^{d_r/2} \leq \epsilon^{-\kappa_s} S_*^{-\rho_g}. \quad (\text{RF6})$$

The second line follows by raising the first floor inequality to ρ_g and substituting $d_r/2 = (1 + h)\rho_g - h\kappa_s$. If $|i_0 - i(\ell)| \leq B$ for one fixed integer B , the bounds only change by the fixed factors $T_g^{\pm B}$ and $\Lambda_g^{\pm B}$. In particular $M^{-p} \leq C_p \epsilon^{p\kappa_s} S_*^{p\rho_g}$.

Define

$$d_a = \log(X/X_a), \quad d_b = \log(X_b/X), \quad \delta = \min(1, d_a, d_b), \quad \zeta = \exp(-a_a/d_a^2 - a_b/d_b^2), \quad a_a, a_b > 0.$$

The space $\mathcal{M}_\alpha$ is the space of compatible smooth torus coefficients with this shell support such that every fixed coefficient derivative I satisfies

$$|\partial^I f| \leq C_I \epsilon^\alpha S_*^{B_I} \zeta \delta^{-C'_I}. \quad (\text{RF7})$$

All constants may depend on the fixed stage, fixed profiles and derivative order, but not on the band. The space $\mathcal{S}_\alpha$ consists of slow functions of (Z, T) with the same estimate without $\zeta \delta^{-C'_I}$. Smooth coefficient derivatives preserve the exponent. Formula (RF5) shows that D_r loses at most κ_s , D_z gains one, and t_* preserves the exponent; on a torus-independent field t_* gains one. Leibniz' rule and $0 < \zeta \leq 1$ prove $\mathcal{M}_a \mathcal{M}_b \subset \mathcal{M}_{a+b}$.

42.2 The full radial exact sequence and its specified splitting

At a fixed slow point let $\mathcal{C}$ denote real C^∞ functions of $(R, y) \in (0, \infty) \times \mathbb{T}^2$ supported in the closed shell and smoothly zero across its edges. For $e \in \{0, 1, 2\}$ define $\mathcal{D}_e = D_r + e/R$ and

$$(\Gamma H)(R, w) = H(R, w + MR^{d_r} v_r).$$

Its inverse uses the opposite sign. The chain rule gives

$$\partial_R \Gamma(R^e \phi) = \Gamma(R^e \mathcal{D}_e \phi). \quad (\text{RF8})$$

Define a continuous linear map from $\mathcal{C}$ into smooth functions on the entire torus by

$$(K_e f)(w) = \int_0^\infty R^e f(R, w + MR^{d_r} v_r) dR. \quad (\text{RF9})$$

Continuity in the smooth topologies follows by differentiating the fixed compact integration interval; for each fixed M the required coefficient bounds are finite.

Proposition 42.1 (All radial obstructions). *The sequence of real vector spaces*

$$0 \longrightarrow \mathcal{C} \xrightarrow{\mathcal{D}_e} \mathcal{C} \xrightarrow{K_e} C^\infty(\mathbb{T}^2; \mathbb{R}) \longrightarrow 0 \quad (\text{RF10})$$

is exact. A compactly supported solution of $\mathcal{D}_e \phi = f$ exists if and only if $K_e f = 0$ as a torus function, and is unique.

Proof. Integrating (RF8) gives necessity, since its primitive is zero at both radial endpoints. Conversely set

$$\phi(R, y) = R^{-e} \int_0^R R'^e f(R', y + M(R'^{d_r} - R^{d_r}) v_r) dR'. \quad (\text{RF11})$$

It vanishes below the shell. Above it the integral equals $(K_e f)(y - MR^{d_r} v_r)$, which vanishes by the specified condition. Inside and at the edges its derivatives exist and match its zero extension, by differentiation of smooth compactly supported data and (RF8). That identity proves $\mathcal{D}_e \phi = f$. If $\mathcal{D}_e \phi = 0$, then $\Gamma(R^e \phi)$ is constant in R ; its value below the shell is zero, proving injectivity and uniqueness. Finally choose $h_m \in C_c^\infty((R_a, R_b))$ of integral one. For any prescribed $F \in C^\infty(\mathbb{T}^2)$, $f(R, y) = R^{-e} h_m(R) F(y - MR^{d_r} v_r)$ satisfies $K_e f = F$, by direct substitution. This proves surjectivity. $\square$

Choose the original slow radial step $\chi_m = \chi_m(X)$, zero near X_a , one near X_b , with $\partial_R \chi_m$ supported in a fixed interior subinterval. For $g = R^e f$, put

$$\begin{aligned} Ig(R, y) &= \int_0^R g(R', y + M(R'^{d_r} - R^{d_r}) v_r) dR', \\ Jg(R, y) &= \int_0^\infty g(R', y + M(R'^{d_r} - R^{d_r}) v_r) dR', \\ T_e f &= R^{-e} (Ig - \chi_m Jg), \quad A_e f = R^{-e} (\partial_R \chi_m) Jg. \end{aligned} \quad (\text{RF12})$$

In particular $Jg(R, y) = (K_e f)(y - MR^{d_r} v_r)$. Equation (RF8) gives $D_r Ig = g$, $D_r Jg = 0$, and hence

$$\mathcal{D}_e T_e = 1 - A_e, \quad K_e A_e = K_e, \quad A_e^2 = A_e, \quad T_e \mathcal{D}_e = 1 \quad \text{on } \mathcal{C}. \quad (\text{RF13})$$

For the second identity, conjugate the integral for $K_e A_e f$: the integrand becomes $\partial_R \chi_m K_e f(w)$, whose integral equals $K_e f(w)$. It then implies idempotence from (RF12). For the last identity $K_e \mathcal{D}_e = 0$, so

the compact primitive $T_e \mathcal{D}_e \phi$ equals ϕ by uniqueness. The same argument proves $T_e A_e = 0$, since the conjugated partial integral of $\partial_R \chi_m K_e f$ equals $\chi_m K_e f$. Thus the specified projector splits the exact sequence.

Let a bar mean Haar average on $\mathbb{T}^2$, with total mass one. Translation invariance and Fubini give

$$\overline{K_e f} = \int_0^\infty R^e \bar{f}(R) dR. \quad (\text{RF14})$$

This scalar condition is strictly weaker than $K_e f = 0$. Indeed, for every nonzero $k \in \mathbb{Z}^2$, the real function

$$f_M(R, y) = e^{-M^2} R^{-e} h_m(R) \cos(2\pi k \cdot (y - MR^{d_r} v_r)) \quad (\text{RF15})$$

has $\bar{f}_M(R) = 0$ at every radius, whereas $K_e f_M(w) = e^{-M^2} \cos(2\pi k \cdot w) \neq 0$. Taking a smoothly scaled fixed interior bump makes this a family on the whole band domain. Every fixed derivative grows by at most a fixed power of M before the factor e^{-M^2} . Since (RF6) implies $M \rightarrow \infty$ exponentially in ℓ , that factor controls every prescribed power of ϵ . Thus even an all-orders flat, zero-Haar source need not admit exact compact radial inversion.

42.3 Weighted primitive bounds with moving shell edges

Set $U = R^{d_r}$ and, on the positive radial interval, define

$$F(U, Z, T, y) = \frac{g(R, Z, T, y)}{d_r R^{d_r-1}}.$$

Extend it by zero to $U \in \mathbb{R}$. This extension is smooth, because its support stays in a compact subset of $U > 0$. The change of variables and its inverse have uniformly bounded fixed derivatives on the radial interval. The integrals (RF12) become

$$A_-(U, y) = \int_{-\infty}^0 F(U + u, y + Muv_r) du, \quad A_+(U, y) = \int_0^\infty F(U + u, y + Muv_r) du, \quad (\text{RF16})$$

with $Ig = A_-$, $Jg = A_- + A_+$. Thus $Ig - \chi_m Jg = (1 - \chi_m)A_- - \chi_m A_+$. The argument shift Muv_r is independent of U, Z, T . Every coefficient derivative in (RF16) therefore differentiates just F ; no extra factor of M is produced. The effective integration length is uniformly bounded. Moving edges give no boundary terms, because the integration is of the smooth zero extension on fixed half-lines.

It remains to retain the prescribed weight. For any $a > 0, B \geq 0$,

$$\frac{d}{ds} \log(e^{-a/s^2} s^{-B}) = 2a/s^3 - B/s > 0 \quad (\text{RF17})$$

on a sufficiently small positive interval (on every positive interval if $B = 0$). Near the inner edge only A_- is present, and its contributing radii satisfy $0 < d_a(R') \leq d_a(R)$. The other edge factor and all its fixed derivatives are bounded there. Apply (RF17) to each differentiated input bound to obtain $C\zeta(R)\delta(R)^{-B'}$. Near the outer edge only A_+ occurs, and its contributing radii satisfy $0 < d_b(R') \leq d_b(R)$; the same calculation applies. On the intervening closed interior region ζ has a positive lower bound, so the unweighted integral bound supplies the desired estimate. Derivatives of χ_m have this interior support. The finite Leibniz and change-of-variable sums now prove

$$T_e : \mathcal{M}_\alpha \longrightarrow \mathcal{M}_\alpha. \quad (\text{RF18})$$

The operator cannot create a slow point outside the projection of the source's closed support, since the entire source integral is zero on every slow neighborhood where the source is zero.

42.4 Every derivative of the projector is flat after Haar cancellation

The directional inverses used here have exact small-divisor bounds. If $k = (k_1, k_2) \neq 0$, then

$$(v_r \cdot k)((k_1 + k_2) + \sqrt{2}k_2) = (k_1 + k_2)^2 - 2k_2^2$$

is a nonzero integer. It cannot vanish unless both integer coordinates are zero, by irrationality of $\sqrt{2}$. The second factor is at most $C|k|$, hence $|v_r \cdot k|^{-1} \leq C(1 + |k|)$. The identical argument with $(k_2 - k_1) + \sqrt{2}k_1 = v_t \cdot k$ proves the bound for N . On zero-mean torus functions define

$$(L^{-p}H)^\wedge(k) = \begin{cases} (2\pi i v_r \cdot k)^{-p} \hat{H}(k), & k \neq 0, \\ 0, & k = 0. \end{cases}$$

Integrate the Fourier coefficient $m + p + 3$ times in a coordinate with $|k_j| \geq |k|/\sqrt{2}$. After an output derivative of order at most m , each summand is bounded by $C\|H\|_{C^{m+p+3}}(1+|k|)^{-3}$. Since that sequence is summable on $\mathbb{Z}^2$,

$$\|L^{-p}H\|_{C^m} \leq C_{m,p}\|H\|_{C^{m+p+3}}. \quad (\text{RF19})$$

The proof also justifies termwise differentiation, preservation of reality, and commutation with coefficient derivatives outside the torus. The same proof gives N^{-1} , with a loss of four derivatives.

Suppose the exact identity $\int R^e \bar{f} dR = 0$ holds at every slow point. For $F^\circ = F - \bar{F}$, integration of

$$\frac{d}{du} L^{-1} F^\circ(U + u, y + Muv_r) = \partial_U L^{-1} F^\circ(U + u, y + Muv_r) + M F^\circ(U + u, y + Muv_r)$$

over the whole line has zero boundary terms. Repeating it p times gives the exact identity, with its retained sign,

$$Jg = (-M^{-1})^p \int_{\mathbb{R}} \partial_U^p L^{-p} F^\circ(U + u, y + Muv_r) du. \quad (\text{RF20})$$

The omitted zero Fourier mode and every one of its slow derivatives integrate to zero by the specified moment identity. Applying (RF19) and the fixed-shift differentiation argument gives, for every fixed coefficient derivative I and every integer $p \geq 1$,

$$|\partial^I A_e f| \leq C_{I,p} \epsilon^{\alpha+p\kappa_s} S_*^{B_{I,p}} \zeta \delta^{-C'_{I,p}}, \quad \overline{A_e f} = 0. \quad (\text{RF21})$$

Only finitely many extra input derivatives are needed for each (I, p) : at most p extra radial derivatives and $p + 3$ extra torus derivatives in the displayed estimate. The output is supported where $\partial_R \chi_m \neq 0$, so the weight there is bounded below. If f is torus independent, then $Jg = \int R'^e f(R') dR' = 0$ exactly; thus $A_e f = 0$.

Here is the quantifier statement needed for a fixed construction. Choose the fixed parameters, band comparison bounds, covering-offset bound and a fixed lower band index $\ell_* \geq 1$ once. Fix a physical field normalization and an output Cartesian derivative order m . In the shell every graph derivative follows (RF2), (RF5), and the derivatives of the cylindrical frame. The latter cost powers of r^{-1} , with $r \asymp Q^{1/2}$. Consequently a finite sum of coefficient derivatives controls the physical derivative, with a loss $Q^{-L_m} \ell^{B_m}$, for some finite L_m independent of p . For desired flat order $N_0 \geq 0$, choose an integer p with $h(\alpha + p\kappa_s) - L_m > N_0$. Formula (RF21) gives

$$|\nabla_{x,t}^{\leq m} (A_e f)_{\text{phys}}| \leq C Q^{N_0+\beta} \ell^B \leq C' q^{N_0}, \quad \beta = h(\alpha + p\kappa_s) - L_m - N_0 > 0. \quad (\text{RF22})$$

The last constant is finite on the same bands $\ell \geq \ell_*$, because $\sup_{u \geq 1} u^B 2^{-\beta u} < \infty$, as follows by differentiating $B \log u - \beta(\log 2)u$. If $B < 0$, the bound is immediate. Thus p and the constants may depend on m, N_0 , while the allowed physical domain does not shrink with m, N_0 . This is all-orders flatness, proved on one fixed domain.

These maps also respect the covering maps. For an integer $n \geq 0$ put $P_n H(y) = H(J_g^n y)$, $M' = \Lambda_g^n M$. Direct substitution gives

$$T_e^M P_n = P_n T_e^{M'}, \quad A_e^M P_n = P_n A_e^{M'}, \quad K_e^M P_n = P_n K_e^{M'}. \quad (\text{RF23})$$

The Haar mean is preserved: a Fourier mode k pulls back to $(J_g^n)^T k$, which is zero only if $k = 0$, since $\det J_g = 14 \neq 0$. Thus noninvertibility of this torus covering does not obstruct the exact compatibility of the radial maps.

42.5 Pressure, temporal inversion, and the actual divergence-free field

Choose the fixed physical bump $\rho_{\text{phys}} = q^{-1/2} \hat{\rho}(r/\sqrt{q})$, $\int \hat{\rho}(x) dx = 1$, with support in a fixed proper interior interval. Its chart representative $\rho = Q^{1/2} \rho_{\text{phys}} = s^{-1/2} \hat{\rho}(R/\sqrt{s})$ has $\int \rho dR = 1$. For a radial source g_r , define

$$P = \int \bar{g}_r dR, \quad p_m = T_0(g_r - \rho P).$$

Then the exact residual, including its flat part, is

$$D_r p_m - g_r = -\rho P - A_0(g_r - \rho P), \quad \partial_R \bar{p}_m = \bar{g}_r - \rho P. \quad (\text{RF24})$$

The first term is a specified slow bump residual. The second has zero Haar mean and is flat by (RF21)–(RF22). Radial integration preserves the exponent into $\mathcal{S}_\alpha$, because every $\zeta \delta^{-B}$ is bounded and the interval has bounded length. Thus (RF18) proves the pressure and pressure-difference estimates.

For a requested axial increment γ_d satisfying $\int R\bar{\gamma}_d dR = 0$, define the actual potential and field

$$\Psi_* = T_1\gamma_d, \quad \Delta\beta = -D_z\Psi_* = -\epsilon\partial_Z\Psi_*, \quad \Delta\gamma = \mathcal{D}_1\Psi_* = \gamma_d - A_1\gamma_d. \quad (\text{RF25})$$

Their divergence vanishes exactly:

$$\mathcal{D}_1\Delta\beta + D_z\Delta\gamma = -\epsilon\mathcal{D}_1\partial_Z\Psi_* + \epsilon\partial_Z\mathcal{D}_1\Psi_* = 0.$$

This uses commutation of differential operators on the constructed potential. It does not commute ∂_Z with T_1 , whose cutoff and shell can depend on Z . Equations (RF18) and (RF21) give $\Psi_*, \Delta\gamma \in \mathcal{M}_\alpha$ and $\Delta\beta \in \mathcal{M}_{\alpha+1}$. Moreover $\overline{\Delta\gamma} = \bar{\gamma}_d$, so the weighted axial moment remains exactly zero. When γ_d is slow, $A_1\gamma_d = 0$ and this realization is pointwise exact.

For zero-Haar tangential errors $E_\theta^\circ, E_z^\circ$, set

$$\Delta v = -c^{-1}N^{-1}E_\theta^\circ, \quad \gamma_d = -c^{-1}N^{-1}E_z^\circ, \quad a = -A_1\gamma_d. \quad (\text{RF26})$$

The inverse Fourier estimate and $c^{-1} \leq CS_*$ give the same coefficient exponents, and the inputs have zero Haar mean at each radius. Realize the axial field by (RF25). Since torus translations and the slow cutoff commute with N ,

$$cN\Delta v = -E_\theta^\circ, \quad cN\Delta\gamma = -E_z^\circ + cNa. \quad (\text{RF27})$$

The term cNa is flat; the terms $-\epsilon\partial_T\Delta v$ and $-\epsilon\partial_T\Delta\gamma$ are the retained slow-time terms with one extra ϵ . The actual velocity contains a . In particular its products contain a^2 and all cross terms; zero Haar mean of a does not eliminate those products. This gives the exact interpretation of the temporal mean step used before the five-equation map.

42.6 The original five rows and every profile-to-chart factor

Let $I_{\text{mean}} \Subset (X_a, X_b)$ be the reserved open power-law interval, and $I_m = \{\sqrt{2X} : X \in I_{\text{mean}}\}$. The original leading profiles on this entire interval are

$$U_0 = 0, \quad E_0 = c_{\text{patch}}(1 + \eta^2)^{-1}X^{-1/2-\lambda}, \quad \lambda > 0, \quad c_{\text{patch}} > 0. \quad (\text{RF28})$$

Hence the q -profile tangential base, under the exact map $x = r/\sqrt{q} = \sqrt{2X}$, is

$$G_q = q^A G_{\text{phys}} = 0, \quad V_q = q^A V_{\text{phys}} = a(\eta)x^{-1-2\lambda}, \quad a(\eta) = 2^{1/2+\lambda}c_{\text{patch}}(1 + \eta^2)^{-1}. \quad (\text{RF29})$$

In particular $a(\eta) \geq a_0 = 2^{-1/2+\lambda}c_{\text{patch}} > 0$ on $|\eta| \leq 1$, and all derivatives of a^{-1} are bounded there. Section below verifies that the positive-order base construction preserves this exact law, not merely its leading term.

For the actual mean velocity (β, v, γ) and fixed full wave covariance W^{ab} , the three slow defects are

$$\begin{aligned} P &= \int \bar{g}_r dR, \\ J_\theta &= \int R^2 \overline{Gv + V\gamma + \gamma v + W^{z\theta}} dR, \\ J_z &= \int R^2 \overline{G\gamma + \gamma^2 + W^{zz}} dR - \frac{1}{2} \int R^2 \bar{g}_r dR. \end{aligned} \quad (\text{RF30})$$

The five requested correction rows in the fixed Q chart are

$$\begin{aligned} \int R^2 \Delta v dR &= 0, & \int R\gamma_d dR &= 0, \\ \int (2V/R)\Delta v dR &= -P, \\ \int R^2 (G\Delta v + V\gamma_d) dR &= -J_\theta, \\ \int (2RG\gamma_d - RV\Delta v) dR &= -J_z. \end{aligned} \quad (\text{RF31})$$

The source $g_{r,*} = Q^{2A+1/2}g_{r,\text{phys}}$ and velocities $u_* = Q^A u_{\text{phys}}$ give

$$P = Q^{2A}P_{\text{phys}}, \quad J_\theta = Q^{2A-3/2}J_{\theta,\text{phys}}, \quad J_z = Q^{2A-1}J_{z,\text{phys}}. \quad (\text{RF32})$$

Indeed $R = Q^{-1/2}r$, $dR = Q^{-1/2}dr$: each product in J_θ has velocity factor Q^{2A} and measure $Q^{-3/2}$, while each term in J_z has factor Q^{2A-1} , including the source integral with its retained $-1/2$. Therefore the exact q -profile targets are

$$P_q = s^{2A}P, \quad J_{\theta,q} = s^{2A-3/2}J_\theta, \quad J_{z,q} = s^{2A-1}J_z. \quad (\text{RF33})$$

If $u(x), d(x)$ are profile increments, their chart fields are

$$\Delta v(R) = s^{-A}u(R/\sqrt{s}), \quad \gamma_d(R) = s^{-A}d(R/\sqrt{s}). \quad (\text{RF34})$$

Substitution $R = \sqrt{s}x$ into every row of (RF31) proves that its left side is its x -profile left side multiplied, respectively, by $s^{3/2-A}, s^{1-A}, s^{-2A}, s^{3/2-2A}, s^{1-2A}$. Equations (RF33) are exactly the matching target factors.

Choose $x_0 \in I_m$ and a fixed $d_0 > 0$ such that $x_0, x_0e^{d_0}, x_0e^{2d_0}$ all lie in I_m . Choose a sufficiently narrow nonnegative nonzero $\eta_0 \in C_c^\infty(I_m)$ around x_0 , and set

$$\eta_j(x) = e^{-jd_0}\eta_0(e^{-jd_0}x), \quad j = 0, 1, 2, \quad \mu_p = \int_0^\infty x^p \eta_0(x) dx > 0.$$

Their supports are disjoint, their closed convex hull is contained in I_m , and direct change of variables gives

$$\int x^p \eta_j(x) dx = \mu_p e^{jd_0 p} \quad (p \in \mathbb{R}). \quad (\text{RF35})$$

The positivity and finite separation of the three centers prove that such a common support width exists.

Use $u(x) = \sum_{j=0}^2 u_j \eta_j(x)$, $d(x) = \sum_{j=0}^1 s_j \eta_j(x)$. Set

$$p_0 = 2, \quad p_1 = -2 - 2\lambda, \quad p_2 = -2\lambda, \quad t_i = e^{d_0 p_i}, \quad C(s'_0) = u_0 + u_1 s'_0 + u_2 (s'_0)^2.$$

The three angular rows are precisely

$$C(t_0) = 0, \quad C(t_1) = -\frac{P_q}{2a\mu_{p_1}} =: b_1, \quad C(t_2) = \frac{J_{z,q}}{a\mu_{p_2}} =: b_2.$$

Their unique solution is the polynomial

$$C(s'_0) = b_1 \frac{(s'_0 - t_0)(s'_0 - t_2)}{(t_1 - t_0)(t_1 - t_2)} + b_2 \frac{(s'_0 - t_0)(s'_0 - t_1)}{(t_2 - t_0)(t_2 - t_1)}. \quad (\text{RF36})$$

Every denominator is nonzero, since $p_1 < p_2 < p_0$. Evaluation at each of the three nodes proves the three equations. Uniqueness follows because a polynomial of degree at most two cannot have three distinct roots: division by their three linear factors would force degree at least three. The signed determinant in row order (angular moment, pressure, axial-flux contribution) is

$$-2a^2 \mu_2 \mu_{-2-2\lambda} \mu_{-2\lambda} (t_1 - t_0)(t_2 - t_0)(t_2 - t_1) \neq 0. \quad (\text{RF37})$$

For the two axial rows set $t_b = e^{d_0}$, $t_a = e^{d_0(1-2\lambda)}$. The exact solution is

$$s_1 = -\frac{J_{\theta,q}}{a\mu_{1-2\lambda}(t_a - t_b)}, \quad s_0 = -t_b s_1. \quad (\text{RF38})$$

Indeed $\mu_1(s_0 + t_b s_1) = 0$ and $a\mu_{1-2\lambda}(s_0 + t_a s_1) = -J_{\theta,q}$. The signed determinant is $a\mu_1 \mu_{1-2\lambda}(t_a - t_b) \neq 0$, since $\lambda > 0$. Equations (RF34)–(RF38) prove all five original rows, including the minus sign of $-RV\Delta v$.

This is a linear map from the three arbitrary target functions into the fixed five-dimensional correction space; two of its five prescribed outputs are zero. Its uniqueness concerns that chosen space. For every fixed $\lambda > 0$, $c_{\text{patch}} > 0$, its inverse coefficients and all fixed slow derivatives are bounded on $|\eta| \leq 1$, because all node differences and μ_p are fixed nonzero constants and a has the explicit positive lower bound. Equations (RF3), (RF33), (RF34), and Leibniz' rule prove the map

$$(\mathcal{S}_\alpha)^3 \longrightarrow \mathcal{M}_\alpha \times \mathcal{M}_\alpha. \quad (\text{RF39})$$

The output weight has a positive lower bound on its fixed interior profile support. The linear solve requires no small target size and no new small- q threshold. Constants are not uniform as $\lambda \downarrow 0$: at $\lambda = 0$, $x^2 V_q = ax$, so $\int x dx = 0$ implies $\int x^2 V_q dx = 0$. A target $J_{\theta,q} \neq 0$ is then impossible for any compact smooth axial correction, not only for the selected bumps. This is the exact degeneracy excluded by the original fixed positive parameter.

The slow input γ_d has zero weighted moment. Equation (RF25) therefore gives $\Delta\gamma = \gamma_d$ exactly. Its potential is supported in the closed interval between the first and last axial bump: below the first bump the integral is zero; above the last it is its zero total. It need not vanish between bumps. Thus the base law and coefficient estimates must hold throughout that interval. Its induced radial component is $\Delta\beta \in \mathcal{M}_{\alpha+1}$.

42.7 Why the Borel base retains the exact reserved power law

The original leading law (RF28) follows from the actual schedule. Retain the fixed step

$$\sigma(v) = \frac{e^{-1/v^2}}{e^{-1/v^2} + e^{-1/(1-v)^2}} \quad (0 < v < 1), \quad \sigma = 0 \quad (v \leq 0), \quad \sigma = 1 \quad (v \geq 1).$$

Its extensions are smooth: every fixed derivative of the vanishing exponential is that exponential times a polynomial in the inverse edge distance, and hence tends to zero at that edge. The parameter order is M_d , then $T_d = e^{M_d} + 10$ and $P_* > e^{T_d}$, then small positive λ , then small positive h relative to λ, e^{-T_d} , then large X_R . The finite restriction $0 < \lambda < e^{-25/60}$ ensures $T_w = 60 \log(1/\lambda) > 25$. Put $X_w = X_R e^{T_d+2}$. The four reserved open intervals are

$$\begin{aligned} I_1 &= X_w \exp((T_w - 25, T_w - 20)), & I_2 &= X_w \exp((T_w - 20, T_w - 15)), \\ I_3 &= X_w \exp((T_w - 14, T_w - 9)), & I_4 &= X_w \exp((T_w - 8, T_w - 3)). \end{aligned} \quad (\text{RF40a})$$

Their assignments are cone repair, heat compensation, positive-order moments and mean correction: $I_{\text{pos}} = I_3$, $I_{\text{mean}} = I_4$. They are proper subintervals of the power-law stage; the gap between I_3, I_4 has logarithmic length one. The initial reference has $U = 4\eta, E = P_* f(X/X_R)^{1/10}$ for $X \leq X_R$, with $f = (1 + \eta^2)^{-1}$. With the logarithmic coordinate reset to zero at each stage, the next three prescriptions are

length	U	$l := \partial_y \log(\sqrt{2XE})$
1	4η	$\frac{3}{5}(1 - \sigma(y))$
T_d	$k(y)\eta$	0
1	0	$-\lambda\sigma(y)$

$$k(y) = 4 \left[1 - \sigma \left(\frac{\log(1+y)}{M_d} \right) \right]. \quad (\text{RF40b})$$

The next stage holds $U = 0, l = -\lambda$ for length T_w . Integrating $\partial_y \log E = l - 1/2$ with the original endpoint amplitudes defines the positive constants

$$\begin{aligned} P_1 &= P_* \exp \left(\int_0^1 \left[\frac{3}{5}(1 - \sigma(v)) - \frac{1}{2} \right] dv \right), \\ e_w &= P_1 e^{-T_d/2} \exp \left(\int_0^1 \left[-\lambda\sigma(v) - \frac{1}{2} \right] dv \right), \quad c_{\text{patch}} = e_w X_w^{1/2+\lambda}. \end{aligned} \quad (\text{RF40c})$$

Thus throughout the power-law stage, including I_4 , $E = e_w f(X/X_w)^{-1/2-\lambda} = c_{\text{patch}} f X^{-1/2-\lambda}$ and $U = 0$.

The subsequent modifications preserve that exact statement on the full interval. The regular-axis join ends before $X_h = X_R e^{-5} < X_R$; cone modulation is followed by repair in I_1 ; heat compensation is in I_2 and changes only E ; and the later pulse and exterior heat edit start after I_4 . None changes E, U on I_4 . For the radial field write $\mathfrak{M}(X, \eta) = \int_0^X U(\xi, \eta) d\xi$. The axis and cone moment repairs restore $\mathfrak{M}$ exactly, while the heat edits have axial increment zero. Above a repair endpoint X_{rep} , equality of the two U 's gives the exact propagation map

$$\mathfrak{M}_{\text{new}}(X, \eta) - \mathfrak{M}_{\text{old}}(X, \eta) = \mathfrak{M}_{\text{new}}(X_{\text{rep}}, \eta) - \mathfrak{M}_{\text{old}}(X_{\text{rep}}, \eta) = 0. \quad (\text{RF40d})$$

It also proves equality of every parameter derivative.

The exact restoration just used is enforced by the actual repair equations. At base $U_0 = 0, E_0 = K(\eta)X^{-1/2-\lambda}$, the increments u_c, e_c change $(\mathfrak{M}, J, I, S, C_p)$, respectively, by

$$\int u_c, \quad \int (\sqrt{2X}E_0 u_c + \sqrt{2X}u_c e_c), \quad \int \sqrt{2X}e_c, \quad \int (u_c^2 - E_0 e_c - e_c^2/2), \quad \int (E_0 e_c/X + e_c^2/(2X)), \quad (\text{RF40e})$$

where the measure is dX . With two axial and three angular bumps this is precisely $B(\eta)c + Q_\eta(c, c)$. The linear blocks have powers $(0, -\lambda)$ and $(1/2, -1/2 - \lambda, -3/2 - \lambda)$, with fixed nonzero amplitude factors. Their matrices are invertible on ordered nonnegative bumps: a nonzero sum of m distinct real powers has at most $m - 1$ positive zeros. Divide by its least power, apply Rolle's theorem, and induct on the derivative's fewer powers to prove this assertion. The evaluation determinant is therefore nonzero with a fixed sign on the connected set of ordered nodes. By multilinearity, the moment determinant is its integral against the product of the ordered nonnegative bumps and cannot vanish. Write $\beta = \sup \|B^{-1}\|$ and $\|Q(c, c')\| \leq \kappa \|c\| \|c'\|$. For discrepancy d_* , the map $c \mapsto B^{-1}(d_* - Q(c, c))$ maps $\|c\| \leq 2\beta \|d_*\|$ into itself and contracts by at most $1/2$ if $8\beta^2 \kappa \|d_*\| \leq 1$. Iteration from zero converges by a geometric-series bound on successive differences, and its limit solves the exact moment equation. Its differentiated matrix

is invertible by the same bound, so differentiating successively supplies every fixed parameter derivative after this one choice. For cone repair $\|d_*\| \leq D_0/N$, and the construction fixes $N \geq 8\beta^2\kappa D_0$ before the physical scale or any band. For the earlier axis repair the fixed invertible row operations $J \mapsto J - 4\eta I$, $S \mapsto S - 8\eta\mathfrak{M}$ give distinct powers $(0, 3/5)$ and $(1/2, 1/10, -9/10)$; the same determinant argument and quadratic iteration implement the construction's chosen finite join tolerance. Thus (RF40d) uses exact solved equations, not estimates on residual moments.

The reference cumulative axial moment can now be calculated. Put

$$\mathfrak{m} = eX_R \left(4 + \int_0^{T_d} k(v)e^v dv \right) > 0. \quad (\text{RF40f})$$

At $X = eX_R$ the reference moment is $4\eta eX_R$. On the next stage $X = eX_R e^v$, $dX = eX_R e^v dv$, so direct integration gives $\mathfrak{M} = \mathfrak{m}\eta$. Every following stage through I_4 has $U = 0$, and (RF40d) proves the same moment for the final leading field. Its exact radial reconstruction then gives

$$V_0 = \frac{2\eta XU_0 - 2D\eta\mathfrak{M} - (1 - \eta^2)\mathfrak{M}\eta}{L_q} = -\mathfrak{m} \frac{2D\eta^2 + 1 - \eta^2}{1 - 2h\eta^2} = -\mathfrak{m}. \quad (\text{RF40g})$$

Here $2D = 1 - 2h$ makes the numerator factor exactly L_q . Consequently the full leading radial field on I_4 is $u_{r,0} = -\mathfrak{m}/r$.

Here the source construction can be checked at the level of its actual coefficients. Its positive-order angular and axial profiles E_n, U_n are cut off before a fixed positive-order patch I_{pos} , then corrected by three angular and two axial bumps in that patch. This lies strictly before the separate mean patch. The imposed axial moment is

$$\int_0^\infty RU_n(R, \eta) dR = \int_0^\infty U_n(X, \eta) dX = 0, \quad R = \sqrt{2X}, \quad dX = R dR. \quad (\text{RF40})$$

Beyond those velocity bump supports, $E_n = U_n = 0$ and $F_n(X, \eta) = \int_0^X U_n(x, \eta) dx = 0$. Their radial profile is the exact formula

$$V_n = \frac{2\eta XU_n - 2\eta(D + 2nh)F_n - (1 - \eta^2)\partial_\eta F_n}{L_q} = 0. \quad (\text{RF41})$$

The zero function F_n is zero for every parameter on this open interval, so its parameter derivatives vanish as well. Notice the coefficient $D + 2nh$, which is retained even though the evaluated terms vanish.

The actual Borel cutoff is applied to the divergence-free potential. Its exact radial and axial components are

$$\begin{aligned} ru_{r,n}^{\text{cut}} &= q^{2nh} \left[\chi(c_n q) V_n - \frac{2\eta}{L_q} (c_n q) \chi'(c_n q) F_n \right], \\ u_{z,n}^{\text{cut}} &= q^{-A+2nh} \chi(c_n q) U_n, \quad u_{\theta,n}^{\text{cut}} = q^{-A+2nh} \chi(c_n q) E_n. \end{aligned} \quad (\text{RF42})$$

The minus sign comes from $ru_r = -\partial_z(q^{1-A+2nh} F_n)$. On the entire mean interval all three components in (RF42) vanish identically, including the derivative-of-cutoff term. Summing any number of these zero functions, or the locally finite Borel sum, changes none of the leading velocities there. Positive-order pressure is not asserted zero on this interval, and is not used in the five matrix rows. This proves the exact passage from the leading law (RF28) to the summed base law (RF29). It also explains why asymptotic closeness alone would have been insufficient for the advertised exact matrix.

The just-proved radial identity and the vanishing of every positive-order cutoff contribution give the exact chart law

$$b = Q^A b_{\text{phys}} = -\epsilon \frac{\mathfrak{m}}{R}, \quad V = a(\eta) s^{\lambda-h} R^{-1-2\lambda}, \quad G = 0. \quad (\text{RF43})$$

The swirl formula follows from the exact coordinate composition, since $-A + (1 + 2\lambda)/2 = \lambda - h$. In particular $\partial_R^j b = (-1)^{j+1} \epsilon \mathfrak{m} j! R^{-j-1}$; every coefficient derivative of b involving Z, T or the torus is zero. The positive lower radial bound proves $\partial^I b = O_I(\epsilon)$ there, while (RF3), (RF43) give $\partial^I V, \partial^I G = O_I(1)$. These estimates hold on the whole induced-potential interval, including its gaps. The constants depend on the fixed profile and requested derivative order. The construction parameters require no choice depending on the band, derivative order or correction stage.

42.8 Exact nonlinear recomputation after the five rows

Keep the base and the actual wave w , including its complete covariance W , fixed in this step. The full radial source is

$$g_r = -t_*\beta - (D_r + 1/R)(2b\beta + \beta^2 + W^{rr}) - D_z(b\gamma + G\beta + \beta\gamma + W^{zr}) + (2Vv + v^2 + W^{\theta\theta})/R + \epsilon(\Delta_0 - R^{-2})\beta. \quad (\text{RF44})$$

Write $B = \Delta\beta, u = \Delta v, d = \Delta\gamma = \gamma_d$ for the actual increments just constructed. Expanding each product with no deletion gives

$$g_{r,\text{new}} - g_r = (2V/R)u + R_g, \quad (\text{RF45})$$

where

$$R_g = -t_*B - (D_r + 1/R)(2bB + 2\beta B + B^2) - D_z(bd + GB + \beta d + \gamma B + Bd) + (2vu + u^2)/R + \epsilon(\Delta_0 - R^{-2})B. \quad (\text{RF46})$$

For example $(\beta + B)^2 - \beta^2 = 2\beta B + B^2$, $(\beta + B)(\gamma + d) - \beta\gamma = \beta d + \gamma B + Bd$, and $(v + u)^2 - v^2 = 2vu + u^2$. These account for every nonlinear term of (RF44); W is unchanged. Integrating (RF45), and expanding the products in (RF30), the five exact rows imply

$$\begin{aligned} P_{\text{new}} &= \int \overline{R_g} dR, \\ (J_\theta)_{\text{new}} &= \int R^2 \overline{\gamma u + vd + du} dR, \\ (J_z)_{\text{new}} &= \int R^2 \overline{\gamma d + d^2} dR - \frac{1}{2} \int R^2 \overline{R_g} dR. \end{aligned} \quad (\text{RF47})$$

In the final row the linear pressure moment is exactly $-\frac{1}{2} \int R^2 (2V/R)u dR = -\int RVu dR$, which explains the retained sign in the fifth matrix row. These formulas cancel the base-linear terms; they do not assert the new defects are zero.

For the Section 9 budgets $v, \gamma \in \mathcal{M}_{0.9}$, $\beta \in \mathcal{M}_{1.9}$, and target order $\alpha \geq 0.9$, (RF39) and (RF25) give $u, d \in \mathcal{M}_\alpha$, $B \in \mathcal{M}_{\alpha+1}$. Every increment is torus independent. With the exact local base bounds just proved, the individual guaranteed exponents in (RF46) are

term	exponent	term	exponent
$-t_*B$	$\alpha + 2$	$-(D_r + 1/R)(2bB)$	$\alpha + 2 - \kappa_s$
$-(D_r + 1/R)(2\beta B)$	$\alpha + 2.9 - \kappa_s$	$-(D_r + 1/R)B^2$	$2\alpha + 2 - \kappa_s$
$-D_z(bd)$	$\alpha + 2$	$-D_z(GB)$	$\alpha + 2$
$-D_z(\beta d)$	$\alpha + 2.9$	$-D_z(\gamma B)$	$\alpha + 2.9$
$-D_z(Bd)$	$2\alpha + 2$	$2vu/R$	$\alpha + 0.9$
u^2/R	2α	$\epsilon D_r^2 B$	$\alpha + 2 - 2\kappa_s$
$\epsilon R^{-1} D_r B$	$\alpha + 2 - \kappa_s$	$\epsilon D_z^2 B$	$\alpha + 4$
$-\epsilon R^{-2} B$	$\alpha + 2$		

(RF48)

The first row uses $t_*B = -\epsilon \partial_T B$, so its coefficient cN is exactly absent. Every listed exponent exceeds or equals $\alpha + 0.9 - 2\kappa_s$: for 2α the difference is $\alpha - 0.9 + 2\kappa_s \geq 2\kappa_s > 0$, and subtraction proves the other entries directly. Thus

$$R_g \in \mathcal{M}_{\alpha+0.9-2\kappa_s}, \quad (P_{\text{new}}, (J_\theta)_{\text{new}}, (J_z)_{\text{new}}) \in (\mathcal{S}_{\alpha+0.9-2\kappa_s})^3. \quad (\text{RF49})$$

For the two product expressions in (RF47), the exponents are $\alpha + 0.9$ and 2α , before the source moment is added. This proves the stated bound for each defect and all fixed derivatives. The first two matrix rows and the exact solenoidal realization preserve both original zero moments of the velocity.

42.9 Use in the initial state and the fixed-domain conclusion

The maps proved here apply to the actual state after temporal correction. They require its computed defects as inputs; those defects include every product with the flat axial remnant a . The fixed prescription is to recompute (RF44), then P and (RF24), then (RF30), using the updated velocity at every step. The

five-equation step preserves the exact two zero moments and divergence, and gives the defect gain (RF49). At the initial target order $H_0 = 1 - \kappa_s$, the gain furnished by this calculation is

$$H_0 + 0.9 - 2\kappa_s = 1.9 - 3\kappa_s = 1.89997 > 1.2 = C_0. \quad (\text{RF50})$$

The increments also satisfy $H_0 = 0.99999 > 0.9$, $H_0 + 1 = 1.99999 > 1.9$, so they preserve the retained velocity class budgets. These exact statements concern the mean map. The estimates supplying the initial wave covariance and the tangential mean residual are separate calculations and are not inferred from (RF50).

For fixed source profile parameters $h, \lambda, c_{\text{patch}}$, fixed shell and bump supports, fixed band comparison and covering bounds, all conclusions above hold on the single inherited positive- q construction domain. No inverse in this fragment imposes a derivative-order dependent threshold or a uniform $\lambda \rightarrow 0$ limit. All-orders flatness is the estimate (RF22) on that same domain. The exact obstruction (RF15) proves why the flat projector remainder cannot be replaced by zero, and the rank loss at $\lambda = 0$ proves why the fixed positive power-law parameter must be retained.

42.10 Signed covariance and original positive-amplitude audit

The covariance matrix, positivity mechanism, exact signed differential inverse, shell weights, and displayed increment exponents are correct. The physical assembly statement at source line 403 needs its domain stated: the physical stress must have admissible local mean-class representatives with the prescribed active-shell support and edge vanishing. An arbitrary auxiliary-independent physical stress is too broad a domain. The precise counterexample and corrected wording appear below.

42.10.1 1. Original objects and input provenance

The source uses

$$A = \frac{1}{2} + h, \quad Q = 2^{-\ell}, \quad \varepsilon = Q^h, \quad S_* = \ell^2, \quad 0 < h < 1/100, \quad \kappa_s = 10^{-5}.$$

The normalized angular average has measure $d\theta/(2\pi)$; the auxiliary torus has normalized Haar measure. For real extended vector fields,

$$\mathcal{C}(w) = \langle \langle w_r w_{\text{tan}} \rangle_\theta \rangle_Y, \quad \mathcal{B}(w, v) = \langle \langle w_r v_{\text{tan}} + v_r w_{\text{tan}} \rangle_\theta \rangle_Y, \quad w_{\text{tan}} = (w_\theta, w_z).$$

Every entry of these maps is a product followed by the two specified linear integrations. Expanding each product gives

$$\mathcal{C}(w + v) = \mathcal{C}(w) + \mathcal{B}(w, v) + \mathcal{C}(v), \quad DC(w)[v] = \mathcal{B}(w, v).$$

In particular, the derivative contains both cross terms and therefore contains a factor of two on a scalar multiple of one wave.

The following inputs are explicitly present outside the assigned interval:

- Source lines 55–90 construct disjoint auxiliary preimages for every pair of simultaneously active distinct labels, including the two signs, and prove preservation of normalized Haar measure under the covering maps. This is the support and measure input used below.
- Source lines 121–143 define the original frozen vectors N, K , the original negative c_0 , and the uniform strict target-direction cone margin. Source lines 673–695 identify the profile endpoint-direction and profile-weight construction imported from the profiles lane. This audit does not independently construct those profile factors.
- Source lines 233–256 and 278–292 supply the two-sided Gaussian bounds for P , the positive radial homogeneous amplitude x , and their derivative bounds. The calculation below gives the complete covariance consequence of these inputs.
- Source lines 94–108 define the shell weight, derivative variables, and coefficient spaces, with all derivative bounds required pointwise. Source lines 417–453 give the exact curl remainder used in the final increment calculation.

In the original tangential component order,

$$N = (N_\theta, N_z), \quad K = (-N_z, N_\theta), \quad |N| = 1.$$

Thus $N \cdot K = 0$, $|K| = 1$, and

$$\det[N \mid K] = N_\theta^2 + N_z^2 = 1.$$

Consequently, expressing the covariance columns in the ordered basis (N, K) preserves the determinant sign in the original (θ, z) order. It is a coordinate expression of the same columns, not a replacement or renormalization of them.

The coefficient derivatives used throughout are, for a five-component multi-index I ,

$$D^I = \partial_R^{I_1} \partial_Z^{I_2} \partial_T^{I_3} \partial_{H_1}^{I_4} \partial_{H_2}^{I_5},$$

where (H_1, H_2) are coordinates on the common auxiliary torus; the covariance matrix H defined below is a different object. Every label, band scale, and discrete harmonic is fixed under these derivatives. For the original radial shell variable X , define

$$\zeta = \exp(-a_a / \log^2(X/X_a) - a_b / \log^2(X_b/X)), \quad \delta = \min\{1, \log(X/X_a), \log(X_b/X)\},$$

on $X_a < X < X_b$, with $a_a, a_b > 0$ the original constants, and extend ζ by zero. The mean class $\mathcal{M}_\alpha$ consists of compatible angular-independent coefficients with smooth zero extension outside this shell and, for every I ,

$$|D^I f| \leq C_I \varepsilon^\alpha S_*^{b_I} \zeta \delta^{-d_I}.$$

It has no auxiliary-rectangle or pulse-envelope requirement and may depend on the auxiliary torus. Its auxiliary-independent subclass is denoted $\mathcal{M}_\alpha^{\text{independent}}$. The wave class $\mathcal{W}_\alpha$ consists of the nonzero-harmonic coefficients $w_{\gamma,m}$ in finite labelled sums $w_{\gamma,m} e^{ikm\Phi_\gamma}$, with the source's prescribed closed slow, transverse-rectangle, and shell supports, satisfying

$$|D^I w_{\gamma,m}| \leq C_I \varepsilon^\alpha S_*^{b_I} \sqrt{\zeta} \delta^{-d_I} P(v).$$

The actual cutoff coefficients additionally have compact pulse support and smooth zero extension at that support. Equal label-harmonic contributions are collected before testing these bounds. The finite constants and nonnegative exponents C_I, b_I, d_I may depend on the derivative order and the fixed finite collection under consideration, but are uniform in the band. Compatibility means equality after the specified torus covering pullbacks on chart overlaps. These are coefficient estimates with the exponential factored out; every coefficient derivative is required to satisfy the stated pointwise envelope bound. No conclusion below differentiates a bare inequality for P .

42.10.2 2. Exact column integral and its constants

For each sign of one slow box, retain

$$b_\sigma = \chi_g(\xi_g) \psi(v) t_\sigma^h \cos(k\Phi_\sigma), \quad H = [\mathcal{C}(b_+) \mid \mathcal{C}(b_-)].$$

The integer kp is nonzero. For every non-angular phase offset c ,

$$\frac{1}{2\pi} \int_0^{2\pi} \cos^2(kp\theta + c) d\theta = \frac{1}{2} + \left[\frac{\sin(2kp\theta + 2c)}{8\pi kp} \right]_0^{2\pi} = \frac{1}{2}.$$

The rectangle map is

$$(\xi_g, \eta_g) \mapsto c_\gamma + \xi_g v_r + \eta_g v_t, \quad v_r = (1, -b_g), \quad v_t = (b_g, 1), \quad b_g = \sqrt{2} - 1.$$

Its absolute Jacobian is $1 + b_g^2$; $\eta_g = c_i v - r_0$ gives $d\eta_g = c_i dv$. Haar preservation means no additional covering degree multiplies or divides the integral. The homogeneous amplitude is independent of ξ_g . Writing its radial component as x , direct substitution therefore gives

$$H_\sigma = D_g c_i \int_0^{L_s} \psi(v)^2 x(v) t_{\sigma, \tan}^h(v) dv,$$

$$h_\sigma = D_g c_i \int_0^{L_s} \psi(v)^2 x(v)^2 dv, \quad D_g = \frac{1+b_g^2}{2} \int_{\mathbb{R}} \chi_g(\xi)^2 d\xi > 0, \quad L_s = \frac{2r_0}{c_i}.$$

All four original factors—the angular half, rectangle Jacobian, transverse integral, and $c_i dv$ —are retained.

Write the proved pulse bounds with positive uniform constants as

$$c_x P(v) \leq x(v) \leq C_x P(v), \quad e^{-C_P(v-L_s/2)^2/L_s} \leq P(v) \leq e^{-c_P(v-L_s/2)^2/L_s}.$$

Put $d = v - L_s/2$. On the central interval $|d| \leq \sqrt{L_s}$, the cutoff equals one when $L_s \geq 25$, since the source cutoff is one for $|d| < L_s/5$. Thus

$$I_0 := \int_0^{L_s} \psi^2 x^2 dv \geq 2c_x^2 e^{-2C_P} \sqrt{L_s} =: C_{0,-} \sqrt{L_s}.$$

Using $0 \leq \psi \leq 1$ and integrating the upper Gaussian over the whole real line gives

$$I_0 \leq C_x^2 \sqrt{\frac{\pi}{2c_P}} \sqrt{L_s} =: C_{0,+} \sqrt{L_s}.$$

The first absolute moment is bounded by

$$I_1 := \int_0^{L_s} \frac{|d|}{L_s} \psi^2 x^2 dv \leq \frac{C_x^2}{L_s} \int_{\mathbb{R}} |d| e^{-2c_P d^2/L_s} dd = \frac{C_x^2}{2c_P}.$$

The source floor estimate is

$$\frac{1}{T_g S_*} < c_i \leq \frac{1}{S_*}, \quad T_g = 4 + \sqrt{2}.$$

Hence $L_s \geq 2r_0 S_*$, so $S_* \geq 25/(2r_0)$ suffices for the central-interval bound. Also

$$c_i \sqrt{L_s} = \sqrt{2r_0 c_i}.$$

Therefore the positive original h_σ obey

$$h_-^{\text{bd}} S_*^{-1/2} \leq h_\sigma \leq h_+^{\text{bd}} S_*^{-1/2},$$

$$h_-^{\text{bd}} = D_g C_{0,-} \sqrt{2r_0/T_g} > 0, \quad h_+^{\text{bd}} = D_g C_{0,+} \sqrt{2r_0}.$$

The strict floor inequality also implies this non-strict lower bound. The symbols $h_-^{\text{bd}}, h_+^{\text{bd}}$ denote bounds, not a replacement for the two different original numbers h_+, h_- .

42.10.3 3. Column directions and the exact determinant

The original frozen constants satisfy $c_0 < 0$, $u_* > 0$, and

$$A_c = -c_0 \sqrt{1+u_*^2} > 0.$$

The homogeneous polarization satisfies the uniform error bound stated in source line 334 relative to

$$f(s) = -sK + c_0 \sqrt{1+s^2} N, \quad s(v) = \sigma(u_*/2 + u_* v/L_s).$$

Here

$$f'(s) = -K + c_0 \frac{s}{\sqrt{1+s^2}} N, \quad |f'(s)| \leq \sqrt{1+c_0^2}, \quad |s(v) - \sigma u_*| = u_* |d|/L_s.$$

The mean-value integral for $f(s(v)) - f(\sigma u_*)$, followed by the probability measure $\psi^2 x^2 dv/I_0$, bounds its average by

$$\sqrt{1+c_0^2} u_* \frac{I_1}{I_0} \leq \sqrt{1+c_0^2} u_* \frac{C_x^2}{2c_P C_{0,-} \sqrt{L_s}}.$$

Adding the source's uniform $O(S_*^{-1})$ polarization error gives the asserted uniform $O(S_*^{-1/2})$ column error without discarding any sign:

$$H_\sigma = h_\sigma(-A_c N - \sigma u_* K + e_\sigma), \quad |e_\sigma| \leq C_e S_*^{-1/2}.$$

Set $e_{\sigma,N} = e_\sigma \cdot N$, $e_{\sigma,K} = e_\sigma \cdot K$. In (N, K) coordinates the full, perturbed original matrix is

$$H = \begin{pmatrix} (-A_c + e_{+,N})h_+ & (-A_c + e_{-,N})h_- \\ (-u_* + e_{+,K})h_+ & (u_* + e_{-,K})h_- \end{pmatrix}.$$

Its exact determinant is

$$\begin{aligned} \det H &= h_+ h_- \Delta, \\ \Delta &= (-A_c + e_{+,N})(u_* + e_{-,K}) - (-A_c + e_{-,N})(-u_* + e_{+,K}) \\ &= -2A_c u_* + A_c(e_{+,K} - e_{-,K}) + u_*(e_{+,N} + e_{-,N}) \\ &\quad + e_{+,N}e_{-,K} - e_{-,N}e_{+,K}. \end{aligned}$$

For a target $T = T_N N + T_K K$, the exact inverse is

$$\begin{aligned} (H^{-1}T)_+ &= \frac{(u_* + e_{-,K})T_N + (A_c - e_{-,N})T_K}{h_+ \Delta}, \\ (H^{-1}T)_- &= \frac{(u_* - e_{+,K})T_N + (-A_c + e_{+,N})T_K}{h_- \Delta}. \end{aligned}$$

At zero error these formulas become exactly

$$h_+(H_{\text{ref}}^{-1}T)_+ = \frac{1}{2}(-T_N/A_c - T_K/u_*), \quad h_-(H_{\text{ref}}^{-1}T)_- = \frac{1}{2}(-T_N/A_c + T_K/u_*).$$

In particular, $\det H_{\text{ref}} = -2A_c u_* h_+ h_-$, with the same sign in the original tangential coordinates.

42.10.4 4. A uniform threshold that proves positivity of the original amplitudes

The strict compact direction gap in source lines 141–143 supplies a number $\mu < 1$. With the original choice

$$\frac{u_*}{\sqrt{1 + u_*^2}} > \mu,$$

the unfrozen direction ratio has a strict margin below $u_*/\sqrt{1 + u_*^2}$. Uniform continuity of the extended normalized target direction, and the fixed mesh diameter $O(S_*^{-3})$, preserves a strictly smaller margin for the frozen box basis. Hence one fixed $\eta_c > 0$ satisfies on every enlarged box after the stated initial band threshold

$$T_N < 0, \quad |T_K|/u_* \leq (1 - \eta_c)(-T_N)/A_c, \quad T = T_{0,*}.$$

No division by T at a zero edge is used: the compact direction is extended first, and the inequalities for nonzero T are applied in the open shell.

To make the perturbation threshold fully quantitative, let $A_{\min} \leq A_c \leq A_{\max}$ be the positive uniform bounds from source line 135 and define

$$H_0 = \begin{pmatrix} -A_c & -A_c \\ -u_* & u_* \end{pmatrix}, \quad \mathcal{E} = [e_+ \mid e_-]_{(N,K)}, \quad D_h = \text{diag}(h_+, h_-).$$

These identities retain $H = (H_0 + \mathcal{E})D_h$ and the original H ; they only factor its two already defined scalar column weights. Direct multiplication gives

$$H_0 H_0^T = \text{diag}(2A_c^2, 2u_*^2), \quad \|H_0^{-1}\| \leq M := \frac{1}{\sqrt{2} \min(A_{\min}, u_*)}.$$

Choose C_E with $\|\mathcal{E}\| \leq C_E S_*^{-1/2}$; for example $C_E = \sqrt{2}C_e$ is valid in the Euclidean operator norm. For every vector z ,

$$|(I + H_0^{-1}\mathcal{E})z| \geq (1 - MC_E S_*^{-1/2})|z|.$$

Thus $MC_E S_*^{-1/2} \leq 1/2$ proves invertibility and

$$\|(H_0 + \mathcal{E})^{-1}\| \leq 2M.$$

This proves the needed estimate directly, with no unproved appeal to a “finite Neumann series.” Along the real path $I + tH_0^{-1}\mathcal{E}$, $0 \leq t \leq 1$, the same lower norm bound prevents a zero determinant, so its determinant remains positive. Both singular values are at least $1/2$. Consequently

$$\Delta < 0, \quad |\Delta| \geq \frac{|\det H_0|}{4} = \frac{A_c u_*}{2},$$

$$|\det H| \geq \frac{A_{\min} u_*}{2} (h_-^{\text{bd}})^2 S_*^{-1}, \quad \|H^{-1}\| \leq \frac{2M}{h_-^{\text{bd}}} \sqrt{S_*}.$$

For coefficient positivity retain $y = H^{-1}T_{0,*}$ and introduce only the comparison coordinates $c = D_h y$, $c^{\text{ref}} = H_0^{-1}T_{0,*}$. Put $p = -T_N/A_c > 0$, $q = T_K/u_*$. Then $|q| \leq (1 - \eta_c)p$, and

$$c_+^{\text{ref}} = \frac{1}{2}(p - q), \quad c_-^{\text{ref}} = \frac{1}{2}(p + q), \quad c_\sigma^{\text{ref}} \geq \frac{\eta_c}{2}p.$$

Moreover

$$|T_{0,*}|^2 = A_c^2 p^2 + u_*^2 q^2 \leq [A_{\max}^2 + (1 - \eta_c)^2 u_*^2] p^2.$$

Hence, with

$$m_0 = \frac{\eta_c}{2\sqrt{A_{\max}^2 + (1 - \eta_c)^2 u_*^2}} > 0,$$

we have $c_\sigma^{\text{ref}} \geq m_0 |T_{0,*}|$. The exact inverse difference identity gives

$$c - c^{\text{ref}} = -(H_0 + \mathcal{E})^{-1} \mathcal{E} H_0^{-1} T_{0,*}, \quad |c - c^{\text{ref}}| \leq 2M^2 C_E S_*^{-1/2} |T_{0,*}|.$$

Therefore one fixed threshold

$$\sqrt{S_*} \geq \max \left\{ 2M C_E, \frac{4M^2 C_E}{m_0} \right\}$$

in addition to the already described pulse and cone thresholds proves

$$\frac{m_0}{2h_+^{\text{bd}}} \sqrt{S_*} |T_{0,*}| \leq y_\sigma \leq \frac{2M}{h_-^{\text{bd}}} \sqrt{S_*} |T_{0,*}|.$$

The original amplitudes are exactly

$$a_\sigma = \sqrt{y_\sigma} > 0$$

on the open shell. They have not been divided by their size, replaced with unit vectors, or signed to absorb a correction. At each shell edge the target and extended amplitudes are zero. The threshold uses only fixed value bounds and direction margins, not the later derivative order or signed correction.

The original local wave at which the covariance is differentiated is

$$W_0 = \sqrt{\varepsilon} \sum_{\sigma=\pm} a_\sigma b_\sigma.$$

42.10.5 5. Derivatives and smooth zero extension with the original shell weight

Retain the source weight

$$\zeta(X) = \exp \left(-\frac{a_a}{\log^2(X/X_a)} - \frac{a_b}{\log^2(X_b/X)} \right), \quad \delta = \min\{1, \log(X/X_a), \log(X_b/X)\},$$

with its original positive constants. The profile-specific source choice is $a_a = t_1^2$, $a_b = 4$; the argument applies to precisely those constants. The profile bounds used here are

$$|T_{0,*}| \geq c_T \zeta, \quad |D^I T_{0,*}| \leq C_I \zeta \delta^{-d_I}.$$

The source’s chart/profile map has bounded derivatives on the dyadic shell and a positive bounded chart multiplier, so those physical profile bounds have the displayed chart form. All differentiations here keep the label and $Q, \varepsilon, S_*, c_i, L_s$ fixed. They therefore do not differentiate a discrete frequency or a moving endpoint.

Each derivative of H is polynomially bounded in S_* by differentiation of the fixed finite pulse integral. To see that this gives every derivative of its inverse with no new smallness condition, differentiate $HH^{-1} = I$. For a nonzero multi-index I ,

$$D^I H^{-1} = -H^{-1} \sum_{0 < J \leq I} \binom{I}{J} (D^J H) (D^{I-J} H^{-1}).$$

Induction on $|I|$, the already established value bound on H^{-1} , and the finite sum prove a polynomial bound at every fixed order. Applying the finite Leibniz expansion to $H^{-1}T_{0,*}$ gives

$$y_\sigma \geq c\sqrt{S_*}\zeta, \quad |D^I y_\sigma| \leq C_I S_*^{b_I} \zeta \delta^{-d_I}.$$

For completeness, for either exponent $\nu = 1/2$ or $\nu = -1/2$, the exact multivariate chain-rule formula on the open shell is

$$D^I (y_\sigma^\nu) = I! \sum_{\substack{(k_J)_{0 < J \leq I} \\ \sum_J k_J J = I}} (\nu)_n y_\sigma^{\nu-n} \prod_{0 < J \leq I} \frac{1}{k_J!} \left(\frac{D^J y_\sigma}{J!} \right)^{k_J}, \quad n = \sum_J k_J,$$

where $(\nu)_n = \nu(\nu-1)\cdots(\nu-n+1)$ and the empty product for $I = 0$ is one. This formula follows by applying the one-variable Taylor expansion to $y(x+h) - y(x)$ and extracting the h^I coefficient; only degrees at most $|I|$ can contribute. It therefore has finitely many terms at each fixed order.

For $\nu = 1/2$ and $|I| > 0$, $n \geq 1$; the factor $y^{1/2-n}$ has a negative exponent and is bounded by the established lower bound on y . The product of the n derivative factors contributes ζ^n , while that inverse power contributes $\zeta^{1/2-n}$. The result is exactly $\sqrt{\zeta}$. For $I = 0$, the upper value bound on y supplies the same weight, with permitted inverse-distance powers. Thus

$$|D^I a_\sigma| \leq C_I S_*^{b_I'} \sqrt{\zeta} \delta^{-d_I'}.$$

For $\nu = -1/2$, the same calculation, including $I = 0$, gives

$$|D^I (a_\sigma^{-1})| = |D^I (y_\sigma^{-1/2})| \leq C_I S_*^{b_I''} \zeta^{-1/2} \delta^{-d_I''}.$$

The inverse amplitude itself is not claimed to extend smoothly; the product with an admissible target will extend.

The flatness statement is rigorous: at either edge, $\sqrt{\zeta} \delta^{-N}$ tends to zero faster than every positive power of the edge distance, because its logarithm has a negative term of order $-c/\delta^2$, which dominates every $N \log(1/\delta)$. The same holds for all the displayed derivatives. In local smooth coordinates consisting of edge distance and tangential variables, the zero-extended value is continuous. Its difference quotient normal to the edge tends to zero by the faster-than-linear value bound; its tangential derivative equals zero on the edge. Repeating this argument for each already identified derivative proves by induction that the zero extension is smooth and all its jets vanish. Smooth chart changes preserve this statement. This proves the shell extension claimed for a_σ and for the signed coefficients below.

42.10.6 6. Exact signed differential inverse, including its full quadratic remainder

The input is exactly the source's real, angular- and auxiliary-independent two-vector $\Sigma(R, Z, T)$ in $\mathcal{M}_\alpha$, with

$$|D^I \Sigma| \leq C_I \varepsilon^\alpha S_*^{b_I} \zeta \delta^{-d_I}.$$

Its values may have arbitrary signs and arbitrary size within that class. Keep the already constructed original $a_\sigma > 0$ fixed and define

$$d_\Sigma = H^{-1}(\Sigma/\varepsilon), \quad \delta a_\sigma = \frac{(d_\Sigma)_\sigma}{2a_\sigma}, \quad \mathcal{L}\Sigma = \sqrt{\varepsilon} \sum_{\sigma=\pm} \delta a_\sigma b_\sigma.$$

The inverse-matrix derivative bounds and Leibniz rule give

$$|D^I d_\Sigma| \leq C_I \varepsilon^{\alpha-1} S_*^{b_I} \zeta \delta^{-d_I}.$$

Multiplication with the proved a_σ^{-1} estimates gives

$$|D^I \delta a_\sigma| \leq C_I \varepsilon^{\alpha-1} S_*^{b'_I} \sqrt{\zeta} \delta^{-d'_I}.$$

The inverse power of the original amplitude cancels exactly one half-power of the target shell weight. Multiplying by $\sqrt{\varepsilon}$ and the original pulse coefficient, whose derivatives have polynomial factors times $P(v)$, proves

$$\mathcal{L} : \mathcal{M}_\alpha^{\text{independent}} \longrightarrow \mathcal{W}_{\alpha-1/2},$$

with the prescribed smooth shell extension, transverse support, pulse cutoff, and later slow cutoff. Both maps in this statement refer to the fixed source chart and labels. Auxiliary independence is required to take δa_σ outside the torus integral; angular independence is required for the same angular calculation.

Because the two sign supports are disjoint,

$$\mathcal{C}(W_0) = \varepsilon \sum_\sigma a_\sigma^2 H_\sigma = \varepsilon H(y_+, y_-)^T = \varepsilon T_{0,*},$$

$$\mathcal{B}(W_0, \mathcal{L}\Sigma) = \varepsilon \sum_\sigma 2a_\sigma \delta a_\sigma H_\sigma = \varepsilon H d_\Sigma = \Sigma.$$

At shell edges both sides have their proved zero extension, so the identity persists there. The map is linear because H, a_σ, b_σ are fixed before Σ is chosen. No positivity or smallness of Σ was used in any equality.

The precise nonlinear remainder can also be written without an unspecified symbol. Set $z = H^{-1}\Sigma$. Then

$$\mathcal{C}(\mathcal{L}\Sigma) = \frac{1}{4\varepsilon} H \begin{pmatrix} z_+^2/y_+ \\ z_-^2/y_- \end{pmatrix}.$$

Thus

$$\mathcal{C}(W_0 + \mathcal{L}\Sigma) = \varepsilon T_{0,*} + \Sigma + \frac{1}{4\varepsilon} H \begin{pmatrix} (H^{-1}\Sigma)_+^2/y_+ \\ (H^{-1}\Sigma)_-^2/y_- \end{pmatrix}.$$

The remainder has ε exponent $2\alpha - 1$; the product of the two wave weights is ζ^{P^2} , and $0 \leq P \leq 1$. After averaging it therefore belongs to $\mathcal{M}_{2\alpha-1}$ with all derivatives, by the finite Leibniz rule. The exact signed inverse is an inverse of the derivative at the fixed original wave; it does not remove this nonzero quadratic term.

42.10.7 7. Physical assembly, chart maps, and the missing domain qualification

The original chart relation is

$$T_{0,*} = (Q/q)^{A+1/2} T_0.$$

Keeping every physical scaling factor gives

$$\begin{aligned} Q^{-2A} \varepsilon T_{0,*} &= Q^{-2A+h} (Q/q)^{A+1/2} T_0 \\ &= Q^{-A+h+1/2} q^{-A-1/2} T_0 \\ &= q^{-A-1/2} T_0, \end{aligned}$$

because $-A + h + 1/2 = 0$. No equality of distinct Q_β is used.

For a physical stress with admissible local representatives, retain the assembled fields exactly as in the source:

$$\begin{aligned} W_{0,\text{phys}}^{\text{as}} &= \sum_\beta Q_\beta^{-A} \eta_\beta W_{0,\beta}, \\ \mathcal{L}_{\text{phys}}^{\text{as}} \sigma &= \sum_\beta Q_\beta^{-A} \eta_\beta \mathcal{L}_\beta (Q_\beta^{2A} \sigma). \end{aligned}$$

At a fixed physical slow point there are finitely many active labels with uniform bounded overlap. Products from distinct active labels vanish by the already proved common-torus support disjointness. The slow η_β and Q_β factors are independent of both averaging variables. Therefore the exact contribution of one label is

$$Q_\beta^{-2A} \eta_\beta^2 \mathcal{B}(W_{0,\beta}, \mathcal{L}_\beta(Q_\beta^{2A} \sigma)) = Q_\beta^{-2A} \eta_\beta^2 Q_\beta^{2A} \sigma = \eta_\beta^2 \sigma.$$

On the region covered by the squared partition, $\sum_\beta \eta_\beta^2 = 1$. Outside the active shell an admissible stress is zero, as are the extended wave fields. These two facts establish the global identity on its proper domain:

$$\mathcal{B}(W_{0,\text{phys}}^{\text{as}}, \mathcal{L}_{\text{phys}}^{\text{as}} \sigma) = \sigma.$$

In a reference chart, the exact representative maps are

$$W_0^{\text{as}} = Q^A W_{0,\text{phys}}^{\text{as}}, \quad \mathcal{L}^{\text{as}} \Sigma = Q^A \mathcal{L}_{\text{phys}}^{\text{as}}(Q^{-2A} \Sigma).$$

Bilinearity gives

$$\mathcal{B}(W_0^{\text{as}}, \mathcal{L}^{\text{as}} \Sigma) = Q^{2A} (Q^{-2A} \Sigma) = \Sigma.$$

For overlapping bands the chart maps are exactly

$$R_\beta = (Q/Q_\beta)^{1/2} R, \quad Z_\beta = (Q/Q_\beta)^D Z, \quad T_\beta = (Q/Q_\beta) T, \quad \varepsilon_\beta = (Q_\beta/Q)^h \varepsilon.$$

Their constants and inverses are uniformly bounded on the allowed band range, as are S_β/S_* and covering multiplicities. The shell variable X and its weights describe the same physical point. Each fixed derivative of these affine scale changes introduces bounded factors, and derivatives of the fixed mesh cutoffs add polynomial powers of S_* . Finite sums and prior collection of equal label-harmonic contributions therefore preserve every stated coefficient exponent.

Necessary qualification at source line 403.

The phrase “For a physical auxiliary-independent stress $\sigma(r,z,t)$ ” must retain the preceding local input class and support. Taken literally for every such physical stress, it is false. Choose any nonzero smooth two-vector stress with compact support in an open set outside the closed active shell. It is auxiliary-independent. Every assembled wave in the displayed formulas vanishes there, so its bilinear covariance is zero there, while the chosen stress is nonzero. Thus the asserted right-inverse identity cannot hold on that enlarged domain. Furthermore an arbitrary nonvanishing stress approaching a shell edge need not cancel $1/a_\sigma$ and need not give any smooth extension.

A precise source replacement is:

For a real, auxiliary-independent physical stress supported in the prescribed active shell whose band representatives $Q_\beta^{2A} \sigma$ satisfy the local $\mathcal{M}_\alpha$ bounds and smooth zero extension used above, use the assembled maps below.

This repairs the domain while retaining all original formulas. The stated local inverse at line 383 and actual increment at line 474 already include the mean-class restriction. No additional sign or smallness requirement should be inserted.

42.10.8 8. Actual divergence-free covariance increment and every exponent

Retain the source hypotheses and fields:

$$U = W_0^{\text{as}} + E, \quad U \in \mathcal{W}_{1/2}, \quad E \in \mathcal{W}_\rho, \quad \Sigma \in \mathcal{M}_\alpha^{\text{independent}}.$$

The exact curl construction applied to the original $\mathcal{L}^{\text{as}} \Sigma$ produces

$$V = \mathcal{L}^{\text{as}} \Sigma + R, \quad R \in \mathcal{W}_{\alpha - \kappa_s},$$

since the input exponent is $\alpha - 1/2$ and the curl correction gains $1/2 - \kappa_s$. With $L = \mathcal{L}^{\text{as}} \Sigma$ only as a temporary notation for this unchanged field, expansion of all products gives

$$\begin{aligned} \mathcal{C}(U + V) - \mathcal{C}(U) &= \mathcal{B}(U, V) + \mathcal{C}(V) \\ &= \mathcal{B}(W_0^{\text{as}} + E, L) + \mathcal{B}(U, R) + \mathcal{C}(V) \\ &= \Sigma + \mathcal{B}(E, L) + \mathcal{B}(U, R) + \mathcal{C}(V). \end{aligned}$$

The first remainder includes precisely the interaction of the already present error with the uncurred signed inverse. The second retains its interactions with both W_0^{as} and E , because U is not replaced. The third retains all three quadratic pieces

$$\mathcal{C}(V) = \mathcal{C}(L) + \mathcal{B}(L, R) + \mathcal{C}(R).$$

Leibniz differentiation and the product weight $\sqrt{\zeta}P\sqrt{\zeta}P = \zeta P^2 \leq \zeta$ give

$$\mathcal{B}(E, L) \in \mathcal{M}_{\rho+\alpha-1/2}, \quad \mathcal{B}(U, R) \in \mathcal{M}_{\alpha+1/2-\kappa_s}.$$

The three quadratic pieces have exponents, respectively,

$$2\alpha - 1, \quad 2\alpha - \frac{1}{2} - \kappa_s, \quad 2\alpha - 2\kappa_s.$$

Because $0 < \varepsilon \leq 1$ and $\kappa_s < 1/2$, the latter two exponents are at least $2\alpha - 1$; hence their estimates imply the same weaker class $\mathcal{M}_{2\alpha-1}$. Equivalently, $R \in \mathcal{W}_{\alpha-\kappa_s}$ implies $R \in \mathcal{W}_{\alpha-1/2}$, so V is in the latter class. This proves exactly the source's third displayed exponent without suppressing a quadratic term.

The original tangential radial-divergence components are $(D_r + 2/R)T_\theta$ and $(D_r + 1/R)T_z$. On the fixed positive radial interval the frame multipliers and all their derivatives are bounded. The established coefficient derivative map is $D_r : \mathcal{M}_\beta \rightarrow \mathcal{M}_{\beta-\kappa_s}$. Consequently the three respective exponents after this conservative radial-divergence estimate are

$$\rho + \alpha - \frac{1}{2} - \kappa_s, \quad \alpha + \frac{1}{2} - 2\kappa_s, \quad 2\alpha - 1 - \kappa_s.$$

For these particular fully averaged remainders one can also observe that they are auxiliary-independent; then the torus component of D_r annihilates them and the ordinary chart derivative need not lose κ_s . The source's weaker estimate remains valid and is the one recorded for stage comparison. The local calculation itself imposes no extra inequality involving α and ρ ; a later residual-improvement claim has to substitute its actual exponents into these proved expressions.

42.10.9 9. Audit disposition

The exact determinant, orientation, reference coefficients, signed inverse, covariance factors, physical chart factors, and all three remainder exponents have been checked. The positivity argument can be made fully explicit by the single value threshold in Section 4, and all fixed derivative orders follow without increasing that threshold. Division by the original positive amplitudes is valid on the prescribed weighted domain and gives smooth zero-extended signed corrections.

The substantive omission is the overly broad physical input wording at line 403; Section 7 provides both its counterexample and the exact corrected domain. The phrase “finite Neumann series estimate” at line 353 is imprecise; Section 4 supplies a direct norm proof and exact inverse-difference identity. The background endpoint factors and pulse evolution remain identifiable imported construction inputs, with their precise source locations recorded in Section 1; this covariance audit does not certify those separate constructions or the later numerical choice of iteration exponents.

43 The actual initial finite state

This calculation uses the objects just constructed, with their original physical definitions. An overbar is normalized auxiliary Haar average. Angular average is denoted by $\langle \cdot \rangle_\theta$ and always averages cylindrical components. The two operations are not identified with physical evaluation. Set $\kappa = 10^{-5}$ and $H = 1 - \kappa$. All the estimates below retain $\varepsilon = Q^h$, $S_* = \ell^2$, and the shell weights in the definitions of $\mathcal{W}_\alpha, \mathcal{M}_\alpha, \mathcal{S}_\alpha$.

43.1 The base expansion supplies the precise chart input

Write the original diagonal sum of the positive-order coefficient potentials as $\sum_{n \geq 1} \chi(c_n q) \mathcal{A}_n$, where $\mathcal{A}_n = q^{1-A+2nh} F_n e_\theta / r$. Its radial and axial components are exactly

$$ru_{r,n}^{\text{cut}} = q^{2nh} \left\{ \chi(c_n q) V_n - \frac{2\eta}{L} (c_n q) \chi'(c_n q) F_n \right\}, \quad u_{z,n}^{\text{cut}} = q^{-A+2nh} \chi(c_n q) U_n.$$

The first equality follows from $ru_r = -\partial_z(r\mathcal{A}_\theta)$, $q_z = 2\eta q^{1-D}/L$, and the original coefficient divergence identity. In particular, no derivative of a cutoff is omitted from the background. The diagonal choice is made for all normalized coefficient seminorms through order n , with each corresponding term at most $2^{-n} q^{nh}$. Such a choice is simultaneous on the original compact profile rectangle: at step n there are finitely many seminorms, each bounded by a fixed constant times q^{2nh} and a fixed power of $|\log q|$ after the scale

derivatives of $\chi(c_n q)$ have been included. On their support $c_n q \leq 1$, and on a nonzero differentiated cutoff $1/2 \leq c_n q \leq 1$; hence $(c_n q)^j \chi^{(j)}(c_n q)$ is bounded independently of c_n . For each of the finitely many constants, $q^{nh}(1+|\log q|)^b$ tends to zero as $q \downarrow 0$. Choose $c_n \geq 2c_{n-1}$ large enough that all the finitely many required bounds hold for $q \leq c_n^{-1}$. This fixes one entire sequence, with no subsequent choices depending on a requested output derivative. The original finite coefficients are unchanged near $q = 0$, where each individual cutoff equals one. For a fixed derivative order m , the finitely many terms with $n \leq m+1$ retain their original powers q^{2nh} ; the tail starting at $n \geq \max(2, m+2)$ is bounded by

$$\sum_{n \geq \max(2, m+2)} 2^{-n} q^{nh} \leq 2q^{2h}, \quad 0 < q \leq 1.$$

This proves the all-fixed-derivative estimate $O(q^{2h})$ for the tangential background difference. Multiplication by the original common radial factor q^h proves $b = O(\varepsilon S_*^B)$ and its positive-order difference $O(\varepsilon^3 S_*^B)$ in a Q chart, because q/Q stays in a fixed compact positive interval. The slow coordinate transition proved above passes every fixed derivative through this estimate.

For the stress, use the same diagonal selection on T_n/ζ for $n \geq 2$. All these quotients are smooth with fixed-order bounds on the compact shell; the single $n = 1$ edge term has the explicitly retained bounds $|\partial^I T_1| \leq C_I \zeta \delta^{-b_I}$. Separating that single term and the finitely many early terms from the tail proves

$$|\partial^I (\hat{T} - T_0)| \leq C_I q^{2h} \zeta \delta^{-b_I}.$$

Here $\hat{T}$ is the actual summed profile stress, not a truncated stress. The original physical stress and its chart value are

$$\Sigma_{\text{phys}} = q^{-A-1/2} \hat{T}, \quad \Sigma = Q^{2A} q^{-A-1/2} \hat{T},$$

so the leading value $\Sigma^{(0)} = \varepsilon T_{0,*}$ satisfies

$$\Sigma - \Sigma^{(0)} \in \mathcal{M}_3. \quad (\text{SI1})$$

Indeed $Q^{2A} q^{-A-1/2} = \varepsilon(Q/q)^{A+1/2}$ because $A = 1/2 + h$, and $q^{2h} = \varepsilon^2(Q/q)^{2h}$. No factor in this identity is replaced by one. The weaker $\mathcal{M}_2$ bound would also suffice below.

43.2 The complete primary curl and its residual

Let w_0^{tan} be the assembled real transverse primary field and let $w = \text{curl}_* \mathcal{A}_0$ be its actual cutoff curl. For each label and nonzero harmonic, its potential coefficient is

$$C_m = \frac{in_\Phi \times t_m}{km|n_\Phi|^2}, \quad n_\Phi = \nabla_* \Phi,$$

with $n_\Phi \cdot t_m = 0$ and the cutoff included in t_m . The vector triple-product identity and the original right-handed cylindrical orientation give

$$\text{curl}_*(C_m e^{ikm\Phi}) = (t_m + r_m) e^{ikm\Phi}, \quad r_m = (-D_z C_{m,\theta}, D_z C_{m,r} - D_r C_{m,z}, (D_r + R^{-1}) C_{m,\theta}).$$

The multiplier $n_\Phi/|n_\Phi|^2$ has polynomial S_* derivative bounds, $k^{-1} \leq \sqrt{\varepsilon}$, and D_r costs at most κ . Consequently

$$w_0^{\text{tan}} \in \mathcal{W}_{1/2}, \quad w - w_0^{\text{tan}} \in \mathcal{W}_{1-\kappa}, \quad w \in \mathcal{W}_{1/2}. \quad (\text{SI2})$$

The full w is divergence free. Expansion of its divergence, for the complete amplitude $a_m = t_m + r_m$, gives the exact longitudinal identity

$$ikm(n_\Phi \cdot a_m) = -\{(D_r + R^{-1})a_{m,r} + D_z a_{m,z}\}. \quad (\text{SI3})$$

Thus $n_\Phi \cdot a_m = n_\Phi \cdot r_m$ has exponent $1 - \kappa$. For two same-label harmonics, the apparently leading transport term is

$$ikm'(a_m \cdot n_\Phi) a_{m'} = -\frac{m'}{m} \{(D_r + R^{-1})a_{m,r} + D_z a_{m,z}\} a_{m'}.$$

It has exponent $1 - \kappa$ for two primary waves. Ordinary amplitude transport and cylindrical connection terms have that exponent or a larger one. Distinct-label derivative products vanish by the support separation proved above. The product weight is ζP^2 , bounded both by ζ and by $\sqrt{\zeta} P$. This proves the stated exponent for both the zero and nonzero angular harmonics, at every fixed coefficient derivative order.

For completeness the primary linear error has the following term bounds. The operator on the transverse amplitude is the homogeneous constrained pulse operator; its temporal cutoff error is kept in the flat residual. The remaining groups and their exponents, for a wave of exponent α , are

$$\begin{array}{l|l} (-\varepsilon\partial_T + bD_r + GD_z)a & \alpha + 1 - \kappa \\ ikmE_{ik}a & \alpha + 1/2 \\ a_rb_Re_r + a_zD_z(b, V, G) + (b/R)a_\theta e_\theta & \alpha + 1 \\ (D_r\pi, 0, D_z\pi) & \alpha + 1/2 - \kappa \\ -\varepsilon(D_r^2 + R^{-1}D_r + D_z^2)a & \alpha + 1 - 2\kappa \\ -2i\varepsilon km(n_rD_r + n_zD_z)a & \alpha + 1/2 - \kappa \\ \text{phase divergence and cylindrical connection terms} & \alpha + 1/2 \end{array}$$

In the last row n_Φ is auxiliary independent in the radial direction of the pulse: $D_r v = 0$; its radial coefficient derivatives therefore have polynomial S_* bounds without an extra $\varepsilon^{-\kappa}$. The phase defect is exactly $E_{ik} = \varepsilon v((H_\Phi)_T - G(H_\Phi)_Z) + bn_{\Phi,r}$. These facts prove every entry by multiplication with the displayed differential operators. Applying the principal pulse velocity operator to r_m preserves its exponent: cN and the coefficient matrix preserve exponents, while $\varepsilon k^2 m^2 |n_\Phi|^2$ is bounded at each fixed m . Taking $\alpha = 1/2$ gives the safe common primary nonzero residual bound

$$G_{\text{wave}}^{\text{pre}} \in \mathcal{W}_{1-3\kappa}. \quad (\text{SI4})$$

43.3 Initial full mean equations and the improved Haar average

Before mean updates put $v = \gamma = \beta = 0$ and retain $W^{ab} = \langle w_a w_b \rangle_\theta$. The actual radial source and pressure are

$$g_r = -(D_r + R^{-1})W^{rr} - D_z W^{zr} + W^{\theta\theta}/R, \quad P = \int \bar{g}_r dR, \quad p_m = \mathcal{T}_0(g_r - \rho P).$$

The product rules give $g_r, p_m \in \mathcal{M}_H$ and $P \in \mathcal{S}_H$. The two residuals are

$$\begin{aligned} E_\theta &= (D_r + 2/R)(W^{r\theta} - \Sigma_\theta) + D_z W^{z\theta}, \\ E_z &= (D_r + 1/R)(W^{rz} - \Sigma_z) + D_z(W^{zz} + p_m). \end{aligned} \quad (\text{SI5})$$

Hence $E_\theta, E_z \in \mathcal{M}_H$. The integrated definitions

$$J_\theta = \int R^2 \overline{W^{z\theta}} dR, \quad J_z = \int R \overline{W^{zz}} dR - \frac{1}{2} \int R^2 \bar{g}_r dR$$

give $(P, J_\theta, J_z) \in \mathcal{S}_H$.

The signed covariance construction at the original positive amplitudes gives $\overline{\langle (w_0^{\text{tan}})_r (w_0^{\text{tan}})_{\text{tan}} \rangle_\theta} = \Sigma^{(0)}$ exactly. Expanding the complete curl $w = w_0^{\text{tan}} + r$ retains both cross terms and $\langle r_r r_{\text{tan}} \rangle_\theta$. Their exponents are $3/2 - \kappa$ and $2 - 2\kappa$. Therefore

$$\overline{W^{r\theta}, W^{rz}} - \Sigma^{(0)} \in \mathcal{M}_{3/2-\kappa}.$$

For Haar averages $\overline{D_r f} = \partial_R \bar{f}$ by integration of the torus derivative. Differentiating the preceding estimate therefore preserves its exponent. Equations (SI1) and (SI5) give

$$\bar{E}_\theta, \bar{E}_z \in \mathcal{M}_{3/2-\kappa} \subset \mathcal{M}_{1.49}. \quad (\text{SI6})$$

The axial flux terms have exponents 2 and $H + 1 = 2 - \kappa$; they do not reduce this bound.

43.4 Temporal cancellation with the entire reconstructed field

Use the fixed common-torus inverse to define

$$\begin{aligned} u &= -c^{-1}N^{-1}(E_\theta - \bar{E}_\theta), \quad \gamma_d = -c^{-1}N^{-1}(E_z - \bar{E}_z), \quad \Psi = \mathcal{T}_1\gamma_d, \\ B &= -\varepsilon\partial_z\Psi, \quad d = (D_r + R^{-1})\Psi = \gamma_d + a, \quad a = -\mathcal{A}_1\gamma_d. \end{aligned} \quad (\text{SI7})$$

These are the actual increments. In particular, a is retained in d in every product. They satisfy $u, d, \Psi \in \mathcal{M}_H$, $B \in \mathcal{M}_{H+1}$, $cNu = -E_\theta + \bar{E}_\theta$, and $cNd = -E_z + \bar{E}_z + cNa$. The projector proof gives flatness of a

and all its physical derivatives. It also gives $\bar{u} = \bar{d} = \bar{\Psi} = \bar{B} = 0$ pointwise in radius. Consequently both original weighted mean moments remain zero exactly.

Recompute the radial source after this update. Direct product expansion gives

$$\Delta g_r = (2V/R)u + R_g^{\text{time}},$$

$$\begin{aligned} R_g^{\text{time}} = & -t_*B - (D_r + R^{-1})(2bB + B^2) \\ & - D_z(bd + GB + Bd) + u^2/R + \varepsilon(\Delta_0 - R^{-2})B. \end{aligned}$$

Here the exponents, in the order written after splitting the finite sums, are

$$H+1, \quad H+2-\kappa, \quad 2H+2-\kappa, \quad H+2, \quad H+2, \quad 2H+2, \quad 2H, \quad H+2-2\kappa.$$

The term t_*B includes cNB ; unlike the later slow five-dimensional correction, this term cannot be assigned exponent $H+2$ in its entirety. Nevertheless $\Delta g_r \in \mathcal{M}_H$. The fixed linear pressure map gives $\Delta p_m \in \mathcal{M}_H$.

Subtracting the complete old and new mean equations now gives exactly

$$\begin{aligned} E_\theta^{\text{time}} - \bar{E}_\theta = & -\varepsilon\partial_T u - \varepsilon(\Delta_0 - R^{-2})u \\ & + (D_r + 2/R)(bu + VB + Bu) + D_z(Gu + Vd + du), \\ E_z^{\text{time}} - \bar{E}_z = & -\varepsilon\partial_T d - \varepsilon\Delta_0 d + cNa \\ & + (D_r + 1/R)(bd + GB + Bd) + D_z(2Gd + d^2 + \Delta p_m). \end{aligned}$$

The slow-time, viscous, linear radial flux, and linear axial flux groups have exponents at least $H+1$, $H+1-2\kappa$, $H+1-\kappa$, and $H+1$, respectively. The radial quadratic flux has exponent $2H+1-\kappa$ and the axial quadratic flux exponent $2H+1$. All of these are at least $H+1-2\kappa = 2-3\kappa$. After the explicitly displayed cNa is placed in the flat residual,

$$E_\theta^{\text{time}}, E_z^{\text{time}} \in \mathcal{M}_{1.49}, \quad (P, J_\theta, J_z)^{\text{time}} \in \mathcal{S}_H. \quad (\text{SI8})$$

This assertion does not say that the Haar averages are unchanged. Their linear changes vanish because the base is auxiliary independent and $\bar{u} = \bar{d} = \bar{B} = 0$. Their quadratic changes need not vanish; the displayed equations retain them. They have sufficient exponents to preserve (SI8).

43.5 The five exact moment rows and the resulting budgets

Apply the proved fixed five-dimensional inverse on the original reserved power-law interval to the recomputed $(P, J_\theta, J_z)^{\text{time}}$. Denote its slow azimuthal, axial, and radial increments by u_2, d_2, B_2 . They have exponents $H, H, H+1$, preserve the first two moments exactly, and have identically zero axial projector remainder. The last three rows cancel the linear contributions in the original fixed base, with their physical and chart factors retained. Put $v = u$, $\gamma = d$, $\beta = B$ only in the following full difference identity:

$$\begin{aligned} g_r^{\text{new}} - g_r = & (2V/R)u_2 + R_g^{\text{slow}}, \\ R_g^{\text{slow}} = & -t_*B_2 - (D_r + R^{-1})(2bB_2 + 2\beta B_2 + B_2^2) \\ & - D_z(bd_2 + GB_2 + \beta d_2 + \gamma B_2 + B_2d_2) \\ & + (2vu_2 + u_2^2)/R + \varepsilon(\Delta_0 - R^{-2})B_2. \end{aligned}$$

Every entry is the difference of the corresponding term in the original radial equation. No wave covariance changes in this update. Since B_2 is auxiliary independent, its time term is $+\varepsilon\partial_T B_2$. Using the deliberately retained bounds $v, \gamma \in \mathcal{M}_{0.9}$, $\beta \in \mathcal{M}_{1.9}$, the respective exponents of the displayed terms are

$$\begin{aligned} & H+2; \quad H+2-\kappa, H+2.9-\kappa, 2H+2-\kappa; \\ & H+2, H+2, H+2.9, H+2.9, 2H+2; \\ & H+0.9, 2H; \quad H+2-2\kappa. \end{aligned}$$

The last common viscous value bounds all four entries of its displayed operator. Subtraction shows that every listed exponent is at least $H+0.9-2\kappa = 1.9-3\kappa$, since $H \geq 0.9$. The exact new defects are

$$P^{\text{new}} = \int \overline{R_g^{\text{slow}}} dR,$$

$$\begin{aligned}
J_\theta^{\text{new}} &= \int R^2 \overline{\gamma u_2 + v d_2 + d_2 u_2} dR, \\
J_z^{\text{new}} &= \int R^2 \overline{\gamma d_2 + d_2^2} dR - \frac{1}{2} \int R^2 \overline{R_g^{\text{slow}}} dR.
\end{aligned} \tag{SI9}$$

In the last equation the radial-source term is indispensable: the linear part $-\frac{1}{2} \int R^2 (2V/R) u_2 = - \int R V u_2$ is canceled by the fifth row, while the displayed nonlinear radial-source moment remains. Thus all three new defects lie in $\mathcal{S}_{1.9-3\kappa}$.

For the tangential residuals the difference equations have exactly the same flux patterns as the temporal difference equations, with (b, V, G) replaced in the products by $(b + \beta, V + v, G + \gamma)$ and (B, u, d) replaced by (B_2, u_2, d_2) . The slow time has no cN part. Linear radial and axial terms have exponents $H + 1 - \kappa$ and $H + 1$, viscosity $H + 1 - 2\kappa$, and the additional existing-mean products have exponent at least $H + 1.9 - \kappa$ in the radial flux and $H + 1.9$ in the axial flux. The pressure change has exponent H , hence its D_z term $H + 1$. All changes lie in $\mathcal{M}_{2-3\kappa}$, so (SI8) persists after the five-dimensional correction.

Neither mean correction changes w . Its interactions with the wave in the full residual do change. For a tangential mean of exponent μ and radial mean of exponent $\mu + 1$, the largest phase term has exponent $1/2 + \mu - 1/2 = \mu$. Indeed n_Φ has bounded polynomial coefficients, $k \leq 2\varepsilon^{-1/2}$, and the radial mean component has one extra ε . The amplitude derivative terms have exponent at least $1/2 + \mu - \kappa$, and the cylindrical connection terms have exponent $1/2 + \mu$. Taking $\mu = H$ for both constructed updates and combining with (SI4) yields $G_{\text{wave}}^{[0]} \in \mathcal{W}_{1-3\kappa}$.

Every claim in the desired initial-state bounds now follows by subtraction of explicit numbers:

$$\begin{aligned}
1 - 3\kappa &= 0.99997 > 0.7, \quad 1.49 > 1.2, \quad 1.9 - 3\kappa = 1.89997 > 1.2, \\
1 - \kappa &= 0.99999 > 0.68, \quad H > 0.9, \quad H + 1 > 1.9.
\end{aligned}$$

Since $0 < \varepsilon \leq 1$, a larger exponent gives the smaller class. The actual state therefore has, with the original budgets unchanged,

$$\begin{aligned}
G_{\text{wave}}^{[0]} &\in \mathcal{W}_{0.7}, \quad E_\theta^{[0]}, E_z^{[0]} \in \mathcal{M}_{1.2}, \quad (P, J_\theta, J_z)^{[0]} \in \mathcal{S}_{1.2}, \\
w &\in \mathcal{W}_{0.5}, \quad w - w_0^{\text{tan}} \in \mathcal{W}_{0.68}, \quad v, \gamma, p_m \in \mathcal{M}_{0.9}, \quad \beta \in \mathcal{M}_{1.9}, \\
\int R^2 \bar{v} dR &= 0, \quad \int R \bar{\gamma} dR = 0.
\end{aligned} \tag{SI10}$$

The residual equations in (SI10) use the wave/mean representative obtained by putting all the explicitly constructed flat terms in the additive flat residual. The velocity and pressure remain the complete constructed fields.

44 One fixed domain and the actual residual source class

Choose the background parameters and original profiles first. Fix the positive compact shell, reserved mean interval, strict cone margins, color centers, auxiliary rectangles, and cutoff functions. Only then choose one integer ℓ_0 large enough for the finite list of value comparisons: nonnegative covering indices; bounded neighboring covering differences; phase length and plane-isomorphism lower bounds; pulse growth perturbation and invariant-interval bounds; the covariance determinant and positive coefficient margins; and the elementary exponential-versus-polynomial bounds used for those value comparisons. The actual mean moment inverse uses the exact power-law shell and a fixed invertible matrix, so adds no source-dependent restriction. Fix

$$0 < q_{\text{big}} \leq \min(q_{\text{base}}, 2^{-(\ell_0+1)}, 1/2). \tag{SI11}$$

An active dyadic band has $Q \leq 2q$, so every band on this domain has $\ell \geq \ell_0$. At a point with $q_0 > 0$, take a slow neighborhood U with $q(U) \subset (q_0/2, 2q_0)$. Its active dyadic scales lie in $(q_0/4, 4q_0)$ and consequently have span at most four integers. This is the needed local restriction for the uniformly bounded common covering changes. Mere finiteness of an arbitrary larger collection of bands would not give that bound.

For clarity, increasing the stage or requested derivative order does not change (SI11). In the pulse equation the exact propagators obey

$$V_m(v, w) = \exp \left(-(m^2 - 1) \int_w^v d(s) ds \right) V_1(v, w), \quad d > 0, \quad |m| \geq 1.$$

Thus the value bound $\|V_m(v, w)\| \leq CP(v)/P(w)$ holds on the same domain for every nonzero integer m . Parameter differentiation of the ODE has the exact triangular form

$$(\partial^I z)' = A_m \partial^I z + \partial^I g + \sum_{0 < J \leq I} \binom{I}{J} (\partial^J A_m) \partial^{I-J} z.$$

At each fixed derivative and finite harmonic set, the already bounded lower derivatives give a new polynomial S_* bound by the same Duhamel integral. No new denominator or smallness comparison is introduced. The torus inverses have finite derivative loss for each requested order; their value definitions are unchanged. The radial inverse's repeated integration by parts estimates the same fixed operator and does not change it with the number of integrations. The signed covariance inverse divides only by the original $2a_\sigma$ and uses the original H^{-1} . The five-dimensional map, pressure map, and support bumps are fixed linear maps. Finite sums, products, and derivatives exist throughout the fixed domain. Induction over the finite directed graph of these operations proves that every finite state exists on (SI11), with constants depending on the stage and derivative order. There is no intersection of a stage-indexed sequence of decreasing positive thresholds.

Here is the precise source invariant for that induction. The nonzero harmonics are finite sums

$$\sum_{\gamma} \sum_{m \in H_j} f_{\gamma, m}^{[j]} e^{ik_{\gamma} m \Phi_{\gamma}}, \quad H_j \in \mathbb{Z} \setminus \{0\},$$

whose equal-label, equal-harmonic coefficients are collected. Each coefficient is defined on the original enlarged pulse rectangle, has the original closed slow and transverse supports, is smooth by zero extension at their boundaries and both radial edges, and satisfies

$$|\partial^I f_{\gamma, m}^{[j]}| \leq C_{j, I} \varepsilon^{\alpha} S_*^{b_{j, I}} \sqrt{\zeta} \delta^{-d_{j, I}} P(v). \quad (\text{SI12})$$

Its forward-pulse source representative need not have zero data at the terminal end; the actual correction is multiplied by the fixed terminal cutoff before taking its curl. The initial sources have (SI12) by the primary amplitude, curl and residual calculations above. A coefficient derivative preserves supports, and shifts only the derivative index. D_r, D_z, t_* have the exact costs already proved. A mean coefficient can have full auxiliary-torus support but its product with a wave keeps the wave support and $P(v)$. A same-label wave product has $P(v)^2 \leq P(v)$; distinct labels have zero products; multiplication adds angular harmonic integers and differentiation preserves them. These facts prove closure under each local polynomial residual operation, not just an abstract membership assertion.

The pulse Duhamel map preserves (SI12) because its integrand has $P(v)/P(w)$ times the source factor $P(w)$; the radial and slow weights are constant along that exact integration path. Its zero-data support property follows from ODE uniqueness, and each boundary jet follows from the differentiated zero-data equations. The signed stress map supplies the exact factor $\varepsilon^{\alpha-1} \sqrt{\zeta}$ on division by the original positive amplitude and hence a wave of exponent $\alpha - 1/2$ with the original envelope. The weighted radial primitives, interior moment subtractions, temporal inverse, and five-dimensional corrections yield smooth shell-supported mean coefficients. Radial integration fixes (z, t) and hence $q(z, t)$, so it never gathers an additional dyadic band. A radial line meets $O(S_*^3)$ boxes, which adds only a polynomial to the estimates. Subsequent products with waves again preserve (SI12).

Finally define the flat class by requiring every fixed physical derivative to be $O(q^N)$ for every N . It is an ideal under multiplication by the present finite-stage fields: if one factor has derivative bound $Cq^{-K}(1 + |\log q|)^b$, use the flat estimate with exponent $N + K + 1$ and the boundedness of $q(1 + |\log q|)^b$. Leibniz's rule has finitely many terms at each order. The physical derivatives of the fields have such polynomial bounds by the exact coordinate and phase maps. The temporal cutoff errors retain the terms $(1 - \psi)f$ and $\psi't$; on their support the Gaussian pulse has $P \leq e^{-cS_*}$. At any fixed physical derivative order their bound has the form

$$CQ^{-M} S_*^b e^{-c\ell^2}.$$

For every N , multiplication by q^{-N} is bounded because $-c\ell^2 + (M + N)\ell \log 2 + 2b \log \ell \rightarrow -\infty$. The radial projector errors are flat by the full-obstruction proof. Their actual fields remain in every product; the resulting flat terms are assigned to $F^{[j]}$, not used as forward-pulse sources. Together with the base's flat residual, these calculations prove on the single fixed domain the actual decomposition, whose nonzero $G^{[j]}$ harmonics satisfy (SI12):

$$\mathcal{R}(u^{[j]}, p^{[j]}) = G^{[j]} + F^{[j]}, \quad |F^{[j]}|_m \leq C_{j, m, N} q^N$$

45 Shrinking-cutoff summation and local realization: exact source audit

45.1 Result and coverage

The shrinking-cutoff argument in Lemma 5.4 is valid under its stated hypotheses. The calculation below verifies its scale selection, every mixed derivative loss, local finiteness, preservation of divergence, the nonlinear differential-polynomial comparison, and its use of one fixed-stage flat remainder. No summability across stages of the flat remainders is required. The specialization to the physical Navier–Stokes residual gives the exact passage from the finite-stage bounds (9.17)–(9.18) to (9.20) and the representation (9.21).

This bounded audit found no concrete defect in that passage. It is not an independent construction of the preceding wave/mean increments, a proof of all earlier support and inverse estimates, a certification of every conclusion of Proposition 9.9, or a Lean/kernel acceptance result. In particular, validating the summation implication does not validate the input estimate (9.18) by itself.

The source is the current 166-page PDF:

`source/navier-stokes-166.pdf`

SHA256: 0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f.

Its preserved Poppler layout text is `downloaded_public_release/current_pdf_extracted.txt`, under the same programme root. The following physical pages and exact text-line ranges were read:

Source	Physical pages	Layout-text lines	Coverage
(4.1), Lemma 4.1 and proof	24–25	1251–1281	Exact similarity map and derivatives used for (5.28).
(5.23), (5.27)	55–56	2906–2910; 2981–3002	Pointwise derivative seminorm, potential convention and tuple types.
Lemma 5.4, full hypotheses, statement and proof	57–59	3005–3158	Complete analytic summation argument, including (5.28)–(5.40).
(6.1)–(6.6), (6.11)–(6.12)	62–64, 66	3295–3374; 3451–3463	Physical graph derivative factors and pulse-coordinate derivatives.
(7.2)–(7.4)	74–75	3899–3920	Fixed label frequencies and exact phase used to check the derivative cost.
Definition 9.4	106	5535–5557	Exact stage gains, residual orders and representation convention.
Lemma 9.7 and proof	111–112	5804–5851	Stated common-domain construction and finite derivative dependence. Earlier inverse proofs are not independently reconstructed here.
(9.15)–(9.19), Lemma 9.8 and proof	112–114	5852–5937	Physical increment tuples, derivative-loss arithmetic and finite-stage residual interface.

Source	Physical pages	Layout-text lines	Coverage
Proposition 9.9, statement and Steps 1–2	114–115	5938–5999	Actual use of Lemma 5.4 and the full potential/direct-field/pressure assembly (9.21).
Proposition 9.9, Steps 3–5	115–116	6000–6055	Read for the scope of its remaining end-point/exterior/growth claims; these earlier dependent constructions are not certified by the summation calculation alone.

Rendered pages 57, 58, 59, 63, 113, 114 and 115 were also inspected. This resolves two potentially misleading layout extractions: the source has

$$d_r = 2((1+h)\rho_g - h\kappa_s), \quad s_x = \frac{1}{2} + \frac{h}{2}, \quad s_t = 1 + \frac{3h}{2},$$

with $\kappa_s = 10^{-5}$. The factor 2 in d_r multiplies both terms. No source equation is changed in this audit.

45.2 1. Original objects, types and derivative convention

Write $x = (x_1, x_2, z)$, $I = (\alpha_1, \alpha_2, \alpha_3, b) \in \mathbb{N}_0^4$, $D^I = \partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \partial_z^{\alpha_3} \partial_t^b$, and $|I| = |\alpha| + b$. The source seminorm (5.23) is the pointwise quantity

$$|V|_m = \max_{|I| \leq m} |D^I V|.$$

The lemma’s domain is Ω , with $0 < q(z, t) < q_0 \leq 1$, q smooth and bounded below on compact subsets. Its exact derivative hypothesis is

$$|\partial_z^a \partial_t^b q| \leq C_{a,b} q^{1-aD-b}, \quad 0 < D < 1. \quad (\text{A1})$$

The base is $U_0 = (u_0, p_0, T_0)$, with $\operatorname{div} u_0 = 0$. The optional stress entries may be omitted. Each increment is represented by

$$\begin{aligned} Z_j &= (A_j, B_j, p_j, T_j), & B_j &= b_j(r, z, t) e_\theta, \\ U_j &= (\operatorname{curl} A_j + B_j, p_j, T_j), & U^{[J]} &= U_0 + \sum_{j=1}^J U_j. \end{aligned} \quad (\text{A2})$$

These are actual physical fields on the same domain, with the Cartesian and zero-extension regularity specified before (5.30). The original bounds are

$$|Z_j|_m \leq C_{j,m} q^{g_j - \ell_m} \Lambda_{\log}(q)^{P_{j,m}}, \quad \Lambda_{\log}(q) = 1 + |\log q|, \quad (\text{A3})$$

where $0 < g_1 \leq g_2 \leq \dots \rightarrow \infty$, and the nonnegative, nondecreasing ℓ_m are independent of j . Constants and logarithmic powers may depend on j . No uniform bound for them in j is used below.

The realization map on one representative tuple is

$$\mathcal{C}_a(A, B, p, T) = (\operatorname{curl}(\chi(aq)A) + \chi(aq)B, \chi(aq)p, \chi(aq)T). \quad (\text{A4})$$

It maps smooth potential/direct-azimuthal/scalar tuples to physical velocity/pressure/stress tuples on Ω . It is linear in the tuple. Its velocity is not defined by merely multiplying $\operatorname{curl} A + B$ by $\chi(aq)$; the exact extra term is retained in Section 3 below.

45.3 2. Exact similarity map proves the required mixed derivatives of q

For the application, the source fixes

$$\tau = 1 - t, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad z = q^D \eta, \quad t = 1 - q(1 - \eta^2), \quad L = 1 - 2h\eta^2, \quad d = 1 - \eta^2. \quad (\text{A5})$$

The map $(q, \eta) \mapsto (z, t)$, from $(0, \infty) \times (-1, 1)$ to $\mathbb{R} \times (-\infty, 1)$, has Jacobian

$$\begin{pmatrix} Dq^{D-1}\eta & q^D \\ -(1 - \eta^2) & 2q\eta \end{pmatrix}, \quad \det = q^D(2D\eta^2 + 1 - \eta^2) = q^D L > 0. \quad (\text{A6})$$

For fixed z and $\tau > 0$, its inverse is the unique solution of $q - z^2 q^{2h} = \tau$ with $q > |z|^{1/D}$, and $\eta = zq^{-D}$. The left side is zero at the stated lower endpoint, tends to infinity, and has positive derivative $1 - 2hz^2 q^{2h-1} \geq 1 - 2h$ there and thereafter. Thus existence and uniqueness follow without choosing a different branch. At $z = 0$, the inverse is $q = \tau, \eta = 0$.

Inverting (A6) gives exactly

$$q_z = \frac{2\eta q^{1-D}}{L}, \quad q_t = -\frac{1}{L}, \quad \eta_z = \frac{d}{q^D L}, \quad \eta_t = \frac{D\eta}{qL}. \quad (\text{A7})$$

For any real β and smooth $F(\eta)$, the two derivative identities are

$$\partial_z(q^\beta F) = q^{\beta-D} \frac{2\beta\eta F + dF'}{L}, \quad \partial_t(q^\beta F) = q^{\beta-1} \frac{-\beta F + D\eta F'}{L}. \quad (\text{A8})$$

Starting with $F_{0,0} = 1$, repeated application of these exact operators gives

$$\partial_z^a \partial_t^b q = q^{1-aD-b} F_{a,b}(\eta). \quad (\text{A9})$$

One concrete recursive construction is first to apply the time operator b times to q , and then the axial operator a times, changing β by -1 and $-D$ respectively at each step. Commutation of the actual smooth derivatives identifies the same mixed derivative for any other order. Every resulting $F_{a,b}$ is a finite expression obtained from polynomials and powers of L^{-1} by differentiation and multiplication. Since $L \geq 1 - 2h > 0$ on $[-1, 1]$, all these functions are smooth there and $C_{a,b} := \max_{[-1,1]} |F_{a,b}| < \infty$. This proves (A1) with the original exponent $1 - aD - b$.

Also $q = \tau + q\eta^2 \geq \tau$. A compact subset of the open time domain $t < 1$ has a positive minimum of τ , so it has a positive lower bound for q . The stronger one-sided bounds at $\tau = 0, q \geq c > 0$ used later in Proposition 9.9 are separate statements about the actual input representatives, not consequences of this compact-subset observation.

45.4 3. Cutoff derivatives, all curl terms and an explicit fixed loss

A fixed smooth cutoff with the stated plateaux exists explicitly. Define $\eta_0(s) = 0$ for $s \leq 0$ and $\eta_0(s) = e^{-1/s}$ for $s > 0$, and set

$$\chi(s) = \frac{\eta_0(1-s)}{\eta_0(1-s) + \eta_0(s-1/2)}. \quad (\text{A10})$$

The denominator is positive for every real s . Each positive-side derivative of η_0 is $e^{-1/s}$ times a polynomial in s^{-1} ; the exponential series shows its limit is zero as $s \downarrow 0$. Hence η_0 , and then χ , are smooth. This cutoff equals one for $s \leq 1/2$, is zero for $s \geq 1$, and lies between zero and one. Every positive-order derivative is supported in $[1/2, 1]$.

For a positive-order mixed derivative $\partial_z^{a'} \partial_t^{b'} \chi(aq)$, the repeated chain rule is a finite sum of terms

$$c_{\mathcal{P}} a^k \chi^{(k)}(aq) \prod_{\nu=1}^k \partial_z^{a_\nu} \partial_t^{b_\nu} q, \quad a_\nu + b_\nu \geq 1, \quad \sum_\nu a_\nu = a', \quad \sum_\nu b_\nu = b', \quad (\text{A11})$$

where $c_{\mathcal{P}}$ is the finite integer multiplicity of the derivative partition. Applying (A1) bounds this term by

$$|c_{\mathcal{P}}| \|\chi^{(k)}\|_\infty \prod_{\nu=1}^k C_{a_\nu, b_\nu} (aq)^k q^{-a'D-b'}. \quad (\text{A12})$$

Its support has $1/2 \leq aq \leq 1$; consequently $(aq)^k \leq 1$. Summing the finite list proves

$$|\partial_z^{a'} \partial_t^{b'} \chi(aq)| \leq C_{a',b'}^\chi q^{-a'D-b'}, \quad (\text{A13})$$

with constants independent of the scale a . Zeroth order uses $|\chi| \leq 1$. Derivatives in x_1, x_2 of the cutoff vanish because $q = q(z, t)$. Thus there is no unrecorded factor of a left in the estimates.

Using the right-handed Cartesian curl and Levi-Civita symbol ϵ_{ikl} , the exact component expansion is

$$(\text{curl}(\chi(aq)A_j))_i = \sum_{k,l=1}^3 \epsilon_{ikl} (\chi(aq)\partial_k(A_j)_l + a\chi'(aq)(\partial_k q)(A_j)_l). \quad (\text{A14})$$

Equivalently the additional velocity is $a\chi'(aq)\nabla q \times A_j$, with its original sign. A derivative D^I of (A14) can instead be written directly as

$$\sum_{k,l} \epsilon_{ikl} \sum_{J \leq I+e_k} \binom{I+e_k}{J} D^J \chi(aq) D^{I+e_k-J}(A_j)_l, \quad (\text{A15})$$

where e_k is the spatial unit multi-index. Every nonzero cutoff derivative in this formula is purely in (z, t) . If it uses a' axial and b' time derivatives, while the derivative of the representative has total order d' , then $d' + a' + b' \leq m + 1$ for $|I| \leq m$, and the total loss is $\ell_{d'} + a'D + b'$. The pressure, direct-field and stress entries have total derivative order at most m , so the same upper list also covers them.

A concrete common choice is therefore

$$\ell'_m := \max_{d', a', b' \in \mathbb{N}_0, d'+a'+b' \leq m+1} (\ell_{d'} + a'D + b'). \quad (\text{A16})$$

It is nonnegative, nondecreasing and independent of j and a . In particular $\ell'_m \leq \ell'_{m+1} + m + 1$, since $0 < D < 1$. The larger explicit choice $\ell_{m+1} + m + 1$ is also valid if desired; no exact identity with (A16) is claimed.

Applying the finite product-rule sums to the original bounds (A3), and retaining their constants and logarithmic powers in their finite maximum/sum, yields finite numbers $\widehat{C}_{j,m}, \widehat{P}_{j,m}$, independent of a , such that

$$|\mathcal{C}_a Z_j|_m \leq \widehat{C}_{j,m} q^{g_j - \ell'_m} \Lambda_{\log}(q) \widehat{P}_{j,m}. \quad (\text{A17})$$

This calculation includes mixed derivatives striking the potential, the cutoff, the extra curl term, and the direct/scalar fields. It does not estimate only the differentiated velocity before applying the cutoff.

45.5 4. Diagonal scales and exact tail estimate

For each j , there are only $j + 1$ requirements in (5.37), one for every integer $0 \leq m \leq j$. Each has positive power $g_j/2$:

$$\widehat{C}_{j,m} \Lambda_{\log}(q) \widehat{P}_{j,m} q^{g_j/2} \leq 2^{-j} \quad (0 < q \leq a_j^{-1}). \quad (\text{A18})$$

These inequalities can be imposed recursively together with $a_j \geq 2a_{j-1}$ and $a_j^{-1} < q_0$. To make the selection explicit, take a_j to be the least positive integer satisfying these conditions and, for every $m \leq j$,

$$1 + \log a_j \geq \frac{2 \max(\widehat{P}_{j,m}, 0)}{g_j}, \quad \widehat{C}_{j,m} a_j^{-g_j/2} (1 + \log a_j) \widehat{P}_{j,m} \leq 2^{-j}. \quad (\text{A19})$$

For $j = 1$, omit the preceding-scale condition and also require $a_1 > q_0^{-1}$. The defining set is nonempty. Indeed, for any fixed positive ε and finite real P , $a^{-\varepsilon}(1 + \log a)^P \rightarrow 0$. One direct proof puts $y = \log a$, chooses an integer $n > \max(P, 0)$, and uses $e^{\varepsilon y/2} \geq (\varepsilon y/2)^n/n!$ for $y > 0$; the remaining exponential and negative power tend to zero. Every condition is eventually true and the list at stage j is finite.

The endpoint test (A19) implies the full interval test (A18). For $0 < q < 1$, direct differentiation gives

$$\frac{q \frac{d}{dq} [q^{g_j/2} \Lambda_{\log}(q)^P]}{q^{g_j/2} \Lambda_{\log}(q)^P} = \frac{g_j}{2} - \frac{P}{\Lambda_{\log}(q)}. \quad (\text{A20})$$

This is nonnegative when $\Lambda_{\log}(q) \geq 2 \max(P, 0)/g_j$. Hence its maximum on $0 < q \leq a_j^{-1}$ is attained at the upper endpoint once (A19) holds. All original logarithmic powers remain present in this check.

Combining (A17) and (A18) gives

$$|\mathcal{C}_{a_j} Z_j|_m \leq 2^{-j} q^{g_j/2-\ell'_m} \quad (m \leq j). \quad (\text{A21})$$

Outside the cutoff support, the summand and all its derivatives are zero, so this bound also applies there. For a compact $K \Subset \Omega$, let $\delta_K = \min_K q > 0$. Since $a_j \geq 2^{j-1} a_1$, all summands with $a_j > \delta_K^{-1}$ vanish on a neighborhood of K . Thus the sum is locally finite; all differentiation on K is differentiation of a finite sum. No convergence assertion at $q = 0$ is needed to obtain smoothness on Ω .

If $J \geq \max(1, m)$ and $0 < q < (2a_J)^{-1}$, then $a_j q < 1/2$ for every $j \leq J$. On a neighborhood of such a point those cutoffs are identically one, including all their derivatives. Therefore

$$U - U^{[J]} = \sum_{j>J} \mathcal{C}_{a_j} Z_j.$$

Because $g_j \geq g_{J+1}$ for $j > J$, $0 < q < 1$, and $\sum_{j>J} 2^{-j} = 2^{-J}$,

$$|U - U^{[J]}|_m \leq \sum_{j>J} 2^{-j} q^{g_j/2-\ell'_m} \leq 2^{-J} q^{g_{J+1}/2-\ell'_m}. \quad (\text{A22})$$

This is exactly the exponent and constant in (5.35). It compares with the uncut finite prefix, not with a differently cut finite prefix.

All added terms vanish on the common collar $a_1^{-1} < q < q_0$, because every $a_j \geq a_1$. They therefore extend smoothly by zero across $q = q_0$. At a lateral support boundary with $q > 0$, local finiteness reduces smooth zero extension to the original finite list of smooth representatives.

45.6 5. Divergence and the coordinate-level representative map

For every smooth Cartesian vector field A ,

$$\operatorname{div}(\operatorname{curl} A) = \sum_{i,k,l} \epsilon_{ikl} \partial_i \partial_k A_l = 0. \quad (\text{A23})$$

The terms for (i, k) and (k, i) cancel because their signs are opposite and mixed derivatives commute. This applies to $\chi(a_j q) A_j$ itself; it therefore includes every cutoff-curl contribution in (A14).

Away from the axis, the direct representative has Cartesian expression

$$B_j = \left(-\frac{b_j x_2}{r}, \frac{b_j x_1}{r}, 0 \right), \quad r = (x_1^2 + x_2^2)^{1/2}.$$

Its divergence is

$$-x_2 \frac{x_1}{r} \partial_r (b_j/r) + x_1 \frac{x_2}{r} \partial_r (b_j/r) = 0. \quad (\text{A24})$$

Moreover $\nabla \chi(a_j q) = (0, 0, a_j \chi'(a_j q) q_z)$, whose dot product with B_j is zero. Thus $\operatorname{div}(\chi(a_j q) B_j) = 0$. The smooth Cartesian extension makes the same identity hold at the axis by continuity. Adding the divergence-free base and using local finiteness proves $\operatorname{div} u = 0$.

For the mean-potential convention actually used in (9.15), the exact right-handed cylindrical map is

$$\Psi_j e_\theta \mapsto \operatorname{curl}(\Psi_j e_\theta) = (-\partial_z \Psi_j, 0, (\partial_r + r^{-1}) \Psi_j). \quad (\text{A25})$$

Multiplying the potential by a function of $q(z, t)$ gives the additional radial component $-\partial_z \chi(a_j q) \Psi_j$, consistent with $\nabla \chi \times \Psi_j e_\theta$, since $e_z \times e_\theta = -e_r$. The direct azimuthal term remains a different entry of the same realization map (A4), with its divergence calculation already proved.

45.7 6. Full differential-polynomial residual comparison

Retain the source polynomial, including its field-independent term:

$$\mathcal{F}(U)^a = c_{a0}(x, t) + \sum_{\nu=1}^{M_a} c_{a\nu}(x, t) \prod_{r=1}^{d_{a\nu}} D^{\beta_{a\nu r}} U^{k_{a\nu r}},$$

with the fixed order/degree bounds $|\beta_{a\nu r}| \leq s$, $1 \leq d_{a\nu} \leq d$. The coefficient derivative bounds are exactly (5.32), on the fixed support regions specified there. No coefficient, support region, order or degree is allowed to depend on the truncation index.

For $V = U^{[J]}$, $e = U - V$, and one monomial, let $a_r = D^{\beta_r}(V^{k_r} + e^{k_r})$ and $b_r = D^{\beta_r}V^{k_r}$. The exact finite product identity is

$$\prod_{r=1}^{d_\nu} a_r - \prod_{r=1}^{d_\nu} b_r = \sum_{r=1}^{d_\nu} \left(\prod_{l < r} a_l \right) (a_r - b_r) \left(\prod_{l > r} b_l \right). \quad (\text{A26})$$

Each term contains one differentiated entry of e . Apply D^I , $|I| \leq m$, distributing its derivatives among the coefficient and all factors with the full multi-index product-rule coefficients. Each field derivative then has order at most $m + s$. Since the number of coefficients, monomials and factors is fixed, all sums have bounds independent of J , apart from the displayed values of V and e .

For this fixed m , set $H_m = 1 + \max(0, \max_{|I| \leq m} K_I)$. The finitely many coefficient logarithms in (5.32) are bounded by a constant times q^{-1} on $0 < q < q_0$, by the exponential-versus-power estimate used above. It follows that their derivatives through order m are bounded by a fixed constant times q^{-H_m} . The resulting comparison is

$$|\mathcal{F}(V + e) - \mathcal{F}(V)|_m \leq C_m q^{-H_m} |e|_{m+s} (1 + |V|_{m+s} + |e|_{m+s})^{d-1}. \quad (\text{A27})$$

Outside the stated coefficient-support region a differentiated monomial vanishes, because the smooth field factors and their derivatives have the original supported extensions. Inside it the coefficient bound applies. Thus no additional support-dependent bound was used. The entire field-independent term c_{a0} cancels from this difference. It is still included in the finite-stage residual hypothesis and is not discarded from $\mathcal{F}(V)$.

There is a J -independent power bound for each finite prefix. One may take $K_m = \max(K_m^{\text{base}}, \ell_{m+1})$, enlarging a negative base exponent to zero if necessary. Indeed, taking the curl of each uncut potential costs at most one derivative, while $q^{g_j} \leq 1$. For fixed J , the finite sum of original constants and the maximum of the finite list of original logarithmic exponents give

$$|U^{[J]}|_m \leq C_{J,m} q^{-K_m} \Lambda_{\log}(q)^{P_{J,m}}. \quad (\text{A28})$$

This does not bound $C_{J,m}$ or $P_{J,m}$ uniformly in J .

Now fix derivative order m and decay order N . Choose one integer $J \geq \max(1, m + s)$ so large that

$$\frac{g_{J+1}}{2} - \ell'_{m+s} \geq N + H_m + (d-1)(K_{m+s} + 1), \quad \rho_J - K_m^{\mathcal{F}} \geq N + 1, \quad (\text{A29})$$

and the first left side is nonnegative. This is possible because both original gain sequences tend to infinity. After fixing J , take q small enough that:

- $q < (2a_J)^{-1}$, so (A22) compares with the actual uncut finite prefix;
- the fixed constants/logarithms in (A28) are bounded by q^{-1} ;
- the fixed constants/logarithms in the primary term of (5.33) are bounded by q^{-1} .

The nonnegative exponent in (A22) gives $|e|_{m+s} \leq 2^{-J} \leq 1$. Equation (A28) then implies $1 + |V|_{m+s} + |e|_{m+s} \leq 3q^{-(K_{m+s}+1)}$. Substitution into (A27) gives

$$|\mathcal{F}(U) - \mathcal{F}(U^{[J]})|_m \leq C_m 3^{d-1} 2^{-J} q^{g_{J+1}/2 - \ell'_{m+s} - H_m - (d-1)(K_{m+s}+1)} \leq C_m 3^{d-1} 2^{-J} q^N. \quad (\text{A30})$$

The same fixed-stage primary residual is at most q^N by (A29). Its original flat term satisfies $E_{J,m} \leq C_{J,m,N} q^N$. Therefore

$$|\mathcal{F}(U)|_m \leq (1 + C_{J,m,N} + C_m 3^{d-1} 2^{-J}) q^N. \quad (\text{A31})$$

The order of choices is exactly (m, N) , then J , then the neighborhood and constants. The only flat remainder in (A31) is $E_{J,m}$ for this one fixed J . Neither $\sum_j E_{j,m}$ nor a bound uniform in j for their constants occurs. This closes the residual part of Lemma 5.4.

45.8 7. Check of the physical derivative losses in (9.17)

The source chart holds $Q = 2^{-\ell}$, $\varepsilon = Q^h$, and $S_* = \ell^2$ fixed while differentiating. Its coordinates are $(R, Z, T) = (r/Q^{1/2}, z/Q^D, \tau/Q)$. On the perturbation annulus $r \asymp Q^{1/2}$. The original auxiliary covering is determined by

$$J_g = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad \Lambda_g = 4 - \sqrt{2}, \quad T_g = 4 + \sqrt{2}, \quad \rho_g = \frac{\log \Lambda_g}{\log T_g},$$

$$\kappa_s = 10^{-5}, \quad d_r = 2((1+h)\rho_g - h\kappa_s), \quad i = \left\lfloor \log_{T_g} \frac{Q^{-1-h}}{S_*} \right\rfloor.$$

From $T_g^{-1}Q^{-1-h}/S_* < T_g^i \leq Q^{-1-h}/S_*$ and $\Lambda_g^i = (T_g^i)^{\rho_g}$,

$$\Lambda_g^i Q^{d_r/2} \asymp Q^{-(1+h)\rho_g} S_*^{-\rho_g} Q^{(1+h)\rho_g - h\kappa_s} = Q^{-h\kappa_s} S_*^{-\rho_g}. \quad (\text{A32})$$

Thus a physical radial derivative of an evaluated coefficient has loss $1/2 + h\kappa_s$; an axial derivative has loss $D = 1/2 - h$; a time derivative has loss $1 + h$. Cartesian differentiation of the cylindrical frame costs $1/2$. Derivatives striking their smooth radial coefficients contribute further factors $Q^{-1/2}$; the larger radial cost already covers each such differentiation. Powers of S_* enter the logarithmic factors and may depend on the fixed derivative order and stage.

For the label phase the exact source expression is

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v(pF + p_z G), \quad k = \lceil \varepsilon^{-1/2} \rceil. \quad (\text{A33})$$

The quantities p, p_z, x_0, k are fixed for the label, including the source's integer rounding rule. The pulse coordinate obeys $\nabla_x v = 0$ and $\partial_t v = Q^{-1-h}$, with higher physical derivatives of this first derivative zero within the fixed chart. These follow from (6.12), not from neglecting the auxiliary phase evaluation. The axial linear phase term has derivative factor $Q^{-D}\varepsilon^{-1} = Q^{-1/2}$. Repeated differentiation of the remaining slow phase factors and the angular coordinate on $r \asymp Q^{1/2}$ gives the phase bound quoted in Lemma 9.8,

$$|\nabla_x^a \partial_t^b \Phi| \leq C_{a,b} Q^{-a/2-b(1+h)} S_*^{P_{a,b}}, \quad a + b \geq 1, \quad (\text{A34})$$

using the stated fixed-order slow-coefficient bounds. Verification of those earlier coefficient estimates is an upstream input; the derivative exponents and auxiliary-coordinate contribution here are checked explicitly.

To avoid confusing the derivative order with the source's harmonic integer in $e^{ikm\Phi}$, write that integer as m_{harm} in the next calculation; it is the same integer, not a changed frequency. A derivative partition with k' nonzero phase-derivative factors gives

$$(ikm_{\text{harm}})^{k'} e^{ikm_{\text{harm}}\Phi} \prod_{\nu=1}^{k'} \nabla_x^{a_\nu} \partial_t^{b_\nu} \Phi, \quad k' \leq a + b, \quad \sum a_\nu = a, \quad \sum b_\nu = b.$$

Since $0 < Q \leq 1$, $k \leq 2Q^{-h/2}$. With (A34), its power of Q is bounded by

$$Q^{-a/2-b(1+h)-k'h/2} \leq Q^{-a(1/2+h/2)-b(1+3h/2)}. \quad (\text{A35})$$

The factor $2^{k'} |m_{\text{harm}}|^{k'}$, derivative-partition coefficients and logarithmic powers are retained in the fixed-stage derivative constant. The source's harmonic set is finite at each fixed stage, independently of band; this permits dependence on the stage in that constant. It does not produce an additional stage-dependent power of Q .

The resulting costs are $s_x = 1/2 + h/2$, $s_t = 1 + 3h/2$. They dominate all four rows of (9.19): $\kappa_s < 1/2$, $D = 1/2 - h$, $1 + h \leq 1 + 3h/2$, and $1/2 \leq s_x$. Also $s_x < s_t$. Physical rescaling has the four factors

$$Q^{-A}, \quad Q^{-2A}, \quad Q^{1/2-A}, \quad Q^{1/2-A},$$

for velocity, pressure, wave potential and mean potential. Each is bounded by Q^{-2A} for $0 < Q \leq 1$ and the original $A = 1/2 + h$. The source keeps the factor $(km_{\text{harm}})^{-1}$ inside the wave potential before this rescaling. One additional spatial derivative recovers velocity from a potential.

Hence the stated common choice

$$\ell_m = 2A + (m+1) \left(1 + \frac{3h}{2} \right) \quad (\text{A36})$$

covers all derivatives of order at most m , including that extra curl derivative. Every retained stage- j increment has the source's class exponent at least $j/10$, so its gain is $g_j = hj/10$. This verifies the loss/gain arithmetic in (9.17). Passing from Q to comparable q may change constants and logarithmic factors, including constants depending on j , but does not change the fixed exponent loss.

The residual conversion uses the fixed factor $Q^{-2A-1/2}$. Together with (9.8), $\sigma_j = 1/5 + j/10$, it gives gain $\rho_j = h\sigma_j$ with a derivative loss independent of j , as claimed in (9.18). The pointwise validity and all-label uniformity of the finite-stage remainder in (9.18) are supplied by the earlier residual decomposition, not by this exponent calculation. No such uniformity is inferred merely from a coefficient bound at one label.

45.9 8. Exact Navier–Stokes specialization and truncation choice

For (9.20), the tuple is $U = (u, p)$, without the optional stress variable. The viscosity in this local construction is exactly one. In Cartesian components,

$$\mathcal{R}_i(u, p) = \partial_t u_i + \sum_{k=1}^3 u_k \partial_k u_i - \sum_{k=1}^3 \partial_k^2 u_i + \partial_i p. \quad (\text{A37})$$

This is a fixed differential polynomial of order $s = 2$ and degree $d = 2$, with constant coefficients. No singular cylindrical coefficients occur in (A37); thus the comparison can take $H_m = 0$.

Put $V = (v, p) = U^{[J]}$, $e = (w, \pi) = U - U^{[J]}$. For every I , the complete mixed-derivative difference is

$$\begin{aligned} D^I [\mathcal{R}_i(v + w, p + \pi) - \mathcal{R}_i(v, p)] &= \partial_t D^I w_i - \sum_{k=1}^3 \partial_k^2 D^I w_i + \partial_i D^I \pi \\ &+ \sum_{k=1}^3 \sum_{J' \leq I} \binom{I}{J'} [(D^{J'} v_k)(D^{I-J'+e_k} w_i) + (D^{J'} w_k)(D^{I-J'+e_k} v_i) \\ &\quad + (D^{J'} w_k)(D^{I-J'+e_k} w_i)]. \end{aligned} \quad (\text{A38})$$

Here J' is a derivative multi-index, not the truncation stage. Every linear, cross and quadratic term is present with its sign. For Euclidean tuple norms, each scalar component is bounded by the tuple norm and the three-component output is at most $\sqrt{3}$ times its maximum component. Since $\sum_{J' \leq I} \binom{I}{J'} = 2^{|I|}$, (A38) gives the explicit sufficient bound

$$\begin{aligned} |\mathcal{R}(V + e) - \mathcal{R}(V)|_m &\leq \sqrt{3} [5 + 6 \cdot 2^m |V|_{m+2} + 3 \cdot 2^m |e|_{m+2}] |e|_{m+2} \\ &\leq C_m (1 + |V|_{m+2} + |e|_{m+2}) |e|_{m+2}, \quad C_m = \sqrt{3} (5 + 9 \cdot 2^m). \end{aligned} \quad (\text{A39})$$

If the source's absolute value convention is taken componentwise instead, the same upper constant remains valid. No norm identification is needed for this inequality.

For prescribed nonnegative integer m, N , a concrete sufficient stage is any integer J satisfying

$$J \geq \max \left\{ 1, m + 2, \left\lceil \frac{20}{h} (N + K_{m+2} + 1 + \ell'_{m+2}) - 1 \right\rceil, \left\lceil \frac{10}{h} (N + 1 + K_m^{\mathcal{R}}) - 2 \right\rceil \right\}. \quad (\text{A40})$$

Indeed, the two gain inequalities become exactly

$$\frac{h(J+1)}{20} - \ell'_{m+2} \geq N + K_{m+2} + 1, \quad \frac{h}{5} + \frac{hJ}{10} - K_m^{\mathcal{R}} \geq N + 1. \quad (\text{A41})$$

Keep this J fixed. On a sufficiently small common neighborhood, (A22), (A28), (A39) and the fixed-stage remainder estimate give

$$|\mathcal{R}(U)|_m \leq (1 + C_{J,m,N} + 3C_m 2^{-J}) q^N. \quad (\text{A42})$$

This proves the exact quantified flatness conclusion (9.20) from the source's stated finite-state estimates. It does not assert that the finite-stage flat pieces were summed, nor that their constants are uniform in the stage.

45.10 9. The actual representation (9.21), initialization and remaining scope

Using the representatives in (9.15), define precisely

$$\begin{aligned} \mathcal{A} &= A_0 + \sum_{j \geq 1} \chi(a_j q) A_j, & \mathcal{B}e_\theta &= B_0 e_\theta + \sum_{j \geq 1} \chi(a_j q) B_j, \\ p_{\text{loc}} &= p_0 + \sum_{j \geq 1} \chi(a_j q) p_j, & u_{\text{loc}} &= \text{curl } \mathcal{A} + \mathcal{B}e_\theta. \end{aligned} \quad (\text{A43})$$

Every sum is locally finite on $q > 0$. Differentiation therefore commutes with these sums locally, and (A43) is exactly (A4) summed over the original representative tuples. It provides the full map to the physical fields, not an identification of potentials with velocities.

The near-axis base potential in (5.27) has Cartesian form $A_n = (S_n/r^2)(-x_2, x_1, 0)$, where S_n/r^2 is smooth in (r^2, z, t) for $q > 0$. The base angular field similarly has the form $b(r^2, z, t)(-x_2, x_1, 0)$. The annular representatives vanish near the axis at each fixed $t < 1$. Their sum has a common such neighborhood locally because only finitely many summands are active there. Thus the Cartesian regularity asserted for the fields in (9.21) is preserved by this summation.

A finite initialization block with a nonpositive gain can be included in the base after one cutoff on its potentials, direct angular fields and pressures. On a fixed neighborhood of $q = 0$, this cutoff is identically one, so every original finite-state residual is unchanged there. Errors supported where that fixed cutoff varies are bounded away from $q = 0$. Once their fixed-stage uniform boundedness on that support is supplied by the stated field bounds, they satisfy a bound $C_N q^N$ there for every fixed N , with C_N allowed to depend on the support's positive lower bound for q . This operation creates no infinite family of errors that would need to be summed.

The source also states that all finite corrections agree with the exact exterior heat field outside one fixed interval of X , so the finite residual is zero there. That is the input which extends the finite-stage estimate from a bounded profile interval to the whole local domain in Proposition 9.9 Step 1. The summation does not establish that exterior agreement independently. Likewise, the one-sided smoothness of the actual finite representatives at $\tau = 0, q \geq c$, fixed discrete label choices, and inner leading-coefficient growth are the further inputs used in Steps 3–5; local finiteness preserves their already established finite-term bounds, but does not replace their proofs.

The completed bounded result is therefore the analytic realization mechanism and the exact matching of its gains, losses, residual polynomial and representative map to (9.17)–(9.21). There is no extra summation gap caused by nonuniform fixed-stage flat-remainder constants. Verification of the global source construction and of the formal endpoint remains outside this result.

45.11 10. Bounded formal-source correspondence and the initial cutoff

All formal paths in this section are under

`formal-source/NavierStokes/`.

The following declarations were located by a bounded read-only search. The summation and cutoff definitions/signatures listed in the first five rows were additionally read directly in this audit. No Lean/Lake/Elan process or proof replay was run.

File and declaration	Exact source lines	Source scope
<code>SmoothCutoffs.lean:</code> <code>cutoffBump, cutoff,</code> <code>scaledCutoff</code>	27–44; 134–146; 161–182	Actual smooth bump with inner radius 1/2, outer radius 1, and <code>scaledCutoff a q = cutoff (a*q)</code> ; finite families equal one on a common neighborhood of zero.
<code>SolenoidalDiagonal.lean:</code> <code>cutStage, potentialSum,</code> <code>eventually_zero_tail,</code> <code>potentialSum_eventuallyEq_</code> <code>partial</code>	32–38; 46–67	Literal cut-potential <code>tsum</code> , local vanishing of a common tail and equality to a finite prefix. Its index set includes zero.

File and declaration	Exact source lines	Source scope
<code>CutStageEstimates.lean:</code> <code>RawStageBounds, cutLoss,</code> <code>exists_diagonal_cut_bounds,</code> <code>exists_finite_diagonal_cut_</code> <code>bounds</code>	199–203; 192; 349–368; 379–417	Raw stage estimates for positive stages; one doubling integer schedule; output gain is exactly $g_j / 2$. One schedule covers a finite family of component spaces.
<code>DiagonalJetBounds.lean:</code> <code>CutStageBounds,</code> <code>potential_tail_jet_bound,</code> <code>exists_uncut_tail_order</code>	196–220; 275–303	Positive stage j controls derivatives through $j+2$; tail after indices $0, \dots, J$ is bounded by $2^{-(J)} q^{(g(J+1)-L_m)}$; a single neighborhood compares all requested jets to the uncut prefix.
<code>MixedDiagonalResidual.lean:</code> <code>velocity, pressure, residual_</code> <code>eq_originalResidual,</code> <code>exists_physical_schedule_</code> <code>residual_zero</code>	26–50; 200–232	Curl of the cut potential sum plus direct cut-field sum, with pressure summed separately; literal nonlinear residual; one physical schedule with gain $g_j / 2$ and vanishing joint residual jets.
<code>ActualIterationLedger.lean:</code> <code>gain, sigma_formula,</code> <code>residualLoss,</code> <code>residualRate_eq</code>	29; 36; 286; 292	Literal gain $h*j/10$, sigma $1/5+j/10$, and a stage-independent physical residual loss.
<code>ActualCycleResidualBounds.lean:</code> <code>fixedLoss,</code> <code>Invariant.residual_jetRate,</code> <code>finite_residual_rates;</code> <code>ActualStageEstimates.lean:</code> <code>stageEstimates_of_</code> <code>representations</code>	1145, 1158, 1190; 347–403	Binding of actual finite-cycle invariants and physical representations to the stage-estimate interface; upstream hypotheses were not traversed.
<code>ActualCandidateAssembly.lean:</code> <code>estimates, Witness,</code> <code>witness, selected_witness</code>	1090; 1121; 1153; 1177	Actual mixed candidate assembly together with later localization/activation. This locator does not identify the final globally modified object with the manuscript’s local field.

The numerical `DiagonalScale.lean` comments refer to “Lemma 11.3,” rather than the present “Lemma 5.4.” Numbering is not used as correspondence evidence.

45.11.1 10.1 Exact half-gain accounting

The formal raw input has gain g_j in `RawStageBounds`. The output of `exists_diagonal_cut_bounds` is explicitly `CutStageBounds ... (fun j => g_j / 2) ...`. The tail theorem’s parameter called g is therefore the already reduced gain

$$\tilde{g}_j := g_j/2.$$

Its displayed output exponent is $\tilde{g}_{J+1} - L_m$, not $\tilde{g}_{J+1}/2 - L_m$. Substituting the literal actual-stage gain gives

$$\tilde{g}_{J+1} = \frac{g_{J+1}}{2} = \frac{1}{2} \frac{h(J+1)}{10} = \frac{h(J+1)}{20}. \quad (\text{A44})$$

Thus the formal sequence spends half the raw gain once, exactly as in (A18)–(A22). There is no further factor of $1/2$ to turn the denominator into 40. The common loss in the formal theorem is `cutLoss` $L_m = \text{finiteBound } L_m + m$; it is a specified sufficient loss, and is not asserted to equal the sharper mixed-coordinate maximum (A16).

The formal derivative budget $m \leq j + 2$ is at least as large as the paper's budget $m \leq j$. For a tail index $j > J$, the formal condition $m \leq J + 3$ implies $m \leq j + 2$, explaining the tail theorem's offset. Recovering velocity still uses one additional spatial derivative of the potential, and the nonlinear residual still uses the needed further physical derivatives. Those operations are not paid for by another halving of the gain; they are handled by the fixed derivative-order losses and the finite derivative ceiling, as in Sections 3, 7 and 8.

45.11.2 10.2 Exact map between cut stage zero and an uncut base

The manuscript's Lemma 5.4 leaves U_0 uncut; formal `potentialSum` cuts every natural-number index, including zero. The precise comparison uses the same original representative sequence and the same schedule. Let the formal zero entries be

$$A(0) = A_0, \quad B(0) = B_0 e_\theta, \quad P(0) = p_0,$$

and let its entries at every $j \geq 1$ be the manuscript's A_j, B_j, p_j , with no index shift. Then `uncutPrefix A (J+1)` is literally $A_0 + \sum_{j=1}^J A_j$, because the range $0, \dots, J$ is the disjoint union of $\{0\}$ and $\{1, \dots, J\}$. The same identity holds for the direct field and pressure. Thus their uncut finite-state objects agree under this representative dictionary.

Use the formal cutoff χ_f , which has the same required plateaux on positive arguments. It need not equal the explicit example (A10); the proof of Lemma 5.4 uses only the stated smoothness/support/plateau properties, so applies to χ_f itself. Let $(\mathcal{A}_{\text{base}}, B_{\text{base}}, p_{\text{base}})$ denote the manuscript-style sum with its zero entries uncut, but these same positive-stage cutoffs. Let $(\mathcal{A}_f, B_f, p_f)$ be the all-indices formal-style sum. Put $c_0 = \chi_f(a_0 q)$. Their exact differences on $q > 0$ are

$$\mathcal{A}_f - \mathcal{A}_{\text{base}} = (c_0 - 1)A_0, \quad B_f - B_{\text{base}} = (c_0 - 1)B_0 e_\theta, \quad p_f - p_{\text{base}} = (c_0 - 1)p_0. \quad (\text{A45})$$

For the physical velocities, with $u_0 = \text{curl } A_0 + B_0 e_\theta$, this gives

$$u_f - u_{\text{base}} = \text{curl}((c_0 - 1)A_0) + (c_0 - 1)B_0 e_\theta = (c_0 - 1)u_0 + \nabla c_0 \times A_0. \quad (\text{A46})$$

No cutoff commutator is omitted. The full residual difference is (A38) with $w = (c_0 - 1)u_0 + \nabla c_0 \times A_0$ and $\pi = (c_0 - 1)p_0$. On the open region

$$0 < q < \frac{1}{2a_0}, \quad (\text{A47})$$

the positive formal schedule has $a_0 > 0$, so c_0 is identically one on a neighborhood of every point. Equations (A45)–(A46) and every one of their mixed derivatives vanish there. Therefore the two representative sums, velocities, pressures and full nonlinear residuals agree exactly there, at every derivative order. Away from this region, (A45)–(A46), rather than equality, are the precise relation.

This proves the local initial-term-cutoff correspondence on the same representatives and schedule. It does not identify two independently selected schedules, different representative sequences, or the later localized/activated candidate. The formal `exists_physical_schedule_residual_zero` conclusion is phrased in terms of joint residual jets at the endpoint; no equivalence of all its filter/domain quantifiers with every clause of the manuscript is claimed by this bounded locator.

45.12 Transcription repair record

The derivative-partition display in Section 7 contained the literal text `quad` between `k'\le a+b`, and `\sum a_nu=a`. It has been corrected to the TeX spacing command `\quad`. This repairs transcription only; every mathematical symbol, coefficient, exponent, inequality and claim is retained. The accompanying `summation_realization_audit_receipt.json` records the final note hash and the completed audit scope.

46 The exact quotient of smooth terminal force extensions

This note extends the explicit terminal-force construction in the accompanying Navier–Stokes reader. It proves the topology of the actual trace quotient and the impossibility of a continuous linear choice of all extensions. It makes no additional claim that the preceding singular fluid construction has completed independent verification. Spatial coordinates, vector components and the right-hand time coordinate $\sigma = t - 1$ are retained.

Fix the compact spatial set $K \subset \mathbb{R}^3$ used for localization, with nonempty interior. For a smooth vector field $g : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ set $\|g\|_{C^m} = \max_{|\alpha| \leq m} \sup_x |\partial_x^\alpha g(x)|_2$. Let V_K consist of such fields with support in K , and put

$$\mathcal{J}_K = \prod_{j=0}^{\infty} V_K, \quad Q_m(F) = \max_{0 \leq j \leq m} \|F_j\|_{C^m}.$$

The topology defined by the increasing seminorms Q_m is exactly the product of the usual smooth topologies: every Q_m uses finitely many coordinate seminorms, while a finite list of coordinate derivative seminorms is bounded by Q_m for m at least all of their indices and derivative orders.

Let $\mathcal{E}_K$ be the vector space of smooth fields $H : \mathbb{R}^3 \times [0, \infty) \rightarrow \mathbb{R}^3$, with all mixed derivatives continuous from the right at zero, and with support contained in $K \times [0, 1]$. Use its increasing seminorms

$$P_m(H) = \max_{|\alpha|+r \leq m} \sup_{x \in \mathbb{R}^3, \sigma \geq 0} |\partial_x^\alpha \partial_\sigma^r H(x, \sigma)|_2.$$

Define the actual linear trace map and its kernel by

$$\mathcal{T}(H) = (\partial_\sigma^j H(\cdot, 0))_{j \geq 0}, \quad \mathcal{N}_K = \{H \in \mathcal{E}_K : \mathcal{T}(H) = 0\}.$$

The derivatives are evaluated in the original three vector components. Their support is contained in K . Evaluation gives

$$Q_m(\mathcal{T}H) \leq P_{2m}(H). \quad (\text{T.1})$$

Thus $\mathcal{T}$ is a continuous linear map and $\mathcal{N}_K$ is closed.

Define $\rho(s) = e^{-1/s}$ for $s > 0$ and $\rho(s) = 0$ for $s \leq 0$, and fix

$$\chi(s) = \frac{\rho(1-s)}{\rho(1-s) + \rho(s-3/4)}.$$

The denominator is positive at every real s : its first term is positive for $s < 1$, and its second for $s > 3/4$. Every right derivative of ρ is a finite polynomial in $1/s$ times $e^{-1/s}$ and tends to zero as $s \downarrow 0$; induction and pasting with zero therefore prove its smoothness. Thus χ is smooth, equals one for $s \leq 3/4$, is zero for $s \geq 1$, and satisfies $0 \leq \chi \leq 1$. Every positive-order derivative is supported in $[3/4, 1]$, so all the derivative supremums used below are finite. For $b \geq 1$ write

$$L_{j,b}(F_j)(x, \sigma) = \chi(b\sigma) \frac{\sigma^j}{j!} F_j(x).$$

For each r the full temporal derivative is

$$\partial_\sigma^r \left(\chi(b\sigma) \frac{\sigma^j}{j!} \right) = \sum_{\substack{0 \leq a \leq r \\ r-a \leq j}} \binom{r}{a} b^a \chi^{(a)}(b\sigma) \frac{\sigma^{j-r+a}}{(j-r+a)!}. \quad (\text{T.2})$$

Every nonzero term has $0 \leq \sigma \leq 1/b$. Consequently, with the finite constant

$$C_{j,r} = \sum_{\substack{0 \leq a \leq r \\ r-a \leq j}} \binom{r}{a} \frac{\|\chi^{(a)}\|_\infty}{(j-r+a)!},$$

one has, for all j, r, b and spatial multi-indices α ,

$$\|\partial_x^\alpha \partial_\sigma^r L_{j,b}(F_j)\|_\infty \leq C_{j,r} b^{r-j} \|F_j\|_{C^{|\alpha|}}. \quad (\text{T.3})$$

All constants and factorials in this estimate are retained. Its exponent is negative whenever $r < j$; no bound uniform in the index j is asserted.

Theorem 46.1. *The trace $\mathcal{T}$ is surjective. The induced map*

$$\overline{\mathcal{T}} : \mathcal{E}_K / \mathcal{N}_K \longrightarrow \mathcal{J}_K, \quad [H] \longmapsto \mathcal{T}H$$

is a linear homeomorphism for the quotient topology and the specified product topology. Nevertheless there is no continuous linear map $B : \mathcal{J}_K \rightarrow \mathcal{E}_K$ satisfying $\mathcal{T}B = \text{id}_{\mathcal{J}_K}$.

Proof. We first construct representatives with an explicit estimate on each quotient seminorm. Fix $F \in \mathcal{J}_K$, an integer $m \geq 0$, and a real $\epsilon > 0$. Keep the first $m + 1$ entries in the finite field

$$H_{\leq m} = \sum_{j=0}^m L_{j,1}(F_j).$$

Equation (T.3) gives the finite constant and estimate

$$A_m = \sum_{j=0}^m \max_{0 \leq r \leq m} C_{j,r}, \quad P_m(H_{\leq m}) \leq A_m Q_m(F). \quad (\text{T.4})$$

Set $b_j = 1$ for $0 \leq j \leq m$. For every $j > m$, choose an integer $b_j \geq 1$, increasing at least by a factor two from its predecessor, such that both of the following finite lists hold:

$$C_{j,r} b_j^{r-j} \|F_j\|_{C^{|\alpha|}} \leq \epsilon 2^{-j} \quad (|\alpha| + r \leq m), \quad (\text{T.5})$$

$$C_{j,r} b_j^{r-j} \|F_j\|_{C^{|\alpha|}} \leq 2^{-j} \quad (|\alpha| + r \leq \lfloor j/2 \rfloor). \quad (\text{T.6})$$

There are finitely many (α, r) in either list, all with $r < j$. Thus every nonzero left side tends to zero as b_j increases, and every zero left side already satisfies its inequality. The set of admissible integers is nonempty; choosing its least member above the previous lower bound specifies the sequence. This construction uses the actual input family F , m , and ϵ .

For each fixed derivative order k , all sufficiently large j have $k \leq \lfloor j/2 \rfloor$. Equations (T.3) and (T.6) show uniform convergence of every differentiated tail in

$$H = H_{\leq m} + \sum_{j>m} L_{j,b_j}(F_j). \quad (\text{T.7})$$

The finitely many remaining terms have finite smooth norms. To identify the sum of first derivatives with the derivative of the sum, integrate each partial sum along a coordinate segment and pass to the limit using uniform convergence of the function and of that derivative. Repeat at every order, using one-sided time segments at $\sigma = 0$. This proves smoothness with all the prescribed mixed right derivatives. All terms vanish off K and for $\sigma \geq 1$; uniform convergence of their derivatives preserves these facts and smoothness on both support boundaries. Hence $H \in \mathcal{E}_K$.

At $\sigma = 0$, each fixed cutoff is constant on a neighborhood. The r th time derivative of its monomial has value zero unless $j = r$, and value one if $j = r$. Uniform convergence of the differentiated series therefore gives $\partial_\sigma^r H(\cdot, 0) = F_r$ for every r . In particular $\mathcal{T}$ is surjective. Summing (T.5) additionally gives

$$P_m(H) \leq A_m Q_m(F) + \epsilon \sum_{j>m} 2^{-j} \leq A_m Q_m(F) + \epsilon. \quad (\text{T.8})$$

For a class $[H] \in \mathcal{E}_K / \mathcal{N}_K$, write

$$\bar{P}_m([H]) = \inf_{N \in \mathcal{N}_K} P_m(H + N).$$

These seminorms generate the quotient topology. Indeed the P_m are increasing, so their single-seminorm balls form a neighborhood basis in $\mathcal{E}_K$. The quotient image of $\{P_m < \delta\}$ is exactly $\{\bar{P}_m < \delta\}$: either membership is witnessed by a representative with $P_m < \delta$. The quotient projection is open, since the full inverse image of the image of an open set O is $O + \mathcal{N}_K = \bigcup_{N \in \mathcal{N}_K} (O + N)$, which is open. This proves the claimed quotient description directly.

The kernel definition makes $\bar{\mathcal{T}}$ well-defined and injective; surjectivity was proved by (T.7). Its linear inverse is well-defined by $F \mapsto [H]$, independently of the chosen realization: two such realizations differ by an element of $\mathcal{N}_K$. Taking the infimum of (T.1) over a class and then using (T.8), with arbitrary $\epsilon > 0$, proves both estimates

$$Q_m(\bar{\mathcal{T}}[H]) \leq \bar{P}_{2m}([H]), \quad \bar{P}_m(\bar{\mathcal{T}}^{-1}F) \leq A_m Q_m(F). \quad (\text{T.9})$$

They prove continuity of the map and its inverse with the stated topologies. In particular the quotient is Hausdorff: if all its seminorms vanish, (T.9) gives all $Q_m(F) = 0$, hence $F = 0$ and the class is zero.

Suppose finally that a continuous linear right inverse B existed. Continuity at zero, the increasing family Q_m , and the target norm P_0 give an integer M and $\delta > 0$ such that

$$Q_M(F) < \delta \implies P_0(BF) < 1.$$

If $Q_M(F) = 0$, this applies to cF for every real c . Linearity and homogeneity give $|c|P_0(BF) < 1$ for all c , whence $P_0(BF) = 0$. Since P_0 is the supremum of the vector length, BF is identically zero. Choose a nonzero $g \in V_K$: a nonzero smooth scalar bump inside a ball whose closure is contained in the interior of K , multiplied by a fixed nonzero coordinate vector, supplies one. Take $F_{M+1} = g$ and all other entries zero. Then $Q_M(F) = 0$, whereas the right-inverse identity forces $\partial_\sigma^{M+1}BF(\cdot, 0) = g \neq 0$, a contradiction. $\square$

For the actual analytical and formal force constructions this identifies the precise relation of two extensions after their full coordinate and vector component maps have been applied. Equal terminal jets imply equality in $\mathcal{E}_K/\mathcal{N}_K$; their pointwise difference is a smooth field flat at the terminal slice. This relation does not identify their positive-time values. If both are pasted to the same already specified past force at $\sigma \leq 0$, the difference extends by zero there and is jointly smooth, since every mixed trace vanishes. Its closed support may meet $\sigma = 0$: flatness does not imply a positive support gap. The fluid and all forcing before the terminal time remain identical in this map of extensions.

Fixing $m = 0$, $\epsilon = 1$, $b_0 = 1$ in the least-integer construction defines one specific right inverse $\mathcal{B} : F \mapsto H$, with the remaining cutoff scales depending on the jets. The theorem explains why the homeomorphism of the quotient is not a continuous linear choice of representatives. No missing fluid estimate or singularity claim is inserted as an assumption in this statement about the explicitly defined trace spaces.

47 Compact radial stress reconstruction with every moment, derivative and chart factor

This is a full independent calculation of the radial stress map in Proposition 9.6 Step 2, physical page 109 of the frozen 166-page manuscript. The source is `programme/downloaded_public_release/navier-stokes.pdf`, SHA256 `0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f`. Page 109 was rendered and visually inspected in this lane; its profile factor, lower integration endpoint, moment, sign and chart power are the ones below. This calculation proves the displayed operator and its full force correspondence. It does not establish the upstream amplitude or finite-stage residual bounds to which the operator is applied.

47.1 1. Original parameters and supported spaces

Keep the source physical variables (r, z, t) , $r > 0$, and its positive smooth function $q = q(z, t)$. Write Ω for its parameter domain. In the source application

$$\tau = 1 - t, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad 0 < h < \frac{1}{100}, \quad q - z^2 q^{2h} = \tau, \quad (\text{R1})$$

on the specified positive branch. The complete inverse and derivative formulas for this branch were proved in Sections 2–3 of `summation_realization_audit.md`; in particular

$$|\partial_z^a \partial_t^b q| \leq C_{a,b} q^{1-aD-b}. \quad (\text{R2})$$

No value of q , Q , h , A , D , viscosity or physical coordinate is set to one in this calculation.

Fix positive constants $0 < a_- < a_+ < \infty$ covering the source annulus and the interior profile supports. Let $\mathcal{E}$ be the vector space of smooth scalar fields $F(r, z, t)$ supported in

$$a_- \sqrt{q(z, t)} \leq r \leq a_+ \sqrt{q(z, t)}. \quad (\text{R3})$$

Smooth zero extensions are part of this space. Thus all derivatives vanish at either boundary when continued outside; on a compact subset of Ω , the radial supports lie in one compact interval bounded away from zero. The source uses $e = 2$ for its angular residual and $e = 1$ for its axial residual. The constructions below are given separately for each $e \in \{1, 2\}$, with that original weight retained.

Choose the source interior profile $\widehat{b}_e \in C_c^\infty((a_-, a_+))$ with

$$\int_0^\infty x^e \widehat{b}_e(x) dx = 1, \quad b_e(r, z, t) = q^{-(e+1)/2} \widehat{b}_e(r/\sqrt{q}). \quad (\text{R4})$$

Such a profile exists explicitly: take a nonnegative nonzero smooth bump η supported inside (a_-, a_+) , keep its positive number $d_e = \int x^e \eta(x) dx$, and set $\widehat{b}_e = \eta/d_e$. This division retains the complete chosen profile and its moment factor. With $x = r/\sqrt{q}$, the exact moment calculation is

$$\int_0^\infty r^e b_e(r, z, t) dr = q^{e/2} q^{-(e+1)/2} q^{1/2} \int_0^\infty x^e \widehat{b}_e(x) dx = 1. \quad (\text{R5})$$

47.2 2. Exact moment complement and radial inverse

Define the moment map, its section, and its complementary projection by

$$M_e F(z, t) = \int_0^\infty r^e F(r, z, t) dr, \quad S_e m = b_e m, \quad P_e F = F - b_e M_e F. \quad (\text{R6})$$

The parameter function m belongs to $C^\infty(\Omega)$. Differentiation under the integral is justified on every compact parameter set by the common radial support described after (R3). Equation (R5) proves

$$M_e S_e = I, \quad P_e^2 = P_e, \quad \text{ran } P_e = \ker M_e, \quad \ker P_e = \text{ran } S_e. \quad (\text{R7})$$

For example $M_e P_e F = M_e F - M_e F = 0$; if $M_e F = 0$, then $P_e F = F$; if $P_e F = 0$, then $F = S_e M_e F$. These establish the displayed range and kernel identities in both directions.

The compact stress operator and radial differential operator are

$$\mathcal{T}_e F(r, z, t) = -r^{-e} \int_0^r s^e (P_e F)(s, z, t) ds, \quad \mathcal{D}_e \sigma = \partial_r \sigma + \frac{e}{r} \sigma. \quad (\text{R8})$$

For brevity write $H = P_e F$, $I(r) = \int_0^r s^e H(s) ds$, $\sigma = \mathcal{T}_e F = -r^{-e} I$. Differentiating with all signs gives

$$\partial_r \sigma = e r^{-e-1} I - H, \quad \mathcal{D}_e \sigma = e r^{-e-1} I - H - e r^{-e-1} I = -H = -F + b_e M_e F. \quad (\text{R9})$$

Below the annulus the integral is zero. Above the annulus it equals $M_e H = 0$. Thus σ has exactly the common support property (R3) and is smooth across both boundaries. This support statement uses the full moment subtraction; it would be false for an arbitrary uncorrected source.

Conversely every $\sigma \in \mathcal{E}$ satisfies

$$M_e(\mathcal{D}_e \sigma) = \int_0^\infty \partial_r (r^e \sigma) dr = 0. \quad (\text{R10})$$

For such a stress,

$$\mathcal{T}_e(-\mathcal{D}_e \sigma) = r^{-e} \int_0^r \partial_s (s^e \sigma(s)) ds = \sigma(r), \quad (\text{R11})$$

because $s^e \sigma(s)$ vanishes near zero. Thus $-\mathcal{D}_e : \mathcal{E} \rightarrow \ker M_e$ is one-to-one and onto, with inverse the restriction of $\mathcal{T}_e$. Injectivity also follows directly: the homogeneous equation gives $\sigma = c(z, t) r^{-e}$, and compact radial support forces $c = 0$.

The complete map retaining the scalar moment is therefore the exact bijection

$$\mathfrak{T}_e : \mathcal{E} \longrightarrow \mathcal{E} \oplus C^\infty(\Omega), \quad F \longmapsto (\mathcal{T}_e F, M_e F), \quad \mathfrak{T}_e^{-1}(\sigma, m) = -\mathcal{D}_e \sigma + b_e m. \quad (\text{R12})$$

To check surjectivity and both inverse identities, apply (R10) and (R5) to the last expression, obtaining moment m and moment-zero part $-\mathcal{D}_e \sigma$. Equation (R11) returns σ . Conversely (R9) reconstructs the original F . Hence a source with nonzero moment has not been replaced by its moment-zero part; its complete missing coordinate is the displayed scalar $M_e F$.

47.3 3. Every moving-q profile derivative

Put $\alpha_e = (e + 1)/2$, $\xi = r q^{-1/2}$, and $\mathcal{E} = -\xi \partial_\xi / 2$. For $k \geq 0$, first differentiate in the original physical radial coordinate:

$$\partial_r^k b_e = q^{-\beta} \widehat{b}_e^{(k)}(\xi), \quad \beta = \alpha_e + k/2. \quad (\text{R13})$$

At fixed r , every derivative with respect to the scalar q obeys the exact identity

$$\frac{d^n}{dq^n} [q^{-\beta} f(r q^{-1/2})] = q^{-\beta-n} \left[\prod_{\ell=0}^{n-1} (\mathcal{E} - \beta - \ell) f \right] (\xi). \quad (\text{R14})$$

For $n = 0$ the product is the identity. For the induction step, differentiating $q^{-\beta-n}$ contributes $-\beta - n$, and differentiating the profile contributes $\mathcal{E}$, both with one additional factor q^{-1} . Their sum is the next factor in (R14), proving the formula at every order.

For nonnegative integers a, b , let $\mathcal{P}_{a,b}$ be the finite set of families of integers $n_{u,v} \geq 0$, indexed by $u, v \geq 0, u + v > 0$, satisfying

$$\sum_{u,v} un_{u,v} = a, \quad \sum_{u,v} vn_{u,v} = b, \quad n = \sum_{u,v} n_{u,v}. \quad (\text{R15})$$

Only $u \leq a, v \leq b$ can occur. For $a = b = 0$, this set has the single empty family with $n = 0$. The full multivariate chain rule is

$$\begin{aligned} \partial_r^k \partial_z^a \partial_t^b b_e &= a!b! \sum_{\mathcal{P}_{a,b}} q^{-\beta-n} \left[\prod_{\ell=0}^{n-1} (\mathcal{E} - \beta - \ell) \widehat{b}_e^{(k)} \right] (\xi) \\ &\times \prod_{u,v} \frac{(\partial_z^u \partial_t^v q)^{n_{u,v}}}{n_{u,v}! (u!v!)^{n_{u,v}}}. \end{aligned} \quad (\text{R16})$$

For justification, distribute the labeled a axial and b time derivatives into nonempty blocks striking q . A block with u axial and v time derivatives gives $\partial_z^u \partial_t^v q$; unordered repetitions give the denominator $n_{u,v}!$, and permutations inside each block give $(u!v!)^{n_{u,v}}$. The outside labels give $a!b!$, and the number of blocks is the order n of the scalar derivative in (R14). This enumerates every product-rule term exactly once.

Combining (R2), (R15) and (R16) preserves the exact power

$$q^{-\beta-n} \prod_{u,v} q^{(1-uD-v)n_{u,v}} = q^{-\alpha_e - k/2 - aD - b}. \quad (\text{R17})$$

An explicit finite derivative constant is

$$\mathfrak{B}_{k,a,b} = a!b! \sum_{\mathcal{P}_{a,b}} \left\| \prod_{\ell=0}^{n-1} (\mathcal{E} - \beta - \ell) \widehat{b}_e^{(k)} \right\|_{\infty} \prod_{u,v} \frac{C_{u,v}^{n_{u,v}}}{n_{u,v}! (u!v!)^{n_{u,v}}}. \quad (\text{R18})$$

The profile norm is on its original compact interval; all derivatives remain supported there. Therefore

$$|\partial_r^k \partial_z^a \partial_t^b b_e| \leq \mathfrak{B}_{k,a,b} q^{-\alpha_e - k/2 - aD - b}. \quad (\text{R19})$$

No moving-boundary term or derivative of q has been dropped.

47.4 4. Full stress derivatives and finite operator estimates

The moment derivative formula and complete derivative of the projected source are

$$\partial_z^a \partial_t^b M_e F = \int_0^\infty r^e \partial_z^a \partial_t^b F dr, \quad (\text{R20})$$

$$\partial_r^k \partial_z^a \partial_t^b H = \partial_r^k \partial_z^a \partial_t^b F - \sum_{a'=0}^a \sum_{b'=0}^b \binom{a}{a'} \binom{b}{b'} (\partial_r^k \partial_z^{a'} \partial_t^{b'} b_e) (\partial_z^{a-a'} \partial_t^{b-b'} M_e F). \quad (\text{R21})$$

The integration limits in (R20) are fixed. The smooth supported extension, rather than a discarded moving-boundary derivative, justifies differentiation.

Let $(e)_k = e(e+1) \cdots (e+k-1)$, with $(e)_0 = 1$, and $e_\ell = e(e-1) \cdots (e-\ell+1)$, with $e_0 = 1$ and value zero when $\ell > e$. Put $I_{a,b}(r) = \int_0^r s^e \partial_z^a \partial_t^b H(s) ds$. The complete radial product rule in (R8) gives

$$\begin{aligned} \partial_r^k \partial_z^a \partial_t^b \sigma &= -(-1)^k (e)_k r^{-e-k} I_{a,b}(r) \\ &- \sum_{j=1}^k \binom{k}{j} (-1)^{k-j} (e)_{k-j} r^{-e-k+j} \sum_{\ell=0}^{j-1} \binom{j-1}{\ell} e_\ell r^{e-\ell} \partial_r^{j-1-\ell} \partial_z^a \partial_t^b H. \end{aligned} \quad (\text{R22})$$

Indeed derivatives of r^{-e} give the first signed rising factors, while for $j \geq 1$, $\partial_r^j I_{a,b} = \partial_r^{j-1} (r^e \partial_z^a \partial_t^b H)$, giving the inner finite sum. When $k = 0$ the second sum is empty. Thus (R22) includes every derivative of the integral and every exterior weight.

Here is a quantitative operator estimate preserving general exponents and logarithms. Write

$$\Lambda(q) = 1 + |\log q|, \quad c_e = \frac{a_+^{e+1} - a_-^{e+1}}{e+1}, \quad (\text{R23})$$

and for real λ, P and integer $m \geq 0$ define the actual weighted seminorm

$$N_{\lambda, P; m}(F) = \max_{k+a+b \leq m} \sup_{r, z, t} q^{-\lambda+k/2+aD+b} \Lambda(q)^{-P} |\partial_r^k \partial_z^a \partial_t^b F|. \quad (\text{R24})$$

For (R25)–(R29), require $a+b \leq m$ when no radial derivative index is present, and $k+a+b \leq m$ otherwise. All inequalities below are valid also as inequalities with an infinite right side. For finite $N = N_{\lambda, P; m}(F)$, the exact annular integral $\int_{a-\sqrt{q}}^{a+\sqrt{q}} r^e dr = c_e q^{\alpha_e}$ gives

$$|\partial_z^a \partial_t^b M_e F| \leq c_e N q^{\lambda+\alpha_e-aD-b} \Lambda(q)^P. \quad (\text{R25})$$

Define the finite constants

$$\mathfrak{H}_{k, a, b} = 1 + c_e \sum_{a'=0}^a \sum_{b'=0}^b \binom{a}{a'} \binom{b}{b'} \mathfrak{B}_{k, a', b'}. \quad (\text{R26})$$

Equations (R19), (R21) and (R25) then give

$$|\partial_r^k \partial_z^a \partial_t^b H| \leq \mathfrak{H}_{k, a, b} N q^{\lambda-k/2-aD-b} \Lambda(q)^P. \quad (\text{R27})$$

The two profile/moment powers cancel their α_e exactly. No logarithmic exponent is changed, since q is constant during the radial integration and derivatives of the complete source already occur in (R24).

On the support (R3), $r \geq a_- \sqrt{q}$. Apply (R27) to (R22), retaining each power, and define

$$\begin{aligned} \mathfrak{S}_{k, a, b} &= (e)_k a_-^{-e-k} c_e \mathfrak{H}_{0, a, b} \\ &+ \sum_{j=1}^k \sum_{\ell=0}^{j-1} \binom{k}{j} (e)_{k-j} \binom{j-1}{\ell} e_{\ell} a_-^{j-k-\ell} \mathfrak{H}_{j-1-\ell, a, b}. \end{aligned} \quad (\text{R28})$$

Every exponent $j-k-\ell$ is nonpositive, so the lower radial bound gives the displayed upper factor. The integral term has power $q^{\lambda+\alpha_e-(e+k)/2-aD-b} = q^{\lambda+1/2-k/2-aD-b}$. Every nonintegral term has power $q^{(j-k-\ell)/2} q^{\lambda-(j-1-\ell)/2-aD-b}$, which is the same power. Thus

$$|\partial_r^k \partial_z^a \partial_t^b \mathcal{T}_e F| \leq \mathfrak{S}_{k, a, b} N q^{\lambda+1/2-k/2-aD-b} \Lambda(q)^P, \quad (\text{R29})$$

and outside the annulus both sides of the differentiated field are zero. Taking the finite maximum proves

$$N_{\lambda+1/2, P; m}(\mathcal{T}_e F) \leq \max_{k+a+b \leq m} \mathfrak{S}_{k, a, b} N_{\lambda, P; m}(F). \quad (\text{R30})$$

Equations (R18), (R26), and (R28) are explicit formulas for every constant. They depend on the original annulus, profile and derivatives of q , not on a correction-stage amplitude. This proves the half-power radial integration gain and full mixed-derivative continuity, without deriving any unavailable estimate for the source F itself.

47.5 5. Full symmetric tensor and all three force components

Use the original right-handed cylindrical frame

$$e_r = (x_1/r, x_2/r, 0), \quad e_{\theta} = (-x_2/r, x_1/r, 0), \quad e_z = (0, 0, 1). \quad (\text{R31})$$

For the source pair (F_2, F_1) , put $\sigma_2 = \mathcal{T}_2 F_2$, $\sigma_1 = \mathcal{T}_1 F_1$. The symmetric tensor with precisely these radial–tangential entries is

$$\mathbb{T} = \sigma_2(e_r \otimes e_{\theta} + e_{\theta} \otimes e_r) + \sigma_1(e_r \otimes e_z + e_z \otimes e_r). \quad (\text{R32})$$

The Cartesian divergence convention is $(\text{div } \mathbb{T})_i = \sum_j \partial_{x_j} \mathbb{T}_{ij}$. For any smooth vectors a, b , the product rule gives $\text{div}(a \otimes b) = (b \cdot \nabla)a + a \text{div } b$. Here $\text{div } e_r = 1/r$, $\text{div } e_{\theta} = \text{div } e_z = 0$, $(e_{\theta} \cdot \nabla)e_r = e_{\theta}/r$, and radial or axial derivatives of both planar unit vectors are zero. These identities follow by differentiating (R31) component by component. The scalar stresses depend only on (r, z, t) , so their angular derivatives vanish. Therefore the four tensor terms have divergences

$$\begin{aligned} \text{div}(\sigma_2 e_r \otimes e_{\theta}) &= (\sigma_2/r) e_{\theta}, \\ \text{div}(\sigma_2 e_{\theta} \otimes e_r) &= (\partial_r \sigma_2 + \sigma_2/r) e_{\theta}, \\ \text{div}(\sigma_1 e_r \otimes e_z) &= (\partial_z \sigma_1) e_r, \\ \text{div}(\sigma_1 e_z \otimes e_r) &= (\partial_r \sigma_1 + \sigma_1/r) e_z. \end{aligned} \quad (\text{R33})$$

Adding and applying (R9) proves the full physical force identity

$$\operatorname{div} \mathbb{T} = (\partial_z \sigma_1) e_r + (-F_2 + b_2 M_2 F_2) e_\theta + (-F_1 + b_1 M_1 F_1) e_z. \quad (\text{R34})$$

In particular the radial force $\partial_z \sigma_1$ is present. The tensor and force extend smoothly across the axis because they vanish in a neighborhood of it at each point of the open parameter domain. This proves a full tensor-to-force map, not just the two selected tangential rows.

47.6 6. Exact physical-to-chart correspondence with Q fixed

Retain a particular original source chart with $Q > 0$ fixed while differentiating:

$$r = \sqrt{Q}R, \quad z = Q^D Z, \quad t = 1 - QT, \quad q_Q(Z, T) = q(Q^D Z, 1 - QT), \quad \tilde{q} = q_Q/Q. \quad (\text{R35})$$

Its inverse is $(R, Z, T) = (r/\sqrt{Q}, z/Q^D, (1-t)/Q)$, and its Jacobian determinant in this order is $-Q^{3/2+D}$. For these three coordinates the absolute measure is $dr dz dt = Q^{3/2+D} dR dZ dT$. With the unchanged angular coordinate retained, the full Cartesian spacetime measure has the additional radial factor: $dx_1 dx_2 dz dt = Q^{2+D} R dR d\theta dZ dT$. The time derivative is $\partial_T = -Q\partial_t$, and the spatial derivatives are $\partial_R = \sqrt{Q}\partial_r$, $\partial_Z = Q^D\partial_z$. These signs and factors are retained.

The source's chart representatives are

$$E_e(R, Z, T) = Q^{2A+1/2} F_e(\sqrt{Q}R, Q^D Z, 1 - QT), \quad \Sigma_e(R, Z, T) = Q^{2A} \sigma_e(\sqrt{Q}R, Q^D Z, 1 - QT), \quad (\text{R36})$$

and define the complete chart moment and profile by

$$\mathcal{M}_e = \int_0^\infty R^e E_e(R, Z, T) dR, \quad \tilde{b}_e = \tilde{q}^{-\alpha_e} \hat{b}_e(R/\sqrt{\tilde{q}}). \quad (\text{R37})$$

Substitution $r = \sqrt{Q}R$, with all powers shown, gives

$$M_e F_e = Q^{(e+1)/2} Q^{-2A-1/2} \mathcal{M}_e = Q^{e/2-2A} \mathcal{M}_e, \quad b_e \circ \Psi_Q = Q^{-\alpha_e} \tilde{b}_e, \quad (\text{R38})$$

where Ψ_Q is the map (R35). Hence

$$Q^{2A+1/2} (b_e M_e F_e) \circ \Psi_Q = Q^{2A+1/2-\alpha_e+e/2-2A} \tilde{b}_e \mathcal{M}_e = \tilde{b}_e \mathcal{M}_e. \quad (\text{R39})$$

The exponent vanishes because $\alpha_e = (e+1)/2$; the original factors are displayed rather than discarded.

Applying the same substitution directly to the lower-endpoint integral in (R8) gives

$$\begin{aligned} \Sigma_e(R) &= -Q^{2A} Q^{-e/2} R^{-e} Q^{(e+1)/2} Q^{-2A-1/2} \int_0^R s^e (E_e(s) - \tilde{b}_e(s) \mathcal{M}_e) ds \\ &= -R^{-e} \int_0^R s^e (E_e(s) - \tilde{b}_e(s) \mathcal{M}_e) ds, \\ (\partial_R + e/R) \Sigma_e &= -E_e + \tilde{b}_e \mathcal{M}_e. \end{aligned} \quad (\text{R40})$$

Thus the moment-complement diagram commutes exactly for the same physical source and its representative, with the full inverse maps in (R12), (R35) and (R36). It does not identify arbitrary independently chosen chart sources.

The previously retained radial force transforms as

$$Q^{2A+1/2} (\partial_z \sigma_1) \circ \Psi_Q = Q^{2A+1/2} Q^{-2A-D} \partial_Z \Sigma_1 = Q^{1/2-D} \partial_Z \Sigma_1 = Q^h \partial_Z \Sigma_1. \quad (\text{R41})$$

This is exactly the source parameter $\varepsilon = Q^h$ multiplying that axial derivative; it accounts for the additional radial contribution's axial gain. In particular it is not a zero force component.

For every k, a, b , differentiation of the same fixed chart gives the exact general identity

$$\partial_R^k \partial_Z^a \partial_T^b \Sigma_e = (-1)^b Q^{2A+k/2+aD+b} (\partial_r^k \partial_z^a \partial_t^b \sigma_e) \circ \Psi_Q, \quad (\text{R42})$$

with an analogous formula replacing $2A$ by $2A + 1/2$ for E_e . Thus all mixed derivative statements and their original time orientation have an explicit inverse correspondence.

47.7 7. The original flat edge weight is preserved by the radial integral

The source class also retains its flat radial-edge weight, which is stronger than the power/logarithm estimates alone. Equations (6.23) and the radial-integral paragraph in Proposition 9.3 give the original objects

$$X = \frac{r^2}{2q}, \quad X_a < X < X_b, \quad L = \log(X_b/X_a) > 0, \quad (\text{R43})$$

$$\zeta(X) = \exp\left(-\frac{a_a}{\log^2(X/X_a)} - \frac{a_b}{\log^2(X_b/X)}\right), \quad \delta(X) = \min\{1, \log(X/X_a), \log(X_b/X)\}, \quad a_a, a_b > 0. \quad (\text{R44})$$

The two positive constants are retained separately. The physical source shell corresponds to $a_- = \sqrt{2X_a}$, $a_+ = \sqrt{2X_b}$ in the preceding formulas. The interior moment profile has support a positive logarithmic distance from either endpoint.

First prove the exact scalar estimate used at an edge. For $a > 0$, $M \geq 0$, and $d > 0$, substitution $v = a/s^2$ gives

$$\int_0^d s^{-M} e^{-a/s^2} ds = \frac{1}{2} a^{(1-M)/2} \int_{a/d^2}^\infty v^{(M-3)/2} e^{-v} dv. \quad (\text{R45})$$

Put $p = (M-3)/2$ and $x = a/d^2$. If $p \leq 0$, then $v^p \leq x^p$ on $v \geq x$, so the right side is at most $d^{3-M} e^{-a/d^2} / (2a)$. If $p > 0$ and $x \geq 2p$, the elementary inequality $\log(v/x) \leq (v-x)/x$ implies

$$v^p e^{-v} \leq x^p e^{-x} e^{-(v-x)/2}, \quad \int_x^\infty v^p e^{-v} dv \leq 2x^p e^{-x}. \quad (\text{R46})$$

Hence in both cases the explicit bound

$$\int_0^d s^{-M} e^{-a/s^2} ds \leq \frac{1}{a} d^{3-M} e^{-a/d^2} \quad (\text{R47})$$

holds for $0 < d \leq \min\{1, \sqrt{a/\max\{1, M-3\}}\}$. The upper restriction is sufficient rather than an asserted largest possible interval.

Every scalar derivative can also be retained exactly. Let $P_0(y) = 1$ and

$$P_{k+1}(y) = 2ay^3 P_k(y) - y^2 P'_k(y). \quad (\text{R48})$$

Induction with $y = 1/s$ gives

$$\frac{d^k}{ds^k} e^{-a/s^2} = e^{-a/s^2} P_k(1/s), \quad \deg P_k \leq 3k. \quad (\text{R49})$$

Thus on $0 < s \leq 1$, the bound is $c_{k,a} s^{-3k} e^{-a/s^2}$, where $c_{k,a}$ is the sum of the absolute values of the finitely many coefficients of the explicitly recursive polynomial P_k . This proves the source's all-order scalar derivative estimate with every parameter retained.

For the left edge set $s = \log(X/X_a)$. On $0 \leq s \leq d_0 < L/2$,

$$\zeta(X_a e^s) = e^{-a_a/s^2} g_a(s), \quad g_a(s) = e^{-a_b/(L-s)^2} > 0. \quad (\text{R50})$$

The function g_a , all of its fixed-order derivatives, and its reciprocal are bounded on this compact interval. Also $dX = X_a e^s ds$, so each original radial weight becomes a smooth bounded positive factor there. The right edge instead uses $s = \log(X_b/X)$, with the roles of a_a , a_b interchanged and $dX = -X_b e^{-s} ds$. The reversed integration limits absorb this last minus sign when bounding an absolute integral. For each fixed exponent M and fixed derivative order being estimated, choose $d_0 > 0$ with $d_0 < L/2$ and

$$d_0 \leq \min\left\{1, \sqrt{\frac{a_a}{\max\{1, M-3\}}}, \sqrt{\frac{a_b}{\max\{1, M-3\}}}\right\}, \quad (\text{R50a})$$

and additionally smaller than the positive logarithmic distance of the moment-profile support from either endpoint. These choices give $\delta(X) = s$ on both edge intervals and permit (R47) with its displayed constant on the whole chosen interval. The number d_0 may depend on the fixed M and derivative order; no one positive edge interval is asserted to meet these restrictions for unbounded M . The profile and all its derivatives vanish on these intervals, so $H = F$ there, including every mixed derivative in (R21).

The original radial integral, in the exact variable X , is

$$\sigma(r, z, t) = -\sqrt{q/2} X^{-e/2} \int_{X_a}^X Y^{(e-1)/2} H(\sqrt{2qY}, z, t) dY. \quad (\text{R51})$$

Indeed $r^{-e} = (2qX)^{-e/2}$ and $s^e ds = 2^{(e-1)/2} q^{(e+1)/2} Y^{(e-1)/2} dY$; their product is the displayed $\sqrt{q/2}$ factor. At the right edge, the moment-zero identity makes (R51) exactly

$$\sigma(r, z, t) = +\sqrt{q/2} X^{-e/2} \int_X^{X_b} Y^{(e-1)/2} H(\sqrt{2qY}, z, t) dY. \quad (\text{R52})$$

Both signs and endpoints are fixed by the same inverse (R8).

Suppose a fixed original derivative of F is bounded by its actual parameter factor times $\zeta(X)\delta(X)^{-M}$; the parameter factor is constant during the radial integration and may contain the original q , ε , S_* and logarithmic powers. Apply (R47), (R50), and the bounded radial factors to the left representation (R51) or right representation (R52). On either edge, the integral is bounded by that same parameter factor times a fixed constant and $\zeta(X)\delta(X)^{3-M}$, with the explicit additional $\sqrt{q/2}$ from (R51)–(R52). For example, the left-edge constant can be taken as

$$\frac{1}{a_a} \left(\sup_{0 \leq s \leq d_0} (X_a e^s)^{(e+1)/2} g_a(s) \right) \left(\inf_{0 \leq s \leq d_0} g_a(s) \right)^{-1} \sup_{X_a \leq X \leq X_a e^{d_0}} X^{-e/2}, \quad (\text{R53})$$

apart from the already displayed $\sqrt{q/2}$ and the source derivative bound. The right constant has the analogous literal $X_b e^{-s}$, a_b , and right-edge positive factor. Both are finite and keep the original endpoints and weights.

Every mixed derivative of σ is already exactly (R22), with (R20)–(R21) for its source. The integral term therefore has the preceding forward/backward edge estimate for its actual differentiated source. The nonintegral terms in (R22) are radial powers times differentiated sources; these preserve ζ and enlarge only a finite power of δ^{-1} . No derivatives of the nonsmooth capped minimum δ are taken: it is only an estimate weight. In the interior, ζ has a positive lower bound, and the general compact radial estimates (R25)–(R30) apply. Thus at each fixed derivative order the full original flat weight is retained, with a finite edge-power loss and the same source amplitude exponent.

To retain uniformity over the source's actual bands, use its Section 6.1 chart property: q is comparable to the fixed band scale Q . Write the original fixed comparison bounds as $0 < c_- \leq q/Q \leq c_+ < \infty$. No numerical value for either comparison bound is introduced. With the same chart and the same $\tilde{q} = (q \circ \Psi_Q)/Q$ as in (R37), differentiation gives

$$\partial_Z^a \partial_T^b \tilde{q} = (-1)^b Q^{aD+b-1} (\partial_Z^a \partial_T^b q) \circ \Psi_Q, \quad |\partial_Z^a \partial_T^b \tilde{q}| \leq C_{a,b} \tilde{q}^{1-aD-b} \leq C_{a,b} \max\{c_-^{1-aD-b}, c_+^{1-aD-b}\}. \quad (\text{R54})$$

Here the first inequality is exactly (R2) with the displayed powers of Q multiplied out. The original physical factor in (R51)–(R52), together with the source and stress representatives (R36), is exactly

$$Q^{2A} \sqrt{q/2} Q^{-2A-1/2} = \sqrt{(q/Q)/2} = \sqrt{\tilde{q}/2}, \quad X = \frac{R^2}{2\tilde{q}}. \quad (\text{R55})$$

In particular the chart shell has inner radius at least $\sqrt{2c_- X_a} > 0$ and outer radius at most $\sqrt{2c_+ X_b}$. The original profile variable and its first derivatives are

$$\partial_R X = R/\tilde{q}, \quad \partial_Z X = -X \partial_Z \tilde{q}/\tilde{q}, \quad \partial_T X = -X \partial_T \tilde{q}/\tilde{q}. \quad (\text{R56})$$

For completeness, all higher inverse-factor derivatives follow from differentiating $\tilde{q} \tilde{q}^{-1} = 1$. For a nonzero two-index $\beta = (a, b)$, the exact recurrence is

$$\partial^\beta (\tilde{q}^{-1}) = -\tilde{q}^{-1} \sum_{0 < \gamma \leq \beta} \binom{\beta}{\gamma} (\partial^\gamma \tilde{q}) \partial^{\beta-\gamma} (\tilde{q}^{-1}), \quad \partial^\beta = \partial_Z^a \partial_T^b. \quad (\text{R57})$$

The initial inverse is bounded by c_-^{-1} . Induction using (R54) and this finite sum bounds every fixed-order inverse derivative by constants independent of Q . Applying the exact product rule to $R^2 \tilde{q}^{-1}/2$ proves the same assertion for every mixed derivative of X on the original shell. The same argument applies to every fixed real power of $\tilde{q}$, by the scalar power derivative formula and the finite composition rule already proved

in (R16)–(R19). Thus every coefficient in (R20)–(R30) and in the edge estimate has a uniform chart bound with the original comparison constants retained.

The source keeps Q , $\varepsilon = Q^h$, and S_* fixed during chart differentiation and radial integration. Applying the proved operator formulas directly to (R40), with its literal ratio $\tilde{q}$, therefore preserves the incoming power of ε and every original S_* derivative factor. It retains ζ with only the finite edge-power losses proved above. The selected averaged source is independent of the auxiliary torus coordinates; the profile and the radial operator are also independent of them. All positive-order auxiliary derivatives of this reconstruction are consequently exactly zero.

This proves the radial integration part of the source’s $\mathcal{M}_\alpha$ -class preservation, including its uniform band factors and the edge behavior needed before division by the fixed primary amplitude. The source’s auxiliary covariance and inter-chart descent are further properties of its input and chosen amplitude map, not consequences of an endpoint integral estimate. They remain separately identified below.

47.8 8. Exact role in the source correction and remaining dependencies

The source defines its physical averaged residuals by $F_2 = Q^{-2A-1/2}\langle E_\theta \rangle_Y$ and $F_1 = Q^{-2A-1/2}\langle E_z \rangle_Y$. Here E_θ, E_z are its original chart fields; the E_e of (R36) are exactly their displayed averaged representatives. The averaging is an actual projection onto the auxiliary-coordinate-independent fields. The radial operators above then act on those physical functions; they do not treat an auxiliary variable as a physical angle or differentiate it as a spatial coordinate.

For the given averaged source, (R9) cancels $F_e - b_e M_e F_e$ and retains $b_e M_e F_e$ as an exact residual coordinate. The higher-order estimate for that retained term in the source uses its separate moment estimate (9.12). Equations (R25), (R38) and (R39) explain its full scaling, but do not prove (9.12) from an unavailable source bound.

The source’s signed amplitude operator prescribes the radial–tangential entries of the symmetrized cross covariance. Its complete covariance has further entries, curl terms and the correction’s quadratic self-interaction. The symmetric tensor (R32) records exactly the force contribution of the prescribed two entries, including its radial component. It does not assert equality of (R32) with the entire newly generated covariance. Those other entries are retained by the full source expansion (9.13), to which the separate four-step gain audit applies.

Consequently this proof establishes a complete compact radial reconstruction, moment-preserving inverse, all original moving- q derivatives and quantitative constants, full tensor divergence and chart correspondence. The source’s actual incoming residual estimates, cross-chart agreement of its independently constructed averaged fields, amplitude covariance identity and later nonlinear residual bounds remain their respective upstream calculations. No source-existence or kernel certificate is inferred. No Lean, Lake, Elan, new publication or parent-source modification was performed for this note.

48 Four-step correction-cycle gain and invariant audit

48.1 Finding and exact scope

The displayed exponent arithmetic in Proposition 9.6 closes. With the original $\kappa_s = 10^{-5}$, the retained estimates after the full cycle give a wave-residual gain of at least 0.39999, a full tangential-mean gain of 0.17, and a defect gain of 0.89996. Each exceeds the proposition’s required $1/10$. The exact covariance, mean-residual and nonlinear defect identities checked below retain all cross products, self-products, pressure changes and the axial reconstruction remainder.

These calculations verify the numerical closure and the algebraic identities used by this particular cycle. They do not, by themselves, construct the pulse inverse, prove the earlier amplitude-class estimates, prove label separation or uniform flatness over infinitely many labels, or certify the whole Navier–Stokes construction. A class membership is attributed below to the source result that supplies it. The conclusion of this audit is not obtained by assuming an unproved additional estimate and calling it a theorem.

The exact source is the frozen 166-page PDF `downloaded_public_release/navier-stokes.pdf` under `programme/`.

PDF SHA256: 0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f.

The full Proposition 9.6 statement and proof were read from `downloaded_public_release/current_pdf_extracted.txt`, lines 5613–5802, and its main formulas were visually checked on PDF pages 108–111. The statement begins on page 107. Additional directly read source:

Source	Physical pages / layout-text lines	Purpose
Definition 6.4 and (6.23)–(6.24)	pp.69–70; 3654–3682	Full mean class, shell-edge weight and smooth zero extension; local base multiplier correction.
Proposition 8.1 and exact equations (8.3)	p.89; 4641–4680	Complete mean balances and pressure source.
Definitions (8.15), Proposition 8.4 and (8.16)	p.94; 4909–4942	Weighted moments, pressure contribution and exact axial product derivative.
Lemma 8.6 and (8.19)–(8.22)	pp.95–96; 4987–5032	Mean inverse, prescribed versus actual axial increment, flat remainder.
Lemma 8.7, (8.24)–(8.25)	p.97; 5061–5089	Five linear rows and their declared supported inverse.
Lemma 8.8 and (8.26)–(8.27)	pp.98–99; 5151–5208	Exact nonlinear remainder after the defect update.
Proposition 9.1 statement and selected proof bounds	pp.101–102; 5284–5350	Linear wave remainder, curl correction and pulse-cutoff tails.
Lemma 9.2, full statement and proof	pp.102–103; 5355–5374	Wave-mean and same-label wave-wave exponents.
Proposition 9.3, statement and relevant full residual/support proof	pp.103–105; 5375–5518	Supported source closure, actual mean equations, finite-stage flat remainder.
Definition 9.4 and Proposition 9.5 conclusion	pp.106–107; 5527–5562; 5593–5612	Exact state, initialization and preserved low-order classes.
Lemmas 9.7–9.8 and (9.15)–(9.19)	pp.111–114; 5804–5937	Common-domain and physical derivative interface to (9.18).

The coordinating task has written the compact radial stress inverse and its moving- q physical rescaling in the separate `radial_stress_reconstruction_audit.md`, including its flat-edge preservation calculation. This note records their exact interface and sign but does not duplicate that inverse construction. No Lean, Lake or Elan was run; no source PDF or parent manuscript was modified.

48.2 1. Original state, parameters and class interpretation

Keep the original fixed quantities

$$0 < h < \frac{1}{100}, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad Q = 2^{-\ell}, \quad \varepsilon = Q^h, \quad \kappa_s = 10^{-5} = \frac{1}{100000}. \quad (\text{C1})$$

The finite stage is $j \in \mathbb{N}_0$, with

$$\sigma_j = \frac{1}{5} + \frac{j}{10}, \quad B = B_j = \frac{1}{2} + \sigma_j = \frac{7}{10} + \frac{j}{10}, \quad C^* = C_j^* = 1 + \sigma_j = B + \frac{1}{2}, \quad H = C^* - 2\kappa_s. \quad (\text{C2})$$

In particular $B \geq 0.7$, $C^* \geq 1.2$, $H \geq 1.19998$. Decimal constants such as 0.68, 0.9 and 0.17 below are the exact finite decimals used in the source, not rounded estimates of a different number.

The original class symbols $\mathcal{W}_\alpha, \mathcal{M}_\alpha, \mathcal{S}_\alpha$ include more than a power of ε : they include amplitude-derivative bounds, logarithmic-scale factors, edge weights, supports and extension conditions. On the fixed range $0 < \varepsilon \leq 1$, an already proved bound with exponent $\alpha' \geq \alpha$ implies the corresponding bound with exponent α , with the same support and derivative properties, because $\varepsilon^{\alpha'} \leq \varepsilon^\alpha$. This elementary comparison is the only class inclusion inferred purely from exponent arithmetic. The product, inverse and support rules themselves remain the earlier analytical claims identified below.

The normalized state is

$$u_* = (b + \beta, V + v, G + \gamma) + w, \quad p_* = p_{B,*} + p_m + p_w, \quad \langle w \rangle_\theta = \langle p_w \rangle_\theta = 0. \quad (\text{C3})$$

Its residual classes are $\mathcal{G}_{*,\text{wave}}^{[j]} \in \mathcal{W}_B$, $E_\theta, E_z \in \mathcal{M}_{C^*}$, and $(\mathcal{P}, J_\theta, J_z) \in \mathcal{S}_{C^*}$. Its cumulative correction bounds are

$$w \in \mathcal{W}_{1/2}, \quad w - w_0^{\text{tan}} \in \mathcal{W}_{0.68}, \quad v, \gamma, p_m \in \mathcal{M}_{0.9}, \quad \beta \in \mathcal{M}_{1.9}. \quad (\text{C4})$$

The two exact constraints are

$$M_\theta := \int_0^\infty R^2 \langle v \rangle_Y dR = 0, \quad M_z := \int_0^\infty R \langle \gamma \rangle_Y dR = 0. \quad (\text{C5})$$

These are not assertions that the three defects $\mathcal{P}, J_\theta, J_z$ vanish. The state allows those defects to be nonzero with the stated increasing decay order.

Angular average $\langle \cdot \rangle_\theta$ holds the auxiliary torus coordinate fixed; auxiliary average $\langle \cdot \rangle_Y$ integrates that independent coordinate before physical evaluation. Their composition is retained explicitly in every covariance below. No assertion that either operation commutes with physical phase evaluation is used.

48.3 2. Full residual and exact recomputation rule

At every update use the actual fields, including all cutoff-curl and mean-reconstruction terms. With $\Delta_0 = D_r^2 + R^{-1}D_r + D_z^2$, the exact source balances are

$$\begin{aligned} E_\theta &= t_* v + (D_r + 2/R)(bv + \beta V + \beta v + W_{r\theta}) \\ &\quad + D_z(Gv + V\gamma + \gamma v + W_{z\theta}) - \varepsilon(\Delta_0 - R^{-2})v - (D_r + 2/R)\Sigma_\theta, \\ E_z &= t_* \gamma + (D_r + 1/R)(b\gamma + \beta G + \beta \gamma + W_{rz}) \\ &\quad + D_z(2G\gamma + \gamma^2 + W_{zz} + p_m) - \varepsilon\Delta_0\gamma - (D_r + 1/R)\Sigma_z, \\ g_r &= -t_*\beta - (D_r + 1/R)(2b\beta + \beta^2 + W_{rr}) \\ &\quad - D_z(b\gamma + G\beta + \beta\gamma + W_{zr}) + (2Vv + v^2 + W_{\theta\theta})/R + \varepsilon(\Delta_0 - R^{-2})\beta, \end{aligned} \quad (\text{C6})$$

where $W_{ab} = \langle w_a w_b \rangle_\theta$ contains the complete waves. The source fixes the radial pressure reconstruction by

$$\mathcal{P} = \int \langle g_r \rangle_Y dR, \quad p_m = \mathcal{T}_0(g_r - \rho\mathcal{P}). \quad (\text{C7})$$

The prescribed ρ and linear map $\mathcal{T}_0$ remain fixed. For each update, $\Delta\mathcal{P} = \int \langle \Delta g_r \rangle_Y dR$ and $\Delta p_m = \mathcal{T}_0(\Delta g_r - \rho\Delta\mathcal{P})$ follow from linearity, but Δg_r itself is recomputed from all the quadratic products in (C6). The improved bounds for Δg_r and Δp_m do not replace the total-field bounds (C4).

In physical Cartesian variables the exact update identity is

$$\mathcal{R}(u + \delta u, p + \delta p) - \mathcal{R}(u, p) = \partial_t \delta u - \Delta \delta u + \nabla \delta p + (u \cdot \nabla) \delta u + (\delta u \cdot \nabla) u + (\delta u \cdot \nabla) \delta u. \quad (\text{C8})$$

This is proved componentwise by expanding $(u_k + \delta u_k) \partial_k (u_i + \delta u_i) - u_k \partial_k u_i$, giving the three displayed transport products with positive signs. The viscosity here is the source's local value one; no rescaling of h, Q, ε or the physical variables has been made.

48.4 3. Step 1: particular supported harmonics

For every supported source coefficient f_m , the source applies the fixed inverse with sign $\mathcal{L}_m(t_m, \pi_m) = -f_m$, and realizes

$$\Delta w_m = \text{curl}_* \left(\frac{i n_\Phi \times (\psi t_m)}{km |n_\Phi|^2} e^{ikm\Phi} \right), \quad \Delta p_{w,m} = \psi \pi_m e^{ikm\Phi}. \quad (\text{C9})$$

Taking the curl after multiplying by ψ retains its derivative terms. Complex conjugate harmonics are paired in the actual real field. Existence of this supported inverse is Proposition 7.2; its remainder and pressure estimates are Proposition 9.1. Applying those results at exponent B supplies velocity class $\mathcal{W}_B$ and pressure class $\mathcal{W}_{B+1/2}$.

The four source rows and their arithmetic are:

New nonzero-harmonic term	Exponent	Gain relative to B	Source of class estimate
Linear remainder, including curl correction	$B + 1/2 - 3\kappa_s$	0.49997	Proposition 9.1.

New nonzero-harmonic term	Exponent	Gain relative to B	Source of class estimate
Cross with old exact wave in $\mathcal{W}_{1/2}$	$B + 1/2 - \kappa_s$	0.49999	Lemma 9.2.
Particular-correction self-interaction	$2B - \kappa_s$	$B - \kappa_s \geq 0.69999$	Lemma 9.2 and the exact self-product in (C8).
Interaction with total means $\mu = 0.9$	$B + 0.9 - 1/2 = B + 0.4$	0.4	Lemma 9.2 with the radial mean one order higher.

The minimum gain is 0.4, since each of the first three gains is at least 0.4 and the fourth equals it. This is the first closing inequality on page 111.

The new covariance has terms bilinear in the old wave and Δw , and quadratic in Δw . Their tensor exponents are at least $B + 1/2 = C^*$, using $2B \geq B + 1/2$ because $B \geq 0.7$. Applying the earlier derivative and mean-reconstruction bounds yields $E_\theta, E_z, \Delta g_r, \Delta p_m \in \mathcal{M}_{C^* - \kappa_s}$, $(\mathcal{P}, J_\theta, J_z) \in \mathcal{S}_{C^* - \kappa_s}$. That class assignment uses (C6)–(C7), Proposition 9.3 and its weighted derivative rules; the numeric inequality alone would not prove it.

The one-extra-order moment gain in (9.12) uses the exact integrated identities

$$\int R^2 \langle E_\theta \rangle_Y dR = \varepsilon \partial_Z J_\theta, \quad \int R \langle E_z \rangle_Y dR = \varepsilon \partial_Z (J_z + c_\rho \mathcal{P}), \quad c_\rho = \frac{1}{2} \int R^2 \rho dR. \quad (\text{C10})$$

To check the pressure sign, the reconstruction gives $\partial_R \langle p_m \rangle_Y = \langle g_r \rangle_Y - \rho \mathcal{P}$ after auxiliary averaging. Integration by parts of the compact pressure gives

$$\int R \langle p_m \rangle_Y dR = -\frac{1}{2} \int R^2 \langle g_r \rangle_Y dR + c_\rho \mathcal{P}. \quad (\text{C11})$$

It is the positive last term in (C11) that appears in (C10). The weighted radial divergence terms integrate to zero, radial viscosity gives coefficients $2 - 1 - 1 = 0$ for the angular equation and $-1 + 1 = 0$ for the axial equation, and time/axial-viscosity terms differentiate the zero moments (C5). Thus (C10) follows from (C6), compact support, the exact constraints and the pressure reconstruction. The full derivative of $c_\rho \mathcal{P}$ is retained: c_ρ can depend on (Z, T) .

The previously proved moment/coefficient bounds then give exponent $C^* + 1 - \kappa_s$, because the displayed factor ε supplies one power while fixed coefficient derivatives retain the class exponent. This gain is not the result of identifying weighted moments with unweighted residuals.

48.5 4. Step 2: signed stress-amplitude change and its exact covariance

The physical radial inverse is not reproved here. Its exact interface in the source is

$$\begin{aligned} F_2 &= Q^{-2A-1/2} \langle E_\theta \rangle_Y, & F_1 &= Q^{-2A-1/2} \langle E_z \rangle_Y, \\ \mathcal{M}_e &= \int_0^\infty r^e F_e(r) dr, & b_e &= q^{-(e+1)/2} \widehat{b}_e(r/\sqrt{q}), & \int x^e \widehat{b}_e(x) dx &= 1, \\ \sigma_e(r) &= -r^{-e} \int_0^r (r')^e [F_e(r') - b_e(r') \mathcal{M}_e] dr', & (\partial_r + e/r) \sigma_e &= -F_e + b_e \mathcal{M}_e, \end{aligned} \quad (\text{C12})$$

for $e = 2, 1$, respectively. The negative sign cancels the supported averaged source, and the positive bump term is retained. The normalized stress pair is $\Sigma = Q^{2A}(\sigma_2, \sigma_1) \in \mathcal{M}_{C^* - \kappa_s}$, independent of auxiliary coordinates. The companion `radial_stress_reconstruction_audit.md` supplies the support and moving-scale proof for this interface. The integral's differential in (C12) was repaired from the draft's literal `],dr'` to `]\,dr'`; this is a transcription repair with no change of integrand, sign or domain.

The fixed signed-amplitude operator uses the original positive primary amplitudes in its denominator. It gives a transverse signed correction with exponent $(C^* - \kappa_s) - 1/2 = B - \kappa_s$. It does not replace the current covariance by a square root. The velocity is again realized as a curl, with the homogeneous pressure coefficient. Lemma 7.7 supplies the curl remainder; Proposition 7.6 and the covariance identity (7.36) supply the prescribed linear stress.

The nonzero-harmonic estimates are:

Term	Original exponent	Gain relative to B
Linear remainder	$B + 1/2 - 4\kappa_s$	0.49996
Mean interaction	$B + 0.4 - \kappa_s$	0.39999
Old-wave cross	$B + 1/2 - 2\kappa_s$	0.49998
Signed self-interaction	$2B - 3\kappa_s$	$B - 3\kappa_s \geq 0.69997$

The minimum is $0.4 - \kappa_s = 0.39999$. Each row is obtained from the Step-1 estimate with input exponent $B - \kappa_s$, except the self-row, which is $2(B - \kappa_s) - \kappa_s$. This explicitly accounts for all three losses in that row.

For the auxiliary-averaged tensor, let $s = t_s + r_s$ be the complete signed increment, decomposed into its transverse and curl pieces. Define the exact bilinear map on the extended angular/auxiliary fields by

$$\mathcal{B}(a, b) = \langle \langle a \otimes b + b \otimes a \rangle_\theta \rangle_Y. \quad (\text{C13})$$

Expanding the tensor product before either average gives

$$\langle \Delta W \rangle_Y = \mathcal{B}(w_0^{\text{tan}}, t_s) + \mathcal{B}(w - w_0^{\text{tan}}, t_s) + \mathcal{B}(w, r_s) + \langle \langle s \otimes s \rangle_\theta \rangle_Y. \quad (\text{C14})$$

Indeed, the first two terms sum to $\mathcal{B}(w, t_s)$, which together with the third equals $\mathcal{B}(w, s)$. Adding $s \otimes s$ is the exact difference between $(w + s) \otimes (w + s)$ and $w \otimes w$. Thus no cross term or signed self-term was dropped. The first term's radial-tangential components equal Σ by the separately proved signed covariance map. Only their specified divergence cancels the source in (C12); the other tensor components and the pressure response remain.

The three remaining tensor exponents before radial divergence, and their resulting radial-flux gains relative to C^* , are:

Exact covariance term	Tensor exponent	After one κ_s loss
Old-wave remainder times transverse signed correction	$B + 0.68 - \kappa_s = C^* + 0.18 - \kappa_s$	$C^* + 0.17998$
Old exact wave times signed curl correction	$C^* + 1/2 - 2\kappa_s$	$C^* + 0.49997$
Signed self-product	$2B - 2\kappa_s = C^* + \sigma_j - 2\kappa_s$	$C^* + \sigma_j - 3\kappa_s \geq C^* + 0.19997$

Other retained mean terms have gains $1 - \kappa_s$ for axial fluxes of the prescribed stress, $1 - 2\kappa_s$ for the pressure effect on E_z , and $1 - \kappa_s$ for the moment bump in (C12), using (C10). Every listed gain exceeds 0.17. Hence the source can retain the weaker common bound

$$\langle E_\theta \rangle_Y, \langle E_z \rangle_Y \in \mathcal{M}_{C^*+0.17}, \quad E_\theta, E_z \in \mathcal{M}_H, \quad (\mathcal{P}, J_\theta, J_z) \in \mathcal{S}_H, \quad H = C^* - 2\kappa_s. \quad (\text{C15})$$

The bounds for the full auxiliary-dependent residual and pressure source at order H use the exact mean formulas and previous mean-operator estimates. They are not consequences of the covariance average alone.

48.6 5. Step 3: nonconstant auxiliary means, with the full axial remainder

For each current exact mean residual, define $E_a^\circ = E_a - \langle E_a \rangle_Y$, $a = \theta, z$. The prescribed update and its actual potential realization are

$$\begin{aligned} \Delta v &= -c_{i_0}^{-1} N_{i_0}^{-1} E_\theta^\circ, & \gamma_d &= -c_{i_0}^{-1} N_{i_0}^{-1} E_z^\circ, \\ \Psi_* &= \mathcal{T}_1 \gamma_d, & \Delta \beta &= -\varepsilon \partial_Z \Psi_*, & \Delta \gamma &= (D_r + R^{-1}) \Psi_*. \end{aligned} \quad (\text{C16})$$

The inverse is the zero-average Fourier inverse of Lemma 8.6. That lemma supplies the stated $\mathcal{M}_H$ bounds for $\Delta v, \gamma_d, \Delta \gamma$, $\mathcal{M}_{H+1}$ for $\Delta \beta$, and preservation of the two exact moments. The actual axial value is $\Delta \gamma = \gamma_d - \mathcal{A}_1 \gamma_d$, so its remainder must remain present:

$$c_{i_0} N_{i_0} \Delta v = -E_\theta^\circ, \quad c_{i_0} N_{i_0} \Delta \gamma = -E_z^\circ + F_{\text{ax}}, \quad F_{\text{ax}} = c_{i_0} N_{i_0} (\Delta \gamma - \gamma_d). \quad (\text{C17})$$

The sign is positive in the axial cancellation equation, despite the axial increment's representation as $\gamma_d - \mathcal{A}_1 \gamma_d$. Here $F_{\text{ax}} = -c_{i_0} N_{i_0} \mathcal{A}_1 \gamma_d$. It is a flat additive remainder by the earlier reconstruction estimate. Its absence is not assumed.

The wave w , hence its complete covariance W , stays fixed during this step. Subtracting the two versions of (C6), using $t_* = -\varepsilon\partial_T + c_{i_0}N_{i_0}$ and (C17), gives exactly

$$\begin{aligned} E_{\theta,\text{new}} - \langle E_{\theta} \rangle_Y &= -\varepsilon\partial_T\Delta v - \varepsilon(\Delta_0 - R^{-2})\Delta v \\ &\quad + (D_r + 2/R)[(b + \beta)\Delta v + (V + v)\Delta\beta + \Delta\beta\Delta v] \\ &\quad + D_z[(G + \gamma)\Delta v + (V + v)\Delta\gamma + \Delta\gamma\Delta v], \\ E_{z,\text{new}} - \langle E_z \rangle_Y &= -\varepsilon\partial_T\Delta\gamma - \varepsilon\Delta_0\Delta\gamma + F_{\text{ax}} \\ &\quad + (D_r + 1/R)[(b + \beta)\Delta\gamma + (G + \gamma)\Delta\beta + \Delta\beta\Delta\gamma] \\ &\quad + D_z[2(G + \gamma)\Delta\gamma + (\Delta\gamma)^2 + \Delta p_m]. \end{aligned} \tag{C18}$$

For example, the radial angular flux difference is $b\Delta v + V\Delta\beta + \beta\Delta v + v\Delta\beta + \Delta\beta\Delta v$, which is exactly the bracket in (C18). The axial square difference is $2\gamma\Delta\gamma + (\Delta\gamma)^2$, together with the base $2G\Delta\gamma$ and pressure Δp_m . All quadratic terms in both equations are retained.

The class arithmetic uses the source's local base estimate

$$|D^I b| \leq C_I \varepsilon S_*^{b_I} \quad \text{on the shell, for every fixed coefficient multi-index } I, \tag{C18a}$$

with V, G of order zero, the cumulative bounds (C4), and the known mean-operator estimates. Equation (C18a), stated on source page 110, does not assert $b \in \mathcal{M}_1$: it does not give b the mean class's shell-edge factor ζ or smooth zero extension. Those properties in $b\Delta v$, $b\Delta\gamma$, and $b\Delta\beta$ are supplied by the actual increments.

More precisely, for an actual increment $f \in \mathcal{M}_\alpha$, the exact coefficient Leibniz formula and Definition 6.4 give

$$\begin{aligned} D^I(bf) &= \sum_{J \leq I} \binom{I}{J} (D^J b)(D^{I-J} f), \\ |D^I(bf)| &\leq \sum_{J \leq I} \binom{I}{J} C_J C_{j,I-J} \varepsilon^{1+\alpha} S_*^{b_J + b_{j,I-J}} \zeta \delta^{-d_{j,I-J}}. \end{aligned} \tag{C18b}$$

Every summand retains the increment's ζ factor and its original edge exponent; none is replaced by a bound for the base's support. The sum is finite for fixed I , and thus has the finite-constant form required by (6.24). On the bounded active shell, any change to a common edge exponent uses only the shell's fixed upper bound for δ and adjusts constants. The product is zero wherever the zero extension of f is zero. At the two shell edges, each displayed derivative tends to zero by (6.23), since $\zeta\delta^{-d}$ tends to zero faster than every power for every fixed d . It therefore has the smooth zero extension required of the product. The base's angular and auxiliary independence and its original chart coherence preserve the corresponding properties of f . This proves the multiplication passage used for these terms from the local estimate, without attributing those support properties to b itself.

Term group in (C18)	Lowest exponent retained
Slow time	$H + 1$
Radial flux	$H + 1 - \kappa_s$
Axial flux, including $\Delta p_m \in \mathcal{M}_H$	$H + 1$
Viscosity with up to two radial losses	$H + 1 - 2\kappa_s$

Every product in the radial bracket has at least order $H + 1$: terms containing b supply 1; terms containing $\Delta\beta$ supply its $H + 1$; old radial β supplies 1.9; and new quadratic terms have at least $2H + 1$. Axial differentiation supplies one to products of order at least H . This verifies the minimum table using the stated input classes.

Thus the supported mean changes, after retaining F_{ax} in the additive flat part, lie in $\mathcal{M}_{H+1-2\kappa_s} = \mathcal{M}_{C^*+1-4\kappa_s}$. With (C15),

$$E_{\theta}, E_z \in \mathcal{M}_{\min(C^*+0.17, C^*+1-4\kappa_s)} = \mathcal{M}_{C^*+0.17}, \tag{C19}$$

because $1 - 4\kappa_s = 0.99996 > 0.17$. Here (C18) is the identity for the actual recomputed mean expressions; (C19) follows the source's supported-mean convention after transferring F_{ax} , together with the specified modified-integral remainders, into the separately retained additive flat residual of Proposition 9.3. In explicit component form this convention preserves the identity

$$E_{a,\text{new}}^{\text{actual}} = E_{a,\text{new}}^{\text{good}} + F_{a,\text{new}}^{\text{mean}}, \quad a \in \{\theta, z\}, \quad (\text{C19a})$$

The exact dictionary to the source's decomposition (9.3) is fixed by its complete mean identity (9.6). Let $\pi_\theta(v_r, v_\theta, v_z) = v_\theta$ and $\pi_z(v_r, v_\theta, v_z) = v_z$. For the recomputed state, retain the source's complete supported vector $G_{*,\text{new}}$, additive flat vector $F_{*,\text{new}}$, and unchanged base error $E_{B,*}$. Then

$$\begin{aligned} (D_r p_{m,\text{new}} - g_{r,\text{new}}, E_{\theta,\text{new}}^{\text{actual}}, E_{z,\text{new}}^{\text{actual}}) &= \langle G_{*,\text{new}} + F_{*,\text{new}} - E_{B,*} \rangle_\theta, \\ E_{a,\text{new}}^{\text{good}} &= \pi_a \langle G_{*,\text{new}} \rangle_\theta, \\ F_{a,\text{new}}^{\text{mean}} &= \pi_a \langle F_{*,\text{new}} - E_{B,*} \rangle_\theta, \quad a \in \{\theta, z\}. \end{aligned} \quad (\text{C19b})$$

Linearity of angular averaging and of the displayed component projections proves (C19a) from (C19b) with the original base-error subtraction retained. In particular, F^{mean} is not being identified with the tangential components of $\langle F_* \rangle_\theta$ alone. The base field has already been subtracted in the exact conservative correction balances (C6), while it remains part of the full physical residual decomposition. This dictionary changes none of the velocity, pressure, or nonlinear products.

Here (C19) is the class estimate for E^{good} , and the axial F^{mean} retains the explicit $+F_{\text{ax}}$ in (C18). Its remaining terms are exactly those transferred by the source's additive residual decomposition. At a fixed stage, all-order q -flatness bounds this additive residual by any prescribed finite power of q ; it does not, by itself, prove the ζ -weighted derivative, shell support, or smooth-extension requirements of $\mathcal{M}_\alpha$. No such class membership for a flat term is inferred here unless the earlier support or interior construction also supplies those properties. In particular F_{ax} remains present in (C18) and (C19a), and the actual $\Delta\gamma$, including its reconstruction remainder, remains in every nonlinear product.

The new nonzero-harmonic interaction uses Lemma 9.2 with old wave exponent $1/2$ and new mean exponent H , yielding $1/2 + H - 1/2 = H$. Therefore its gain relative to B is

$$H - B = \frac{1}{2} - 2\kappa_s = 0.49998. \quad (\text{C20})$$

The defects remain in $\mathcal{S}_H$ after this mean step; this estimate uses their recomputed definitions, not an assertion that the temporal inverse corrects the three scalar defects.

48.7 6. Step 4: two exact moment constraints and three nonlinear residual defects

The five-equation inverse is applied to the current triple $(\mathcal{P}, J_\theta, J_z)$, with its original right-hand signs:

$$\begin{aligned} \int R^2 \Delta v \, dR &= 0, \quad \int R \gamma_d \, dR = 0, \\ \int (2V/R) \Delta v \, dR &= -\mathcal{P}, \quad \int R^2 (G \Delta v + V \gamma_d) \, dR = -J_\theta, \\ \int (2RG \gamma_d - RV \Delta v) \, dR &= -J_z. \end{aligned} \quad (\text{C21})$$

Lemma 8.7 supplies its fixed supported linear inverse under the stated reserved-patch base conditions, with $\Delta\gamma = \gamma_d$ exactly, $\Delta v, \Delta\gamma \in \mathcal{M}_H$ and $\Delta\beta \in \mathcal{M}_{H+1}$. These increments are independent of both angular and auxiliary variables. Their existence, boundedness and the equality $\Delta\gamma = \gamma_d$ do not follow just from displaying five equations; they are the earlier construction used at this step.

Keeping the wave and base fixed, define $R_g = g_{r,\text{new}} - g_r - (2V/R)\Delta v$. Expansion of (C6) gives

$$\begin{aligned} R_g &= -t_* \Delta\beta - (D_r + 1/R)[2b\Delta\beta + 2\beta\Delta\beta + (\Delta\beta)^2] \\ &\quad - D_z[b\Delta\gamma + G\Delta\beta + \beta\Delta\gamma + \gamma\Delta\beta + \Delta\beta\Delta\gamma] \\ &\quad + [2v\Delta v + (\Delta v)^2]/R + \varepsilon(\Delta_0 - R^{-2})\Delta\beta. \end{aligned} \quad (\text{C22})$$

In particular the centrifugal contribution beyond $(2V/R)\Delta v$ has a positive sign. The wave covariance has no change because this is a pure mean update.

The three defects use the original definitions

$$J_\theta = \int R^2 \langle Gv + V\gamma + \gamma v + W_{z\theta} \rangle_Y \, dR,$$

$$J_z = \int R \langle 2G\gamma + \gamma^2 + W_{zz} \rangle_Y dR - \frac{1}{2} \int R^2 \langle g_r \rangle_Y dR. \quad (\text{C23})$$

Substitute $g_{r,\text{new}} = g_r + (2V/R)\Delta v + R_g$ and expand all products. The last three rows of (C21), with $\Delta\gamma = \gamma_d$, cancel the old triple plus its base-linear changes. What remains exactly is

$$\begin{aligned} \mathcal{P}_{\text{new}} &= \int \langle R_g \rangle_Y dR, \\ (J_\theta)_{\text{new}} &= \int R^2 \langle \gamma \Delta v + v \Delta \gamma + \Delta \gamma \Delta v \rangle_Y dR, \\ (J_z)_{\text{new}} &= \int R \langle 2\gamma \Delta \gamma + (\Delta \gamma)^2 \rangle_Y dR - \frac{1}{2} \int R^2 \langle R_g \rangle_Y dR. \end{aligned} \quad (\text{C24})$$

The coefficient $-RV\Delta v$ in the last linear row of (C21) is exactly $-\frac{1}{2}R^2(2V/R)\Delta v$ from the pressure-source moment in (C23). Therefore the minus sign in that row and the minus-half remainder in (C24) agree. The final defects are not identically zero: (C24) retains every nonlinear remainder.

The estimates are checked term by term from the same supported base and cumulative classes. Since the update is slow, $t_*\Delta\beta = -\varepsilon\partial_T\Delta\beta$. The principal bounds in (C22) are

Term	Exponent
$-t_*\Delta\beta$	$H + 2$
$(D_r + 1/R)(2b\Delta\beta)$	$H + 2 - \kappa_s$
$D_z(b\Delta\gamma + G\Delta\beta)$	$H + 2$
$2v\Delta v/R$	$H + 0.9$
$(\Delta v)^2/R$	$2H$
$\varepsilon(\Delta_0 - R^{-2})\Delta\beta$	$H + 2 - 2\kappa_s$

The transport terms with old means have no lower exponent, using $v, \gamma \in \mathcal{M}_{0.9}$, $\beta \in \mathcal{M}_{1.9}$. Since $H \geq 1.19998 > 0.9$, also $2H \geq H + 0.9$. Thus every term is at least $H + 0.9 - 2\kappa_s$. The remaining moment products in (C24) have exponents $H + 0.9$ or $2H$, and the earlier normalized moment estimates preserve their powers. Therefore

$$(\mathcal{P}, J_\theta, J_z)_{\text{new}} \in \mathcal{S}_{H+0.9-2\kappa_s} = \mathcal{S}_{C^*+0.9-4\kappa_s} = \mathcal{S}_{C^*+0.89996}. \quad (\text{C25})$$

The changes in wave and tangential residuals obey the same bounds as Step 3; there is no fast-time cancellation term in the slow update. Hence its added mean residual has order $C^* + 0.99996$, and its added nonzero wave residual has gain $H - B = 0.49998$.

48.8 7. Exact invariant checks and the completed exponent ledger

The structural invariants are preserved for the following precise reasons:

1. **Divergence and real fields.** Wave updates in Steps 1–2 are curls of the complete cutoff potentials. In Cartesian coordinates, $\partial_i \epsilon_{ikl} \partial_k A_l = 0$ by antisymmetry and commuting mixed derivatives. Mean meridional updates are curls of azimuthal potentials, and a direct angular field independent of θ has zero divergence. Pairing conjugate harmonic coefficients preserves reality. These arguments require the stated smooth representatives, which Proposition 9.3 supplies for the generated sources and updates.
2. **Two zero moments.** Steps 1–2 change w and its pressure harmonics; every nonzero angular harmonic has zero angular average, so the direct mean components v, γ and (C5) are unchanged. Step 3's prescribed increments have zero auxiliary mean. Its actual axial reconstruction preserves the axial-flux moment by the identity in Lemma 8.6; it is not identified with the prescribed field without the remainder. Equivalently, for the smooth supported azimuthal potential, averaging $(D_r + R^{-1})\Psi_*$ gives $(\partial_R + R^{-1})\langle \Psi_* \rangle_Y$, whose R -weighted integral is the zero boundary term $[R\langle \Psi_* \rangle_Y]_0^\infty$. Step 4 preserves both moments by the first two rows of (C21), because $\Delta\gamma = \gamma_d$ there.
3. **Complete pressure and nonlinear defects.** Each step uses (C7) on the complete new fields. The radial residual is the reconstructed one, $-\rho\mathcal{P}$ plus the recorded flat reconstruction remainder, not an omitted radial equation. The three nonlinear defects are estimated by (C24) rather than declared zero.

4. **Cumulative low-order bounds.** Particular wave increments begin at $B \geq 0.7$; signed increments begin at $B - \kappa_s \geq 0.69999 > 0.68 > 1/2$. They therefore preserve both wave bounds in (C4) by finite addition. Mean updates have tangential and pressure order at least $H \geq 1.19998 > 0.9$, while radial updates have order $H + 1 \geq 2.19998 > 1.9$. Earlier wave-pressure increments have order at least $C^* - \kappa_s \geq 1.19999$. All improved $\Delta g_r, \Delta p_m$ estimates concern increments; the total g_r, p_m need not acquire those high orders.
5. **Supports, common torus and harmonic closure.** These do not follow from scalar exponent inequalities. Proposition 9.3 supplies closure under differentiation, products, curl, pressure reconstruction, mean maps and fixed inverses, with smooth zero extensions. Same-label harmonics add integer multipliers; a quadratic product can double the finite harmonic range at one fixed stage. The range may grow with the stage but is finite and independent of the band at that stage. Distinct-label products vanish only by the earlier support-separation proof. Mean inverses may fill the auxiliary torus, but a nonzero harmonic product retains a wave factor and its source support. These are the specific upstream facts needed to use the next inverse.

Taking the minimum of the new nonzero-harmonic gains from all four steps gives

$$\min\{0.4, 0.4 - \kappa_s, 1/2 - 2\kappa_s, 1/2 - 2\kappa_s\} = 0.39999 > 0.1. \quad (\text{C26})$$

For the tangential mean after Steps 2–4,

$$\min\{0.17, 1 - 4\kappa_s, 1 - 4\kappa_s\} = 0.17 > 0.1. \quad (\text{C27})$$

For defects the final gain is $0.9 - 4\kappa_s = 0.89996 > 0.1$. Hence the already established class bounds imply the weaker state requirements

$$\mathcal{G}_{*,\text{wave}}^{[j+1]} \in \mathcal{W}_{B+1/10}, \quad E_\theta, E_z \in \mathcal{M}_{C^*+1/10}, \quad (\mathcal{P}, J_\theta, J_z) \in \mathcal{S}_{C^*+1/10}. \quad (\text{C28})$$

These indices are exactly B_{j+1} and C_{j+1}^* from (C2). No altered stage increment or smaller κ_s is needed.

48.9 8. Flat remainders and the use in the physical estimate (9.18)

The actual finite residual is decomposed in Proposition 9.3 as

$$\mathcal{R}(u^{[j]}, p^{[j]}) = G^{[j]} + F^{[j]}, \quad |F^{[j]}|_m \leq C_{j,m,N} Q^N \quad \text{for every fixed } j, m, N. \quad (\text{C29})$$

The terms in $F^{[j]}$ include the slow-base error, pulse-cutoff tails, modified radial-integral remainders and the axial F_{ax} from (C17). The full velocity, including its reconstruction remainders, is still used in every nonlinear term of (C8), (C14), (C18) and (C22). Thus isolating an additive flat residual does not authorize deleting the associated velocity from later products.

The old additive flat term enters (C8) once. Newly generated flat terms are added; a term actually canceled by a fixed mean inverse is removed from that decomposition after cancellation. At any fixed stage the source proves there are finitely many types, uniformly estimated over labels. This is the role of Proposition 9.3; the exponent closure table alone cannot establish it.

For pulse tails the source bound has the form $CQ^{-M}S_*^Pe^{-cS_*}$, with $S_* = \ell^2$, $Q = 2^{-\ell}$. Dividing by a target Q^N leaves $C\ell^{2P} \exp[-c\ell^2 + (M+N)\ell \log 2]$. Its exponent tends to minus infinity because the negative quadratic dominates the fixed linear term and logarithmic power; explicitly, beyond a fixed ℓ , the latter contributions total at most $c\ell^2/2$. It is therefore bounded by $Ce^{-c\ell^2/2}$, which is bounded on the remaining half-line. Polynomially many labels alter P but retain this argument. Modified radial-integral flatness instead uses its separately proved all-order integration estimate; it is not replaced by this Gaussian argument.

In a fixed chart, the supported normalized residual at stage j has nonzero-wave order $B_j = 1/2 + \sigma_j$, tangential mean order $C_j^* = 1 + \sigma_j$, and radial defect order C_j^* . Each is at least σ_j . Keeping the original physical residual factor,

$$\mathcal{R} = Q^{-2A-1/2} \mathcal{R}_*. \quad (\text{C30})$$

Lemma 9.8 supplies mixed derivative costs independent of stage. In the actual coordinates these are dominated by $s_x = 1/2 + h/2$ per spatial derivative and $s_t = 1 + 3h/2$ per time derivative, including evaluated phase factors and frame derivatives. For total order m , a sufficient fixed loss for a supported residual coefficient is $2A+1/2+m(1+3h/2)$, enlarged if needed to a common loss for the stated background

and supported terms. Its coefficient constants and powers of $1 + |\log q|$ may depend on j, m . Converting $Q \asymp q$ changes constants, not the stage independence of the loss.

Consequently the source’s analytic class estimates plus this fixed conversion yield precisely the form

$$|\mathcal{R}(u^{[j]}, p^{[j]})|_m \leq C_{j,m} q^{h\sigma_j - K_m} (1 + |\log q|)^{P_{j,m}} + E_{j,m}, \quad E_{j,m} \leq C_{j,m,N} q^N, \quad (\text{C31})$$

with $h\sigma_j = h/5 + hj/10 \rightarrow \infty$ and K_m independent of j . This is (9.18). Exponent arithmetic proves that the gain tends to infinity; the coefficient estimates, actual residual decomposition, uniformity over labels and physical phase differentiation are the preceding analytical work required for the inequality itself. No uniformity in j of the constants is inferred.

The companion `summation_realization_audit.md` proves how one fixed-stage comparison uses (C31), including its fixed $E_{j,m}$, to obtain all-order flatness of the realized residual. The flat remainders are not summed over j . That later summation is not repeated here.

48.10 9. Verification boundary

The following have been checked directly in this bounded audit: all displayed four-step exponent substitutions and closing inequalities; the covariance identity (C14); exact mean-update identities (C18); exact nonlinear radial and defect remainders (C22)–(C24); the pressure sign and retained product $\partial_Z(c_\rho \mathcal{P})$; the distinction between the two exactly preserved zero moments and the three nonzero improved defects; preservation of low-order classes by the specified finite increments; and the arithmetic passage to a gain $h\sigma_j$ with a fixed derivative loss.

The following remain upstream source assertions, with their precise supplying results identified above: existence and uniform bounds for the pulse inverse; the full wave-amplitude estimates behind Propositions 9.1 and Lemma 9.2; the signed covariance operator; the temporal and five-equation inverse constructions; all-label support/flatness closure in Proposition 9.3; and the common domain in Lemma 9.7. The companion `radial_stress_reconstruction_audit.md` now proves the Step-2 compact radial inverse, its complete retained moment coordinate, mixed derivatives, full tensor force and chart correspondence, and flat-edge and uniform-band preservation. The actual incoming residual estimates, covariance realization, and inter-chart descent remain their respective source dependencies. This report does not replace the remaining claims with new assumptions or claim to prove the whole construction without them. It records exactly what the audited calculation establishes and where the actual earlier proof content must be verified.

No concrete exponent or algebraic sign defect was found in the audited cycle. The material interpretation is that Step 4 removes the old defects and their linear changes while retaining the nonlinear expressions (C24), whose higher order closes the next stage. Calling all three final defects identically zero would misstate the source.

48.11 10. Static formal-source correspondence

The following bounded six-file locator was completed by the subsidiary source-audit worker. The declaration bodies around `CorrectionAnalyticStep.step` and `ActualCyclePreservation.state_runInvariant` were also read directly in this lane. All line numbers refer to the exact hashed files below, under `formal/released_ns_validation/NavierStokes/`. No Lean was executed and no transitive proof-closure audit was made.

The scalar dictionary is literal: take Lean’s stage index n to be the paper’s j , its real argument σ to be $\sigma_j = 1/5 + j/10$, and its κ to be $\kappa_s = 1/100000$. Then `waveExponent` σ , `meanExponent` σ , and `meanUpdateExponent` σ κ unfold to $1/2 + \sigma_j = B$, $1 + \sigma_j = C^*$, and $1 + \sigma_j - 2\kappa_s = H$. Under this dictionary the stage successor $\sigma + 1/10$ is exactly σ_{j+1} , because $1/5 + j/10 + 1/10 = 1/5 + (j+1)/10$. Thus the following arithmetic declarations address the same scalar expressions, without changing the source’s parameters.

File and exact declarations	What the inspected source contains
<code>ExponentLedger.lean</code> : <code>waveExponent 24</code> , <code>meanExponent 27</code> , <code>meanUpdateExponent 30</code> ; <code>particular_gain_eq 80</code> ; <code>signed_gain_eq 114</code>	Definitions B, C^*, H ; particular gain $2/5$; signed gain $2/5 - \kappa$ for the displayed admissible parameters.

File and exact declarations	What the inspected source contains
ExponentLedger.lean: signedBarGain 44; signed_bar_gain_exceeds_seventeen_hundredths 178	The complete five-term minimum $\min\{9/50 - 2\kappa, 1/2 - 3\kappa, \sigma - 3\kappa, 1 - \kappa, 1 - 2\kappa\}$, and its strict comparison with $17/100$. These are exactly the three tensor remainders, axial/moment contribution and pressure contribution retained in Section 4.
ExponentLedger.lean: mean_update_wave_margin 193; completed_mean_gain_eq 203; completed_mean_margin 208; completed_defect_margin 220; stageParameter 286; all_stage_arithmetic 304	The mean-update wave margin, $\min(17/100, 1 - 4\kappa) = 17/100$, the mean and defect stage margins, and their package at $1/5 + n/10$. These declarations state comparisons of real numbers, without asserting the existence of fields.
ActualIterationLedger.lean: gain 29, kappa 32, sigma_formula 36, all_cycle_margins 119; residualWave_formula 231, residualMean_formula 235, cycle_wave_output 239, cycle_mean_output 243, residual_physical_gap 252	The fixed $\kappa = 1/100000$, $\sigma_j = 1/5 + j/10$, $B_j = j/10 + 7/10$, $C_j^* = j/10 + 6/5$, successor comparisons, and the exact identity $hB_j = hj/10 + 7h/10$. gain is $hj/10$. This ledger does not by itself establish an estimate for an actual residual.
CorrectionState.lean: MeanResidualBounds 216; radialMoment 226; pressureDefect 229; thetaDefect 232; axialDefect 236; debt 242; DefectBounds 246; ZeroMasses 250	Mean angular/axial residual membership at $1 + \sigma$; weighted radial moments; the stored defect vector in the order $(\mathcal{P}, J_\theta, J_z)$; its class membership at $1 + \sigma$; and the two exactly vanishing masses.
CorrectionStep.lean: CycleParameters.afterParticular 5042, afterSigned 5066, afterTemporal 5073, afterRank 5079, next 5084; CycleState.step 6947	The four successive actual updates and subsequent radial-pressure alias refresh, including the represented state, coefficients and axisymmetric alias.
CorrectionStep.lean: CycleParameters.next_zeroMassesOn 9144; CycleAnalyticInvariant 9408, especially 9440–9448 and 9457–9459	The analytic invariant stores harmonic residual membership at $1/2 + \sigma$, mean/debt membership at $1 + \sigma$, the reconstructed-state identity, exact zero masses on its slow carrier, zero Gaussian angular mean, zero alias coefficients and all-power axis-alias bounds. The zero-mass preservation theorem uses the primitive regularity, moving covariance, rank geometry, matching-length and radial-containment data; it is not an arithmetic declaration.
CorrectionAnalyticStep.lean: StepData 470, cross_tail 500, StepResult 506, step 521	step produces the invariant at $\sigma + 1/10$, temporal/rank increments and pressure difference at $1 + \sigma - 2\kappa$, velocity-coefficient difference at $1/2 + \sigma - \kappa$, and pressure-coefficient difference at $1 + \sigma - \kappa$. Its actual inputs include StaticData , incoming invariant and StepData : wave estimates and regularity, supports, assembly, rank geometry and the exact tail cross-stress identity.
ActualCyclePreservation.lean: RunInvariant 768; next_runInvariant_of_particular 800; particularData 812; next_runInvariant 817; state_runInvariant 826; state_result 882; state_afterTemporal_debt 894	The actual iteration combines the analytic invariant, chart coherence and periodicity. particularData invokes the construction of particular-solver data internally. Accordingly state_runInvariant B N O h N j has the fixed construction indices, the displayed geometric-threshold premise and stage index as its arguments; it does not leave particular-solver data as a separately supplied argument. state_result retains the corresponding StepResult .

The exact component dictionary for the stored defects can be read without identifying any unexamined function spaces. At line 226, `radialMoment k f` is `fun n p => PressureStream.pressureMass (fun x => x.1 ^ k * f n x) p`. Substitution into the three definitions gives:

$$\begin{aligned}
\text{pressureDefect} &= \text{radialMoment } 0 \ (u.g_r), \\
\text{thetaDefect} &= \text{radialMoment } 2 \ (\text{thetaAxial}(c.\text{base}, u.\text{mean}) + u.\text{covariance}_{2,1}), \\
\text{axialDefect} &= \text{radialMoment } 1 \ (\text{axialAxial}(c.\text{base}, u.\text{mean}) + u.\text{covariance}_{2,2}) \\
&\quad - \frac{1}{2} \text{radialMoment } 2 \ (u.g_r).
\end{aligned} \tag{C32}$$

Here formal indices 0, 1, 2 encode the radial, angular and axial component slots, respectively. Formula (C32) preserves the paper’s weighted powers 0, 2, 1, 2, the covariance positions (z, θ) , (z, z) , and the negative 1/2 pressure-source term. The **ZeroMasses** fields are the weighted angular and axial mean moments of orders 2 and 1. This is a declaration-level component correspondence; this bounded lookup does not assert that all carrier definitions, evaluations, regularity classes and reconstruction operators are fully equivalent to the manuscript presentations. The exact construction/proof content for those operators remains in the upstream source identified in the table.

In particular, reading a generic analytic theorem with structured inputs does not justify claiming that the released actual iteration merely assumes those inputs: the inspected **particularData** and **state_runInvariant** declarations explicitly supply that interface at the actual iteration. Conversely, seeing those declarations in text does not establish that the complete dependency closure passes a kernel or configured comparator check. This report makes neither claim.

48.11.1 Frozen source identities

Relative file under the research root	Bytes	SHA256
downloaded_public_releases/ navier-stokes.pdf	2959204	0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f
downloaded_public_releases/ current_pdf_extracted.txt	683841	f83414ea543ffd89910b9b8a35d1680a2e3295246c91cbef54e719a7bc316d0
formal/released_ns_validation/ NavierStokes/ExponentLedger.lean	11470	06b9d8d6439b0bdd72f44ce404b1ce986e68f367c03c7dfd5edca8e2d2a0140a
formal/released_ns_validation/ NavierStokes/ActualIterationLedger.lean	19306	25a1b7e920cfe0af7cdda488c4d0407ee47c526abcfa3a6491a8fb335a1a6a31
formal/released_ns_validation/ NavierStokes/CorrectionState.lean	23856	22eeff8985d8e546ad4fac61a9f87f778c0f152413bbb8e0e2b55fe7d06b7a1d
formal/released_ns_validation/ NavierStokes/CorrectionStep.lean	575619	a5ab01cd25d186deec3f136cc15d2d31032151ae7ddb5213a7f3d6b244ec8f8f
formal/released_ns_validation/ NavierStokes/CorrectionAnalyticStep.lean	39528	cdb8d393c22b2d11e39e7589bcb3cbfa0ed8cc8837835dea69e17f4863821ab2
formal/released_ns_validation/ NavierStokes/ActualCyclePreservation.lean	47390	f6d26b802063ee803bcc2e7923555481144e6919f64ddcb4d265cdd4d0d4674c

The historical **stage_cycle_gain_audit_receipt.json** records its earlier note snapshot, exact source identities and hashes of the four rendered pages inspected. The later **stage_cycle_precision_review.md** records the amended note hash and the precise added mean-residual dictionary and companion coverage. The audited bounded task is complete; no whole-construction or Navier–Stokes proof conclusion is asserted.

49 Constructed leading profiles, complete background and pulse inputs

49.1 1. Exact input and its provenance

Retain

$$0 < h < 1/100, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \\ z = q^D \eta, \quad 1 - t = q(1 - \eta^2), \quad X = \frac{r^2}{2q}, \quad d = 1 - \eta^2, \quad L = 1 - 2h\eta^2.$$

The particular h is the one selected in the completed leading-profile assembly. It is not a new arbitrary member of the displayed interval. In particular, $L \geq 1 - 2h > 0$ on $-1 \leq \eta \leq 1$.

The constructed leading profiles have physical interpretation

$$u_{\theta,0} = q^{-A} E_0, \quad u_{z,0} = q^{-A} U_0, \quad r u_{r,0} = V_0, \quad p_0 = q^{-2A} \Pi_0^{\text{profile}}, \quad E_0 = \sqrt{2X} \phi_0 / C.$$

Here the notation Π_0^{profile} in this audit distinguishes the complete zero-order pressure profile from the axis datum also called Π_0 in the assembly; it denotes exactly the zero-index Π_n of the base recursion. Their exact relationship is

$$\Pi_0^{\text{profile}}(X, \eta) = \Pi_{\text{ax}}(\eta) + \int_0^X \frac{E_0(x, \eta)^2}{2x} dx.$$

The full proof at reader 5676–5751 produces, for the same fixed object:

- one interval $[0, X_{\text{an}}]$, extending beyond the stress-free endpoint X_a , with smooth radial coefficients and one complex parameter neighborhood on which every fixed radial derivative is holomorphic;
- the exact totals $M(\infty) = J(\infty) = S(\infty) = 0$, canonical pressure, and zero renormalized angular total;
- strictly positive compact cone margins for the leading shear and stress direction on $[X_a, X_b] \times [-1, 1]$;
- the original weight $\zeta = \exp(-t_1^2/y_a^2 - 4/y_b^2)$ and distance $\delta = \min\{1, y_a, y_b\}$, with $y_a = \log(X/X_a)$, $y_b = \log(X_b/X)$, and all fixed weighted stress bounds;
- the exact heat exterior and $U_0 = V_0 = 0$ for $X \geq X_v < X_b$;
- the unchanged reserved intervals $I_{\text{pos}} = I_3 < I_4 = I_{\text{mean}}$, lying before X_v , with

$$U_0 = 0, \quad E_0 = c_{\text{patch}}(1 + \eta^2)^{-1} X^{-1/2-\lambda}, \quad \lambda > 0.$$

These are actual outputs of the assembly, not hypotheses imported from the later base construction. The common holomorphic collar is proved at reader 7106 (the original continuation proof at 713); the nonlinear five-moment restorations preserve that entire collar and preserve the outer fields pointwise. The exact exterior morphism is reader 7593–7603. Neither restoration is based only on an asymptotic comparison. The finite parameter sequence and all leading-profile choices precede physical q , band number, correction stage, and requested derivative order.

49.2 2. The common-radius analytic solution map

Choose once

$$X_a < X_- < X_{\text{keep}} < X_{\text{cut}} < a^2 \leq X_{\text{an}} < \inf I_{\text{pos}}.$$

The first five radii can all be chosen inside the unchanged analytic collar; no later positive-order coefficient changes their values. At order $n \geq 1$, put $\lambda_n = 2nh$. The finalized lower orders are used in the equations at reader 1020–1033, rather than the uncut inner versions of those lower orders.

For every real b , the exact physical differentiation maps are

$$\partial_t(q^b f) = q^{b-1} T_b f, \quad T_b f = \frac{-b f + D \eta f_\eta + X f_X}{L}, \\ \partial_z(q^b f) = q^{b-D} Z_b f, \quad Z_b f = \frac{2b \eta f + d f_\eta - 2 \eta X f_X}{L}.$$

Thus two axial derivatives are $Z_{b-D} Z_b$, in this order. Incompressibility with the original powers gives

$$F_n(X, \eta) = \int_0^X U_n(x, \eta) dx, \quad V_n = \frac{2 \eta X U_n - 2 \eta (D + \lambda_n) F_n - d(F_n)_\eta}{L}. \quad (\text{B1})$$

In particular the factor $D + \lambda_n$ cannot be replaced by D .

Set $X = \xi^2$, $K_n = F_n/X - U_n$, and

$$W_n = (\phi_n, U_n, K_n, \Pi_n, \partial_\xi \phi_n, \partial_\xi U_n)^T.$$

The exact system is

$$\partial_\xi W_n + \xi^{-1} \text{diag}(0, 0, 2, 0, 3, 1)W_n = A_0 W_n + A_1 \partial_\eta W_n + f_n, \quad W_n(0) = 0. \quad (\text{B2})$$

Here f_n contains only the finalized lower orders. The current-index products are precisely the pairs $(0, n), (n, 0)$, so (B2) is linear in the current unknown. Every apparent axis quotient is smooth: (B1) gives $V_n = X v_n$, and every term of Ω_{n-1} has a factor X . The pressure row is

$$\Pi_{n,\xi} = 4\xi C^{-2} \phi_0 \phi_n + f_{n,4}.$$

Using that row in the axial equation eliminates the $X\Pi_{n,X}$ occurrence without differentiating an unknown pressure in the radial direction again.

For $H_c = D\eta + dU_0$, the only potentially nonzero entries of A_1 are

$$\begin{aligned} (A_1)_{51} &= 2H_c/L, & (A_1)_{52} &= (A_1)_{53} = -2d(\phi_0 + X\phi_{0,X})/L, \\ (A_1)_{62} &= 2(H_c - dXU_{0,X})/L, & (A_1)_{63} &= -2dXU_{0,X}/L, & (A_1)_{64} &= 2d/L. \end{aligned} \quad (\text{B3})$$

These entries follow directly from the η derivative terms in the original coefficient equations; the factor two comes from the transformed radial second-order operators. They remain unchanged as a list of positions under every parameter derivative. Therefore A_1 maps the first four coordinate subspace into the last two and kills the last two, including after parameter differentiation.

Let $c_i = (0, 0, 2, 0, 3, 1)_i$ and define

$$(Gg)_i(\xi) = \int_0^\xi (s/\xi)^{c_i} g_i(s) ds, \quad K = G(A_0 + A_1 \partial_\eta).$$

The diagonal kernel preserves both subspaces above. In a composition, two adjacent derivative choices vanish: when the left derivative hits the right matrix the product is $A_1 D_0 \partial_\eta A_1 = 0$, and when it hits the remaining function the product is $A_1 D_0 A_1 = 0$. Consequently a nonzero word of k factors has at most $p_k = \lceil k/2 \rceil$ parameter differentiations. This statement counts differentiations on coefficients as well as on the terminal source.

For each fixed n , choose a sufficiently small bounded complex neighborhood S_ρ of $[-1, 1]$ on which L has no zero and the coefficients and source are bounded uniformly for $0 \leq \xi \leq a$. This is possible because f_n uses only finitely many previously constructed orders and finitely many derivatives. For $0 < \rho' < \rho$, split $\Delta = \rho - \rho'$ among the at most p_k derivative operators. Applying Cauchy's estimate to the successive holomorphic operator compositions bounds them by $\max(1, p_k/\Delta)^{p_k}$, without adding uncounted Leibniz terms. The ordered $k+1$ radial integrals have volume $a^{k+1}/(k+1)!$, and the at most 2^k words are absorbed into C_n^{k+1} . Hence

$$\|K^k G f_n\|_{S_{\rho'}} \leq \frac{C_n^{k+1} a^{k+1}}{(k+1)!} \max(1, p_k/\Delta)^{p_k}. \quad (\text{B4})$$

The logarithm of this bound divided by k is $-\frac{1}{2} \log k + O_n(1)$. Its k -th root tends to zero. Thus $\sum_{k \geq 0} K^k G f_n$ converges for the entire chosen a , whatever the finite size of C_n . Applying the same estimate with k radial integrals to a homogeneous solution $W = K^k W$ proves uniqueness in the uniformly bounded holomorphic class.

The identities

$$(Gg)_i = \xi \int_0^1 t^{c_i} g_i(t\xi) dt, \quad \partial_\xi (Gg)_i = g_i(\xi) - c_i \int_0^1 t^{c_i} g_i(t\xi) dt$$

bootstrap radial regularity, after decreasing the parameter neighborhood for each fixed derivative if needed. The reflected equations have even first four components and odd last two. Uniqueness gives these parities. Taylor's formula for an even smooth function of ξ , to order $2m+2$, proves smoothness through $X = \xi^2 = 0$ with m derivatives, for every fixed m .

This proves an exact map from the finalized lower coefficients on the common radial rectangle to the unique current inner coefficient on that same radial rectangle. The parameter neighborhood is allowed to depend on n and the fixed radial derivative. The frozen source says this explicitly at extracted-text lines 2874–2882. The argument makes no claim of one complex radius uniform in every order, and no such claim is needed by subsequent summation.

49.3 3. Exact extension, five rows, and common support

Multiply the inner ϕ_n, U_n by a fixed radial cutoff equal to one through X_{keep} and zero from X_{cut} . On I_{pos} , add two axial bumps and three angular bumps in $R = \sqrt{2X}$, with $dX = R dR$. The bumps are nonnegative with unit dR integral and ordered disjoint supports. The exact leading law there is

$$U_0 = 0, \quad E_0 = e_* f(\eta) R^{-1-2\lambda}, \quad f = (1 + \eta^2)^{-1}, \quad e_* = 2^{1/2+\lambda} c_{\text{patch}} > 0.$$

The last factor is the exact conversion from the unchanged X power law.

The five moments are

$$\begin{aligned} m_1 &= \int_0^\infty R U_n dR, & m_2 &= \int_0^\infty R^2 E_n dR, & m_3 &= \int_0^\infty \Pi_{n,R} dR, \\ m_4 &= \int_0^\infty R^2 \sum_{i+j=n} U_i E_j dR, & m_5 &= \int_0^\infty \left(R \sum_{i+j=n} U_i U_j - \frac{1}{2} R^2 \Pi_{n,R} \right) dR. \end{aligned}$$

Reconstruct V_n by (B1) and pressure with zero axis value by

$$\Pi_{n,R} = \frac{2E_0 E_n + \sum_{i=1}^{n-1} E_i E_{n-i} - \Omega_{n-1}}{R}.$$

With the lower orders fixed, the exact current-order increments are

$$\begin{aligned} \delta m_1 &= \int R \delta U_n dR, & \delta m_4 &= e_* f \int R^{1-2\lambda} \delta U_n dR, \\ \delta m_2 &= \int R^2 \delta E_n dR, & \delta m_3 &= 2e_* f \int R^{-2-2\lambda} \delta E_n dR, \\ & & \delta m_5 &= -e_* f \int R^{-2\lambda} \delta E_n dR. \end{aligned} \tag{B5}$$

There is no current-order quadratic remainder because $n > 0$ and $i + j = n$. Let B_U, B_E be the bump moment matrices with row powers $(1, 1 - 2\lambda), (2, -2 - 2\lambda, -2\lambda)$. Their determinants do not vanish: a sum of m distinct real powers has at most $m - 1$ positive zeros, by division by the least power and induction using Rolle's theorem. Thus the evaluation determinant on ordered nodes is continuous, never zero, and has one sign. Its integral against the ordered nonnegative bump product is the bump moment determinant, by multilinearity and Fubini, and has that same nonzero sign.

For pre-correction moments m^0 , the coefficient vectors are exactly

$$\alpha = -B_U^{-1}(m_1^0, m_4^0/(e_* f))^T, \quad \beta = -B_E^{-1}(m_2^0, m_3^0/(2e_* f), -m_5^0/(e_* f))^T. \tag{B6}$$

Thus every moment vanishes for every η . The inverse works for arbitrary finite discrepancies; no smallness condition is being assumed. The correction coefficients are smooth because $f \geq 1/2$ on the parameter interval. They do not affect parameter analyticity on $[0, a^2]$, since their supports lie after that interval and the earlier cutoff is independent of η . Reconstruction by forward integration leaves the original inner solution exactly unchanged through X_- . This closes induction on n .

Past the velocity bump supports, $F_n = m_1 = 0$, so $V_n = 0$. Choose a common X_+ after both those supports and the leading axial pulse, with $X_v < X_+ < X_b$. Every term of every Ω_k then vanishes there, including its shifted axial viscosity term. At positive total order every $\phi_i \phi_j$ has a positive-order factor, so $\Pi_{n,X} = 0$ there and $\Pi_n = m_3 = 0$. This proves common compact support of every reconstructed positive-order velocity, pressure and potential. Merely choosing X_+ after I_{pos} would not prove this pressure statement; the actual proof at reader 1280–1287 explicitly retains the leading-pulse requirement.

49.4 4. Exact integrated residual map and the edge term

The conservative angular and axial residual identities, including their cylindrical connection terms, are reader 1299–1303. Their integrated current-order terms are respectively

$$q^{3/2-A+\lambda_n} m_2, \quad q^{3/2-2A+\lambda_n} m_4, \quad q^{1-A+\lambda_n} m_1, \quad q^{1-2A+\lambda_n} m_5.$$

For the pressure contribution, integration by parts gives

$$\int_0^\infty R \Pi_n dR = -\frac{1}{2} \int_0^\infty R^2 \Pi_{n,R} dR.$$

Its two boundary terms are zero by axis smoothness and the proved compact pressure support. These moments vanish as functions of the parameter before any physical differentiation, so their t, z derivatives vanish as well. Axis parity and compact support eliminate the radial flux boundaries.

At order one the shifted angular viscosity uses the leading field. The exact needed integral is

$$\int_0^\infty r^2 (u_{\theta,0} - c_\infty (r^2/2)^{-A}) dr = 0. \quad (\text{B7})$$

It follows from the exact assembled renormalized moment by $r = \sqrt{q}R$. The subtracted power has integrable weight r^{1-2h} at zero, and the heat asymptotic difference and time derivative have integrable weight $O_h(r^{-1-2h})$ at infinity. The exterior heat field is independent of z ; all its remaining z derivatives have bounded radial support on compact physical parameter neighborhoods. Thus the required two axial differentiations of (B7) are legitimate and zero. The leading axial moment is already zero. For higher orders the same assertion uses $m_{n-1,2} = m_{n-1,1} = 0$.

Consequently the exact weighted residual totals are zero, and the stress is both the negative forward primitive and positive backward primitive:

$$\begin{aligned} T_{n,\theta} &= -R^{-2} \int_0^R \rho^2 r_{\theta,n} d\rho = R^{-2} \int_R^\infty \rho^2 r_{\theta,n} d\rho, \\ T_{n,z} &= -R^{-1} \int_0^R \rho r_{z,n} d\rho = R^{-1} \int_R^\infty \rho r_{z,n} d\rho. \end{aligned}$$

The support is $[X_-, X_+]$ for $n \geq 2$, and $[X_-, X_b]$ for $n = 1$. The latter reaches the exterior collar because of the leading angular axial-viscosity term, not because of an uncontrolled pressure constant.

For this single edge term, the exact flat primitive is

$$\int_0^\delta e^{-c/u^2} u^{-j} b(u, \eta) du = \frac{1}{2} e^{-c/\delta^2} \delta^{3-j} \int_0^\infty e^{-cv} (1 + \delta^2 v)^{(j-3)/2} b\left(\frac{\delta}{\sqrt{1 + \delta^2 v}}, \eta\right) dv. \quad (\text{B8})$$

It is obtained by $u = \delta(1 + \delta^2 v)^{-1/2}$. Every fixed derivative of the integral coefficient is dominated by an exponential times a polynomial in v , uniformly on the closed parameter rectangle. The coefficient is smooth, with boundary value $b(0, \eta)/c$. In the actual angular order-one source, the two terms carry $c = 4, j = 6$ and $c = 4, j = 3$; (B8) yields the exact permitted inverse-distance powers after the radial integral. The physical power is

$$q^{-A-2D+1/2} = q^{-A-1/2+2h}.$$

Hence $|\partial^I T_1| \leq C_I \zeta \delta^{-N_I}$. All T_n/ζ , $n \geq 2$, are smooth and have finite fixed derivative bounds because they have common compact support strictly within the stress annulus. The proof uses reader 2296–2318, not a generic claim that every flat function has a bounded quotient by ζ .

49.5 5. Exact Borel map and all fixed derivative bounds

The positive-order streamfunction and potential are

$$S_n = q^{1-A+2nh} F_n, \quad \mathcal{A}_n = (S_n/r) e_\theta = \frac{1}{2} q^{-A+2nh} (F_n/X)(-x_2, x_1, 0).$$

The equality $F_n/X = \int_0^1 U_n(tX, \eta) dt$ proves Cartesian axis smoothness. Their curls give the original radial and axial velocity exactly. Apply the cutoff to this potential. Direct differentiation, retaining the original sign $ru_r = -\partial_z S$, gives

$$\begin{aligned} ru_{r,n}^{\text{cut}} &= q^{2nh} \left[\chi(c_n q) V_n - \frac{2\eta}{L} (c_n q) \chi'(c_n q) F_n \right], \\ u_{z,n}^{\text{cut}} &= q^{-A+2nh} \chi(c_n q) U_n, \quad u_{\theta,n}^{\text{cut}} = q^{-A+2nh} \chi(c_n q) E_n. \end{aligned} \quad (\text{B9})$$

Every potential curl is divergence free, and every axisymmetric angular cutoff field is divergence free. Local finiteness at $q > 0$ therefore proves exact incompressibility even in cutoff transition regions.

The derivative loss is independent of n . A fixed physical derivative of a term $q^b g$ costs $q^{-|\beta|/2-Dk-\ell}$ for $\partial_{x_\perp}^\beta \partial_z^k \partial_t^\ell$; powers of b enter only its finite constant. A differentiated cutoff contributes $c_n^j \chi^{(j)}(c_n q)$ together with q^j from the differentiated q 's. On its support $1/2 \leq c_n q \leq 1$, these factors are bounded uniformly in c_n . Thus the extra curl derivative also has an order-independent physical loss.

At step n , choose $c_n \geq 2c_{n-1}$ large enough for the finite list of physical and normalized coefficient seminorms through order n , including the bracket in (B9) and T_n/ζ for $n \geq 2$. After retaining the original

q^{2nh} , make each such term at most $2^{-n}q^{nh}$. This is possible because q^{nh} dominates every fixed power of $|\log q|$ near zero. The choices fix one sequence; no choice is made after a derivative order is requested.

For any fixed derivative order m , the finitely many terms below the tail retain their original powers at least q^{2h} . Start the tail at $n \geq \max(2, m+2)$. For $0 < q \leq 1$,

$$\sum_{n \geq \max(2, m+2)} 2^{-n} q^{nh} \leq 2q^{2h}.$$

This proves all fixed normalized derivative estimates

$$\begin{aligned} \widehat{E} - E_0 &= O(q^{2h}), & \widehat{U} - U_0 &= O(q^{2h}), \\ q^A u_r - \frac{q^h}{\sqrt{2X}} V_0 &= O(q^{3h}), & |\partial^I(\widehat{T} - T_0)| &\leq C_I q^{2h} \zeta \delta^{-N_I}. \end{aligned} \quad (\text{B10})$$

In the radial estimate, the additional cutoff term in (B9) is included. Separating the single order-one edge term proves the stress estimate. The diagonal choice is made on one rectangle containing all common correction supports; larger rectangles add only the exact leading heat field. This is why the construction is simultaneous on every compact profile range.

For completeness, the retained coefficient equations give finite augmented residual bound

$$|F_{\text{slow}}(U^{[N]})|_m \leq C_{N,m} q^{2h(N+1)-(2A+1/2+m)}.$$

The Borel tail after $J \geq \max(1, m)$ has bound $2^{-J} q^{h(J+1)-\ell'_m}$ near zero, with ℓ'_m independent of J . For a fixed desired derivative and decay power, choose one finite J so both exponents dominate all losses in the finite differential-polynomial residual comparison. Then decrease the neighborhood of zero to absorb the fixed finite constants and logarithms. This proves flat augmented residual. It does not exchange the $J \rightarrow \infty$ limit with an uncontrolled family of fixed-stage remainders.

49.6 6. The exact chart morphism into the pulse hypotheses

For a fixed band, retain

$$Q = 2^{-\ell}, \quad \varepsilon = Q^h, \quad S_* = \ell^2, \quad R = r/\sqrt{Q}, \quad Z = z/Q^D, \quad T = (1-t)/Q.$$

Set $s_Q = q/Q$. The exact chart/profile map is

$$Z = s_Q^D \eta, \quad T = s_Q(1 - \eta^2), \quad X = R^2/(2s_Q), \quad R_{\text{profile}} = R/\sqrt{s_Q}. \quad (\text{B11})$$

On the fixed enlarged annulus and $1/2 \leq s_Q \leq 2$, its first derivatives have denominators only in s_Q, L, R , bounded below by $1/2, 1 - 2h, \sqrt{X_a}$. Repeated differentiation gives finite sums of the same type, so all fixed derivative bounds are uniform in the band. The complete calculation is reader 7713–7756. It proves a map of derivative classes, not equality of the two radius coordinates.

Let b, V, G be the actual chart velocity $Q^A u$, in radial, angular, axial order. The exact leading tangential chart coefficients are

$$V^{(0)} = s_Q^{-A} E_0(X, \eta), \quad G^{(0)} = s_Q^{-A} U_0(X, \eta),$$

while

$$b^{(0)} = \varepsilon V_0(X, \eta)/R.$$

Inserting (B9)–(B10) in these formulas yields, at every fixed slow derivative,

$$b = O(\varepsilon), \quad b - b^{(0)} = O(\varepsilon^3), \quad V - V^{(0)} = O(\varepsilon^2), \quad G - G^{(0)} = O(\varepsilon^2). \quad (\text{B12})$$

The auxiliary derivatives are zero because the background is auxiliary independent. These bounds do not need a polynomial S_* loss: the profile and compact chart maps are fixed. The weaker allowance $O(\varepsilon S_*^B)$ at reader 10077 is therefore consistent but does not weaken the $O(\varepsilon)$ input used by the phase proof at 8213.

For stress, the exact scale is

$$\Sigma = Q^{2A} q^{-A-1/2} \widehat{T}, \quad Q^{2A} q^{-A-1/2} = \varepsilon(Q/q)^{A+1/2}.$$

Since $q^{2h} = \varepsilon^2 s_Q^{2h}$, (B10) gives the actual normalized stress difference of mean class $\mathcal{M}_3$, including the ζ, δ factors. No scale ratio is replaced by one. The leading compact cone and its derivative bounds pass through (B11); this is the concrete input needed for the frozen representative, the plane isomorphism and both covariance columns.

49.7 7. Exact preservation of the reserved interval

The entire reserved mean interval, including gaps between the later bumps, matters because the induced streamfunction can be nonzero in those gaps. The actual outer schedule gives

$$\mathfrak{m} = eX_R \left(4 + \int_0^{T_d} k(v)e^v dv \right) > 0, \quad M(X, \eta) = \mathfrak{m}\eta$$

throughout I_4 . This follows by integrating $U = 4\eta$ to eX_R , then $U = k(v)\eta$ with $X = eX_R e^v$, and then $U = 0$ to I_4 . The two exact moment restorations preserve M and all its parameter derivatives at every later radius; the heat edits have zero axial increment. Thus (B1) at zero order gives

$$V_0 = -\mathfrak{m} \frac{2D\eta^2 + 1 - \eta^2}{1 - 2h\eta^2} = -\mathfrak{m}.$$

The final equality uses $2D = 1 - 2h$. Every positive-order E_n, U_n, F_n, V_n is identically zero throughout this interval, by (B5)–(B6) and the strict ordering $I_3 < I_4$. Every term in (B9), including its derivative-of-cutoff term, is therefore zero there, for all parameter values.

Accordingly the actual summed chart background has the exact full-interval law

$$b = -\varepsilon\mathfrak{m}/R, \quad V = a(\eta)s_Q^{\lambda-h}R^{-1-2\lambda}, \quad G = 0,$$

where $a(\eta) = 2^{1/2+\lambda}c_{\text{patch}}(1+\eta^2)^{-1}$. This proves the advertised five-dimensional mean matrix is evaluated at the actual Borel background. Positive-order pressure has not been asserted zero on I_4 ; it is not part of the matrix input. The exact proof occupies reader 9042–9254.

49.8 8. Bounded disposition

The base-side chain closes for the actual constructed profile: the analytic collar, zero totals, strict cone, reserved intervals, heat exterior, and flat edge factors are supplied by the completed leading-profile assembly; the recursion and five-row correction construct every positive order on a common radial interval; the cutoff potential preserves exact incompressibility and the reserved mean law; and the diagonal estimates give every fixed derivative required by the phase/pulse proof.

There is one precise pulse interpretation repair, described in `pulse_check.md`: the actual shear g differs by $O(\varepsilon^2)$ from its leading value at the representative. Therefore the production term after the exact energy identity is $-|g_0|T_N - (g - g_0) \cdot T$, not literally $-|g_0|T_N$. This is a missing term in the explanatory equality, not in the displayed energy equation or the pulse ODE. Its controlled contribution must be retained when asserting positivity. The independent pulse note proves that contribution does not destroy the positive lower production bound and verifies the pulse estimates entering covariance.

No conditional theorem, unresolved analytic assumption, or Lean result is used to supply this bounded base-to-pulse chain. The parent retains responsibility for the later stages outside the scope stated at the beginning.

50 Full phase, pulse, inverse and covariance audit

50.1 1. Frozen labels and rounded carrier

Keep $Q = 2^{-\ell}$, $\varepsilon = Q^h$, $S_* = \ell^2$, $0 < h < 1/100$, and each label fixed under the continuous derivatives. The geometry gives

$$\frac{1}{T_g S_*} < c_i \leq \frac{1}{S_*}, \quad L_s = \frac{2r_0}{c_i}, \quad 2r_0 S_* \leq L_s < 2r_0 T_g S_*.$$

The original $F = V/R$, $g = (RF_R, G_R)$, and compact positive annulus make all fixed slow derivatives of F, g bounded. Let the leading shear at the representative be $g_0 = F_0(-a, b_s) = |g_0|N$, and retain $K = (-N_z, N_\theta)$, $t_s = -b_s/a$, $v_s = a(1 + t_s^2)$. Substituting

$$N = -\frac{(1, t_s)}{\sqrt{1 + t_s^2}}, \quad K = \frac{(t_s, -1)}{\sqrt{1 + t_s^2}}$$

in the original formula gives

$$\lambda_0^2 = 2aF_0^2(1 - 2/v_s) > 0, \quad c_0 = \frac{\lambda_0}{2F_0N_\theta} < 0, \quad c_0^2 = (v_s - 2)/2.$$

All denominator and spectral lower bounds therefore follow from the given compact leading cone. The quantities $F_0, g_0, N, K, \lambda_0, c_0, B_s, p, p_z, L_s$ are label constants; none is differentiated through a moving representative or a rounding rule.

For $k = \lceil \varepsilon^{-1/2} \rceil$, $1 \leq \varepsilon k^2 \leq 4$ and $k^{-1} \leq \sqrt{\varepsilon}$. A nearest nonzero integer j to $k\tilde{p}$ satisfies $|j - k\tilde{p}| \leq 1$, including $k\tilde{p} = 0$. Thus $p = j/k \neq 0$ and $|p - \tilde{p}| \leq k^{-1}$. This choice both controls the perturbation and supplies the exact angular identity used below.

With the source's original constants,

$$B_s^2 = \frac{\lambda_0}{\varepsilon k^2(1 + u_*^2)^{3/2}}, \quad (\tilde{p}/R_0, p_z) = B_s \left(K - \frac{\sigma u_* g_0}{L_s |g_0|^2} \right),$$

the scalar product of the last vector with g_0 is exactly $-\sigma B_s u_*/L_s$. At a varying slow point,

$$(H_\Phi)_R = (p/R, p_z) \cdot g, \quad H_\Phi = pF + p_z G.$$

The three differences are: $R^{-1} - R_0^{-1} = O(S_*^{-3})$, the leading shear variation $O(S_*^{-3})$ plus actual shear difference $O(\varepsilon^2)$, and the integer rounding $O(k^{-1})$. All multiplying factors are uniformly bounded. Consequently

$$\left| (H_\Phi)_R + \frac{\sigma B_s u_*}{L_s} \right| \leq C(S_*^{-3} + \varepsilon^2 + k^{-1}).$$

50.2 2. Exact gradient and transport defect

The affine pulse coordinate obeys $D_r v = D_z v = 0$, $\mathbf{t}_* v = 1$, while $\mathbf{t}_* = -\varepsilon \partial_T + c_i N_i$. Directly differentiating the full phase, with its axial inverse power retained, gives

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v H_\Phi, \quad n_\Phi = (x_0 - v(H_\Phi)_R, p/R, p_z - \varepsilon v(H_\Phi)_Z).$$

The target radial component is $B_s s(v) = \sigma B_s(u_*/2 + u_* v/L_s)$. The radial error is bounded by $CL_s(S_*^{-3} + \varepsilon^2 + k^{-1})$. The tangential errors are bounded by $C(S_*^{-1} + S_*^{-3} + k^{-1} + \varepsilon L_s)$. One fixed lower bound on ℓ makes

$$S_*^2(\varepsilon + \varepsilon^2 + k^{-1}) \leq 1,$$

because every power of ℓ multiplied by $2^{-h\ell/2}$ tends to zero. Hence

$$|n_\Phi - B_s(s(v), K)| \leq C/S_*, \quad n'_\Phi = (-(H_\Phi)_R, 0, -\varepsilon(H_\Phi)_Z) = O(S_*^{-1}).$$

Each fixed coefficient derivative of these expressions is polynomially bounded in S_* . The term $p_z Z/\varepsilon$ cancels exactly with the $\varepsilon \partial_Z$ in the gradient; it has not been removed from the phase or the physical exponential.

The term-by-term transport identity is

$$\mathbf{t}_* \Phi = -H_\Phi + \varepsilon v(H_\Phi)_T, \quad F \partial_\theta \Phi + G D_z \Phi = H_\Phi - \varepsilon v G(H_\Phi)_Z,$$

and therefore

$$E_{ik} = \varepsilon v((H_\Phi)_T - G(H_\Phi)_Z) + b n_{\Phi, r}.$$

The precise product estimate is

$$D^I(b n_{\Phi, r}) = \sum_{J \leq I} \binom{I}{J} (D^J b)(D^{I-J} n_{\Phi, r}) = O_I(\varepsilon S_*^{d_I}).$$

For example, $\partial_R n_{\Phi, r} = -v(H_\Phi)_{RR}$ can be $O(S_*)$; a derivative-free $O_I(\varepsilon)$ claim for the entire product is not justified by the stated hypotheses. The exact product estimate, together with the other displayed term, proves the required $D^I E_{ik} = O_I(\varepsilon S_*^{b_I})$, on the same fixed domain.

50.3 3. Pressure and the exact plane morphism

For the real normal $n = n_\Phi$, retain

$$\mathcal{K} = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad d = \varepsilon k^2 |n|^2, \quad \mathcal{A} = -\mathcal{K} + \frac{n(n^T \mathcal{K} - (n')^T)}{|n|^2}.$$

Normal multiplication of the full equation, using $(n \cdot t)' = 0$, gives the unique pressure

$$\pi_m = \frac{i}{km} \frac{n \cdot \mathcal{K} t_m - n' \cdot t_m + n \cdot f_m}{|n|^2}.$$

Its sign is $+i/(km)$: multiplication by $ikmn$ produces minus the displayed normal vector. Substitution gives

$$t'_m = \mathcal{A} t_m - m^2 dt_m - \text{proj}_{n^\perp} f_m, \quad (n \cdot t_m)' = -m^2 d(n \cdot t_m).$$

The initially zero constraint therefore persists and the projected/full equations are equivalent, with normal forcing included.

The uniform lower bound for $|n_{\text{tan}}|$ follows from the phase comparison and $|B_s K| = B_s \geq c > 0$. Put $K_a = n_{\text{tan}}/|n_{\text{tan}}|$, $N_a = ((K_a)_z, -(K_a)_\theta)$, $s_a = n_r/|n_{\text{tan}}|$,

$$U = [e_r - s_a K_a, N_a], \quad M = \begin{pmatrix} 1 & 1 \\ c_0 \sqrt{1+s^2} & -c_0 \sqrt{1+s^2} \end{pmatrix}, \quad B = UM, \quad B^\ell = M^{-1} \begin{pmatrix} e_r^T \\ N_a^T \end{pmatrix}.$$

Both columns of U lie in $n^\perp$, the indicated row matrix times U is I_2 , and $\det M = -2c_0 \sqrt{1+s^2} \neq 0$. Thus $B : \mathbb{C}^2 \rightarrow n^\perp$ is an exact isomorphism, $B^\ell B = I_2$, and $BB^\ell t = t$ for $t \in n^\perp$.

For completeness the coordinate converse also follows exactly. Differentiate $n^T B = 0$ to obtain $n^T B' = -n'^T B$. Since $n^T \mathcal{A} B = -n'^T B$, every column of $\mathcal{A} B - B'$ lies in $n^\perp$. The damping and projected forcing lie in this plane too. Applying BB^ℓ therefore recovers their original vectors. Thus solving the two-dimensional coordinate equation and setting $t = Bz$ really solves the three-dimensional constrained equation; no right-inverse assertion outside $n^\perp$ is used.

At the reference normal the matrix in the $(e_r - sK, N)$ basis is

$$\begin{pmatrix} 0 & 2F_0 N_\theta / (1+s^2) \\ -(2F_0 N_\theta + |g_0|) & 0 \end{pmatrix}.$$

Multiplication verifies that the columns of M have respective eigenvalues $\lambda_0/\sqrt{1+s^2}$ and its negative. The actual-frame and normal-motion perturbations are $O(S_*^{-1})$, giving precisely

$$B^\ell (\mathcal{A} B - B') = \text{diag}(\lambda, -\lambda) + E, \quad |E| \leq C/S_*, \quad |d - d_{\text{ref}}| \leq C/S_*.$$

Their fixed derivatives are polynomially bounded by ordinary product and quotient rules, with the already established uniform denominators.

50.4 4. Forward inverse and every fixed derivative

The original envelope has $P' = a_{\text{net}} P$, $P(L_s/2) = 1$, with

$$a_{\text{net}} = \frac{\lambda_0}{\sqrt{1+s^2}} - \frac{\lambda_0(1+s^2)}{(1+u_*^2)^{3/2}} = \lambda - d_{\text{ref}}.$$

Direct differentiation gives $-C/L_s \leq a'_{\text{net}} \leq -c/L_s$ on the full pulse interval. Both $\log P$ and its first derivative vanish at the midpoint. Integrating the second derivative on either side proves

$$e^{-C_P(v-L_s/2)^2/L_s} \leq P(v) \leq e^{-c_P(v-L_s/2)^2/L_s} \leq 1.$$

For $A_m = \text{diag}(\lambda, -\lambda) + E - m^2 dI_2$, the real part of $z^* A_1 z$ is at most $(\lambda - d_{\text{ref}} + C/S_*)|z|^2$. Hence its forward propagator satisfies $\|V_1(v, w)\| \leq CP(v)/P(w)$. The scalar commutation gives exactly

$$V_m(v, w) = \exp \left(-(m^2 - 1) \int_w^v d(a) da \right) V_1(v, w).$$

This is verified by differentiation and the initial identity matrix at $v = w$. For every nonzero integer m the scalar exponential is at most one; the value estimate is therefore uniform over all nonzero harmonics.

Existence of the propagator can, without a further hypothesis, be obtained by the time-ordered series $I + \sum_{j \geq 1} \int_{w < a_j < \dots < a_1 < v} A_m(a_1) \cdots A_m(a_j) da_j \cdots da_1$. On a fixed label and finite m , boundedness of A_m by K_m bounds term j by $K_m^j (v - w)^j / j!$. Uniform convergence and the same bound after one v -derivative prove the differential equation. The norm inequality above gives uniqueness, so this series is the propagator used in the formulas.

For $g_m = -B^\ell \text{proj}_{n^\perp} f_m$, the zero-data solution is

$$z_m(v) = \int_0^v V_m(v, w) g_m(w) dw.$$

At fixed slow and transverse point, X, ζ, δ are constant in this integral. The $P(w)$ in the source cancels $P(w)^{-1}$ in the propagator. The only interval-length factor is $L_s = O(S_*)$, proving the stated value weight.

Use slow variables and the initial transverse coordinate as parameters $\mathbf{p}$, with v held fixed. Differentiation gives exactly

$$(\partial_{\mathbf{p}}^I z_m)' = A_m \partial_{\mathbf{p}}^I z_m + \partial_{\mathbf{p}}^I g_m + \sum_{0 < J \leq I} \binom{I}{J} (\partial_{\mathbf{p}}^J A_m) \partial_{\mathbf{p}}^{I-J} z_m.$$

All initial parameter derivatives are zero. For fixed $|m| \leq M$, $|\partial^J A_m| \leq C_{J,M} S_*^{c_J}$. Induction on $|I|$, the same forward estimate, and $0 < \delta \leq 1$ give a bound $C_{I,M} \varepsilon^\alpha S_*^{b'_I} \sqrt{\zeta} \delta^{-a'_I} P(v)$, choosing a'_I at least every exponent already present and $b'_I \geq 1 + \max\{b_I, \max_{0 < J \leq I} (c_J + b'_{I-J})\}$. Thus no exponential propagator is differentiated in isolation and no positive power of ε is lost.

Pulse derivatives follow by differentiating $z' = A_m z + g_m$; each fixed mixed derivative is a finite sum of coefficient/source derivatives and already controlled derivatives. The common-torus map is affine:

$$H_w = H + T_g^{-\Delta} c_i (w - v(H)) v_t, \quad D_H v = T_g^\Delta \lambda_t / c_i = O(S_*), \quad D_H H_w = I - v_t \lambda_t,$$

and higher derivatives vanish. Consequently conversion to the stated coefficient derivatives in (R, Z, T, H) introduces only polynomial powers. Multiplication by B preserves the required weights. The pressure formula introduces the additional $1/(km) \leq 1/k \leq \sqrt{\varepsilon}$, while every other factor has the proved fixed-derivative bounds. This establishes exactly the velocity exponent α and pressure exponent $\alpha + 1/2$.

These inductions impose no additional smallness inequalities. The single threshold is determined by the finite value comparisons, tangential-normal and frame denominators, and the fixed invariant-band threshold below.

50.5 5. Support, smoothness, and descent

The path fixes the slow and transverse coordinates. Outside the specified closed slow/transverse support the source is zero for the whole path; the Duhamel integral is zero. Its parameter-differentiated equations give zero jets at the boundaries. At the active-shell edges every bound $\sqrt{\zeta} \delta^{-a}$ tends to zero faster than every power of edge distance, and the same is true after each fixed derivative, proving smooth zero extension. This argument concerns exactly the three support variables specified in the original proposition; it does not assert that a source's temporal support is preserved by a forward ODE.

The covering deck translations reindex the lifted rectangle copies and commute with the path. Their pullbacks obey the same ODE with the same pulled-back data, so uniqueness proves descent to the common torus even when the source differs at distinct preimages of a band point. On the prescribed fixed-factor rectangle enlargement, $|v| \leq C_{\text{enl}} L_s$. The tangential phase comparison above consequently remains $n_{\Phi, \text{tan}} = B_s K + O(S_*^{-1})$: replacing $v \leq L_s$ by this bound changes only the fixed constant in the axial error $\varepsilon v (H_\Phi)_Z$. Hence the normal and frame denominators stay uniformly nonzero there after the same type of finite threshold choice. The coefficients and source are smooth on that enlargement by the stated input, and the finite-dimensional integral equation extends the solution there. For $v < 0$, its successive-integral construction uses the signed integrals from zero to v ; their absolute values are bounded by the same factorial series with $|v|$ in place of v . Its differentiated equations prove smooth dependence on the whole enlargement. At compact one-sided slow endpoints with $q > 0$, these parameter equations transfer the prescribed background/source derivatives to the solution and pressure.

50.6 6. Homogeneous lower bound and exact energy production

For real $m = 1, f = 0$, take $z_+(0) = P(0) > 0, z_-(0) = 0$. The actual ratio satisfies

$$r' = E_{21} + (-2\lambda + E_{22} - E_{11})r - E_{12}r^2.$$

At the boundaries $r = \pm K_b/S_*$, the outward derivative is at most

$$\frac{-2c_\lambda K_b + C(1 + K_b/S_* + K_b^2/S_*^2)}{S_*}.$$

A fixed sufficiently large K_b , followed by a fixed large- S_* threshold, makes both outward derivatives negative. Thus $|r| \leq K_b/S_*$ until a potential zero of z_+ . On this invariant band,

$$\left(\log \frac{z_+}{P}\right)' = -(d - d_{\text{ref}}) + E_{11} + E_{12}r = O(S_*^{-1}).$$

Integration on $0 \leq v \leq L_s$ gives $cP \leq z_+ \leq CP$, which excludes that potential zero. Taking the threshold also so that $K_b/S_* \leq 1/2$ gives

$$x = z_+(1 + r), \quad y = c_0 \sqrt{1 + s^2} z_+(1 - r), \quad c_x P \leq x \leq C_x P, \quad \frac{y}{x} = c_0 \sqrt{1 + s^2} + O(S_*^{-1}).$$

The nonzero datum $P(0)$ is fixed with the label, so its positive-order parameter derivatives vanish. The order-zero term is controlled by $\|V_1(v, 0)\|P(0) \leq CP(v)$. The derivative induction above therefore applies without change to the homogeneous solution.

The exact actual energy equation is already correct in the source:

$$\frac{1}{2} \frac{d}{dv} |t|^2 = -g \cdot T - \varepsilon k^2 |n_\Phi|^2 |t|^2, \quad T = t_r(t_\theta, t_z).$$

Indeed the normal part of $\mathcal{A}t$ is orthogonal to t , and

$$t \cdot \mathcal{K}t = -2Ft_r t_\theta + (2F + RF_R)t_r t_\theta + G_R t_r t_z = g \cdot T.$$

At the representative the actual g , not merely its leading term, enters this identity. Writing $T = T_N N + T_K K$ gives the exact correction to the prose following that identity:

$$-g \cdot T = -|g_0|T_N - (g - g_0) \cdot T.$$

Here $g - g_0 = O(\varepsilon^2)$ at the representative, and $O(S_*^{-3} + \varepsilon^2)$ throughout its enlarged slow box. No equality $g = g_0$ was supplied or derived. For a prescribed stress in the compact cone, $-T_N \geq c_T |T|$ and $|g_0| \geq g_- > 0$. Consequently, for the fixed large-band threshold,

$$-g \cdot T \geq (g_- c_T - C(S_*^{-3} + \varepsilon^2))|T| \geq \frac{1}{2}g_- c_T |T|.$$

For the actual homogeneous pulse there is a still more specific relation. Since $t = x(e_r - s_a K_a) + yN_a$,

$$T = x^2 \left(-s_a K_a + \frac{y}{x} N_a \right) = x^2 \left[-sK + c_0 \sqrt{1 + s^2} N + O(S_*^{-1}) \right].$$

Thus, using the actual shear variation,

$$-g \cdot T = \left[-|g_0|c_0 \sqrt{1 + s^2} + O(S_*^{-1}) \right] x^2.$$

The leading coefficient is uniformly positive, since $c_0 < 0, |c_0| \geq c_- > 0$, and $|g_0| \geq g_- > 0$. It follows that $cP(v)^2 \leq -g \cdot T \leq CP(v)^2$ after a fixed threshold. Also

$$|t|^2 = (1 + s_a^2)x^2 + y^2,$$

so $cP(v)^2 \leq |t(v)|^2 \leq CP(v)^2$. Positive shear production is compatible with decay after the midpoint because the actual damping remains in the exact energy equation. Neither the equality repair nor these estimates require deleting a part of the actual background.

50.7 7. Exact transfer of the lower bound to covariance mass

The positive covariance mass used downstream comes from the proved $x \geq c_x P$, not from discarding damping in the energy equation. The angular frequency $kp \neq 0$ gives exactly $\langle \cos^2(k\Phi) \rangle_\theta = 1/2$. Haar preservation and the original rectangle Jacobian give

$$h_\sigma = D_g c_i I_0, \quad I_0 = \int_0^{L_s} \psi^2 x^2 dv, \quad D_g = \frac{1+b_g^2}{2} \int_{\mathbb{R}} \chi_g(\xi)^2 d\xi > 0.$$

The expanded reader at lines 9503–9554 explicitly proves the constants. For $L_s \geq 25$, the interval $|v - L_s/2| \leq \sqrt{L_s}$ lies where $\psi = 1$, giving

$$2c_x^2 e^{-2C_P} \sqrt{L_s} \leq I_0 \leq C_x^2 \sqrt{\frac{\pi}{2c_P}} \sqrt{L_s}.$$

The floor bound and $c_i \sqrt{L_s} = \sqrt{2r_0 c_i}$ therefore give the strictly positive original mass bounds

$$D_g (2c_x^2 e^{-2C_P}) \sqrt{2r_0/T_g} S_*^{-1/2} \leq h_\sigma \leq D_g C_x^2 \sqrt{\frac{\pi}{2c_P}} \sqrt{2r_0} S_*^{-1/2}.$$

Furthermore

$$\int_0^{L_s} \frac{|v - L_s/2|}{L_s} \psi^2 x^2 dv \leq \frac{C_x^2}{2c_P}.$$

Dividing by the positive I_0 bounds the corresponding mean absolute scaled displacement by $CL_s^{-1/2}$. Applying this probability measure to the exact flux polarization above, and using $|s(v) - \sigma u_*| = u_* |v - L_s/2|/L_s$, gives

$$H_\sigma = h_\sigma (-A_c N - \sigma u_* K + e_\sigma), \quad A_c = -c_0 \sqrt{1 + u_*^2} > 0, \quad |e_\sigma| \leq CS_*^{-1/2}.$$

This is the precise original-variable map from the homogeneous pulse lower bound to the two positive covariance columns. The reader includes its complete integral constants, retained angular half, retained Jacobian, retained $c_i dv$, and both original signs. All fixed slow derivatives of these finite integrals follow from the proved homogeneous derivative bounds, with label-fixed endpoints and cutoffs.

51 Actual shear and the full force-pressure correspondence

51.1 1. Original chart and the two contributions to the shear difference

Retain the source parameters and coordinates

$$0 < h < 1/100, \quad A = 1/2 + h, \quad D = 1/2 - h, \quad (AS1)$$

$$Q = 2^{-\ell}, \quad \varepsilon = Q^h, \quad S_* = \ell^2, \quad R = r/\sqrt{Q}, \quad Z = z/Q^D, \quad T = (1-t)/Q.$$

The value of h is the one selected by the preceding profile construction. Constants below may depend on that value and the fixed source profiles. No uniform threshold as $h \downarrow 0$ is asserted. Work on the original compact enlarged chart annulus, whose inner radius is denoted by $R_- > 0$. The label and its representative $p_0 = (R_0, Z_0, T_0)$ are held fixed under every continuous derivative.

Write $s_Q = q/Q$. The exact leading tangential chart fields and the actual Borel fields are

$$V^{(0)} = s_Q^{-A} E_0(X, \eta), \quad G^{(0)} = s_Q^{-A} U_0(X, \eta), \quad X = R^2/(2s_Q), \quad Z = s_Q^D \eta, \quad T = s_Q(1 - \eta^2), \quad (AS2)$$

$$\delta V = V - V^{(0)}, \quad \delta G = G - G^{(0)}. \quad (AS3)$$

The constructed base estimate (B12), on this fixed enlarged annulus, gives for each fixed slow multi-index I actual constants

$$|\partial^I \delta V| \leq C_{V,I} \varepsilon^2, \quad |\partial^I \delta G| \leq C_{G,I} \varepsilon^2. \quad (AS4)$$

The base is independent of the auxiliary torus variables. The source supplies (AS4) through its Borel and chart constructions; this note does not replace that construction by a hypothesis about arbitrary fields.

Define the actual, varying leading, and frozen shear coordinates by

$$\begin{aligned} F &= V/R, \quad F^{(0)} = V^{(0)}/R, \quad F_0 = F^{(0)}(p_0), \\ g &= (R\partial_R F, \partial_R G), \quad g^{(0)} = (R\partial_R F^{(0)}, \partial_R G^{(0)}), \quad g_0 = g^{(0)}(p_0). \end{aligned} \quad (\text{AS5})$$

Their exact difference is

$$\begin{aligned} \delta g &:= g - g^{(0)} = (\partial_R \delta V - \delta V/R, \partial_R \delta G), \\ \Delta g &:= g - g_0 = \delta g + g^{(0)}(p) - g^{(0)}(p_0), \\ \Delta F &:= F - F_0 = \delta V/R + F^{(0)}(p) - F^{(0)}(p_0). \end{aligned} \quad (\text{AS6})$$

At the representative the varying-leading differences in the second terms vanish exactly. The Borel differences in the first terms have not been shown to vanish, and are retained.

For $I = (i_R, i_Z, i_T)$, the full derivative formula is

$$\begin{aligned} \partial^I \delta g_\theta &= \partial^{I+e_R} \delta V - \sum_{j=0}^{i_R} \binom{i_R}{j} (-1)^j j! R^{-j-1} \partial^{I-j e_R} \delta V, \\ \partial^I \delta g_z &= \partial^{I+e_R} \delta G. \end{aligned} \quad (\text{AS7})$$

Indeed $\partial_R^j (R^{-1}) = (-1)^j j! R^{-j-1}$; applying the full product rule to the first line of (AS6) gives exactly (AS7). Consequently the respective constants in the bounds for the two entries are

$$C_{\theta, I} = C_{V, I+e_R} + \sum_{j=0}^{i_R} \binom{i_R}{j} j! R^{-j-1} C_{V, I-j e_R}, \quad C_{z, I} = C_{G, I+e_R}, \quad (\text{AS8})$$

and the Euclidean vector bound is $|\partial^I \delta g| \leq (C_{\theta, I}^2 + C_{z, I}^2)^{1/2} \varepsilon^2$. Every factor of the original radius and every derivative is present. The frozen difference Δg has a different derivative identity: for $|I| > 0$,

$$\partial^I \Delta g = \partial^I \delta g + \partial^I g^{(0)}. \quad (\text{AS9})$$

Thus its higher derivatives are not inferred to be $O(\varepsilon^2)$ from a small value difference on a narrow box.

Let the original box satisfy $|p - p_0| \leq C_{\text{box}} S_*^{-3}$, with the connecting segment inside the enlarged domain. Let L_g be a bound for the Euclidean operator norm of $D_p g^{(0)}$ there, and set $C_B = (C_{\theta, 0}^2 + C_{z, 0}^2)^{1/2}$. Integration along that segment gives

$$|\Delta g| \leq C_B \varepsilon^2 + L_g C_{\text{box}} S_*^{-3}. \quad (\text{AS10})$$

This proves the precise value bound used by the pulse audit. It retains the independent Borel and box-variation contributions.

51.2 2. Complete first-jet coordinate map and both inverses

For each original $R > 0$, regard the first radial jet difference as the four-coordinate vector $j = (v, v_R, w, w_R) \in \mathbb{C}^4$. The shear map and its retained fibre coordinates are

$$L_R j = \begin{pmatrix} -1/R & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} j, \quad \mathcal{J}_R j = (L_R j, v, w) \in \mathbb{C}^2 \oplus \mathbb{C}^2. \quad (\text{AS11})$$

They apply to $j = (\delta V, \partial_R \delta V, \delta G, \partial_R \delta G)$, without identifying a field value with its derivative. The complete inverse is

$$\mathcal{J}_R^{-1}(a, b, v, w) = (v, a + v/R, w, b). \quad (\text{AS12})$$

Substitution gives $(a + v/R) - v/R = a$, leaves the fourth entry b , and recovers v, w . Conversely, substituting $a = v_R - v/R$ and $b = w_R$ into (AS12) returns the original four entries. This proves both inverse compositions. In particular L_R is onto, has right inverse $(a, b) \mapsto (0, a, 0, b)$, and has exactly the kernel

$$\ker L_R = \{(v, v/R, w, 0) : v, w \in \mathbb{C}\}. \quad (\text{AS13})$$

All these maps are smooth in the original positive radius. The entire jet has been retained in the bijection; a small shear difference does not erase the two fibre coordinates.

51.3 3. The same actual phase in the two background operators

Use the source's actual phase Φ , actual nonzero normal $n = n_\Phi$, and its derivative $n' = \partial_v n$. They are fixed as the same functions on both sides of the following comparison. Retain the original harmonic $m \in \mathbb{Z} \setminus \{0\}$, source choice $k = \lceil \varepsilon^{-1/2} \rceil \geq 1$, and damping $d = \varepsilon k^2 |n|^2$. In particular $km \neq 0$ on the original domain of every pressure formula below. Define

$$\mathcal{K}(F, g) = \begin{pmatrix} 0 & -2F & 0 \\ 2F + g_\theta & 0 & 0 \\ g_z & 0 & 0 \end{pmatrix}, \quad \mathcal{K}_0 = \mathcal{K}(F_0, g_0), \quad \Pi_n = I - \frac{nn^T}{|n|^2}. \quad (\text{AS14})$$

Their exact matrix difference is

$$\Delta\mathcal{K} = \begin{pmatrix} 0 & -2\Delta F & 0 \\ 2\Delta F + \Delta g_\theta & 0 & 0 \\ \Delta g_z & 0 & 0 \end{pmatrix}. \quad (\text{AS15})$$

The comparison operator uses frozen background entries and the actual phase. The ideal reference phase $B_s(s(v), K)$ is not silently substituted for n .

The complex constrained equation is

$$t' + \mathcal{K}t + m^2 dt + ikmn\pi = -f, \quad n^T t = 0. \quad (\text{AS16})$$

Differentiating the constraint gives $n^T t' = -n'^T t$. Multiplying the first equation by n^T and solving the scalar equation gives the unique pressure

$$\pi = \frac{i}{km} \frac{n^T \mathcal{K}t - n'^T t + n^T f}{|n|^2}. \quad (\text{AS17})$$

Substituting this value, including $i^2 = -1$, gives

$$t' = \mathcal{A}t - m^2 dt - \Pi_n f, \quad \mathcal{A} = -\mathcal{K} + \frac{n(n^T \mathcal{K} - n'^T)}{|n|^2}. \quad (\text{AS18})$$

The same formula with $\mathcal{K}_0$ gives $\mathcal{A}_0$, and direct subtraction proves

$$\mathcal{A} - \mathcal{A}_0 = -\Pi_n \Delta\mathcal{K}. \quad (\text{AS19})$$

The normal-motion term cancels because it is the same actual term, not because it is zero.

51.4 4. Exact force–pressure bijection on the full constrained solution data

For each datum (t, f, π) satisfying (AS16), set

$$t_0 = t, \quad f_0 = f + \Pi_n \Delta\mathcal{K}t, \quad \pi_0 = \pi - \frac{i}{km} \frac{n^T \Delta\mathcal{K}t}{|n|^2}. \quad (\text{AS20})$$

The data satisfy the equation with $\mathcal{K}_0$, the same normal and damping, and right side $-f_0$. To check the complete equation, calculate

$$\begin{aligned} t'_0 + \mathcal{K}_0 t_0 + m^2 dt_0 + ikmn\pi_0 &= -f - \Delta\mathcal{K}t + n \frac{n^T \Delta\mathcal{K}t}{|n|^2} \\ &= -f - \Pi_n \Delta\mathcal{K}t = -f_0. \end{aligned} \quad (\text{AS21})$$

The constraint remains $n^T t_0 = 0$. Conversely, the exact inverse on the full solution data is

$$t = t_0, \quad f = f_0 - \Pi_n \Delta\mathcal{K}t_0, \quad \pi = \pi_0 + \frac{i}{km} \frac{n^T \Delta\mathcal{K}t_0}{|n|^2}. \quad (\text{AS22})$$

Both compositions return every component because the unchanged vector t appears in equal and opposite increments. Substitution in (AS21) also proves the converse equation, so this is a bijection of the constrained solution-data spaces, including their pressure. For the actual homogeneous pulse $f = 0$, the comparison force is exactly $\Pi_n \Delta\mathcal{K}t$; it is not asserted zero. Smoothness follows from the explicit expressions and the source's positive normal denominator. The force increment can have support wherever t and the background difference overlap; no unsupported support equality is claimed.

51.5 5. Exact real energy production and an explicit finite positivity threshold

Restrict now to the original real homogeneous pulse, $m = 1$ and $f = 0$. Put

$$T = t_r(t_\theta, t_z) \in \mathbb{R}^2. \quad (\text{AS23})$$

Multiplication of the original matrix gives

$$t^T \mathcal{K} t = -2F t_r t_\theta + (2F + g_\theta) t_r t_\theta + g_z t_r t_z = g \cdot T. \quad (\text{AS24})$$

Because $n^T t = 0$, the normal term in (AS18) has zero inner product with t . Thus the exact energy identity is

$$\frac{1}{2} \partial_v |t|^2 = -g \cdot T - d|t|^2. \quad (\text{AS25})$$

Let $N = g_0/|g_0|$, with the nonzero frozen shear supplied by the compact cone, and $K = (-N_z, N_\theta)$. Then (N, K) is an orthonormal ordered basis in the original (θ, z) plane. Writing $T = T_N N + T_K K$, the exact replacement for the disputed sentence is

$$-g \cdot T = -|g_0| T_N - \Delta g \cdot T. \quad (\text{AS26})$$

In particular (AS26) is still the formula at p_0 , where $\Delta g = \delta g$. Equation (AS24) with the matrix difference also gives $t^T \Delta \mathcal{K} t = \Delta g \cdot T$: both $2\Delta F$ contributions cancel algebraically. No component of $\Delta \mathcal{K}$ is removed from (AS20).

For the actual prescribed source stress cone, write its positive compact margin as $-T_N \geq c_T |T|$, and write $|g_0| \geq g_- > 0$. The Cauchy-Schwarz inequality and (AS10) give

$$-g \cdot T \geq [g_- c_T - C_B 2^{-2h\ell} - L_g C_{\text{box}} \ell^{-6}] |T|. \quad (\text{AS27})$$

This includes $T = 0$, without division by its norm. One explicit sufficient integer threshold is

$$\ell \geq \left\lceil \max \left\{ 1, \left(\frac{4L_g C_{\text{box}}}{g_- c_T} \right)^{1/6}, \frac{\log \max\{1, 4C_B/(g_- c_T)\}}{2h \log 2} \right\} \right\rceil. \quad (\text{AS28})$$

The first power bound makes the box term at most $g_- c_T/4$; the logarithmic bound makes the Borel term at most the same number. Hence (AS27) gives the proved bound $-g \cdot T \geq (g_- c_T/2) |T|$. The threshold is added to the already selected finite source value thresholds, never substituted for them.

51.6 6. Quantitative production for the actual homogeneous pulse

For that pulse use its actual frame

$$K_a = n_{\text{tan}}/|n_{\text{tan}}|, \quad N_a = ((K_a)_z, -(K_a)_\theta), \quad s_a = n_r/|n_{\text{tan}}|, \quad t = x(e_r - s_a K_a) + y N_a. \quad (\text{AS29})$$

The source's fixed-label reference has $B_s \geq B_- > 0$, $s(v) = \sigma(u_*/2 + u_* v/L_s)$, and $|s(v)| \leq s_{\text{max}} = 3u_*/2$ on $0 \leq v \leq L_s$. Write the actual proved phase and ratio bounds with constants as

$$|n - B_s(s, K)| \leq E_n/S_*, \quad |y/x - \gamma_0| \leq E_\gamma/S_*, \quad \gamma_0 = c_0 \sqrt{1 + s^2}, \quad c_0 < 0, \quad c_x P \leq x \leq C_x P. \quad (\text{AS30})$$

These are the constructed pulse bounds; their invariant-band and propagator proofs were read in the complete pulse audit. Choose $S_* \geq 2E_n/B_-$, so $|n_{\text{tan}}| \geq B_-/2$. For any nonzero vectors a, b , direct addition and subtraction of $a/|a| - b/|b| \leq 2|a - b|/|b|$. Applying this to $a = n_{\text{tan}}$, $b = B_s K$, and using (AS30), gives

$$|K_a - K| = |N_a - N| \leq \frac{2E_n}{B_- S_*}, \quad |s_a - s| \leq \frac{2(1 + s_{\text{max}})E_n}{B_- S_*}. \quad (\text{AS31})$$

The second inequality follows by writing the numerator as $n_r - s|n_{\text{tan}}| = (n_r - B_s s) + s(B_s - |n_{\text{tan}}|)$, with denominator $|n_{\text{tan}}|$.

Let c_+ bound $|c_0|$, let $c_- > 0$ bound it below, and set

$$\Gamma_{\text{max}} = c_+ \sqrt{1 + s_{\text{max}}^2}, \quad C_T = \frac{2E_n}{B_-} (1 + 2s_{\text{max}} + \Gamma_{\text{max}}) + E_\gamma. \quad (\text{AS32})$$

The exact stress has the form

$$T = x^2[-sK + \gamma_0 N + e_T], \quad e_T = -(s_a K_a - sK) + (y/x)N_a - \gamma_0 N, \quad |e_T| \leq C_T/S_*. \quad (\text{AS33})$$

For this bound, expand $s_a K_a - sK = (s_a - s)K_a + s(K_a - K)$ and $(y/x)N_a - \gamma_0 N = (y/x - \gamma_0)N_a + \gamma_0(N_a - N)$, then use (AS30)–(AS32). Every summand is retained.

Let g_+ bound $|g_0|$. The exact production is

$$\frac{-g \cdot T}{x^2} = -|g_0|\gamma_0 - g_0 \cdot e_T - \Delta g \cdot (-sK + \gamma_0 N + e_T). \quad (\text{AS34})$$

The last two terms have total absolute value at most

$$\mathcal{E}_\ell = g_+ C_T \ell^{-2} + (C_B 2^{-2h\ell} + L_g C_{\text{box}} \ell^{-6}) (s_{\max} + \Gamma_{\max} + C_T \ell^{-2}). \quad (\text{AS35})$$

This is a concrete nonnegative expression decreasing to zero. Set $W = s_{\max} + \Gamma_{\max} + C_T > 0$. In addition to the prior source and (AS31) thresholds, it suffices to choose

$$\ell \geq \left\lceil \max \left\{ 1, \left(\frac{4g_+ C_T}{g_- c_-} \right)^{1/2}, \left(\frac{8W L_g C_{\text{box}}}{g_- c_-} \right)^{1/6}, \frac{\log \max\{1, 8W C_B/(g_- c_-)\}}{2h \log 2} \right\} \right\rceil. \quad (\text{AS36})$$

The three resulting error contributions are at most $g_- c_-/4$, $g_- c_-/8$, and $g_- c_-/8$. Thus $\mathcal{E}_\ell \leq g_- c_-/2$. The leading coefficient in (AS34) is at least $g_- c_-$, because $-\gamma_0 = |c_0|\sqrt{1+s^2} \geq c_-$. Consequently

$$\frac{1}{2} g_- c_- c_x^2 P(v)^2 \leq -g \cdot T \leq (g_+ \Gamma_{\max} + g_- c_-/2) C_x^2 P(v)^2. \quad (\text{AS37})$$

This proves the two-sided production bound with every remainder retained. It does not assert that $\partial_v |t|^2$ is positive: the original damping term in (AS25) is still present. The exact norm identity

$$|t|^2 = (1 + s_a^2)x^2 + y^2 \quad (\text{AS38})$$

also follows from the actual orthonormal frame; no ideal frame is substituted into this identity.

51.7 7. The derivative bound for the radial transport product

For the actual phase let

$$H_\Phi = pF + p_z G, \quad n_{\Phi,r} = x_0 - v \partial_R H_\Phi, \quad E_{\text{ik}} = \varepsilon v (\partial_T H_\Phi - G \partial_Z H_\Phi) + b n_{\Phi,r}. \quad (\text{AS39})$$

Here p, p_z, x_0 and the label are fixed; v is the original affine pulse coordinate. Let D^I be the ordinary coefficient derivatives in (R, Z, T, H_1, H_2) , not the evaluated physical derivative operator. The actual source gives $|D^J b| \leq C_{b,J} \varepsilon$, and the full phase formula gives $|D^J n_{\Phi,r}| \leq C_{n,J} S_*^{d_J}$ for each fixed J . Their exact product rule and resulting bound are

$$D^I (b n_{\Phi,r}) = \sum_{J \leq I} \binom{I}{J} (D^J b) (D^{I-J} n_{\Phi,r}), \quad |D^I (b n_{\Phi,r})| \leq \varepsilon \sum_{J \leq I} \binom{I}{J} C_{b,J} C_{n,I-J} S_*^{d_{I-J}}. \quad (\text{AS40})$$

For fixed I the sum is finite; since $S_* \geq 1$, it is bounded by $\varepsilon S_*^{\max_{J \leq I} d_{I-J}} \sum_{J \leq I} \binom{I}{J} C_{b,J} C_{n,I-J}$. Already $\partial_R n_{\Phi,r} = -v \partial_R^2 H_\Phi$ may carry the interval-length factor L_s . The bounds do not supply a uniform derivative-free $C_I \varepsilon$ estimate for the entire product. Applying the same full finite product rule to the first term in (AS39), with the retained v and its affine derivatives, gives the source's stated $C_I \varepsilon S_*^{b_I}$ transport-defect bound. Thus the corrected derivative sentence proves exactly the coefficient class used subsequently.

52 Complete downstream closure of the constructed finite states

52.1 1. The exact physical mean-stress domain is attained

Retain

$$0 < h < 1/100, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad \kappa_s = 10^{-5}, \quad Q = 2^{-\ell}, \quad \varepsilon = Q^h, \quad S_* = \ell^2.$$

The active shell is $X_a < X < X_b$, where $X = r^2/(2q)$, with

$$\zeta(X) = \exp\left(-\frac{a_a}{\log^2(X/X_a)} - \frac{a_b}{\log^2(X_b/X)}\right), \quad \delta = \min\{1, \log(X/X_a), \log(X_b/X)\},$$

and the same original positive constants a_a, a_b, X_a, X_b . The mean class used by the physical signed inverse consists of compatible, real, angular-independent, shell-supported coefficients satisfying every fixed coefficient derivative estimate

$$|D^I f| \leq C_I \varepsilon^\alpha S_*^{b_I} \zeta \delta^{-d_I}$$

and smooth zero extension. Its auxiliary-independent subspace is the domain needed for the physical stress inverse. These requirements, not auxiliary independence alone, are indispensable. The proof and the explicit counterexample for an enlarged domain occur under “Physical assembly, chart maps, and the missing domain qualification” (observed lines 9876–9962).

At the beginning of step (ii) of the actual stage cycle, put $B = B_j = 1/2 + \sigma_j$, $C = C_j = 1 + \sigma_j$, with $\sigma_j = 1/5 + j/10$. Step (i) and the full integrated mean identities have established

$$E_\theta, E_z \in \mathcal{M}_{C-\kappa_s}, \quad \int R^2 \overline{E_\theta} dR, \int R \overline{E_z} dR \in \mathcal{S}_{C+1-\kappa_s}.$$

Bars here are normalized auxiliary Haar averages; angular averaging has already been performed when defining the mean residual. For $e = 2$ take $E_e = \overline{E_\theta}$, and for $e = 1$ take $E_e = \overline{E_z}$. The actual physical quantities in step (ii) are

$$\begin{aligned} F_e(r, z, t) &= Q^{-2A-1/2} E_e(R, Z, T), \quad R = r/\sqrt{Q}, \\ b_e(r, z, t) &= q^{-(e+1)/2} \widehat{b}_e(r/\sqrt{q}), \quad \int_0^\infty x^e \widehat{b}_e(x) dx = 1, \\ M_e &= \int_0^\infty r^e F_e(r, z, t) dr, \quad \sigma_e = -r^{-e} \int_0^r (r')^e (F_e(r') - b_e(r') M_e) dr'. \end{aligned}$$

The bump is fixed strictly inside the shell. Every radial integral keeps (z, t) , and hence q , fixed. The change of variable $r = \sqrt{Q}R$, with its original measure, proves

$$m_e := \int_0^\infty R^e E_e(R) dR, \quad M_e = Q^{-2A+e/2} m_e. \quad (\text{GC1})$$

Define the exact chart representative of the unit-moment bump by

$$b_{e,Q}(R) = Q^{(e+1)/2} b_e(\sqrt{Q}R) = \left(\frac{Q}{q}\right)^{(e+1)/2} \widehat{b}_e\left(\sqrt{\frac{Q}{q}} R\right). \quad (\text{GC2})$$

With $x = \sqrt{Q/q} R$, the integral of $R^e b_{e,Q} dR$ equals

$$\left(\frac{Q}{q}\right)^{(e+1)/2} \left(\frac{q}{Q}\right)^{e/2} \left(\frac{q}{Q}\right)^{1/2} \int x^e \widehat{b}_e(x) dx = 1.$$

Substitution in the definition of σ_e gives the exact, unaltered chart stress

$$Q^{2A} \sigma_e = -R^{-e} \int_0^R (R')^e (E_e(R') - b_{e,Q}(R') m_e) dR'. \quad (\text{GC3})$$

Indeed the factors in the first source term are $Q^{-e/2} Q^{e/2} Q^{1/2} Q^{-2A-1/2} = Q^{-2A}$. For the second source term, (GC1) and (GC2) give the same factor. No ratio q/Q or neighboring band scale has been set equal to one.

Let $f_{e,Q} = E_e - b_{e,Q} m_e$. Equations (GC1)–(GC2) prove

$$\int_0^\infty R^e f_{e,Q} dR = m_e - m_e = 0. \quad (\text{GC4})$$

The chart/profile map is smooth with every fixed derivative bounded on the original compact rectangle $X_a \leq X \leq X_b$, $1/2 \leq q/Q \leq 2$, $-1 \leq \eta \leq 1$. Its denominators are positive powers of q/Q , R , and $L = 1 - 2h\eta^2$, all bounded away from zero. The support of $b_{e,Q}$ stays in the original fixed interior profile

interval, where ζ has a positive lower bound. Thus the bump has mean exponent zero with all required weighted derivatives. Since m_e has exponent $C + 1 - \kappa_s$, the product $b_{e,Q} m_e$ has that exponent. Because $0 < \varepsilon \leq 1$, it also has the weaker exponent $C - \kappa_s$. Consequently $f_{e,Q} \in \mathcal{M}_{C-\kappa_s}^{\text{independent}}$.

The actual weighted radial inverse RF12–RF18 was read in full. On an auxiliary-independent source its characteristic torus moment is its ordinary weighted integral. Equation (GC4) therefore makes that entire obstruction zero, not merely its Haar average. In this case the cutoff correction term in RF12 is identically zero and (GC3) is exactly $-T_e f_{e,Q}$. Its lower and upper primitives vanish below and above the shell, respectively.

The weighted estimate uses the same edge exponents. For an inner edge coordinate $s > 0$, differentiation of the input gives a finite bound $C e^{-a_a/s^2} s^{-M}$ times a smooth bounded outer factor. Substitution $u = a_a/s^2$ gives

$$\int_0^\delta s^{-M} e^{-a_a/s^2} ds = \frac{1}{2} a_a^{(1-M)/2} \int_{a_a/\delta^2}^\infty u^{(M-3)/2} e^{-u} du \leq C_{a_a, M} \delta^{3-M} e^{-a_a/\delta^2}.$$

For $u = u_0 + y$, $u_0 \geq 1$, a nonpositive exponent of u is at most its value at u_0 ; a positive exponent is at most $u_0^{(M-3)/2} (1+y)^{(M-3)/2}$. Integrating that bound proves the displayed estimate. The corresponding upper primitive uses the same argument at the outer edge with a_b . A fixed derivative is a finite sum of such integrals and differentiated endpoint factors. The fixed-shift RF16 formula makes the moving support produce no boundary terms. On the remaining compact interior, the positive lower bound for ζ supplies the weighted estimate. Every fixed inverse-distance power multiplying either exponential vanishes with all derivatives at its endpoint. It follows that

$$\Sigma := Q^{2A}(\sigma_2, \sigma_1) \in \mathcal{M}_{C-\kappa_s}^{\text{independent}} \quad (\text{GC5})$$

with precisely the shell support and smooth zero extension required by the corrected signed covariance domain.

The construction is independent of the chosen band chart. The quantities F_e, b_e, M_e, σ_e above are physical functions; on an overlap with scale Q' , applying the same change of variables gives $\Sigma' = Q'^{2A} \sigma = (Q'/Q)^{2A} \Sigma$. Their chart arguments are

$$R' = (Q/Q')^{1/2} R, \quad Z' = (Q/Q')^D Z, \quad T' = (Q/Q') T.$$

Haar averaging agrees under the covering pullbacks because $(J_g^i)^T k = 0$ exactly when $k = 0$. This proves that the averaged residual used to define F_e is itself one compatible physical field. The radial integration and the unit-moment subtraction therefore define that same σ in every chart. This proves the remaining domain-and-compatibility interface of stage (ii).

52.2 2. Covariance inversion and the actual residual gain

The signed inverse uses the original positive amplitudes. Its local map is

$$d\Sigma = H^{-1}(\Sigma/\varepsilon), \quad \delta a_\sigma = (d\Sigma)_\sigma / (2a_\sigma),$$

and sends $\mathcal{M}_\alpha^{\text{independent}}$ to $\mathcal{W}_{\alpha-1/2}$. The proved lower bound for a_σ , including its shell factor, gives

$$|D^I \delta a_\sigma| \leq C_I \varepsilon^{\alpha-1} S_*^{P_I} \sqrt{\zeta} \delta^{-M_I}.$$

The fixed original envelope restores the required $P(v)$ factor. The map never takes a square root of a newly perturbed stress. Under the actual squared partition, one label contributes

$$Q_\beta^{-2A} \eta_\beta^2 \mathcal{B}(W_{0,\beta}, \mathcal{L}_\beta(Q_\beta^{2A} \sigma)) = \eta_\beta^2 \sigma.$$

Every product of distinct simultaneously active labels is zero before and after evaluation, including derivative products: their closed supports are disjoint on the common torus. Summing and using $\sum_\beta \eta_\beta^2 = 1$ gives the exact physical covariance target σ . On the complement of the shell both sides are zero by (GC5). This proves the identity on its full actual domain.

For the complete signed curl increment $s = t_s + r_s$, put $\alpha = C - \kappa_s$, $\rho = 0.68$. The uncurled signed field has exponent $\alpha - 1/2 = B - \kappa_s$, while its curl remainder has exponent $\alpha - \kappa_s = C - 2\kappa_s$. The complete covariance identity is

$$\Delta \mathcal{C} = \Sigma + \mathcal{B}(w - w_0^{\text{tan}}, t_s) + \mathcal{B}(w, r_s) + \mathcal{C}(s). \quad (\text{GC6})$$

The three error exponents before divergence are

$$C + 0.18 - \kappa_s, \quad C + \frac{1}{2} - 2\kappa_s, \quad 2C - 1 - 2\kappa_s = C + \sigma_j - 2\kappa_s.$$

The last expression includes all three terms $\mathcal{C}(t_s) + \mathcal{B}(t_s, r_s) + \mathcal{C}(r_s)$. The weaker radial derivative map costs at most κ_s , so the three gains above C are

$$0.18 - 2\kappa_s = 0.17998, \quad \frac{1}{2} - 3\kappa_s = 0.49997, \quad \sigma_j - 3\kappa_s \geq 0.19997.$$

All exceed 0.17. The uncanceled compact bump in the integrated residual has exponent $C + 1 - \kappa_s$, also higher. These are the exact exponents of the corrected physical target, so the scope qualification in the covariance audit causes no loss of the stage-(ii) conclusion.

After this step the actual temporal inverse uses

$$\begin{aligned} \Delta v &= -c^{-1}N^{-1}E_\theta^\circ, \quad \gamma_d = -c^{-1}N^{-1}E_z^\circ, \quad \Psi = T_1\gamma_d, \\ \Delta\beta &= -\varepsilon\partial_Z\Psi, \quad \Delta\gamma = (D_r + R^{-1})\Psi = \gamma_d - A_1\gamma_d. \end{aligned}$$

The radial obstruction term is retained in $\Delta\gamma$, all new quadratic products, and recomputed pressure. Its flatness follows from RF20–RF22 on the same fixed domain. Zero Haar mean implies both original moments stay zero, but is never used as a false assertion that the full characteristic obstruction vanishes. The remaining non-flat residual gains are the displayed stage-(iii) values $H+1$, $H+1-\kappa_s$, and $H+1-2\kappa_s$, where $H = C - 2\kappa_s$. The five-row correction is auxiliary independent, has identically zero axial projector, and leaves defect exponent

$$H + 0.9 - 2\kappa_s = C + 0.9 - 4\kappa_s.$$

The exact expressions MC28–MC30 and SI9 retain the pressure-source contribution $-\frac{1}{2} \int R^2 \overline{R}_g dR$ in the new J_z . This is necessary for the fifth-row sign $-\int RV\Delta v dR$. It is present in the included proof.

Combining the complete actual substeps yields $B_{j+1} = B_j + 0.1$, $C_{j+1} = C_j + 0.1$. The fixed retained field budgets remain $\mathcal{W}_{0.5}$, $\mathcal{W}_{0.68}$, $\mathcal{M}_{0.9}$, and $\mathcal{M}_{1.9}$, because the smallest signed increment is $B_j - \kappa_s \geq 0.69999 > 0.68$, the temporal increment has $H \geq 1.19998 > 0.9$, and the radial increments have one additional ε . The initial values are actually constructed in SI1–SI10; they are not an additional assumed stage.

52.3 3. The finite stage residual is the physical residual used by summation

The exact evaluation map is

$$(\mathcal{E}F)(r, \theta, z, t) = F(r, \theta, z, t, v_r r^{d_r} + v_t t \bmod \mathbb{Z}^2).$$

It preserves products and intertwines $\partial_t, \partial_r, \partial_z, \partial_\theta$ with $\mathbf{t}, \mathbf{r}, \partial_z, \partial_\theta$. The cylindrical frame derivatives are included in the residual and curl formulas. Iterating the chain rule therefore sends the extended residual identity to the actual Cartesian Navier–Stokes residual. Auxiliary averaging is a separate linear map: the identity $\mathcal{E}F = \mathcal{P}F + \mathcal{E}(F - \mathcal{J}\mathcal{P}F)$ retains the nonzero auxiliary part. No physical residual is replaced by its Haar mean.

The actual coefficient invariant SI12 includes the original supports, zero extension, $\sqrt{\zeta}\delta^{-d_t}P(v)$, every fixed derivative, and finite harmonics at every stage. Products of same-label waves use $P^2 \leq P$; distinct-label products vanish; multiplication by a mean keeps the wave support and envelope. The pulse inverse holds slow and transverse coordinates fixed, so its $P(v)/P(w)$ kernel cancels exactly the source’s $P(w)$. Radial integration keeps q fixed, and its count of boxes adds only a polynomial in S_* . These proved operations establish the stated invariant for the actual finite residual graph.

One fixed threshold SI11 serves every stage. The later harmonic propagators satisfy

$$V_m(v, w) = \exp\left(-(m^2 - 1) \int_w^v d(s) ds\right) V_1(v, w), \quad |m| \geq 1, \quad d > 0.$$

The prefactor has modulus at most one for forward time. Differentiated inhomogeneous ODEs have lower-order parameter derivatives as sources, so fixed derivative estimates change polynomial degrees and constants, not the threshold. The signed covariance map divides only by original amplitudes, and the radial, temporal, pressure, and five-row maps are fixed. This verifies that the physical domain is one positive q_{big} , not an unverified intersection of shrinking stage domains.

At fixed physical derivative order, the exact conversion costs at most

$$\partial_r : \frac{1}{2} + h\kappa_s, \quad \partial_z : \frac{1}{2} - h, \quad \partial_t : 1 + h, \quad \text{frame} : \frac{1}{2}.$$

Phase differentiation is dominated by $s_x = 1/2 + h/2$, $s_t = 1 + 3h/2$, since $k = \lceil \varepsilon^{-1/2} \rceil \leq 2\varepsilon^{-1/2}$. Every fixed coefficient derivative obeys the same coefficient exponent; the additional input derivative counts in an inverse affect only the fixed-stage constant and polynomial S_* power. Thus a finite-stage-dependent derivative graph does not create a stage-dependent physical power loss. With

$$\ell_m = 2A + (m+1)(1+3h/2), \quad g_j = hj/10,$$

the representatives obey (9.17), and the residual conversion has the fixed factor $Q^{-2A-1/2}$. The wave exponent $B_j = 1/2 + \sigma_j$, the mean exponent $C_j = 1 + \sigma_j$, and the compact radial defect all dominate the common retained exponent σ_j . Consequently one obtains exactly (9.18):

$$|\mathcal{R}(u^{[j]}, p^{[j]})|_m \leq C_{j,m} q^{h\sigma_j - K_m} (1 + |\log q|)^{P_{j,m}} + E_{j,m}, \quad |E_{j,m}| \leq C_{j,m,N} q^N, \quad (\text{GC7})$$

where K_m is independent of j . Here the logarithmic factor multiplies the preceding q -term, exactly as in (9.18); the displayed line break is not an additional summand. Every flat contribution comes from the actual base residual, terminal pulse cutoff, radial projector, or a product with one such factor. At fixed order its physical derivative has the form $CQ^{-M}S_*^b e^{-c\ell^2}$, or the arbitrary-order RF22 estimate. The exponent identity $-c\ell^2 + (M+N)\ell \log 2 + 2b \log \ell \rightarrow -\infty$ proves the first kind is $O(q^N)$ for every N ; the flat ideal proof in SI12 handles all products. Thus (GC7) is supplied by the included actual construction, rather than only by an exponent calculation at one label.

52.4 4. The exact representative sum and terminal residual

For $j \geq 1$, use the actual physical representatives

$$A_j = A_j^w + \Psi_j e_\theta, \quad B_j = b_j e_\theta, \quad p_j = p^{[j]} - p^{[j-1]}, \quad \Delta u_j = \text{curl } A_j + B_j.$$

The direct scalar b_j is independent of θ . Let $\chi = 1$ for positive arguments at most $1/2$, $\chi = 0$ for arguments at least one, and retain the actual diagonal scales a_j . The actual cut increment is

$$V_j = \text{curl}(\chi(a_j q) A_j) + \chi(a_j q) B_j = \chi(a_j q) \Delta u_j + a_j \chi'(a_j q) \nabla q \times A_j, \quad (\text{GC8})$$

The exact similarity map gives $\partial_z^a \partial_t^b q = q^{1-aD-b} F_{a,b}(\eta)$, with $F_{a,b}$ smooth on $[-1, 1]$ because $L \geq 1 - 2h > 0$. Every nonzero differentiated cutoff term contains

$$a_j^k \chi^{(k)}(a_j q) \prod_{\nu=1}^k \partial_z^{a_\nu} \partial_t^{b_\nu} q, \quad \sum a_\nu = a, \quad \sum b_\nu = b.$$

On its support $a_j q \in [1/2, 1]$; the product's factor q^k pairs with a_j^k . Therefore the exact derivative loss is $aD + b$, independent of a_j . Leibniz applied to (GC8) supplies a common loss

$$L_m = \ell_{m+1} + m + 1$$

for $|(V_j, \Pi_j)|_m$. The finite scale inequalities at index j give

$$|(V_j, \Pi_j)|_m \leq 2^{-j} q^{g_j/2 - L_m}, \quad m \leq j. \quad (\text{GC9})$$

Their existence is proved by $q^\gamma (1 + |\log q|)^P \rightarrow 0$ for each fixed $\gamma > 0$, imposing finitely many conditions at each j , and $a_j \geq 2a_{j-1}$. These are one selected sequence of scales, not a sequence reselected for each output derivative.

On every compact positive- q set only finitely many summands survive. The sum is therefore a smooth actual physical field. Divergence of each curl is zero. Also $\nabla \chi(a_j q)$ has only an axial component and B_j has only an angular component; hence its divergence is $r^{-1} \partial_\theta (\chi(a_j q) b_j) = 0$. Smooth Cartesian extension proves the identity at the axis.

For fixed $J \geq m$ and $q < (2a_J)^{-1}$, the first J cutoffs and all their derivatives equal the uncut prefix. Thus

$$e_J := (u_{\text{loc}}, p_{\text{loc}}) - (u^{[J]}, p^{[J]}), \quad |e_J|_m \leq 2^{-J} q^{g_{J+1}/2 - L_m}. \quad (\text{GC10})$$

Write $e_J = (v_J, \pi_J)$. The exact nonlinear residual difference is

$$\partial_t v_J - \Delta v_J + \nabla \pi_J + (u^{[J]} \cdot \nabla) v_J + (v_J \cdot \nabla) u^{[J]} + (v_J \cdot \nabla) v_J. \quad (\text{GC11})$$

Its order- m bound is $C_m |e_J|_{m+2} (1 + |u^{[J]}|_{m+2} + |e_J|_{m+2})$. For given m, N , choose one finite $J \geq m+2$ so that

$$g_{J+1}/2 - L_{m+2} > N + K_{m+2} + 2, \quad h\sigma_J - K_m > N + 1.$$

Both left sides tend to infinity with J . Keep that stage fixed thereafter. Its finite logarithmic powers are $O(q^{-1})$. Equations (GC7), (GC10), and (GC11) then give

$$|\mathcal{R}(u_{\text{loc}}, p_{\text{loc}})|_m = O(q^N). \quad (\text{GC12})$$

Only the flat estimate of that one selected stage is used. No summation of its stage residuals or uniform bound in j on the flat constants is required.

52.5 5. Representatives, endpoint jets, localization, and the classical equation

The actual base potential is Se_θ/r , where

$$S_n = q^{1-A+2nh} \int_0^X U_n(X', \eta) dX'.$$

The zero full radial moment of each U_n makes this potential zero beyond its common axial support. Near the axis it is $a(r^2, z, t)(-x_2, x_1, 0)$, hence smooth Cartesian. Wave potentials are annular; mean potentials vanish outside the radial source and the fixed interior subtraction region. The diagonal stage sum is locally finite on $q \geq c > 0$. Each of these actual representatives, not just its curl, has every fixed one-sided derivative on $-1 \leq \eta \leq 1$, after the proved base/pulse regularity and the fixed linear mean maps. Coordinate conversion has no divisor $1 - \eta^2$. Bounding one additional derivative and integrating it gives uniform Cauchy convergence of each derivative as $t \uparrow 1$ on such compact sets. Spatial segment integration identifies the limiting spatial derivatives; time integration identifies successive time jets.

For $r > 0$ and small q , the exact exterior is

$$K(r, \tau) = r^{-1-2h} H_{\text{ext}}(\tau/r^2), \quad H_{\text{ext}}(s) = 2^A c_\infty H(4s),$$

$$H(Z) = \Gamma(1+h)^{-1} \int_0^\infty e^{-v} v^h (1+Zv)^{-h} dv, \quad p_{\text{ext}} = - \int_r^\infty K(\rho, \tau)^2 d\rho/\rho.$$

The m -th derivative integrand has factor $(-1)^m (h)_m v^m (1+Zv)^{-m}$, so its common integrable bound gives

$$\sup_{s \geq 0} |H_{\text{ext}}^{(m)}(s)| \leq 2^A c_\infty 4^m (h)_m (1+h)_m.$$

The full pressure product rule is bounded by $C_j \rho^{-3-4h-2j}$, integrable for positive lower radius. It permits all time derivatives through the pressure integral, uniformly including $\tau = 0$. The lower endpoint gives $\partial_r p_{\text{ext}} = K^2/r$, cancelling the radial convection $-K^2/r$; the exact heat equation cancels the angular residual; the axial residual is zero. Thus the exact representative endpoint coverage includes $z = 0, r > 0$, where $q \rightarrow 0$, as well as $z \neq 0$, where $q \geq |z|^{1/D} > 0$.

With the actual compact cutoffs $c = \chi_x \chi_t$, localization is

$$u = \text{curl}(c\mathcal{A}) + cBe_\theta = cu_{\text{loc}} + \nabla c \times \mathcal{A}, \quad p = cp_{\text{loc}}. \quad (\text{GC13})$$

The support lies strictly inside the positive local-domain margin. Define the actual force by

$$f = \partial_t u + (u \cdot \nabla) u - \Delta u + \nabla p. \quad (\text{GC14})$$

Every cutoff derivative and quadratic interaction in (GC13) is part of this identity. Near the singular point $c = 1$; (GC12) applies in the bounded profile region and the exterior residual is exactly zero. The exact elementary bound $q \leq C_0(\tau + |z|^{1/D})$ therefore makes every force derivative tend to zero at $(0, 1)$. On the rest of the fixed compact support, the proved one-sided representative bounds give uniform limits. Hence every terminal jet $F_j = \lim_{t \uparrow 1} \partial_t^j f$ belongs to $C_c^\infty(K; \mathbb{R}^3)$, and mixed spatial derivatives are the limits of the corresponding derivatives of f .

The actual extension is

$$f(x, 1 + \sigma) = \sum_{j \geq 0} \chi_0(b_j \sigma) \frac{\sigma^j}{j!} F_j(x), \quad \sigma \geq 0.$$

The exact differentiated cutoff-monomial estimate is

$$\left\| \partial_x^\alpha \partial_\sigma^m \left(\chi_0(b\sigma) \frac{\sigma^j}{j!} F_j \right) \right\|_\infty \leq \left(\sum_{\substack{0 \leq a \leq m \\ m-a \leq j}} \binom{m}{a} \frac{\|\chi_0^{(a)}\|_\infty}{(j-m+a)!} \right) b^{m-j} \|F_j\|_{C^{|\alpha|}}.$$

Choosing the least increasing integers satisfying the finite bounds E.11 makes every differentiated tail uniformly summable. At $\sigma = 0$ the m -th derivative has exactly the $j = m$ contribution F_m . This proves joint smooth gluing and $f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3)$. The trace theorem T.1–T.9 proves the exact relation to any other extension with these jets: the difference is in the closed kernel of the trace, smooth and flat at the terminal slice. It does not assert equality of the later forces or a positive gap in their supports. The continuous linear isomorphism is the quotient map, and the included scaling argument correctly rules out a continuous linear section of the full jet space.

The compact support before time one justifies the exact energy identity

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_2^2 + \|\nabla u(t)\|_2^2 = \langle f(t), u(t) \rangle.$$

Writing $F(t) = \int_0^t \|f(s)\|_2 ds$ and dividing first by $(\|u\|_2^2 + \delta^2)^{1/2}$, then letting $\delta \downarrow 0$, gives $\|u(t)\|_2 \leq F(t)$. Integrating the original identity gives

$$\|u(t)\|_2^2 + 2 \int_0^t \|\nabla u(s)\|_2^2 ds \leq F(t)^2.$$

Thus both energy and total dissipation are bounded up to time one. The whole-space comparison proof was read through E.22: its harmonic pressure-gradient ambiguity is removed in H^{-3} , its localized commutator kernel has $L^{4/3}$ -norm $O(R^{-3/4})$, and its pressure and convection fluxes are absorbed into localized dissipation. The final estimate $E_R(t) \leq C_T/R$ proves equality with every smooth finite-energy competitor on $[0, T]$, $T < 1$, without an added pressure condition at infinity.

The actual unchanged inner path has

$$x_\tau = (\sqrt{2X_{\text{in}}\tau}, 0, 0), \quad u_\theta(x_\tau, 1 - \tau) = \tau^{-A}(e_0 + O(\tau^{2h})), \quad e_0 > 0.$$

Both localization cutoffs are one there and all annular corrections vanish. The positive-order base cutoff sum is included in the error. For small τ , the bracket is at least $e_0/2$, proving divergence along points approaching the origin. The comparison estimate would force any global smooth finite-energy competitor to have these values, contradicting its continuity on a compact neighborhood of $(0, 1)$.

The viscosity map E.24 and parabolic map E.26 were checked on the complete triple. In particular $\mathcal{T}_\nu = \mathcal{S}_{\sqrt{\nu}} \circ \mathcal{A}_\nu$ has prefactors (ν, ν^2, ν^2) , time argument νt , and energy factor ν^2 . Periodization E.29–E.30 has disjoint compact translated supports, so every convection cross term is zero. It preserves the original unit periods of velocity, pressure, and force and transfers the actual divergent path with factor $\lambda\sqrt{\nu}$. No relation between independently chosen analytical and formal representative sequences is inferred from these exact adapters.

52.6 6. Audited conclusion and remaining integration record

The corrected physical signed-stress domain is satisfied by the actual stage-(ii) stress; (GC1)–(GC5) are a complete proof of that interface. The downstream finite-stage, physical evaluation, shrinking-cutoff, representative endpoint, compact force extension, comparison, and periodization interfaces close with the original objects. No still-unproved new analytic assertion was found in those interfaces. Standard analytical background results explicitly used by the reader (the inverse-function theorem, the classical three-dimensional Sobolev inequality, and the classical Riesz-transform bounds) remain visible uses of established theorems; this statement does not claim that their foundational proofs are included in the 162-page reader.

The independent base/pulse conclusion and any concrete wording repairs are to be read in `base_pulse/`; the final combined audit records that result separately. The analytical proof-map assessment is not a Lean kernel certificate, a validation of every source revision, or a theorem inferred from symbolic replay counts.

53 Complete ambient projection and both directions of the coordinate equation

Retain the actual nonzero real normal n_Φ , the actual tangential frame K_a, N_a , the original reference coordinate $s(v)$, and the same matrix $\mathcal{A}_\Phi$ and damping d from the full pulse audit. With

$$U = [e_r - s_a K_a, N_a], \quad M = \begin{pmatrix} 1 & 1 \\ c_0 \sqrt{1+s^2} & -c_0 \sqrt{1+s^2} \end{pmatrix}, \quad B = UM, \quad B^\ell = M^{-1}(e_r^T; N_a^T),$$

the determinant $-2c_0\sqrt{1+s^2}$ is nonzero by the original compact cone. The full equation and its pressure have already been proved with the original $m \in \mathbb{Z} \setminus \{0\}$, $k \geq 1$, and right side $-f_m$. The following calculation proves the ambient map and the converse of the coordinate transformation without extending a right-inverse identity beyond its exact domain.

For clarity the ambient map also has an exact formula. Regard K_a as the three-vector with zero radial component, and let $W_a = (e_r^T; N_a^T)$. Direct multiplication gives $W_a U = I_2$. For any $w = w_r e_r + w_K K_a + w_N N_a \in \mathbb{C}^3$,

$$BB^\ell w = UW_a w = w_r(e_r - s_a K_a) + w_N N_a = w - K_a(w_K + s_a w_r).$$

Since $n_\Phi = |n_{\Phi, \tan}|(s_a e_r + K_a)$, this proves

$$B^\ell B = I_2, \quad BB^\ell = I_3 - \frac{K_a n_\Phi^T}{|n_{\Phi, \tan}|}.$$

Thus BB^ℓ is the projection onto $n_\Phi^\perp$ along $\text{span}_{\mathbb{C}}\{K_a\}$. Its restriction to $n_\Phi^\perp$ is the identity; its action on K_a is zero. In particular the inverse assertion is not made on all of $\mathbb{C}^3$.

The moving-plane differential equation has the exact coordinate form

$$z'_m = B^\ell(\mathcal{A}_\Phi B - B')z_m - m^2 dz_m - B^\ell \text{proj}_{n_\Phi^\perp} f_m.$$

For its converse, differentiate $n_\Phi^T B = 0$ to obtain $n_\Phi^T B' = -(n'_\Phi)^T B$. The definition of $\mathcal{A}_\Phi$ gives $n_\Phi^T \mathcal{A}_\Phi B = -(n'_\Phi)^T B$. Therefore each column of $\mathcal{A}_\Phi B - B'$ lies in $n_\Phi^\perp$. Both Bz_m and the projected force lie there as well. Multiplying the coordinate equation by B and using $BB^\ell = I$ on these three terms yields

$$Bz'_m = (\mathcal{A}_\Phi B - B')z_m - m^2 dBz_m - \text{proj}_{n_\Phi^\perp} f_m.$$

Adding $B'z_m$ proves the projected equation for $t_m = Bz_m$, and $n_\Phi^T t_m = 0$ holds by construction. The displayed pressure then recovers the full equation with its actual normal forcing. Conversely, a constrained solution has $z_m = B^\ell t_m$ and $t_m = Bz_m$. Differentiating the latter identity and multiplying the projected equation by B^ℓ gives exactly the coordinate equation above. Both inverse compositions, including their initial data and the unique pressure, are therefore proved.

54 Combined analytic interface disposition

The original leading profiles, positive-order background, auxiliary geometry, pulse, covariance, correction cycle, representative sum, and force extension are the constructed objects retained in the accompanying full derivations. The original source is the 165-page edition with SHA-256 8c8a94ad9ac824c8b605b9827cadf7beaca48bd10b380de3cfc872a2c37afa81. The 166-page revision is a distinct source. Neither the interfaces proved here nor the preservation of an earlier reader identifies those editions.

The full base audit proves the common-radius solution map from finalized lower coefficients to each positive order, its exact five moments and common compact support, the integrated stress with its original edge weight, the Borel potential cutoff with its radial derivative term, the exact chart powers, and the full reserved interval. The full pulse audit proves the phase and transport formulas, pressure sign, the constrained evolution and harmonic propagator, all fixed derivative classes and supports, the homogeneous lower bound, and the covariance integral with its angular half and original Jacobian. The full downstream audit proves the exact physical stress-domain morphism, complete signed-curl errors and stage gains, physical residual evaluation, the one-selected-stage realization comparison, endpoint representative and force jets, and the original comparison, viscosity and periodization maps. These derivations are reproduced in full, rather than replaced by this disposition.

The literal frozen-shear production equality has been corrected in every affected live component copy. The retained term is $-(g - g_0) \cdot T$, which may remain nonzero even at the representative. The appended complete actual-shear calculation establishes the first-radial-jet coordinate isomorphism and both inverses, the full force-pressure bijection using the same actual phase, and the explicit positivity bounds from the actual homogeneous pulse polarization. Positive shear production is not a claim that the damped energy derivative is positive. The exact polynomial coefficient-product estimate for $bn_{\Phi,r}$ is propagated at the same time. The completed coordinate converse uses the explicitly computed projection $I_3 - K_a n_{\Phi}^T / |n_{\Phi,\tan}|$, whose kernel is the original tangential direction and whose restriction to the constrained plane is the identity.

The first-jet and actual-shear force-pressure derivations (AS1)-(AS40) are reproduced with attribution to the independent source-comparison lane's note, *Actual shear, frozen leading shear, and the exact pulse force-pressure correspondence*. The present continuation checked those formulas, propagated their correction, and added the full ambient-projection and coordinate converse. Independent reviews identify their exact source hashes. This attribution distinguishes mathematical derivation from later source propagation.

No additional unresolved analytical assertion was identified in the bounded interfaces actually audited. The concrete corrections above are proved and integrated into the live components. This disposition does not turn a source ledger, symbolic execution count, or earlier publication into a proof. In particular, the inverse-function theorem, the classical three-dimensional Sobolev inequality, and the classical Riesz-transform bounds remain named established analytical inputs of the cumulative construction; their foundational proofs are not supplied by this audit. No Lean certificate is asserted, and this standalone interface document is not a substitute for the complete original-component proofs included in the cumulative reader.

The published earlier edition remains immutable. Its historical sentence is an identified correction for a later edition, not a retroactive edit. The durable source-change receipt records the six live component repairs, their exact preimages and postimages, and the unchanged protected objects. Cumulative integration and verification of the new edition remain distinct recorded tasks until their own build and review receipts are complete.